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Bilinear Maps and Tensor Products in Operator Theory

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Bilinear Maps and Tensor Products in Operator Theory

Springer

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To Jessica and Alan

Istigkeit—wasn't that the word Meister Eckhart liked to use? 'Is-ness.' The Being of Platonic philosophy—except that Plato seems to have made the enormous, the grotesque mistake of separating Being from becoming, and identifying it with the mathematical abstraction of the Idea. He could never, poor fellow, have seen a bunch of flowers shining with their own inner light and all but quivering under the pressure of the significance with which they were charged; could never have perceived that what rose and iris and carnation so intensely signified was nothing more, and nothing less, than what they were—a transience that was yet eternal life, a perpetual perishing that was at the same time pure Being, a bundle of minute, unique particulars in which, by some unspeakable and yet self-evident paradox, was to be seen the divine source of all existence.

Aldous Huxley's *The Doors of Perception*

Preface

This book, intended as a first course in bilinear maps and tensor products, introduces tensor products in Banach spaces to graduate students, helping them to learn the subject from the beginning. Tensor products, particularly in infinite-dimensional normed spaces, are heavily based on bilinear maps. The book brings these issues together by using bilinear maps as an auxiliary, yet fundamental, tool for accomplishing a consistent, useful, and straightforward theory of tensor products. Bearing in mind that graduate students typically find this subject complex and challenging, I have presented the material in ways that should make grasping the concepts simpler and easier.

Nine chapters, each consisting of four sections, make up the book. Although the text is not formally split into parts, it has been naturally divided into two implicit portions. The first offers a purely algebraic introduction to bilinear maps and tensor products of linear spaces, while the second brings these concepts to the realm of normed spaces and therefore to Banach spaces, with a few ventures into Hilbert spaces. In order to keep the prerequisites as modest as possible, there are two introductory chapters, one on linear spaces (Chapter 1) and another on normed spaces (Chapter 5), summarizing to a large extent the background material required for a thorough understanding and efficient learning strategies—and also establishing a fixed scheme of notation and terminology. However, an introductory course in functional analysis will naturally be assumed.

Chapter 1 supplies those essential algebraic notions necessary in subsequent chapters, and only those. These are all related to linear spaces only, mainly focusing on extension of linear transformations and on quotient spaces of a linear space modulo a linear manifold. Algebraic aspects of bilinear maps are presented in Chapter 2. These first two chapters lay the groundwork required to introduce an axiomatic theory of algebraic tensor products of linear spaces, which is done in the subsequent chapters. Chapter 3 introduces a unified approach to algebraic tensor products, where the universal property is taken as an axiomatic starting point, rather than as a theorem for a specific

realization. This leads to an abstract notion of algebraic tensor products of linear spaces. Concrete realizations of tensor product spaces, as interpretations of the axiomatic formulation, are considered in Chapter 4. Two approaches to offer such interpretations are investigated, namely, the quotient space approach and the linear maps of bilinear maps approach, and in these approaches, examples and embeddings of tensor product spaces are exhibited. These first four chapters on algebraic concepts comprise the first part of the book.

Chapter 5 contains the essential normed-space theorems required in the book. It may be thought of as the normed-space counterpart to the linear-space outline of Chapter 1. Chapter 5 deals mainly with extension of bounded linear transformations and with normed quotient spaces, whose purpose is to put together most of the results (and only those results) that will be necessary farther along in the book. Normed-space aspects of bilinear maps are treated in Chapter 6, where continuous bilinear maps are considered, emphasizing extension theorems for bounded bilinear maps. Chapters 5 and 6 together allow us to advance an axiomatic theory of tensor products of Banach spaces.

Chapter 7 contains a detailed account of basic results on tensor products of Banach spaces, featuring suitable norms that equip a tensor product space of normed spaces. By suitable norms, we mean what are commonly referred to as *reasonable crossnorms*. The chapter concentrates on the notions of two extreme instances of reasonable crossnorms, viz., the *injective* and the *projective* norms, leading to the definitions of injective and projective tensor products.

Chapter 8 focuses on the central idea of a uniform crossnorm, which is essentially a reasonable crossnorm that makes the tensor product of bounded linear operators a bounded linear operator itself, satisfying an operator-norm-like property. A feasible theory of tensor products of operators is based on the concept of uniform crossnorm. This is dealt with in Chapter 9, and includes the Hilbert-space tensor product of Hilbert spaces, and the associated Hilbert-space tensor product operator made up of Hilbert-space operators. Some aspects of their spectral properties are also discussed in this final chapter.

Each chapter has a common closing section entitled *Additional Propositions*. It consists of (i) auxiliary results required in the sequel, and (ii) additional results extending the subject beyond the bounds of the main text. These are followed by a set of *Notes*, where each proposition is briefly discussed and references are provided indicating proofs for all of them. Such Additional Propositions also play the role of a section of proposed problems, and their respective Notes can be viewed as hints for solving them. This section is followed by a list of *Suggested Readings* that offer alternate approaches, different proofs, and further questions related to the subject of the chapter.

The contents of the book have been devised to be covered in a one-semester graduate course (but as usual the pace of lectures will vary according to specific needs). Since the prerequisites are kept to a minimum, the book is rather self-contained, and so, while it is directly addressed to mathematicians, it

may be accessible to a wider audience of students in mathematics, statistics, economics, engineering, and physics. It may also be a useful resource for mathematicians working in areas that deal with tensor products, as well as for scientists wishing to apply tensor products to their fields. In order to attend to a continual request from students, many proofs here are given in great detail, and I do hope this will not be overly boring for the experienced mathematician. I believe complete and detailed proofs, some even emerging from first principles, will help students who are having their first contact with the subject; and this also includes occasional discussions on delicate points, trying to explain some usually hidden features. The logical dependence of the various sections and chapters is linear. An updated bibliography is given at the end of the book, listing only references cited within the text, which contains most of the classics and also recent texts reflecting some of the present-day techniques.

That being said, let me also say that the book was not designed to provide a comprehensive treatment on all aspects of tensor products of Banach spaces; much less is it intended to cover the celebrated Grothendieck's theory. On the contrary, as already mentioned, it is a book for a first course, and it intends to open the doors for further readings. For instance, it may open the way towards a reading of the excellent texts by Ryan [83], Diestel, Fourie and Swart [17], and Defant and Floret [15], perhaps in this order, whose works have been crucial for compiling the present text. Along these lines, it is equally fair to register here that I benefited from the help of highly qualified referees who suggested significant improvements for the final version of the book, and I am really grateful to them. Thanks are due to the copyeditor Brian Treadway as well, who did an admirable job. I also wish to thank Elizabeth Loew from Springer New York for a pleasant and lasting partnership.

All in all, the book tries to take the students by the hand to the doors of perception, so they can push the doors open. I wish them a nice trip.

Rio de Janeiro
April 2023

Carlos S. Kubrusly

Contents

Preface . . . . . **ix**

1 Linear-Space Results **1**

1.1 Notation, Terminology and Definitions 1
1.2 Extension of Linear Transformations 7
1.3 Quotient Space 9
1.4 Additional Propositions 14

2 Bilinear Maps: Algebraic Aspects **19**

2.1 The Linear Space of Bilinear Maps 19
2.2 Extension of Bilinear Maps 23
2.3 Identification with Linear Transformations 26
2.4 Additional Propositions 28

3 Algebraic Tensor Product **35**

3.1 Tensor Product of Linear Spaces 35
3.2 Further Properties of Tensor Product Spaces 39
3.3 Tensor Product of Linear Transformations 44
3.4 Additional Propositions 52

4 Interpretations **57**

4.1 Interpretation via Quotient Space 57
4.2 Interpretation via Linear Maps of Bilinear Maps 62
4.3 Variants of the Linear-Bilinear Approach 65
4.4 Additional Propositions 72

5 Normed-Space Results 79
5.1 Notation, Terminology and Definitions 79
5.2 Extension of Bounded Linear Transformations 82
5.3 Normed Quotient Space 93
5.4 Additional Propositions 101

6 Bounded Bilinear Maps 113
6.1 Boundedness and Continuity 113
6.2 Identification with Bounded Linear Transformations 117
6.3 Extension of Bounded Bilinear Maps 119
6.4 Additional Propositions 128

7 Norms on Tensor Products 135
7.1 Reasonable Crossnorms 135
7.2 Projective Tensor Product 146
7.3 Injective Tensor Product 159
7.4 Additional Propositions 167

8 Operator Norms 179
8.1 Uniform Crossnorms 179
8.2 Tensor Norms 185
8.3 Dual Norms 192
8.4 Additional Propositions 199

9 Tensor Product Operators 203
9.1 Tensor Product of Bounded Linear Transformations 203
9.2 A Hilbert-Space Setting 209
9.3 Some Spectral Properties 217
9.4 Additional Propositions 228

References 239

List of Symbols 245

Index 249



1

Linear-Space Results

The chapter outlines essential algebraic notions required in the book. These are all related to linear spaces — in particular, to linear transformations and to quotient spaces of a linear space modulo a linear manifold. The purpose of such an introductory chapter is to put together only those results necessary in the forthcoming chapters. Algebraic aspects of bilinear maps will be discussed in Chapter 2. These first two chapters enable us to advance an axiomatic theory of algebraic tensor products of linear spaces in Chapter 3.

1.1 Notation, Terminology and Definitions

This section summarizes the fundamental algebraic concepts, establishes the notation and terminology, and sets forth the pertinent definitions used in the text. The term *algebraic concepts* essentially means concepts related to linear space. The initial entries of this section focus on set-theoretic notions, mainly on the basic attributes of functions between abstract sets. The subsequent entries are all related to linear algebra: linear spaces and linear transformations.

Image and range. The *range* of a function $F : X \to Y$ is the subset of the *codomain* Y of F consisting of all *images* $F(x)$ for every $x \in X$, that is, $\text{range}(F) = \{y \in Y : y = F(x) \text{ for some } x \in X\} = F(X)$.

Injective, surjective and invertible. A function $F : X \to Y$ is *surjective* if $F(X) = Y$ and *injective* (or *one-to-one*) if $x_1 \neq x_2$ in X implies $F(x_1) \neq F(x_2)$ in Y, and it is *invertible* (or *bijective*) if it is injective and surjective.

Extension and restriction. Let S be a subset of X. An *extension* of a function $F : S \to Y$ over a larger *domain* X is another function $G : X \to Y$ whose *restriction* to the smaller *domain* S coincides with the original function, that is, $G|_S = F$, which means $G(s) = F(s)$ for every $s \in S$, and so $\text{range}(F) = G(S)$.

Composition. We use both notations HG or $H \circ G$ for *composition* of two functions H and G, say $H(G(x))$ for every $x \in X$ where the sets X, Y, Z are

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such that $G:X \to Y$ and $H:Y \to Z$. If F is a function of X into itself, then $F^n : X \to X$ stands for the composition of $F:X \to X$ with itself n times.

Idempotent and involution. A function $F:X \to X$ of a set X into itself is *idempotent* if $F^2 = F$ (i.e., $F(F(x)) = F(x)$ for every $x \in X$). Hence the range of an idempotent function F is precisely the set of all its fixed points. In fact, $F = F^2$ if and only if $F(X) = \{x \in X : F(x) = x\}$. Thus a function is idempotent if and only if it acts as the identity function I on its range. An *involution* is a function $F:X \to X$ such that $F^2 = I$.

Cartesian product. Let $\varnothing$ be the empty set. The *Cartesian product* $\prod_{\gamma\in\Gamma} X_\gamma$ of an indexed family $\{X_\gamma\}_{\gamma\in\Gamma}$ of sets is the set consisting of all indexed families $\{x_\gamma\}_{\gamma\in\Gamma}$ such that $x_\gamma \in X_\gamma$ for each $\gamma \in \Gamma \neq \varnothing$. In particular, if $X_\gamma = X$ for all $\gamma \in \Gamma$, then $\prod_{\gamma\in\Gamma} X = X^\Gamma$, where X^Γ is the collection of all functions of Γ to X. More particularly, if $I_n = \{i \in \mathbb{N} : i \le n\} \subseteq \mathbb{N}$ for some $n \in \mathbb{N}$ ($\mathbb{N}$ is the set of positive integers), then write X^n for X^{I_n} so that the Cartesian product of n copies of an arbitrary set X is written as $X^n = X^{I_n} = \prod_{i\in I_n} X = \prod_{i=1}^n X$. The Cartesian product of a pair of sets (X_1, X_2) is denoted by $X_1 \times X_2 = \prod_{i=1}^2 X_i$.

Throughout the book $\mathbb{F}$ denotes either the complex field $\mathbb{C}$ or the real field $\mathbb{R}$. Let $(\mathcal{X}, +, \cdot, \mathbb{F})$ be a *linear space*, where $\mathcal{X}$ is an underlying set upon which a binary operation $(\cdot + \cdot) : \mathcal{X} \times \mathcal{X} \to \mathcal{X}$, called *vector addition*, and a function $(\cdot \cdot \cdot) : \mathbb{F} \times \mathcal{X} \to \mathcal{X}$, called *scalar multiplication*, are defined. In general, we refer to a linear space by writing simply $\mathcal{X}$, and the scalar multiplication symbol $\cdot$ will always be omitted. The rest of this section is related to notation, terminology and definitions of basic linear-space concepts that will be required later.

Linear manifold. A *linear manifold* $\mathcal{M}$ of a linear space $\mathcal{X}$ over $\mathbb{F}$ is a subset of $\mathcal{X}$ which is itself a linear space over $\mathbb{F}$, with operations inherited from $\mathcal{X}$. Linear manifolds are also called *linear subspaces*. Given a nonempty set S, if $\mathcal{Y}$ is a linear space over $\mathbb{F}$, then so is the collection of all $\mathcal{Y}$-valued functions on S, denoted by $\mathcal{Y}^S$. In particular, $\mathbb{F}^S$ is the linear space of all $\mathbb{F}$-valued functions on S. Let $\mathcal{L}at(\mathcal{X})$ denote the *lattice of all linear manifolds* of a linear space $\mathcal{X}$.

Hamel basis. Let S be a nonempty subset of a linear space $\mathcal{X}$. The *span* of S, denoted by $\operatorname{span} S$, is the smallest linear manifold of $\mathcal{X}$ including S (i.e., the intersection of all linear manifolds of $\mathcal{X}$ that include S), which coincides with the set of all (finite) linear combinations of vectors in S. If $\mathcal{M} = \operatorname{span} S$, then $\mathcal{M}$ is said to be spanned by the set S, and the set S is said to span the linear manifold $\mathcal{M}$. A *Hamel basis* B for a linear space $\mathcal{X}$ is a linearly independent set spanning $\mathcal{X}$ (i.e., $\operatorname{span} B = \mathcal{X}$). Every linear space has a Hamel basis, the cardinality of all Hamel bases for $\mathcal{X}$ is an invariant, and this constant cardinality is the *dimension* of $\mathcal{X}$; notation: $\dim \mathcal{X}$. If $\dim \mathcal{X} = n$ for some $n \in \mathbb{N}_0$ ($\mathbb{N}_0$ is the set of nonnegative integers), then the linear space $\mathcal{X}$ is *finite-dimensional*; otherwise it is *infinite-dimensional* (cf. Proposition 1.B in Section 1.4).

Linear transformation. Let $\mathcal{X}$ and $\mathcal{Y}$ be linear spaces over the same field $\mathbb{F}$. A linear transformation $L : \mathcal{X} \to \mathcal{Y}$ is an additive and homogeneous function.

Let $\mathcal{L}[\mathcal{X},\mathcal{Y}]$ be the linear space over $\mathbb{F}$ of all linear transformations from $\mathcal{X}$ to $\mathcal{Y}$. If $\mathcal{Y}=\mathcal{X}$, write $\mathcal{L}[\mathcal{X}]=\mathcal{L}[\mathcal{X},\mathcal{X}]$, the algebra of all linear transformations of $\mathcal{X}$ into itself. The *range* (or *image space*) and *kernel* (or *null space*) of L in $\mathcal{L}[\mathcal{X},\mathcal{Y}]$ are denoted by $\mathcal{R}(L)=\text{range}(L)=L(\mathcal{X})=\{y\in\mathcal{Y}\colon y=Lx \text{ for } x\in\mathcal{X}\}$, a linear manifold of $\mathcal{Y}$, and $\mathcal{N}(L)=\text{kernel}(L)=L^{-1}(\{0\})=\{x\in\mathcal{X}\colon Lx=0\}$, a linear manifold of $\mathcal{X}$, where $\{0\}$ is the linear space containing the *origin* only. The *null transformation* is denoted by $O\colon\mathcal{X}\to\mathcal{Y}$, and the *identity transformation* of $\mathcal{X}$ onto itself by $I\colon\mathcal{X}\to\mathcal{X}$. The restriction $L|_{\mathcal{M}}$ of a linear transformation L to a linear manifold $\mathcal{M}$ of $\mathcal{X}$ is a linear transformation in $\mathcal{L}[\mathcal{M},\mathcal{Y}]$.

Linear functionals, algebraic dual and algebraic adjoint. For the particular case of $\mathcal{Y}=\mathbb{F}$ the elements of $\mathcal{L}[\mathcal{X},\mathbb{F}]$ are referred to as *linear functionals* or *linear forms*, and the linear space $\mathcal{L}[\mathcal{X},\mathbb{F}]$ of all linear functionals on $\mathcal{X}$ is the *algebraic dual* of $\mathcal{X}$, denoted by $\mathcal{X}^\sharp$ (i.e., $\mathcal{X}^\sharp=\mathcal{L}[\mathcal{X},\mathbb{F}]$). The value $f(x)$ of $f\in\mathcal{X}^\sharp$ at $x\in\mathcal{X}$ is also denoted by $\langle x\,;f\rangle$. For every linear transformation $L\in\mathcal{L}[\mathcal{X},\mathcal{Y}]$ consider the linear transformation $L^\sharp\in\mathcal{L}[\mathcal{Y}^\sharp,\mathcal{X}^\sharp]$ defined by $L^\sharp g=gL\in\mathcal{X}^\sharp$ for every $g\in\mathcal{Y}^\sharp$ (i.e., $(L^\sharp g)(x)=g(Lx)\in\mathbb{F}$ for every $g\in\mathcal{Y}^\sharp$ and every $x\in\mathcal{X}$ — this is also written as $\langle x\,;L^\sharp g\rangle=\langle Lx\,;g\rangle$). The linear transformation $L^\sharp\in\mathcal{L}[\mathcal{Y}^\sharp,\mathcal{X}^\sharp]$ is the *algebraic adjoint* of $L\in\mathcal{L}[\mathcal{X},\mathcal{Y}]$.

Projections. A *projection* $E:\mathcal{X}\to\mathcal{X}$ is an idempotent linear transformation of a linear space $\mathcal{X}$ into itself. If E is a projection then so is $I-E$ (this is the *complementary projection* of E), where I stands for the identity on $\mathcal{X}$. Their range and kernel are related by $\mathcal{R}(E)=\mathcal{N}(I-E)$ and $\mathcal{N}(E)=\mathcal{R}(I-E)$.

Isomorphisms. A linear transformation $L\in\mathcal{L}[\mathcal{X},\mathcal{Y}]$ is injective (i.e., it is one-to-one) if and only if $\mathcal{N}(L)=\{0\}$. By an *isomorphism* (or an *algebraic isomorphism*, or a *linear-space isomorphism*) we mean an invertible linear transformation between linear spaces. (The inverse of a linear transformation is linear, so the inverse of an isomorphism is an isomorphism.) We stick to the above terminology although it is not unique. Alternate jargons are:

(a) linear transformation = *homomorphism* (or simply *morphism*),

(b) linear transformation of $\mathcal{X}$ into itself = *endomorphism*,

(c) injective linear transformation = *monomorphism*,

(d) surjective linear transformation = *epimorphism*,

(e) bijective linear transformation = *isomorphism* (same as above),

(f) invertible linear transformation of $\mathcal{X}$ into itself = *automorphism*.

Isomorphic linear spaces. Two linear spaces $\mathcal{X}$ and $\mathcal{Y}$ (over the same field) are *isomorphic* if there is an isomorphism of one onto the other. In this case (i.e., if $\mathcal{X}$ and $\mathcal{Y}$ are isomorphic) we write $\mathcal{X}\cong\mathcal{Y}$ — isomorphism is an *equivalence relation*: reflexive, transitive, and symmetric. Two linear spaces are isomorphic if and only if they have the same dimension.

Isomorphic equivalence — Isomorphic linear transformations. Let $\mathcal{X},\widetilde{\mathcal{X}},\mathcal{Y},\widetilde{\mathcal{Y}}$ be linear spaces over the same field, where $\mathcal{X}$ and $\widetilde{\mathcal{X}}$, as well as $\mathcal{Y}$

and $\widetilde{\mathcal{Y}}$, are isomorphic (i.e., $\widetilde{\mathcal{X}} \cong \mathcal{X}$ and $\widetilde{\mathcal{Y}} \cong \mathcal{Y}$). Two linear transformations $L \in \mathcal{L}[\mathcal{X}, \mathcal{Y}]$ and $\widetilde{L} \in \mathcal{L}[\widetilde{\mathcal{X}}, \widetilde{\mathcal{Y}}]$ are *isomorphically equivalent* (notation: $\widetilde{L} \cong L$) if there exist isomorphisms $J \in \mathcal{L}[\mathcal{X}, \widetilde{\mathcal{X}}]$ and $K \in \mathcal{L}[\mathcal{Y}, \widetilde{\mathcal{Y}}]$ such that

$$KL = \widetilde{L}J,$$

that is, $\widetilde{L} = KLJ^{-1}$. This means the following diagram commutes:

$$\begin{array}{ccc} \mathcal{X} & \xrightarrow{\ L\ } & \mathcal{Y} \\ {\scriptstyle J}\big\downarrow & & \big\downarrow{\scriptstyle K} \\ \widetilde{\mathcal{X}} & \xrightarrow{\ \widetilde{L}\ } & \widetilde{\mathcal{Y}}\,. \end{array}$$

For the isomorphically equivalent case we get

$$\mathcal{N}(\widetilde{L}) \cong \mathcal{N}(L) \quad \text{and} \quad \mathcal{R}(\widetilde{L}) \cong \mathcal{R}(L).$$

(Indeed, with $\widetilde{x} = Jx$, $J(\mathcal{N}(L)) = \{Jx \in \widetilde{\mathcal{X}}\colon Lx = 0\} = \{\widetilde{x} \in \widetilde{\mathcal{X}}\colon LJ^{-1}\widetilde{x} = 0\}$ $= \{\widetilde{x} \in \widetilde{\mathcal{X}}\colon K^{-1}\widetilde{L}JJ^{-1}\widetilde{x} = 0\} = \{\widetilde{x} \in \widetilde{\mathcal{X}}\colon \widetilde{L}\widetilde{x} = 0\} = \mathcal{N}(\widetilde{L})$ and $\mathcal{R}(\widetilde{L}) = \widetilde{L}(\widetilde{\mathcal{X}})$ $= \widetilde{L}(J(\mathcal{X})) = KLJ^{-1}J(\mathcal{X}) = KL(\mathcal{X}) = K\mathcal{R}(L)$.)

Particular cases.

If $\mathcal{X} = \mathcal{Y}$ and $JL = \widetilde{L}J$, then $L \in \mathcal{L}[\mathcal{X}]$ and $\widetilde{L} \in \mathcal{L}[\widetilde{\mathcal{X}}]$ are *similar*.

If $\widetilde{\mathcal{X}} = \mathcal{X}$ and $L = K^{-1}\widetilde{L}$, then $\widetilde{L} \cong L$ for $L \in \mathcal{L}[\mathcal{X}, \mathcal{Y}]$ and $\widetilde{L} \in \mathcal{L}[\mathcal{X}, \widetilde{\mathcal{Y}}]$.

If $\widetilde{\mathcal{Y}} = \mathcal{Y}$ and $L = \widetilde{L}J$, then $\widetilde{L} \cong L$ for $L \in \mathcal{L}[\mathcal{X}, \mathcal{Y}]$ and $\widetilde{L} \in \mathcal{L}[\widetilde{\mathcal{X}}, \mathcal{Y}]$.

Embeddings. An *embedding* (or a *linear-space embedding*) is an isomorphism onto its range. A linear space $\mathcal{X}$ is *embedded* (or *algebraically embedded*) in a linear space $\mathcal{Y}$ if there exists an embedding (i.e., an injective linear transformation) $J\colon \mathcal{X} \to \mathcal{Y}$ or, equivalently, if $J\colon \mathcal{X} \to J(\mathcal{X}) \subseteq \mathcal{Y}$ is an isomorphism between the linear space $\mathcal{X}$ and its range $\mathcal{R}(J) = J(\mathcal{X})$. In this case J *embeds* $\mathcal{X}$ into $\mathcal{Y}$. Notation: $J\colon \mathcal{X} \hookrightarrow \mathcal{Y}$; if J either is clear in the context or is immaterial, then we write $\mathcal{X} \hookrightarrow \mathcal{Y}$. (If $\mathcal{X}$ is a linear manifold of $\mathcal{Y}$ then the identity on $\mathcal{X}$, viz., $I\colon \mathcal{X} \to \mathcal{X} \subseteq \mathcal{Y}$, trivially embeds $\mathcal{X}$ in $\mathcal{Y}$.)

Ordinary sum of linear manifolds. Let $\mathcal{M}$ and $\mathcal{N}$ be linear manifolds of a linear space $\mathcal{X}$. The *sum* (or *ordinary sum*) of $\mathcal{M}$ and $\mathcal{N}$ is the linear space of $\mathcal{X}$ consisting of all sums of vectors from $\mathcal{M}$ with vectors from $\mathcal{N}$:

$$\mathcal{M} + \mathcal{N} = \big\{x \in \mathcal{X}\colon x = u + v \text{ for arbitrary } u \in \mathcal{M} \text{ and } v \in \mathcal{N}\big\}.$$

The ordinary sum of an arbitrary family of linear manifolds is similarly defined.

Algebraic complement. Let $\mathcal{M}$ be a linear manifold of a linear space $\mathcal{X}$. An *algebraic complement* of $\mathcal{M}$ is any linear manifold $\mathcal{N}$ of $\mathcal{X}$ such that $\mathcal{M} + \mathcal{N} = \mathcal{X}$ and $\mathcal{M} \cap \mathcal{N} = \{0\}$. Every linear manifold has an algebraic complement. Two linear manifolds $\mathcal{M}$ and $\mathcal{N}$ satisfying the above identities are referred to as *algebraic complements* or simply as *complementary*. Two linear manifolds $\mathcal{M}$ and $\mathcal{N}$ are *algebraically disjoint* if $\mathcal{M} \cap \mathcal{N} = \{0\}$. Thus a pair

of algebraic complements is a pair of algebraically disjoint linear manifolds that sum to $\mathcal{X}$. The dimension of any algebraic complement of $\mathcal{M}$ is another invariant for $\mathcal{M}$, referred to as the *codimension* of $\mathcal{M}$; notation $\operatorname{codim}\mathcal{M}$. The range and the kernel of a projection are complementary linear manifolds. In fact, $\mathcal{M}$ and $\mathcal{N}$ are complementary linear manifolds of a linear space $\mathcal{X}$ if and only if there exists a projection $E \in \mathcal{L}[\mathcal{X},\mathcal{X}]$ with $\mathcal{R}(E) = \mathcal{M}$ and $\mathcal{N}(E) = \mathcal{N}$.

Direct sum of linear spaces. Let $\mathcal{X}$ and $\mathcal{Y}$ be linear spaces over the same field $\mathbb{F}$. The set consisting of all pairs (x, y) with $x \in \mathcal{X}$ and $y \in \mathcal{Y}$ is the Cartesian product $\mathcal{X}\times\mathcal{Y}$ of $\mathcal{X}$ and $\mathcal{Y}$. This is not a linear space by itself. However, it may be equipped with a vector addition $\oplus$ called direct sum and a scalar multiplication that make it into a linear space $(\mathcal{X}\times\mathcal{Y}, \oplus, \centerdot, \mathbb{F})$ under the common field $\mathbb{F}$. This linear space, whose underlying set is the Cartesian product $\mathcal{X}\times\mathcal{Y}$ of the linear spaces $\mathcal{X}$ and $\mathcal{Y}$, is denoted by $\mathcal{X} \oplus \mathcal{Y} = (\mathcal{X}\times\mathcal{Y}, \oplus, \centerdot, \mathbb{F})$ and is referred to as the *direct sum*, or the *full algebraic direct sum*, or the *external direct sum* of $\mathcal{X}$ and $\mathcal{Y}$. The operations of addition (direct sum) and scalar multiplication are defined by

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1 + x_2, y_1 + y_2) \quad \text{and} \quad \alpha(x, y) = (\alpha x, \alpha y)$$

for every (x, y), (x_1, y_1), (x_2, y_2) in $\mathcal{X} \oplus \mathcal{Y}$ and every α in $\mathbb{F}$. In other words, the direct sum $\mathcal{X} \oplus \mathcal{Y}$ of a pair of linear spaces $\mathcal{X}$ and $\mathcal{Y}$ is a linear space consisting of all pairs (x, y) in the Cartesian product $\mathcal{X}\times\mathcal{Y}$ when vector addition and scalar multiplication are defined as above. A direct sum of an arbitrary family of linear spaces (e.g., $\bigoplus_{\gamma\in\Gamma}\mathcal{X}_\gamma$) is defined accordingly, having the Cartesian product of such a family of linear spaces (e.g., $\prod_{\gamma\in\Gamma}\mathcal{X}_\gamma$) as the underlying set. In particular, for a linear space $\mathcal{X}$ over $\mathbb{F}$ let $\mathcal{X}^n = \bigoplus_{i=1}^n \mathcal{X} = \big(\prod_{i=1}^n \mathcal{X}, \oplus, \centerdot, \mathbb{F}\big)$ be the linear space consisting of the direct sum of n copies of $\mathcal{X}$. Equivalently, write $\mathbb{F}^n(\mathcal{X}) = \mathcal{X}^{I_n} = \big\{(x_1, \cdots, x_n)\colon\, x_i \in \mathcal{X},\, i \in I_n = \{1, \ldots, n\}\big\}$ for the collection of all n-tuples of vectors in $\mathcal{X}$, equipped with the direct sum algebra. Thus these are different notations for the same linear space, and so we write

$$\mathcal{X}^n = \mathbb{F}^n(\mathcal{X}) = \bigoplus\nolimits_{i=1}^n \mathcal{X}.$$

In particular, $\mathbb{F}^n = \mathbb{F}^n(\mathbb{F}) = \bigoplus_{i=1}^n \mathbb{F}$. Another important particular case refers to the direct sum of a pair of linear manifolds of the same linear space (sometimes called *internal direct sum*). If $\mathcal{M}$ and $\mathcal{N}$ are linear manifolds of a linear space $\mathcal{X}$, then the ordinary sum $\mathcal{M} + \mathcal{N}$ is isomorphic to the (internal) direct sum $\mathcal{M} \oplus \mathcal{N}$ if and only if $\mathcal{M}$ and $\mathcal{N}$ are algebraically disjoint,

$$\big\{\mathcal{M} + \mathcal{N} \;\cong\; \mathcal{M} \oplus \mathcal{N}\big\} \iff \mathcal{M} \cap \mathcal{N} = \{0\}.$$

Thus we identify ordinary and direct sums of algebraic complements. Again, direct sums of more than a pair of linear manifolds are similarly defined.

Formal linear combination. Let S be an arbitrary set and let $\mathbb{F}$ be a field. Suppose $S \neq \varnothing$ to avoid trivialities. Consider the linear space $\mathbb{F}^S$ of all functions from S to $\mathbb{F}$ (i.e., of all scalar-valued functions $\boldsymbol{f}\colon S \to \mathbb{F}$ on S),

where vector addition and scalar multiplication are pointwise defined (see, e.g., [52, Example 2.E]): for every $\boldsymbol{f}, \boldsymbol{g} \in \mathbb{F}^S$ and every $\alpha \in \mathbb{F}$,

$$(\boldsymbol{f}+\boldsymbol{g})(s) = \boldsymbol{f}(s) + \boldsymbol{g}(s) \quad \text{and} \quad (\alpha\boldsymbol{f})(s) = \alpha\boldsymbol{f}(s) \quad \text{for every } \ s \in S.$$

Let # stand for *cardinality* and consider the set

$$\mathscr{F} = \big\{\boldsymbol{f} \in \mathbb{F}^S : \boldsymbol{f}(S\backslash A) = 0 \text{ for some } A \subseteq S \text{ with } \#A < \infty\big\}$$

of all functions $\boldsymbol{f}\colon S \to \mathbb{F}$ that vanish everywhere on the complement of some finite subset A of S (which depends on $\boldsymbol{f}$). This $\mathscr{F}$ is a linear manifold of $\mathbb{F}^S$.

(Indeed, $\alpha\boldsymbol{f}$ clearly lies in $\mathscr{F}$ for every $\alpha \in \mathbb{F}$ and every $\boldsymbol{f} \in \mathscr{F}$. Take an arbitrary pair $\boldsymbol{f}, \boldsymbol{g} \in \mathscr{F}$ and let A_f and A_g be finite subsets of S — so that $A_f \cup A_g \subseteq S$ is finite — for which $\boldsymbol{f}(S\backslash A_f) = \boldsymbol{g}(S\backslash A_g) = 0$. If $s \notin A_f \cup A_g$, then $\boldsymbol{f}(s) = \boldsymbol{g}(s) = 0$ and so $(\boldsymbol{f}+\boldsymbol{g})(s) = \boldsymbol{f}(s) + \boldsymbol{g}(s) = 0$, which implies $\boldsymbol{f}+\boldsymbol{g} \in \mathscr{F}$. Therefore $\mathscr{F}$ is itself a linear space over $\mathbb{F}$, whose origin is the zero function $0 \in \mathscr{F}$ — defined by $0(s) = 0 \in \mathbb{F}$ for all $s \in S$.)

Now for each $s \in S$ take the *characteristic function* $e_s = \chi_{\{s\}} : S \to \mathbb{F}$ of the singleton $\{s\} \subseteq S$ (i.e., $e_s(t) = 1 \in \mathbb{F}$ if $t = s \in S$ and zero otherwise).

$$\text{The set } \mathbb{S} = \{e_s\}_{s\in S} \text{ is a Hamel basis for the linear space } \mathscr{F}.$$

In fact, each function e_s lies in $\mathscr{F}$ since $e_s(S\backslash\{s\}) = 0$ for each $s \in S$. Moreover, if $s, t \in S$ and $t \neq s$, then $e_s(t) = e_t(s) = 0$, so that $\{e_s\}_{s\in S}$ is linearly independent. Also, since an arbitrary vector $\boldsymbol{f} \in \mathscr{F}$ is a scalar-valued function that takes nonzero values only over a finite subset of S, say on $\{s_i\}_{i=1}^n$, then

$$\boldsymbol{f} = \sum\nolimits_{i=1}^{n} \alpha_i e_{s_i} \ \in \mathscr{F},$$

with $\alpha_i \in \mathbb{F}$. So the vectors $\boldsymbol{f} \in \mathscr{F}$ are (finite) linear combinations of vectors in $\{e_s\}_{s\in S} \subset \mathscr{F}$. In other words, $\operatorname{span}\{e_s\}_{s\in S} = \mathscr{F}$. Thus $\mathbb{S} = \{e_s\}_{s\in S} \subset \mathscr{F} \subseteq \mathbb{F}^S$ is a Hamel basis for $\mathscr{F}$. This linear space $\mathscr{F}$ is referred to as the *free linear space generated by* S. Since $\#\{e_s\}_{s\in S} = \#S$, we get $\dim \mathscr{F} = \#\mathbb{S} = \#S$. This sets up a *natural identification* $\approx$ such that $s \approx e_s$ and so $S \approx \mathbb{S}$, which in turn leads to a natural identification for an arbitrary linear combination in $\mathscr{F}$,

$$\sum\nolimits_{i=1}^{n} \alpha_i e_{s_i} \approx \sum\nolimits_{i=1}^{n} \alpha_i s_i,$$

where $\sum_{i=1}^n \alpha_i s_i$ is referred to as a *formal linear combination of points* $s_i \in S$ (although addition or scalar multiplication is not directly defined on the set S), the collection of which is the *linear space of formal linear combinations from* S. So any function $\boldsymbol{f}$ in the linear space $\mathscr{F}$ is identified with a formal linear combination of points in S, and the set S that generates the free linear space $\mathscr{F}$ is identified with the Hamel basis $\mathbb{S}$ for $\mathscr{F}$. In this sense the set S may be regarded as a subset of $\mathscr{F}$, and a function $\boldsymbol{f}$ in $\mathscr{F}$ may be regarded as a formal linear combination. Thus write

$$\boldsymbol{f} = \sum\nolimits_{i=1}^{n} \alpha_i s_i \quad \text{for} \quad \sum\nolimits_{i=1}^{n} \alpha_i s_i \approx \boldsymbol{f} \in \mathscr{F} \quad \text{and} \quad S \subset \mathscr{F} \quad \text{for} \quad S \approx \mathbb{S} \subset \mathscr{F}.$$

Summing up: *Every set S generates a linear space $\mathscr{S}$ over $\mathbb{F}$, which is a linear manifold of the linear space $\mathbb{F}^S$, where every vector in $\mathscr{S}$ is identified with a formal linear combination $\sum_{i=1}^{n} \alpha_i s_i$ of elements s_i of S.*

1.2 Extension of Linear Transformations

Extension of linear transformations will play a crucial role throughout the book. An algebraic extension result for linear transformations goes as follows.

Theorem 1.1. *Let $\mathcal{X}$ and $\mathcal{Y}$ be linear spaces over the same field $\mathbb{F}$ and let $\mathcal{M}$ be a linear manifold of $\mathcal{X}$. If $L\colon \mathcal{M} \to \mathcal{Y}$ is a linear transformation, then there exists a linear extension $\widehat{L}\colon \mathcal{X} \to \mathcal{Y}$ of L defined on the whole space $\mathcal{X}$.*

Proof. We borrow the proof from [52, Theorem 2.9]. Let $\mathcal{L}at(\mathcal{X})$ stand for the lattice of all linear manifolds of a linear space $\mathcal{X}$. Take $L \in \mathcal{L}[\mathcal{M}, \mathcal{Y}]$ and set

$$\boldsymbol{L} = \big\{\widehat{L} \in \mathcal{L}[\mathcal{N}, \mathcal{Y}]\colon\ \mathcal{N} \in \mathcal{L}at(\mathcal{X}),\ \mathcal{M} \subseteq \mathcal{N} \text{ and } L = \widehat{L}|_{\mathcal{M}}\big\},$$

the collection of all linear transformations from linear manifolds of $\mathcal{X}$ to $\mathcal{Y}$ which are extensions of L. This $\boldsymbol{L}$ is nonempty since at least $L \in \boldsymbol{L}$ — it will be clear below that the set $\boldsymbol{L}$ contains plenty of elements. Also, since $\boldsymbol{L}$ is a subcollection of the collection $\boldsymbol{F} = \bigcup_{S \subseteq \mathcal{X}} \mathcal{Y}^S$ of all functions into $\mathcal{Y}$ whose domain S is included in $\mathcal{X}$, the set $\boldsymbol{L}$ is partially ordered in the extension ordering $\leq$, and every chain $\{\widehat{L}_\gamma\}$ in $\boldsymbol{L}$ has a supremum $\bigvee_\gamma \widehat{L}_\gamma$ in $\boldsymbol{F}$ with domain $\mathcal{D}\big(\bigvee_\gamma \widehat{L}_\gamma\big) = \bigcup_\gamma \mathcal{D}(\widehat{L}_\gamma)$ and range $\mathcal{R}\big(\bigvee_\gamma \widehat{L}_\gamma\big) = \bigcup_\gamma \mathcal{R}(\widehat{L}_\gamma)$ (see, e.g., [52, Problem 1.17]). Since $\mathcal{D}(\widehat{L}_\gamma) \in \mathcal{L}at(\mathcal{X})$, because each $\widehat{L}_\gamma$ is a linear transformation defined on a linear manifold of $\mathcal{X}$, and since $\mathcal{L}at(\mathcal{X})$ is a complete lattice (i.e., every subset of it has a supremum and an infimum), the domain $\mathcal{D}\big(\bigvee_\gamma \widehat{L}_\gamma\big) = \bigcup_\gamma \mathcal{D}(\widehat{L}_\gamma)$ lies in $\mathcal{L}at(\mathcal{X})$. Note as well that $\mathcal{R}\big(\bigvee_\gamma \widehat{L}_\gamma\big) = \bigcup_\gamma \mathcal{R}(\widehat{L}_\gamma) \in \mathcal{L}at(\mathcal{Y})$.

Claim. The supremum $\bigvee_\gamma \widehat{L}_\gamma$ lies in $\boldsymbol{L}$.

Proof of Claim. Take arbitrary vectors $u, v \in \mathcal{D}\big(\bigvee_\gamma \widehat{L}_\gamma\big) = \bigcup_\gamma \mathcal{D}(\widehat{L}_\gamma) \in \mathcal{L}at(\mathcal{X})$. Thus $u \in \mathcal{D}(\widehat{L}_\lambda)$ for some $\widehat{L}_\lambda \in \{\widehat{L}_\gamma\}$ and $v \in \mathcal{D}(\widehat{L}_\mu)$ for some $\widehat{L}_\mu \in \{\widehat{L}_\gamma\}$. As $\{\widehat{L}_\gamma\}$ is a chain, we get $\widehat{L}_\lambda \leq \widehat{L}_\mu$ (or vice versa), and so $\mathcal{D}(\widehat{L}_\lambda) \subseteq \mathcal{D}(\widehat{L}_\mu)$. Then $\alpha u + \beta v \in \mathcal{D}(\widehat{L}_\mu)$ and hence $\widehat{L}_\mu(\alpha u + \beta v) = \alpha \widehat{L}_\mu u + \beta \widehat{L}_\mu v$ for every $\alpha, \beta \in \mathbb{F}$ (since each $\widehat{L}_\gamma$ is linear). Then $\big(\bigvee_\gamma \widehat{L}_\gamma\big)|_{\mathcal{D}(\widehat{L}_\mu)} = \widehat{L}_\mu$. Thus $\big(\bigvee_\gamma \widehat{L}_\gamma\big)(\alpha u + \beta v) = \alpha\big(\bigvee_\gamma \widehat{L}_\gamma\big)u + \beta\big(\bigvee_\gamma \widehat{L}_\gamma\big)v$, and so $\bigvee_\gamma \widehat{L}_\gamma\colon \mathcal{D}\big(\bigvee_\gamma \widehat{L}_\gamma\big) \to \mathcal{Y}$ is linear. Also, $\widehat{L}_\gamma|_{\mathcal{M}} = L$ and $\{\widehat{L}_\gamma\}$ is a chain, so that $\big(\bigvee_\gamma \widehat{L}_\gamma\big)|_{\mathcal{M}} = L$. Conclusion: $\bigvee_\gamma \widehat{L}_\gamma \in \boldsymbol{L}$. □

Then every chain in $\boldsymbol{L}$ has a supremum (hence an upper bound) in $\boldsymbol{L}$. Thus Zorn's Lemma ensures that $\boldsymbol{L}$ contains a maximal element, say $\widehat{L}_0\colon \mathcal{N}_0 \to \mathcal{Y}$. *The program is to show that $\mathcal{N}_0 = \mathcal{X}$, and so $\widehat{L}_0$ is a linear extension of L over $\mathcal{X}$.* A proof by contradiction goes as follows. Suppose $\mathcal{N}_0 \neq \mathcal{X}$. Take $x_1 \in \mathcal{X} \backslash \mathcal{N}_0$ (and so $x_1 \neq 0$ because $\mathcal{N}_0$ is a linear manifold of $\mathcal{X}$) and consider the sum of $\mathcal{N}_0$ and the one-dimensional linear manifold of $\mathcal{X}$ spanned by $\{x_1\}$,

$$\mathcal{N}_1 = \mathcal{N}_0 + \operatorname{span}\{x_1\}.$$

This is a linear manifold of $\mathcal{X}$ properly including $\mathcal{M}$ (as $\mathcal{M} \subseteq \mathcal{N}_0 \subset \mathcal{N}_1$). Since $\mathcal{N}_0 \cap \operatorname{span}\{x_1\} = \{0\}$, every x in $\mathcal{N}_1$ has a unique representation as a sum of a vector in $\mathcal{N}_0$ and a vector in $\operatorname{span}\{x_1\}$. Thus for each $x \in \mathcal{N}_1$ there is a unique pair (x_0, α) in $\mathcal{N}_0 \times \mathbb{F}$ for which $x = x_0 + \alpha x_1$.

(In fact, if $x = x_0 + \alpha x_1 = x_0' + \alpha' x_1$, then $x_0 - x_0' = (\alpha' - \alpha)x_1$ lies in $\mathcal{N}_0 \cap \operatorname{span}\{x_1\} = \{0\}$ so that $x_0' = x_0$ and $\alpha' = \alpha$ — since $x_1 \neq 0$.)

Take any $y \in \mathcal{Y}$ and consider the map $\widehat{L}_1 \colon \mathcal{N}_1 \to \mathcal{Y}$ defined for each $x \in \mathcal{N}_1$ by

$$\widehat{L}_1 x = \widehat{L}_0 x_0 + \alpha y,$$

which is linear (inherits linearity from $\widehat{L}_0$) and $\widehat{L}_1|_{\mathcal{N}_0} = \widehat{L}_0$ (and so $\widehat{L}_0 \leq \widehat{L}_1$). Since $\mathcal{M} \subseteq \mathcal{N}_0 \subset \mathcal{N}_1$, we get $L = \widehat{L}_0|_{\mathcal{M}} = \widehat{L}_1|_{\mathcal{M}}$. Thus $\widehat{L}_1 \in \boldsymbol{L}$, which contradicts the fact that $\widehat{L}_0$ is maximal in $\boldsymbol{L}$ (for $\widehat{L}_0 \neq \widehat{L}_1$). Hence $\mathcal{N}_0 = \mathcal{X}$. □

Remark 1.2. *Two Basic Properties of Linear Extension.*

(a) Linear extension is not unique. For instance, with $\{0\} \neq \mathcal{M} \subset \mathcal{X} = \mathcal{Y}$, the identity transformation $I \colon \mathcal{M} \to \mathcal{M} \subset \mathcal{X}$ acts as the natural embedding of $\mathcal{M}$ into $\mathcal{X}$, and any projection $E \in \mathcal{L}[\mathcal{X}, \mathcal{X}]$ with $\mathcal{R}(E) = \mathcal{M}$ linearly extends I over $\mathcal{X}$. Thus take arbitrary linear spaces $\mathcal{X}$, $\mathcal{Y}$ over $\mathbb{F}$, an arbitrary (but fixed) linear manifold $\mathcal{M}$ of $\mathcal{X}$, and consider a relation $\sim$ (or $\sim_{\mathcal{M}}$) on $\mathcal{L}[\mathcal{X}, \mathcal{Y}]$ given for $L, L' \in \mathcal{L}[\mathcal{X}, \mathcal{Y}]$ by

$$L' \sim L \iff L'|_{\mathcal{M}} = L|_{\mathcal{M}}.$$

This $\sim$ is an *equivalence relation* (reflexive, transitive and symmetric). For each $L \in \mathcal{L}[\mathcal{X}, \mathcal{Y}]$ take the *equivalence class* $[L] = \{L' \in \mathcal{L}[\mathcal{X}, \mathcal{Y}] \colon L' \sim L\} = \{L' \in \mathcal{L}[\mathcal{X}, \mathcal{Y}] \colon L'|_{\mathcal{M}} = L|_{\mathcal{M}}\}$, and consider the *quotient space* $\mathcal{L}[\mathcal{X}, \mathcal{Y}]/\sim = \mathcal{L}[\mathcal{X}, \mathcal{Y}]/\mathcal{M} = \{[L] \subseteq \mathcal{L}[\mathcal{X}, \mathcal{Y}] \colon L \in \mathcal{L}[\mathcal{X}, \mathcal{Y}]\}$ consisting of all equivalence classes $[L]$ for every $L \in \mathcal{L}[\mathcal{X}, \mathcal{Y}]$. Such an equivalence relation $\sim$ is linear, which means that for every $L_1, L_1', L_2, L_2' \in \mathcal{L}[\mathcal{X}, \mathcal{Y}]$ and every $\alpha \in \mathbb{F}$,

$$L_1' \sim L_1 \text{ and } L_2' \sim L_2 \quad \text{imply} \quad L_1' + L_2' \sim L_1 + L_2 \text{ and } \alpha L_1' \sim \alpha L_1$$

(i.e., if $L_1'|_{\mathcal{M}} = L_1|_{\mathcal{M}}$ and $L_2'|_{\mathcal{M}} = L_2|_{\mathcal{M}}$, then $(L_1' + L_2')|_{\mathcal{M}} = L_1'|_{\mathcal{M}} + L_2'|_{\mathcal{M}} = L_1|_{\mathcal{M}} + L_2|_{\mathcal{M}} = (L_1 + L_2)|_{\mathcal{M}}$, and so $L_1' + L_2' \sim L_1 + L_2$ — also, $\alpha L_1' \sim \alpha L_1$).

Thus (see, e.g., [52, Example 2G]) addition and scaling are defined by

$$[L_1] + [L_2] = [L_1 + L_2] \quad \text{and} \quad \alpha[L_1] = [\alpha L_1]$$

in the linear space $\mathcal{L}[\mathcal{X}, \mathcal{Y}]/\mathcal{M}$ over $\mathbb{F}$ (this sort of linear space is the subject of the next section). Take the *extension operation* $\widehat{\ } \colon \mathcal{L}[\mathcal{M}, \mathcal{Y}] \to \mathcal{L}[\mathcal{X}, \mathcal{Y}]/\mathcal{M}$ between linear spaces over the same field given by

$$\widehat{L} = [\widehat{L}] \quad \text{for every} \quad L \in \mathcal{L}[\mathcal{M}, \mathcal{Y}],$$

where $\widehat{L} \in \mathcal{L}[\mathcal{X},\mathcal{Y}]$ on the right-hand side (inside square brackets) means *any* individual extension of $L \in \mathcal{L}[\mathcal{M},\mathcal{Y}]$ as in Theorem 1.1 and, using the same notation, $\widehat{L} \in \mathcal{L}[\mathcal{X},\mathcal{Y}]\,/\mathcal{M}$ on the left-hand side (i.e., $\widehat{L} = [\widehat{L}] \in \mathcal{L}[\mathcal{X},\mathcal{Y}]\,/\mathcal{M}$) is the value of the extension operation $\widehat{\ }$ at a given $L \in \mathcal{L}[\mathcal{M},\mathcal{Y}]$.

(b) The extension operation $\widehat{\ }$ is linear. Indeed, for arbitrary L_1, L_2 in $\mathcal{L}[\mathcal{M},\mathcal{Y}]$ and for arbitrary individual extensions of them (same notation for both), say $\widehat{L_1}, \widehat{L_2}$ in $\mathcal{L}[\mathcal{X},\mathcal{Y}]$, we get $(\widehat{L_1 + L_2})|_{\mathcal{M}} = L_1 + L_2 = \widehat{L_1}|_{\mathcal{M}} + \widehat{L_2}|_{\mathcal{M}} = (\widehat{L_1} + \widehat{L_2})|_{\mathcal{M}}$, and $[\widehat{L_1 + L_2}] = [\widehat{L_1} + \widehat{L_2}]$ so that (see argument in item (a))

$$L_1 \widehat{+} L_2 = [\widehat{L_1 + L_2}] = [\widehat{L_1} + \widehat{L_2}] = [\widehat{L_1}] + [\widehat{L_2}] = \widehat{L_1} + \widehat{L_2}.$$

Similarly, $\widehat{\alpha L} = \alpha \widehat{L}$ (i.e., $[\alpha \widehat{L}] = \alpha[\widehat{L}]$) for every $L \in \mathcal{L}[\mathcal{M},\mathcal{Y}]$ and $\alpha \in \mathbb{F}$.

1.3 Quotient Space

Let $\mathcal{M}$ be a linear manifold of a linear space $\mathcal{X}$ over a field $\mathbb{F}$, and consider a relation $\sim$ on $\mathcal{X}$ defined as follows. If x and x' are vectors in $\mathcal{X}$, then

$$x' \sim x \iff x' - x \in \mathcal{M}.$$

That is, $x' \sim x$ if x' is *congruent to* x *modulo* $\mathcal{M}$ — notation: $x' \equiv x (\mathrm{mod}\ \mathcal{M})$. Since $\mathcal{M}$ is a linear manifold of $\mathcal{X}$, the relation $\sim$ in fact is an *equivalence relation* on $\mathcal{X}$ (reason: $0 \in \mathcal{M}$ — *reflexivity*, $x' - x'' = (x' - x) + (x - x'') \in \mathcal{M}$ whenever $x' - x$ and $x - x''$ lie in $\mathcal{M}$ — *transitivity*, and $x - x' \in \mathcal{M}$ whenever $x' - x \in \mathcal{M}$ — *symmetry*). The *equivalence class* (with respect to $\sim$)

$$[x] = \{x' \in \mathcal{X} \colon\ x' \sim x\} = \{x' \in \mathcal{X} \colon\ x' = x + z \text{ for some } z \in \mathcal{M}\} = x + \mathcal{M}$$

of a vector x in $\mathcal{X}$ is called the *coset of* x *modulo* $\mathcal{M}$. The *quotient space* $\mathcal{X}/\mathcal{M}$ of $\mathcal{X}$ modulo $\mathcal{M}$ is the collection of all cosets $[x]$ modulo $\mathcal{M}$ for every $x \in \mathcal{X}$. In other words, $\mathcal{X}/\mathcal{M}$ is precisely the collection of all equivalence classes $[x]$ with respect to the equivalence relation $\sim$ for every x in $\mathcal{X}$. Equivalently, the quotient space $\mathcal{X}/\mathcal{M}$ of $\mathcal{X}$ modulo $\mathcal{M}$ is the collection of all *translations* of the linear manifold $\mathcal{M}$ (also called *affine spaces* or *linear varieties*), and so for each vector $x \in \mathcal{X}$ the coset $[x] = x + \mathcal{M} \in \mathcal{X}/\mathcal{M}$ is the translation of $\mathcal{M}$ by x.

The equivalence relation $\sim$ is a *linear equivalence relation* on the linear space $\mathcal{X}$ in the sense that if $x' \sim x$ and $y' \sim y$, then $x' + y' \sim x + y$ and $\alpha x' \sim \alpha x$. Indeed, if $x' - x \in \mathcal{M}$ and $y' - y \in \mathcal{M}$, then $(x' + y') - (x + y) = (x' - x) + (y' - y) \in \mathcal{M}$ and $\alpha x' - \alpha x = \alpha(x' - x) \in \mathcal{M}$ for every scalar α in $\mathbb{F}$. Therefore, with vector addition and scalar multiplication defined by

$$[x] + [y] = [x + y] \quad \text{and} \quad \alpha[x] = [\alpha x],$$

the quotient space $\mathcal{X}/\mathcal{M}$ of the linear space $\mathcal{X}$ modulo a linear manifold $\mathcal{M}$ is made into a linear space over the same scalar field. The origin $[0]$ of $\mathcal{X}/\mathcal{M}$ is

$$[0] = \mathcal{M} \in \mathcal{X}/\mathcal{M} \qquad (\text{i.e., } [y] = [0] \iff y \in \mathcal{M}).$$

So, as $[x] = x + \mathcal{M}$ for $x \in \mathcal{X}$, the trivial cases are ($\cong$ denotes isomorphic to):

$$\mathcal{M} = \{0\} \iff [0] = \{0\} \iff [x] = \{x\}\ \forall\, x \in \mathcal{X} \iff \mathcal{X}/\mathcal{M} = \mathcal{X}/\{0\} \cong \mathcal{X},$$

$$\mathcal{M} = \mathcal{X} \iff [0] = \mathcal{X} \iff [x] = [0]\ \forall\, x \in \mathcal{X} \iff \mathcal{X}/\mathcal{M} = \mathcal{X}/\mathcal{X} = \{[0]\} \cong \{0\}.$$

The *quotient map* (or the *natural quotient map*) $\pi\colon \mathcal{X} \to \mathcal{X}/\mathcal{M}$ of the linear space $\mathcal{X}$ onto the linear space $\mathcal{X}/\mathcal{M}$ is defined by

$$\pi(x) = [x] = x + \mathcal{M} \quad \text{for every} \quad x \in \mathcal{X},$$

which is a *surjective linear transformation* according to the definition of addition and scalar multiplication in $\mathcal{X}/\mathcal{M}$. Indeed, for every $x \in \mathcal{X}$ and $\alpha \in \mathbb{F}$

$$\pi(x+y) = [x+y] = [x]+[y] = \pi(x)+\pi(y) \quad \text{and} \quad \pi(\alpha x) = [\alpha x] = \alpha[x] = \alpha\pi(x).$$

The kernel $\mathcal{N}(\pi)$ and range $\mathcal{R}(\pi)$ of the quotient map π are given by

$$\begin{aligned}\mathcal{N}(\pi) &= \pi^{-1}([0]) = \big\{x \in \mathcal{X}\colon\ \pi(x) = [0]\big\}\\ &= \big\{x \in \mathcal{X}\colon\ [x] = [0]\big\} = \big\{x \in \mathcal{X}\colon\ x \in \mathcal{M}\big\} = \mathcal{M} \subseteq \mathcal{X},\end{aligned}$$

$$\begin{aligned}\mathcal{R}(\pi) &= \pi(\mathcal{X}) = \big\{[y] \in \mathcal{X}/\mathcal{M}\colon\ [y] = \pi(x) \text{ for some } x \in \mathcal{X}\big\}\\ &= \big\{[y] \in \mathcal{X}/\mathcal{M}\colon\ y + \mathcal{M} = x + \mathcal{M} \text{ for some } x \in \mathcal{X}\big\} = \mathcal{X}/\mathcal{M}.\end{aligned}$$

So $\mathcal{X}/\mathcal{M} = \mathcal{X}/\mathcal{N}(\pi) = \mathcal{R}(\pi)$ (which actually is a property that, up to an isomorphism, holds for every linear transformation — cf. Theorem 1.4(b) below), and the quotient map $\pi \in \mathcal{L}[\mathcal{X}, \mathcal{X}/\mathcal{M}]$ is a linear transformation such that

(i) π is surjective,

(ii) π is injective if and only if $\mathcal{M} = \{0\}$,

(iii) $\pi = 0$ if and only if $\mathcal{M} = \mathcal{X}$.

Remark 1.3. *Linear Manifolds and Restrictions.* The notions of subsets and linear manifolds of a quotient space and of restrictions of functions on quotient spaces require some attention, and it is advisable to refine the notation.

(a) Write $[x]_{\mathcal{M}} = x + \mathcal{M} \in \mathcal{X}/\mathcal{M}$ for the coset of $x \in \mathcal{X}$ modulo a linear manifold $\mathcal{M} \subseteq \mathcal{X}$ of the linear space $\mathcal{X}$. An element of $\mathcal{X}/\mathcal{M}$ is of the form $x + \mathcal{M}$ for a fixed linear manifold $\mathcal{M}$ of $\mathcal{X}$ and for an arbitrary $x \in \mathcal{X}$, and therefore a subset $V \subseteq \mathcal{X}/\mathcal{M}$ of the quotient space $\mathcal{X}/\mathcal{M}$ consists of elements $s + \mathcal{M}$ for arbitrary vectors $s \in S$ in an arbitrary subset $S \subseteq \mathcal{X}$ of the linear space $\mathcal{X}$:

$$V \text{ is a subset of } \mathcal{X}/\mathcal{M} \iff V = S/\mathcal{M} \text{ for a subset } S \text{ of } \mathcal{X}.$$

Thus, as is readily verified,

$$\mathcal{V} \text{ is a linear manifold of } \mathcal{X}/\mathcal{M} \iff \mathcal{V} = \mathcal{S}/\mathcal{M} \text{ for a linear manifold } \mathcal{S} \text{ of } \mathcal{X}.$$

Therefore, if $\mathcal{S}$ is a linear manifold of $\mathcal{X}$, then $\mathcal{S}/\mathcal{M}$ is a linear manifold of $\mathcal{X}/\mathcal{M}$ made up of all cosets $[s]_{\mathcal{M}} = s + \mathcal{M}$ such that $s \in \mathcal{S} \subseteq \mathcal{X}$. Thus

$$\mathcal{S} \subseteq \mathcal{X} \implies \mathcal{S}/\mathcal{M} \subseteq \mathcal{X}/\mathcal{M}.$$

(b) Consider the case of linear manifolds of $\mathcal{M}$ rather than linear manifolds of $\mathcal{X}$. Fix an arbitrary $x \in \mathcal{X}$ and let $\mathcal{R}$ be an arbitrary linear manifold of the linear manifold $\mathcal{M}$, and consider the coset $[x]_{\mathcal{R}} = x + \mathcal{R} \in \mathcal{X}/\mathcal{R}$ as an element of the quotient space $\mathcal{X}/\mathcal{R}$. Since $\mathcal{R} \subseteq \mathcal{M}$, we get $[x]_{\mathcal{R}} = x + \mathcal{R} \subseteq x + \mathcal{M} = [x]_{\mathcal{M}}$. However, the coset $[x]_{\mathcal{R}} = x + \mathcal{R} \in \mathcal{X}/\mathcal{R}$ is not an element of the quotient space $\mathcal{X}/\mathcal{M}$ which consists of elements of the form $x + \mathcal{M}$ only, for each $x \in \mathcal{X}$ and for a fixed $\mathcal{M} \subseteq \mathcal{X}$. In other words, although $\mathcal{R} \subseteq \mathcal{M} \Longrightarrow x + \mathcal{R} \subseteq x + \mathcal{M}$ for each $x \in \mathcal{X}$, the inclusion of quotient spaces fails:

$$\mathcal{R} \subseteq \mathcal{M} \not\Longrightarrow \mathcal{X}/\mathcal{R} \subseteq \mathcal{X}/\mathcal{M}.$$

Actually, every $[x]_{\mathcal{R}} \in \mathcal{X}/\mathcal{R}$ does lie in $\mathcal{X}/\mathcal{M}$.

(c) In particular, for $\mathcal{R} \subseteq \mathcal{M} \subseteq \mathcal{X}$,

$$\mathcal{R}/\{0\} \subseteq \mathcal{M}/\{0\} \subseteq \mathcal{X}/\{0\},$$

although, for $\{0\} \subset \mathcal{R} \subset \mathcal{M} \subset \mathcal{X}$,

$$\mathcal{X}/\{0\} \not\subseteq \mathcal{X}/\mathcal{R} \not\subseteq \mathcal{X}/\mathcal{M}.$$

(d) We may naturally identify each $x \in \mathcal{X}$ with $[x]_{\{0\}} = x + \{0\} \in \mathcal{X}/\{0\} \cong \mathcal{X}$, and so we may regard each $x \in \mathcal{X}$ as an element of $\mathcal{X}/\{0\}$. Analogously, we may identify $\mathcal{X}$ with $\mathcal{X}/\{0\}$. However, if $\mathcal{M} \neq \{0\}$, then $\mathcal{X}/\{0\} \not\subseteq \mathcal{X}/\mathcal{M}$, and we may not regard each x in $\mathcal{X}$ when identified with $[x]_{\{0\}}$ as an element of $\mathcal{X}/\mathcal{M}$, as we may not regard $\mathcal{X}/\{0\}$ as a linear manifold of $\mathcal{X}/\mathcal{M}$.

(e) Now take a function $F\colon \mathcal{X}/\mathcal{M} \to \mathcal{Z}$ on the quotient space $\mathcal{X}/\mathcal{M}$. It is clear that restrictions $F|_V$ of it only make sense if over a subset V of its domain $\mathcal{X}/\mathcal{M}$. As we saw in (a), this holds only if $V = S/\mathcal{M}$ for some subset S of $\mathcal{X}$.

Theorem 1.4. *Let $\mathcal{X}$ and $\mathcal{Z}$ be linear spaces over the same field, let $\mathcal{M}$ be a linear manifold of $\mathcal{X}$, consider the quotient space $\mathcal{X}/\mathcal{M}$, and take the natural quotient map $\pi\colon \mathcal{X} \to \mathcal{X}/\mathcal{M}$.*

(a) (Universal Property): *If $L \in \mathcal{L}[\mathcal{X}, \mathcal{Z}]$ and $\mathcal{M} \subseteq \mathcal{N}(L)$, then there exists a unique $\widehat{L} \in \mathcal{L}[\mathcal{X}/\mathcal{M}, \mathcal{Z}]$ such that*

$$L = \widehat{L} \circ \pi.$$

In other words, the diagram

$$\begin{array}{ccc} \mathcal{X} & \xrightarrow{\ L\ } & \mathcal{Z} \\ & {\scriptstyle\pi}\searrow & \uparrow{\scriptstyle\widehat{L}} \\ & & \mathcal{X}/\mathcal{M} \end{array}$$

commutes, which means the quotient map π factors the linear transformation L through $\mathcal{X}/\mathcal{M}$. Moreover, $\mathcal{N}(\widehat{L}) = \mathcal{N}(L)/\mathcal{M}$ and $\mathcal{R}(\widehat{L}) = \mathcal{R}(L)$.

(b) (First Isomorphism Theorem): *$\mathcal{R}(L) \cong \mathcal{X}/\mathcal{N}(L)$ for every $L \in \mathcal{L}[\mathcal{X}, \mathcal{Z}]$.*

Proof. (a) Let $\mathcal{X}$ and $\mathcal{Z}$ be linear spaces over the same field, let $\mathcal{M}$ be a linear manifold of $\mathcal{X}$, and consider the quotient space $\mathcal{X}/\mathcal{M}$. Let $L\colon \mathcal{X} \to \mathcal{Z}$ be a linear transformation, and consider the natural quotient map $\pi\colon \mathcal{X} \to \mathcal{X}/\mathcal{M}$, which is given by $\pi(x) = [x]$ for every $x \in \mathcal{X}$.

Claim. If $\mathcal{M} \subseteq \mathcal{N}(L)$, then there exists a unique linear transformation $\widehat{L}\colon \mathcal{X}/\mathcal{M} \to \mathcal{Z}$ for which $L = \widehat{L} \circ \pi$.

Proof of Claim. Define a transformation $\widehat{L}\colon \mathcal{X}/\mathcal{M} \to \mathcal{Z}$ by

$$\widehat{L}[x] = Lx \quad \text{for every} \quad [x] \in \mathcal{X}/\mathcal{M}.$$

As a function, such a definition makes sense if and only if

$$[x_1], [x_2] \in \mathcal{X}/\mathcal{M} \ \text{ and } \ \widehat{L}[x_1] \neq \widehat{L}[x_2] \implies [x_1] \neq [x_2],$$

which means if $[x_1 - x_2] = [0]$ for arbitrary $[x_1], [x_2] \in \mathcal{X}/\mathcal{M}$, then $\widehat{L}[x_1] = \widehat{L}[x_2]$. Since L is linear and since $[0] = \mathcal{M}$, this is equivalent to

$$x_1, x_2 \in \mathcal{X} \ \text{ and } \ x_1 - x_2 \in \mathcal{M} \implies L(x_1 - x_2) = 0,$$

which in turn (as $\mathcal{M}$ is a linear space) is equivalent to $Lx = 0$ for every $x \in \mathcal{M}$, that is, $\mathcal{M} \subseteq \mathcal{N}(L)$. Thus the definition of $\widehat{L}$, which is equivalent to

$$\widehat{L} \circ \pi = L,$$

makes sense (i.e., $\widehat{L}$ is well-defined) if and only if

$$\mathcal{M} \subseteq \mathcal{N}(L).$$

Moreover, $\widehat{L}$ is linear since linearity of L is readily transferred to $\widehat{L}$, and uniqueness follows at once.

(Indeed, $\widehat{L}([x_1]+[x_2]) = \widehat{L}[x_1+x_2] = L(x_1+x_2) = Lx_1+Lx_2 = \widehat{L}[x_1]+\widehat{L}[x_1]$ and $\widehat{L}(\alpha[x]) = \widehat{L}[\alpha x] = L(\alpha x) = \alpha Lx = \alpha\widehat{L}[x]$, ensuring linearity, and if $L = \widehat{L}' \circ \pi = \widehat{L} \circ \pi$, then $\widehat{L}'[x] = \widehat{L}[x]$ and so $\widehat{L}' = \widehat{L}$, ensuring uniqueness.) □

Also, since $\pi\colon \mathcal{X} \to \mathcal{X}/\mathcal{M}$ is surjective (i.e., $\pi(\mathcal{X}) = \mathcal{X}/\mathcal{M}$), we get

$$\mathcal{R}(\widehat{L}) = \widehat{L}(\mathcal{X}/\mathcal{M}) = \widehat{L}(\pi(\mathcal{X})) = (\widehat{L} \circ \pi)(\mathcal{X}) = L(\mathcal{X}) = \mathcal{R}(L),$$

and locally $\widehat{L}[x] = \widehat{L}(\pi(x)) = (\widehat{L} \circ \pi)(x) = Lx$ for every $[x] \in \mathcal{X}/\mathcal{M}$, so that $\widehat{L}[x] = 0$ if and only if $x \in \mathcal{N}(L)$, which implies

$$\mathcal{N}(\widehat{L}) = \big\{[x] \in \mathcal{X}/\mathcal{M}\colon\ \widehat{L}[x] = 0\big\} = \big\{[x] \in \mathcal{X}/\mathcal{M}\colon\ x \in \mathcal{N}(L)\big\} = \mathcal{N}(L)/\mathcal{M}.$$

(b) In particular, for $\mathcal{M} = \mathcal{N}(L)$ we get $\mathcal{N}(\widehat{L}) = \mathcal{M}/\mathcal{M} = \mathcal{M} = [0]$, which means the linear transformation $\widehat{L}\colon \mathcal{X}/\mathcal{N}(L) \to \mathcal{R}(\widehat{L}) \subseteq \mathcal{Z}$ is injective, and so it is an isomorphism of $\mathcal{X}/\mathcal{N}(L)$ onto $\mathcal{R}(\widehat{L})$. But $\mathcal{R}(\widehat{L}) = \mathcal{R}(L)$. □

If S is an arbitrary subset of $\mathcal{X}$, then

$$\begin{aligned}
\pi(S) &= \big\{[y] \in \mathcal{X}/\mathcal{M}\colon\ [y] = \pi(s) \text{ for some } s \in S\big\} \\
&= \big\{[y] \in \mathcal{X}/\mathcal{M}\colon\ [y] = [s] \text{ for some } s \in S\big\} \\
&= \big\{[y] \in \mathcal{X}/\mathcal{M}\colon\ [y] - [s] = [0] \text{ for some } s \in S\big\} \\
&= \big\{[y] \in \mathcal{X}/\mathcal{M}\colon\ [y - s] = [0] \text{ for some } s \in S\big\} \\
&= \big\{[y] \in \mathcal{X}/\mathcal{M}\colon\ y - s \in \mathcal{M} \text{ for some } s \in S\big\} \\
&= \big\{[y] \in \mathcal{X}/\mathcal{M}\colon\ y \in s + \mathcal{M} \text{ for some } s \in S\big\} \\
&= \big\{[y] \in \mathcal{X}/\mathcal{M}\colon\ y \in S + \mathcal{M}\big\}.
\end{aligned}$$

In particular, $\pi(\mathcal{X}) = \{[y] \in \mathcal{X}/\mathcal{M}\colon\ y \in \mathcal{X}\} = \mathcal{X}/\mathcal{M}$ since $\mathcal{X} + \mathcal{M} = \mathcal{X}$ (i.e., π is surjective). Hence the inverse image of $\pi(S)$ under π is

$$\begin{aligned}
\pi^{-1}(\pi(S)) &= \big\{x \in \mathcal{X}\colon\ \pi(x) \in \pi(S)\big\} = \big\{x \in \mathcal{X}\colon\ [x] \in \pi(S)\big\} \\
&= \big\{x \in \mathcal{X}\colon\ x + \mathcal{M} \in \mathcal{M} + S\big\} = S + \mathcal{M}.
\end{aligned}$$

If $\mathcal{N}$ is a linear manifold of $\mathcal{X}$ for which $\mathcal{X} = \mathcal{M} + \mathcal{N}$, then

$$\pi(\mathcal{N}) = \big\{[y] \in \mathcal{X}/\mathcal{M}\colon\ y \in \mathcal{N} + \mathcal{M}\big\} = \big\{[y] \in \mathcal{X}/\mathcal{M}\colon\ y \in \mathcal{X}\big\} = \mathcal{X}/\mathcal{M},$$

and in this case the restriction $\pi|_{\mathcal{N}}\colon \mathcal{N} \to \mathcal{X}/\mathcal{M}$ of π to $\mathcal{N}$ is surjective as well. Moreover, if $\mathcal{M} \cap \mathcal{N} = \varnothing$, then

$$\begin{aligned}
\mathcal{N}(\pi|_{\mathcal{N}}) &= {\pi|_{\mathcal{N}}}^{-1}([0]) = \big\{v \in \mathcal{N}\colon \pi(v) = [0]\big\} \\
&= \big\{v \in \mathcal{N}\colon [v] = [0]\big\} = \big\{v \in \mathcal{N}\colon v \in \mathcal{M}\big\} = \{0\},
\end{aligned}$$

and in this case the restriction $\pi|_{\mathcal{N}}\colon \mathcal{N} \to \mathcal{X}/\mathcal{M}$ of π to $\mathcal{N}$ is injective. Since π is linear, its restriction to a linear manifold is again linear.

Thus if $\mathcal{N}$ is an algebraic complement of $\mathcal{M}$, then $\pi|_{\mathcal{N}}\colon \mathcal{N} \to \mathcal{X}/\mathcal{M}$ is an algebraic isomorphism (referred to as the *natural quotient isomorphism*) between the linear spaces $\mathcal{N}$ and $\mathcal{X}/\mathcal{M}$: every algebraic complement of $\mathcal{M}$ is isomorphic to the quotient space $\mathcal{X}/\mathcal{M}$. Again, with $\cong$ standing for isomorphism,

$$\mathcal{N} \cong \mathcal{X}/\mathcal{M} \quad \text{for every algebraic complement } \mathcal{N} \text{ of } \mathcal{M}.$$

Then every algebraic complement of $\mathcal{M}$ has the same (constant) dimension:

$$\dim \mathcal{N} = \dim \mathcal{X}/\mathcal{M} \quad \text{for every algebraic complement } \mathcal{N} \text{ of } \mathcal{M},$$

and so two algebraic complements of $\mathcal{M}$ are algebraically isomorphic (cf. Section 1.1). Thus (see, e.g., [52, Lemma 2.17, Theorem 2.18]).

$$\operatorname{codim} \mathcal{M} = \dim \mathcal{X}/\mathcal{M}.$$

Remark 1.5. *Hamel Basis.* Let $\mathcal{M}$, $\mathcal{N}$, and $\mathcal{R}$ be linear manifolds of a linear space $\mathcal{X}$. A set $\{[\mathrm{e}_\delta]\}_{\delta\in\Delta}$ of cosets modulo $\mathcal{M}$ (with each e_δ in $\mathcal{X}$) is a Hamel basis for the linear space $\mathcal{X}/\mathcal{M}$ if and only if $\{\mathrm{e}_\delta\}_{\delta\in\Delta}$ is a Hamel basis for some algebraic complement of $\mathcal{M}$ (see, e.g., [56, Remark A.1(b)]). Hence every Hamel basis $B_\mathcal{X} = \{e_\gamma\}_{\gamma\in\Gamma}$ for $\mathcal{X}$ is such that $\{[e_\gamma]\}_{\gamma\in\Gamma}$ includes a Hamel basis $B_{\mathcal{X}/\mathcal{M}}$ for $\mathcal{X}/\mathcal{M}$. Since the quotient map $\pi\colon\mathcal{X}\to\mathcal{X}/\mathcal{M}$ is surjective, the image $\pi(B_\mathcal{X})$ of an arbitrary Hamel basis $B_\mathcal{X}$ for $\mathcal{X}$ includes a Hamel basis $B_{\mathcal{X}/\mathcal{M}}$ for $\mathcal{X}/\mathcal{M}$ (i.e., $B_{\mathcal{X}/\mathcal{M}}\subseteq\pi(B_\mathcal{X})$). Therefore

$$\operatorname{span}\pi(B_\mathcal{X}) = \mathcal{X}/\mathcal{M} \quad \text{for every Hamel basis } B_\mathcal{X} \text{ for } \mathcal{X},$$

and so every element $[x]$ of $\mathcal{X}/\mathcal{M}$ can be written as a (finite) linear combination of images under π of elements of an arbitrary Hamel basis for $\mathcal{X}$. (This result plays an important role in Section 4.1 in the forthcoming Chapter 4.)

1.4 Additional Propositions

As before, $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ denote nonzero linear spaces over the same field $\mathbb{F}$, $\mathcal{M}$ and $\mathcal{N}$ stand for linear manifolds of a fixed linear space, $\mathcal{L}[\mathcal{X},\mathcal{Y}]$ denotes the linear space over $\mathbb{F}$ of all linear transformations of $\mathcal{X}$ into $\mathcal{Y}$, and $\mathcal{X}^\sharp = \mathcal{L}[\mathcal{X},\mathbb{F}]$ denotes the algebraic dual of $\mathcal{X}$. Also, as everywhere in this chapter, $\cong$ stands for (algebraic) isomorphism. A subset S of a linear space is *linearly independent* if each vector s in S is not a (finite) linear combination of vectors in $S\backslash\{s\}$.

Proposition 1.A. *Take any $\varnothing \neq S \subset \mathcal{X}$. The assertions below are equivalent.*

(a) *S is linearly independent.*

(b) *Each nonzero vector in* span S *has a unique representation as a linear combination of vectors in S.*

(c) *Every finite subset of S is linearly independent.*

(d) *There is no proper subset of S whose span coincides with* span S.

Proposition 1.B. *Every linear space $\mathcal{X}$ has a Hamel basis, the empty set $\varnothing$ is the Hamel basis for the trivial space $\mathcal{X} = \{0\}$ (note that $\{0\}$ is not linearly independent), and every Hamel basis for a linear space has the same cardinality. (Such an invariant — i.e., this constant cardinality — is the dimension of $\mathcal{X}$, denoted by* $\dim\mathcal{X}$, *and so* $\dim\mathcal{X} = 0$ *if and only if* $\mathcal{X} = \{0\}$.)

Proposition 1.C. *Two linear spaces over the same field are isomorphic if and only if they have the same dimension.*

Proposition 1.D. *$L \in \mathcal{L}[\mathcal{X},\mathcal{Y}]$ is injective if and only if $\mathcal{N}(L) = \{0\}$.*

Proposition 1.E. *This is the* rank and nullity identity. *If $L \in \mathcal{L}[\mathcal{X},\mathcal{Y}]$, then*

$$\dim\mathcal{N}(L) + \dim\mathcal{R}(L) = \dim\mathcal{X}.$$

Proposition 1.F. *$f \in \mathcal{X}^\sharp$ is injective if and only if $\mathcal{X}$ is isomorphic to $\mathbb{F}$.*

Proposition 1.G. *If $x \in \mathcal{X} \cap \mathcal{N}(f)$ for all $f \in \mathcal{X}^\sharp$, then $x = 0$.*

Three equivalent ways of stating Proposition 1.G go as follows.

(i) *If $x \in \mathcal{X}$ is such that $f(x) = 0$ for all $f \in \mathcal{X}^\sharp$, then $x = 0$.*

(ii) *If $x \in \mathcal{X}$ is nonzero, then there exists $f \in \mathcal{X}^\sharp$ for which $f(x) \neq 0$.*

(iii) *If $A \subseteq \mathcal{X}$ is such that $f(A) = 0$ for all $f \in \mathcal{X}^\sharp$, then $A = \{0\}$.*

Clearly, all converses trivially hold true.

Proposition 1.H. *If $0 \neq f \in \mathcal{X}^\sharp$ and $x \notin \mathcal{N}(f)$, then the one-dimensional linear manifold* $\operatorname{span}\{x\}$ *and $\mathcal{N}(f)$ are algebraic complements:*

$$\operatorname{span}\{x\} \cap \mathcal{N}(f) = \{0\} \quad \text{and} \quad \operatorname{span}\{x\} + \mathcal{N}(f) = \mathcal{X}.$$

Proposition 1.I. *If $\mathcal{N}(f) = \mathcal{N}(g)$ for $f, g \in \mathcal{X}^\sharp$, then $f = \alpha g$ for $0 \neq \alpha \in \mathbb{F}$.*

Proposition 1.J. *Let $\mathbb{F}$ stand for either $\mathbb{R}$ or $\mathbb{C}$, let $\ell^\#$ be the collection of all $\mathbb{F}$-valued sequences with a finite number of nonzero entries, and let $\mathcal{P}[0,1]$ be the collection of all polynomials in the variable $t \in [0,1]$ with coefficients in $\mathbb{F}$ of any degree. Then $\mathcal{P}[0,1]$ and $\ell^\#$ are isomorphic linear spaces over $\mathbb{F}$.*

Proposition 1.K. *Every linear manifold has an algebraic complement.*

Proposition 1.L. *Every algebraic complement of a linear manifold $\mathcal{M}$ of a linear space $\mathcal{X}$ is isomorphic to the quotient space $\mathcal{X}/\mathcal{M}$.*

Proposition 1.M. *The dimension of any algebraic complement of a linear manifold is invariant (called codimension of $\mathcal{M}$ and denoted by* $\operatorname{codim}\mathcal{M}$*).*

Proposition 1.N. *Two linear manifolds $\mathcal{M}$ and $\mathcal{N}$ of a linear space $\mathcal{X}$ are algebraic complements of each other if and only if there exists a projection $E \in \mathcal{L}[\mathcal{X}, \mathcal{X}]$ such that $\mathcal{R}(E) = \mathcal{M}$ and $\mathcal{N}(E) = \mathcal{N}$.*

Proposition 1.O. *If $\mathcal{M}$ and $\mathcal{R}$ are linear manifolds of a linear space $\mathcal{X}$, then*

$$\mathcal{M} + \mathcal{R} = \mathcal{X} \text{ and } \dim\mathcal{R} < \infty \implies \operatorname{codim}\mathcal{M} < \infty$$

even if $\mathcal{R}$ is not algebraically disjoint from $\mathcal{M}$.

Proposition 1.P. *If $\mathcal{M}$ and $\mathcal{N}$ are algebraic complements in $\mathcal{X}$, then*

$$\dim\mathcal{X} = \dim\mathcal{M} + \dim\mathcal{N}.$$

Thus if $\mathcal{M}$ is a linear manifold of a linear space $\mathcal{X}$, then

$$\dim\mathcal{X} = \dim\mathcal{M} + \dim\mathcal{X}/\mathcal{M} = \dim\mathcal{M} + \operatorname{codim}\mathcal{M}$$

according to Propositions 1.C, 1.L, 1.M and 1.P.

Proposition 1.Q. *Let $\mathcal{M}$ and $\mathcal{N}$ be linear manifolds of a linear space $\mathcal{X}$. The following assertions are pairwise equivalent.*

(a) $\mathcal{M} \cap \mathcal{N} = \{0\}$ (*i.e., $\mathcal{M}$ and $\mathcal{N}$ are algebraically disjoint*).

(b) *For each x in $\mathcal{M} + \mathcal{N}$ there exists a unique u in $\mathcal{M}$ and a unique v in $\mathcal{N}$ such that $x = u + v$.*

(c) *The natural mapping $\Phi\colon \mathcal{M} \oplus \mathcal{N} \to \mathcal{M} + \mathcal{N}$ (which is given by $\Phi(u, v) = u + v$ for every $u \in \mathcal{M}$ and $v \in \mathcal{N}$) is an isomorphism.*

Proposition 1.R. *Let $\mathcal{M}$ and $\mathcal{N}$ be linear manifolds of a linear space.*

(a) *If $\mathcal{M}$ and $\mathcal{N}$ are algebraically disjoint, then*

$$\dim(\mathcal{M} \oplus \mathcal{N}) = \dim(\mathcal{M} + \mathcal{N}) = \dim \mathcal{M} + \dim \mathcal{N}.$$

(b) *If $\mathcal{M}$ and $\mathcal{N}$ are finite-dimensional, then*

$$\dim(\mathcal{M} + \mathcal{N}) = \dim \mathcal{M} + \dim \mathcal{N} - \dim(\mathcal{M} \cap \mathcal{N}).$$

Proposition 1.S. *Let $\mathcal{X}$ be a linear space and consider its decomposition*

$$\mathcal{X} = \mathcal{M} \oplus \mathcal{N}$$

as a direct sum of algebraically disjoint linear manifolds $\mathcal{M}$ and $\mathcal{N}$ of $\mathcal{X}$. Let E in $\mathcal{L}[\mathcal{X}, \mathcal{X}]$ be the projection on $\mathcal{M}$ along $\mathcal{N}$ (i.e., $\mathcal{R}(E) = \mathcal{M}$ and $\mathcal{N}(E) = \mathcal{N}$) and let $P = I - E$ be the complementary projection on $\mathcal{N}$ along $\mathcal{M}$. With respect to the decomposition $\mathcal{X} = \mathcal{M} \oplus \mathcal{N}$ every linear transformation L in $\mathcal{L}[\mathcal{X}, \mathcal{X}]$ can be written as a 2×2 matrix with linear transformation entries

$$L = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

where A, B, C, D are linear transformations given by $A = EL|_{\mathcal{M}}\colon \mathcal{M} \to \mathcal{M}$, $B = EL|_{\mathcal{N}}\colon \mathcal{N} \to \mathcal{M}$, $C = PL|_{\mathcal{M}}\colon \mathcal{M} \to \mathcal{N}$, and $D = PL|_{\mathcal{N}}\colon \mathcal{N} \to \mathcal{N}$. The linear transformation $L \in \mathcal{L}[\mathcal{X}, \mathcal{X}]$ is said to be the direct sum of the linear transformations $A \in \mathcal{L}[\mathcal{M}, \mathcal{M}]$ and $D \in \mathcal{L}[\mathcal{N}, \mathcal{N}]$, written as

$$L = A \oplus D$$

if B and C are null (i.e., $B = O \in \mathcal{L}[\mathcal{N}, \mathcal{M}]$ and $C = O \in \mathcal{L}[\mathcal{M}, \mathcal{N}]$). Moreover, with respect to the same decomposition the projections E and P are written as (where I and O stand for the identity and null transformations)

$$E = I \oplus O \quad \text{and} \quad P = O \oplus I.$$

Proposition 1.T. *If $\{\mathcal{M}_\gamma\}$ is an indexed family of pairwise algebraically disjoint linear manifolds of a linear space, say $\mathcal{X}$, then the dimension of the direct sum $\bigoplus_\gamma \mathcal{M}_\gamma$ is the sum of the dimensions of each $\mathcal{M}_\gamma$.*

Proposition 1.U. *Let $\mathcal{M}$ and $\mathcal{N}$ be linear manifolds of a linear space $\mathcal{X}$, and let $L\colon \mathcal{X} \to \mathcal{Z}$ be a linear transformation of $\mathcal{X}$ into any linear space $\mathcal{Z}$.*

(a) $\mathcal{R}(L) \cong \mathcal{X}/\mathcal{N}(L)$
(*this is the First Isomorphism Theorem — as in Theorem 1.4*).

(b) $(\mathcal{M}+\mathcal{N})/\mathcal{N} \cong \mathcal{M}/(\mathcal{M}\cap\mathcal{N})$
(*this is referred to as the Second Isomorphism Theorem*).

(c) *If $\mathcal{M} \subseteq \mathcal{N}$, then* $(\mathcal{X}/\mathcal{M})/(\mathcal{N}/\mathcal{M}) \cong \mathcal{X}/\mathcal{N}$
(*this is referred to as the Third Isomorphism Theorem*).

(d) *If $\mathcal{M}$ and $\mathcal{N}$ are algebraic complements and $\mathcal{M}'$ and $\mathcal{N}'$ are linear manifolds of $\mathcal{M}$ and $\mathcal{N}$, then* $(\mathcal{M}\oplus\mathcal{N})/(\mathcal{M}'\oplus\mathcal{N}') \cong \mathcal{M}/\mathcal{M}' \oplus \mathcal{N}/\mathcal{N}'$ (*where the direct sums on the left-hand side are internal direct sums, and the direct sum on the right-hand side is an external direct sum*).

Proposition 1.V. *Let $\mathcal{X}^\sharp$ be the algebraic dual of a linear space $\mathcal{X}$. Then*

$$\dim \mathcal{X} \le \dim \mathcal{X}^\sharp,$$

and

$$\dim \mathcal{X} = \dim \mathcal{X}^\sharp \quad \Longleftrightarrow \quad \dim \mathcal{X} < \infty.$$

Notes: The above propositions summarize some of the results discussed in the main body of this chapter, and introduce some results that will be required in the upcoming chapters. These are all standard results whose proofs can be found in essentially every text dealing with linear space. The books listed below are among the many possible sources. In particular, for Propositions 1.A to 1.E see, e.g., [52, Proposition 2.3, Theorems 2.5 to 2.8, 2.12, Problem 2.17]. In addition to being well-known useful results, Propositions 1.F to 1.I will often be used in the remaining portions of the present book, and so their proofs are sketched below.

Proof of Proposition 1.F. By Propositions 1.D and 1.E, if f is injective, then $\dim \mathcal{R}(f) = \dim \mathcal{X}$. Since $\mathcal{X} \neq \{0\}$ and $\mathcal{R}(f) \subseteq \mathbb{F}$, it follows that $0 < \dim \mathcal{X} = \dim \mathcal{R}(f) \le \dim \mathbb{F} = 1$. Thus $\dim \mathcal{X} = 1$, which means $\mathcal{X}$ is isomorphic to $\mathbb{F}$ by Proposition 1.C. The converse is trivially verified. □

Proof of Proposition 1.G. Let $\{e_\gamma\}$ be a Hamel basis for $\mathcal{X}$. Take any $x \in \mathcal{X}$. Let $x = \sum_\gamma \alpha_\gamma(x) e_\gamma$ be its unique (finite) expansion in terms of $\{e_\gamma\}$. Consider the functional $\alpha_\gamma\colon \mathcal{X} \to \mathbb{F}$ associating to each x the coefficient $\alpha_\gamma(x)$ in its expansion. As is readily verified, it is linear. So $\alpha_\gamma \in \mathcal{X}^\sharp$. If $f(x) = 0$ for all $f \in \mathcal{X}^\sharp$, then $\alpha_\gamma(x) = 0$ for all γ and $x = \sum_\gamma \alpha_\gamma(x) e_\gamma = 0$. The converse is trivial. □

Proof of Proposition 1.H. If $f \neq 0$, take $x \in \mathcal{X}\backslash N(f)$. If $y \in \operatorname{span}\{x\} \cap \mathcal{N}(f)$, then $y = \alpha x$ for $\alpha \in \mathbb{F}$ and $0 = f(y) = \alpha f(x)$. Thus $\alpha = 0$ (as $x \notin \mathcal{N}(f)$) so $y = 0$. Hence $\operatorname{span}\{x\} \cap \mathcal{N}(f) \subseteq \{0\}$. Now take $y \in \mathcal{X}$. Since $f(x) \neq 0$, write $y = \frac{f(y)}{f(x)}x + (y - \frac{f(y)}{f(x)}x)$, where $\frac{f(y)}{f(x)}x \in \operatorname{span}\{x\}$ and $f(y - \frac{f(y)}{f(x)}x) = 0$ (i.e., $y = \frac{f(y)}{f(x)}x \in \mathcal{N}(f)$). Then $y \in \operatorname{span}\{x\} + \mathcal{N}(f)$. So $\mathcal{X} \subseteq \operatorname{span}\{x\} + \mathcal{N}(f)$. □

Proof of Proposition 1.I. Let $\mathcal{N}(f) = \mathcal{N}(g)$. Set $\mathcal{N} = \mathcal{N}(f) = \mathcal{N}(g)$. To avoid trivialities, $\mathcal{N} \neq \mathcal{X}$. Take any $x \in \mathcal{X} \backslash \mathcal{N}$. By Proposition 1.H span $\{x\} \cap \mathcal{N} = \{0\}$ and span $\{x\} + \mathcal{N} = \mathcal{X}$. Take an arbitrary $y \in \mathcal{X}$. Write $y = \gamma x + u$ for γ in $\mathbb{F}$ and u in $\mathcal{N}$. Thus $f(y) = \gamma f(x)$ and $g(y) = \gamma g(x)$ for every $y \in \mathcal{X}$. Since $x \notin \mathcal{N}$, $\frac{f(y)}{f(x)} = \gamma = \frac{g(y)}{g(x)}$. Then set $0 \neq \alpha = \frac{f(x)}{g(x)} \in \mathbb{F}$ so that $f(y) = \alpha g(y)$. □

Also see [81, Theorem 3.10] for Propositions 1.F to 1.I. For Propositions 1.J to 1.N see, e.g., [52, Example 2.M, Theorems 2.16 to 2.20], and for Proposition 1.O, [56, Remark A.1(a)]. Proposition 1.P is a straightforward consequence of the rank and nullity identity of Proposition 1.E with a little help from Proposition 1.N (also see [91, Theorem 6.2]). For Propositions 1.Q to 1.S see, e.g., [52, Theorem 2.14, Problem 2.25, Proposition 2.22]. Proposition 1.T says that

$$\mathcal{M}_\alpha \cap \mathcal{M}_\beta = \{0\} \quad \forall\, \alpha \neq \beta \quad \Longrightarrow \quad \dim\left(\bigoplus\nolimits_\gamma \mathcal{M}_\gamma\right) = \sum\nolimits_\gamma \dim \mathcal{M}_\gamma$$

(see, e.g., [5, Problem 2.F]). Recall from Section 1.1 that if $\mathcal{M}_\alpha \cap \mathcal{M}_\beta = \{0\}$ for all $\alpha \neq \beta$, then $\sum_\gamma \mathcal{M}_\gamma \cong \bigoplus_\gamma \mathcal{M}_\gamma$. Thus, in this case, we also get

$$\dim\left(\sum\nolimits_\gamma \mathcal{M}_\gamma\right) = \sum\nolimits_\gamma \dim \mathcal{M}_\gamma$$

by Proposition 1.C. For further issues on direct sum as defined here see, e.g., [5, Chapter 2] and [52, Section 2.8]). For Proposition 1.U see, e.g., [81, Theorems 3.5, 3.7, 3.8, 3.9]. For Proposition 1.V see, e.g., [81, Theorem 3.12].

We close this first chapter with a remark on notation. As usual, $\subset$ means proper inclusion — $A \subset B$ means that the set A is a proper subset of another set B (A is included in B but is not equal to B), and $A \subseteq B$ means that a set A is either a proper subset of a set B or is equal to B (i.e., $\subseteq$ means $\subset$ or $=$). Equality between two sets A and B means $A \subseteq B$ and $B \subseteq A$. Thus $\not\subseteq$ simply means that a set is not subset of another set (it is not included — not properly included and not equal), and hence $A \subset B$ means $A \subseteq B$ and $B \not\subseteq A$. The symbol $\subsetneqq$ is somewhat similar to $\subset$, it means an inclusion that may in certain cases be a proper inclusion.

Suggested Readings

Berberian [3]
Brown and Pearcy [5]
Halmos [30, 32]
Kubrusly [52, 56]
Lang [66]
MacLane and Birkhoff [69]
Roman [81]
Taylor and Lay [91]



2

Bilinear Maps: Algebraic Aspects

Let $\mathcal{X}$, $\mathcal{Y}$, $\mathcal{Z}$ be nonzero linear spaces over a field $\mathbb{F}$ and consider the Cartesian product $\mathcal{X}\times\mathcal{Y}$ (without imposing any algebraic structure on $\mathcal{X}\times\mathcal{Y}$ besides the fact that both $\mathcal{X}$ and $\mathcal{Y}$ are linear spaces).

A *bilinear map* $\phi\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{Z}$ is a function from the Cartesian product of linear spaces to a linear space whose sections are linear transformations.

Precisely, for each $y \in \mathcal{Y}$ let $\phi^y = \phi(\cdot, y) = \phi|_{\mathcal{X}\times\{y\}}\colon \mathcal{X} \to \mathcal{Z}$ be the *y-section* of ϕ, and for each $x \in \mathcal{X}$ let $\phi_x = \phi(x, \cdot) = \phi|_{\{x\}\times\mathcal{Y}}\colon \mathcal{Y} \to \mathcal{Z}$ be the *x-section* of ϕ. The map ϕ is bilinear if ϕ^y and ϕ_x are linear transformations, that is, if $\phi^y = \phi(\cdot, y) \in \mathcal{L}[\mathcal{X}, \mathcal{Z}]$ and $\phi_x = \phi(x, \cdot,) \in \mathcal{L}[\mathcal{Y}, \mathcal{Z}]$. In other words, ϕ is bilinear if for every $x, x_1, x_2 \in \mathcal{X}$, $y, y_1, y_2 \in \mathcal{Y}$, and $\alpha, \alpha_1, \alpha_2, \beta_1, \beta_2 \in \mathbb{F}$,

$$\phi(x_1 + x_2, y) = \phi(x_1, y) + \phi(x_2, y), \qquad \phi(x, y_1 + y_2) = \phi(x, y_1) + \phi(x, y_2),$$

$$\phi(\alpha x, y) = \alpha\phi(x, y) = \phi(x, \alpha y).$$

Equivalently,

$$\begin{aligned}\phi(\alpha_1 x_1 + \alpha_2 x_2,\, \beta_1 y_1 + \beta_2 y_2) &= \phi(\alpha_1 x_1,\, \beta_1 y_1 + \beta_2 y_2) + \phi(\alpha_2 x_2,\, \beta_1 y_1 + \beta_2 y_2)\\ &= \phi(\alpha_1 x_1,\, \beta_1 y_1) + \phi(\alpha_1 x_1,\, \beta_2 y_2) + \phi(\alpha_2 x_2,\, \beta_1 y_1) + \phi(\alpha_2 x_2,\, \beta_2 y_2)\\ &= \alpha_1\beta_1\phi(x_1, y_1) + \alpha_1\beta_2\phi(x_1, y_2) + \alpha_2\beta_1\phi(x_2, y_1) + \alpha_2\beta_2\phi(x_2, y_2).\end{aligned}$$

For the particular case of $\mathcal{Z} = \mathbb{F}$ we will refer to a scalar-valued bilinear map $\phi\colon \mathcal{X}\times\mathcal{Y} \to \mathbb{F}$ either as a *bilinear functional* or as a *bilinear form*.

2.1 The Linear Space of Bilinear Maps

As usual, let $\mathcal{Z}^S$ denote the collection of all functions on a set S with values in a linear space $\mathcal{Z}$, which is itself a linear space over the same field $\mathbb{F}$. Let

$$b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] = \big\{\phi \in \mathcal{Z}^{\mathcal{X}\times\mathcal{Y}}\colon \phi \text{ is bilinear}\big\}$$

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Universitext, https://doi.org/10.1007/978-3-031-34093-2_2

denote the collection of all $\mathcal{Z}$-valued bilinear maps on $\mathcal{X}\times\mathcal{Y}$. Bilinearity of elements $\phi\colon \mathcal{X}\times\mathcal{Y}\to\mathcal{Z}$ in $b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$ implies that $b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$ is a linear manifold of the linear space $\mathcal{Z}^{\mathcal{X}\times\mathcal{Y}}$ of all $\mathcal{Z}$-valued functions on the set $S=\mathcal{X}\times\mathcal{Y}$. Indeed, take arbitrary ϕ and ϕ' from $b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$. Since ϕ and ϕ' are bilinear, it follows that for every $x_1,x_2\in\mathcal{X}$ and $y\in\mathcal{Y}$,

$$\begin{aligned}(\phi+\phi')(x_1+x_2,y)&=\phi(x_1+x_2,y)+\phi'(x_1+x_2,y)\\&=\phi(x_1,y)+\phi(x_2,y)+\phi'(x_1,y)+\phi'(x_2,y)\\&=(\phi+\phi')(x_1,y)+(\phi+\phi')(x_2,y).\end{aligned}$$

Similarly, for every $x\in\mathcal{X}$, every $y,y_1,y_2\in\mathcal{Y}$, and every $\alpha\in\mathbb{F}$,

$$(\phi+\phi')(x,y_1+y_2)=(\phi+\phi')(x,y_1)+(\phi+\phi')(x,y_2),$$

$$(\phi+\phi')(\alpha x,y)=\alpha(\phi+\phi')(x,y)=(\phi+\phi')(x,\alpha y).$$

Thus $\phi+\phi'$ is a bilinear map, which means $\phi+\phi'$ lies in $b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$. Also, since ϕ is bilinear, then for every $x_1,x_2\in\mathcal{X}$, $y\in\mathcal{Y}$. and $\beta\in\mathbb{F}$,

$$\begin{aligned}(\beta\phi)(x_1+x_2,y)&=\phi(x_1+x_2,\beta y)=\phi(x_1,\beta y)+\phi(x_2,\beta y)\\&=(\beta\phi)(x_1,y)+(\beta\phi)(x_2,y).\end{aligned}$$

Similarly, for every $x\in\mathcal{X}$, every $y,y_1,y_2\in\mathcal{Y}$, and every $\alpha,\beta\in\mathbb{F}$,

$$(\beta\phi)(x,y_1+y_2)=(\beta\phi)(x,y_1)+(\beta\phi)(x,y_2),$$

$$(\beta\phi)(\alpha x,y)=\alpha(\beta\phi)(x,y)=(\beta\phi)(x,\alpha y).$$

So $\beta\phi$ is again a bilinear map, which means $\beta\phi$ lies in $b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$. Hence $b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$ is a linear manifold of the linear space $\mathcal{Z}^{\mathcal{X}\times\mathcal{Y}}$. Therefore

$$b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]\ \text{ is a linear space over }\mathbb{F}.$$

Let $R(\phi)=\text{range}(\phi)=\phi(\mathcal{X}\times\mathcal{Y})$ denote the range of $\phi\in b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$, a subset of the linear space $\mathcal{Z}$ which in general is not a linear manifold of $\mathcal{Z}$.

Remark 2.1. *On Restrictions of Bilinear Maps.* Consider the restriction $\phi|_{M\times N}\colon M\times N\to\mathcal{Z}$ of a bilinear map $\phi\colon\mathcal{X}\times\mathcal{Y}\to\mathcal{Z}$ to a Cartesian product of subsets M and N of the linear spaces $\mathcal{X}$ and $\mathcal{Y}$. Two cases deserve attention.

(a) Restriction to the Cartesian product of linear manifolds $\mathcal{M}$ and $\mathcal{N}$ of $\mathcal{X}$ and $\mathcal{Y}$. A straightforward consequence of the definitions of linear manifold and bilinear map leads to the following result. *The restriction of a bilinear map to a Cartesian product of linear manifolds is bilinear*. That is, if $\mathcal{M}$ and $\mathcal{N}$ are linear manifolds of $\mathcal{X}$ and $\mathcal{Y}$, then $\phi\in b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$ implies $\phi|_{\mathcal{M}\times\mathcal{N}}\in b[\mathcal{M}\times\mathcal{N},\mathcal{Z}]$.

(b) Restriction to the Cartesian product of a singleton with either $\mathcal{X}$ or $\mathcal{Y}$. This goes back to the definition of bilinear maps, leading to linear sections $\phi^y=\phi|_{\mathcal{X}\times\{y\}}$ in $\mathcal{L}[\mathcal{X},\mathcal{Z}]$ and $\phi_x=\phi|_{\{x\}\times\mathcal{Y}}$ in $\mathcal{L}[\mathcal{Y},\mathcal{Z}]$. If the singleton is the

origin of one of the linear spaces, then it becomes the null linear manifold. Let 0 stand for the origin of $\mathcal{X}$ and also for the origin of $\mathcal{Y}$. If ϕ is bilinear, then $\phi^0 = \phi|_{\mathcal{X}\times\{0\}} = O$ is the null transformation in $\mathcal{L}[\mathcal{X},\mathcal{Z}] = b[\mathcal{X}\times\{0\},\mathcal{Z}]$ and $\phi_0 = \phi|_{\{0\}\times\mathcal{Y}} = O$ is the null transformation in $\mathcal{L}[\mathcal{Y},\mathcal{Z}] = b[\{0\}\times\mathcal{Y},\mathcal{Z}]$.

Remark 2.2. *On Composition of Bilinear Maps*. As composition of a pair of linear transformations is a linear transformation, composition of a bilinear map with a linear transformation from both sides is a bilinear map.

(a) On the one hand, composition of a bilinear map with a linear transformation is a bilinear map. That is, if $\mathcal{X},\mathcal{Y},\mathcal{Z},\mathcal{Z}'$ are linear spaces, then

$$\phi \in b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}] \quad \text{and} \quad L \in \mathcal{L}[\mathcal{Z},\mathcal{Z}'] \quad \Longrightarrow \quad L\circ\phi \in b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}'].$$

Indeed, since ϕ is bilinear and L is linear, then

$$\begin{aligned}(L\circ\phi)\,(x_1+x_2,y) &= L\big(\phi(x_1+x_2,y)\big) = L\big(\phi(x_1,y)+\phi(x_2,y)\big)\\ &= L\big(\phi(x_1,y)\big)+L\big(\phi(x_2,y)\big) = (L\circ\phi)\,(x_1,y)+(L\circ\phi)\,(x_2,y)\end{aligned}$$

for every $x_1,x_2\in\mathcal{X}$ and $y\in\mathcal{Y}$. Similarly, for every $x\in\mathcal{X}$, $y_1,y_2\in\mathcal{Y}$ and $\alpha\in\mathbb{F}$,

$$(L\circ\phi)\,(x,y_1+y_2) = (L\circ\phi)\,(x,y_1)+(L\circ\phi)\,(x,y_2),$$

$$(L\circ\phi)\,(\alpha x,y) = \alpha(L\circ\phi)\,(x,y) = (L\circ\phi)\,(x,\alpha y).$$

(b) On the other hand, composition of a pair of linear transformations with a bilinear map is a bilinear map. That is, for linear spaces $\mathcal{X},\mathcal{X}',\mathcal{Y},\mathcal{Y}',\mathcal{Z}$,

$$\phi\in b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}],\ L\in\mathcal{L}[\mathcal{X}',\mathcal{X}],\ K\in\mathcal{L}[\mathcal{Y}',\mathcal{Y}] \implies \phi(L(\cdot),K(\cdot))\in b[\mathcal{X}'\times\mathcal{Y}',\mathcal{Z}].$$

In fact, as is readily verified, for every $u,x\in\mathcal{X}'$, $v,y\in\mathcal{Y}'$ and $\alpha,\beta,\gamma,\delta\in\mathbb{F}$,

$$\begin{aligned}&\phi(L(\alpha u+\beta x),K(\gamma v+\delta y)\\ &\qquad = \alpha\gamma\,\phi(Lu,Kv)+\alpha\delta\,\phi(Lu,Ky)+\beta\gamma\,\phi(Lx,Kv)+\beta\delta\,\phi(Lx,Ky).\end{aligned}$$

Let $\mathcal{X}$, $\mathcal{Y}$, $\mathcal{Z}$ be linear spaces over $\mathbb{F}$ and take the Cartesian product $\mathcal{X}\times\mathcal{Y}$, which is not itself a linear space. Consider the linear space direct sum $\mathcal{X}\oplus\mathcal{Y} = (\mathcal{X}\times\mathcal{Y},\oplus,\centerdot,\mathbb{F})$ whose underlying set is $\mathcal{X}\times\mathcal{Y}$. Take $\psi\colon\mathcal{X}\oplus\mathcal{Y}\to\mathcal{Z}$. Observe that ψ acts on the underlying set $\mathcal{X}\times\mathcal{Y}$ of the linear space $\mathcal{X}\oplus\mathcal{Y}$. Suppose $\psi\colon\mathcal{X}\times\mathcal{Y}\to\mathcal{Z}$ is a bilinear map (i.e., $\psi\in b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$). Set

$$b[\mathcal{X}\oplus\mathcal{Y},\mathcal{Z}] = \big\{\psi\in\mathcal{Z}^{\mathcal{X}\oplus\mathcal{Y}}\colon \psi \text{ is bilinear}\big\} = \big\{\psi\in\mathcal{Z}^{\mathcal{X}\oplus\mathcal{Y}}\colon \psi\in b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]\big\},$$

the linear space of all $\mathcal{Z}$-valued bilinear maps on the linear space $\mathcal{X}\oplus\mathcal{Y}$. In other words, since bilinearity only requires the algebraic structure of each linear space $\mathcal{X}$ and $\mathcal{Y}$ individually, and since the map ψ actually acts on the underlying set $\mathcal{X}\times\mathcal{Y}$ of $\mathcal{X}\oplus\mathcal{Y}$ independently of the algebra equipping $\mathcal{X}\oplus\mathcal{Y}$, its bilinearity does not depend on the algebra of the linear space $\mathcal{X}\oplus\mathcal{Y}$. Thus

$$b[\mathcal{X}\oplus\mathcal{Y},\mathcal{Z}] = b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}].$$

Such an identity deserves a few words. The domain of a map in $b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$ has no algebraic structure, while the domain of a map in $b[\mathcal{X}\oplus\mathcal{Y},\mathcal{Z}]$ has. Proposition 2.C in Section 2.4 will discuss this point. So if some algebraic property, say, translation, is required in the domain of bilinear maps, then $b[\mathcal{X}\oplus\mathcal{Y},\mathcal{Z}]$ is the proper setup. This is illustrated in the proof of Proposition 6.F in Section 6.4.

Remark 2.3. *Nonlinearity and Quadratic Forms*. Bilinear forms are closer to quadratic forms than to linear forms.

(a) Even if a bilinear map ψ is regarded as an element of $b[\mathcal{X}\oplus\mathcal{Y},\mathcal{Z}]$, it is not a linear transformation on the linear space $\mathcal{X}\oplus\mathcal{Y}$. In other words,

$$\psi\in b[\mathcal{X}\oplus\mathcal{Y},\mathcal{Z}]\backslash\mathcal{L}[\mathcal{X}\oplus\mathcal{Y},\mathcal{Z}],$$

where $\mathcal{L}[\mathcal{X}\oplus\mathcal{Y},\mathcal{Z}]$ is the linear space of all linear transformations from the direct sum $\mathcal{X}\oplus\mathcal{Y}$ to the linear space $\mathcal{Z}$. For instance, take any $(x,y)\in\mathcal{X}\oplus\mathcal{Y}$ for which $\psi(x,y)\neq 0$ in $\mathcal{Z}$ (i.e., $(x,y)\in(\mathcal{X}\oplus\mathcal{Y})\backslash\mathcal{N}(\psi)$) and compute

$$\psi((x,y)\oplus(x,y))=\psi(2x,2y)=4\psi(x,y)\neq 2\psi(x,y)=\psi(x,y)+\psi(x,y).$$

So even when acting on the linear space $\mathcal{X}\oplus\mathcal{Y}$ bilinear maps are not additive.

(b) Moreover, the product of linear forms acting on $\mathcal{X}$ and $\mathcal{Y}$ is a bilinear form acting on $\mathcal{X}\times\mathcal{Y}$. In other words, if $f\in\mathcal{X}^\sharp=\mathcal{L}[\mathcal{X},\mathbb{F}]$ and $g\in\mathcal{Y}^\sharp=\mathcal{L}[\mathcal{Y},\mathbb{F}]$ are linear forms, then $\phi\colon\mathcal{X}\times\mathcal{Y}\to\mathbb{F}$ given by

$$\phi(x,y)=f(x)\,g(y)\ \in\ \mathbb{F}\quad\text{ for every }\quad(x,y)\in\mathcal{X}\times\mathcal{Y}$$

is clearly a bilinear form (i.e., $\phi\in b[\mathcal{X}\times\mathcal{Y},\mathbb{F}]$). In particular, if $\mathcal{Y}=\mathcal{X}$ and $f=g$, then $\phi(x,y)=\phi(y,x)$ and $\phi(x,x)=f(x)^2$ for every $x,y\in\mathcal{X}$. There are, however, bilinear forms which are not the product of linear forms. Indeed, *if a bilinear form on $\mathcal{X}\times\mathcal{X}$ induces a positive quadratic form on a space $\mathcal{X}$ of dimension greater than one, then it is not the product of linear forms*. For example, set $\mathcal{X}=\mathcal{Y}$, let $\mathbb{F}=\mathbb{R}$, and take an inner product $\langle\cdot\,;\cdot\rangle\colon\mathcal{X}\times\mathcal{X}\to\mathbb{R}$ in the real linear space $\mathcal{X}$. This is a (symmetric) bilinear form on $\mathcal{X}\times\mathcal{X}$ which induces a positive quadratic form when evaluated at (x,x), namely, the norm squared,

$$\langle x\,;x\rangle=\|x\|^2\quad\text{ for every }\quad x\in\mathcal{X}.$$

(Quadratic forms and inner product are discussed in Section 2.4). Suppose there exist $f,g\in\mathcal{X}^\sharp$ such that $\|x\|^2=\langle x\,;x\rangle=f(x)\,g(x)$ for every x in $\mathcal{X}$. Since $x\neq 0$ if and only if $f(x)\,g(x)\neq 0$, then $\mathcal{N}(f)\cup\mathcal{N}(g)=\{0\}$. This implies $\mathcal{N}(f)=\mathcal{N}(g)=\{0\}$ or, equivalently, f and g are injective. But there is no injective linear functional on a linear space of dimension greater than 1 (i.e., a linear functional on a nonzero space is injective if and only if it acts on a one-dimensional space — cf. Proposition 1.F in Section 1.4). So if $\dim\mathcal{X}\geq 2$, then a real inner product (which is bilinear) is never the product of two linear forms.

(c) There are, however, relations between bilinear maps and linear transformations. This is investigated in Theorem 2.6 below, and then again in Theorem 3.7, wherein tensor products will somehow rescue linearity from bilinearity.

2.2 Extension of Bilinear Maps

Bilinear maps as products of linear transformations extend bilinearly from Cartesian products of linear manifolds to Cartesian products of linear spaces, as linear transformations extend linearly from linear manifolds to linear spaces.

Theorem 2.4. *Let $\mathcal{X}$ and $\mathcal{Y}$ be linear spaces, let $\mathcal{Z}$ be an algebra, all over the same field, and let $\mathcal{M}$ and $\mathcal{N}$ be linear manifolds of $\mathcal{X}$ and $\mathcal{Y}$ respectively. If $\phi \in b[\mathcal{M}\times\mathcal{N}, \mathcal{Z}]$ is a bilinear map on $\mathcal{M}\times\mathcal{N}$ of the form $\phi(u,v) = Lu\,Kv$ for every $(u,v) \in \mathcal{M}\times\mathcal{N}$ with $L \in \mathcal{L}[\mathcal{M}, \mathcal{Z}]$ and $K \in \mathcal{L}[\mathcal{N}, \mathcal{Z}]$, then there exists a bilinear extension $\widehat{\phi} \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ of ϕ on $\mathcal{X}\times\mathcal{Y}$ of the form $\widehat{\phi}(x,y) = \widehat{L}x\,\widehat{K}y$ for every $(x,y) \in \mathcal{X}\times\mathcal{Y}$ with $\widehat{L} \in \mathcal{L}[\mathcal{X}, \mathcal{Z}]$ and $\widehat{K} \in \mathcal{L}[\mathcal{Y}, \mathcal{Z}]$.*

Proof. Suppose $\mathcal{Z}$ is an algebra. Then take a bilinear map $\phi \in b[\mathcal{M}\times\mathcal{N}, \mathcal{Z}]$ such that $\phi(u,v) = Lu\,Kv$ for every $(u,v) \in \mathcal{M}\times\mathcal{N}$. Since $L \in \mathcal{L}[\mathcal{M}, \mathcal{Z}]$ and $K \in \mathcal{L}[\mathcal{N}, \mathcal{Z}]$, they have linear extensions over $\mathcal{X}$ and $\mathcal{Y}$, say, $\widehat{L} \in \mathcal{L}[\mathcal{X}, \mathcal{Z}]$ and $\widehat{K} \in \mathcal{L}[\mathcal{Y}, \mathcal{Z}]$ such that $\widehat{L}|_{\mathcal{M}} = L$ and $\widehat{K}|_{\mathcal{N}} = K$ (cf. Theorem 1.1). Thus take $\widehat{\phi}\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{Z}$ given by $\widehat{\phi}(x,y) = \widehat{L}x\,\widehat{K}y$ for every $(x,y) \in \mathcal{X}\times\mathcal{Y}$, which is bilinear (i.e., $\widehat{\phi} \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$). Moreover, $\widehat{\phi}|_{\mathcal{M}\times\mathcal{N}}\colon \mathcal{M}\times\mathcal{N} \to \mathcal{Z}$ is such that $\widehat{\phi}|_{\mathcal{M}\times\mathcal{N}}(u,v) = \widehat{L}|_{\mathcal{M}}u\,\widehat{K}|_{\mathcal{N}}v = Lu\,Kv = \phi(u,v)$ for every $(u,v) \in \mathcal{M}\times\mathcal{N}$. □

The above extension is restricted to bilinear maps that are the product of two linear transformations with values in an algebra. In particular, it applies to bilinear forms that are the product of two linear forms (i.e., two linear functionals). The proof is an application of Theorem 1.1 on extensions of linear transformations. The next result, whose proof follows the argument in the proof of Theorem 1.1, generalizes Theorem 2.4.

Theorem 2.5. *Let $\mathcal{X}$, $\mathcal{Y}$ and $\mathcal{Z}$ be linear spaces over the same field $\mathbb{F}$ and let $\mathcal{M}$ and $\mathcal{N}$ be nonzero linear manifolds of $\mathcal{X}$ and $\mathcal{Y}$. If $\phi\colon \mathcal{M}\times\mathcal{N} \to \mathcal{Z}$ is a bilinear map, then there exists a bilinear extension $\widehat{\phi}\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{Z}$ of ϕ defined on the Cartesian product $\mathcal{X}\times\mathcal{Y}$ of the whole spaces $\mathcal{X}$ and $\mathcal{Y}$.*

Proof. Again, let $\mathcal{L}at(\mathcal{X})$ stand for the lattice of all linear manifolds of a linear space $\mathcal{X}$. Take an arbitrary $\phi \in b[\mathcal{M}\times\mathcal{N}, \mathcal{Z}]$ and set

$$\boldsymbol{\Phi} = \big\{\widehat{\phi} \in b[\mathcal{U}\times\mathcal{V}, \mathcal{Z}]\colon \mathcal{U}\times\mathcal{V} \in \mathcal{L}at(\mathcal{X})\times\mathcal{L}at(\mathcal{Y}),\ \mathcal{M}\subseteq\mathcal{U}, \mathcal{N}\subseteq\mathcal{V},\ \phi = \widehat{\phi}|_{\mathcal{M}\times\mathcal{N}}\big\},$$

the collection of all bilinear maps $\widehat{\phi}$ from the Cartesian product of linear manifolds of $\mathcal{X}$ and $\mathcal{Y}$ to $\mathcal{Z}$ which are extensions of ϕ. Note that $\boldsymbol{\Phi}$ is nonempty (at least ϕ is there). Also, as a subcollection of $\boldsymbol{C} = \bigcup_{M\times N\subseteq\mathcal{X}\times\mathcal{Y}} \mathcal{Z}^{M\times N}$ (the collection of all maps into $\mathcal{Z}$ whose domain $M\times N$ is included in $\mathcal{X}\times\mathcal{Y}$ with M and N being subsets of $\mathcal{X}$ and $\mathcal{Y}$), $\boldsymbol{\Phi}$ is partially ordered in the extension ordering $\leq$ and every chain $\{\widehat{\phi}_\gamma\}$ in $\boldsymbol{\Phi}$ has a supremum $\bigvee_\gamma \widehat{\phi}_\gamma$ in $\boldsymbol{C}$ with domain $D\big(\bigvee_\gamma \widehat{\phi}_\gamma\big) = \bigcup_\gamma D(\widehat{\phi}_\gamma)$ and range $R\big(\bigvee_\gamma \widehat{\phi}_\gamma\big) = \bigcup_\gamma R(\widehat{\phi}_\gamma)$ (see, e.g., [52,

Problem 1.17]). Since $D(\widehat{\phi}_\gamma) \in \mathcal{L}at(\mathcal{X})\times\mathcal{L}at(\mathcal{Y})$ (each $\widehat{\phi}_\gamma$ is a bilinear map on the Cartesian product of linear manifolds of $\mathcal{X}$ and $\mathcal{Y}$), and since $\mathcal{L}at(\mathcal{X})$ and $\mathcal{L}at(\mathcal{Y})$ are complete lattices (i.e., every subset of each of them has a supremum and an infimum), it follows that $D\big(\bigvee_\gamma \widehat{\phi}_\gamma\big) = \bigcup_\gamma D(\widehat{\phi}_\gamma)$ lies in $\mathcal{L}at(\mathcal{X})\times\mathcal{L}at(\mathcal{Y})$. Note: $R\big(\bigvee_\gamma \widehat{\phi}_\gamma\big) = \bigcup_\gamma R(\widehat{\phi}_\gamma)$ is a subset of $\mathcal{Z}$ (not in $\mathcal{L}at(\mathcal{Z})$).

Claim. The supremum $\bigvee_\gamma \widehat{\phi}_\gamma$ lies in $\boldsymbol{\Phi}$.

Proof of Claim. Take any (u_1, v_1) and (u_2, v_2) in $D\big(\bigvee_\gamma \widehat{\phi}_\gamma\big) \in \mathcal{L}at(\mathcal{X})\times\mathcal{L}at(\mathcal{Y})$. Then $(u_1, v_1) \in D(\widehat{\phi}_1)$ for some $\widehat{\phi}_1 \in \{\widehat{\phi}_\gamma\}$ and $(u_2, v_2) \in D(\widehat{\phi}_2)$ for some $\widehat{\phi}_2 \in \{\widehat{\phi}_\gamma\}$. Since $\{\widehat{\phi}_\gamma\}$ is a chain we get $\widehat{\phi}_1 \le \widehat{\phi}_2$ (or vice versa), which implies $D(\widehat{\phi}_1) \subseteq D(\widehat{\phi}_2)$. Thus $\big(\bigvee_\gamma \widehat{\phi}_\gamma\big)|_{D(\widehat{\phi}_2)} = \widehat{\phi}_2$. So $\bigvee_\gamma \widehat{\phi}_\gamma \colon D\big(\bigvee_\gamma \widehat{\phi}_\gamma\big) \to \mathcal{Z}$ is bilinear.

(Indeed, since $D(\widehat{\phi}_1) \subseteq D(\widehat{\phi}_2)$, $(\alpha_1 u_1 + \alpha_2 u_2,\ \beta_1 v_1 + \beta_2 v_2) \in D(\widehat{\phi}_2)$. Since each $\widehat{\phi}_\gamma$ is a bilinear map, $\widehat{\phi}_2(\alpha_1 u_1 + \alpha_2 u_2,\ \beta_1 v_1 + \beta_2 v_2) = \alpha_1\beta_1\widehat{\phi}_2(u_1, v_1) + \alpha_1\beta_2\widehat{\phi}_2(u_1, v_2) + \alpha_2\beta_1\widehat{\phi}_2(u_2, v_1) + \alpha_2\beta_2\widehat{\phi}_2(u_2, v_2)$ for every $\alpha_1, \alpha_2, \beta_1, \beta_2 \in \mathbb{F}$. However, since $\big(\bigvee_\gamma \widehat{\phi}_\gamma\big)|_{D(\widehat{\phi}_2)} = \widehat{\phi}_2$, we get $\big(\bigvee_\gamma \widehat{\phi}_\gamma\big)(\alpha_1 u_1 + \alpha_2 u_2, \beta_1 v_1 + \beta_2 v_2) = \alpha_1\beta_1\big(\bigvee_\gamma \widehat{\phi}_\gamma\big)(u_1, v_1) + \alpha_1\beta_2\big(\bigvee_\gamma \widehat{\phi}_\gamma\big)(u_1, v_2) + \alpha_2\beta_1\big(\bigvee_\gamma \widehat{\phi}_\gamma\big)(u_2, v_1) + \alpha_2\beta_2\big(\bigvee_\gamma \widehat{\phi}_\gamma\big)(u_2, v_2)$ because $\widehat{\phi}_2$ is bilinear. Since (u_1, v_1) and (u_2, v_2) were arbitrarily taken in $D\big(\bigvee_\gamma \widehat{\phi}_\gamma\big) \in \mathcal{L}at(\mathcal{X})\times\mathcal{L}at(\mathcal{Y})$, thus $\bigvee_\gamma \widehat{\phi}_\gamma \colon D\big(\bigvee_\gamma \widehat{\phi}_\gamma\big) \to \mathcal{Z}$ is bilinear.)

Moreover, since each $\widehat{\phi}_\gamma$ is such that $\widehat{\phi}_\gamma|_{\mathcal{M}\times\mathcal{N}} = \phi$, and since $\{\widehat{\phi}_\gamma\}$ is a chain, it also follows that $\big(\bigvee_\gamma \widehat{\phi}_\gamma\big)|_{\mathcal{M}\times\mathcal{N}} = \phi$. Conclusion: $\bigvee_\gamma \widehat{\phi}_\gamma \in \boldsymbol{\Phi}$. □

Therefore every chain in $\boldsymbol{\Phi}$ has a supremum (and so an upper bound) in $\boldsymbol{\Phi}$. Thus according to Zorn's Lemma, $\boldsymbol{\Phi}$ contains a maximal element, say $\widehat{\phi}_0 \colon \mathcal{U}_0\times\mathcal{V}_0 \to \mathcal{Y}$, where $\mathcal{U}_0$ and $\mathcal{V}_0$ are linear manifolds of $\mathcal{X}$ and $\mathcal{Y}$ such that $\mathcal{M} \subseteq \mathcal{U}_0 \subseteq \mathcal{X}$ and $\mathcal{N} \subseteq \mathcal{V}_0 \subseteq \mathcal{Y}$. We will show that $\mathcal{U}_0\times\mathcal{V}_0 = \mathcal{X}\times\mathcal{Y}$, and hence $\widehat{\phi}_0$ is a bilinear extension of ϕ over $\mathcal{X}\times\mathcal{Y}$. The proof goes by contradiction.

$$\text{Suppose } \mathcal{U}_0\times\mathcal{V}_0 \neq \mathcal{X}\times\mathcal{Y} \text{ so that } \mathcal{U}_0 \neq \mathcal{X}, \text{ or } \mathcal{V}_0 \neq \mathcal{Y}, \text{ or both.}$$

(a) To begin with, suppose $\mathcal{U}_0 \neq \mathcal{X}$. Take $x_1 \in \mathcal{X}\backslash\mathcal{U}_0$ so that $x_1 \neq 0$. Consider the sum of $\mathcal{U}_0$ and the one-dimensional linear manifold of $\mathcal{X}$ spanned by $\{x_1\}$,

$$\mathcal{U}_1 = \mathcal{U}_0 + \text{span}\,\{x_1\}, \qquad (*)$$

which properly includes $\mathcal{M}$ (as $\mathcal{M} \subseteq \mathcal{U}_0 \subset \mathcal{U}_1$). Since $\mathcal{U}_0 \cap \text{span}\,\{x_1\} = \{0\}$, every x in $\mathcal{U}_1$ has a unique representation as a sum of vectors in $\mathcal{U}_0$ and in $\text{span}\,\{x_1\}$: for each $x \in \mathcal{U}_1$ there is a unique pair (x_0, α) in $\mathcal{U}_0\times\mathbb{F}$ such that

$$x = x_0 + \alpha x_1.$$

(Indeed, if $x = x_0 + \alpha x_1 = x_0' + \alpha' x_1$, then $x_0 - x_0' = (\alpha' - \alpha)x_1$ lies in $\mathcal{N}_0 \cap \text{span}\,\{x_1\} = \{0\}$ so that $x_0' = x_0$ and $\alpha' = \alpha$ — recall: $x_1 \neq 0$.)

Now let $\mathcal{V}_1$ be a linear manifold of $\mathcal{Y}$ such that $\mathcal{N} \subseteq \mathcal{V}_0 \subseteq \mathcal{V}_1 \subseteq \mathcal{Y}$.

(a_1) First suppose $\mathcal{V}_1 = \mathcal{V}_0$. Consider the map $\widehat{\phi}_1 : \mathcal{U}_1 \times \mathcal{V}_1 \to \mathcal{Z}$ defined by

$$\widehat{\phi}_1(x, y) = \widehat{\phi}_0(x_0, y) \quad \text{for every} \quad (x, y) \in \mathcal{U}_1 \times \mathcal{V}_1 = \mathcal{U}_1 \times \mathcal{V}_0.$$

Since $\widehat{\phi}_0 : \mathcal{U}_0 \times \mathcal{V}_0 \to \mathcal{Z}$ is bilinear and since $\mathcal{U}_0 \subset \mathcal{U}_1$ (proper inclusion), then

$$\widehat{\phi}_1 \text{ is a bilinear map and } \widehat{\phi}_0 = \widehat{\phi}_1|_{\mathcal{U}_0 \times \mathcal{V}_0}, \text{ and therefore } \widehat{\phi}_0 \leq \widehat{\phi}_1.$$

Also, since $\mathcal{M} \subseteq \mathcal{U}_0 \subset \mathcal{U}_1 \subseteq \mathcal{X}$, $\mathcal{N} \subseteq \mathcal{V}_0 = \mathcal{V}_1 \subseteq \mathcal{Y}$, and $\phi = \widehat{\phi}_0|_{\mathcal{M} \times \mathcal{N}}$, then

$$\phi = \widehat{\phi}_0|_{\mathcal{M} \times \mathcal{N}} = \widehat{\phi}_1|_{\mathcal{M} \times \mathcal{N}}.$$

So $\widehat{\phi}_1 \in \boldsymbol{\Phi}$, which contradicts the fact that $\widehat{\phi}_0$ is maximal in $\boldsymbol{\Phi}$ (as $\widehat{\phi}_0 \neq \widehat{\phi}_1$).

(a_2) Next suppose $\mathcal{V}_0 \neq \mathcal{V}_1$ (i.e., $\mathcal{V}_0 \subset \mathcal{V}_1$). Take $y_1 \in \mathcal{V}_1 \backslash \mathcal{V}_0$ (again $y_1 \neq 0$). Then in addition to $(*)$ consider the linear manifold $\mathcal{V}_1{}'$ defined by

$$\mathcal{V}_1{}' = \mathcal{V}_0 + \text{span}\,\{y_1\}, \tag{$**$}$$

which satisfies $\mathcal{N} \subseteq \mathcal{V}_0 \subset \mathcal{V}_1{}' \subseteq \mathcal{V}_1 \subseteq \mathcal{Y}$. In $(*)$ we saw that every $x \in \mathcal{U}_1$ has a unique representation as $x = x_0 + \alpha x_1$ for some $x_0 \in \mathcal{U}_0$ and $\alpha \in \mathbb{F}$. Similarly, from $(**)$ we see that every $y \in \mathcal{V}_1{}'$ has a unique representation as

$$y = y_0 + \beta y_1$$

for some $y_0 \in \mathcal{V}_0$ and $\beta \in \mathbb{F}$. Consider the map $\widehat{\phi}_1' : \mathcal{U}_1 \times \mathcal{V}_1{}' \to \mathcal{Z}$ defined by

$$\widehat{\phi}_1'(x, y) = \widehat{\phi}_0(x_0, y_0) + \alpha\beta z \quad \text{for every} \quad (x, y) \in \mathcal{U}_1 \times \mathcal{V}_1{}'$$

for an arbitrary $z \in \mathcal{Z}$. Again, since $\mathcal{U}_0 \subset \mathcal{U}_1$ and $\mathcal{V}_0 \subset \mathcal{V}_1'$ (proper inclusions),

$$\widehat{\phi}_1' \text{ is bilinear and } \widehat{\phi}_0 = \widehat{\phi}_1'|_{\mathcal{U}_0 \times \mathcal{V}_0}, \text{ and hence } \widehat{\phi}_0 \leq \widehat{\phi}_1',$$

as $\widehat{\phi}_1'$ inherits the bilinearity of $\widehat{\phi}_0$.

(Indeed,

(i) $\widehat{\phi}_1'(x + x', y) = \widehat{\phi}_0(x_0 + x_0', y) + (\alpha + \alpha')\beta z$
$= (\widehat{\phi}_0(x_0, y) + \alpha\beta z) + (\widehat{\phi}_0(x_0', y) + \alpha'\beta z) = \widehat{\phi}_1'(x', y) + \widehat{\phi}_1'(x', y)$,

(ii) $\widehat{\phi}_1'(\delta x, y) = \widehat{\phi}_0(\delta x_0, y_0) + \delta\alpha\beta z = \delta(\widehat{\phi}_0(x_0, y_0) + \alpha\beta z) = \delta\widehat{\phi}_1'(x, y))$,

(iii) $\widehat{\phi}_1'(x, y + y') = \widehat{\phi}_0(x_0, y_0 + y_0') + \alpha(\beta + \beta')z$
$= (\widehat{\phi}_0(x_0, y_0) + \alpha\beta z) + (\widehat{\phi}_0(x_0, y_0') + \alpha\beta' z) = \widehat{\phi}_1'(x, y) + \widehat{\phi}_1'(x, y')$,

(vi) $\widehat{\phi}_1'(x, \delta y) = \widehat{\phi}_0(x_0, \delta y_0) + \alpha\delta\beta z = \delta(\widehat{\phi}_0(x_0, y_0) + \alpha\beta z) = \delta\widehat{\phi}_1'(x, y)$,

and hence $\widehat{\phi}_1' : \mathcal{U}_1 \times \mathcal{V}_1{}' \to \mathcal{Z}$ is bilinear.)

Since $\phi = \widehat{\phi}_0|_{\mathcal{M} \times \mathcal{N}}$ is a bilinear map, and since $\mathcal{M} \subseteq \mathcal{U}_0 \subset \mathcal{U}_1 \subseteq \mathcal{X}$ and $\mathcal{N} \subseteq \mathcal{V}_0 \subset \mathcal{V}_1{}' \subseteq \mathcal{V}_1 \subseteq \mathcal{Y}$,

$$\phi = \widehat{\phi}_0|_{\mathcal{M}\times\mathcal{N}} = \widehat{\phi}'_1|_{\mathcal{M}\times\mathcal{N}}.$$

So $\widehat{\phi}'_1 \in \boldsymbol{\Phi}$, contradicting the fact that $\widehat{\phi}_0$ is maximal in $\boldsymbol{\Phi}$ (as $\widehat{\phi}_0 \neq \widehat{\phi}'_1$).

(b) On the other hand, suppose $\mathcal{V}_0 \neq \mathcal{Y}$. Let $\mathcal{U}_1$ be any linear manifold of $\mathcal{X}$ such that $\mathcal{M} \subseteq \mathcal{U}_0 \subseteq \mathcal{U}_1 \subseteq \mathcal{X}$. If $\mathcal{U}_0 = \mathcal{U}_1$, then we get a case symmetrical to (a_1). If $\mathcal{U}_0 \subset \mathcal{U}_1$, then we get a case symmetrical to (a_2). Both cases then lead to the same consequence as before.

(c) Thus by (a) and (b) $\mathcal{U}_0\times\mathcal{V}_0 \neq \mathcal{X}\times\mathcal{Y}$ leads to a contradiction. □

2.3 Identification with Linear Transformations

Bilinear maps can also be extended by factoring them by the natural bilinear map through a tensor product space. This will be shown in Remark 3.8(a) (as a consequence of Theorem 3.7 — next chapter). Remark 3.8(a) will ensure bilinear extension out of linear extension, yielding a shorter proof for Theorem 2.5. It is, however, advisable to have a proof for bilinear extension independently of any further result, as was shown in Theorem 2.5. Yet there is still another way to prove Theorem 2.5. This will be shown in Remark 2.7(a) below, which depends on the following pivotal theorem.

Theorem 2.6. *Take an arbitrary triple $(\mathcal{X}, \mathcal{Y}, \mathcal{Z})$ of linear spaces over the same field $\mathbb{F}$. The linear spaces $b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ and $\mathcal{L}[\mathcal{X}, \mathcal{L}[\mathcal{Y}, \mathcal{Z}]]$ are isomorphic:*

$$b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] \cong \mathcal{L}[\mathcal{X}, \mathcal{L}[\mathcal{Y}, \mathcal{Z}]].$$

Proof. Take an arbitrary $\phi \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$. For each $x \in \mathcal{X}$ take the x-section $\phi_x \in \mathcal{L}[\mathcal{Y}, \mathcal{Z}]$, which determines a map $L_\phi\colon \mathcal{X} \to \mathcal{L}[\mathcal{Y}, \mathcal{Z}]$ given by

$$L_\phi\, x = \phi_x \quad \text{for every} \quad x \in \mathcal{X},$$

which is clearly linear:

$$L_\phi \in \mathcal{L}[\mathcal{X}, \mathcal{L}[\mathcal{Y}, \mathcal{Z}]] \quad \text{for each} \quad \phi \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}].$$

(Indeed, $L_\phi(x_1+x_2) = \phi_{x_1+x_2} = \phi(x_1+x_2, \cdot) = \phi(x_1, \cdot)+\phi(x_2, \cdot) = \phi_{x_1}+\phi_{x_2} = L_\phi(x_1+x_2)$ and $L_\phi(\alpha x) = \phi_{\alpha x} = \alpha\phi_x = \alpha L_\phi\, x$ for $x, x_1, x_2 \in \mathcal{X}$ and $\alpha \in \mathbb{F}$.)

Now consider the map $\Im\colon b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] \to \mathcal{L}[\mathcal{X}, \mathcal{L}[\mathcal{Y}, \mathcal{Z}]]$ defined by

$$\Im(\phi) = L_\phi \quad \text{for every} \quad \phi \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}].$$

It is easy to verify that this is again a linear transformation:

$$\Im \in \mathcal{L}\big[b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}], \mathcal{L}[\mathcal{X}, \mathcal{L}[\mathcal{Y}, \mathcal{Z}]]\big].$$

(In fact, $L_{\phi+\phi'}\, x = (\phi+\phi')_x = (\phi+\phi')(x, \cdot) = \phi(x, \cdot)+\phi'(x, \cdot) = \phi_x+\phi'_x = L_\phi\, x + L_{\phi'}\, x = (L_\phi + L_{\phi'})\, x$ and $L_{\alpha\phi}\, x = (\alpha\phi)_x = \alpha\phi_x = \alpha L_\phi\, x$ for every

$x \in \mathcal{X}$ and $\alpha \in \mathbb{F}$, and so $\Im(\phi+\phi') = L_{\phi+\phi'} = L_\phi + L_{\phi'} = \Im(\phi)+\Im(\phi')$ and $\Im(\alpha\phi) = L_{\alpha\phi} = \alpha L_\phi = \alpha\Im(\phi)$, for every $\phi, \phi \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ and $\alpha \in \mathbb{F}$.)

It is also readily verified that $\Im$ is injective (for $\Im(\phi) = 0$ implies $L_\phi(x) = 0$ and so $\phi_x = 0$, for every $x \in \mathcal{X}$, which leads to $\phi = 0$, thus ensuring $\mathcal{N}(\Im) = \{0\}$). To show that $\Im$ is surjective, take an arbitrary $L \in \mathcal{L}[\mathcal{X}, \mathcal{L}[\mathcal{Y}, \mathcal{Z}]]$ and define $\phi\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{Z}$ by $\phi(x, y) = (Lx)\, y \in \mathcal{Z}$, which is bilinear (as $\phi^y(\cdot) = (L(\cdot))\, y$ lies in $\mathcal{L}[\mathcal{X}, \mathcal{Z}]$ for each y in $\mathcal{Y}$ and $\phi_x(\cdot) = (Lx)(\cdot)$ lies in $\mathcal{L}[\mathcal{Y}, \mathcal{Z}]$ for each x in $\mathcal{X}$), and hence $L_\phi(x) = \phi_x = (Lx)(\cdot)$ so that $\Im(\phi) = L_\phi = L_\phi(\cdot) = L(\cdot)(\cdot) = L$, and therefore $L \in \mathcal{R}(\Im)$, which implies $\mathcal{R}(\Im) = \mathcal{L}[\mathcal{X}, \mathcal{L}[\mathcal{Y}, \mathcal{Z}]]$. Thus

$$\Im\colon b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] \to \mathcal{L}[\mathcal{X}, \mathcal{L}[\mathcal{Y}, \mathcal{Z}]] \text{ is an isomorphism.} \qquad \square$$

Remark 2.7. *Two Important Consequences of Theorem 2.6.*

(a) A second proof of Theorem 2.5 (based on Theorems 1.1 and 2.6). Take an arbitrary $\phi \in b[\mathcal{M}\times\mathcal{N}, \mathcal{Z}]$. By Theorem 2.6 there exists a linear-space isomorphism $\Im$ of $b[\mathcal{M}\times\mathcal{N}, \mathcal{Z}]$ onto $\mathcal{L}[\mathcal{M}, \mathcal{L}[\mathcal{N}, \mathcal{Z}]]$ whose inverse takes each $L = \Im(\phi) \in \mathcal{L}[\mathcal{M}, \mathcal{L}[\mathcal{N}, \mathcal{Z}]]$ to $\phi = \Im^{-1}(L)$ such that $\phi(u, v) = (Lu)\, v$ for every (u, v) in $\mathcal{M}\times\mathcal{N}$ where $(Lu) \in \mathcal{L}[\mathcal{N}, \mathcal{Z}]$ for each $u \in \mathcal{M}$.

(i) Apply Theorem 1.1 to extend L over $\mathcal{X}$ and get $L^\wedge \in \mathcal{L}[\mathcal{X}, \mathcal{L}[\mathcal{N}, \mathcal{Z}]]$ such that $(L^\wedge x) \in \mathcal{L}[\mathcal{N}, \mathcal{Z}]$ for each $x \in \mathcal{X}$ and $(L^\wedge u) = L^\wedge|_{\mathcal{M}}\, u = (Lu) \in \mathcal{L}[\mathcal{N}, \mathcal{Z}]$ for every $u \in \mathcal{M} \subseteq \mathcal{X}$, and so $\phi(u, v) = (L^\wedge u)v$ for every $(u, v) \in \mathcal{M}\times\mathcal{N}$.

(ii) Next for each $x \in \mathcal{X}$ apply Theorem 1.1 again to extend $(L^\wedge x) \in \mathcal{L}[\mathcal{N}, \mathcal{Z}]$ over $\mathcal{Y}$ and get $\widehat{L}x = \widehat{(L^\wedge x)} \in \mathcal{L}[\mathcal{Y}, \mathcal{Z}]$. This defines a linear transformation $\widehat{L} \in \mathcal{L}[\mathcal{X}, \mathcal{L}[\mathcal{Y}, \mathcal{Z}]]$. Indeed, the map $L^\wedge\colon \mathcal{X} \to \mathcal{L}[\mathcal{N}, \mathcal{Z}]$ is clearly a linear transformation, and the extension operation is also linear (regarded as a map $\widehat{\ }$ of $\mathcal{L}[\mathcal{N}, \mathcal{Z}]$ into the quotient space $\mathcal{L}[\mathcal{Y}, \mathcal{Z}]/\mathcal{M}$ consisting of the equivalence classes $[\widehat{L}]$ of all possible linear extensions $\widehat{L}$ of each linear transformation L in $\mathcal{L}[\mathcal{Y}, \mathcal{Z}]$ as in Remark 1.2). Then $L^\wedge(x_1 + x_2) = (L^\wedge x_1) + (L^\wedge x_2) \in \mathcal{L}[\mathcal{N}, \mathcal{Z}]$, and so $\widehat{L}(x_1 + x_2) = \widehat{(L^\wedge(x_1 + x_2))} = \widehat{(L^\wedge x_1) + (L^\wedge x_2)} = \widehat{(L^\wedge x_1)} + \widehat{(L^\wedge x_2)} = \widehat{L}x_1 + \widehat{L}x_2$ for $x_1, x_2 \in \mathcal{X}$. Similarly, $\widehat{L}(\alpha x) = \alpha\widehat{L}x$ for $x \in \mathcal{X}$ and $\alpha \in \mathbb{F}$.

(iii) So for $\widehat{L} \in \mathcal{L}[\mathcal{X}, \mathcal{L}[\mathcal{Y}, \mathcal{Z}]]$ we get $(\widehat{L}x)(y) \in \mathcal{Z}$ for every $(x, y) \in \mathcal{X}\times\mathcal{Y}$ and also $(\widehat{L}u)(v) = (\widehat{L^\wedge u})(v) = (\widehat{L^\wedge u})|_{\mathcal{N}}(v) = (L^\wedge u)(v) = (Lu)(v) \in \mathcal{Z}$ for every $(u, v) \in \mathcal{M}\times\mathcal{N} \subseteq \mathcal{X}\times\mathcal{Y}$. Thus $\phi(u, v) = (\widehat{L}u)(v)$ for every $(u, v) \in \mathcal{M}\times\mathcal{N}$.

(iv) Finally, set $\widehat{\phi}(x, y) = (\widehat{L}x)(y)$ for every (x, y) in $\mathcal{X}\times\mathcal{Y}$. This defines a map $\widehat{\phi}\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{Z}$ which is clearly bilinear (the same argument as in the proof of Theorem 2.6 once $\widehat{L}\colon \mathcal{X} \to \mathcal{L}[\mathcal{N}, \mathcal{Z}]$ is linear). Therefore $\widehat{\phi} \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ is an extension of $\phi \in b[\mathcal{M}\times\mathcal{N}, \mathcal{Z}]$ over $\mathcal{X}\times\mathcal{Y}$ since $\widehat{\phi}|_{\mathcal{M}\times\mathcal{N}} = \phi$. $\square$

(b) Another consequence (straightforward but helpful) of Theorem 2.6 is this. Take the *permutation map* $\Pi\colon \mathcal{Y}\times\mathcal{X} \to \mathcal{X}\times\mathcal{Y}$ such that $\Pi(y, x) = (x, y)$ and consider a transformation $\wp\colon b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] \to b[\mathcal{Y}\times\mathcal{X}, \mathcal{Z}]$ given by $\wp(\phi) = \phi \circ \Pi$.

In other words, $\psi = \wp\,\phi$ in $b[\mathcal{Y}\times\mathcal{X}, \mathcal{Z}]$ is such that $\psi(y,x) = \phi(\Pi(y,x)) = \phi(x,y)$ for every ϕ in $b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$. As is readily verified, $\wp$ is an isomorphism between the linear spaces $b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ and $b[\mathcal{Y}\times\mathcal{X}, \mathcal{Z}]$. Thus by Theorem 2.6

$$\mathcal{L}[\mathcal{Y}, \mathcal{L}[\mathcal{X}, \mathcal{Z}]] \cong b[\mathcal{Y}\times\mathcal{X}, \mathcal{Z}] \cong b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] \cong \mathcal{L}[\mathcal{X}, \mathcal{L}[\mathcal{Y}, \mathcal{Z}]].$$

Now let $\mathcal{X}^\sharp = \mathcal{L}[\mathcal{X}, \mathbb{F}]$ and $\mathcal{Y}^\sharp = \mathcal{L}[\mathcal{Y}, \mathbb{F}]$ be the algebraic duals of $\mathcal{X}$ and $\mathcal{Y}$. Set $\mathcal{Z} = \mathbb{F}$ above and get an isomorphic representation of the bilinear forms:

$$\mathcal{L}[\mathcal{Y}, \mathcal{X}^\sharp] \cong b[\mathcal{Y}\times\mathcal{X}, \mathbb{F}] \cong b[\mathcal{X}\times\mathcal{Y}, \mathbb{F}] \cong \mathcal{L}[\mathcal{X}, \mathcal{Y}^\sharp].$$

2.4 Additional Propositions

Let $\mathcal{X}$, $\mathcal{Y}$, $\mathcal{Z}$ be nonzero linear spaces over the same field $\mathbb{F}$, let $\cong$ stand for isomorphism, and let $b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ and $\mathcal{L}[\mathcal{X}, \mathcal{Z}]$ denote the linear spaces of all bilinear maps from $\mathcal{X}\times\mathcal{Y}$ to $\mathcal{Z}$ and of all linear transformations of $\mathcal{X}$ into $\mathcal{Z}$.

Proposition 2.A. *$b[\mathcal{X}\times\mathcal{Y}, \mathbb{F}] = \{0\}$ if and only if either $\mathcal{X} = \{0\}$ or $\mathcal{Y} = \{0\}$.*

Under what circumstances is a bilinear map injective? [15, Exercise 1.10].

For an arbitrary $\phi \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ consider the following subset of $\mathcal{X}\times\mathcal{Y}$:

$$N(\phi) = \{(x,y) \in \mathcal{X}\times\mathcal{Y} \colon \phi(x,y) = 0\}.$$

Let $\mathcal{N}(\phi^y)$ and $\mathcal{N}(\phi_x)$ be the kernels of the sections ϕ^y in $\mathcal{L}[\mathcal{X}, \mathcal{Z}]$ and ϕ_x in $\mathcal{L}[\mathcal{Y}, \mathcal{Z}]$, which are linear manifolds of the linear spaces $\mathcal{X}$ and $\mathcal{Y}$, respectively. A bilinear map $\phi \colon \mathcal{X}\times\mathcal{Y} \to \mathcal{Z}$ is injective (i.e., one-to-one) if for every z in the range $R(\phi) \subseteq \mathcal{Z}$ of ϕ there is a unique pair (x,y) in $\mathcal{X}\times\mathcal{Y}$ such that $z = \phi(x,y)$.

Proposition 2.B. *Take an arbitrary $\phi \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$.*

(a) $N(\phi) = \bigcup_{x\in\mathcal{X}} \big(\{x\}\times\mathcal{N}(\phi_x)\big) \cup \bigcup_{y\in\mathcal{Y}} \big(\mathcal{N}(\phi^y)\times\{y\}\big)$.

(b) *If ϕ is injective, then $N(\phi) = \{(0,0)\}$.*

(c) *If one of the linear spaces $\mathcal{X}$ or $\mathcal{Y}$ is nonzero, then $N(\phi) \neq \{(0,0)\}$.*

Proposition 2.C. *Take an arbitrary bilinear map $\phi \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$. Consider the linear space direct sum $\mathcal{X}\oplus\mathcal{Y} = (\mathcal{X}\times\mathcal{Y}, \oplus, \cdot, \mathbb{F})$. Let $\Pi_X : \mathcal{X}\oplus\mathcal{Y} \to \mathcal{X}\oplus\mathcal{Y}$ and $\Pi_Y : \mathcal{X}\oplus\mathcal{Y} \to \mathcal{X}\oplus\mathcal{Y}$ be given by $\Pi_X(x,y) = (0,y)$ and $\Pi_Y(x,y) = (x,0)$ for every $(x,y) \in \mathcal{X}\oplus\mathcal{Y}$. These are linear projections with $\mathcal{R}(\Pi_X) = \{0\}\oplus\mathcal{Y}$ and $\mathcal{R}(\Pi_Y) = \mathcal{X}\oplus\{0\}$. Since $\mathcal{X}\oplus\{0\}$ and $\{0\}\oplus\mathcal{Y}$ are naturally isomorphic to $\mathcal{X}$ and $\mathcal{Y}$, write (i.e., identify) $\Pi_X(x,y) = y$ and $\Pi_Y(x,y) = x$ for every $(x,y) \in \mathcal{X}\oplus\mathcal{Y}$. The map $\psi \colon \mathcal{X}\oplus\mathcal{Y} \to \mathcal{Z}$ given by*

$$\psi(x,y) = \phi(\Pi_Y(x,y), \Pi_X(x,y)) = \phi(x,y) \in \mathcal{Z},$$

for every $x \in \mathcal{X}$ and $y \in \mathcal{Y}$, is bilinear (i.e., $\psi \in b[\mathcal{X}\oplus\mathcal{Y}, \mathcal{Z}]$). Moreover,

$$\psi((x_1,y_1)\oplus(x_2,y_2))+\psi((x_1,y_1)\ominus(x_2,y_2))=2\psi(x_1,y_1)+2\psi(x_2,y_2),$$
$$\psi(\alpha(x,y))=\alpha^2\psi(x,y),$$

for every $x,x_1,x_2\in\mathcal{X}$, *every* $y,y_1,y_2\in\mathcal{Y}$, *and every* $\alpha\in\mathbb{F}$, *where the symbol* $\ominus$ *means* $(x_1,y_1)\ominus(x_2,y_2)=(x_1,y_1)\oplus(-x_2,-y_2)\in\mathcal{X}\oplus\mathcal{Y}$. *In particular,*

$$\psi((x,y)\oplus(x,y))=\psi(2(x,y))=4\psi(x,y).$$

Proposition 2.D. *Take* $\phi\in b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$, *let* $\mathcal{M}$ *and* $\mathcal{N}$ *be linear manifolds of* $\mathcal{X}$ *and* $\mathcal{Y}$, *consider the quotient spaces* $\mathcal{X}/\mathcal{M}$ *and* $\mathcal{Y}/\mathcal{N}$, *and take the quotient maps* $\pi_{_X}\in\mathcal{L}[\mathcal{X},\mathcal{X}/\mathcal{M}]$ *and* $\pi_{_Y}\in\mathcal{L}[\mathcal{Y},\mathcal{Y}/\mathcal{N}]$. *Now consider both sections* $\phi^v=\phi(\cdot,v)$ *in* $\mathcal{L}[\mathcal{X},\mathcal{Z}]$ *for each* $v\in\mathcal{N}$ *and* $\phi_u=\phi(u,\cdot)$ *in* $\mathcal{L}[\mathcal{Y},\mathcal{Z}]$ *for each* $u\in\mathcal{M}$. *Next take the linear manifold* $\mathcal{R}=\operatorname{span}R(\phi)\subseteq\mathcal{Z}$ *and the quotient map* $\pi_{_Z}$ *in* $\mathcal{L}[\mathcal{Z},\mathcal{Z}/\mathcal{R}]$. *Since* $\pi_{_Z}$ *is surjective, define* $\psi\colon\mathcal{X}/\mathcal{M}\times\mathcal{Y}/\mathcal{N}\to\mathcal{Z}/\mathcal{R}$ *by*

$$\psi(\pi_{_X}x,\pi_{_Y}y)=\pi_{_Z}\phi(x,y)$$

for every $([x],[y])=(\pi_{_X}x,\pi_{_Y}y)\in\mathcal{X}/\mathcal{M}\times\mathcal{Y}/\mathcal{N}$. *The map* ψ *is bilinear.*

Take $\phi\in b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$. Let $\mathcal{N}(\phi^y)$ and $\mathcal{N}(\phi_x)$ be the kernels of the linear transformations $\phi^y\in\mathcal{L}[\mathcal{X},\mathcal{Z}]$ and $\phi_x\in\mathcal{L}[\mathcal{Y},\mathcal{Z}]$. Consider the linear manifolds

$$\mathcal{N}^{\mathcal{Y}}(\phi)=\bigcap\nolimits_{y\in\mathcal{Y}}\mathcal{N}(\phi^y)\subseteq\mathcal{X}\quad\text{ and }\quad\mathcal{N}_{\mathcal{X}}(\phi)=\bigcap\nolimits_{x\in\mathcal{X}}\mathcal{N}(\phi_x)\subseteq\mathcal{Y}.$$

Proposition 2.E. *Take* $\phi\in b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$. *If* $\mathcal{M}=\mathcal{N}^{\mathcal{Y}}(\phi)$, $\mathcal{N}=\mathcal{N}_{\mathcal{X}}(\phi)$, *and if* $\psi\colon\mathcal{X}/\mathcal{M}\times\mathcal{Y}/\mathcal{N}\to\mathcal{Z}/\mathcal{R}$ *is the bilinear map defined in Proposition 2.D, then*

$$\mathcal{N}^{\mathcal{Y}}(\psi)=\{0\}\quad\textit{and}\quad\mathcal{N}_{\mathcal{X}}(\psi)=\{0\}.$$

The next propositions are isolated restatements of part of the isometric chain in Remark 2.7(b) as they will be required in the sequel.

Proposition 2.F. $b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]\cong b[\mathcal{Y}\times\mathcal{X},\mathcal{Z}]$.

Proposition 2.G. $\mathcal{L}[\mathcal{X},\mathcal{L}[\mathcal{Y},\mathcal{Z}]]\cong\mathcal{L}[\mathcal{Y},\mathcal{L}[\mathcal{X},\mathcal{Z}]]$.

We have already seen some examples of bilinear maps and bilinear forms. For instance, the product of linear forms (i..e., $\phi(x,y)=f(x)\,g(y)$) and the inner product in any real inner product space (i.e., $(x,y)\mapsto\langle x\,;y\rangle\in\mathbb{R}$) are common examples of bilinear forms, as we saw in Remark 2.3. The real inner product offers a large source of concrete examples of bilinear forms. The product of a linear transformation with a linear form is a bilinear map (e.g., $(x,y)\mapsto f(x)\,Ky$ is a bilinear map whenever f is a linear form and K is a linear transformation). If $\mathcal{Z}$ is an algebra, then the product of linear transformations is a bilinear map (i.e., $\phi(x,y)=Kx\,Ky$ is a bilinear map if L,K are linear transformations taking values on the same algebra $\mathcal{Z}$ — as in Theorem 2.4). Also, the composition of a bilinear map with a linear transformation

is a bilinear map (i.e., $(x,y) \mapsto L\big(\phi(x,y)\big)$ is bilinear if ϕ is a $\mathcal{Z}$-valued bilinear map and L is a linear transformation on $\mathcal{Z}$), and the composition of two linear transformations with a bilinear map is a bilinear map (i.e., $(x,y) \mapsto \phi\big(Lx, Ky\big)$ is bilinear if ϕ is bilinear and L, K are linear transformations), as we saw in Remark 2.2. Moreover, since $b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ is a linear space, every (finite) linear combination of bilinear maps yields a bilinear map $(x,y) \mapsto \sum_{i=1}^{n} \alpha_i \phi_i(x,y)$. The following propositions exhibit four more classical examples.

Proposition 2.H. *The* evaluation *map* $\phi\colon \mathcal{L}[\mathcal{X},\mathcal{Y}]\times\mathcal{X} \to \mathcal{Y}$ *defined by*

$$\phi(L,x) = Lx$$

for every $L \in \mathcal{L}[\mathcal{X},\mathcal{Y}]$ *and every* $x \in \mathcal{X}$ *is bilinear.*

Proposition 2.I. *The* convolution *map*

$$(f,g) \mapsto f \star g$$

is bilinear. In other words, if L^1 *denotes the linear space of all integrable real-valued functions on the real line, then the map* $\phi\colon L^1\times L^1 \to L^1$ *given by* $\phi(f,g)(\tau) = (f\star g)(\tau) = \int f(\tau - t)\, g(t)\, dt$ *for every* $f, g \in L^1$ *is a bilinear.*

Proposition 2.J. *The* composition *map*

$$(L,F) \mapsto L \circ F$$

is bilinear. Precisely, let $\mathcal{X}, \mathcal{Y}$ *be linear spaces* (*over the same field, of course*) *and let* S *be a nonempty set. Take the linear space* $\mathcal{L}[\mathcal{X},\mathcal{Y}]$ *of all linear transformations of* $\mathcal{X}$ *into* $\mathcal{Y}$*, and the linear spaces* $\mathcal{X}^S$ *and* $\mathcal{Y}^S$ *of all functions from* S *to* $\mathcal{X}$ *and to* $\mathcal{Y}$*, respectively. The map* $\phi\colon \mathcal{L}[\mathcal{X},\mathcal{Y}]\times\mathcal{X}^S \to \mathcal{Y}^S$ *such that* $\phi(L,F) = L\circ F$ *for every* $L \in \mathcal{L}[\mathcal{X},\mathcal{Y}]$ *and* $F \in \mathcal{X}^S$ *is bilinear.*

Proposition 2.K. *Take the linear space* $\mathbb{F}^S$ *of all* $\mathbb{F}$*-valued functions on a nonempty set* S*. Let* $\mathcal{Y}$ *be a linear space* (*over the field* $\mathbb{F}$) *and take the linear space* $\mathcal{Y}^S$ *of all* $\mathcal{Y}$*-valued functions on* S*. The map* $\phi\colon \mathbb{F}^S\times\mathcal{Y} \to \mathcal{Y}^S$ *defined by*

$$\phi(f,y) = f(\cdot)y$$

for every $f \in \mathbb{F}^S$ *and every* $y \in \mathcal{Y}$ *is bilinear. In particular, by setting* $S = \mathcal{X}$ (*a linear space over* $\mathbb{F}$) *and replacing* $\mathbb{F}^S$ *by* $\mathcal{X}^\sharp \subset \mathbb{F}^{\mathcal{X}}$*, then the bilinear map* $\phi\colon \mathcal{X}^\sharp\times\mathcal{Y} \to \mathcal{Y}^{\mathcal{X}}$ *such that* $\phi(f,y) = f(\cdot)y$ *for* $f \in \mathcal{X}^\sharp$ *and* $y \in \mathcal{Y}$ *is such that* $R(\phi) \subseteq \mathcal{L}[\mathcal{X},\mathcal{Y}]$ (*i.e.,* $x \in \mathcal{X} \mapsto f(x)y \in \operatorname{span}\{y\} \subseteq \mathcal{Y}$ *is linear for every* $y \in \mathcal{Y}$ *and every* $f \in \mathcal{X}^\sharp$)*. In other words,*

$$\phi \in b[\mathcal{X}^\sharp\times\mathcal{Y}, \mathcal{L}[\mathcal{X},\mathcal{Y}]].$$

The next propositions exhibit four special examples required in Chapter 4.

Proposition 2.L. *For each pair* (x,y) *in the Cartesian product* $\mathcal{X}\times\mathcal{Y}$*, consider the transformation* $L_{x,y}\colon b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] \to \mathcal{Z}$ *defined by*

$$L_{x,y}(\psi) = \psi(x,y) \in \mathcal{Z} \quad \text{for every} \quad \psi \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}].$$

(a) *Each* $L_{x,y}$ *is linear*: $L_{x,y} \in \mathcal{L}[\,b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}],\mathcal{Z}]$ *for each* $(x,y)\in\mathcal{X}\times\mathcal{Y}$.

Now consider the map $\theta\colon \mathcal{X}\times\mathcal{Y}\to\mathcal{L}[\,b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}],\mathcal{Z}]$ *given by*

$$\theta(x,y) = L_{x,y} \in \mathcal{L}[\,b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}],\mathcal{Z}] \quad \text{for every} \quad (x,y)\in\mathcal{X}\times\mathcal{Y}.$$

(b) *This* θ *is bilinear*: $\theta \in b[\mathcal{X}\times\mathcal{Y},\, \mathcal{L}[\,b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}],\mathcal{Z}]]$.

The particular case of $\mathcal{Z}=\mathbb{F}$ *leads to* $\theta \in b[\mathcal{X}\times\mathcal{Y},\, b[\mathcal{X}\times\mathcal{Y},\mathbb{F}]^\sharp]$.

Proposition 2.M. *For each pair* (x,y) *in the Cartesian product* $\mathcal{X}\times\mathcal{Y}$*, consider the map* $\vartheta'_{x,y}\colon \mathcal{X}^\sharp\times\mathcal{Y}^\sharp\to\mathbb{F}$ *defined by*

$$\vartheta'_{x,y}(f,g) = f(x)\,g(y) \in \mathbb{F} \quad \text{for every} \quad (f,g)\in\mathcal{X}^\sharp\times\mathcal{Y}^\sharp.$$

(a) *Each* $\vartheta'_{x,y}$ *is bilinear*: $\vartheta'_{x,y}\in b[\mathcal{X}^\sharp\times\mathcal{Y}^\sharp,\mathbb{F}]$ *for each* $(x,y)\in\mathcal{X}\times\mathcal{Y}$.

Now consider the map $\theta'\colon \mathcal{X}\times\mathcal{Y}\to b[\mathcal{X}^\sharp\times\mathcal{Y}^\sharp,\mathbb{F}]$ *given by*

$$\theta'(x,y) = \vartheta'_{x,y} \in b[\mathcal{X}^\sharp\times\mathcal{Y}^\sharp,\mathbb{F}] \quad \text{for every} \quad (x,y)\in\mathcal{X}\times\mathcal{Y}.$$

(b) *This* θ' *is bilinear*: $\theta'\in b[\mathcal{X}\times\mathcal{Y},\, b[\mathcal{X}^\sharp\times\mathcal{Y}^\sharp,\mathbb{F}]]$.

Proposition 2.N. *For each pair* (x,y) *in the Cartesian product* $\mathcal{X}\times\mathcal{Y}$*, consider the map* $L''_{x,y}\colon \mathcal{X}^\sharp\to\mathcal{Y}$ *defined by*

$$L''_{x,y}(f) = f(x)y \quad \text{for every} \quad f\in\mathcal{X}^\sharp.$$

(a) *Each* $L''_{x,y}$ *is linear*: $L''_{x,y}\in\mathcal{L}[\mathcal{X}^\sharp,\mathcal{Y}]$ *for each* $f\in\mathcal{X}^\sharp$.

Now consider the map $\theta''\colon \mathcal{X}\times\mathcal{Y}\to\mathcal{L}[\mathcal{X}^\sharp,\mathcal{Y}]$ *given by*

$$\theta''(x,y) = L''_{x,y} \in \mathcal{L}[\mathcal{X}^\sharp,\mathcal{Y}] \quad \text{for every} \quad (x,y)\in\mathcal{X}\times\mathcal{Y}.$$

(b) *This* θ'' *is bilinear*: $\theta''\in b[\mathcal{X}\times\mathcal{Y},\, \mathcal{L}[\mathcal{X}^\sharp,\mathcal{Y}]]$.

Proposition 2.O. *For each pair* (x,y) *in the Cartesian product* $\mathcal{X}\times\mathcal{Y}$*, consider the map* $L'''_{x,y}\colon \mathcal{Y}^\sharp\to\mathcal{X}$ *defined by*

$$L'''_{x,y}(g) = g(y)x \quad \text{for every} \quad g\in\mathcal{Y}^\sharp.$$

(a) *Each* $L'''_{x,y}$ *is linear*: $L'''_{x,y}\in\mathcal{L}[\mathcal{Y}^\sharp,\mathcal{X}]$ *for each* $g\in\mathcal{Y}^\sharp$.

Now consider the map $\theta'''\colon \mathcal{X}\times\mathcal{Y}\to\mathcal{L}[\mathcal{Y}^\sharp,\mathcal{X}]$ *given by*

$$\theta'''(x,y) = L'''_{x,y} \in \mathcal{L}[\mathcal{Y}^\sharp,\mathcal{X}] \quad \text{for every} \quad (x,y)\in\mathcal{X}\times\mathcal{Y}.$$

(b) *This* θ''' *is bilinear*: $\theta\in b[\mathcal{X}\times\mathcal{Y},\, \mathcal{L}[\mathcal{Y}^\sharp,\mathcal{X}]]$.

We know from Theorem 2.5 (or Remark 2.7(a)) that bilinear maps have a bilinear extension. This ensures that conditions (a) and (b) in the next proposition hold when the sets M and N are replaced by linear manifolds $\mathcal{M}$ and $\mathcal{N}$.

Proposition 2.P. *Let M and N be nonempty subsets of $\mathcal{X}$ and $\mathcal{Y}$, respectively. Take an arbitrary map $\phi\colon M\times N \to \mathcal{Z}$ of the Cartesian product of M and N into a linear space $\mathcal{Z}$. There exists a bilinear map $\widehat{\phi} \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ such that $\widehat{\phi}|_{M\times N} = \phi$ if and only if the following two conditions hold true:*

(a) $\sum_i \alpha_i u_i = 0$ *for a finite subset $\{u_i\}$ of M and a similarly indexed set of scalars $\{\alpha_i\}$ implies $\sum_i \alpha_i \phi(u_i, v) = 0$ for every $v \in N$, and*

(b) $\sum_j \beta_j v_j = 0$ *for a finite subset $\{v_j\}$ of N and a similarly indexed set of scalars $\{\beta_j\}$ implies $\sum_j \beta_j \phi(u, v_j) = 0$ for every $u \in M$.*

A bilinear form $\phi\colon \mathcal{X}\times\mathcal{X} \to \mathbb{F}$ from the Cartesian product of a linear space $\mathcal{X}$ with itself to a field $\mathbb{F}$ (in our case $\mathbb{F}$ is either $\mathbb{R}$ or $\mathbb{C}$) is *symmetric* if $\phi(x, y) = \phi(y, x)$ for every $x, y \in \mathcal{X}$. A *quadratic form of a bilinear form* is a functional $\varrho_\phi\colon \mathcal{X} \to \mathbb{F}$ obtained from a symmetric bilinear form ϕ as follows:

$$\varrho_\phi(x) = \phi(x, x) \quad \text{for every} \quad x \in \mathcal{X}.$$

A symmetric bilinear form ϕ can be written in terms of the corresponding quadratic form ϱ_ϕ as (see, e.g., [30, Exercise 6, Section 23], [6, Problem 3.P])

$$\phi(x, y) = \tfrac{1}{4}\big(\varrho_\phi(x + y) - \varrho_\phi(x - y)\big) \quad \text{for every} \quad x, y \in \mathcal{X}.$$

If $\mathbb{F} = \mathcal{R}$, then the above identity is referred to as the *real polarization identity*.

A *sesquilinear form* $\sigma\colon \mathcal{X}\times\mathcal{Y} \to \mathbb{F}$ is a functional from the Cartesian product of linear spaces $\mathcal{X}$ and $\mathcal{Y}$ to a field $\mathbb{F}$ ($\mathbb{R}$ or $\mathbb{C}$) such that one section is linear and the other is conjugate linear — hence the terminology: "sesqui" means "one-and-a-half". Thus we define a sesquilinear form as being additive in both arguments, homogeneous in the first argument, and conjugate homogeneous in the second argument. Therefore the y-section $\sigma^y = \sigma(\cdot, y)\colon \mathcal{X} \to \mathbb{F}$ is a linear form for every $y \in \mathcal{Y}$, and the complex conjugate of the x-section $\overline{\sigma}_x = \overline{\sigma(x, \cdot)}\colon \mathcal{Y} \to \mathbb{F}$ is a linear form for every $x \in \mathcal{X}$. That is,

$$\sigma^y \in \mathcal{X}^\sharp = \mathcal{L}[\mathcal{X}, \mathbb{F}] \quad \text{and} \quad \overline{\sigma}_x \in \mathcal{Y}^\sharp = \mathcal{L}[\mathcal{Y}, \mathbb{F}].$$

A sesquilinear form $\sigma\colon \mathcal{X}\times\mathcal{X} \to \mathbb{F}$ from the Cartesian product of a linear space $\mathcal{X}$ with itself to a field $\mathbb{F}$ is *Hermitian symmetric* if $\sigma(x, y) = \overline{\sigma(y, x)}$ for every $x, y \in \mathcal{X}$. A *quadratic form of a sesquilinear form*, $\varrho_\sigma\colon \mathcal{X} \to \mathbb{F}$, is a functional obtained from a Hermitian symmetric sesquilinear form σ as follows:

$$\varrho_\sigma(x) = \sigma(x, x) \quad \text{for every} \quad x \in \mathcal{X}.$$

If $\mathbb{F} = \mathbb{C}$, then the Hermitian symmetric sesquilinear form σ can be written in terms of the corresponding quadratic form ϱ_σ according to the expression

$$\sigma(x, y) = \tfrac{1}{4}\big(\varrho_\sigma(x+y) - \varrho_\sigma(x-y) + i\varrho_\sigma(x+iy) - i\varrho_\sigma(x-iy)\big) \quad \text{for every} \quad x, y \in \mathcal{X}.$$

In this case the above identity is called the *complex polarization identity*. A quadratic form of a Hermitian symmetric sesquilinear form is real-valued (i.e., $\varrho_\sigma(x) = \sigma(x, x) \in \mathbb{R}$ for every $x \in \mathcal{X}$ — see, e.g., [6, Problem 3.P]).

If $\mathbb{F}=\mathbb{R}$, then the notions of bilinear and sesquilinear forms, and of symmetric bilinear and Hermitian symmetric sesquilinear forms, coincide.

An *inner product* $\langle\cdot\,;\cdot\rangle\colon \mathcal{X}\times\mathcal{X}\to\mathbb{F}$ defined on the Cartesian product $\mathcal{X}\times\mathcal{X}$ of a linear space $\mathcal{X}$ with itself to a field $\mathbb{F}$ (either $\mathbb{R}$ or $\mathbb{C}$) is a Hermitian symmetric sesquilinear form that induces a *positive quadratic form* (i.e., $\varrho_\sigma(x)=\sigma(x,x)=\langle x\,;x\rangle>0$ for every $0\neq x\in\mathcal{X}$), and in this case the square root of the positive quadratic form generated by the inner product is a norm on $\mathcal{X}$ (i.e., $\varrho_\sigma(x)=\sigma(x,x)=\langle x\,;x\rangle=\|x\|^2$ for every $x\in\mathcal{X}$).

The above paragraphs on symmetric bilinear forms, quadratic forms, sesquilinear forms, Hermitian symmetric sesquilinear forms, positive quadratic forms, and inner product constitute a basis for the discussion in Remark 2.3(b). We will return to inner product in Section 5.4. The next proposition deals with restrictions of sesquilinear forms in light of the results for restrictions of bilinear maps in Proposition 2.P.

Proposition 2.Q. *Let S be a subset of $\mathcal{X}$ spanning $\mathcal{X}$ (i.e.,* $\operatorname{span} S=\mathcal{X}$*) and take an $\mathbb{F}$-valued function $\sigma\colon\mathcal{X}\times\mathcal{X}\to\mathbb{F}$ on the Cartesian product of $\mathcal{X}$ with itself. Take the subset $S\times S$ of $\mathcal{X}\times\mathcal{X}$. There exists a sesquilinear form $\widehat{\sigma}\colon\mathcal{X}\times\mathcal{X}\to\mathbb{F}$ such that $\widehat{\sigma}|_{S\times S}=\sigma$ if and only if the following condition holds*:

> $\sum_i\alpha_i s_i=0$ *for a finite subset* $\{s_i\}$ *of* S *and a finite set of scalars* $\{\alpha_i\}$ *implies* $\sum_i\alpha_i\sigma(s_i,s)=0$ *and* $\sum_i\overline{\alpha}_i\sigma(s_i,s)=0$ *for every* $s\in S$.

In this case, $\widehat{\sigma}$ is the unique extension of σ. Also, $\widehat{\sigma}$ is Hermitian symmetric if and only if σ is (i.e., if and only if $\sigma(r,s)=\overline{\sigma(s,r)}$ for every $r,s\in S$).

Notes: Bilinear maps play *the* central role in defining tensor products, the topic of the next chapter. The present section focuses on basic additional results and offers a collection of classical examples of bilinear maps acting on the Cartesian product of linear spaces. Proposition 2.A, whose proof follows easily from Proposition 1.G in Section 1.4, is fundamental. Proposition 2.B shows that there is no nontrivial injective bilinear map.

Proof of Proposition 2.A. Suppose $b[\mathcal{X}\times\mathcal{Y},\mathbb{F}]=\{0\}$. Set $\phi_{f,g}=f\cdot g\in b[\mathcal{X}\times\mathcal{Y},\mathbb{F}]$ for each pair $f\in\mathcal{X}^\sharp$ and $g\in\mathcal{Y}^\sharp$. Fix an arbitrary $f_0\in\mathcal{X}^\sharp$. If $\mathcal{Y}\neq\{0\}$, then for every $0\neq y\in\mathcal{Y}$ there exists $g_y\in\mathcal{Y}^\sharp$ such that $g_y(y)\neq 0$ by Proposition 1.G(ii). If $\phi_{f_0,g_y}(x,y)=f_0(x)g_y(y)=0$ for every $y\in\mathcal{Y}$, then $f_0(x)=0$ for every $x\in\mathcal{X}$ so that $f_0=0$. Since f_0 is arbitrary, this implies that every $f\in\mathcal{X}^\sharp$ is null, and so $\mathcal{X}=\{0\}$ by Proposition 1.G(ii) again. The converse is trivial. □

Proof of Proposition 2.B. Recall: $\phi^y(x)=\phi(x,y)=\phi_x(y)$ for every (x,y) in $\mathcal{X}\times\mathcal{Y}$. Take an arbitrary $x\in\mathcal{X}$. If $y\in\mathcal{N}(\phi_x)$, then $\phi_x(y)=0$. So $\phi(x,y)=0$ and hence $(x,y)\in N(\phi)$. Thus $\{x\}\times\mathcal{N}(\phi_x)\subseteq N(\phi)$ for every $x\in\mathcal{X}$. Therefore $\bigcup_{x\in\mathcal{X}}\big(\{x\}\times\mathcal{N}(\phi_x)\big)\subseteq N(\phi)$. Similarly, $\bigcup_{y\in\mathcal{Y}}\big(\mathcal{N}(\phi^y)\times\{y\}\big)\subseteq N(\phi)$. Then

$$\bigcup\nolimits_{x\in\mathcal{X}}\big(\{x\}\times\mathcal{N}(\phi_x)\big)\cup\bigcup\nolimits_{y\in\mathcal{Y}}\big(\mathcal{N}(\phi^y)\times\{y\}\big)\subseteq N(\phi).$$

Conversely, if $(x,y) \in N(\phi)$, then for this $x \in \mathcal{X}$ we get $y \in \mathcal{N}(\phi_x)$ so that (x,y) lies in $\{x\}\times\mathcal{N}(\phi_x)$, which trivially is a subset of $\bigcup_{x\in\mathcal{X}} \big(\{x\}\times\mathcal{N}(\phi_x)\big)$. Symmetrically, (x,y) also lies in $\bigcup_{y\in\mathcal{Y}} \big(\mathcal{N}(\phi^y)\times\{y\}\big)$. This proves (a). Item (b) is readily verified since $\phi(0,0) = 0$. Item (c) follows from (a) since $N(\phi) = \{(0,0)\}$ implies $\mathcal{X} = \mathcal{Y} = \{0\}$ by (a). In fact, if $N(\phi) = \{(0,0)\}$, then $\phi(x,y) \neq 0$ for every $y \in \mathcal{Y}$ whenever $x \neq 0$. So $\phi(x,0) = \phi_x(0) \neq 0$ for every $0 \neq x \in \mathcal{X}$, which is a contradiction since ϕ_x is a linear transformation for each $x \in \mathcal{X}$. Thus $N(\phi) = \{(0,0)\}$ implies $\mathcal{X} = \{0\}$. Symmetrically, $N(\phi) = \{(0,0)\}$ implies $\mathcal{Y} = \{0\}$. This gives another proof for (c). □

Propositions 2.C and 2.D deal with composition of bilinear maps with respect to direct sum (extending the discussion in Remark 2.3) and quotient spaces, respectively (see, e.g., [27, Problem 1.2 and Section 1.2]). (The expression $\psi(x,y) = \phi(x,y)$ in Proposition 2.C may seem tautological, but it isn't: $\phi((x_1,y_1)+(x_2,y_2))$ is meaningless as there is no addition in $\mathcal{X}\times\mathcal{Y}$, but $\psi((x_1,y_1)\oplus(x_2,y_2))$ is well-defined in $\mathcal{X}\oplus\mathcal{Y}$ — compare the last expression in Proposition 2.C with Remark 2.3(a)). For Proposition 2.E see, e.g., [27, Problem 1.5]. As said before, Propositions 2.F and 2.G are isolated restatements picked up from the chain of isomorphisms in Remark 2.7(b) as they will be required in the sequel, which are immediate consequences of using the permutation $\Pi\colon \mathcal{Y}\times\mathcal{X} \to \mathcal{X}\times\mathcal{Y}$ in connection with Theorem 2.6. The examples in Propositions 2.H, 2.I and 2.K (see, e.g., [27, Problem 1.3.3] and [15, Sections 1.3(2), 1.3(4), 1.3(5), p. 9]), and in Proposition 2.J, are also readily verified. Propositions 2.L to 2.O will be verified in Section 4.3 and applied in Theorem 4.3 and Corollaries 4.5, 4.7, and 4.8. Proposition 2.P is a counterpart for the Cartesian product of sets of the extension of bilinear maps for the Cartesian product of linear manifolds in Theorem 2.5 (cf. [5, Problem 2.R]). Proposition 2.Q is a version of it concerning sesquilinear forms (cf. [5, Problem 2.R]).

Suggested Readings

Brown and Pearcy [5]
Defant and Floret [15]
Greub [27]
Halmos [30]
MacLane and Birkhoff [69]
Roman [81]



3

Algebraic Tensor Product

This chapter presents a unified view of algebraic tensor products wherein the so-called universal property is taken as an axiomatic starting point, rather than as a theorem for a specific realization, and this approach leads to an abstract notion of algebraic tensor products of linear spaces. The concrete standard forms will be shown in the next chapter to be interpretations of the axiomatic formulation. Tensor products are axiomatically defined as follows.

3.1 Tensor Product of Linear Spaces

Definition 3.1. Let $\mathcal{X}$ and $\mathcal{Y}$ be nonzero linear spaces over the same field $\mathbb{F}$. A *tensor product of* $\mathcal{X}$ *and* $\mathcal{Y}$ is a pair $(\mathcal{T}, \theta)$ consisting of a linear space $\mathcal{T}$ over $\mathbb{F}$ and a bilinear map $\theta\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{T}$ fulfilling the following two axioms.

(a) The range $R(\theta) = \theta(\mathcal{X}\times\mathcal{Y})$ of the bilinear map $\theta \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{T}]$ spans $\mathcal{T}$:

$$\operatorname{span} R(\theta) = \mathcal{T} \qquad (\text{i.e., } \operatorname{span} \theta(\mathcal{X}\times\mathcal{Y}) = \mathcal{T}).$$

(b) If $\phi \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ is an arbitrary bilinear map into an arbitrary linear space $\mathcal{Z}$ over $\mathbb{F}$, then there is a linear transformation $\Phi \in \mathcal{L}[\mathcal{T}, \mathcal{Z}]$ for which

$$\phi = \Phi \circ \theta.$$

That is, the diagram

$$\begin{array}{ccc} \mathcal{X}\times\mathcal{Y} & \xrightarrow{\ \phi\ } & \mathcal{Z} \\ & {\scriptstyle\theta}\searrow \quad \uparrow{\scriptstyle\Phi} & \\ & \mathcal{T} & \end{array}$$

commutes, and so the map θ factors every bilinear map through $\mathcal{T}$.

A word on terminology. The linear space $\mathcal{T}$ is referred to as a *tensor product space of* $\mathcal{X}$ *and* $\mathcal{Y}$ associated with the bilinear map θ, which in turn is referred

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to as the *natural bilinear map* (or simply the *natural map*) associated with the linear space $\mathcal{T} = \operatorname{span} R(\theta) = \operatorname{span} \theta(\mathcal{X}\times\mathcal{Y})$.

Consider the natural bilinear map $\theta\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{T} = \operatorname{span} \theta(\mathcal{X}\times\mathcal{Y})$ associated with $\mathcal{T}$. The value $\theta(x,y)$ of θ at a pair (x,y) in $\mathcal{X}\times\mathcal{Y}$ is denoted by $x \otimes y$,

$$x \otimes y = \theta(x,y) \in \mathcal{T} \quad \text{for every} \quad (x,y) \in \mathcal{X}\times\mathcal{Y},$$

and $x \otimes y$ is called a *single tensor* (or a *simple tensor*, or an *elementary tensor*, or a *decomposable element*, or a *single tensor product of x and y*) in the tensor product space $\mathcal{T}$. Take $(x,y) \in \mathcal{X}\times\mathcal{Y}$ and $\alpha \in \mathbb{F}$ arbitrarily. Since θ is a bilinear map, $\alpha\theta(x,y) = \theta(\alpha x, y) = \theta(x, \alpha y)$; in particular, $\theta(x,y) = \theta(\alpha x, \frac{1}{\alpha} y) = \theta(\frac{1}{\alpha} x, \alpha y)$ for $\alpha \neq 0$. Thus a multiple of a single tensor is again a single tensor,

$$\alpha(x \otimes y) = (\alpha x) \otimes y = x \otimes (\alpha y),$$

and the representation of any single tensor is not unique. For instance,

$$x \otimes y = (\alpha x) \otimes (\tfrac{1}{\alpha} y) = (\tfrac{1}{\alpha} x) \otimes (\alpha y)$$

for every nonzero $\alpha \in \mathbb{F}$. In fact, finite sums of single tensors have uncountably many representations. For example, $x \otimes y = \alpha x \otimes \frac{1}{\alpha} y + u \otimes v - \beta u \otimes \frac{1}{\beta} v$, where $u \otimes v - \beta u \otimes \frac{1}{\beta} v = 0 \otimes 0$, for every $x, u \in \mathcal{X}$, every $y, v \in \mathcal{Y}$, and every nonzero $\alpha, \beta \in \mathbb{F}$. So the origin $0 \in R(\theta) \subseteq \operatorname{span} R(\theta) = \mathcal{T}$ of the linear space $\mathcal{T}$ can be represented by $0 \otimes 0 = x \otimes 0 = 0 \otimes y$ for every $x \in \mathcal{X}$ and every $y \in \mathcal{Y}$. Also

$$(x_1 + x_2) \otimes (y_1 + y_2) = (x_1 \otimes y_1) + (x_1 \otimes y_2) + (x_2 \otimes y_1) + (x_2 \otimes y_2)$$

for every $x_1, x_2 \in \mathcal{X}$ and $y_1, y_2 \in \mathcal{Y}$, since $x \otimes y$ is the value of a bilinear map $\theta\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{Z}$ at a pair $(x,y) \in \mathcal{X}\times\mathcal{Y}$, where the above sums are linear-space additions, in $\mathcal{X}$ and $\mathcal{Y}$ on the left-hand side and in $\mathcal{Z}$ on the right-hand side.

Remark 3.2. *Existence and Interpretations.*

(a) Given a pair $(\mathcal{X}, \mathcal{Y})$ of linear spaces over the same field, there exists a linear space $\mathcal{T}$ and a bilinear map $\theta \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{T}]$ such that the pair $(\mathcal{T}, \theta)$ is a tensor product of $\mathcal{X}$ and $\mathcal{Y}$ as in Definition 3.1. In other words, for every pair $(\mathcal{X}, \mathcal{Y})$ of linear spaces over the same field there are examples (or interpretations) of tensor product spaces $\mathcal{T}$ of $\mathcal{X}$ and $\mathcal{Y}$.

(b) Some of these will be exhibited in Chapter 4. So the proof of the existence of tensor products will be postponed till then. (The simplest example is presented in Proposition 4.B.) If $(\mathcal{T}, \theta)$ and $(\mathcal{T}', \theta')$ are interpretations of tensor products for a given pair $(\mathcal{X}, \mathcal{Y})$ of linear spaces, then these interpretations are isomorphic, as will be shown in Theorem 3.5 and Corollary 3.6.

(c) For every bilinear map $\phi \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ there exists a linear transformation $\Phi \in \mathcal{L}[\mathcal{T}, \mathcal{Z}]$ such that $\phi = \Phi \circ \theta$ whenever $(\mathcal{T}, \theta)$ is a tensor product of $\mathcal{X}$ and $\mathcal{Y}$ (Definition 3.1). If there is a bilinear map ϕ for which the associated linear transformation Φ is injective, then $\Phi\colon \mathcal{T} \hookrightarrow \mathcal{Z}$ embeds the tensor product space $\mathcal{T}$ in the linear space $\mathcal{Z}$. (This will be shown in Corollary 4.10; next chapter.)

The following two theorems and their corollaries establish the first properties of tensor products of linear spaces, viz., representation and uniqueness.

Theorem 3.3. *An element of a tensor product space is represented as a finite sum of single tensors* (*and such a representation is not unique*)*:*

$$F \in \mathcal{T} \quad \Longleftrightarrow \quad F = \sum\nolimits_i x_i \otimes y_i \quad (\textit{a finite sum}).$$

Proof. This follows from axiom (a) in Definition 3.1. Indeed, $\sum_{i=1}^n x_i \otimes y_i = \sum_{i=1}^n \theta(x_i, y_i)$ lies in $\mathcal{T}$ for every finite collection $\{(x_i, y_i)\}_{i=1}^n$ of pairs (x_i, y_i) in $\mathcal{X}\times\mathcal{Y}$ since $R(\theta) \subseteq \mathcal{T}$ and $\mathcal{T}$ is a linear space. Conversely, if F lies in $\mathcal{T} = \operatorname{span} R(\theta)$, then $F = \sum_{i=1}^n \alpha_i \theta(x_i, y_i) = \sum_{i=1}^n \theta(\alpha_i x_i, y_i) = \sum_{i=1}^n (\alpha_i x_i) \otimes y_i$ because θ is bilinear and $\theta(\alpha_i x_i, y_i)$ is the single tensor $(\alpha_i x_i) \otimes y_i$. As we saw before, the representation of a finite sum of single tensors is not unique. □

Corollary 3.4. *If $\mathcal{T}$ is a tensor product space, then the linear transformation $\Phi \in \mathcal{L}[\mathcal{T}, \mathcal{Z}]$ associated with each bilinear map $\phi \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ in axiom* (*b*) *of Definition 3.1 is unique.*

Proof. Consider Definition 3.1. If for a given ϕ in $b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ there are Φ_1 and Φ_2 in $\mathcal{L}[\mathcal{T}, \mathcal{Z}]$ such that $\Phi_1 \circ \theta = \Phi_2 \circ \theta = \phi$, then $(\Phi_1 - \Phi_2) \circ \theta = 0$, so that $(\Phi_1 - \Phi_2)\big(\theta(x, y)\big) = 0$ for every $x, y \in \mathcal{X}\times\mathcal{Y}$. Consequently $(\Phi_1 - \Phi_2)(F) = (\Phi_1 - \Phi_2)\big(\sum_i \theta(x_i, y_i)\big) = \sum_i (\Phi_1 - \Phi_2)\,\theta(x_i, y_i) = 0$ for every $F \in \mathcal{T}$ (since $\Phi_1 - \Phi_2$ is linear and $F = \sum_i \theta(x_i, y_i)$ by Theorem 3.3). Hence $\Phi_1 = \Phi_2$. □

The natural bilinear map θ associated with a tensor product space $\mathcal{T}$ is unique and, conversely, the tensor product space $\mathcal{T}$ associated with a natural bilinear map θ is unique. Thus θ is the unique bilinear map that factors every bilinear map through a tensor product space $\mathcal{T}$. This is shown in the next theorem and its corollary. All linear spaces are over the same field.

Theorem 3.5. *If $(\mathcal{T}, \theta)$ and $(\mathcal{T}', \theta')$ are tensor products of linear spaces $\mathcal{X}$ and $\mathcal{Y}$, then there is a unique isomorphism $\Theta \in \mathcal{L}[\mathcal{T}', \mathcal{T}]$ such that*

$$\Theta \circ \theta' = \theta.$$

That is, Θ is such that the following diagram commutes:

$$\begin{array}{ccc} \mathcal{X}\times\mathcal{Y} & \xrightarrow{\ \theta\ } & \mathcal{T} \\ & {\scriptstyle\theta'}\searrow & \uparrow{\scriptstyle\Theta} \\ & & \mathcal{T}'. \end{array}$$

Proof. Let $(\mathcal{T}, \theta)$ and $(\mathcal{T}', \theta')$ be tensor products of linear spaces $\mathcal{X}$ and $\mathcal{Y}$. Take an arbitrary bilinear map $\phi\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{Z}$ into any linear space $\mathcal{Z}$. Thus according to Definition 3.1 there are linear transformations $\Phi\colon \mathcal{T} \to \mathcal{Z}$ and $\Phi'\colon \mathcal{T}' \to \mathcal{Z}$ such that $\phi = \Phi \circ \theta = \Phi' \circ \theta'$, which means the diagrams

$$\begin{array}{ccc} \mathcal{X}\times\mathcal{Y} & \xrightarrow{\phi} & \mathcal{Z} \\ & {\scriptstyle\theta}\searrow \quad \uparrow{\scriptstyle\Phi} & \\ & \mathcal{T} & \end{array} \qquad \text{and} \qquad \begin{array}{ccc} \mathcal{X}\times\mathcal{Y} & \xrightarrow{\phi} & \mathcal{Z} \\ & {\scriptstyle\theta'}\searrow \quad \uparrow{\scriptstyle\Phi'} & \\ & \mathcal{T}' & \end{array}$$

commute. Since $\theta' \colon \mathcal{X}\times\mathcal{Y} \to \mathcal{T}'$ and $\theta \colon \mathcal{X}\times\mathcal{Y} \to \mathcal{T}$ are bilinear maps, Definition 3.1 again ensures the existence of linear transformations $\Theta' \colon \mathcal{T} \to \mathcal{T}'$ and $\Theta \colon \mathcal{T}' \to \mathcal{T}$ such that $\theta' = \Theta' \circ \theta$ and $\theta = \Theta \circ \theta'$, which means the diagrams

$$\begin{array}{ccc} \mathcal{X}\times\mathcal{Y} & \xrightarrow{\theta'} & \mathcal{T}' \\ & {\scriptstyle\theta}\searrow \quad \uparrow{\scriptstyle\Theta'} & \\ & \mathcal{T} & \end{array} \qquad \text{and} \qquad \begin{array}{ccc} \mathcal{X}\times\mathcal{Y} & \xrightarrow{\theta} & \mathcal{T} \\ & {\scriptstyle\theta'}\searrow \quad \uparrow{\scriptstyle\Theta} & \\ & \mathcal{T}' & \end{array}$$

commute. Hence

$$\theta' = (\Theta' \circ \Theta) \circ \theta' \quad \text{and} \quad \theta = (\Theta \circ \Theta') \circ \theta.$$

Let I_S be the identity function on an arbitrary set S. By the above equations,

$$\Theta' \circ \Theta|_{R(\theta')} = I_{R(\theta')} \colon R(\theta') \to R(\theta') \quad \text{and} \quad \Theta \circ \Theta'|_{R(\theta)} = I_{R(\theta)} \colon R(\theta) \to R(\theta).$$

Since Θ and Θ' are linear transformations (and so are their compositions), and since $\operatorname{span} R(\theta) = \mathcal{T}$ and $\operatorname{span} R(\theta') = \mathcal{T}'$ by Definition 3.1, then we get

$$\Theta' \circ \Theta = \Theta' \circ \Theta|_{\operatorname{span} R(\theta')} = I_{\operatorname{span} R(\theta')} = I_{\mathcal{T}'},$$

$$\Theta \circ \Theta' = \Theta \circ \Theta'|_{\operatorname{span} R(\theta)} = I_{\operatorname{span} R(\theta)} = I_{\mathcal{T}}.$$

Hence each linear transformation Θ and Θ' is the inverse of the other (thus isomorphisms), and Θ (and so Θ') is unique according Corollary 3.4. □

Corollary 3.6. *A tensor product of linear spaces is unique up to an isomorphism in the following sense. If $(\mathcal{T}, \theta)$ and $(\mathcal{T}', \theta')$ are tensor products of the same pair of linear spaces, then $(\mathcal{T}, \theta) = (\Theta\mathcal{T}', \Theta\,\theta')$ for an isomorphism Θ in $\mathcal{L}[\mathcal{T}', \mathcal{T}]$. In particular, two tensor product spaces of the same pair of linear spaces coincide if and only if the natural bilinear maps coincide.*

Proof. This is a restatement of Theorem 3.5. Uniqueness up to isomorphism is immediate from Theorem 3.5. By uniqueness of the isomorphism Θ in Theorem 3.5, if $\mathcal{T} = \mathcal{T}'$ then Θ is the identity map and so $\theta = \theta'$, and vice versa. □

There exists a tensor product for every pair of linear spaces (Remark 3.2). A tensor product for a pair of linear spaces is unique up to an isomorphism (Corollary 3.6). Then for any pair $(\mathcal{X}, \mathcal{Y})$ of linear spaces it is common to write

$$\mathcal{T} = \mathcal{X} \otimes \mathcal{Y}.$$

So one refers to $\mathcal{X} \otimes \mathcal{Y}$ as *the* tensor product space (and to $(\mathcal{X} \otimes \mathcal{Y}, \theta)$ as *the* tensor product) of any pair of linear spaces $\mathcal{X}$ and $\mathcal{Y}$. Thus if $\mathcal{M}$ and $\mathcal{N}$ are linear manifolds of $\mathcal{X}$ and $\mathcal{Y}$, then $\mathcal{M} \otimes \mathcal{N}$ is a tensor product space. Actually,

$$\mathcal{M} \otimes \mathcal{N} \text{ is a linear manifold of } \mathcal{X} \otimes \mathcal{Y}.$$

(Indeed, if $F = \sum_i u_i \otimes v_i \in \mathcal{M} \otimes \mathcal{N} \subseteq \mathcal{X} \otimes \mathcal{Y}$ for $u_i \in \mathcal{M} \subseteq \mathcal{X}$ and $v_i \in \mathcal{N} \subseteq \mathcal{Y}$, then so does the sum of two such elements, as well as its product by a scalar.) Moreover, it will be shown in Theorem 3.13 that $\mathcal{M} \otimes \mathcal{N} = \text{span } R(\theta|_{\mathcal{M}\times\mathcal{N}})$.

3.2 Further Properties of Tensor Product Spaces

Each bilinear map on the Cartesian product $\mathcal{X}\times\mathcal{Y}$ is in a one-to-one correspondence with a linear transformation on the tensor product space $\mathcal{X} \otimes \mathcal{Y}$. In fact, the linear space of all bilinear maps $b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ is isomorphic to the linear space of all linear transformations $\mathcal{L}[\mathcal{T}, \mathcal{Z}]$. Thus a crucial property of a tensor product $(\mathcal{T}, \theta)$ is to *linearize bilinear maps* as in Theorem 3.7 below (and this is done via factorization by θ through $\mathcal{T}$ as in Definition 3.1).

Theorem 3.7. *Take an arbitrary triple $(\mathcal{X}, \mathcal{Y}, \mathcal{Z})$ of linear spaces. The linear spaces $b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ and $\mathcal{L}[\mathcal{T}, \mathcal{Z}]$ are isomorphic. That is, with $\mathcal{T} = \mathcal{X} \otimes \mathcal{Y}$,*

$$b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] \cong \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{Z}].$$

Proof. Take an arbitrary $\Phi \in \mathcal{L}[\mathcal{T}, \mathcal{Z}]$. The composition $\Phi \circ \theta \colon \mathcal{X}\times\mathcal{Y} \to \mathcal{Z}$ lies in $b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ since θ is bilinear and Φ is linear (composition of a bilinear map with a linear transformation is a bilinear map). Then consider the transformation $C_\theta \colon \mathcal{L}[\mathcal{T}, \mathcal{Z}] \to b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ defined by

$$C_\theta(\Phi) = \Phi \circ \theta \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] \quad \text{for every} \quad \Phi \in \mathcal{L}[\mathcal{T}, \mathcal{Z}].$$

According to Corollary 3.4, for every ϕ in $b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ there exists one and only one Φ in $\mathcal{L}[\mathcal{T}, \mathcal{Z}]$ for which $\phi = \Phi \circ \theta$. Then C_θ is injective and surjective, thus invertible. Moreover, C_θ is linear. In fact,

$$C_\theta(\Phi_1 + \Phi_2) = (\Phi_1 + \Phi_2) \circ \theta = (\Phi_1 \circ \theta) + (\Phi_2 \circ \theta) = C_\theta(\Phi_1) + C_\theta(\Phi_2),$$

$$C_\theta(\alpha\Phi) = (\alpha\Phi) \circ \theta = \alpha(\Phi \circ \theta) = \alpha C_\theta(\Phi),$$

for every $\Phi, \Phi_1, \Phi_2 \in \mathcal{L}[\mathcal{T}, \mathcal{Z}]$ and $\alpha \in \mathbb{F}$. Therefore C_θ is an isomorphism. □

Remark 3.8. *Two Relevant Consequences of Theorem 3.7.*

(a) A third proof of Theorem 2.5 (based on Theorem 1.1 and Theorem 3.7). Theorem 2.5 says that every bilinear map on the Cartesian product of linear manifolds of a pair of linear spaces can be extended to a bilinear map on the Cartesian product of the linear spaces. Remark 2.7(a) gave a second proof of Theorem 2.5, and it was also suggested after proving Theorem 2.5 that still another proof can be obtained via Theorem 3.7. Here it is.

Let $\mathcal{M}$ and $\mathcal{N}$ be linear manifolds of linear spaces $\mathcal{X}$ and $\mathcal{Y}$. Let $\phi \colon \mathcal{M}\times\mathcal{N} \to \mathcal{Z}$ be a bilinear map into a linear space $\mathcal{Z}$. Consider the isomorphism C_θ

of $\mathcal{L}[\mathcal{M} \otimes \mathcal{N}, \mathcal{Z}]$ onto $b[\mathcal{M}\times\mathcal{N}, \mathcal{Z}]$ such that ϕ in $b[\mathcal{M}\times\mathcal{N}, \mathcal{Z}]$ and Φ in $\mathcal{L}[\mathcal{M} \otimes \mathcal{N}, \mathcal{Z}]$ are uniquely related by $\phi = C_\theta(\Phi) = \Phi \circ \theta$, as in the proof of Theorem 3.7, where θ in $b[\mathcal{M}\times\mathcal{N}, \mathcal{M} \otimes \mathcal{N}]$ is the natural bilinear map associated with $\mathcal{M} \otimes \mathcal{N}$. Since $\mathcal{M} \otimes \mathcal{N}$ is a linear manifold of the linear space $\mathcal{X} \otimes \mathcal{Y}$, extend Φ over $\mathcal{X} \otimes \mathcal{Y}$ using Theorem 1.1 to get $\widehat{\Phi}$ in $\mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{Z}]$ such that $\widehat{\Phi}|_{\mathcal{M}\otimes\mathcal{N}} = \Phi$. Set $\widehat{\phi} = C_{\widehat{\theta}}(\widehat{\Phi}) = \widehat{\Phi} \circ \widehat{\theta}$ where $\widehat{\theta}$ in $b[\mathcal{X}\times\mathcal{Y}, \mathcal{X} \otimes \mathcal{Y}]$ is the natural bilinear map associated with $\mathcal{X} \otimes \mathcal{Y}$. Since restrictions of bilinear maps to Cartesian products of linear manifolds are bilinear (Remark 2.1), and since there is a unique natural bilinear map associated with a tensor product space (Corollary 3.6), then $\theta = \widehat{\theta}|_{\mathcal{M}\times\mathcal{N}}$. Thus $\widehat{\phi}$ extends ϕ over $\mathcal{X}\times\mathcal{Y}$ since

$$\widehat{\phi}|_{\mathcal{M}\times\mathcal{N}} = (\widehat{\Phi} \circ \widehat{\theta})|_{\mathcal{M}\times\mathcal{N}} = \widehat{\Phi} \circ (\widehat{\theta}|_{\mathcal{M}\times\mathcal{N}}) = \widehat{\Phi}|_{\mathcal{M}\otimes\mathcal{N}} \circ \theta = \Phi \circ \theta = \phi.$$

(b) For a second consequence, take a tensor product $(\mathcal{T}, \theta)$ of $\mathcal{X}$ and $\mathcal{Y}$. Let $\mathcal{T}^\sharp = \mathcal{L}[\mathcal{T}, \mathbb{F}]$ be the algebraic dual of the tensor product space $\mathcal{T}$. Set $\mathcal{Z} = \mathbb{F}$ in Theorem 3.7 and get an isomorphic representation of the algebraic dual $\mathcal{T}^\sharp = (\mathcal{X} \otimes \mathcal{Y})^\sharp$ of the tensor product space $\mathcal{T} = \mathcal{X} \otimes \mathcal{Y}$ of $\mathcal{X}$ and $\mathcal{Y}$, viz.,

$$(\mathcal{X} \otimes \mathcal{Y})^\sharp \cong b[\mathcal{X}\times\mathcal{Y}, \mathbb{F}] \qquad (\cong \mathcal{L}[\mathcal{X}, \mathcal{Y}^\sharp] \cong \mathcal{L}[\mathcal{Y}, \mathcal{X}^\sharp] \text{ by Remark 2.7(b)}).$$

Next take the Cartesian product $\mathcal{X}^\sharp\times\mathcal{Y}^\sharp$ of the dual spaces $\mathcal{X}^\sharp = \mathcal{L}[\mathcal{X}, \mathbb{F}]$ and $\mathcal{Y}^\sharp = \mathcal{L}[\mathcal{Y}, \mathbb{F}]$ of $\mathcal{X}$ and $\mathcal{Y}$, and let $(\mathcal{T}_\sharp, \theta_\sharp)$ be a tensor product of $\mathcal{X}^\sharp$ and $\mathcal{Y}^\sharp$. Thus applying Theorem 3.7, again with $\mathcal{Z} = \mathbb{F}$ but now for the tensor product space $\mathcal{T}_\sharp = \mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp$ of $\mathcal{X}^\sharp$ and $\mathcal{Y}^\sharp$, we get (as above)

$$(\mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp)^\sharp \cong b[\mathcal{X}^\sharp\times\mathcal{Y}^\sharp, \mathbb{F}].$$

Theorem 3.9. *Let $\mathcal{T} = \mathcal{X} \otimes \mathcal{Y}$ be the tensor product space of $\mathcal{X}$ and $\mathcal{Y}$, let E and D be nonempty subsets of $\mathcal{X}$ and $\mathcal{Y}$ respectively, and set*

$$\top_{E,D} = \{x \otimes y \in \mathcal{X} \otimes \mathcal{Y}\colon\ x \in E \text{ and } y \in D\}.$$

(a) *If* $\operatorname{span} E = \mathcal{X}$ *and* $\operatorname{span} D = \mathcal{Y}$, *then* $\operatorname{span} \top_{E,D} = \mathcal{X} \otimes \mathcal{Y}$.

(b) *If E and D are linearly independent, then $\top_{E,D}$ is linearly independent.*

Therefore if E is a Hamel basis for $\mathcal{X}$ and D is a Hamel basis for $\mathcal{Y}$, then $\top_{E,D}$ is a Hamel basis for $\mathcal{T} = \mathcal{X} \otimes \mathcal{Y}$.

Proof. (a) Take an arbitrary element F from $\mathcal{X} \otimes \mathcal{Y}$ (cf. Theorem 3.3),

$$F = \sum\nolimits_{i=1}^{n} x_i \otimes y_i.$$

If $\operatorname{span} E = \mathcal{X}$ and $\operatorname{span} D = \mathcal{Y}$ then consider any expansion of each $x_i \in \mathcal{X}$ in terms of vectors $e_{i,j}$ from E and any expansion of each $y_i \in \mathcal{Y}$ in terms of vectors $d_{i,k}$ from D. Since $x \otimes y = \theta(x, y)$ for each pair $(x, y) \in \mathcal{X}\times\mathcal{Y}$, and since the map $\theta\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{X} \otimes \mathcal{Y}$ is bilinear, we get

$$F = \sum\nolimits_{i=1}^{n} \Big(\sum\nolimits_{j=1}^{m} \beta_j e_{i,j} \otimes \sum\nolimits_{k=1}^{\ell} \gamma_k d_{i,k} \Big) = \sum\nolimits_{i,j,k}^{n,m,\ell} \beta_j \gamma_k (e_{i,j} \otimes d_{i,k}).$$

Thus F lies in $\operatorname{span} \top_{E,D}$. Then $\mathcal{X} \otimes \mathcal{Y} \subseteq \operatorname{span} \top_{E,D}$. So $\operatorname{span} \top_{E,D} = \mathcal{X} \otimes \mathcal{Y}$.

(b) Let $E' = \{e_j\}_{j=1}^{m}$ and $D' = \{d_k\}_{k=1}^{\ell}$ be arbitrary nonempty finite linearly independent subsets of E and D respectively. Consider the linear manifolds

$$\mathcal{M} = \operatorname{span} E' = \operatorname{span} \{e_j\}_{j=1}^{m} \subseteq \mathcal{X} \quad \text{and} \quad \mathcal{N} = \operatorname{span} D' = \operatorname{span} \{d_k\}_{k=1}^{\ell} \subseteq \mathcal{Y}$$

spanned by them. Set $\mathcal{Z} = \mathcal{L}[\mathbb{F}^m, \mathbb{F}^\ell]$, which is identified with the linear space of all $\ell \times m$ matrices of entries in $\mathbb{F}$. Take the map $\phi\colon \mathcal{M} \times \mathcal{N} \to \mathcal{Z}$ given for every $u = \sum_{j=1}^{m} \beta_j e_j$ in $\mathcal{M}$ and every $v = \sum_{k=1}^{\ell} \gamma_k d_k$ in $\mathcal{N}$ by

$$\phi(u, v) = \big(\beta_j \gamma_k\big) \in \mathcal{Z} \quad \text{for} \quad j = 1, \ldots, m \ \text{ and } \ k = 1, \ldots, \ell,$$

where $\big(\beta_j \gamma_k\big)$ is the $\ell \times m$ matrix whose entries are the products $\beta_j \gamma_k$ of the coefficients of the unique expansion of arbitrary vectors $u \in \mathcal{M}$ and $v \in \mathcal{N}$ in terms of the linearly independent sets E' and D'. This $\phi\colon \mathcal{M} \times \mathcal{N} \to \mathcal{Z}$ is bilinear.

(Indeed, $\phi(u + u', v) = \phi(u, v) + \phi(u', v)$, $\phi(u, v + v') = \phi(u, v) + \phi(u, v')$, $\alpha\phi(u, v) = \phi(\alpha u, v) = \phi(u, \alpha v)$ for every $u, u' \in \mathcal{M}$, $v, v' \in \mathcal{N}$ and $\alpha \in \mathbb{F}$).

So, according to axiom (b) in Definition 3.1, consider the linear transformation $\Phi\colon \mathcal{M} \otimes \mathcal{N} \to \mathcal{Z}$ such that $\phi = \Phi \circ \theta$. Then $\phi(u, v) = \Phi\,(\theta(u, v)) = \Phi(u \otimes v)$ for every $u = \sum_{j=1}^{m} \beta_j e_j \in \mathcal{M}$ and $v = \sum_{k=1}^{\ell} \gamma_k d_k \in \mathcal{N}$. In particular,

$$\Phi(e_j \otimes d_k) = \phi(e_j, d_k) = \Pi_{j,k},$$

where $\Pi_{j,k} \in \mathcal{Z}$ is the $\ell \times m$ matrix whose entry at position j, k is 1 and all other entries are 0. These matrices form a linearly independent set in $\mathcal{Z}$. (In fact, $\{\Pi_{j,k}\}_{j,k=1}^{m,\ell}$ is the canonical Hamel basis for $\mathcal{Z}$.) Take any pair of integers k', j' and suppose $e_{j'} \otimes d_{k'}$ is a linear combination of the remaining single tensors $\{e_j \otimes d_k\}_{j,k \in I'}$ with $I' = \{j, k = 1 \text{ to } m, \ell\colon\ j \neq j', k \neq k'\}$, say

$$e_{j'} \otimes d_{k'} = \sum\nolimits_{j,k \in I'} \delta_{j,k}\, (e_j \otimes d_k).$$

Then, as $\Phi\colon \mathcal{M} \otimes \mathcal{N} \to \mathcal{Z}$ is linear,

$$\Pi_{j',k'} = \Phi(e_{j'} \otimes d_{k'}) = \sum\nolimits_{j,k \in I'} \delta_{j,k} \Phi(e_j \otimes d_k) = \sum\nolimits_{j,k \in I'} \delta_{j,k} \Pi_{j,k},$$

and so $\delta_{j,k} = 0$ for every $j, k \in I'$ since $\{\Pi_{j,k}\}_{j,k=1}^{m,\ell}$ is linearly independent in $\mathcal{Z}$. Thus $\{e_j \otimes d_k\}_{j,k=1}^{m,\ell}$ is linearly independent in $\top_{E,D}$. In other words,

$$\top_{E',D'} = \{x \otimes y \in \mathcal{X} \otimes \mathcal{Y}\colon\ x \in E' \text{ and } y \in D'\}$$

is a finite linearly independent subset of $\top_{E,D}$ whenever E' and D' are finite linearly independent subsets of E and D. Therefore if E and D are linearly independent subsets of $\mathcal{X}$ and $\mathcal{Y}$, then every finite subset of each of them is

trivially linearly independent, and so is every finite subset of $\top_{E,D}$ as we saw above. But if every finite subset of $\top_{E,D}$ is linearly independent, then so is $\top_{E,D}$ (cf. Proposition 1.A in Section 1.4). □

Corollary 3.10. $\dim(\mathcal{X} \otimes \mathcal{Y}) = \dim \mathcal{X} \cdot \dim \mathcal{Y}$.

Proof. Let E and D be Hamel bases for $\mathcal{X}$ and $\mathcal{Y}$. Thus $\top_{E,D}$ is a Hamel basis for $\mathcal{T}$ by Theorem 3.9. Therefore $\top_{E,D} = \{x \otimes y \in \mathcal{X} \otimes \mathcal{Y}\colon\ x \in E \text{ and } y \in D\}$ is in a one-to-one correspondence with the Cartesian product $E \times D$.

(In fact, if $x \otimes y \in \top_{E,D}$, then (i) x in E is unique for a fixed y in D as E is linearly independent and, (ii) y in D is unique for a fixed x in E as D is linearly independent — cf. Propositions 3.B and 3.C in Section 3.4.)

As $\#(E \times D) = \#E \cdot \#D$ by definition of product of cardinal numbers (where $\#$ stands for cardinality; see, e.g., [52, Problem 1.30]), $\#\top_{E,D} = \#E \cdot \#D$. So we get the claimed the identity (by definition of dimension). □

Corollary 3.11. (a) *Tensor product is commutative up to an isomorphism,*

$$\mathcal{X} \otimes \mathcal{Y} \cong \mathcal{Y} \otimes \mathcal{X}.$$

(b) *Isomorphic linear spaces yield isomorphic tensor product spaces:*

$$\mathcal{X}' \cong \mathcal{X} \quad \textit{and} \quad \mathcal{Y}' \cong \mathcal{Y} \quad \Longrightarrow \quad \mathcal{X}' \otimes \mathcal{Y}' \cong \mathcal{X} \otimes \mathcal{Y}.$$

(c) *The converse holds if $\mathcal{X}'$ and $\mathcal{Y}'$ are embedded in $\mathcal{X}$ and $\mathcal{Y}$, respectively.*

$$\mathcal{X}' \otimes \mathcal{Y}' \cong \mathcal{X} \otimes \mathcal{Y} \quad \Longrightarrow \quad \mathcal{X}' \cong \mathcal{X} \quad \textit{and} \quad \mathcal{Y}' \cong \mathcal{Y}.$$

Proof. These are straightforward consequences of the dimension identity of Corollary 3.10, since two linear spaces are isomorphic if and only if they have the same dimension (cf. Proposition 1.C in Section 1.4), and the dimension of an embedded linear space is clearly not greater than the dimension of the linear space it is embedded in. □

Since we are identifying isomorphic tensor products after Corollary 3.6, we might write $\mathcal{X} \otimes \mathcal{Y} = \mathcal{Y} \otimes \mathcal{X}$ and $\mathcal{X}' \otimes \mathcal{Y}' = \mathcal{X} \otimes \mathcal{Y}$ in Corollary 3.11, but interpreting $\cong$ as $=$ in this particular case should perhaps be avoided.

Remark 3.12. *Isomorphism, Duality, and Finite-Dimensional Spaces.*

(a) $\mathcal{X} \cong \mathcal{X}^\sharp,\ \mathcal{Y} \cong \mathcal{Y}^\sharp,\ \mathcal{X} \otimes \mathcal{Y} \cong (\mathcal{X} \otimes \mathcal{Y})^\sharp \Longrightarrow \mathcal{X} \otimes \mathcal{Y} \cong \mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp \cong (\mathcal{X} \otimes \mathcal{Y})^\sharp$.

by Corollary 3.11(b). In this particular case Remark 3.8(b) ensures that

$$\mathcal{X} \otimes \mathcal{Y} \cong \mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp \cong (\mathcal{X} \otimes \mathcal{Y})^\sharp \cong b[\mathcal{X}, \mathcal{Y}, \mathbb{F}].$$

(b) If $\mathcal{X}$ and $\mathcal{Y}$ are finite-dimensional, then so are their algebraic duals $\mathcal{X}^\sharp$ and $\mathcal{Y}^\sharp$, and a linear space and its dual have the same finite dimension. Moreover, dimensions coincide if and only if the linear spaces are isomorphic (cf. Propositions 1.C and 1.V in Section 1.4). Then

$\dim \mathcal{X} < \infty, \dim \mathcal{Y} < \infty \Rightarrow \dim \mathcal{X} = \dim \mathcal{X}^\sharp$, $\dim \mathcal{Y} = \dim \mathcal{Y}^\sharp \Leftrightarrow \mathcal{X} \cong \mathcal{X}^\sharp, \mathcal{Y} \cong \mathcal{Y}^\sharp$.

Thus Corollary 3.10 ensures that

$$\dim \mathcal{X} \otimes \mathcal{Y} = \dim \mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp = \dim(\mathcal{X} \otimes \mathcal{Y})^\sharp = \dim(\mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp)^\sharp,$$

or equivalently,

$$\mathcal{X} \otimes \mathcal{Y} \cong \mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp \cong (\mathcal{X} \otimes \mathcal{Y})^\sharp \cong (\mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp)^\sharp.$$

Hence, according to Remark 3.8(b),

$$\dim \mathcal{X} < \infty, \quad \dim \mathcal{Y} < \infty \implies$$

$$\mathcal{X} \otimes \mathcal{Y} \cong \mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp \cong (\mathcal{X} \otimes \mathcal{Y})^\sharp \cong (\mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp)^\sharp \cong b[\mathcal{X}^\sharp \times \mathcal{Y}^\sharp, \mathbb{F}] \cong b[\mathcal{X} \times \mathcal{Y}, \mathbb{F}].$$

The following result identifies a special type of linear manifold of a tensor product space which will be referred to as a *regular linear manifold*. It shows that the tensor product space $\mathcal{M} \otimes \mathcal{N}$ of linear manifolds $\mathcal{M}$ and $\mathcal{N}$ of linear spaces $\mathcal{X}$ and $\mathcal{Y}$, which is a linear manifold of $\mathcal{X} \otimes \mathcal{Y}$, coincides with the linear manifold span $R(\theta|_{\mathcal{M} \times \mathcal{N}})$ of $\mathcal{X} \otimes \mathcal{Y}$, so that $\mathcal{M} \otimes \mathcal{N}$ has as its natural bilinear map the restriction to $\mathcal{M} \times \mathcal{N}$ of the natural bilinear map θ of $\mathcal{X} \otimes \mathcal{Y}$.

Theorem 3.13. *Let $\mathcal{M}$ and $\mathcal{N}$ be linear manifolds of linear spaces $\mathcal{X}$ and $\mathcal{Y}$, respectively. Consider the tensor product $(\mathcal{X} \otimes \mathcal{Y}, \theta)$ of $\mathcal{X}$ and $\mathcal{Y}$, and take the tensor product space $\mathcal{M} \otimes \mathcal{N}$. Then*

(a) $\mathcal{M} \otimes \mathcal{N} = \operatorname{span} R(\theta|_{\mathcal{M} \times \mathcal{N}})$ (*i.e.,* $\mathcal{M} \otimes \mathcal{N} = \operatorname{span} \theta(\mathcal{M} \times \mathcal{N})$),

(b) *and so $(\mathcal{M} \otimes \mathcal{N}, \theta')$ is a tensor product of $\mathcal{M}$ and $\mathcal{N}$ with $\theta' = \theta|_{\mathcal{M} \times \mathcal{N}}$.*

Proof. Let $\mathcal{X}$ and $\mathcal{Y}$ be linear spaces and take a tensor product $(\mathcal{X} \otimes \mathcal{Y}, \theta)$. According to Definition 3.1 this means: $\mathcal{X} \otimes \mathcal{Y}$ is a linear space such that $\mathcal{X} \otimes \mathcal{Y} = \operatorname{span} R(\theta)$ for some bilinear map $\theta\colon \mathcal{X} \times \mathcal{Y} \to R(\theta) \subseteq \mathcal{X} \otimes \mathcal{Y}$ with the property that for every bilinear map $\phi\colon \mathcal{X} \times \mathcal{Y} \to \mathcal{Z}$ into an arbitrary linear space $\mathcal{Z}$ there exists a linear transformation $\Phi\colon \mathcal{X} \otimes \mathcal{Y} \to \mathcal{Z}$ such that

$$\phi = \Phi \circ \theta.$$

Now let $\mathcal{M}$ and $\mathcal{N}$ be linear manifolds of $\mathcal{X}$ and $\mathcal{Y}$ respectively. Since a restriction of a bilinear map to a Cartesian product of linear manifolds is bilinear (cf. Remark 2.1), then take the restriction $\theta|_{\mathcal{M} \times \mathcal{N}}$ of the natural bilinear map θ of $\mathcal{X} \otimes \mathcal{Y}$ to the Cartesian product $\mathcal{M} \times \mathcal{N} \subseteq \mathcal{X} \times \mathcal{Y}$ so that

$$\theta|_{\mathcal{M} \times \mathcal{N}}\colon \mathcal{M} \times \mathcal{N} \to R(\theta|_{\mathcal{M} \times \mathcal{N}}) \subseteq \operatorname{span} R(\theta|_{\mathcal{M} \times \mathcal{N}}) \subseteq \operatorname{span} R(\theta) = \mathcal{X} \otimes \mathcal{Y}$$

is a bilinear map into the linear space span $R(\theta|_{\mathcal{M} \times \mathcal{N}})$, which is a linear manifold of the linear space $\mathcal{X} \otimes \mathcal{Y} = \operatorname{span} R(\theta)$. Take an arbitrary bilinear map $\phi\colon \mathcal{X} \times \mathcal{Y} \to \mathcal{Z}$ into an arbitrary linear space $\mathcal{Z}$ and consider its restriction $\phi|_{\mathcal{M} \times \mathcal{N}}$ to $\mathcal{M} \times \mathcal{N}$, which is again a bilinear map. Thus

$$\phi|_{\mathcal{M}\times\mathcal{N}} = (\Phi\circ\theta)|_{\mathcal{M}\times\mathcal{N}} = \Phi\circ\theta|_{\mathcal{M}\times\mathcal{N}} = \Phi|_{\operatorname{span} R(\theta|_{\mathcal{M}\times\mathcal{N}})}\circ\theta|_{\mathcal{M}\times\mathcal{N}},$$

where $\Phi|_{\operatorname{span} R(\theta|_{\mathcal{M}\times\mathcal{N}})}\colon \operatorname{span} R(\theta|_{\mathcal{M}\times\mathcal{N}}) \to \mathcal{Z}$ is a linear transformation, the restriction of the associated linear transformation $\Phi\colon \mathcal{X}\otimes\mathcal{Y}\to\mathcal{Z}$. By Theorem 2.5, every bilinear map on $\mathcal{M}\times\mathcal{N}$ has a bilinear extension over $\mathcal{X}\times\mathcal{Y}$. Then every bilinear map $\phi'\colon \mathcal{M}\times\mathcal{N}\to\mathcal{Z}$ is the restriction of a bilinear map $\phi\colon \mathcal{X}\times\mathcal{Y}\to\mathcal{Z}$. Equivalently, every bilinear map $\phi'\colon \mathcal{M}\times\mathcal{N}\to\mathcal{Z}$ is of the form $\phi' = \phi|_{\mathcal{M}\times\mathcal{N}}\colon \mathcal{M}\times\mathcal{N}\to\mathcal{Z}$ for some bilinear map $\phi\colon \mathcal{X}\times\mathcal{Y}\to\mathcal{Z}$. Then, by the above displayed identity, for every bilinear map $\phi'\colon \mathcal{M}\times\mathcal{N}\to\mathcal{Z}$ there exists a linear transformation $\Phi|_{\operatorname{span} R(\theta|_{\mathcal{M}\times\mathcal{N}})}\colon \operatorname{span} R(\theta|_{\mathcal{M}\times\mathcal{N}})\to\mathcal{Z}$ such that

$$\phi' = \Phi|_{\operatorname{span} R(\theta|_{\mathcal{M}\times\mathcal{N}})}\circ\theta|_{\mathcal{M}\times\mathcal{N}}.$$

So $\mathcal{M}\otimes\mathcal{N} = \operatorname{span} R(\theta|_{\mathcal{M}\times\mathcal{N}})$ by Definition 3.1, and hence $\theta' = \theta|_{\mathcal{M}\times\mathcal{N}}$ (the bilinear restriction of θ), is the natural bilinear map of $\mathcal{M}\otimes\mathcal{N}$. Therefore

$$(\mathcal{M}\otimes\mathcal{N}, \theta|_{\mathcal{M}\times\mathcal{N}}) \text{ is a tensor product.}$$

This proves (a) and (b). □

Definition 3.14. A linear manifold Υ of a tensor product space $\mathcal{T} = \mathcal{X}\otimes\mathcal{Y}$ is *regular* if $\Upsilon = \mathcal{M}\otimes\mathcal{N}$ for some linear manifolds $\mathcal{M}$ and $\mathcal{N}$ of the linear spaces $\mathcal{X}$ and $\mathcal{Y}$. Otherwise Υ is called *irregular*.

For a collection of properties of regular linear manifolds of tensor product spaces see [50, 51, 53]. Regular linear manifolds are characterized as follows.

Corollary 3.15. *A linear manifold Υ of $\mathcal{T} = \mathcal{X}\otimes\mathcal{Y}$ is regular if and only if either it is zero or there exist Hamel bases E for $\mathcal{X}$ and D for $\mathcal{Y}$ such that*

$$\Upsilon = \operatorname{span}\top_{E',D'},$$

where $\top_{E',D'} = \{x\otimes y\in\mathcal{X}\otimes\mathcal{Y}\colon x\in E' \text{ and } y\in D'\}$ for some nonempty subsets $E'\subseteq E$ and $D'\subseteq D$.

Proof. Suppose $\Upsilon\neq\{0\}$. Let $\mathcal{M}$ and $\mathcal{N}$ be linear manifolds of $\mathcal{X}$ and $\mathcal{Y}$. If $\Upsilon = \mathcal{M}\otimes\mathcal{N}$, then $\Upsilon = \operatorname{span}\top_{E',D'}$ for some Hamel bases E' and D' for $\mathcal{M}$ and $\mathcal{N}$ by Theorem 3.9, and so there are Hamel bases E and D for $\mathcal{X}$ and $\mathcal{Y}$ such that $E'\subseteq E$ and $D'\subseteq D$ (see, e.g., [52, Theorem 2.5]). On the other hand, suppose $\Upsilon = \operatorname{span}\top_{E',D'}$ with $E'\subseteq E$ and $D'\subseteq D$ for Hamel bases E and D for $\mathcal{X}$ and $\mathcal{Y}$. Then E' and D' are Hamel bases for $\mathcal{M} = \operatorname{span} E'\subseteq\mathcal{X}$ and $\mathcal{N} = \operatorname{span} D'\subseteq\mathcal{Y}$. Therefore $\operatorname{span}\top_{E',D'} = \mathcal{M}\otimes\mathcal{N}$ by Theorem 3.9. □

3.3 Tensor Product of Linear Transformations

Now we focus on the concept of tensor product of linear transformations. Let $\mathcal{X},\mathcal{Y},\mathcal{V},\mathcal{W}$ be linear spaces, and let $\mathcal{L}[\mathcal{X},\mathcal{V}]$ and $\mathcal{L}[\mathcal{Y},\mathcal{W}]$ be the linear spaces

of all linear transformations of $\mathcal{X}$ into $\mathcal{V}$ and of $\mathcal{Y}$ into $\mathcal{W}$, respectively, all over the same field $\mathbb{F}$. Consider the tensor products $(\mathcal{X} \otimes \mathcal{Y}, \vartheta)$ of the linear spaces $\mathcal{X}$ and $\mathcal{Y}$, and $(\mathcal{V} \otimes \mathcal{W}, \vartheta')$ of the linear spaces $\mathcal{V}$ and $\mathcal{W}$.

Definition 3.16. Let $\mathcal{X}, \mathcal{Y}, \mathcal{V}, \mathcal{W}$ be linear spaces and consider the tensor product spaces $\mathcal{X} \otimes \mathcal{Y}$ and $\mathcal{V} \otimes \mathcal{W}$. Let $A \in \mathcal{L}[\mathcal{X}, \mathcal{V}]$ and $B \in \mathcal{L}[\mathcal{Y}, \mathcal{W}]$ be linear transformations. For each $\sum_{i=1}^{n} x_i \otimes y_i$ in $\mathcal{X} \otimes \mathcal{Y}$ set

$$(A \otimes B)\sum\nolimits_{i=1}^{n} x_i \otimes y_i = \sum\nolimits_{i=1}^{n} Ax_i \otimes By_i \ \text{ in } \ \mathcal{V} \otimes \mathcal{W}.$$

It is shown below that this defines a linear transformation $A \otimes B$ of the linear space $\mathcal{X} \otimes \mathcal{Y}$ into the linear space $\mathcal{V} \otimes \mathcal{W}$, referred to as the *tensor product of the transformations* A *and* B, or the *tensor product transformation* $A \otimes B$.

Theorem 3.17. *Let* $\mathcal{X} \otimes \mathcal{Y}$ *and* $\mathcal{V} \otimes \mathcal{W}$ *be tensor product spaces of linear spaces* $\mathcal{X}, \mathcal{Y}, \mathcal{V}, \mathcal{W}$. *Take* $A \in \mathcal{L}[\mathcal{X}, \mathcal{V}]$ *and* $B \in \mathcal{L}[\mathcal{Y}, \mathcal{W}]$.

(a) *In fact,* $A \otimes B$ *in Definition 3.16 defines a linear transformation,*

$$A \otimes B \in \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{V} \otimes \mathcal{W}],$$

and the value of $A \otimes B$ *at* $F \in \mathcal{X} \otimes \mathcal{Y}$ *(i.e.,* $(A \otimes B)F$*) does not depend on the representation* $\sum_{i=1}^{n} x_i \otimes y_i$ *of* F.

(b) *The map* $\theta\colon \mathcal{L}[\mathcal{X}, \mathcal{V}] \times \mathcal{L}[\mathcal{Y}, \mathcal{W}] \to \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{V} \otimes \mathcal{W}]$ *defined by*

$$\theta(A, B) = A \otimes B \quad \text{for every} \quad (A, B) \in \mathcal{L}[\mathcal{X}, \mathcal{V}] \times \mathcal{L}[\mathcal{Y}, \mathcal{W}],$$

with $A \otimes B \in \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{V} \otimes \mathcal{W}]$ *as in* (a)*, is bilinear.*

(c) *Set* $\mathcal{L}[\mathcal{X}, \mathcal{V}] \otimes \mathcal{L}[\mathcal{Y}, \mathcal{W}] = \operatorname{span} R(\theta)$. *Then* $(\mathcal{L}[\mathcal{X}, \mathcal{V}] \otimes \mathcal{L}[\mathcal{Y}, \mathcal{W}], \theta)$ *is a tensor product of the linear spaces* $\mathcal{L}[\mathcal{X}, \mathcal{V}]$ *and* $\mathcal{L}[\mathcal{Y}, \mathcal{W}]$*, and so*

$$\mathcal{L}[\mathcal{X}, \mathcal{V}] \otimes \mathcal{L}[\mathcal{Y}, \mathcal{W}] \subseteq \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{V} \otimes \mathcal{W}].$$

(d) *The transformation* $A \otimes B \in \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{V} \otimes \mathcal{W}]$ *in* (a) *coincides with a single tensor in the tensor product space* $\mathcal{L}[\mathcal{X}, \mathcal{V}] \otimes \mathcal{L}[\mathcal{Y}, \mathcal{W}]$*:*

$$A \otimes B \in \mathcal{L}[\mathcal{X}, \mathcal{V}] \otimes \mathcal{L}[\mathcal{Y}, \mathcal{W}] \subseteq \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{V} \otimes \mathcal{W}].$$

Proof. (a) Take the tensor product $(\mathcal{X} \otimes \mathcal{Y}, \vartheta)$. Since A and B are linear transformations in $\mathcal{L}[\mathcal{X}, \mathcal{Y}]$ and $\mathcal{L}[\mathcal{V}, \mathcal{W}]$, the map ψ from the Cartesian product $\mathcal{X} \times \mathcal{Y}$ to the linear space $\mathcal{V} \otimes \mathcal{W}$ given by $\psi(x, y) = Ax \otimes By$ for every (x, y) in $\mathcal{X} \times \mathcal{Y}$ is bilinear. By Definition 3.1 and Corollary 3.4, there is a unique $\Psi \in \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{V} \otimes \mathcal{W}]$ such that $\psi = \Psi \circ \vartheta$; that is, such that the diagram

$$\begin{array}{ccc} \mathcal{X} \times \mathcal{Y} & \xrightarrow{\ \psi\ } & \mathcal{V} \otimes \mathcal{W} \\ & {\scriptstyle\vartheta}\searrow \quad \uparrow{\scriptstyle\Psi} & \\ & \mathcal{X} \otimes \mathcal{Y} & \end{array}$$

commutes. Take an arbitrary $F = \sum_{i=1}^{n} x_i \otimes y_i \in \mathcal{X} \otimes \mathcal{Y}$. Thus by Definition 3.16,

$$\begin{aligned}(A \otimes B)(F) &= \sum\nolimits_{i=1}^{n} Ax_i \otimes By_i = \sum\nolimits_{i=1}^{n} \psi(x_i, y_i) \\ &= \sum\nolimits_{i=1}^{n} \Psi(\vartheta(x_i, y_i)) = \Psi \sum\nolimits_{i=1}^{n} x_i \otimes y_i = \Psi(F).\end{aligned}$$

Then $A \otimes B = \Psi$. So $A \otimes B$ is linear and $(A \otimes B)(F)$ does not depend on the representation of F as Ψ is linear and does not depend on the representation of F.

(b) Take $A, A' \in \mathcal{L}[\mathcal{X}, \mathcal{V}]$, $B, B' \in \mathcal{L}[\mathcal{Y}, \mathcal{W}]$, $F = \sum_{i=1}^{n} x_i \otimes y_i \in \mathcal{X} \otimes \mathcal{Y}$, and also $\alpha \in \mathbb{F}$, all arbitrary. For the map θ defined in item (b) we get $\theta(A + A', B)F = ((A + A') \otimes B)F = \sum_{i=1}^{n} (A + A')x_i \otimes By_i = \sum_{i=1}^{n} (Ax_i + A'x_i) \otimes By_i = \sum_{i=1}^{n} (Ax_i \otimes By_i) + (A'x_i \otimes By_i) = \sum_{i=1}^{n} (Ax_i \otimes By_i) + \sum_{i=1}^{n} (A'x_i \otimes By_i) = (A \otimes B)F + (A' \otimes B)F = \theta(A, B)F + \theta(A', B)F$. Since this holds for every F, $\theta(A + A', B) = \theta(A, B) + \theta(A', B)$. Similarly, $\theta(A, B + B') = \theta(A, B) + \theta(A, B')$ and $\alpha\theta(A, B) = \theta(\alpha A, B) = \theta(A, \alpha B)$. Therefore

$$\theta \colon \mathcal{L}[\mathcal{X}, \mathcal{V}] \times \mathcal{L}[\mathcal{Y}, \mathcal{W}] \to R(\theta) \subseteq \operatorname{span} R(\theta) \subseteq \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{V} \otimes \mathcal{W}] \text{ is bilinear.}$$

(c) According to the definition of θ,

$$\begin{aligned}&\operatorname{span} R(\theta) = \operatorname{span}\big\{A \otimes B \in \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{V} \otimes \mathcal{W}] \colon A \in \mathcal{L}[\mathcal{X}, \mathcal{V}],\ B \in \mathcal{L}[\mathcal{Y}, \mathcal{W}]\big\} \\ &= \big\{\textstyle\sum_{j=1}^{n} A_j \otimes B_j \in \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{V} \otimes \mathcal{W}] \colon\ A \in \mathcal{L}[\mathcal{X}, \mathcal{V}],\ B \in \mathcal{L}[\mathcal{Y}, \mathcal{W}],\ n \in \mathbb{N}\big\}.\end{aligned}$$

If $\mathcal{Z}$ is an arbitrary linear space over $\mathbb{F}$, and if $\phi \colon \mathcal{L}[\mathcal{X}, \mathcal{V}] \times \mathcal{L}[\mathcal{Y}, \mathcal{W}] \to \mathcal{Z}$ is an arbitrary bilinear map on $\mathcal{L}[\mathcal{X}, \mathcal{V}] \times \mathcal{L}[\mathcal{Y}, \mathcal{W}]$ with values in $\mathcal{Z}$, then associate with ϕ a transformation $\Phi \colon \operatorname{span} R(\theta) \to \mathcal{Z}$ defined by

$$\Phi\Big(\sum\nolimits_{j=1}^{n} A_j \otimes B_j\Big) = \sum\nolimits_{j=1}^{n} \phi(A_j, B_j) \in \mathcal{Z}$$

for every $\sum_{j=1}^{n} A_j \otimes B_j \in \operatorname{span} R(\theta) \subseteq \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{V} \otimes \mathcal{W}]$. Since ϕ is a bilinear map, then it is readily verified that Φ is a linear transformation,

$$\Phi \in \mathcal{L}[\operatorname{span} R(\theta), \mathcal{Z}].$$

(Indeed, for any elements $\sum_j A_j \otimes B_j$ and $\sum_k A'_k \otimes B'_k$ in $\mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{V} \otimes \mathcal{W}]$ it follows that $\Phi(\sum_j A_j \otimes B_j + \sum_k A'_k \otimes B'_k) = \sum_j \phi(A_j, B_j) + \sum_k \phi(A'_k, B'_k) = \Phi(\sum_j A_j \otimes B_j) + \Phi(\sum_k A'_k \otimes B'_k)$, and $\Phi(\alpha \sum_j A_j \otimes B_j) = \Phi(\sum_j \alpha A_j \otimes B_j) = \sum_j \phi(\alpha A_j, B_j) = \alpha \sum_j \phi(A_j, B_j) = \alpha \Phi(\sum_j A_j \otimes B_j)$ for $\alpha \in \mathbb{F}$. So Φ is linear.)

Furthermore, for every $(A, B) \in \mathcal{L}[\mathcal{X}, \mathcal{V}] \times \mathcal{L}[\mathcal{Y}, \mathcal{W}]$,

$$(\Phi \circ \theta)(A, B) = \Phi\big(\theta(A, B)\big) = \Phi(A \otimes B) = \phi(A, B).$$

Thus $\phi = \Phi \circ \theta$; equivalently, the diagram

$$\begin{array}{ccc} \mathcal{L}[\mathcal{X}, \mathcal{V}] \times \mathcal{L}[\mathcal{Y}, \mathcal{W}] & \xrightarrow{\ \phi\ } & \mathcal{Z} \\ & {\scriptstyle\theta}\searrow \quad \uparrow{\scriptstyle\Phi} & \\ & \operatorname{span} R(\theta) & \end{array}$$

commutes. Hence $(\operatorname{span} R(\theta), \theta)$ satisfies the axioms of Definition 3.1, and therefore $(\mathcal{L}[\mathcal{X}, \mathcal{V}] \otimes \mathcal{L}[\mathcal{Y}, \mathcal{W}], \theta)$ is the tensor product of $\mathcal{L}[\mathcal{X}, \mathcal{V}]$ and $\mathcal{L}[\mathcal{Y}, \mathcal{W}]$.

(d) A single tensor is by definition the value of the natural bilinear map θ at a pair. Thus by definition of θ the value of it at a pair (A, B) is precisely the tensor product transformation $A \otimes B$ in (a). □

Remark 3.18. *Tensor Product of Duals and Dual of Tensor Products.*

(a) It was proved in Theorem 3.17(a) that $A \otimes B$, defined in Definition 3.16, is a function on $\mathcal{X} \otimes \mathcal{Y}$, and as such $(A \otimes B)F$ cannot depend on the representation of F in $\mathcal{X} \otimes \mathcal{Y}$ (which is the case for every transformation $L \in \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{Z}]$ into any linear space $\mathcal{Z}$). There are two particular cases worth noticing.

(i) $\mathcal{V} = \mathcal{X},\ \mathcal{W} = \mathcal{Y} \implies A \in \mathcal{L}[\mathcal{X}],\quad B \in \mathcal{L}[\mathcal{Y}]$ and $A \otimes B \in \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}]$.

(ii) $\mathcal{V} = \mathcal{W} = \mathbb{F} \implies \mathcal{L}[\mathcal{X}, \mathcal{V}] = \mathcal{L}[\mathcal{X}, \mathbb{F}] = \mathcal{X}^\sharp,\ \mathcal{L}[\mathcal{Y}, \mathcal{W}] = \mathcal{L}[\mathcal{Y}, \mathbb{F}] = \mathcal{Y}^\sharp$ and

$$\mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{V} \otimes \mathcal{W}] = \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathbb{F} \otimes \mathbb{F}] = \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathbb{F}] = (\mathcal{X} \otimes \mathcal{Y})^\sharp.$$

Case (i) is a trivial particularization; case (ii) deserves a few words.

(b) Consider the tensor product $(\mathcal{V} \otimes \mathcal{W}, \vartheta')$ with $\mathcal{V} = \mathcal{W} = \mathbb{F}$. First note by Corollary 3.10 that $\dim(\mathbb{F} \otimes \mathbb{F}) = \dim \mathbb{F} = 1$, thus $\mathbb{F} \otimes \mathbb{F} \cong \mathbb{F}$ and, as usual, identification with a field will be interpreted as identity and we write $\mathbb{F} \otimes \mathbb{F} = \mathbb{F}$, which justifies writing $\mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathbb{F} \otimes \mathbb{F}] = \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathbb{F}] = (\mathcal{X} \otimes \mathcal{Y})^\sharp$. In this case the natural bilinear map $\vartheta' \colon \mathbb{F} \times \mathbb{F} = \mathbb{F}^2 \to \mathbb{F} \otimes \mathbb{F} = \mathbb{F}$ is given by

$$\vartheta'(\alpha, \beta) = \alpha \otimes \beta = \alpha\,\beta \quad \text{for every} \quad \alpha, \beta \in \mathbb{F}.$$

So for every $f \in \mathcal{X}^\sharp$ and $g \in \mathcal{Y}^\sharp$,

$$(f \otimes g)F = \sum_{i=1}^{n} f(x_i) \otimes g(y_i) = \sum_{i=1}^{n} f(x_i)\, g(y_i) \in \mathbb{F} \otimes \mathbb{F} = \mathbb{F} \quad \text{for} \quad F \in \mathcal{X} \otimes \mathcal{Y}$$

and, in particular,

$$(f \otimes g)(x \otimes y) = f(x)\, g(y) \quad \text{for every} \quad x \in \mathcal{X} \text{ and } y \in \mathcal{Y},$$

where $f \otimes g$ is a tensor product transformation in $\mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathbb{F} \otimes \mathbb{F}] = (\mathcal{X} \otimes \mathcal{Y})^\sharp$, and a single tensor in the tensor product space $\mathcal{L}[\mathcal{X}, \mathbb{F}] \otimes \mathcal{L}[\mathcal{Y}, \mathbb{F}] = \mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp$. Thus, since $\mathcal{L}[\mathcal{X}, \mathbb{F}] \otimes \mathcal{L}[\mathcal{Y}, \mathbb{F}] \subseteq \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathbb{F} \otimes \mathbb{F}]$ by Theorem 3.17(c),

$$\mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp \subseteq (\mathcal{X} \otimes \mathcal{Y})^\sharp.$$

Theorem 3.19. *Let $\mathcal{V}, \mathcal{W}, \mathcal{X}, \mathcal{Y}, \mathcal{X}', \mathcal{Y}'$ be linear spaces over the same field $\mathbb{F}$. If $\alpha, \beta \in \mathbb{F}$, $A, A_1, A_2 \in \mathcal{L}[\mathcal{X}, \mathcal{V}]$, $B, B_1, B_2 \in \mathcal{L}[\mathcal{Y}, \mathcal{W}]$, $C \in \mathcal{L}[\mathcal{X}', \mathcal{X}]$ and $D \in \mathcal{L}[\mathcal{Y}', \mathcal{Y}]$, then the following identities hold true.*

(a) $\alpha\,\beta\,(A \otimes B) = \alpha A \otimes \beta B = (\alpha\,\beta A) \otimes B = A \otimes (\alpha\,\beta B)$.

(b) $(A_1 + A_2) \otimes (B_1 + B_2) = A_1 \otimes B_1 + A_2 \otimes B_1 + A_1 \otimes B_2 + A_2 \otimes B_2$.

(c) $AC \otimes BD = (A \otimes B)\,(C \otimes D)$.

(d) *A,B are invertible if and only if* $A \otimes B$ *is, and* $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$.

(e) $\mathcal{R}(A) \otimes \mathcal{R}(B) = \mathcal{R}(A \otimes B)$.

(f) $A \otimes B$ *is surjective if and only if both* A *and* B *are surjective:* $\mathcal{R}(A \otimes B) = \mathcal{V} \otimes \mathcal{W} \iff \mathcal{R}(A) = \mathcal{V}$ *and* $\mathcal{R}(B) = \mathcal{W}$.

(g) $A \otimes B$ *is injective if and only if both* A *and* B *are injective:* $\mathcal{N}(A \otimes B) = \{0\} \iff \mathcal{N}(A) = \{0\}$ *and* $\mathcal{N}(B) = \{0\}$.

(h) $\mathcal{N}(A) \otimes \mathcal{N}(B) \subsetneqq \mathcal{N}(A \otimes B) = \big(\mathcal{N}(A) \otimes \mathcal{Y}\big) + \big(\mathcal{X} \otimes \mathcal{N}(B)\big)$.

(i) $A \otimes B \cong B \otimes A$.

(j) $(A \otimes B)^\sharp = A^\sharp \otimes B^\sharp$ (*as transformations in* $\mathcal{L}[(\mathcal{V} \otimes \mathcal{W})^\sharp, (\mathcal{X} \otimes \mathcal{Y})^\sharp]$).

Proof. If a linear space is null, then the above results are trivial or meaningless. So suppose they are nonzero. Recall: when dealing with tensor products, identities are interpreted up to isomorphisms, and inclusions mean embeddings.

(a,b) The results in (a) and (b) are straightforward since $A \otimes B$ is a single tensor in the tensor product space $\mathcal{L}[\mathcal{X}, \mathcal{V}] \otimes \mathcal{L}[\mathcal{Y}, \mathcal{W}]$ and so $A \otimes B$ is the value of a bilinear map θ at the pair (A, B) in $\mathcal{L}[\mathcal{X}, \mathcal{V}] \times \mathcal{L}[\mathcal{Y}, \mathcal{W}]$.

(c,d) Take $A \in \mathcal{L}[\mathcal{X}, \mathcal{V}]$, $B \in \mathcal{L}[\mathcal{Y}, \mathcal{W}]$, $C \in \mathcal{L}[\mathcal{X}', \mathcal{X}]$, and $D \in \mathcal{L}[\mathcal{Y}', \mathcal{Y}]$, consider the tensor product transformations $A \otimes B$ in $\mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{V} \otimes \mathcal{W}]$ and $C \otimes D$ in $\mathcal{L}[\mathcal{X}' \otimes \mathcal{Y}', \mathcal{X} \otimes \mathcal{Y}]$, and take an arbitrary $F = \sum_i v_i \otimes w_i$ in $\mathcal{V} \otimes \mathcal{W}$. Then

$$\begin{aligned}\big[(A \otimes B)\,(C \otimes D)\big]F &= (A \otimes B)\big[(C \otimes D)F\big] = (A \otimes B)\sum\nolimits_i Cv_i \otimes Dw_i \\ &= \sum\nolimits_i ACv_i \otimes BDw_i = (AC \otimes BD)F,\end{aligned}$$

which proves (c). Setting $\mathcal{X}' = \mathcal{V}$, $\mathcal{Y}' = \mathcal{W}$, $C = A^{-1}$ in $\mathcal{L}[\mathcal{V}, \mathcal{X}]$ and $D = B^{-1}$ in $\mathcal{L}[\mathcal{W}, \mathcal{Y}]$, then (c) implies (d):

$$(A \otimes B)\,(A^{-1} \otimes B^{-1}) = I_\mathcal{V} \otimes I_\mathcal{W} \quad \text{and} \quad (A^{-1} \otimes B^{-1})\,(A \otimes B) = I_\mathcal{X} \otimes I_\mathcal{Y},$$

where $I_\mathcal{X}$, $I_\mathcal{Y}$, $I_\mathcal{V}$ and $I_\mathcal{W}$ stand for the identities on $\mathcal{X}$, $\mathcal{Y}$, $\mathcal{V}$ and $\mathcal{W}$, and $I_\mathcal{X} \otimes I_\mathcal{Y}$ and $I_\mathcal{V} \otimes I_\mathcal{W}$ are the identities on $\mathcal{X} \otimes \mathcal{Y}$ and $\mathcal{V} \otimes \mathcal{W}$.

(e,f) An element G lies in $\mathcal{R}(A \otimes B)$ if and only if $\mathrm{G} = (A \otimes B)F$ for some $F = \sum_i x_i \otimes y_i \in \mathcal{X} \otimes \mathcal{Y}$, which means $\mathrm{G} = \sum_i Ax_i \otimes By_i = \sum_i u_i \otimes v_i$ for $u_i \in \mathcal{R}(A)$ and $v_i \in \mathcal{R}(B)$ or, equivalently, $\mathrm{G} \in \mathcal{R}(A) \otimes \mathcal{R}(B)$. So we get (e):

$$\mathcal{R}(A \otimes B) = \mathcal{R}(A) \otimes \mathcal{R}(B).$$

On the other hand (e) implies (f). In fact, since $\mathcal{R}(A) \otimes \mathcal{R}(B) \subseteq \mathcal{V} \otimes \mathcal{W}$, then $\mathcal{R}(A) = \mathcal{V}$ and $\mathcal{R}(B) = \mathcal{W}$ implies $\mathcal{R}(A) \otimes \mathcal{R}(B) = \mathcal{V} \otimes \mathcal{W}$ by (e), and the converse is straightforward by (e) and Corollary 3.11(c). This proves (f).

(g) Suppose one of A or B is not injective. With no loss of generality, suppose A is not injective (i.e., $\mathcal{N}(A) \neq \{0\}$) and take $0 \neq x \in \mathcal{N}(A)$ (i.e., $Ax = 0$ and

$x \neq 0$) so that $(A \otimes B)(x \otimes y) = Ax \otimes By = 0$ for an arbitrary $0 \neq y \in \mathcal{Y}$. As $x \otimes y \neq 0$ (cf. Proposition 3.C), then there exists $x \otimes y \neq 0$ in $\mathcal{N}(A \otimes B)$. So $A \otimes B$ is not injective (i.e., $\mathcal{N}(A \otimes B) \neq \{0\}$). Equivalently, if $A \otimes B$ is injective, then so are both A and B. For the converse, suppose A and B are injective. If $A \otimes B$ is not injective, then there exists $\digamma = \sum_i x_i \otimes y_i \neq 0$ in $\mathcal{X} \otimes \mathcal{Y}$ such that $(A \otimes B) \sum_i x_i \otimes y_i = \sum_i Ax_i \otimes By_i = 0$. Thus $\{Ax_i \otimes By_i\}$ is not linearly independent, and so one of $\{Ax_i\}$ or $\{By_i\}$ is not linearly independent (cf. Theorem 3.9). With no loss of generality suppose $\{Ax_i\}$ is not linearly independent. But we can take a representation of the same $\digamma$ with $\{x_i\}$ being linearly independent (cf. Proposition 3.A). Hence $\{Ax_i\}$ is linearly independent because A is injective. This leads to a contradiction. So if A and B are injective, then $A \otimes B$ is injective. This proves (g).

(h) Take the linear manifold $\mathcal{N}(A) \otimes \mathcal{N}(B)$ of $\mathcal{X} \otimes \mathcal{Y}$. If $\digamma \in \mathcal{N}(A) \otimes \mathcal{N}(B)$, then $\digamma = \sum_i x_i \otimes y_i$ with x_i in $\mathcal{N}(A)$ and y_i in $\mathcal{N}(B)$, and so $Ax_i = 0$ and $By_i = 0$, which implies $(A \otimes B)\digamma = \sum_i Ax_i \otimes By_i = 0$. Thus $\digamma \in \mathcal{N}(A \otimes B)$. Hence $\mathcal{N}(A) \otimes \mathcal{N}(B) \subseteq \mathcal{N}(A \otimes B)$ for every A, B. The reverse inclusion fails if one of A or B is not injective and the other is not null. Indeed, take $x \notin \mathcal{N}(A)$ and $0 \neq y \in \mathcal{N}(B)$ so that $0 \neq x \otimes y \notin \mathcal{N}(A) \otimes \mathcal{N}(B)$ but $x \otimes y \in \mathcal{N}(A \otimes B)$ as $(A \otimes B)(x \otimes y) = Ax \otimes By = Ax \otimes 0 = 0$. So there are A and B for which

$$\mathcal{N}(A \otimes B) \not\subseteq \mathcal{N}(A) \otimes \mathcal{N}(B).$$

To prove the identity $\mathcal{N}(A \otimes B) = \mathcal{N}(A) \otimes \mathcal{Y} + \mathcal{X} \otimes \mathcal{N}(B)$, suppose A and B are not null (otherwise the identity is trivial). Let $\mathcal{M}$ and $\mathcal{N}$ be the nonzero algebraic complements of $\mathcal{N}(A)$ and $\mathcal{N}(B)$ in $\mathcal{X}$ and $\mathcal{Y}$, that is,

$$\mathcal{X} = \mathcal{N}(A) + \mathcal{M} \quad \text{and} \quad \mathcal{Y} = \mathcal{N}(B) + \mathcal{N}.$$

Then $A|_{\mathcal{M}} \in \mathcal{L}[\mathcal{M}, \mathcal{V}]$ and $B|_{\mathcal{N}} \in \mathcal{L}[\mathcal{N}, \mathcal{W}]$ are injective (i.e., $\mathcal{N}(A|_{\mathcal{M}}) = \{0\}$ and $\mathcal{N}(B|_{\mathcal{N}}) = \{0\}$). Take an arbitrary $\digamma = \sum_i x_i \otimes y_i \in \mathcal{X} \otimes \mathcal{Y}$. Write $x_i = a_i + u_i$ and $y_i = b_i + v_i$ with $a_i \in \mathcal{N}(A)$, $u_i \in \mathcal{M}$, $b_i \in \mathcal{N}(B)$, $v_i \in \mathcal{N}$ so that

$$x_i \otimes y_i = (a_i + u_i) \otimes (b_i + v_i) = (a_i \otimes b_i) + (a_i \otimes v_i) + (u_i \otimes b_i) + (u_i \otimes v_i),$$

and therefore (since $Aa_i = 0$ and $Bb_i = 0$)

$$\begin{aligned}(A \otimes B)\digamma &= (A \otimes B)\Big(\textstyle\sum_i a_i \otimes b_i + \sum_i a_i \otimes v_i + \sum_i u_i \otimes b_i + \sum_i u_i \otimes v_i\Big) \\ &= \textstyle\sum_i Aa_i \otimes Bb_i + \sum_i Aa_i \otimes Bv_i + \sum_i Au_i \otimes Bb_i + \sum_i Au_i \otimes Bv_i \\ &= \textstyle\sum_i Au_i \otimes Bv_i = \sum_i A|_{\mathcal{M}} u_i \otimes B|_{\mathcal{N}} v_i.\end{aligned}$$

Suppose $\digamma \in \mathcal{N}(A \otimes B)$. Thus $(A \otimes B)\digamma = 0$ and hence $\sum_i A|_{\mathcal{M}} u_i \otimes B|_{\mathcal{N}} v_i = (A|_{\mathcal{M}} \otimes B|_{\mathcal{N}}) \sum_i u_i \otimes v_i = 0$ so that $\sum_i u_i \otimes v_i \in \mathcal{N}(A|_{\mathcal{M}} \otimes B|_{\mathcal{N}})$. But the restrictions $A|_{\mathcal{M}} \in \mathcal{L}[\mathcal{M}, \mathcal{V}]$ and $B|_{\mathcal{N}} \in \mathcal{L}[\mathcal{N}, \mathcal{W}]$ are injective (by their definition according to the decomposition $\mathcal{X} = \mathcal{N}(A) + \mathcal{M}$ and $\mathcal{Y} = \mathcal{N}(B) + \mathcal{N}$), and so $A|_{\mathcal{M}} \otimes B|_{\mathcal{N}}$ is injective (i.e., $\mathcal{N}(A|_{\mathcal{M}} \otimes B|_{\mathcal{N}}) = \{0\}$) by (g). Hence

$$\sum\nolimits_i u_i \otimes v_i = 0.$$

Thus

$$\begin{aligned} F &= \textstyle\sum_i x_i \otimes y_i = \sum_i a_i \otimes b_i + \sum_i a_i \otimes v_i + \sum_i u_i \otimes b_i \\ &= \textstyle\sum_i a_i \otimes (b_i + v_i) + \sum_i u_i \otimes b_i \ \subseteq\ \mathcal{N}(A) \otimes \mathcal{Y} + \mathcal{X} \otimes \mathcal{N}. \end{aligned}$$

Then $\mathcal{N}(A \otimes B) \subseteq \mathcal{N}(A) \otimes \mathcal{Y} + \mathcal{X} \otimes \mathcal{N}(B)$. The reverse inclusion is straightforward. Indeed, if $F \in \mathcal{N}(A) \otimes \mathcal{Y} + \mathcal{X} \otimes \mathcal{N}(B) \subseteq \mathcal{X} \otimes \mathcal{Y}$, then $(A \otimes B)F \subseteq A(\mathcal{N}(A)) \otimes B(\mathcal{Y}) + A(\mathcal{X}) \otimes B(\mathcal{N}(B)) = \{0\} \otimes \mathcal{R}(B) + \mathcal{R}(A) \otimes \{0\} = \{0\}$, and so $F \in \mathcal{N}(A \otimes B)$. Thus $\mathcal{N}(A) \otimes \mathcal{Y} + \mathcal{X} \otimes \mathcal{N}(B) \subseteq \mathcal{N}(A \otimes B)$. Therefore

$$\mathcal{N}(A) \otimes \mathcal{Y} + \mathcal{X} \otimes \mathcal{N}(B) = \mathcal{N}(A \otimes B).$$

(i) By Corollary 3.11(a), $\mathcal{X} \otimes \mathcal{Y} \cong \mathcal{Y} \otimes \mathcal{X}$ and $\mathcal{V} \otimes \mathcal{W} \cong \mathcal{W} \otimes \mathcal{V}$. It is readily verified that $\Pi_1 \colon \mathcal{X} \otimes \mathcal{Y} \to \mathcal{Y} \otimes \mathcal{X}$ and $\Pi_2 \colon \mathcal{V} \otimes \mathcal{W} \to \mathcal{W} \otimes \mathcal{V}$ given by $\Pi_1 F = \sum_i y_i \otimes x_i$ and $\Pi_2 \mathrm{G} = \sum_i w_i \otimes v_v$ for every $F = \sum_i x_i \otimes y_i \in \mathcal{X} \otimes \mathcal{Y}$ and $\mathrm{G} = \sum_i v_i \otimes w_i \in \mathcal{V} \otimes \mathcal{W}$ are invertible linear transformations (called the *natural permutation isomorphisms*). Thus for an arbitrary $F = \sum_i x_i \otimes y_i \in \mathcal{X} \otimes \mathcal{Y}$,

$$\begin{aligned} \Pi_2(A \otimes B)F &= \Pi_2 \textstyle\sum_i Ax_i \otimes By_i = \sum_i By_i \otimes Ax_i \\ &= (B \otimes A) \textstyle\sum_i y_i \otimes x_i = (B \otimes A)\Pi_1 \sum_i x_i \otimes y_i = (B \otimes A)\Pi_1 F. \end{aligned}$$

Hence $\Pi_2(A \otimes B) = (B \otimes A)\Pi_1$. That is, tensor product of linear transformations is commutative up to isomorphisms, which means $A \otimes B$ and $B \otimes A$ are isomorphically equivalent: $A \otimes B \cong B \otimes A$. In particular, if $\mathcal{V} = \mathcal{X}$ and $\mathcal{W} = \mathcal{Y}$ so that $A \in \mathcal{L}[\mathcal{X}]$ and $B \in \mathcal{L}[\mathcal{Y}]$, then $A \otimes B$ and $B \otimes A$ are similar.

(j) Take A in $\mathcal{L}[\mathcal{X}, \mathcal{V}]$, B in $\mathcal{L}[\mathcal{Y}, \mathcal{W}]$, $A^\sharp$ in $\mathcal{L}[\mathcal{V}^\sharp, \mathcal{X}^\sharp]$, $B^\sharp$ in $\mathcal{L}[\mathcal{W}^\sharp, \mathcal{Y}^\sharp]$, $A \otimes B$ in $\mathcal{L}[\mathcal{X}, \mathcal{V}] \otimes \mathcal{L}[\mathcal{Y}, \mathcal{W}] \subseteq \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{V} \otimes \mathcal{W}]$, its algebraic adjoint $(A \otimes B)^\sharp$ in $\mathcal{L}[(\mathcal{V} \otimes \mathcal{W})^\sharp, (\mathcal{X} \otimes \mathcal{Y})^\sharp]$, and the tensor product of algebraic adjoints $A^\sharp \otimes B^\sharp$ in

$$\mathcal{L}[\mathcal{V}^\sharp, \mathcal{X}^\sharp] \otimes \mathcal{L}[\mathcal{W}^\sharp, \mathcal{Y}^\sharp] \subseteq \mathcal{L}[\mathcal{V}^\sharp \otimes \mathcal{W}^\sharp, \mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp] \subseteq \mathcal{L}[(\mathcal{V} \otimes \mathcal{W})^\sharp, (\mathcal{X} \otimes \mathcal{Y})^\sharp].$$

Take any single tensor $f \otimes g \in \mathcal{V}^\sharp \otimes \mathcal{W}^\sharp \subseteq (\mathcal{V} \otimes \mathcal{W})^\sharp$. By definition of tensor product transformation and of algebraic adjoint,

$$(A^\sharp \otimes B^\sharp)(f \otimes g) = A^\sharp f \otimes B^\sharp g = fA \otimes gB \ \in\ \mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp \subseteq (\mathcal{X} \otimes \mathcal{Y})^\sharp, \qquad (*)$$

$$(A \otimes B)^\sharp(f \otimes g) = (f \otimes g)(A \otimes B) \ \in\ (\mathcal{X} \otimes \mathcal{Y})^\sharp. \qquad (**)$$

Cliam. $\quad (f \otimes g)(A \otimes B) = fA \otimes gB \ \in\ \mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp \subseteq (\mathcal{X} \otimes \mathcal{Y})^\sharp.$

Proof of Claim. Take any single tensor $x \otimes y$ in $\mathcal{X} \otimes \mathcal{Y}$, and an arbitrary single tensor $A \otimes B$ in $\mathcal{L}[\mathcal{X}, \mathcal{V}] \otimes \mathcal{L}[\mathcal{Y}, \mathcal{W}]$. By definition of tensor product transformation, $(A \otimes B)(x \otimes y) = Ax \otimes By$, again a single tensor now in $\mathcal{V} \otimes \mathcal{W}$. Next take an arbitrary single tensor $f \otimes g$ in $\mathcal{V}^\sharp \otimes \mathcal{W}^\sharp = \mathcal{L}[\mathcal{V}, \mathbb{F}] \otimes \mathcal{L}[\mathcal{W}, \mathbb{F}]$ so that, by definition of tensor product transformation again, $(f \otimes g)(Ax \otimes By) = fAx \otimes gBy \in \mathbb{F} \otimes \mathbb{F} = \mathbb{F}$. On the other hand, since $fA \in \mathcal{X}^\sharp = \mathcal{L}[\mathcal{X}, \mathbb{F}]$ and $gB \in \mathcal{Y}^\sharp = \mathcal{L}[\mathcal{Y}, \mathbb{F}]$, we get by the definition of tensor product transformation once again that $(fA \otimes gB)(x \otimes y) = fAx \otimes gBy \in \mathbb{F} \otimes \mathbb{F} = \mathbb{F}$. Summing up:

$(f \otimes g)(A \otimes B)(x \otimes y) = (f \otimes g)(Ax \otimes By) = fAx \otimes gBy = (fA \otimes gB)(x \otimes y)$

for every $x \otimes y \in \mathcal{X} \otimes \mathcal{Y}$. So $(f \otimes g)(A \otimes B)\digamma = (f \otimes g)(A \otimes B)\sum_i x_i \otimes y_i = \sum_i (f \otimes g)(A \otimes B)(x_i \otimes y_i) = \sum_i (fA \otimes gB)(x_i \otimes y_i) = (fA \otimes gB)\sum_i x_i \otimes y_i = (fA \otimes gB)\digamma$ for every $\digamma \in \mathcal{X} \otimes \mathcal{Y}$, and hence

$$(f \otimes g)(A \otimes B) = fA \otimes gB$$

in $\mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp = \mathcal{L}[\mathcal{X}, \mathbb{F}] \otimes \mathcal{L}[\mathcal{Y}, \mathbb{F}] \subseteq \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathbb{F}] = (\mathcal{X} \otimes \mathcal{Y})^\sharp$. □

Then by $(*)$, $(**)$ and the above claim

$$(A^\sharp \otimes B^\sharp)(f \otimes g) = (A \otimes B)^\sharp(f \otimes g) \quad \text{for every} \quad f \otimes g \in \mathcal{V}^\sharp \otimes \mathcal{W}^\sharp \subseteq (\mathcal{V} \otimes \mathcal{W})^\sharp.$$

Thus for an arbitrary finite sum $\Omega = \sum_i f_i \otimes g_i$ in $\mathcal{V}^\sharp \otimes \mathcal{W}^\sharp \subseteq (\mathcal{V} \otimes \mathcal{W})^\sharp$ we get $(A^\sharp \otimes B^\sharp)\Omega = (A \otimes B)^\sharp\Omega$ for every $\Omega \in \mathcal{V}^\sharp \otimes \mathcal{W}^\sharp$. Hence

$$A^\sharp \otimes B^\sharp = (A \otimes B)^\sharp \quad \text{on} \quad \mathcal{V}^\sharp \otimes \mathcal{W}^\sharp \subseteq (\mathcal{V} \otimes \mathcal{W})^\sharp,$$

in $\mathcal{L}[\mathcal{V}^\sharp, \mathcal{X}^\sharp] \otimes \mathcal{L}[\mathcal{W}^\sharp, \mathcal{Y}^\sharp] \subseteq \mathcal{L}[\mathcal{V}^\sharp \otimes \mathcal{W}^\sharp, \mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp] \subseteq \mathcal{L}[(\mathcal{V} \otimes \mathcal{W})^\sharp, (\mathcal{X} \otimes \mathcal{Y})^\sharp]$. This concludes the proof of (j) and, consequently, of Theorem 3.19. □

Remark 3.20. *Basic Properties and Tensor Sums of Transformations.*

(a) Take $A \in \mathcal{L}[\mathcal{X}, \mathcal{V}]$ and $B \in \mathcal{L}[\mathcal{Y}, \mathcal{W}]$ arbitrarily. Take any $0 \neq \alpha \in \mathbb{F}$. Thus

$$A \otimes B = \alpha A \otimes \tfrac{1}{\alpha} B$$

by Theorem 3.19(a). Similarly, using the same notation for the null transformation in the linear spaces $\mathcal{L}[\mathcal{X}, \mathcal{V}]$ and $\mathcal{L}[\mathcal{Y}, \mathcal{W}]$, it follows that the null transformation in $\mathcal{L}[\mathcal{X}, \mathcal{V}] \otimes \mathcal{L}[\mathcal{Y}, \mathcal{W}] \subseteq \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{V} \otimes \mathcal{W}]$ can be written as

$$A \otimes O = O \otimes B = O \otimes O.$$

Products are not uniquely represented, and so single tensors do not have a unique representation (as we saw before). This tautologically extends to transformations (as we saw above): a tensor product of linear transformations does not have a unique representation. However, as happens with products in general (and with every sort of single tensors in particular),

$$A \neq O \ \text{ and } \ A \otimes B_1 = A \otimes B_2 \quad \Longrightarrow \quad B_1 = B_2$$

by Proposition 3.C in Section 3.4. Symmetrically,

$$B \neq O \ \text{ and } \ A_1 \otimes B = A_2 \otimes B \quad \Longrightarrow \quad A_1 = A_2.$$

If $\mathcal{X} = \mathcal{V}$ and $\mathcal{Y} = \mathcal{W}$ and if $I_\mathcal{X} \colon \mathcal{X} \to \mathcal{X}$ and $I_\mathcal{Y} \colon \mathcal{Y} \to \mathcal{Y}$ stand for the identity transformations on $\mathcal{X}$ and on $\mathcal{Y}$, then the identity $I \colon \mathcal{X} \otimes \mathcal{Y} \to \mathcal{X} \otimes \mathcal{Y}$ on $\mathcal{X} \otimes \mathcal{Y}$ is such that for any $0 \neq \alpha \in \mathbb{F}$

$$I = I_\mathcal{X} \otimes I_\mathcal{Y} = \alpha I_\mathcal{X} \otimes \tfrac{1}{\alpha} I_\mathcal{Y}.$$

Also, according to Theorem 3.19(c), for every $A \in \mathcal{L}[\mathcal{X}]$ and $B \in \mathcal{L}[\mathcal{Y}]$ we get

$$A \otimes B = (A \otimes I_{\mathcal{Y}})\,(I_{\mathcal{X}} \otimes B) = (I_{\mathcal{X}} \otimes B)\,(A \otimes I_{\mathcal{Y}}).$$

(b) Motivated by the above identity, the *tensor sum* of $A \in \mathcal{L}[\mathcal{X}]$ and $B \in \mathcal{L}[\mathcal{Y}]$ is defined as the transformation $A \boxplus B \colon \mathcal{X} \otimes \mathcal{Y} \to \mathcal{X} \otimes \mathcal{Y}$ given by

$$A \boxplus B = (A \otimes I_{\mathcal{Y}}) + (I_{\mathcal{X}} \otimes B),$$

which lies in $\mathcal{L}[\mathcal{X} \otimes \mathcal{Y}]$. (It is sometimes written $\oplus$ instead of $\boxplus$, but we reserve the symbol $\oplus$ for direct sum, as usual.) It is worth noticing that the tensor sum is not commutative even if $\mathcal{X} = \mathcal{Y}$ with O, A, B, I lying in $\mathcal{L}[\mathcal{X}]$. Indeed,

$$A \boxplus O = A \otimes I, \qquad O \boxplus B = I \otimes B,$$

$$A \boxplus I = A \otimes I + I \otimes I, \qquad I \boxplus B = I \otimes I + I \otimes B.$$

In particular, if $A = B$, then

$$A \boxplus O = A \otimes I \neq I \otimes A = O \boxplus A,$$

$$A \boxplus I = A \otimes I + I \otimes I \neq I \otimes I + I \otimes A = I \boxplus A.$$

3.4 Additional Propositions

All linear spaces are nonzero and over the same field $\mathbb{F}$, all sums of single tensors $\sum_i x_i \otimes y_i$ are finite, and $\cong$ stands for isomorphism. As agreed at the end of Section 3.1, and according to Corollary 3.6, equality of tensor products is regarded up to an isomorphism and so the equality or non-equality signs referring to tensor product spaces may be interpreted as $\cong$ or $\ncong$.

Proposition 3.A. *Let F be an arbitrary element in the tensor product space $\mathcal{T} = \mathcal{X} \otimes \mathcal{Y}$ of linear spaces $\mathcal{X}$ and $\mathcal{Y}$. There exists a representation for F such that $F = \sum_i x_i \otimes y_i$ with linearly independent sets $\{x_i\}$ and $\{y_i\}$.*

Proposition 3.B. *Let $\sum_{i=1}^n x_i \otimes y_i$ be a representation with n single tensors for an arbitrary element F in the tensor product space $\mathcal{T} = \mathcal{X} \otimes \mathcal{Y}$ of linear spaces $\mathcal{X}$ and $\mathcal{Y}$. If the set $\{x_i\}_{i=1}^n$ is linearly independent, then the set $\{y_i\}_{i=1}^n$ is uniquely determined by F in the following sense: if $\sum_{i=1}^n x_i \otimes y_i = \sum_{i=1}^n x_i \otimes y_i'$, then $y_i' = y_i$ for every $i = 1, \dots, n$. Symmetrically, if the set $\{y_i\}_{i=1}^n$ is linearly independent, then $\{x_i\}_{i=1}^n$ is uniquely determined by F.*

Proposition 3.C. *Consider the above setup. Suppose $x, u \in \mathcal{X}$ and $y, v \in \mathcal{Y}$.*

(a) *$x \otimes y = 0$ if and only if one of x or y is zero.*

(b) *$x \otimes y = u \otimes v \neq 0$ if and only if $u = \alpha x$ and $v = \frac{1}{\alpha} y$ for some $0 \neq \alpha \in \mathbb{F}$.*

Proposition 3.D. *For every triple $(\mathcal{X}, \mathcal{Y}, \mathcal{Z})$ of linear spaces,*

(a) $\mathcal{X} \otimes (\mathcal{Y} \oplus \mathcal{Z}) \cong (\mathcal{X} \otimes \mathcal{Y}) \oplus (\mathcal{X} \otimes \mathcal{Z})$,

(b) $(\mathcal{X} \oplus \mathcal{Y}) \otimes \mathcal{Z} \cong (\mathcal{X} \otimes \mathcal{Z}) \oplus (\mathcal{Y} \otimes \mathcal{Z})$,

(c) $(\mathcal{X} \otimes \mathcal{Y}) \otimes \mathcal{Z} \cong \mathcal{X} \otimes (\mathcal{Y} \otimes \mathcal{Z})$.

Proposition 3.E. *The following assertions hold for arbitrary linear manifolds $\mathcal{M}, \mathcal{M}_1, \mathcal{M}_2$ of a linear space $\mathcal{X}$ and $\mathcal{N}, \mathcal{N}_1, \mathcal{N}_2$ of a linear space $\mathcal{Y}$.*

(a) $(\mathcal{M}_1 \otimes \mathcal{Y}) \cap (\mathcal{M}_2 \otimes \mathcal{Y}) = (\mathcal{M}_1 \cap \mathcal{M}_2) \otimes \mathcal{Y}$.

(b) $(\mathcal{X} \otimes \mathcal{N}_1) \cap (\mathcal{X} \otimes \mathcal{N}_2) = \mathcal{X} \otimes (\mathcal{N}_1 \cap \mathcal{N}_2)$.

(c) $(\mathcal{M} \otimes \mathcal{Y}) \cap (\mathcal{X} \otimes \mathcal{N}) = \mathcal{M} \otimes \mathcal{N}$.

(d) $(\mathcal{M}_1 \otimes \mathcal{N}_1) \cap (\mathcal{M}_2 \otimes \mathcal{N}_2) = (\mathcal{M}_1 \cap \mathcal{M}_2) \otimes (\mathcal{N}_1 \cap \mathcal{N}_2)$.

Proposition 3.F. *If $\mathcal{X}$ and $\mathcal{Y}$ are linear spaces over $\mathbb{F}$ and $m, n \in \mathbb{N}$, then*

$$\mathcal{X} \otimes \mathbb{F}^n \cong \mathcal{X}^n \text{ and } \mathbb{F}^m \otimes \mathcal{Y} \cong \mathcal{Y}^m, \text{ and so } \mathbb{F}^m \otimes \mathbb{F}^n \cong \mathbb{F}^{mn}.$$

In particular, $\mathcal{X} \otimes \mathbb{F} \cong \mathcal{X}$ and $\mathbb{F} \otimes \mathcal{Y} \cong \mathcal{Y}$, and so $\mathbb{F} \otimes \mathbb{F} \cong \mathbb{F}$.

Proposition 3.G. *In general, for linear manifolds $\mathcal{M}$ and $\mathcal{N}$ of $\mathcal{X}$ and $\mathcal{Y}$,*

$$(\mathcal{X}/\mathcal{M}) \otimes (\mathcal{Y}/\mathcal{N}) \ncong (\mathcal{X} \otimes \mathcal{Y})/(\mathcal{M} \otimes \mathcal{N}).$$

Proposition 3.H. *If $A \in \mathcal{L}[\mathcal{X}, \mathcal{V}]$ and $B \in \mathcal{L}[\mathcal{Y}, \mathcal{W}]$, then*

$$(\mathcal{X}/\mathcal{N}(A)) \otimes (\mathcal{Y}/\mathcal{N}(B)) \cong (\mathcal{X} \otimes \mathcal{Y})/\mathcal{N}(A \otimes B).$$

Proposition 3.I. *If $\mathcal{M}$ and $\mathcal{N}$ are linear manifolds of $\mathcal{X}$ and $\mathcal{Y}$, then*

$$\mathcal{M} \otimes \mathcal{N} = \mathcal{X} \otimes \mathcal{Y} \iff \mathcal{M} = \mathcal{X} \text{ and } \mathcal{N} = \mathcal{Y}.$$

The converse of Corollary 3.10 reads as follows.

Proposition 3.J. *Let $\mathcal{X}, \mathcal{Y}, \mathcal{T}$ be linear spaces over the same field and take a bilinear map $\theta\colon \mathcal{X} \times \mathcal{Y} \to \mathcal{T}$. If $\dim \mathcal{T} = \dim \mathcal{X} \cdot \dim \mathcal{Y}$ and $\operatorname{span} \theta(\mathcal{X} \times \mathcal{Y}) = \mathcal{T}$, then $(\mathcal{T}, \theta)$ is a tensor product of $\mathcal{X}$ and $\mathcal{Y}$.*

Proposition 3.K. *Let $(\mathcal{T}, \theta)$ be a tensor product of linear spaces $\mathcal{X}$ and $\mathcal{Y}$ and let $\Upsilon \subseteq \mathcal{T}$ be a linear manifold of the tensor product space $\mathcal{T} = \mathcal{X} \otimes \mathcal{Y}$. If Υ is regular, then (Υ, θ') is a tensor product of linear manifolds $\mathcal{M}$ and $\mathcal{N}$ of $\mathcal{X}$ and $\mathcal{Y}$ such that Υ is isomorphic to $\operatorname{span} \theta(\mathcal{M} \times \mathcal{N})$. Moreover, there is a unique isomorphism $\Theta\colon \operatorname{span} \theta(\mathcal{M} \times \mathcal{N}) \to \Upsilon$ for which $\theta' = \Theta \circ \theta|_{\mathcal{M} \times \mathcal{N}}$.*

The next result is the converse of Theorem 3.19(d).

Proposition 3.L. *Consider the setup of Theorem 3.19. Suppose*

$$A \otimes B \in \mathcal{L}[\mathcal{X}, \mathcal{V}] \otimes \mathcal{L}[\mathcal{Y}, \mathcal{W}] \subseteq \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{V} \otimes \mathcal{W}]$$

is invertible. If its inverse $(A \otimes B)^{-1} \in \mathcal{L}[\mathcal{V} \otimes \mathcal{W}, \mathcal{X} \otimes \mathcal{Y}]$ is such that

$$(A \otimes B)^{-1} = C \otimes D \in \mathcal{L}[\mathcal{V}, \mathcal{X}] \otimes \mathcal{L}[\mathcal{W}, \mathcal{Y}] \subseteq \mathcal{L}[\mathcal{V} \otimes \mathcal{W}, \mathcal{X} \otimes \mathcal{Y}]$$

for some $C \in \mathcal{L}[\mathcal{V}, \mathcal{X}]$ and $D \in \mathcal{L}[\mathcal{W}, \mathcal{Y}]$ with one of them invertible, then $A \in \mathcal{L}[\mathcal{X}, \mathcal{V}]$ and $B \in \mathcal{L}[\mathcal{Y}, \mathcal{W}]$ are invertible and $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$.

Some additional properties related to Theorem 3.19(h) are given below.

Proposition 3.M. *Consider the setup of Theorem 3.19. Take* $\lambda, \mu \in \mathbb{F}$. *Let* I *stand for the identity in* $\mathcal{L}[\mathcal{X}]$, *in* $\mathcal{L}[\mathcal{Y}]$ *and in* $\mathcal{L}[\mathcal{X} \otimes \mathcal{Y}]$. *The following holds.*

(a) $\big(\mathcal{N}(A) \otimes \mathcal{Y}\big) \cup \big(\mathcal{X} \otimes \mathcal{N}(B)\big) \subsetneqq \mathcal{N}(A \otimes B)$.

(b) *If* $\dim \mathcal{X} = \infty$ *and* A *is not injective, or if* $\dim \mathcal{Y} = \infty$ *and* B *is not injective, then* $\dim \mathcal{N}(A \otimes B) = \infty$.

(c) $\mathcal{N}(\lambda I - A) \otimes \mathcal{N}(\mu I - B) \subseteq \mathcal{N}(\lambda\mu I - A \otimes B)$.

The inclusions in Proposition 3.M(a,c) extend the inclusion in Theorem 3.19(h) since $\mathcal{N}(A) \otimes \mathcal{N}(B) \subseteq \big(\mathcal{N}(A) \otimes \mathcal{Y}\big) \cap \big(\mathcal{X} \otimes \mathcal{N}(B)\big)$.

Basic operations with the tensor sums of Remark 3.20 are listed below.

Proposition 3.N. *For every* $\alpha, \beta \in \mathbb{F}$, $A, A_1, A_2 \in \mathcal{L}[\mathcal{X}]$ *and* $B, B_1, B_2 \in \mathcal{L}[\mathcal{Y}]$,

(a) $(\alpha + \beta)(A \boxplus B) = \alpha A \boxplus \beta B + \beta A \boxplus \alpha B$,

(b) $(A_1 + A_2) \boxplus (B_1 + B_2) = A_1 \boxplus B_1 + A_2 \boxplus B_2$,

(c) $(A_1 \boxplus B_1)(A_2 \boxplus B_2) = A_1 \otimes B_2 + A_2 \otimes B_1 + A_1 A_2 \boxplus B_1 B_2$,

(d) $(A \boxplus B)^\sharp = A^\sharp \boxplus B^\sharp$ (*as transformations in* $\mathcal{L}[(\mathcal{X} \otimes \mathcal{Y})^\sharp]$).

Notes: An organized exposition of tensor products in book form dates back to Schatten's 1950 monograph [85], where the notion of *direct products* of linear spaces was given in terms of *formal products* (which match what we call *tensor product space* and *single tensors*). This came after a Kronecker-product-like notion on finite-dimensional spaces given in Weyl's 1931book [93, Chapter V].

Definition 3.1 provides an axiomatic starting point to introduce tensor products. This leads to the abstract notion of algebraic tensor products of linear spaces, where the concrete standard forms are shown to be interpretations of the axiomatic formulation. Such concrete standard forms (namely, the quotient space and the linear maps of bilinear maps formulations) will be the subject of the next chapter. A similar abstract presentation has been considered, for instance, in [66, Section XVI.1] and [81, Chapter 14] for the quotient space formulation, in [94, Section I.4] and [27, Chapter 1] for both formulations, and in [42, Section 1.6] and [15, Section 2.2] for the linear maps of bilinear maps formulation. See also the recent expository paper in [57].

The above propositions are basic properties resulting from such axiomatic presentation. Proposition 3.A is a useful property whose proof goes as follows.

Proof of Proposition 3.A. Take $F = \sum_{i=1}^n x_i \otimes y_i \in \mathcal{X} \otimes \mathcal{Y}$ with $n > 1$ to avoid trivialities. Consider the subsets $\{x_i\}_{i=1}^n$ and $\{y_i\}_{i=1}^n$ of $\mathcal{X}$ and $\mathcal{Y}$, respectively,

(a) If $\{x_i\}_{i=1}^n$ is not linearly independent, then there exists x_k such that $x_k = \sum_{i \neq k} \alpha_i x_i$. So $F = x_k \otimes y_k + \sum_{i \neq k} x_i \otimes y_i = \sum_{i \neq k} \alpha_i x_i \otimes y_k + \sum_{i \neq k} x_i \otimes y_i = \sum_{i \neq k} x_i \otimes \alpha_i y_k + \sum_{i \neq k} x_i \otimes y_i = \sum_{i \neq k} x_i \otimes (\alpha_i y_k + y_i)$. Set $y_i' = (\alpha_i y_k + y_i)$ so that $F = \sum_{i \neq k} x_i \otimes y_i'$ is a representation of F with a smaller number of

single tensors (with $n-1$ single tensors), where $\{x_i\}_{i\neq k}$ is a proper subset of $\{x_i\}_{i=1}^n$. If $\{x_i\}_{i\neq k}$ is still linearly dependent, then repeat the same procedure to get another representation for the same F with an even smaller number of single tensors in terms of a proper subset of $\{x_i\}_{i\neq k}$. Since the representation of F has a finite number of single tensors, this process will eventually reach a representation with a minimum number of terms in x_i, and so the remaining subset of $\{x_i\}_{i=1}^n$ must be linearly independent.

(b) Now suppose $\{x_i\}_{i=1}^m$ is already linearly independent. Apply the same process to $\{y_i'\}_{i=1}^m$ until reaching a representation of F with a linearly independent subset of $\{y_i'\}_{i=1}^m$. Since this finite process involves only successive linear combinations of the already linearly independent set $\{x_i\}_{i=1}^m$, and since sets obtained by taking linear combinations of subsets of a linearly independent set remain linearly independent, then the resulting minimal representation of $F = \sum_j x_j' \otimes y_j'$ consists of a minimum number of single tensors $x_j' \otimes y_j'$ obtained from finite linearly independent sets $\{x_j'\}$ and $\{y_j'\}$. □

For Proposition 3.B, see, e.g., [94, Lemma 1.1, p. 9], which leads to Proposition 3.C (see, e.g., [5, Problem 2.U]). Proposition 3.D presents standard associative and distributive properties (see, e.g., [94, Exercise Section 5, p. 11 and Proposition 1.8, p. 13]). The intersections of regular subspaces in Proposition 3.E can be found in, for instance, [27, Section 1.15]. Proposition 3.F is an application of Corollary 3.10 and Proposition 1.C (since $\dim \mathcal{X}^n = n \dim \mathcal{X}$ in light of Proposition 1.T). To verify Proposition 3.G set $\mathcal{X} = \mathbb{F}$ and $\mathcal{M} = \{0\}$, and apply Proposition 3.F. Proposition 3.H can be verified by using the First Isomorphism Theorem (Theorem 1.4(b)) and Theorem 3.19(e) on the range of a tensor product transformation. Proposition 3.I is straightforward from Corollary 3.11(c). For Proposition 3.J see, e.g., [94, Lemma 1.2, p. 19].

Proof of Proposition 3.K. Let $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ be nonzero linear spaces over the same field. Let $(\mathcal{T}, \theta)$ be a tensor product of $\mathcal{X}$ and $\mathcal{Y}$. By Definition 3.1, for every bilinear map $\phi: \mathcal{X}\times\mathcal{Y} \to \mathcal{Z}$ there is a linear transformation $\Phi: \mathcal{T} \to \mathcal{Z}$ such that $\phi = \Phi \circ \theta$. If Υ is a regular linear manifold of $\mathcal{T}$, then (cf. Definition 3.14) (Υ, θ') is a tensor product of linear manifolds $\mathcal{M}$ and $\mathcal{N}$ of $\mathcal{X}$ and $\mathcal{Y}$. Take any bilinear map $\phi': \mathcal{M}\times\mathcal{N} \to \mathcal{Z}$. By Theorem 2.5, there is a bilinear extension $\phi: \mathcal{X}\times\mathcal{Y} \to \mathcal{Z}$ of ϕ' over $\mathcal{X}\times\mathcal{Y}$. Thus

$$\phi' = \phi|_{\mathcal{M}\times\mathcal{N}} = (\Phi \circ \theta)|_{\mathcal{M}\times\mathcal{N}} = \Phi \circ \theta|_{\mathcal{M}\times\mathcal{N}} = \Phi_{\operatorname{span}\theta(\mathcal{M}\times\mathcal{N})} \circ \theta|_{\mathcal{M}\times\mathcal{N}},$$

where $\Phi|_{\operatorname{span}\theta(\mathcal{M}\times\mathcal{N})} : \operatorname{span}\theta(\mathcal{M}\times\mathcal{N}) \to \mathcal{Z}$ is a linear transformation (the restriction of a linear transformation to a linear manifold is linear) associated with the bilinear map $\theta|_{\mathcal{M}\times\mathcal{N}}: \mathcal{M}\times\mathcal{N} \to \Upsilon$ (the restriction of a bilinear map to the Cartesian product of linear manifolds is bilinear). So (cf. Definition 3.1)

$$(\operatorname{span}\theta(\mathcal{M}\times\mathcal{N}),\ \theta|_{\mathcal{M}\times\mathcal{N}}) \text{ is a tensor product of } \mathcal{M} \text{ and } \mathcal{N}.$$

Thus (Υ, θ') and $(\operatorname{span}\theta(\mathcal{M}\times\mathcal{N}), \theta|_{\mathcal{M}\times\mathcal{N}})$ are tensor products of the same pair of linear spaces $\mathcal{M}$ and $\mathcal{N}$, and so Theorem 3.5 says that there is a unique isomorphism $\Theta \in \mathcal{L}[\operatorname{span}\theta(\mathcal{M}\times\mathcal{N}), \Upsilon]$ for which $\theta' = \Theta \circ \theta|_{\mathcal{M}\times\mathcal{N}}$). □

Proof of Proposition 3.L. If there is $(A\otimes B)^{-1}\subseteq\mathcal{L}[\mathcal{V}\otimes\mathcal{W},\mathcal{X}\otimes\mathcal{Y}]$ such that

$$(A\otimes B)^{-1}(A\otimes B)=I_{\mathcal{X}\otimes\mathcal{Y}}\quad\text{and}\quad(A\otimes B)(A\otimes B)^{-1}=I_{\mathcal{V}\otimes\mathcal{W}},$$

where $I_{\mathcal{X}\otimes\mathcal{Y}}=I_{\mathcal{X}}\otimes I_{\mathcal{Y}}$ and $I_{\mathcal{V}\otimes\mathcal{W}}=I_{\mathcal{V}}\otimes I_{\mathcal{W}}$. These are the identities on $\mathcal{X}\otimes\mathcal{Y}$, on $\mathcal{V}\otimes\mathcal{W}$, and on $\mathcal{X},\mathcal{Y},\mathcal{V},\mathcal{W}$, respectively. If, in addition,

$$(A\otimes B)^{-1}=C\otimes D$$

with $C\in\mathcal{L}[\mathcal{V},\mathcal{X}]$ and $D\in\mathcal{L}[\mathcal{W},\mathcal{Y}]$, so that $C\otimes D\neq 0$, then by Theorem 3.19(c)

$$I_{\mathcal{X}}\otimes I_{\mathcal{Y}}=(C\otimes D)(A\otimes B)=CA\otimes DB,\quad I_{\mathcal{V}}\otimes I_{\mathcal{W}}=(A\otimes B)(C\otimes D)=AC\otimes BD.$$

Thus Proposition 3.C(b) ensures that there are nonzero $\alpha,\beta\in\mathbb{F}$ for which

$$CA=\alpha I_{\mathcal{X}},\quad DB=\tfrac{1}{\alpha}I_{\mathcal{Y}},\quad\text{and}\quad AC=\beta I_{\mathcal{V}},\quad BD=\tfrac{1}{\beta}I_{\mathcal{W}}.$$

If one of C or D is invertible, say, if C is invertible, then $A=\alpha C^{-1}=\beta C^{-1}$. Hence (i) $\beta=\alpha$ and (ii) A is invertible with $A^{-1}=\frac{1}{\alpha}C$. Since $\beta=\alpha$ we get $(\alpha D)B=I_{\mathcal{Y}}$ and $B(\alpha D)=I_{\mathcal{W}}$ so that B is invertible with $B^{-1}=\alpha D$. As A and B are invertible, Theorem 3.19(d) ensures $(A\otimes B)^{-1}=A^{-1}\otimes B^{-1}$. □

Proof of Proposition 3.M. (a) This is immediate by the equation in Theorem 3.19(h) — union of subspaces is included in their sum — and so the inclusion is proper whenever both subspaces are nonzero.

(b) This comes from item (a) and Corollary 3.10.

(c) If $\mathcal{N}(\lambda I-A)=\{0\}$ or $\mathcal{N}(\mu I-B)=\{0\}$, then $\mathcal{N}(\lambda I-A)\otimes\mathcal{N}(\mu I-B)=\{0\}$ and the inclusion is trivial. So suppose $\mathcal{N}(\lambda I-A)\neq\{0\}$ and $\mathcal{N}(\mu I-B)\neq\{0\}$. If $F\in\mathcal{N}(\lambda I-A)\otimes\mathcal{N}(\mu I-B)$ is nonzero, then consider a representation $F=\sum_i x_i\otimes y_i$ of it with all single tensors $x_i\otimes y_i$ nonzero. Then $0\neq x_i\in\mathcal{N}(\lambda I-A)$ and $0\neq y_i\in\mathcal{N}(\mu I-B)$, and so $\lambda x_i=Ax_i$ and $\mu y_i=Ay_i$, for every integer i. Then $\lambda\mu(x_i\otimes y_i)=\lambda x_i\otimes\mu y_i=Ax_i\otimes By_i=(A\otimes B)(x_i\otimes y_i)$, and hence $\lambda\mu F=\lambda\mu\sum_i x_i\otimes y_i=\sum_i\lambda x_i\otimes\mu y_i=\sum_i Ax_i\otimes By_i=(A\otimes B)\sum_i x_i\otimes y_i=(A\otimes B)F$. Therefore the nonzero F lies in $\mathcal{N}(\lambda\mu I-A\otimes B)$. □

Proposition 3.N is readily verified by using Theorem 3.19 and the definition of tensor sum in Remark 3.20(b).

Suggested Readings

Brown and Pearcy [5]
Defant and Floret [15]
Greub [27]
Halmos [30]
Jarchow [42]
Lang [66]
MacLane and Birkhoff [69]
Roman [81]
Ryan [83]
Yokonuma [94]



4

Interpretations

Tensor products do exist. The next sections exhibit concrete examples of tensor products as promised in Remark 3.2(b). These represent different, but isomorphic, means of constructing tensor products out of a pair of linear spaces.

4.1 Interpretation via Quotient Space

Let $\mathcal{X}$ and $\mathcal{Y}$ be nonzero linear spaces over a field $\mathbb{F}$, and take the Cartesian product $S = \mathcal{X}\times\mathcal{Y}$ of $\mathcal{X}$ and $\mathcal{Y}$. Now consider the notation and terminology of Section 1.1. Thus let $\mathscr{F} \subset \mathbb{F}^S$ be the free linear space generated by S (i.e., the linear manifold of the linear space $\mathbb{F}^S$ of all $\mathbb{F}$-valued functions on S consisting of all functions $\boldsymbol{f}\colon S \to \mathbb{F}$ that vanish everywhere on the complement of some finite subset of S, so that $\mathscr{F}$ is itself a linear space over $\mathbb{F}$). Consider the Hamel basis $\mathbb{S} = \{e_{(x,y)}\}_{(x,y)\in S}$ for the linear space $\mathscr{F}$ consisting of characteristic functions $e_{(x,y)} = \chi_{\{(x,y)\}} = \chi_{\{x\}}\chi_{\{y\}}\colon \mathcal{X}\times\mathcal{Y} \to \mathbb{F}$ of all singletons at each pair of vectors (x, y) in the Cartesian product $S = \mathcal{X}\times\mathcal{Y}$. Being indexed by the set S, the Hamel basis $\mathbb{S}$ is naturally identified with S. Thus a formal linear combination $\sum_{i=1}^n \alpha_i(x_i, y_i)$ of elements (x_i, y_i) in $S = \mathcal{X}\times\mathcal{Y}$ is identified (by definition) with a linear combination $\sum_{i=1}^n \alpha_i e_{(x_i,y_i)}$ of functions $e_{(x_i,y_i)}$ from the Hamel basis $\mathbb{S}$ for $\mathscr{F}$. In this sense the free linear space $\mathscr{F}$ generated by S is identified with the so-called linear space of all formal linear combinations $\sum_{i=1}^n \alpha_i(x_i, y_i)$ of elements (x_i, y_i) of the set $S = \mathcal{X}\times\mathcal{Y}$. In particular, recall that the pairs of vectors (x, y) in the Cartesian product $S = \mathcal{X}\times\mathcal{Y}$ are in a one-to-one correspondence with the functions $e_{(x,y)}$ in the Hamel basis $\mathbb{S}$ for the function space $\mathscr{F}$. Thus let now $\approx$ stand for such a natural identification:

$$\sum\nolimits_{i=1}^n \alpha_i(x_i, y_i) \approx \sum\nolimits_{i=1}^n \alpha_i e_{(x_i,y_i)} \quad \text{and so} \quad (x, y) \approx e_{(x,y)}.$$

The Cartesian product $S = \mathcal{X} \times \mathcal{Y}$, being identified with the Hamel basis $\mathbb{S}$ for the linear space $\mathscr{F}$, is then regarded as a subset of $\mathscr{F}$, and so we write

$$S \subset \mathscr{F} \quad \text{for} \quad S \approx \mathbb{S} \subset \mathscr{F}.$$

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C. S. Kubrusly, *Bilinear Maps and Tensor Products in Operator Theory*,
Universitext, https://doi.org/10.1007/978-3-031-34093-2_4

With the identification $\approx$ still in force, consider the vector addition and the scalar multiplication in the linear space $\mathcal{F}$, and their respective formal addition and formal scalar multiplication of elements in $S = \mathcal{X}\times\mathcal{Y}$ as follows. For every $x, x_1, x_2 \in \mathcal{X}$, every $y, y_1, y_2 \in \mathcal{Y}$, and every $\alpha \in \mathbb{F}$,

$$(x_1, y_1) + (x_2, y_2) \approx e_{(x_1,y_1)} + e_{(x_2,y_2)} = \chi_{\{(x_1,y_1)\}} + \chi_{\{(x_2,y_2)\}},$$

$$\alpha\,(x\,, y) \approx \alpha\, e_{(x,y)} = \alpha \chi_{\{(x,y)\}},$$

where each $e_{(x,y)} = \chi_{\{(x,y)\}} = \chi_{\{x\}}\chi_{\{y\}}$ is the product of characteristic functions of singletons at $x \in \mathcal{X}$ and $y \in \mathcal{Y}$. Then consider the differences

(i) $e_{(x_1+x_2,y)} - e_{(x_1,y)} - e_{(x_2,y)} \approx (x_1 + x_2\,, y) - (x_1\,, y) - (x_2\,, y)$,

(ii) $e_{(x,y_1+y_2)} - e_{(x,y_1)} - e_{(x,y_2)} \approx (x\,, y_1 + y_2) - (x\,, y_1) - (x\,, y_2)$,

(iii) $e_{(\alpha x,y)} - \alpha\, e_{(x,y)} \approx (\alpha x, y) - \alpha(x, y)$,

(iv) $e_{(x,\alpha y)} - \alpha\, e_{(x,y)} \approx (x, \alpha y) - \alpha(x, y)$.

It is clear that the above differences are not null. If they were, then we might identify a bilinear rule functioning for the ordered pairs $(x, y) \in \mathcal{X}\times\mathcal{Y}$. Roughly speaking, this is what we would like to have. Thus we will, in some fashion, identify the ordered pairs (x, y) so as to make those differences be included in some equivalence class of elements in S that is associated with some equivalence class in $\mathcal{F}$ that contains the origin of $\mathcal{F}$. To begin approaching this identification scheme, we will look at the cosets $[\boldsymbol{f}]$ of functions $\boldsymbol{f}$ in $\mathcal{F}$ or, equivalently, at an equivalence class $[(x, y)]$ of pairs (x, y) in $S = \mathcal{X}\times\mathcal{Y}$ with respect to some suitable equivalence relation. A convenient equivalence relation gathering those differences at the origin of some appropriate further linear space generated by $\mathcal{F}$ is naturally obtained through quotient spaces.

Thus take the Cartesian product $\mathcal{X}\times\mathcal{Y}$ of the linear spaces $\mathcal{X}$ and $\mathcal{Y}$. With $\mathcal{F}$ standing for the free linear space generated by $S = \mathcal{X}\times\mathcal{Y}$, consider the linear manifold $\mathcal{M}$ of $\mathcal{F}$ generated by the differences in (i) to (iv) above,

$$\mathcal{M} = \text{span}\,\big\{\, e_{(x_1+x_2,y)} - e_{(x_1,y)} - e_{(x_2,y)}\,,\quad e_{(x,y_1+x_2)} - e_{(x,y_1)} - e_{(x,y_2)}\,,$$
$$e_{(\alpha x,y)} - \alpha\, e_{(x,y)}\,,\quad e_{(x,\alpha y)} - \alpha\, e_{(x,y)}\,\big\}.$$

Since a pair (x, y) is identified with the element $e_{(x,y)}$ of the Hamel basis for $\mathcal{F}$ (i.e., $(x, y) \approx e_{(x,y)}$), as formal linear combinations of pairs in $\mathcal{X}\times\mathcal{Y}$ are identified with linear combinations of functions in $\mathcal{F}$, we may identify the above differences of functions in $\mathcal{F}$ with formal differences of pairs in S:

$$\mathcal{M} \approx \text{span}\,\big\{\, (x_1 + x_2, y) - (x_1, y) - (x_2, y)\,,\quad (\alpha x, y) - \alpha\,(x, y)\,,$$
$$(x, y_1 + y_2) - (x, y_1) - (x, y_2)\,,\quad (x, \alpha y) - \alpha(x, y)\,\big\}.$$

Now consider the quotient space $\mathcal{F}/\mathcal{M}$ of $\mathcal{F}$ modulo $\mathcal{M}$. It will be shown in Theorem 4.1 that the quotient space $\mathcal{F}/\mathcal{M}$ is a tensor product space (sometimes called the *algebraic tensor product*) of $\mathcal{X}$ and $\mathcal{Y}$ associated with a natural

bilinear map θ (which will be defined shortly). Take an arbitrary pair (x, y) in the Cartesian product $\mathcal{X}\times\mathcal{Y}$, let $e_{(x,y)} = \chi_{\{(x,y)\}}$ be the corresponding characteristic function in the linear space $\mathcal{F}$, and consider the natural quotient map (which is surjective)

$$\pi\colon \mathcal{F} \to \mathcal{F}/\mathcal{M}$$

such that $\pi(\boldsymbol{f}) = [\boldsymbol{f}] = \boldsymbol{f} + \mathcal{M}$ in $\mathcal{F}/\mathcal{M}$ for every $\boldsymbol{f}$ in $\mathcal{F}$. Define the map

$$\theta\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{F}/\mathcal{M}$$

as follows. For every $(x, y) \in \mathcal{X}\times\mathcal{Y}$, set $\theta(x, y) = \pi(e_{(x,y)})$. Therefore

$$\theta(S) = R(\theta) = R(\pi|_{\mathbb{S}}) = \pi(\mathbb{S}).$$

Since the Hamel basis $\mathbb{S} = \{e_{(x,y)}\}_{(x,y)\in S}$ for the free linear space $\mathcal{F}$ generated by the set S is naturally identified with S itself, $\mathbb{S} = \{e_{(x,y)}\}_{(x,y)\in S} \approx S = \mathcal{X}\times\mathcal{Y}$, then the domain S of θ is identified with the domain $\mathbb{S}$ of $\pi|_{\mathbb{S}}$. Also, since their values coincide, $\theta(x, y) = \pi(e_{(x,y)})$, then θ is naturally identified with the restriction $\pi|_{\mathbb{S}}$ of the natural quotient map π to the Hamel basis $\mathbb{S}$. So write

$$\theta = \pi|_{\mathbb{S}} \quad \text{for} \quad \theta \approx \pi|_{\mathbb{S}}.$$

Denote by $x \otimes y$ the image of $e_{(x,y)} \in \mathbb{S} \subset \mathcal{F} \approx (x, y) \in \mathcal{X}\times\mathcal{Y} = S$ under π,

$$x \otimes y = \pi(e_{(x,y)}) = [e_{(x,y)}] = e_{(x,y)} + \mathcal{M} \approx [(x, y)],$$

where $[(x, y)]$ is the equivalence class of all pairs (x, y) for which $e_{(x,y)}$ belongs to the equivalence class $[e_{(x,y)}]$. So $x \otimes y$ is identified with $[(x, y)]$.

> *Each pair* (x, y) *in* $\mathcal{X}\times\mathcal{Y}$ *regarded as an element of* $\mathcal{F}$ (i.e., identified with an element of the Hamel basis $\mathbb{S} = \{e_{(x,y)}\}_{(x,y)\in\mathcal{X}\times\mathcal{Y}}$ for the linear space $\mathcal{F}$) *is projected* (through the quotient map $\pi\colon \mathcal{F} \to \mathcal{F}/\mathcal{M}$) *onto an element of* $\mathcal{F}/\mathcal{M}$ *denoted by* $x \otimes y$.

Elements of $\mathcal{F}/\mathcal{M}$ of the form $x \otimes y$ are the images of $\theta\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{F}/\mathcal{M}$, and are again referred to as *single tensors* or *decomposables*:

$$x \otimes y = \theta(x, y) \quad \text{for every} \quad (x, y) \in \mathcal{X}\times\mathcal{Y}.$$

With $\mathcal{F}$ standing for the free linear space generated by the Cartesian product $\mathcal{X}\times\mathcal{Y}$ of linear spaces $\mathcal{X}$ and $\mathcal{Y}$, and $\mathcal{M}$ standing for the linear manifold generated by the differences in (i) to (iv), we show that $\mathcal{F}/\mathcal{M}$ is in fact a tensor product space of $\mathcal{X}$ and $\mathcal{Y}$ associated with the bilinear map $\theta\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{F}/\mathcal{M}$.

Theorem 4.1. $(\mathcal{F}/\mathcal{M}, \theta)$ *is a tensor product of* $\mathcal{X}$ *and* $\mathcal{Y}$.

Proof. Consider the axioms (a) and (b) in Definition 3.1, where $\mathcal{X} \times \mathcal{Y}$ is the Cartesian product of linear spaces $\mathcal{X}$ and $\mathcal{Y}$.

(a_1) By the definition of $\mathcal{M}$, the differences in (i) to (iv) lie in $\mathcal{M} = [0] \in \mathcal{F}/\mathcal{M}$. Thus for every $x, x_1, x_2 \in \mathcal{X}$, every $y, y_1, y_2 \in \mathcal{Y}$, and every $\alpha \in \mathbb{F}$,

$$\theta(x_1 + x_2, y) = [e_{(x_1+x_2,y)}] = [e_{(x_1,y)}] + [e_{(x_2,y)}] = \theta(x_1, y) + \theta(x_2, y),$$

$$\theta(x, y_1 + y_2) = [e_{(x,y_1+y_2)}] = [e_{(x,y_1)}] + [e_{(x,y_2)}] = \theta(x, y_1) + \theta(x, y_2),$$

$$\theta(\alpha x, y) = [e_{(\alpha x,y)}] = \alpha[e_{(x,y)}] = \alpha\theta(x, y),$$

$$\theta(x, \alpha y) = [e_{(x,\alpha y)}] = \alpha[e_{(x,y)}] = \alpha\theta(x, y).$$

Hence $\theta\colon \mathcal{X}\times\mathcal{Y} \to \mathscr{F}/\mathcal{M}$ is a bilinear map (and so $\alpha x \otimes y = \alpha(x \otimes y) = x \otimes \alpha y$).

(a$_2$) Since $\mathbb{S} = \{e_{(x,y)}\}_{(x,y)\in\mathcal{X}\times\mathcal{Y}}$ is a Hamel basis for the linear space $\mathscr{F}$, then $\operatorname{span}\pi(\mathbb{S}) = \mathscr{F}/\mathcal{M}$ (Remark 1.5). Thus, as the ranges coincide, $R(\theta) = R(\pi|_{\mathbb{S}})$,

$$\operatorname{span} R(\theta) = \operatorname{span} R(\pi|_{\mathbb{S}}) = \operatorname{span}\pi(\mathbb{S}) = \mathscr{F}/\mathcal{M}.$$

(b) Take a bilinear map $\phi\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{Z}$ into a linear space $\mathcal{Z}$ and consider the transformation $\widetilde{\Phi}$ on the free linear space $\mathscr{F}$ generated by $S = \mathcal{X}\times\mathcal{Y}$,

$$\widetilde{\Phi}\colon \mathscr{F} \to \mathcal{Z},$$

associating a bona fide linear combination of vectors from $\mathcal{Z}$ with each $\boldsymbol{f}\in\mathscr{F}$ as follows. For each $\boldsymbol{f} = \sum_{i=1}^{n}\alpha_i e_{(x_i,y_i)} \in \mathscr{F}$, set

$$\widetilde{\Phi}(\boldsymbol{f}) = \sum\nolimits_{i=1}^{n} \alpha_i\, \phi(x_i, y_i) \;\in\; \mathcal{Z},$$

which defines a linear transformation $\widetilde{\Phi}$ between the linear spaces $\mathscr{F}$ and $\mathcal{Z}$.

(Indeed, for every $\boldsymbol{f} = \sum_{i=1}^{n}\alpha_i(x_i, y_i)$ and $\boldsymbol{g} = \sum_{i=1}^{m}\beta_i(u_i, v_i)$ in $\mathscr{F}$ we get $\widetilde{\Phi}(\boldsymbol{f} + \boldsymbol{g}) = \sum_{i=1}^{n}\alpha_i\,\phi(x_i, y_i) + \sum_{i=1}^{m}\beta_i\,\phi(x_i, y_i) = \widetilde{\Phi}(\boldsymbol{f}) + \widetilde{\Phi}(\boldsymbol{g})$ and $\widetilde{\Phi}(\alpha\boldsymbol{f}) = \sum_{i=1}^{n}\alpha\alpha_i\,\phi(x_i, y_i) = \alpha\,\widetilde{\Phi}(\boldsymbol{f})$ — regardless of whether ϕ is bilinear or not.)

Moreover, since $\widetilde{\Phi}(e_{(x_i,y_i)}) = \phi(x, y)$, then

$$\widetilde{\Phi}(\mathbb{S}) = \phi(S).$$

Since $\mathbb{S} \approx S = \mathcal{X}\times\mathcal{Y}$, then $\widetilde{\Phi}|_{\mathbb{S}}$ is naturally identified with ϕ. So write

$$\widetilde{\Phi}|_{\mathbb{S}} = \phi.$$

Furthermore, since $\phi\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{Z}$ is a bilinear map, the linear transformation $\widetilde{\Phi}\colon \mathscr{F} \to \mathcal{Z}$ evaluated at the differences in (i) to (iv) is null, and hence by definition of the linear manifold $\mathcal{M}$ of $\mathscr{F}$ we get $\widetilde{\Phi}(\mathcal{M}) = 0$. That is, $\mathcal{M} \subseteq \mathcal{N}(\widetilde{\Phi})$. Then according to Theorem 1.4(a) there is a unique linear transformation

$$\Phi\colon \mathscr{F}/\mathcal{M} \to \mathcal{Z}$$

such that $\widetilde{\Phi} = \Phi \circ \pi$. Now restricting to $\mathrm{S} \subset \mathscr{F}$, and since $\operatorname{span}\pi(\mathbb{S}) = \mathscr{F}/\mathcal{M}$, we get $\phi = \widetilde{\Phi}|_{\mathbb{S}} = (\Phi\circ\pi)|_{\mathbb{S}} = \Phi|_{\operatorname{span}\mathcal{R}(\pi)} \circ \pi|_{\mathbb{S}} = \Phi \circ \theta$. Equivalently, the diagrams

$$\begin{array}{ccc} \mathscr{F} & \xrightarrow{\widetilde{\Phi}} & \mathcal{Z} \\ & \searrow^{\pi} \quad \uparrow_{\Phi} & \\ & \mathscr{F}/\mathcal{M} & \end{array} \qquad \text{and} \qquad \begin{array}{ccc} \mathbb{S} & \xrightarrow{\phi} & \mathcal{Z} \\ & \searrow^{\theta} \quad \uparrow_{\Phi} & \\ & \mathscr{F}/\mathcal{M} & \end{array}$$

commute. If we again identify $S \approx \mathbb{S}$ (thus regarding $S = \mathcal{X}\times\mathcal{Y}$ as a subset of $\mathscr{F}$ and writing $\theta = \pi|_S$ for $\theta = \pi|_{\mathbb{S}}$ and $\phi = \widetilde{\Phi}|_S$ for $\phi = \widetilde{\Phi}|_{\mathbb{S}}$), then the diagram

$$\begin{array}{ccc} \mathcal{X}\times\mathcal{Y} & \xrightarrow{\phi} & \mathcal{Z} \\ & \searrow^{\theta} \quad \uparrow_{\Phi} & \\ & \mathscr{F}/\mathcal{M} & \end{array}$$

commutes, and so the pair $(\mathscr{F}/\mathcal{M}, \theta)$ satisfies the axioms of Definition 3.1. □

Thus consider the tensor product space $\mathcal{X} \otimes \mathcal{Y} = \mathscr{F}/\mathcal{M}$ of two linear spaces $\mathcal{X}$ and $\mathcal{Y}$. Let F denote an arbitrary element in $\mathscr{F}/\mathcal{M}$ so that $F = [\boldsymbol{f}] \in \mathscr{F}/\mathcal{M}$, where $\boldsymbol{f} \in \mathscr{F}$ is any representative of the equivalence class $[\boldsymbol{f}]$. Since $\mathbb{S} = \{e_{(x,y)}\}_{(x,y)\in\mathcal{X}\times\mathcal{Y}}$ is a Hamel basis for $\mathscr{F}$, then take the expansion $\boldsymbol{f} = \sum_{i=1}^{n} \alpha_i e_{(x_i,y_i)}$ of $\boldsymbol{f}$ in terms of $\{e_{(x,y)}\}_{(x,y)\in\mathcal{X}\times\mathcal{Y}}$ to get

$$F = [\boldsymbol{f}] = \Big[\sum\nolimits_{i=1}^{n} \alpha_i e_{(x_i,y_i)}\Big] = \sum\nolimits_{i=1}^{n} \alpha_i [e_{(x_i,y_i)}] = \sum\nolimits_{i=1}^{n} \alpha_i (x_i \otimes y_i).$$

Recall from Remark 1.5 that a Hamel basis for $\mathscr{F}/\mathcal{M}$ is in a one-to-one correspondence with a Hamel basis for any algebraic complement of $\mathcal{M}$ in $\mathscr{F}$, and therefore a Hamel basis for $\mathscr{F}/\mathcal{M}$ is included in a Hamel basis for $\mathscr{F}$. As we saw above, the set $\{x \otimes y\}_{(x,y)\in\mathcal{X}\times\mathcal{Y}} = \{[e_{(x,y)}]\}_{(x,y)\in\mathcal{X}\times\mathcal{Y}}$, which is in a one-to-one correspondence with the Hamel basis $\mathbb{S} = \{e_{(x,y)}\}_{(x,y)\in\mathcal{X}\times\mathcal{Y}}$ for $\mathscr{F}$, spans $\mathscr{F}/\mathcal{M}$. So it includes a Hamel basis for $\mathscr{F}/\mathcal{M}$. Thus there exists a Hamel basis for the tensor product space $\mathscr{F}/\mathcal{M}$ consisting of single tensors only, say $\{x_\gamma \otimes y_\gamma\}_{\gamma\in\Gamma} \subseteq \{x \otimes y\}_{(x,y)\in\mathcal{X}\times\mathcal{Y}}$, in accordance with Theorem 3.9. In fact, since $\alpha_i(x_i \otimes y_i) = \alpha_i x_i \otimes y_i = x_i \otimes \alpha_i y_i$, write the above expansion as

$$F = [\boldsymbol{f}] = \sum x \otimes y = \sum x_\gamma \otimes y_\gamma,$$

both sums being finite, the first one drawn from the collection of all single tensors $\{x \otimes y\}_{(x,y)\in\mathcal{X}\times\mathcal{Y}}$ and the second from a Hamel basis $\{x_\gamma \otimes y_\gamma\}_{\gamma\in\Gamma}$ for $\mathscr{F}/\mathcal{M}$, as in Corollary 3.4. Also, $\dim \mathscr{F}/\mathcal{M} = \dim \mathcal{X} \cdot \dim \mathcal{Y}$ by Corollary 3.10.

This interpretation where single tensors are defined as equivalence classes of characteristic functions of singletons in the Cartesian product of linear spaces is referred to as the *quotient space approach*.

Remark 4.2. *A Summary of the Quotient Space Approach.*

◇ $S = \mathcal{X}\times\mathcal{Y}$: the Cartesian product of linear spaces $\mathcal{X}$ and $\mathcal{Y}$.

- $\diamond$ $\mathscr{F} \subset \mathbb{F}^S$: the linear space of functions vanishing outside finite subsets of S.
- $\diamond$ $e_{(x,y)} = \chi_{\{(x,y)\}} : \mathbb{F}^{\mathcal{X}\times\mathcal{Y}} \to \mathbb{F}$.
- $\diamond$ $\mathbb{S} = \{e_{(x,y)}\}_{(x,y)\in\mathcal{X}\times\mathcal{Y}} \subset \mathscr{F}$: a Hamel basis for $\mathscr{F}$.
- $\diamond$ $\mathbb{S} \approx S \subset \mathscr{F}$.
- $\diamond$ $\mathcal{M}$: the linear manifold of $\mathscr{F}$ spanned by specific differences of $e_{(x,y)} \in \mathscr{F}$; those differences that prevent the rule $(x, y) \mapsto e_{(x,y)}$ to be bilinear.
- $\diamond$ $\mathscr{F}/\mathcal{M}$: the quotient space where the equivalence classes $[e_{(x,y)}] \in \mathscr{F}/\mathcal{M}$ are such that the rule $(x, y) \mapsto [e_{(x,y)}]$ is bilinear.
- $\diamond$ $\mathcal{X} \otimes \mathcal{Y} = \mathscr{F}/\mathcal{M}$.
- $\diamond$ $\pi(\boldsymbol{f}) = [\boldsymbol{f}]$: the natural quotient map $\pi : \mathscr{F} \to \mathscr{F}/\mathcal{M}$.
- $\diamond$ $\theta = \pi|_{\mathbb{S}}$: the natural bilinear map $\theta : \mathcal{X}\times\mathcal{Y} \to \mathscr{F}/\mathcal{M}$.
- $\diamond$ $x \otimes y = \theta(x, y) = \pi(e_{(x,y)}) = [e_{(x,y)}] \in \mathscr{F}/\mathcal{M}$ (a single tensor).
- $\diamond$ $F = [\boldsymbol{f}] \in \mathscr{F}/\mathcal{M} \iff F = [\boldsymbol{f}] = \sum x \otimes y$ (a finite sum).

Such a realization of the tensor product via quotient space has naturally been extended from linear spaces over fields to modules over rings (see, e.g., [66, Chapter XVI]), where a tensor product is regarded as a universal object in the realm of category theory.

4.2 Interpretation via Linear Maps of Bilinear Maps

Let $\mathcal{X}$ and $\mathcal{Y}$ be nonzero linear spaces over the same field $\mathbb{F}$. Given an arbitrary but fixed linear space $\mathcal{F}$ over $\mathbb{F}$, consider the linear space $b[\mathcal{X}\times\mathcal{Y}, \mathcal{F}]$ of all bilinear maps $\psi : \mathcal{X}\times\mathcal{Y} \to \mathcal{F}$ of the Cartesian product $\mathcal{X}\times\mathcal{Y}$ into $\mathcal{F}$ (in particular, ψ may be acting on the direct sum $\mathcal{X} \oplus \mathcal{Y}$). Associated with each pair $(x, y) \in \mathcal{X}\times\mathcal{Y}$, consider a transformation $x \otimes y : b[\mathcal{X}\times\mathcal{Y}, \mathcal{F}] \to \mathcal{F}$ defined by

$$(x \otimes y)(\psi) = \psi(x, y) \in \mathcal{F} \quad \text{for every} \quad \psi \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{F}].$$

This again is referred to as a *single tensor*, which is a linear transformation.

(In fact, since $b[\mathcal{X}\times\mathcal{Y}, \mathcal{F}]$ is a linear space, the sum of bilinear maps is a bilinear map, and so it makes sense to evaluate $(x \otimes y)(\psi_1 + \psi_2) = (\psi_1 + \psi_2)(x, y) = \psi_1(x, y) + \psi_2(x, y) = (x \otimes y)(\psi_1) + (x \otimes y)(\psi_2)$ for every $\psi_1, \psi_2 \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{F}]$. Also, $\alpha(x \otimes y)(\psi) = (x \otimes y)(\alpha\psi)$ for every $\psi \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{F}]$ and $\alpha \in \mathbb{F}$.)

Thus

$$x \otimes y \in \mathcal{L}[\, b[\mathcal{X}\times\mathcal{Y}, \mathcal{F}], \mathcal{F}],$$

a linear map of bilinear maps, where $\mathcal{L}[\, b[\mathcal{X}\times\mathcal{Y}, \mathcal{F}], \mathcal{F}]$ is the linear space of all linear transformations from $b[\mathcal{X}\times\mathcal{Y}, \mathcal{F}]$ to $\mathcal{F}$.

> *Hence a single tensor $x \otimes y$ associated with a pair (x, y) is defined as a* linear transformation of bilinear maps *given by evaluation at the pair*

(i.e., $(x \otimes y)(\psi) = \psi(x, y)$), *and therefore this approach to tensor product focuses explicitly and precisely on the* linearization of bilinear maps.

Now take the collection $\top_{\mathcal{X},\mathcal{Y},\mathcal{F}}$ of all single tensors as defined above,

$$\top_{\mathcal{X},\mathcal{Y},\mathcal{F}} = \big\{x \otimes y \in \mathcal{L}[\,b[\mathcal{X}\times\mathcal{Y},\mathcal{F}],\mathcal{F}]\colon\ x \in \mathcal{X} \text{ and } y \in \mathcal{Y}\big\},$$

consider its span, which is a linear manifold of the linear space $\mathcal{L}[\,b[\mathcal{X}\times\mathcal{Y},\mathcal{F}],\mathcal{F}]$,

$$\operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}} \subseteq \mathcal{L}[\,b[\mathcal{X}\times\mathcal{Y},\mathcal{F}],\mathcal{F}],$$

and define the map

$$\theta\colon \mathcal{X}\times\mathcal{Y} \to \top_{\mathcal{X},\mathcal{Y},\mathcal{F}} \subseteq \operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}} \subseteq \mathcal{L}[\,b[\mathcal{X}\times\mathcal{Y},\mathcal{F}],\mathcal{F}]$$

as follows: for each pair $(x, y) \in \mathcal{X}\times\mathcal{Y}$ set

$$\theta(x, y) = x \otimes y.$$

Thus $\theta(x, y) \in \mathcal{L}[b[\mathcal{X}\times\mathcal{Y},\mathcal{F}],\mathcal{F}]$ is such that $\theta(x, y)(\psi) = \psi(x, y) \in \mathcal{F}$ for every $\psi \in b[\mathcal{X}\times\mathcal{Y},\mathcal{F}]$. The linear space spanned by the single tensors, $\operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}}$, is a tensor product space of $\mathcal{X}$ and $\mathcal{Y}$ associated with the natural bilinear map $\theta\colon \mathcal{X}\times\mathcal{Y} \to R(\theta) = \top_{\mathcal{X},\mathcal{Y},\mathcal{F}} \subseteq \operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}} = \operatorname{span} R(\theta)$. This is shown below.

Theorem 4.3. $(\operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}}, \theta)$ *is a tensor product of* $\mathcal{X}$ *and* $\mathcal{Y}$.

Proof. Take an arbitrary linear space $\mathcal{F}$. Consider the axioms (a) and (b) in Definition 3.1 where $\mathcal{X}\times\mathcal{Y}$ is the Cartesian product of linear spaces $\mathcal{X}$ and $\mathcal{Y}$.

(a_1) The map $\theta\colon \mathcal{X}\times\mathcal{Y} \to \operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}}$ is bilinear (i.e., $\theta \in b[\mathcal{X}\times\mathcal{Y}, \operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}}]$ — this is the bilinear map $\theta\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{L}[\,b[\mathcal{X}\times\mathcal{Y},\mathcal{F}],\mathcal{F}]$ in Proposition 2.L.)

Indeed, take $x, x_1, x_2 \in \mathcal{X}$, $y, y_1, y_2 \in \mathcal{Y}$, and $\alpha \in \mathbb{F}$ arbitrarily. Also take an arbitrary bilinear map $\psi \in b[\mathcal{X}\times\mathcal{Y},\mathcal{F}]$. By the definition of the single tensor $x \otimes y$, and since ψ is bilinear, we get $((x_1 + x_2) \otimes y)(\psi) = \psi(x_1 + x_2, y) = \psi(x_1, y) + \psi(x_2, y) = (x_1 \otimes y)(\psi) + (x_2 \otimes y)(\psi)$, Therefore $(x_1 + x_2) \otimes y = (x_1 \otimes y) + (x_2 \otimes y)$. Similarly, $x \otimes (y_1 + y_2) = (x \otimes y_1) + (x \otimes y_2)$. Moreover, $\alpha(x \otimes y)(\psi) = \alpha\psi(x, y) = \psi(\alpha x, y) = \psi(x, \alpha y) = (\alpha x \otimes y)(\psi) = (x \otimes \alpha y)(\psi)$, and hence $\alpha(x \otimes y) = \alpha x \otimes y = x \otimes \alpha y$. Thus by the definition of θ,

$$\theta(x_1 + x_2, y) = \theta(x_1, y) + \theta(x_2, y), \qquad \theta(x, y_1 + y_2) = \theta(x, y_1) + \theta(x, y_2),$$

$$\alpha\theta(x, y) = \theta(\alpha x, y) = \theta(x, \alpha y).$$

(a_2) Since $\theta(x, y) = x \otimes y$, then $R(\theta) = \top_{\mathcal{X},\mathcal{Y},\mathcal{F}}$. So $\operatorname{span} R(\theta) = \operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}}$.

(b) An arbitrary element F of the linear space $\operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}}$ is a (finite) linear combination of single tensors, and therefore it is a linear transformation in $\mathcal{L}[\,b[\mathcal{X}\times\mathcal{Y},\mathcal{F}],\mathcal{F}]$. Since $\alpha(x \otimes y) = \alpha x \otimes y = x \otimes \alpha y$ for every $\alpha \in \mathbb{F}$, every F in $\operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}}$ is a finite sum of single tensors:

$$F = \sum\nolimits_i x_i \otimes y_i \in \operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}} \subseteq \mathcal{L}[\,b[\mathcal{X}\times\mathcal{Y},\mathcal{F}],\mathcal{F}].$$

Given an arbitrary bilinear map $\phi \in b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$ into any linear space $\mathcal{Z}$ over $\mathbb{F}$, consider the transformation

$$\Phi\colon \operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}} \to \mathcal{Z}$$

associating a sum of vectors in the linear space $\mathcal{Z}$ with each sum of single tensors in the linear space $\operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}}$ as follows.

$$\Phi(F) = \sum\nolimits_i \phi(x_i,y_i) \in \mathcal{Z} \quad \text{for every} \quad F = \sum\nolimits_i x_i \otimes y_i \in \operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}}.$$

(Note: $\phi(x_i,y_i)$ lies in $\mathcal{Z}$ for a given bilinear map ϕ of $\mathcal{X}\times\mathcal{Y}$ into any $\mathcal{Z}$, while $(x_i\otimes y_i)(\psi) = \psi(x_i,y_i)$ lies in $\mathcal{F}$ for every bilinear map ψ of $\mathcal{X}\times\mathcal{Y}$ into a fixed $\mathcal{F}$.) As is readily verified, Φ is a linear transformation:

$$\Phi \in \mathcal{L}[\operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}},\mathcal{Z}].$$

(Indeed, for $F = \sum_i x_i\otimes y_i$, $F' = \sum_k x'_k\otimes y'_k$ and $\alpha\in\mathbb{F}$ we get $\Phi(F+F') = \Phi(\sum_i x_i\otimes y_i + \sum_k x'_k\otimes y'_k) = \sum_i\phi(x_i,y_i) + \sum_k\phi(x'_k,y'_k) = \Phi(F)+\Phi(F')$ and $\Phi(\alpha F) = \Phi(\sum_i\alpha(x_i\otimes y_i)) = \Phi(\sum_i\alpha x_i\otimes y_i) = \sum_i\phi(\alpha x_i,y_i) = \alpha\sum_i\phi(x_i,y_i) = \alpha\Phi(F)$ — since ϕ is a bilinear map.)

Moreover, for every $(x,y)\in\mathcal{X}\times\mathcal{Y}$,

$$(\Phi\circ\theta)(x,y) = \Phi\big(\theta(x,y)\big) = \Phi(x\otimes y) = \phi(x,y)$$

and so $\phi = \Phi\circ\theta$, leading to the commutative diagram

$$\begin{array}{ccc} \mathcal{X}\times\mathcal{Y} & \xrightarrow{\ \phi\ } & \mathcal{Z} \\ & {\scriptstyle\theta}\searrow \quad \uparrow{\scriptstyle\Phi} & \\ & \operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}}, & \end{array}$$

and the pair $(\operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}},\theta)$ satisfies the axioms of Definition 3.1. □

This interpretation where single tensors are defined as linear transformations of bilinear maps is referred to as the *linear-bilinear approach*, and it is finely tailored to highlight the central property of tensor products as a tool to linearize bilinear maps according to Theorem 3.7. This is the reason why the proof of Theorem 4.3 is simpler than the proof of its quotient space counterpart in Theorem 4.1.

Remark 4.4. *A Summary of the Linear-Bilinear Approach.*

- $b[\mathcal{X}\times\mathcal{Y},\mathcal{F}] = \{\psi\in\mathcal{F}^{\mathcal{X}\times\mathcal{Y}}\colon \psi \text{ is a bilinear map}\}$: a linear space.
- $(x\otimes y)(\psi) = \psi(x,y)$: a single tensor $x\otimes y\colon b[\mathcal{X}\times\mathcal{Y},\mathcal{F}]\to\mathcal{F}$.

- $x \otimes y \in \mathcal{L}[\, b[\mathcal{X}\times\mathcal{Y}, \mathcal{F}], \mathcal{F}]$ ($x \otimes y$ is a linear transformation).
- $\top_{\mathcal{X},\mathcal{Y},\mathcal{F}} = \{x \otimes y \in \mathcal{L}[\, b[\mathcal{X}\times\mathcal{Y}, \mathcal{F}], \mathcal{F}] \colon x \in \mathcal{X} \text{ and } y \in \mathcal{Y}\}$.
- $\mathcal{X} \otimes \mathcal{Y} = \operatorname{span} \top_{\mathcal{X},\mathcal{Y},\mathcal{F}} \subseteq \mathcal{L}[\, b[\mathcal{X}\times\mathcal{Y}, \mathcal{F}], \mathcal{F}]$.
- $\theta \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{X} \otimes \mathcal{Y}]$: the natural bilinear map $\theta \colon \mathcal{X}\times\mathcal{Y} \to \mathcal{X} \otimes \mathcal{Y}$.
- $\theta(x, y) = x \otimes y \in \top_{\mathcal{X},\mathcal{Y},\mathcal{F}} \subseteq \operatorname{span} \top_{\mathcal{X},\mathcal{Y},\mathcal{F}} = \mathcal{X} \otimes \mathcal{Y} \subseteq \mathcal{L}[\, b[\mathcal{X}\times\mathcal{Y}, \mathcal{F}], \mathcal{F}]$.
- $F \in \operatorname{span} \top_{\mathcal{X},\mathcal{Y},\mathcal{F}} \iff F = \sum x \otimes y$ (a finite sum),

and so an element F of the tensor product space $\mathcal{X} \otimes \mathcal{Y} = \operatorname{span} \top_{\mathcal{X},\mathcal{Y},\mathcal{F}}$ lies in the linear space $\mathcal{L}[\, b[\mathcal{X}\times\mathcal{Y}, \mathcal{F}], \mathcal{F}]$ of all linear transformations of bilinear maps,

$$F \in \mathcal{X} \otimes \mathcal{Y} = \operatorname{span} \top_{\mathcal{X},\mathcal{Y},\mathcal{F}} \subseteq \mathcal{L}[\, b[\mathcal{X}\times\mathcal{Y}, \mathcal{F}], \mathcal{F}].$$

4.3 Variants of the Linear-Bilinear Approach

An important particular case of Theorem 4.3 refers to linear and bilinear forms by setting $\mathcal{F} = \mathbb{F}$ so that single tensors are linear forms of bilinear forms,

$$x \otimes y \in \top_{\mathcal{X},\mathcal{Y},\mathbb{F}} \subseteq \mathcal{L}[b[\mathcal{X}\times\mathcal{Y}, \mathbb{F}], \mathbb{F}] = b[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]^\sharp,$$

with $b[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]^\sharp$ standing for the dual of the linear space $b[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$.

Now consider a slightly different procedure, still referring to the particular case of $\mathcal{F} = \mathbb{F}$, where bilinear forms ψ are restricted to a subset of $b[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$ consisting of bilinear forms which are products of linear forms f and g,

$$\psi(x, y) = f(x)\, g(y) \quad \text{for some} \quad f \in \mathcal{X}^\sharp = \mathcal{L}[\mathcal{X}, \mathbb{F}] \text{ and } g \in \mathcal{Y}^\sharp = \mathcal{L}[\mathcal{Y}, \mathbb{F}].$$

Such a subset of the linear space $b[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$, namely,

$$b_{\mathcal{X}^\sharp\times\mathcal{Y}^\sharp}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}] = \big\{\psi \in b[\mathcal{X}\times\mathcal{Y}, \mathbb{F}] \colon \psi(x, y) = f(x)\, g(y) \text{ for } (f, g) \in \mathcal{X}^\sharp\times\mathcal{Y}^\sharp\big\},$$

is not a linear manifold of $b[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$ (indeed, a sum of two bilinear forms in $b_{\mathcal{X}^\sharp\times\mathcal{Y}^\sharp}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$ may not lie in $b_{\mathcal{X}^\sharp\times\mathcal{Y}^\sharp}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$). Define single tensors as before: to each pair (x, y) in the Cartesian product $\mathcal{X}\times\mathcal{Y}$ associate a function $x \otimes y \colon b_{\mathcal{X}^\sharp\times\mathcal{Y}^\sharp}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}] \to \mathbb{F}$ defined for every $\psi \in b_{\mathcal{X}^\sharp\times\mathcal{Y}^\sharp}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$ by

$$(x\otimes y)(f, g) = (x\otimes y)(\psi) = \psi(x, y) = f(x)\, g(y) \quad \text{for every} \quad (f, g) \in \mathcal{X}^\sharp\times\mathcal{Y}^\sharp.$$

The difference between this and the previous procedure is due to the fact that single tensors are not linear transformations (or linear forms) any longer since their domain $b_{\mathcal{X}^\sharp\times\mathcal{Y}^\sharp}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$ is not a linear manifold of the linear space $b[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$, thus not a linear space. However, they can now be regarded as bilinear forms on the Cartesian product of the linear spaces $\mathcal{X}^\sharp$ and $\mathcal{Y}^\sharp$.

(In fact, we get $(x \otimes y)(f_1 + f_2, g) = (f_1 + f_2)(x)\, g(y) = (f_1(x) + f_2(x))\, g(y) = f_1(x)\, g(y) + f_2(x)\, g(y) = (x \otimes y)(f_1, g) + (x \otimes y)(f_2, g)$ and, on the other hand,

$(x \otimes y)(f, g_1+g_2) = (x \otimes y)(f, g_1) + (x \otimes y)(f, g_2)$; moreover $(x \otimes y)(\alpha f, g) = \alpha(x \otimes y)(f, g) = (x \otimes y)(f, \alpha g)$, for every $f_1, f_2, f \in \mathcal{X}^\sharp$, every $g_1, g_2, g \in \mathcal{Y}^\sharp$, and every $\alpha \in \mathbb{F}$; for each pair $(x, y) \in \mathcal{X} \times \mathcal{Y}$.)

Hence

$$x \otimes y \in b[\mathcal{X}^\sharp \times \mathcal{Y}^\sharp, \mathbb{F}] = b[\mathcal{L}[\mathcal{X}, \mathbb{F}] \times \mathcal{L}[\mathcal{Y}, \mathbb{F}], \mathbb{F}].$$

Thus take the collection $\top'_{\mathcal{X},\mathcal{Y}}$ of all these single tensors

$$\top'_{\mathcal{X},\mathcal{Y}} = \big\{x \otimes y \in b[\mathcal{X}^\sharp \times \mathcal{Y}^\sharp, \mathbb{F}] \colon\ x \in \mathcal{X} \text{ and } y \in \mathcal{Y}\big\},$$

and consider its span, $\operatorname{span} \top'_{\mathcal{X},\mathcal{Y}}$, which is now a linear manifold of the linear space $b[\mathcal{X}^\sharp \times \mathcal{Y}^\sharp, \mathbb{F}]$ of all bilinear forms of pairs of linear forms,

$$\operatorname{span} \top'_{\mathcal{X},\mathcal{Y}} \subseteq b[\mathcal{X}^\sharp \times \mathcal{Y}^\sharp, \mathbb{F}].$$

As before, define the map $\theta' \colon \mathcal{X} \times \mathcal{Y} \to \top'_{\mathcal{X},\mathcal{Y}}$ for each pair $(x, y) \in \mathcal{X} \times \mathcal{Y}$ by

$$\theta'(x, y) = x \otimes y.$$

Then the value of θ' at $(x, y) \in \mathcal{X} \times \mathcal{Y}$ is a bilinear form, $\theta'(x, y) \in b[\mathcal{X}^\sharp \times \mathcal{Y}^\sharp, \mathbb{F}]$, which is given by $\theta'(x, y)(f, g) = f(x)\, g(y) \in \mathbb{F}$ for every $(f, g) \in \mathcal{X}^\sharp \times \mathcal{Y}^\sharp$ and, as before, $\theta' \colon \mathcal{X} \times \mathcal{Y} \to \top'_{\mathcal{X},\mathcal{Y}} \subseteq \operatorname{span} \top'_{\mathcal{X},\mathcal{Y}} \subseteq b[\mathcal{X}^\sharp \times \mathcal{Y}^\sharp, \mathbb{F}]$ is a bilinear map.

Corollary 4.5. $(\operatorname{span} \top'_{\mathcal{X},\mathcal{Y}}, \theta')$ *is a tensor product of* $\mathcal{X}$ *and* $\mathcal{Y}$.

Proof. Replace $\mathcal{F}$ by $\mathbb{F}$ in the proof of Theorem 4.3 so that $\mathcal{L}[\, b[\mathcal{X} \times \mathcal{Y}, \mathcal{F}], \mathcal{F}]$ is replaced by $\mathcal{L}[\, b[\mathcal{X} \times \mathcal{Y}, \mathbb{F}], \mathbb{F}]$. Then replace the linear space $b[\mathcal{X} \times \mathcal{Y}, \mathbb{F}]$ by the subset $b_{\mathcal{X}^\sharp \times \mathcal{Y}^\sharp}[\mathcal{X} \times \mathcal{Y}, \mathbb{F}]$, still keeping the same definition of single tensors, and replace $\top_{\mathcal{X},\mathcal{Y},\mathcal{F}} \subseteq \mathcal{L}[\, b[\mathcal{X} \times \mathcal{Y}, \mathcal{F}], \mathcal{F}]$ by $\top'_{\mathcal{X},\mathcal{Y}} \subseteq b[\mathcal{L}[\mathcal{X}, \mathbb{F}] \times \mathcal{L}[\mathcal{Y}, \mathbb{F}], \mathbb{F}]$. Again, $\theta' \colon \mathcal{X} \times \mathcal{Y} \to \operatorname{span} \top'_{\mathcal{X},\mathcal{Y}}$ is a bilinear map with $\operatorname{span} R(\theta') = \operatorname{span} \top'_{\mathcal{X},\mathcal{Y}}$ — this is the bilinear map $\theta' \colon \mathcal{X} \times \mathcal{Y} \to b[\mathcal{X}^\sharp \times \mathcal{Y}^\sharp, \mathbb{F}]$ in Proposition 2.M. These modifications do not alter the argument in the proof of Theorem 4.3, which still works when to each bilinear map $\phi \colon \mathcal{X} \times \mathcal{Y} \to \mathcal{Z}$ there is associated the same linear transformation Φ into $\mathcal{Z}$ now acting on $\operatorname{span} \top'_{\mathcal{X},\mathcal{Y}}$. □

Remark 4.6. *Single Tensors in Hilbert Space*. A common and useful example of such a particular case goes as follows. Let $\mathcal{X}$ and $\mathcal{Y}$ be Hilbert spaces (either both real or both complex) with inner products $\langle \cdot\,; \cdot \rangle_{\mathcal{X}}$ and $\langle \cdot\,; \cdot \rangle_{\mathcal{Y}}$, which are sesquilinear forms (but not bilinear forms in the complex case). Here algebraic duals $\mathcal{X}^\sharp$ and $\mathcal{Y}^\sharp$ are naturally replaced by topological duals $\mathcal{X}^*$ and $\mathcal{Y}^*$ of continuous linear functionals. (Hilbert spaces are discussed in Section 5.4 and the present example is revisited at the beginning of Section 9.4). A single tensor $x \otimes y$ associated with a pair (x, y) in $\mathcal{X} \times \mathcal{Y}$ is usually defined in this case as

$$(x \otimes y)(u, v) = \langle x\,; u \rangle_{\mathcal{X}}\, \langle y\,; v \rangle_{\mathcal{Y}} \quad \text{for every} \quad (u, v) \in \mathcal{X} \times \mathcal{Y}.$$

The Riesz Representation Theorem for Hilbert spaces (stated in Proposition 5.U) ensures that f lies in $\mathcal{X}^*$ and g lies in $\mathcal{Y}^*$ if and only if $f(\cdot) = \langle \cdot\,; u \rangle_{\mathcal{X}}$ and

$g(\cdot) = \langle \cdot\,; v\rangle_{\mathcal{Y}}$ for some u in $\mathcal{X}$ and v in $\mathcal{Y}$. Thus identify $f \in \mathcal{X}^*$ and $g \in \mathcal{Y}^*$ with $u \in \mathcal{X}$ and $v \in \mathcal{Y}$ such that the pair $(u, v) \in \mathcal{X}\times\mathcal{Y}$ is identified with the pair $(f, g) \in \mathcal{X}^*\times\mathcal{Y}^*$. Then a single tensor $x \otimes y$ associated with a pair (x, y) in $\mathcal{X}\times\mathcal{Y}$ is a bilinear form $x \otimes y\colon \mathcal{X}^*\times\mathcal{Y}^* \to \mathbb{F}$ on $\mathcal{X}^*\times\mathcal{Y}^*$, not on $\mathcal{X}\times\mathcal{Y}$ — i.e., $x \otimes y \in b[\mathcal{X}^*\times\mathcal{Y}^*, \mathbb{F}]$, which is equivalently written (as in Corollary 4.5) by

$$(x \otimes y)(f, g) = f(x)\, g(y) \quad \text{for every} \quad (f, g) \in \mathcal{X}^*\times\mathcal{Y}^*.$$

By Theorem 4.3 with $\mathcal{F} = \mathbb{F}$ we get $\operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathbb{F}} \subseteq b[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]^\sharp$. Moreover, according to Corollary 4.5 (which assumes $\mathcal{F} = \mathbb{F}$ *a priori*) we got $\operatorname{span}\top'_{\mathcal{X},\mathcal{Y}} \subseteq b[\mathcal{X}^\sharp\times\mathcal{Y}^\sharp, \mathbb{F}]$. So there are realizations of tensor product spaces of linear spaces $\mathcal{X}$ and $\mathcal{Y}$ which can be included in the algebraic dual of the linear space of all bilinear forms on the Cartesian product $\mathcal{X}\times\mathcal{Y}$, or in the linear space of all bilinear forms on the Cartesian product $\mathcal{X}^\sharp\times\mathcal{Y}^\sharp$ of their algebraic duals. We consider further possibilities as follows.

For each $(x, y) \in \mathcal{X}\times\mathcal{Y}$ take a transformation $x \otimes y\colon \mathcal{X}^\sharp \to \mathcal{Y}$ defined by

$$(x \otimes y)(f) = f(x)y \;\in\; \mathcal{Y} \quad \text{for every} \quad f \in \mathcal{X}^\sharp.$$

This is a linear transformation of the linear space $\mathcal{X}^\sharp$ (i.e., of the algebraic dual of the linear space $\mathcal{X}$) onto the linear space $\mathcal{Y}$.

(In fact, $(x \otimes y)(f_1 + f_2) = (f_1 + f_2)(x)y = (f_1(x) + f_2(x))y = f_1(x)y + f_2(x)y$ $= (x \otimes y)(f_1) + (x \otimes y)(f_2)$ and $(x \otimes y)(\alpha f) = \alpha f(x)y = \alpha(x \otimes y)(f)$ for every $f, f_1, f_2 \in \mathcal{X}^\sharp$ and every $\alpha \in \mathbb{F}$ — for each pair $(\mathcal{X}, \mathcal{Y}) \in \mathcal{X}\times\mathcal{Y}$.)

Therefore

$$(x \otimes y) \in \mathcal{L}[\mathcal{X}^\sharp, \mathcal{Y}].$$

As before, set $\top''_{\mathcal{X},\mathcal{Y}} = \big\{x \otimes y \in \mathcal{L}[\mathcal{X}^\sharp, \mathcal{Y}] : x \in \mathcal{X} \text{ and } y \in \mathcal{Y}\big\}$, take its span

$$\operatorname{span}\top''_{\mathcal{X},\mathcal{Y}} \;\subseteq\; \mathcal{L}[\mathcal{X}^\sharp, \mathcal{Y}],$$

and define the map $\theta''\colon \mathcal{X}\times\mathcal{Y} \to \operatorname{span}\top''_{\mathcal{X},\mathcal{Y}}$ as follows: for every $(x, y) \in \mathcal{X}\times\mathcal{Y}$,

$$\theta''(x, y) = x \otimes y.$$

Now the value of θ'' at $(x, y) \in \mathcal{X}\times\mathcal{Y}$ is a linear transformation, $\theta''(x, y)$ in $\mathcal{L}[\mathcal{X}^\sharp, \mathcal{Y}]$, which is given by $\theta''(x, y)(f) = f(x)y \in \mathcal{Y}$ for every $f \in \mathcal{X}^\sharp$.

Corollary 4.7. $(\operatorname{span}\top''_{\mathcal{X},\mathcal{Y}}, \theta'')$ *is a tensor product of* $\mathcal{X}$ *and* $\mathcal{Y}$.

Proof. Consider the argument in the proof of Theorem 4.3. (a) Since f is a linear functional and $\mathcal{Y}$ is a linear space, $\theta''\colon \mathcal{X}\times\mathcal{Y} \to \operatorname{span}\top''_{\mathcal{X},\mathcal{Y}} \subseteq \mathcal{L}[\mathcal{X}^\sharp, \mathcal{Y}]$ is a bilinear map whose range satisfies the identity $\operatorname{span} R(\theta'') = \operatorname{span}\top''_{\mathcal{X},\mathcal{Y}}$ (the bilinear map in Proposition 2.N). Also, (b) for an arbitrary bilinear map $\phi\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{Z}$ into any linear space $\mathcal{Z}$ over $\mathbb{F}$, take the same linear transformation $\Phi\colon \operatorname{span}\top''_{\mathcal{X},\mathcal{Y}} \to \mathcal{Z}$ defined in the proof of Theorem 4.3, now acting on $\operatorname{span}\top''_{\mathcal{X},\mathcal{Y}}$. Again, $\phi = \Phi \circ \theta''$ and the axioms of Definition 3.1 are fulfilled. □

Symmetrically, for each $(x,y) \in \mathcal{X}\times\mathcal{Y}$ define $x \otimes y\colon \mathcal{Y}^\sharp \to \mathcal{X}$ by

$$(x \otimes y)(g) = g(y)x \in \mathcal{X} \quad \text{for every} \quad g \in \mathcal{Y}^\sharp.$$

Again this is a linear transformation, now of $\mathcal{Y}^\sharp$ onto $\mathcal{X}$, and so

$$(x \otimes y) \in \mathcal{L}[\mathcal{Y}^\sharp, \mathcal{X}].$$

Thus set $\top'''_{\mathcal{X},\mathcal{Y}} = \big\{x \otimes y \in \mathcal{L}[\mathcal{Y}^\sharp, \mathcal{X}]\colon x \in \mathcal{X} \text{ and } y \in \mathcal{Y}\big\}$, take its span

$$\operatorname{span} \top'''_{\mathcal{X},\mathcal{Y}} \subseteq \mathcal{L}[\mathcal{Y}^\sharp, \mathcal{X}],$$

and define the map $\theta'''\colon \mathcal{X}\times\mathcal{Y} \to \operatorname{span} \top'''_{\mathcal{X},\mathcal{Y}}$ for every $(x,y) \in \mathcal{X}\times\mathcal{Y}$ by

$$\theta'''(x,y) = x \otimes y.$$

This time the value of θ''' at $(x,y) \in \mathcal{X}\times\mathcal{Y}$ is the linear transformation $\theta'''(x,y)$ in $\mathcal{L}[\mathcal{Y}^\sharp, \mathcal{X}]$ given by $\theta'''(x,y)(g) = g(y)x \in \mathcal{X}$ for every $g \in \mathcal{Y}^\sharp$.

Corollary 4.8. $(\operatorname{span} \top'''_{\mathcal{X},\mathcal{Y}}, \theta''')$ *is a tensor product of* $\mathcal{X}$ *and* $\mathcal{Y}$.

Proof. A symmetrical argument as in the proof of Corollary 4.7 (the bilinear map $\theta'''\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{L}[\mathcal{Y}^\sharp, \mathcal{X}]$ is the one in Proposition 2.O). □

Let $\mathcal{X}$ and $\mathcal{Y}$ be linear spaces, let $\mathcal{T}$ be a tensor product space of $\mathcal{X}$ and $\mathcal{Y}$, and let $\mathcal{L}$ be another linear space, all over the same field. Since tensor product spaces for the same pair of linear spaces are isomorphic (Corollary 3.6), they are unique up to an isomorphism, and we have referred to *the* tensor product space of $\mathcal{X}$ and $\mathcal{Y}$ by denoting it by $\mathcal{X} \otimes \mathcal{Y}$. This means that $\mathcal{X} \otimes \mathcal{Y} \cong \mathcal{T}$ for every tensor product space $\mathcal{T}$ for $\mathcal{X}$ and $\mathcal{Y}$. Thus if $\mathcal{T} \subseteq \mathcal{L}$, as we had in Theorem 4.3 and Corollaries 4.5 and 4.7, then

$$\mathcal{X} \otimes \mathcal{Y} \cong \mathcal{T} \subseteq \mathcal{L},$$

and so every tensor product space of $\mathcal{X}$ and $\mathcal{Y}$ is embedded in $\mathcal{L}$, and we write

$$\mathcal{X} \otimes \mathcal{Y} \subseteq \mathcal{L} \quad \text{for} \quad \mathcal{X} \otimes \mathcal{Y} \hookrightarrow \mathcal{L}.$$

Remark 4.9. *Embeddings of Tensor Products*. Some inclusions (embeddings) of tensor product spaces into well-known linear spaces are summarized below.

(a) $\mathcal{X} \otimes \mathcal{Y} \subseteq b[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]^\sharp$
(with $(x \otimes y)(\psi) = \psi(x,y)$ in Theorem 4.3 for $\mathcal{F} = \mathbb{F}$).

(b) $\mathcal{X} \otimes \mathcal{Y} \subseteq b[\mathcal{X}^\sharp\times\mathcal{Y}^\sharp, \mathbb{F}]$
(with $(x \otimes y)(f,g) = f(x)\,g(y)$ in Corollary 4.5).

(c) $\mathcal{X} \otimes \mathcal{Y} \subseteq \mathcal{L}[\mathcal{X}^\sharp, \mathcal{Y}]$
(with $(x \otimes y)(f) = f(x)y$ in Corollary 4.7).

(d) $\mathcal{X} \otimes \mathcal{Y} \subseteq \mathcal{L}[\mathcal{Y}^\sharp, \mathcal{X}]$
(with $(x \otimes y)(g) = g(y)x$ in Corollary 4.8).

(e) $\mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp \subseteq (\mathcal{X} \otimes \mathcal{Y})^\sharp$
(Remark 3.18).

(f) $\mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp \subseteq b[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$
(by item (e) since $(\mathcal{X} \otimes \mathcal{Y})^\sharp \cong b[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$ according to Remark 3.8(b)).

(g) $\mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp \subseteq \mathcal{L}[\mathcal{X}, \mathcal{Y}^\sharp]$
(set $(f \otimes g)(x) = f(x)g$, argument of Corollary 4.7;
or by item (f) since $b[\mathcal{X}\times\mathcal{Y}, \mathbb{F}] \cong \mathcal{L}[\mathcal{X}, \mathcal{Y}^\sharp]$ according to Remark 2.7(b)).

(h) $\mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp \subseteq \mathcal{L}[\mathcal{Y}, \mathcal{X}^\sharp]$
(set $(f \otimes g)(y) = g(y)f$, argument of Corollary 4.8);
or by item (g) since $\mathcal{L}[\mathcal{X}, \mathcal{Y}^\sharp] \cong \mathcal{L}[\mathcal{Y}, \mathcal{X}^\sharp]$ according to Remark 2.7(b)).

(i) $\mathcal{X}^\sharp \otimes \mathcal{Y} \subseteq \mathcal{L}[\mathcal{X}, \mathcal{Y}]$
(set $(f \otimes y)(x) = f(x)y$, argument of Corollary 4.7).

(j) $\mathcal{X}^\sharp \otimes \mathcal{Y} \subseteq \mathcal{L}[\mathcal{Y}^\sharp, \mathcal{X}^\sharp]$
(replace $\mathcal{X}$ by $\mathcal{X}^\sharp$ in item (d)).

(k) $\mathcal{X}^\sharp \otimes \mathcal{Y} \subseteq b[\mathcal{X}\times\mathcal{Y}^\sharp, \mathbb{F}]$
(by item (j) since $\mathcal{L}[\mathcal{Y}^\sharp, \mathcal{X}^\sharp] \cong b[\mathcal{X}\times\mathcal{Y}^\sharp, \mathbb{F}]$ according to Remark 2.7(b)).

(ℓ) $\mathcal{X} \otimes \mathcal{Y}^\sharp \subseteq \mathcal{L}[\mathcal{Y}, \mathcal{X}]$
(set $(x \otimes g)(y) = g(y)x$, argument of Corollary 4.8).

(m) $\mathcal{X} \otimes \mathcal{Y}^\sharp \subseteq \mathcal{L}[\mathcal{X}^\sharp, \mathcal{Y}^\sharp]$
(replace $\mathcal{Y}$ by $\mathcal{Y}^\sharp$ in item (c)).

(n) $\mathcal{X} \otimes \mathcal{Y}^\sharp \subseteq b[\mathcal{X}^\sharp\times\mathcal{Y}, \mathbb{F}]$
(by item (m) since $\mathcal{L}[\mathcal{X}^\sharp, \mathcal{Y}^\sharp] \cong b[\mathcal{X}^\sharp\times\mathcal{Y}, \mathbb{F}]$ according to Remark 2.7(b)).

By Corollary 4.5 the tensor product space $\operatorname{span}\top'_{\mathcal{X},\mathcal{Y}}$ of $\mathcal{X}$ and $\mathcal{Y}$ is included in the linear space $b[\mathcal{X}^\sharp\times\mathcal{Y}^\sharp, \mathbb{F}]$,

$$\operatorname{span}\top'_{\mathcal{X},\mathcal{Y}} \subseteq b[\mathcal{X}^\sharp\times\mathcal{Y}^\sharp, \mathbb{F}].$$

Thus according to Corollary 3.6 every tensor product space of $\mathcal{X}$ and $\mathcal{Y}$ is isomorphic to $\operatorname{span}\top'_{\mathcal{X},\mathcal{Y}}$, and therefore is embedded in $b[\mathcal{X}^\sharp\times\mathcal{Y}^\sharp, \mathbb{F}]$ (cf. Remark 4.9(b)). In particular, the tensor product space of Theorem 4.3,

$$\operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}} \subseteq \mathcal{L}[\,b[\mathcal{X}\times\mathcal{Y}, \mathcal{F}], \mathcal{F}],$$

(and so $\operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathbb{F}} \subseteq b[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]^\sharp$) is embedded in $b[\mathcal{X}^\sharp\times\mathcal{Y}^\sharp, \mathbb{F}]$:

$$\operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}} \hookrightarrow b[\mathcal{X}^\sharp\times\mathcal{Y}^\sharp, \mathbb{F}]$$

for every linear space $\mathcal{F}$. And all this comes from Theorem 3.5 which, through Corollary 3.6, established the embeddings summarized in Remark 4.9.

There is, however, a special configuration of Theorem 4.3 (by considering a particular concrete realization for the linear space $\mathcal{Z}$) which is able to supply an alternate way of exhibiting the above embedding without using the isomorphism between all tensor product spaces of $\mathcal{X}$ and $\mathcal{Y}$ established in Theorem 3.5. The construction of such an alternate procedure is not as neat as the isomorphism procedure of Remark 4.9 that uses Theorem 3.5, but it has an interest by itself due to the technique applied in its proof. This is shown below.

Corollary 4.10. *Let $\mathcal{X}, \mathcal{Y}, \mathcal{F}$ be nonzero linear spaces over the same field $\mathbb{F}$. Consider the tensor product $(\operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}}, \theta)$ of $\mathcal{X}$ and $\mathcal{Y}$ as in Theorem 4.3. There exist a particular linear space $\mathcal{Z}$, viz.,*

$$\mathcal{Z} = b[\mathcal{X}^\sharp \times \mathcal{Y}^\sharp, \mathbb{F}],$$

and a bilinear map ϕ in $b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ whose associated linear transformation Φ in $\mathcal{L}[\operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}}, \mathcal{Z}]$ is injective. Thus Φ embeds $\operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}}$ into $\mathcal{Z}$:

$$\Phi\colon \operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}} \hookrightarrow b[\mathcal{X}^\sharp \times \mathcal{Y}^\sharp, \mathbb{F}].$$

Proof. Fix a pair $(x, y) \in \mathcal{X}\times\mathcal{Y}$. Take arbitrary linear forms $f \in \mathcal{X}^\sharp$ and $g \in \mathcal{Y}^\sharp$. Consider the functional $\phi(x, y)\colon \mathcal{X}^\sharp \times \mathcal{Y}^\sharp \to \mathbb{F}$ given for every $(f, g) \in \mathcal{X}^\sharp \times \mathcal{Y}^\sharp$ by

$$\phi(x, y)(f, g) = f(x)\, g(y) \ \in \ \mathbb{F}.$$

As we saw before (related to Corollary 4.5), this is a bilinear form:

$$\phi(x, y) \in b[\mathcal{X}^\sharp \times \mathcal{Y}^\sharp, \mathbb{F}].$$

Since $b[\mathcal{X}^\sharp \times \mathcal{Y}^\sharp, \mathbb{F}]$ is a linear space, set

$$\mathcal{Z} = b[\mathcal{X}^\sharp \times \mathcal{Y}^\sharp, \mathbb{F}]$$

and consider the map

$$\phi\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{Z} = b[\mathcal{X}^\sharp \times \mathcal{Y}^\sharp, \mathbb{F}]$$

with $\phi(x, y) \in b[\mathcal{X}^\sharp \times \mathcal{Y}^\sharp, \mathbb{F}]$ defined above. As in Corollary 4.5, ϕ is bilinear:

$$\phi \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] = b[\mathcal{X}\times\mathcal{Y}, b[\mathcal{X}^\sharp \times \mathcal{Y}^\sharp, \mathbb{F}]].$$

Take an arbitrary linear space $\mathcal{F}$ over $\mathbb{F}$ and consider the tensor product $(\operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}}, \theta)$ of linear spaces $\mathcal{X}$ and $\mathcal{Y}$ as in Theorem 4.3. Thus by Definition 3.1 and Corollary 3.4 there is a unique linear transformation

$$\Phi \in \mathcal{L}[\operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}}, \mathcal{Z}] = \mathcal{L}[\operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}}, b[\mathcal{X}^\sharp \times \mathcal{Y}^\sharp, \mathbb{F}]]$$

for which the diagram

$$\begin{array}{ccc} \mathcal{X}\times\mathcal{Y} & \xrightarrow{\ \phi\ } & \mathcal{Z} = b[\mathcal{X}^\sharp\times\mathcal{Y}^\sharp,\mathbb{F}] \\ & {\scriptstyle\theta}\searrow \quad \uparrow{\scriptstyle\Phi} & \\ & \operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}} & \end{array}$$

commutes, where the tensor product space $\operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}}$ of $\mathcal{X}$ and $\mathcal{Y}$ is included in $\mathcal{L}[\,b[\mathcal{X}\times\mathcal{Y},\mathcal{F}],\mathcal{F}]$ and the natural bilinear map θ in $b[\mathcal{X}\times\mathcal{Y},\operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}}]$ is given by $\theta(x,y) = x\otimes y \in \mathcal{L}[b[\mathcal{X}\times\mathcal{Y},\mathcal{F}],\mathcal{F}]$ for each $(x,y)\in\mathcal{X}\times\mathcal{Y}$, with the single tensor $x\otimes y$ defined by $(x\otimes y)(\psi) = \psi(x,y)\in\mathcal{F}$ for $\psi\in b[\mathcal{X}\times\mathcal{Y},\mathcal{F}]$. Hence $\theta(x,y)(\psi) = \psi(x,y)\in\mathcal{F}$ for every $\psi\in b[\mathcal{X}\times\mathcal{Y},\mathcal{F}]$. Moreover, the linear transformation Φ is such that for every $F = \sum_i x_i\otimes y_i \in \operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}}$

$$\Phi(F) = \sum\nolimits_i \phi(x_i,y_i) \ \in\ \mathcal{Z} = b[\mathcal{X}^\sharp\times\mathcal{Y}^\sharp,\mathbb{F}]$$

(cf. proof of Theorem 4.3), and so for every $(f,g)\in\mathcal{X}^\sharp\times\mathcal{Y}^\sharp$

$$\Phi(F)(f,g) = \sum\nolimits_i \phi(x_i,y_i)(f,g) = \sum\nolimits_i f(x_i)\,g(y_i) \ \in\ \mathbb{F}.$$

Observe that $F\ne 0 \iff F = \sum_{i=1}^n x_i\otimes y_i \ne 0$ with $x_i\ne 0$ and $y_i\ne 0$ for all $i = 1,\ldots,n$ for some $n\ge 1$ (since $x\otimes y = 0$ if either $x = 0$ or $y = 0$ because $\alpha x\otimes y = \alpha(x\otimes y) = x\otimes\alpha y$). Also observe that for $F = \sum_{i=1}^n x_i\otimes y_i$

$$\Phi(F) = 0 \iff \Phi(F)(f,g) = 0 \ \ \forall f,g \iff \sum\nolimits_{i=1}^n f(x_i)\,g(y_i) = 0 \ \ \forall f,g.$$

Claim. $\quad \sum_{i=1}^n x_i\otimes y_i \ne 0 \quad\Longrightarrow\quad \sum_{i=1}^n f(x_i)\,g(y_i)\ne 0 \quad$ for some f,g.

Proof of Claim. Suppose $\sum_{i=1}^n x_i\otimes y_i\ne 0$. Since $\theta(x,y) = x\otimes y$ and θ is bilinear, we may assume the finite sets $\{x_i\}_{i=1}^n$ and $\{y_i\}_{i=1}^n$ are linearly independent (cf. Proposition 3.A in Section 3.4), so all single tensors $x_i\otimes y_i$ in the above sum are nonzero and $n\le\min\{\dim\mathcal{X},\dim\mathcal{Y}\}$ is the least integer yielding the corresponding nonzero finite sum of single tensors. Recall: *for each* $0\ne x\in\mathcal{X}$ *and each* $0\ne y\in\mathcal{Y}$ *there exist* $f\in\mathcal{X}^\sharp$ *and* $g\in\mathcal{Y}^\sharp$ *for which* $f(x)\ne 0$ *and* $g(y)\ne 0$ (cf. Proposition 1.G). Also, if $\sum_{i=1}^n g(y_i)x_i = 0$, then $g_i(y) = 0$ for every $i = 1,\ldots,n$, as $\{x_i\}_{i=1}^n$ is linearly independent. Thus by the above italicized result there exists g such that $g(y_i)\ne 0$ for some y_i in $\{y_j\}_{j=1}^n$ (as $\{y_i\}_{i=1}^n$ is linearly independent and so has only nonzero vectors). Hence $\sum_{i=1}^n g(y_i)x_i\ne 0$. So by the same italicized result there is an f for which $f\big(\sum_{i=1}^n g(y_i)x_i\big)\ne 0$, which means $\sum_{i=1}^n f(x_i)g(y_i)\ne 0$ since f is linear. □

Thus $F\ne 0\Longrightarrow\Phi(F)\ne 0$. Then Φ is injective. So the linear transformation

$$\Phi\colon \operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}} \hookrightarrow \mathcal{Z} = b[\mathcal{X}^\sharp\times\mathcal{Y}^\sharp,\mathbb{F}]$$

embeds the linear space $\operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}}\subseteq\mathcal{L}[b[\mathcal{X}\times\mathcal{Y},\mathcal{F}],\mathcal{F}]$ into the linear space $\mathcal{Z} = b[\mathcal{X}^\sharp\times\mathcal{Y}^\sharp,\mathbb{F}]$ (cf. Remark 3.2(c)). □

Remark 4.11. *Multiple Tensor Products.* It is readily verified that the preceding arguments (in Chapters 3 and 4) can be naturally extended to build up the concept of an algebraic tensor product of a finite collection $\{\mathcal{X}_i\}_{i=1}^n$ of linear spaces over the same field, yielding a tensor product space $\bigotimes_{i=1}^n \mathcal{X}_i$ of a finite number of linear spaces. The construction is based on the notions of multiple Cartesian products $\prod_{i=1}^n \mathcal{X}_i$, n-tuples, and multilinear maps, as natural extensions of Cartesian products of two linear spaces, ordered pairs, and bilinear maps. Actually, the results in Chapters 3 and 4 remain true (essentially with the same statement, following similar arguments) if extended to such multiple tensor product cases. It may be a significant endeavor to extend a result concerning tensor products from a pair of linear spaces to an n-tuple (or to an ∞-tuple). Sometimes this is a simple task (achieved by induction, when necessary), but not always. On the other hand, what may be not always simple is working the other way around: when a notion is primarily defined for an n-tuple, it may be wise to reduce the multiplicity down to a pair to see clearly what is really happening.

4.4 Additional Propositions

Let $\mathcal{X}$ and $\mathcal{Y}$ be linear spaces over the same field $\mathbb{F}$. Take their Cartesian product $\mathcal{X}\times\mathcal{Y}$. Equip $\mathcal{X}\times\mathcal{Y}$ with the direct sum $\oplus$ (which simply means coordinatewise addition) and with standard coordinatewise scalar multiplication as usual: for every (x_1, y_1), (x_2, y_2), (x, y) in $\mathcal{X} \times \mathcal{Y}$ and every α in $\mathbb{F}$,

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1 + x_2\,,\, y_1 + y_2) \quad \text{and} \quad \alpha(x, y) = (\alpha x, \alpha y).$$

This leads to the notion of the linear space direct sum $\mathcal{X} \oplus \mathcal{Y} = (\mathcal{X}\times\mathcal{Y}, \oplus, \cdot\,, \mathbb{F})$ whose underlying space is the Cartesian product $\mathcal{X}\times\mathcal{Y}$ of linear spaces. As seen in Section 2.1, a bilinear map can be equally defined on $\mathcal{X}\times\mathcal{Y}$ or on $\mathcal{X} \oplus \mathcal{Y}$, as the algebra acquired by $\mathcal{X}\times\mathcal{Y}$ when equipping it with the direct sum plays no role when dealing with bilinear maps: $b[\mathcal{X} \oplus \mathcal{Y}, \mathcal{Z}] = b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ for any linear space $\mathcal{Z}$. Now since a single tensor $x \otimes y$ is precisely the image $\theta(x, y)$ of a bilinear map θ defined on $\mathcal{X}\times\mathcal{Y}$ (or on $\mathcal{X} \oplus \mathcal{Y}$ — in this context it makes no difference if an algebra equips the Cartesian product of $\mathcal{X}$ and $\mathcal{Y}$) into a linear space $\mathcal{T}$, say $x \otimes y = \theta(x, y)$ for every $(x, y) \in \mathcal{X}\times\mathcal{Y}$, we get

$$\alpha(x \otimes y) = (\alpha x) \otimes y = x \otimes (\alpha y),$$

$$(x_1 \otimes y_1) + (x_2 \otimes y_2) = (x_1 + x_2) \otimes (y_1 + y_2) - (x_1 \otimes y_2) - (x_2 \otimes y_1),$$

as in Section 3.1, for every $x, x_1, x_2 \in \mathcal{X}$, every $y, y_1, y_2 \in \mathcal{Y}$, and every $\alpha \in \mathbb{F}$ (so addition of single tensors is not necessarily a single tensor as bilinear maps are not linear), and yet the collection of all linear combinations of single tensors coincides with the collection of all (finite) sums of single tensors (cf. Theorem 3.3), leading to the notion of tensor product linear space of Section 3.1,

$$\begin{aligned}\mathcal{X}\otimes\mathcal{Y} = \mathcal{T} &= \operatorname{span} R(\theta) = \operatorname{span}\theta(\mathcal{X}\times\mathcal{Y})\\ &= \{\textstyle\sum_{i=1}^{N}\alpha_i(x_i\otimes y_i)\colon\ \alpha_i\in\mathbb{F},\ \ (x_i,y_i)\in\mathcal{X}\times\mathcal{Y},\ \ N\in\mathbb{N}\}\\ &= \{\textstyle\sum_{i=1}^{N}x_i\otimes y_i\colon\ \ (x_i,y_i)\in\mathcal{X}\times\mathcal{Y},\quad N\in\mathbb{N}\}.\end{aligned}$$

Now particularize to finite-dimensional spaces. Set $\mathcal{X}=\mathbb{F}^m$ and $\mathcal{Y}=\mathbb{F}^n$. Consider the direct sum $\bigoplus_{j=1}^{m}\mathbb{F}^n$ of m copies of $\mathbb{F}^n$. Equivalently, the collection of all m-tuples of n-vectors, $\{(y_1,\cdots,y_m)\colon\ y_j\in\mathbb{F}^n,\ j=1,\ldots,m\}=\mathbb{F}^m(\mathbb{F}^n)$, equipped with the usual direct sum algebra. Thus regard $(\mathbb{F}^n)^m=\bigoplus_{j=1}^{m}\mathbb{F}^n$ and $\mathbb{F}^m(\mathbb{F}^n)=\bigoplus_{j=1}^{n}\mathbb{F}^m$ as different notations for the same (i.e., isomorphic) linear space, which in turn are isomorphic to $\mathbb{F}^{mn}=\mathbb{F}^{nm}$. That is,

$$\mathbb{F}^m(\mathbb{F}^n)=\bigoplus_{j=1}^{m}\mathbb{F}^n\cong\mathbb{F}^{mn}=\mathbb{F}^{nm}\cong\bigoplus_{j=1}^{n}\mathbb{F}^m=\mathbb{F}^n(\mathbb{F}^m).$$

The *Kronecker product* $x\circledast y$ of two vectors $x=(\xi_1,\ldots,\xi_m)$ in $\mathbb{F}^m$ and $y=(\upsilon_1,\ldots,\upsilon_n)$ in $\mathbb{F}^n$ is the vector $x\circledast y$ in $\mathbb{F}^m(\mathbb{F}^n)$ given by

$$x\circledast y=(\xi_1 y,\ldots,\xi_m y)=(\xi_1\upsilon_1,\ldots,\xi_1\upsilon_n,\cdots,\xi_m\upsilon_1,\ldots,\xi_m\upsilon_n).$$

There are vectors in $\mathbb{F}^m(\mathbb{F}^n)$ that are not Kronecker products (e.g., $(0,1,1,0)$ in $\mathbb{R}^2(\mathbb{R}^2)\cong\mathbb{R}^4$ is not a Kronecker product for any pair of vectors (x,y) in $\mathbb{R}^2\times\mathbb{R}^2$), and the sum of Kronecker products is not necessarily a Kronecker product. The next propositions show how the notion of Kronecker product for a pair (x,y) in $\mathbb{F}^m\times\mathbb{F}^n$ fits into the abstract notion of tensor products.

Proposition 4.A. *The operation $\circledast$ is distributive with respect to addition in $\mathbb{F}^{mn}$ and bilaterally homogeneous with respect to scalar multiplication*:

$$\alpha(x\circledast y)=(\alpha x)\circledast y=x\circledast(\alpha y),$$

$$(x_1\circledast y_1)+(x_2\circledast y_2)=(x_1+x_2)\circledast(y_1+y_2)-(x_1\circledast y_2)-(x_2\circledast y_1),$$

for every $x,x_1,x_2\in\mathcal{X}$, every $y,y_1,y_2\in\mathcal{Y}$, and every $\alpha\in\mathbb{F}$.

Proposition 4.B. *The collection $\mathbb{F}^m\circledast\mathbb{F}^n$ of all linear combinations of Kronecker products is such that*

$$\begin{aligned}\mathbb{F}^m\circledast\mathbb{F}^n &= \{\textstyle\sum_{i=1}^{N}\alpha_i(x_i\circledast y_i)\colon\ \alpha_i\in\mathbb{F},\ \ (x_i,y_i)\in\mathbb{F}^m\times\mathbb{F}^n,\ \ N\in\mathbb{N}\}\\ &= \{\textstyle\sum_{i=1}^{N}x_i\circledast y_i\colon\ \ (x_i,y_i)\in\mathbb{F}^m\times\mathbb{F}^n,\quad N\in\mathbb{N}\},\end{aligned}$$

and $\mathbb{F}^m\circledast\mathbb{F}^n$ is a linear space over $\mathbb{F}$.

The linear space $\mathbb{F}^m\circledast\mathbb{F}^n$ is referred to as the *Kronecker product space*.

Let $\{e_j\}_{j=1}^{m}$ be the canonical basis for $\mathbb{F}^m$ (made up of unit vectors). For each $j=1,\ldots,m$ take $y_j=(\upsilon_{j,1},\ldots,\upsilon_{j,n})$ in $\mathbb{F}^n$ and set $y=(y_1,\ldots,y_m)=(\upsilon_{1,1},\ldots,\upsilon_{1,n},\cdots,\upsilon_{m,1},\ldots,\upsilon_{m,n})$ in $\mathbb{F}^m(\mathbb{F}^n)\cong\mathbb{F}^{mn}$, which is naturally identified with $\sum_{j=1}^{m}e_j\circledast y_j$. Thus $\mathbb{F}^m(\mathbb{F}^n)\subseteq\mathbb{F}^m\circledast\mathbb{F}^n$. The converse identification

is clear by the definition of Kronecker product: $\mathbb{F}^m \circledast \mathbb{F}^n \subseteq \mathbb{F}^{mn}$. These inclusions stand for embeddings so that

$$\mathbb{F}^m \circledast \mathbb{F}^n \cong \mathbb{F}^m(\mathbb{F}^n) \cong \mathbb{F}^{mn}.$$

Proposition 4.C. *Let* $\langle \cdot\,;\cdot \rangle_{\mathbb{F}^{mn}}$, $\langle \cdot\,;\cdot \rangle_{\mathbb{F}^m}$, *and* $\langle \cdot\,;\cdot \rangle_{\mathbb{F}^n}$ *be the usual inner products equipping the linear spaces* $\mathbb{F}^{mn}$, $\mathbb{F}^m$ *and* $\mathbb{F}^n$, *respectively. Then*

$$\langle x \circledast y\,; u \circledast v\rangle_{\mathbb{F}^{mn}} = \langle x\,; u\rangle_{\mathbb{F}^m} \langle y\,; v\rangle_{\mathbb{F}^n} \quad \textit{for every} \quad x, u \in \mathbb{F}^m \ \textit{ and } \ y, v \in \mathbb{F}^n.$$

Recall from Remark 4.6 that single tensors as in Corollary 4.5 are given by

$$(x \otimes y)(u, v) = \langle x\,; u\rangle_{\mathbb{F}^m}\, \langle y\,; v\rangle_{\mathbb{F}^n} \quad \text{for every} \quad (x, y), (u, v) \in \mathbb{F}^m \times \mathbb{F}^n.$$

So the proposition below shows that Kronecker products $x \circledast y$ in $\mathbb{F}^m \circledast \mathbb{F}^n$ are identified with such single tensors $x \otimes y$ in the tensor product space $\mathbb{F}^m \otimes \mathbb{F}^n$.

Proposition 4.D. *The mapping* $\Theta\colon \mathbb{F}^m \circledast \mathbb{F}^n \to \mathbb{F}^m \otimes \mathbb{F}^n$ *defined by*

$$\Theta\Big(\sum\nolimits_{i=1}^{N} x_i \circledast y_i\Big) = \sum\nolimits_{i=1}^{N} \langle x_i\,; \cdot\,\rangle_{\mathbb{F}^m} \langle y_i\,; \cdot\,\rangle_{\mathbb{F}^n} = \sum\nolimits_{i=1}^{N} x_i \otimes y_i$$

is an isomorphism between the linear spaces $\mathbb{F}^m \circledast \mathbb{F}^n$ *and* $\mathbb{F}^m \otimes \mathbb{F}^n$.

Therefore $\mathbb{F}^m \circledast \mathbb{F}^n$ is isomorphic to $\mathbb{F}^m \otimes \mathbb{F}^n$,

$$\mathbb{F}^m \circledast \mathbb{F}^n \ \cong\ \mathbb{F}^m \otimes \mathbb{F}^n.$$

Indeed, the natural bilinear map $\theta'\colon \mathbb{F}^m \times \mathbb{F}^n \to \mathbb{F}^m \otimes \mathbb{F}^n$ of Corollary 4.5 for the particular case of Remark 4.6 is given by $\theta'(x, y)(u, v) = (x \otimes y)(u, v) = \langle x\,; u\rangle_{\mathbb{F}^m} \langle y\,; v\rangle_{\mathbb{F}^n}$ for every $(u, v) \in \mathbb{F}^m \times \mathbb{F}^n$, and so the natural bilinear map $\widetilde{\theta}'\colon \mathbb{F}^m \times \mathbb{F}^n \to \mathbb{F}^m \circledast \mathbb{F}^n$ is given by $\widetilde{\theta}' = \Theta\theta'$ as in Corollary 3.6. Thus consider the tensor product $(\mathbb{F}^m \circledast \mathbb{F}^n, \widetilde{\theta}')$ of the linear spaces $\mathbb{F}^m$ and $\mathbb{F}^n$, and consequently the *Kronecker product space is a tensor product space of* $\mathbb{F}^m$ *and* $\mathbb{F}^n$. Since $\mathbb{F}^m \circledast \mathbb{F}^n$ has been identified with $\mathbb{F}^m(\mathbb{F}^n)$ we get by transitivity

$$\mathbb{F}^m \otimes \mathbb{F}^n \cong \mathbb{F}^m(\mathbb{F}^n).$$

Summing up:

$$\mathbb{F}^m \otimes \mathbb{F}^n \cong \mathbb{F}^m \circledast \mathbb{F}^n \cong \mathbb{F}^m(\mathbb{F}^n) \cong \mathbb{F}^{mn}.$$

Compare the above relations with Proposition 3.F in Section 3.4.

Next take arbitrary linear transformations A in $\mathcal{L}[\mathbb{F}^m]$ and B in $\mathcal{L}[\mathbb{F}^n]$, and let them be represented with respect to the canonical basis for $\mathbb{F}^m$ and $\mathbb{F}^n$ by the square ($m \times m$ and $n \times n$) matrices

$$A = \begin{pmatrix} \alpha_{11} & \dots & \alpha_{1m} \\ \vdots & & \vdots \\ \alpha_{m1} & \dots & \alpha_{mm} \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} \beta_{11} & \dots & \beta_{1n} \\ \vdots & & \vdots \\ \beta_{n1} & \dots & \beta_{nn} \end{pmatrix}.$$

The *Kronecker product* $A \circledast B$ *of the matrices* A *and* B in $\mathcal{L}[\mathbb{F}^m]$ in $\mathcal{L}[\mathbb{F}^n]$ is

$$A \circledast B = \begin{pmatrix} \alpha_{11}B & \dots & \alpha_{1m}B \\ \vdots & & \vdots \\ \alpha_{m1}B & \dots & \alpha_{mm}B \end{pmatrix} \quad \text{in} \quad \mathcal{L}\big[\bigoplus_{i=1}^{m} \mathbb{F}^n\big] \cong \mathcal{L}[\mathbb{F}^{mn}],$$

a square $(mn \times mn)$ matrix.

Proposition 4.E. *For each* $A \in \mathcal{L}[\mathbb{F}^m]$ *and* $B \in \mathcal{L}[\mathbb{F}^n]$,

$$(A \circledast B)\,(x \circledast y) = \Big(\sum\nolimits_{j=1}^{m} \alpha_{1,j}\,\xi_j\,By, \cdots, \sum\nolimits_{j=1}^{m} \alpha_{m,j}\,\xi_j\,By\Big) = Ax \circledast By,$$

and so

$$(A \circledast B) \sum\nolimits_{i=1}^{N} x_i \circledast y_i = \sum\nolimits_{i=1}^{N} (A \circledast B)\,(x_i \circledast y_i) = \sum\nolimits_{i=1}^{N} Ax_i \circledast By_i,$$

for every $\sum_{i=1}^{N} x_i \circledast y_i \in \mathbb{F}^m \circledast \mathbb{F}^n$.

Consider the notion of tensor product of linear transformations $A \otimes B$ as in Definition 3.16 and Theorem 3.17. Let Θ be as in Proposition 4.D.

Proposition 4.F. *For each* $A \in \mathcal{L}[\mathbb{F}^m]$ *and* $B \in \mathcal{L}[\mathbb{F}^n]$,

$$\Theta(A \circledast B)\Theta^{-1} = A \otimes B,$$

where $\Theta \in \mathcal{L}[\mathbb{F}^m \circledast \mathbb{F}^n, \mathbb{F}^m \otimes \mathbb{F}^n]$ *is an isomorphism such that*

$$\Theta(\mathbb{F}^m \circledast \mathbb{F}^n) = \mathbb{F}^m \otimes \mathbb{F}^n.$$

Thus the Kronecker product of matrices is similar to the tensor product of linear transformations,

$$A \circledast B \cong A \otimes B.$$

As before, equality of tensor products is regarded up to isomorphism, so that all these identifications, viz., $\mathbb{F}^m \circledast \mathbb{F}^n \cong \mathbb{F}^m \otimes \mathbb{F}^n$, $x \circledast y \cong x \otimes y$ and $A \circledast B \cong A \otimes B$ are usually expressed by writing $=$ for $\cong$. So it is also usual to write

$$\mathbb{F}^m \circledast \mathbb{F}^n = \mathbb{F}^m \otimes \mathbb{F}^n, \qquad x \circledast y = x \otimes y, \qquad A \circledast B = A \otimes B.$$

The linear space $\mathcal{L}[\mathbb{F}^m, \mathbb{F}^n]$ of all linear transformations of $\mathbb{F}^m$ into $\mathbb{F}^m$ is naturally identified with the linear space $\mathbb{F}_{m,n}$ of all $m \times n$ matrices (representing transformations in $\mathcal{L}[\mathbb{F}^m, \mathbb{F}]$ with respect to the canonical basis for $\mathbb{F}^m$ and $\mathbb{F}^n$),

$$\mathcal{L}[\mathbb{F}^m, \mathbb{F}^n] \cong \mathbb{F}_{m,n}.$$

Proposition 4.G. $\mathbb{F}^m \otimes \mathbb{F}^n \cong \mathbb{F}_{m,n}$.

Proposition 4.H. *If* $\mathcal{P}_n[0,1]$ *is the linear space of all polynomials in the variable* $t \in [0,1]$ *with coefficients in* $\mathbb{F}$ *of degree not greater than* $n \in \mathbb{N}$,

$$\mathcal{P}_n[0,1] = \big\{p \in \mathbb{F}^{[0,1]}\colon\ p(t) = \textstyle\sum_{i=0}^{n} \alpha_i t^i, t \in [0,1], \textit{with each } \alpha_i \textit{ in } \mathbb{F}\big\},$$

then for every $m, n \in \mathbb{N}$

$$\mathcal{P}_n[0,1] \otimes \mathcal{P}_m[0,1] \cong \mathcal{P}_{(n+1)(m+1)}[0,1].$$

Let $\mathcal{X}$ and $\mathcal{Y}$ be finite-dimensional linear spaces over the same field $\mathbb{F}$ with $\dim \mathcal{X} = m$ and $\dim \mathcal{Y} = n$ so that $\mathcal{X} \cong \mathbb{F}^m$, $\mathcal{Y} \cong \mathbb{F}^n$, and $\mathcal{X} \otimes \mathcal{Y} \cong \mathbb{F}^m \otimes \mathbb{F}^m$ (cf. Corollary 3.11). Let $\sigma(A)$ denote the spectrum of $A \in \mathcal{L}[\mathcal{X}]$ (which in this finite-dimensional case coincides with the set of all eigenvalues of A). The trace and the determinant of $A \in \mathcal{L}[\mathcal{X}]$ are defined as the sum and the product of its eigenvalues (including multiplicity), denoted by $\operatorname{tr}(A)$ and $\det(A)$. If Λ and Δ are nonempty sets of scalars (i.e., nonempty subsets of $\mathbb{F}$), then the *set product* $\Lambda \cdot \Delta$ is defined as the subset of $\mathbb{F}$ consisting of the products of all elements from Λ with all elements from Δ. To avoid empty spectrum set $\mathbb{F} = \mathbb{C}$.

Proposition 4.I. *If* $\mathcal{X}$ *and* $\mathcal{Y}$ *are complex finite-dimensional linear spaces with* $\dim \mathcal{X} = m$ *and* $\dim \mathcal{Y} = n$*, and if* $A \in \mathcal{L}[\mathcal{X}]$ *and* $B \in \mathcal{L}[\mathcal{Y}]$*, then*

(a) $\sigma(A \otimes B) = \sigma(A) \cdot \sigma(B)$,

(b) $\operatorname{tr}(A \otimes B) = \operatorname{tr}(A) \cdot \operatorname{tr}(B)$,

(c) $\det(A \otimes B) = \det(A)^m \cdot \det(B)^n$.

All the above results dealt with finite-dimensional spaces where tensor products are identified with Kronecker products. The following two results are not restricted to the finite-dimensional case.

Proposition 4.J. *If— neither* $\mathcal{X}$ *nor* $\mathcal{Y}$ *are infinite-dimensional, then the embedding in Remark 4.9*(f) *is proper:* $\mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp \subset b[\mathcal{X} \times \mathcal{Y}, \mathbb{F}]$.

The next proposition extends the claim in the proof of Corollary 4.10.

Proposition 4.K. *Let* $\sum_i x_i \otimes y_i$ *be an element in the tensor product space of linear spaces* $\mathcal{X}$ *and* $\mathcal{Y}$*. The following assertions are pairwise equivalent.*

(a) $\sum_i x_i \otimes y_i = 0$.

(b) $\sum_i f(x_i) g(y_i) = 0$ *for every* $f \in \mathcal{X}^\sharp$ *and every* $g \in \mathcal{Y}^\sharp$.

(c) $\sum_i f(x_i) y_i = 0$ *for every* $f \in \mathcal{X}^\sharp$.

(d) $\sum_i g(y_i) x_i = 0$ *for every* $g \in \mathcal{Y}^\sharp$.

Notes: Grothendieck's fundamental work in the 1950's on the metric theory of tensor products has been unified and updated by Diestel, Fourie and Swart in 2008 [17]. See also Pisier's 2012 exposition [77]. In Grothendieck's pioneering work, the notion of tensor product space was essentially given in terms of the dual of the linear space of bilinear forms (or, more generally, the linear space of linear maps of bilinear maps). The same sort of tensor product definition also

appears in Halmos's 1958 book on finite-dimensional vector spaces [30, Section 24]. Another representative work along this line (dual of the linear space of bilinear forms) is Ryan's 2002 book on tensor products of Banach spaces [83]. From the algebraic point of view, this is the approach discussed in Sections 4.2 and 4.3 (and also in Section 4.4) in this chapter. Tensor products of normed spaces and norms on tensor product spaces will be considered in Chapter 7.

On the other hand, the alternative (but still usual) approach for defining tensor products as in Section 4.1 relies on quotient spaces of free linear spaces (which are equivalent to the linear space of formal linear combinations) of Cartesian products of linear spaces. Such a quotient space approach has been sometimes referred to as the *algebraic tensor product* (although the linear-bilinear approach is equally algebraic). See, for instance, [5, pp. 22–25], [92, Section 3.4] and [81, Chapter 14] for a linear space version, and [69, Section IX.8] and [66, Section XVI.1]) for a module version.

Chapter 4 has dealt with interpretations of the axiomatic approach of Chapter 3, focusing on two distinct lines of action to exhibit concrete examples of tensor product spaces of a pair of linear spaces. Most propositions here in Section 4.4 aim at examples and properties for the finite-dimensional case, which boils down to the notion of Kronecker products. Propositions 4.A to 4.F are standard basic properties of Kronecker products that can be found in many texts dealing with this topic. For instance, see the unified introductory exposition in [48, Section 3]. For Proposition 4.G see, e.g., [5, Problem 2.T]. Proposition 4.H is again an application of Corollary 3.10 and Proposition 1.C (as $\dim \mathcal{P}_n[0,1] = n+1$; see, e.g., [52, Example 2.M]). The finite-dimensional spectral properties in Proposition 4.I are standard; see, e.g., [94, Proposition 1.10 and its Corollary]. For Proposition 4.J see, e.g., [83, Exercise 1.10 and Proposition 1.2]. Proposition 4.K is an extended version of the claim in the proof of Corollary 4.10 (see, e.g., [83, Proposition 1.2]).

Suggested Readings

Brown and Pearcy [5]
Defant and Floret [15]
Greub [27]
Halmos [30]
Jarchow [42]
Lang [66]
MacLane and Birkhoff [69]
Roman [81]
Ryan [83]
Yokonuma [94]



5

Normed-Space Results

This chapter outlines the essential normed-space notions required in the book. It focuses on bounded (i.e., continuous) linear transformations and normed quotient spaces. As in Chapter 1, the purpose here is to put together only those results necessary in the forthcoming chapters. Normed-space aspects of bilinear maps will be discussed in Chapter 6. Chapters 5 and 6 enable us to advance an axiomatic theory of tensor products of Banach spaces.

5.1 Notation, Terminology and Definitions

Again, all linear spaces are over the same field $\mathbb{F}$. A normed space is a linear space $\mathcal{X}$ equipped with a norm $\|\cdot\|\colon \mathcal{X} \to \mathbb{R}$. A norm generates a metric, and so normed spaces are metric spaces. A normed space $\mathcal{X}$ may also be denoted by $(\mathcal{X}, \|\cdot\|)$, and a subscript may be inserted if necessary, for instance $\|\cdot\|_{\mathcal{X}}$.

Linear manifold. By a *linear manifold* $\mathcal{M}$ of a normed space $\mathcal{X}$ we mean a linear manifold of the linear space $\mathcal{X}$ which inherits its norm from the normed space $\mathcal{X}$; the norm on the normed space $(\mathcal{M}, \|\cdot\|_{\mathcal{M}})$ is the restriction of the norm on $(\mathcal{X}, \|\cdot\|_{\mathcal{X}})$, that is, $\|\cdot\|_{\mathcal{M}} = \|\cdot\|_{\mathcal{X}}|_{\mathcal{M}} \colon \mathcal{M} \to \mathbb{R}$, which is indeed a norm on $\mathcal{M}$. A *subspace* of a normed space is a closed linear manifold (closed with respect to the metric generated by the norm). Every finite-dimensional linear manifold of any normed space is a subspace. The closure $\mathcal{M}^-$ of a linear manifold $\mathcal{M}$ of a normed space is a subspace. A *Banach space* is a complete normed space (complete with respect to the metric generated by the norm). Subspaces (closed linear manifolds) of Banach spaces are Banach spaces. A *Hilbert space* is a Banach space whose norm is induced by an inner product.

Topological isomorphism. In this context, a *homeomorphism* is an invertible map between metric spaces which is continuous and has a continuous inverse. A *topological isomorphism* is an isomorphism between normed spaces which is also a homeomorphism, that is, an invertible continuous linear transformation between normed spaces whose inverse is continuous. If there exists a topological isomorphism of a normed space $\mathcal{X}$ onto a normed space $\mathcal{Y}$, then we say $\mathcal{X}$ and $\mathcal{Y}$ are *topologically isomorphic* — same notation: $\mathcal{X} \cong \mathcal{Y}$.

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Isometric isomorphism. This is a particular case of a topological isomorphism that preserves distance. An *isometry* between metric spaces is a map that preserves distance. An *isometric isomorphism* is a surjective linear isometry or, equivalently, an invertible linear isometry. (Recall: (i) the inverse of an invertible linear transformation is linear, (ii) the inverse of an invertible isometry is an isometry, (iii) an isometry is always injective, (iv) the term isometry in this context will from now on mean *linear isometry*: a linear transformation between normed spaces that preserves distance.) If there exists an isometric isomorphism of a normed space $\mathcal{X}$ onto a normed space $\mathcal{Y}$ then we say that $\mathcal{X}$ and $\mathcal{Y}$ are *isometrically isomorphic* — still the same notation $\mathcal{X} \cong \mathcal{Y}$.

Topology on Cartesian products and direct sums. Take a pair, say $(\mathcal{X}, \|\cdot\|_{\mathcal{X}})$ and $(\mathcal{Y}, \|\cdot\|_{\mathcal{Y}})$, of normed spaces and let $\mathcal{X}\times\mathcal{Y}$ be the Cartesian product of the linear spaces $\mathcal{X}$ and $\mathcal{Y}$. Consider the following equivalent alternatives (which naturally extend to any countable family of normed spaces).

(i) Either equip the Cartesian product $\mathcal{X}\times\mathcal{Y}$ with any product metric d generated by those norms, say, d_p for $p \geq 1$ or d_∞, from $(\mathcal{X}\times\mathcal{Y})\times(\mathcal{X}\times\mathcal{Y})$ to $\mathbb{R}$, given by $d_p((x,y),(x',y')) = (\|x-x'\|_{\mathcal{X}}^p + \|y-y'\|_{\mathcal{Y}}^p)^{1/p}$ and $d_\infty((x,y),(x',y')) = \max\{\|x-x'\|_{\mathcal{X}}, \|y-y'\|_{\mathcal{Y}}\}$ for $(x,y),(x',y') \in \mathcal{X}\times\mathcal{Y}$, and this leads to a metric space $(\mathcal{X}\times\mathcal{Y}, d)$ where the set $\mathcal{X}\times\mathcal{Y}$ is equipped with a product metric d;

(ii) or equip the direct sum $\mathcal{X} \oplus \mathcal{Y}$ (a linear space whose underlying set is $\mathcal{X}\times\mathcal{Y}$) with any of its standard norms $\|\cdot\|$, say, $\|(x,y)\|_p = (\|x\|_{\mathcal{X}}^p + \|y\|_{\mathcal{Y}}^p)^{1/p}$ for $p \geq 1$ or $\|(x,y)\|_\infty = \max\{\|x\|_{\mathcal{X}}, \|y\|_{\mathcal{Y}}\}$ for $(x,y) \in \mathcal{X} \oplus \mathcal{Y}$, and this leads to a normed space $(\mathcal{X} \oplus \mathcal{Y}, \|\cdot\|) = (\mathcal{X}\times\mathcal{Y}, \oplus, \centerdot, \mathbb{F}, \|\cdot\|)$, where the norm $\|\cdot\|$ on $\mathcal{X} \oplus \mathcal{Y}$ generates a product metric d on the Cartesian product $\mathcal{X}\times\mathcal{Y}$.

Bounded linear transformation. If $(\mathcal{X}, \|\cdot\|_{\mathcal{X}})$ and $(\mathcal{Y}, \|\cdot\|_{\mathcal{Y}})$ are normed spaces, then let $\mathcal{B}[\mathcal{X},\mathcal{Y}]$ denote the linear space of all *bounded linear transformations* of $\mathcal{X}$ into $\mathcal{Y}$, equivalently, of all *continuous linear transformations* of $\mathcal{X}$ into $\mathcal{Y}$, with respect to the norms $\|\cdot\|_{\mathcal{X}}$ and $\|\cdot\|_{\mathcal{Y}}$. The kernel $\mathcal{N}(T)$ of a bounded linear transformation $T \in \mathcal{B}[\mathcal{X},\mathcal{Y}]$ is a subspace of $\mathcal{X}$ and the range $\mathcal{R}(T)$ is a linear manifold of $\mathcal{Y}$. The *induced uniform norm* $\|\cdot\|_{\mathcal{B}[\mathcal{X},\mathcal{Y}]}$ of $T \in \mathcal{B}[\mathcal{X},\mathcal{Y}]$ is a norm on $\mathcal{B}[\mathcal{X},\mathcal{Y}]$ defined by $\|T\|_{\mathcal{B}[\mathcal{X},\mathcal{Y}]} = \sup_{x\neq 0} \frac{\|Tx\|_{\mathcal{Y}}}{\|x\|_{\mathcal{X}}}$, and so $\|Tx\|_{\mathcal{Y}} \leq \|T\|_{\mathcal{B}[\mathcal{X},\mathcal{Y}]}\|x\|_{\mathcal{X}}$ for every $x \in \mathcal{X}$ (i.e., T is a *bounded transformation*). When equipped with this norm, the normed space $(\mathcal{B}[\mathcal{X},\mathcal{Y}], \|\cdot\|_{\mathcal{B}[\mathcal{X},\mathcal{Y}]})$ becomes a Banach space if $(\mathcal{Y}, \|\cdot\|_{\mathcal{Y}})$ is a Banach space (the converse holds if $\mathcal{X} \neq \{0\}$). We will drop subscripts from norms whenever this is clear in the context and will write $\|x\|$, $\|Tx\|$, $\|T\|$ all with the same norm notation. An *isometry* is a transformation $T \in \mathcal{B}[\mathcal{X},\mathcal{Y}]$ such that $\|Tx\| = \|x\|$ for every $x \in \mathcal{X}$, which implies $\|T\| = 1$. Thus T is an isometric isomorphism if and only if it is surjective and $\|Tx\| = \|x\|$ for every $x \in \mathcal{X}$. Moreover, $T \in \mathcal{B}[\mathcal{X},\mathcal{Y}]$ is a *contraction* if $\|Tx\| \leq \|x\|$ for every $x \in \mathcal{X}$, which means $\|T\| \leq 1$. By an *operator* we mean a bounded linear transformation of a normed space into itself: T is an operator if $T \in \mathcal{B}[\mathcal{X},\mathcal{X}]$. We use the notation $\mathcal{B}[\mathcal{X}]$ as a short for the normed algebra $\mathcal{B}[\mathcal{X},\mathcal{X}]$ of all operators on $\mathcal{X}$ (so $\mathcal{B}[\mathcal{X}]$ is a Banach algebra if and only

if $\mathcal{X}$ is a Banach space). If S and T lie in the normed algebra $\mathcal{B}[\mathcal{X}]$, then their product ST (i.e., composition $S \circ T$) also lies in $\mathcal{B}[\mathcal{X}]$ and $\|ST\| \leq \|S\|\,\|T\|$.

Bounded linear functionals, dual and normed-space adjoint. It is clear that $\mathcal{B}[\mathcal{X},\mathcal{Y}] \subseteq \mathcal{L}[\mathcal{X},\mathcal{Y}]$. Let $\mathcal{X}^* = \mathcal{B}[\mathcal{X},\mathbb{F}] \subseteq \mathcal{L}[\mathcal{X},\mathbb{F}] = \mathcal{X}^\sharp$ be the *dual* of the normed space $\mathcal{X}$, consisting of all linear functionals $f\colon \mathcal{X} \to \mathbb{F}$ from $\mathcal{X}^\sharp$ which are continuous (i.e., bounded) with respect to the norm $\|\cdot\|_{\mathcal{X}}$ on $\mathcal{X}$ and the usual norm $|\cdot|$ on $\mathbb{F}$. Since $(\mathbb{F}, |\cdot|)$ is a Banach space, it follows that the dual $\mathcal{X}^* = (\mathcal{X}, \|\cdot\|_{\mathcal{X}})^* = (\mathcal{X}^*, \|\cdot\|_{\mathcal{X}^*}) = (\mathcal{B}[\mathcal{X},\mathbb{F}], \|\cdot\|_{\mathcal{B}[\mathcal{X},\mathbb{F}]})$ of any normed space $\mathcal{X}$ is a Banach space, where the norm on $\mathcal{X}^*$ is the induced uniform norm on $\mathcal{B}[\mathcal{X},\mathbb{F}]$ which is given by $\|f\|_{\mathcal{X}^*} = \|f\|_{\mathcal{B}[\mathcal{X},\mathbb{F}]} = \sup_{x\neq 0} \frac{|f(x)|}{\|x\|_{\mathcal{X}}}$ for every $f \in \mathcal{X}^*$. So $|f(x)| \leq \|f\|_{\mathcal{X}^*}\|x\|_{\mathcal{X}}$ for every $x \in \mathcal{X}$. Let $T^* \in \mathcal{B}[\mathcal{Y}^*, \mathcal{X}^*]$ be the *normed-space adjoint* of $T \in \mathcal{B}[\mathcal{X},\mathcal{Y}]$, which is the unique transformation in $\mathcal{B}[\mathcal{Y}^*, \mathcal{X}^*]$ for which $T^*g = g\,T \in \mathcal{X}^*$ for every $g \in \mathcal{Y}^*$ (i.e., $(T^*g)(x) = g(Tx) \in \mathbb{F}$ for every $g \in \mathcal{Y}^*$ and every $x \in \mathcal{X}$ — this is also written as $\langle x\,; T^*g\rangle = \langle Tx\,; g\rangle$). The usual basic properties of Hilbert-space adjoints are transferred to normed-space adjoints. For instance, $\|T^*\| = \|T\|$, $(T+S)^* = T^* + S^*$ for $S \in \mathcal{B}[\mathcal{X},\mathcal{Y}]$ and, slightly different, $(\alpha T)^* = \alpha T^*$ for $\alpha \in \mathbb{F}$. If $S, T \in \mathcal{B}[\mathcal{X}]$, then $(TS)^* = S^*T^*$, and I^* is the identity in $\mathcal{B}[\mathcal{X}^*]$ where I is the identity in $\mathcal{B}[\mathcal{X}]$. (See, e.g., [70, Section 3.1] or [87, Section 3.2].) When a norm either is clear or is immaterial, we refer to the normed spaces $\mathcal{X}$, $\mathcal{B}[\mathcal{X},\mathcal{Y}]$, $\mathcal{B}[\mathcal{X}]$ or $\mathcal{X}^*$ without specifying which norm is being considered, and write norms without subscripts, such as

$$\|T\| = \sup_{x\neq 0} \tfrac{\|Tx\|}{\|x\|} = \sup_{\|x\|=1} \|Tx\| = \sup_{\|x\|\leq 1} \|Tx\| = \sup_{\|x\|<1} \|Tx\|,$$
$$\|f\| = \sup_{x\neq 0} \tfrac{|f(x)|}{\|x\|} = \sup_{\|x\|=1} |f(x)| = \sup_{\|x\|\leq 1} |f(x)| = \sup_{\|x\|<1} |f(x)|,$$

where x, Tx, T and f lie in the normed spaces $\mathcal{X}$, $\mathcal{Y}$, $\mathcal{B}[\mathcal{X},\mathcal{Y}]$ and $\mathcal{X}^*$. Recall:

$$\|x\| = \sup_{f\neq 0} \tfrac{|f(x)|}{\|f\|} = \sup_{\|f\|=1} |f(x)| = \sup_{\|f\|\leq 1} |f(x)| = \sup_{\|f\|<1} |f(x)|.$$

(The above expression will be reviewed in Remark 5.2(d) in Section 5.2 ahead.)

Isometrically equivalent transformations. Let $\mathcal{X}, \widetilde{\mathcal{X}}, \mathcal{Y}, \widetilde{\mathcal{Y}}$ be normed spaces where $\mathcal{X}$ and $\widetilde{\mathcal{X}}$ as well as $\mathcal{Y}$ and $\widetilde{\mathcal{Y}}$ are isometrically isomorphic (or topologically isomorphic). Two transformations $T \in \mathcal{B}[\mathcal{X},\mathcal{Y}]$ and $\widetilde{T} \in \mathcal{B}[\widetilde{\mathcal{X}}, \widetilde{\mathcal{Y}}]$ are *isometrically equivalent* (or just *equivalent*) if there are isometric isomorphisms (or topological isomorphisms) $J \in \mathcal{B}[\mathcal{X}, \widetilde{\mathcal{X}}]$ and $K \in \mathcal{B}[\mathcal{Y}, \widetilde{\mathcal{Y}}]$ such that

$$KT = \widetilde{T}J.$$

This means the following diagram commutes (compare with Section 1.1):

$$\begin{array}{ccc} \mathcal{X} & \xrightarrow{\;T\;} & \mathcal{Y} \\ {\scriptstyle J}\big\downarrow & & \big\downarrow{\scriptstyle K} \\ \widetilde{\mathcal{X}} & \xrightarrow{\;\widetilde{T}\;} & \widetilde{\mathcal{Y}}. \end{array}$$

Notation for both cases: $\widetilde{T} \cong T$. For the isometrically equivalent case we get

$$\|\widetilde{T}\| = \|KTJ^{-1}\| = \|T\|, \quad \mathcal{N}(\widetilde{T}) \cong \mathcal{N}(T) \quad \text{and} \quad \mathcal{R}(\widetilde{T}) \cong \mathcal{R}(T),$$

where $\cong$ stands for isometrically isomorphic normed spaces. Particular cases:

(i) If $\widetilde{\mathcal{X}} = \mathcal{X}$ and $T = K^{-1}\widetilde{T}$, then $\widetilde{T} \cong T$ for $T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ and $\widetilde{T} \in \mathcal{B}[\mathcal{X}, \widetilde{\mathcal{Y}}]$.

(ii) If $\widetilde{\mathcal{Y}} = \mathcal{Y}$ and $T = \widetilde{T}J$, then $\widetilde{T} \cong T$ for $T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ and $\widetilde{T} \in \mathcal{B}[\widetilde{\mathcal{X}}, \mathcal{Y}]$.

In other words, $T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ and $\widetilde{T} \in \mathcal{B}[\mathcal{X}, \widetilde{\mathcal{Y}}]$ are isometrically equivalent (or equivalent), and similarly, $T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ and $\widetilde{T} \in \mathcal{B}[\widetilde{\mathcal{X}}, \mathcal{Y}]$ are isometrically equivalent (or equivalent), if the diagram (i) or the diagram (ii) below commutes:

(i)
$$\begin{array}{ccc} \mathcal{X} & \xrightarrow{\ T\ } & \mathcal{Y} \\ & {\scriptstyle \widetilde{T}}\searrow \quad \uparrow{\scriptstyle K^{-1}} & \\ & \widetilde{\mathcal{Y}} & \end{array}, \qquad \text{(ii)} \quad \begin{array}{ccc} \mathcal{X} & \xrightarrow{\ T\ } & \mathcal{Y} \\ {\scriptstyle J}\downarrow & \nearrow{\scriptstyle \widetilde{T}} & \\ \widetilde{\mathcal{X}} & & \end{array}.$$

The only relevant algebraic structure in the above two particular cases concern the isomorphism (either isometric or topological) $K : \mathcal{Y} \to \widetilde{\mathcal{Y}}$ between the normed spaces $\mathcal{Y}$ and $\widetilde{\mathcal{Y}}$ as soon as the composition $T = K^{-1}\widetilde{T}$ holds in (i), and the isomorphism (either isometric or topological) $J : \mathcal{X} \to \widetilde{\mathcal{X}}$ between the normed spaces $\mathcal{X}$ and $\widetilde{\mathcal{X}}$ as soon as the composition $T = \widetilde{T}J$ holds in (ii), independently of linearity or continuity of T and $\widetilde{T}$. Thus when the above commutative diagrams hold, then just the isomorphisms (either isometric or topological) $\widetilde{\mathcal{Y}} \cong \mathcal{Y}$ in (i) and $\widetilde{\mathcal{X}} \cong \mathcal{X}$ in (ii) are enough to define T and $\widetilde{T}$ as being isometrically equivalent (or just equivalent — denoted by $\widetilde{T} \cong T$), independently of any algebraic or topological properties T and $\widetilde{T}$ may possess or not.

Completion. For every normed space $\mathcal{X}$ there exist a complete normed space $\widehat{\mathcal{X}}$ and a linear isometry $J : \mathcal{X} \to \widehat{\mathcal{X}}$ such that its range $\widetilde{\mathcal{X}} = J(\mathcal{X})$ is a dense linear manifold of $\widehat{\mathcal{X}}$ (and so $J \in \mathcal{B}[\mathcal{X}, \widetilde{\mathcal{X}}]$ is an isometric isomorphism between the normed spaces $\mathcal{X}$ and $\widetilde{\mathcal{X}}$, where the norm in $\widetilde{\mathcal{X}}$ is that inherited from $\widehat{\mathcal{X}}$). Thus every normed space $\mathcal{X}$ is isometrically isomorphic to a dense linear manifold $\widetilde{\mathcal{X}}$ of a Banach space $\widehat{\mathcal{X}}$ (i.e., $\mathcal{X}$ is isometrically densely embedded in $\widehat{\mathcal{X}}$). In this case $\widehat{\mathcal{X}}$ is called a *completion* of $\mathcal{X}$. So every $x \in \mathcal{X}$ has a correspondent (or an extension) $\widehat{x} \in \widehat{\mathcal{X}}$ such that $\|x\|_{\mathcal{X}} = \|\widehat{x}\|_{\widehat{\mathcal{X}}}$. Summing up, every normed space has a completion, and completions of a normed space are all isometrically isomorphic (see, e.g., [52, Theorems 4.40 and 4.41]). In particular, a Banach space is trivially a completion of every dense linear manifold of it.

5.2 Extension of Bounded Linear Transformations

The following are well-known results on completion of normed spaces and extension of bounded linear transformations over completion that will be frequently required in the sequel. If $\widehat{\mathcal{X}}$ is a completion of a normed space $\mathcal{X}$ and

$\widehat{\mathcal{Y}}$ is a completion of a normed space $\mathcal{Y}$, then a transformation $\widehat{T} \in \mathcal{B}[\widehat{\mathcal{X}}, \widehat{\mathcal{Y}}]$ is called an *extension over completion* of a transformation $T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ if the restriction of $\widehat{T}$ to a dense linear manifold of $\widehat{\mathcal{X}}$ into a dense linear manifold of $\widehat{\mathcal{Y}}$ is isometrically equivalent to T. Precisely, if $\widehat{\mathcal{X}}$ and $\widehat{\mathcal{Y}}$ are completions of $\mathcal{X}$ and $\mathcal{Y}$, then $\widehat{T} \in \mathcal{B}[\widehat{\mathcal{X}}, \widehat{\mathcal{Y}}]$ is an extension over completion of $T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ if $\widetilde{T} = \widehat{T}|_{J(\mathcal{X})} = KTJ^{-1}$ for some isometric isomorphisms $J \colon \mathcal{X} \to J(\mathcal{X}) = \widetilde{\mathcal{X}} \subseteq \widehat{\mathcal{X}}$ and $K \colon \mathcal{Y} \to K(\mathcal{Y}) = \widetilde{\mathcal{Y}} \subseteq \widehat{\mathcal{Y}}$ with $\widetilde{\mathcal{X}}^- = \widehat{\mathcal{X}}$ and $\widetilde{\mathcal{Y}}^- = \widehat{\mathcal{Y}}$.

Theorem 5.1. *Let $\widehat{\mathcal{X}}$ and $\widehat{\mathcal{Y}}$ be completions of normed spaces $\mathcal{X}$ and $\mathcal{Y}$.*

(a) *Every $T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ has an extension $\widehat{T} \in \mathcal{B}[\widehat{\mathcal{X}}, \widehat{\mathcal{Y}}]$ over the completion $\widehat{\mathcal{X}}$ of $\mathcal{X}$ into the completion $\widehat{\mathcal{Y}}$ of $\mathcal{Y}$.*

(b) $\|\widehat{T}\| = \|T\|$.

(c) *$\widehat{T}$ is unique up to isometric isomorphisms.*

(d) *If $T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ is an isometric isomorphism, then so is $\widehat{T} \in \mathcal{B}[\widehat{\mathcal{X}}, \widehat{\mathcal{Y}}]$.*

(e) *If $\mathcal{Y}$ is a Banach space, then $\mathcal{B}[\mathcal{X}, \mathcal{Y}] \cong \mathcal{B}[\widehat{\mathcal{X}}, \mathcal{Y}]$.*

(f) $\widehat{\mathcal{X}}^* \cong \mathcal{X}^* = \widehat{\mathcal{X}^*}$ *and* $\widehat{T}^* \cong T^* = \widehat{T^*}$.

(g) *If $\mathcal{M}$ is a linear manifold of a normed space $\mathcal{X}$, then there is a completion $\widehat{\mathcal{M}}$ of $\mathcal{M}$ included in a completion $\widehat{\mathcal{X}}$ of $\mathcal{X}$ (thus $\widehat{\mathcal{M}}$ is a subspace of $\widehat{\mathcal{X}}$).*

(h) $\mathcal{R}(T) \subseteq \mathcal{R}(\widehat{T}) \subseteq \mathcal{R}(\widehat{T})^- \cong \mathcal{R}(T)^-$ *(first $\subseteq$ means isometric embedding).*

(i) $\mathcal{N}(T) \subseteq \mathcal{N}(\widehat{T})$ *(isometric embedding: $\mathcal{N}(T) \cong \mathcal{N}(\widetilde{T}) \subseteq \mathcal{N}(\widehat{T})$);*
if $\mathcal{R}(T)$ is closed, then $\mathcal{N}(T) = \{0\}$ if and only if $\mathcal{N}(\widehat{T}) = \{0\}$;
if $\dim \mathcal{R}(T) < \infty$, then $\mathcal{N}(T)^- \cong \mathcal{N}(\widehat{T})$ (closure of $\mathcal{N}(\widetilde{T}) \cong \mathcal{N}(T)$ in $\widehat{\mathcal{X}}$).

Proof. (a) *Existence.* Let $\widehat{\mathcal{X}}$ be a completion of a normed space $\mathcal{X}$ and let $\widehat{\mathcal{Y}}$ be a completion of a normed space $\mathcal{Y}$. Thus there are dense linear manifolds $\widetilde{\mathcal{X}}$ of $\widehat{\mathcal{X}}$ and $\widetilde{\mathcal{Y}}$ of $\widehat{\mathcal{Y}}$, and isometric isomorphisms $J \colon \mathcal{X} \to \widetilde{\mathcal{X}}$ of $\mathcal{X}$ onto $J(\mathcal{X}) = \widetilde{\mathcal{X}}$ with $\widetilde{\mathcal{X}}^- = \widehat{\mathcal{X}}$ and $K \colon \mathcal{Y} \to \widetilde{\mathcal{Y}}$ of $\mathcal{Y}$ onto $K(\mathcal{Y}) = \widetilde{\mathcal{Y}}$ with $\widetilde{\mathcal{Y}}^- = \widehat{\mathcal{Y}}$. Suppose

$$T \colon \mathcal{X} \to \mathcal{Y}$$

is a bounded linear transformation of $\mathcal{X}$ into $\mathcal{Y}$. Consider the composition $KTJ^{-1} \colon \widetilde{\mathcal{X}} \to \widetilde{\mathcal{Y}} \subseteq \widehat{\mathcal{Y}}$, which is again bounded and linear, and therefore uniformly continuous. Thus since $\widetilde{\mathcal{X}}$ is dense in $\widehat{\mathcal{X}}$ and $\widehat{\mathcal{Y}}$ is complete, there is a unique continuous extension

$$\widehat{T} = \widehat{KTJ^{-1}} \colon \widehat{\mathcal{X}} \to \widehat{\mathcal{Y}}$$

over $\widehat{\mathcal{X}}$ of the uniformly continuous map

$$\widetilde{T} = KTJ^{-1} \colon \widetilde{\mathcal{X}} \to \widetilde{\mathcal{Y}} \subseteq \widehat{\mathcal{Y}},$$

which is uniformly continuous (see, e.g., [52, Theorem 3.45]), so that

$$\widehat{T}|_{\widetilde{\mathcal{X}}} = \widetilde{T}.$$

The transformation $\widehat{T}$ is the *extension by continuity* of $\widetilde{T}$, and also the *extension over completion* of T (all are linear and bounded). The diagram below (where $\subseteq^{-}$ denotes dense inclusion) illustrates extension over completion.

$$\begin{array}{ccccc} \mathcal{Y} & \xrightarrow{K} & \widetilde{\mathcal{Y}} = K(\mathcal{Y}) & \subseteq^{-} & \widehat{\mathcal{Y}} \\ {\scriptstyle T}\uparrow & & \uparrow{\scriptstyle \widetilde{T}=\widehat{T}|_{\widetilde{\mathcal{X}}}=KTJ^{-1}} & & \uparrow{\scriptstyle \widehat{T}} \\ \mathcal{X} & \xrightarrow{J} & \widetilde{\mathcal{X}} = J(\mathcal{X}) & \subseteq^{-} & \widehat{\mathcal{X}}. \end{array}$$

(b) *Norm invariance.* Extension by continuity preserves norm (see, e.g., [52, Theorem 4.35]). So $\|\widehat{T}\| = \|\widetilde{T}\|$, where $\widehat{T} = \widehat{KTJ^{-1}}: \widehat{\mathcal{X}} \to \widehat{\mathcal{Y}}$ is the extension by continuity of $\widetilde{T} = KTJ^{-1}: \widetilde{\mathcal{X}} \to \widetilde{\mathcal{Y}} \subseteq \widehat{\mathcal{Y}}$ over $\widehat{\mathcal{X}}$. Since K are J^{-1} are isometric isomorphisms between normed spaces, $\|KTJ^{-1}\| = \|T\|$, and so $\|\widehat{T}\| = \|T\|$.

(c) *Uniqueness.* Let $\widehat{T}': \widehat{\mathcal{X}}' \to \widehat{\mathcal{Y}}'$ be a bounded linear extension over completion of $T: \mathcal{X} \to \mathcal{Y}$ defined on a completion $\widehat{\mathcal{X}}'$ of $\mathcal{X}$ into a completion $\widehat{\mathcal{Y}}'$ of $\mathcal{Y}$. So $\widehat{T}'|_{\widetilde{\mathcal{X}}'} = K'TJ'^{-1}: \widetilde{\mathcal{X}}' \to \widetilde{\mathcal{Y}}'$, where $\widetilde{\mathcal{X}}'^{-} = \widehat{\mathcal{X}}'$ and $\widetilde{\mathcal{Y}}'^{-} = \widehat{\mathcal{Y}}'$ with isometric isomorphisms $J': \mathcal{X} \to \widetilde{\mathcal{X}}'$ and $K': \mathcal{Y} \to \widetilde{\mathcal{Y}}'$. Set $\widetilde{J} = J'J^{-1}: \widetilde{\mathcal{X}} \to \widetilde{\mathcal{X}}' \subseteq \widehat{\mathcal{X}}'$ and $\widetilde{K} = K'K^{-1}: \widetilde{\mathcal{Y}} \to \widetilde{\mathcal{Y}}' \subseteq \widehat{\mathcal{Y}}'$ (composition of isometric isomorphisms, thus isometric isomorphisms themselves onto $\widetilde{\mathcal{X}}'$ and $\widetilde{\mathcal{Y}}'$, and hence uniformly continuous into $\widehat{\mathcal{X}}'$ and $\widehat{\mathcal{Y}}'$). Let $\widehat{J}: \widehat{\mathcal{X}} \to \widehat{\mathcal{X}}'$ and $\widehat{K}: \widehat{\mathcal{Y}} \to \widehat{\mathcal{Y}}'$ be the extensions (by continuity) of them over $\widehat{\mathcal{X}} = \widetilde{\mathcal{X}}^{-}$ and $\widehat{\mathcal{Y}} = \widetilde{\mathcal{Y}}^{-}$ (as in the proof of item (a)), and so $\widehat{J}|_{\widetilde{\mathcal{X}}} = \widetilde{J}: \widetilde{\mathcal{X}} \to \widetilde{\mathcal{X}}'$ and $\widehat{K}|_{\widetilde{\mathcal{Y}}} = \widetilde{K}: \widetilde{\mathcal{Y}} \to \widetilde{\mathcal{Y}}'$. Similarly, $\widehat{J}^{-1}|_{\widetilde{\mathcal{X}}'} = \widetilde{J}^{-1}: \widetilde{\mathcal{X}}' \to \widetilde{\mathcal{X}}$ and $\widehat{K}^{-1}|_{\widetilde{\mathcal{Y}}'} = \widetilde{K}^{-1}: \widetilde{\mathcal{Y}}' \to \widetilde{\mathcal{Y}}$. Since $\widehat{T}|_{\widetilde{\mathcal{X}}} = \widetilde{T} = KTJ^{-1}: \widetilde{\mathcal{X}} \to \widetilde{\mathcal{Y}}$,

$$\begin{aligned} (\widehat{K}\,\widehat{T}\,\widehat{J}^{-1})|_{\widetilde{\mathcal{X}}'} &= \widehat{K}|_{\widetilde{\mathcal{Y}}}\,\widehat{T}|_{\widetilde{\mathcal{X}}}\,\widehat{J}^{-1}|_{\widetilde{\mathcal{X}}'} = \widetilde{K}\,\widehat{T}|_{\widetilde{\mathcal{X}}}\,\widetilde{J}^{-1} = K'K^{-1}\,\widehat{T}|_{\widetilde{\mathcal{X}}}\,JJ'^{-1} \\ &= K'K^{-1}KTJ^{-1}JJ'^{-1} = K'TJ'^{-1} = \widehat{T}'|_{\widetilde{\mathcal{X}}'}. \end{aligned}$$

Thus the bounded linear transformations $\widehat{K}\widehat{T}\widehat{J}^{-1}: \widehat{\mathcal{X}}' \to \widehat{\mathcal{Y}}'$ (composition of bounded linear transformations) and $\widehat{T}': \widehat{\mathcal{X}}' \to \widehat{\mathcal{Y}}'$ coincide when acting between dense linear manifolds $\widetilde{\mathcal{X}}'$ and $\widetilde{\mathcal{Y}}'$ of the complete normed spaces $\widehat{\mathcal{X}}'$ and $\widehat{\mathcal{Y}}'$, and therefore $\widehat{T}' = \widehat{K}\widehat{T}\widehat{J}^{-1}$ (uniqueness of extension by continuity as seen in the proof of item (a)). Conclusion: if $\widehat{T}': \widehat{\mathcal{X}}' \to \widehat{\mathcal{Y}}'$ is another uniformly continuous extension of T between some completions $\widehat{\mathcal{X}}'$ and $\widehat{\mathcal{Y}}'$ of $\mathcal{X}$ and $\mathcal{Y}$, then there exist isometric isomorphisms $\widehat{J}: \widehat{\mathcal{X}} \to \widehat{\mathcal{X}}'$ and $\widehat{K}: \widehat{\mathcal{Y}} \to \widehat{\mathcal{Y}}'$ such that $\widehat{T}' = \widehat{K}\widehat{T}\widehat{J}^{-1}$. In other words, a continuous linear extension of T over a completion of $\mathcal{X}$ is unique up to isometric isomorphisms.

(d) *Isometric isomorphisms.* Since $T: \mathcal{X} \to \mathcal{Y}$ and $\widehat{T}: \widehat{\mathcal{X}} \to \widehat{\mathcal{Y}}$ are linear, they are isometric isomorphisms if and only if they are surjective isometries. Suppose T is a surjective isometry, which means T is surjective and $\|Tx\| = \|x\|$ for every $x \in \mathcal{X}$. Since $J: \mathcal{X} \to \widetilde{\mathcal{X}}$, $J^{-1}: \widetilde{\mathcal{X}} \to \mathcal{X}$ and $K: \mathcal{Y} \to \widetilde{\mathcal{Y}}$ are surjective isometries,

so is $KTJ^{-1}\colon \widetilde{\mathcal{X}} \to \widetilde{\mathcal{Y}}$. (Indeed, KTJ^{-1} is a surjective since composition of surjective maps is surjective, and $\|KTJ^{-1}\widetilde{x}\| = \|TJ^{-1}\widetilde{x}\| = \|Tx\| = \|x\| = \|J^{-1}\widetilde{x}\| = \|\widetilde{x}\|$ for every $\widetilde{x} \in \widetilde{\mathcal{X}}$.) Thus $\widetilde{T} = KTJ^{-1}\colon \widetilde{\mathcal{X}} \subseteq \widehat{\mathcal{X}} \to \widetilde{\mathcal{Y}} \subseteq \widehat{\mathcal{Y}}$ extends by continuity to a surjective isometry $\widehat{T}'\colon \widehat{\mathcal{X}} \to \widehat{\mathcal{Y}}$, so that $\widehat{T}'_{\widetilde{\mathcal{X}}} = \widetilde{T}$ (which is unique for these $\widetilde{\mathcal{X}}$ and $\widetilde{\mathcal{Y}}$ — see, e.g., [52, Corollary 3.47]). Thus $\widehat{T}'$ and $\widehat{T}$ are isometrically isomorphic by (c), and hence $\widehat{T}$ is a surjective isometry.

(e) *Isometrically isomorphic*. Recall: $\mathcal{B}[\widehat{\mathcal{X}}, \widehat{\mathcal{Y}}]$ is complete because $\widehat{\mathcal{Y}}$ is, and so if $\mathcal{B}[\mathcal{X}, \mathcal{Y}]$ is isometrically isomorphic to $\mathcal{B}[\widehat{\mathcal{X}}, \widehat{\mathcal{Y}}]$, then $\mathcal{B}[\mathcal{X}, \mathcal{Y}]$ must be complete (see, e.g., [52, Theorem 3.44]), and this means $\mathcal{Y}$ is complete. Thus suppose $\mathcal{Y}$ is complete and take it as its own natural completion so that $\mathcal{Y} = \widetilde{\mathcal{Y}} = \widehat{\mathcal{Y}}$. Let K be the identity in the above diagram. Take a fixed isometric isomorphism J, and an arbitrary T in $\mathcal{B}[\mathcal{X}, \mathcal{Y}]$. Thus by items (a), (b) and (c)

$$\widehat{T}|_{\widetilde{\mathcal{X}}} = \widetilde{T} = TJ^{-1}\colon J(\mathcal{X}) = \widetilde{\mathcal{X}} \subseteq \widehat{\mathcal{X}} \to \mathcal{Y} \quad \text{with} \quad \|\widehat{T}\| = \|\widetilde{T}\| = \|T\|.$$

Here $\widetilde{\mathcal{X}}$ is the range of J and $\widetilde{T}$ is the composition $TJ^{-1} \in \mathcal{B}[\widetilde{\mathcal{X}}, \mathcal{Y}]$. Since $\widetilde{\mathcal{X}}$ is dense in $\widehat{\mathcal{X}}$ and $\mathcal{Y}$ is a Banach space, $\widehat{T} \in \mathcal{B}[\widehat{\mathcal{X}}, \mathcal{Y}]$ is the *unique* extension by continuity of $\widetilde{T} \in \mathcal{B}[\widetilde{\mathcal{X}}, \mathcal{Y}]$ over $\widehat{\mathcal{X}}$ (see, e.g., [52, Theorem 4.35]). Thus (for a fixed J) consider the extension-over-completion map $T \mapsto \widehat{T}$ taking each T in $\mathcal{B}[\mathcal{X}, \mathcal{Y}]$ to its extension over completion $\widehat{T}$ in $\mathcal{B}[\widehat{\mathcal{X}}, \mathcal{Y}]$. This is an isometric isomorphism between $\mathcal{B}[\mathcal{X}, \mathcal{Y}]$ and $\mathcal{B}[\widehat{\mathcal{X}}, \mathcal{Y}]$. We split the proof into four parts.

(1) First we verify surjectivity. Take any $S' \in \mathcal{B}[\widehat{\mathcal{X}}, \mathcal{Y}]$, so that $S'|_{\widetilde{\mathcal{X}}} \in \mathcal{B}[\widetilde{\mathcal{X}}, \mathcal{Y}]$. Set $S = S'|_{\widetilde{\mathcal{X}}} J \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ and let $\widehat{S} \in \mathcal{B}[\widehat{\mathcal{X}}, \mathcal{Y}]$ be the extension over completion of S. Hence $\widehat{S}|_{\widetilde{\mathcal{X}}} = SJ^{-1} = S'|_{\widetilde{\mathcal{X}}} JJ^{-1} = S'|_{\widetilde{\mathcal{X}}}$. Thus take their *unique* extension by continuity over $\widehat{\mathcal{X}}$ and get $S' = \widehat{S}$. So the map $T \mapsto \widehat{T}$ is surjective.

(2) Next we show that it is linear. Indeed, for arbitrary $T, S \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$,

$$\widehat{(T+S)}|_{\widetilde{\mathcal{X}}} = (T+S)J^{-1} = TJ^{-1} + SJ^{-1} = \widetilde{T} + \widetilde{S} = \widehat{T}|_{\widetilde{\mathcal{X}}} + \widehat{S}|_{\widetilde{\mathcal{X}}} = (\widehat{T} + \widehat{S})|_{\widetilde{\mathcal{X}}}.$$

Again, take their *unique* extension by continuity over $\widehat{\mathcal{X}}$ and get $\widehat{T+S} = \widehat{T} + \widehat{S}$. In other words, since these are Banach-space-valued continuous linear transformations defined on a dense linear manifold $\widetilde{\mathcal{X}}$ of $\widehat{\mathcal{X}}$, they all have a *unique* bounded linear extension by continuity over $\widehat{\mathcal{X}}$ (as we saw above) so that $\widehat{T+S} = \widehat{T} + \widehat{S}$. Similarly, $\widehat{\alpha T} = \alpha \widehat{T}$ for every $T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ and $\alpha \in \mathbb{F}$.

(3) The transformation $T \mapsto \widehat{T}$ is norm-preserving as well (i.e., $\|\widehat{T}\| = \|T\|$).

(4) Since $T \mapsto \widehat{T}$ is linear and norm-preserving, it is injective.

This establishes a norm-preserving one-to-one linear correspondence

$$T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}] \iff \widehat{T} \in \mathcal{B}[\widehat{\mathcal{X}}, \mathcal{Y}].$$

So $\mathcal{B}[\mathcal{X}, \mathcal{Y}]$ and $\mathcal{B}[\widehat{\mathcal{X}}, \mathcal{Y}]$ are isometrically isomorphic: $\mathcal{B}[\mathcal{X}, \mathcal{Y}] \cong \mathcal{B}[\widehat{\mathcal{X}}, \mathcal{Y}]$.

(f) *Duals and adjoints.* As $\mathcal{X}^*$ is complete, it is its own completion: $\mathcal{X}^* = \widehat{\mathcal{X}^*}$. Also, $\mathcal{X}^* = \mathcal{B}[\mathcal{X}, \mathbb{F}] \cong \mathcal{B}[\widehat{\mathcal{X}}, \mathbb{F}] = \widehat{\mathcal{X}}^*$ by (e). Thus T^* is its own extension over completion: $\widehat{T^*} = T^* \in \mathcal{B}[\mathcal{Y}^*, \mathcal{X}^*] = \mathcal{B}[\widehat{\mathcal{Y}^*}, \widehat{\mathcal{X}^*}]$. Now take $\widehat{T} \in \mathcal{B}[\widehat{\mathcal{X}}, \widehat{\mathcal{Y}}]$ so that $(\widehat{T}^* g)(\widehat{x}) = g(\widehat{T}\widehat{x})$ for every $g \cong \widehat{g} \in \widehat{\mathcal{Y}}^* \cong \mathcal{Y}^*$ and $\widehat{x} \in \widehat{\mathcal{X}}$. Therefore for every $x \in \mathcal{X} \cong \widetilde{\mathcal{X}}$ and its correspondent $\widehat{x} \in \widehat{\mathcal{X}}$ with $x \cong \widetilde{x} \in \widetilde{\mathcal{X}} \subseteq \widehat{\mathcal{X}}$, it follows that $(\widehat{T}^* g)(\widehat{x}) = g(\widehat{T}\widehat{x}) = g(\widetilde{T}\widetilde{x}) = g(Tx) = (T^* g)(x) \in \mathbb{F}$, which implies $\widehat{T}^* \cong T^*$.

(g) *Included completion.* Let $\mathcal{M}$ be a linear manifold of a normed space $\mathcal{X}$ (so that $\mathcal{M}$ inherits the norm of $\mathcal{X}$). If $\widehat{\mathcal{X}}$ is a completion of $\mathcal{X}$, then there exists an isometric isomorphism $J\colon \mathcal{X} \to \widetilde{\mathcal{X}} = J(\mathcal{X}) \subseteq \widehat{\mathcal{X}}$, where $\widetilde{\mathcal{X}}$ is a dense linear manifold of $\widehat{\mathcal{X}}$. Since restrictions of linear isometries are linear isometries, set

$$J' = J|_{\mathcal{M}} : \mathcal{M} \to \widetilde{\mathcal{M}} = J'(\mathcal{M}) = J(\mathcal{M}) \subseteq J(\mathcal{X}) = \widetilde{\mathcal{X}} \subseteq \widehat{\mathcal{X}},$$

an isometric isomorphism of $\mathcal{M}$ onto $\widetilde{\mathcal{M}} = J(\mathcal{M})$, which is a linear manifold of $\widehat{\mathcal{X}}$. Thus the closure of $\widetilde{\mathcal{M}}$ in $\widehat{\mathcal{X}}$, say $\widehat{\mathcal{M}} = \widetilde{\mathcal{M}}^- \subseteq \widehat{\mathcal{X}}$, is a subspace of $\widehat{\mathcal{X}}$ which is a completion of $\mathcal{M}$ included in a completion of $\mathcal{X}$: $\mathcal{M} \subseteq \mathcal{X} \Rightarrow \widehat{\mathcal{M}} \subseteq \widehat{\mathcal{X}}$.

(h) *Ranges.* With $\widetilde{T} = \widehat{T}|_{\widetilde{\mathcal{X}}} = KTJ^{-1}\colon \widetilde{\mathcal{X}} \subseteq \widehat{\mathcal{X}} \to \widetilde{\mathcal{Y}} \subseteq \widehat{\mathcal{Y}}$, we show that

$$\mathcal{R}(T) \cong \mathcal{R}(\widetilde{T}) \subseteq \mathcal{R}(\widehat{T}) \subseteq \mathcal{R}(\widehat{T})^- = \mathcal{R}(\widetilde{T})^- \cong \mathcal{R}(T)^-.$$

The first isometric equivalence comes from Section 5.1, the first inclusion is trivial (i.e., $\mathcal{R}(\widehat{T}|_{\widetilde{\mathcal{X}}}) \subseteq \mathcal{R}(\widehat{T})$) and so is the second. For the identity, note that if $\widehat{y} \in \mathcal{R}(\widehat{T})$, then $\widehat{y} = \widehat{T}\widehat{x} = \widehat{T}(\lim_n \widetilde{x}_n) = \lim_n \widehat{T}(\widetilde{x}_n) = \lim_n \widetilde{T}(\widetilde{x}_n) \in \mathcal{R}(\widetilde{T})^-$ (since $\widetilde{\mathcal{X}}$ is dense in $\widehat{\mathcal{X}}$ and $\widehat{T}$ is continuous), and so $\mathcal{R}(\widehat{T}) \subseteq \mathcal{R}(\widetilde{T})^-$, which implies $\mathcal{R}(\widehat{T})^- \subseteq \mathcal{R}(\widetilde{T})^-$. The converse is trivial since $\mathcal{R}(\widetilde{T}) \subseteq \mathcal{R}(\widehat{T})$ — by the first inclusion. The final isomorphic equivalence is a consequence of the first.

(i) *Kernels.* $\mathcal{N}(\widetilde{T}) \subseteq \mathcal{N}(\widehat{T})$ trivially (i.e., $\mathcal{N}(\widehat{T}|_{\widetilde{\mathcal{X}}}) \subseteq \mathcal{N}(\widehat{T})$) and $\mathcal{N}(\widetilde{T}) \cong \mathcal{N}(T)$ by isometric equivalence of Section 5.1. So $\mathcal{N}(\widehat{T}) = \{0\}$ implies $\mathcal{N}(T) = \{0\}$. Conversely, suppose $\mathcal{N}(T) = \{0\}$ and $\mathcal{R}(T)$ is closed. Equivalently, suppose $\mathcal{N}(\widetilde{T}) = \{0\}$ and $\mathcal{R}(\widetilde{T})$ is closed. Then $\widetilde{T}\colon \widetilde{\mathcal{X}} \to \mathcal{R}(\widetilde{T})$ has a bounded inverse on its range, and hence it is a uniform homeomorphism, and so is its extension by continuity, $\widehat{T}\colon \widehat{\mathcal{X}} \to \mathcal{R}(\widehat{T}) \subseteq \widehat{\mathcal{Y}}$ (see, e.g., [52, Corollaries 4.24 and 4.36]). Thus $\widehat{T}$ is injective, and so $\mathcal{N}(\widehat{T}) = \{0\}$. Moreover, since $\mathcal{R}(T)^- \cong \mathcal{R}(\widetilde{T})^- = \mathcal{R}(\widehat{T})^-$ from (h), $\dim \mathcal{R}(T) = \dim \mathcal{R}(\widetilde{T}) = \dim \mathcal{R}(\widehat{T})$. Also, $\dim \widetilde{\mathcal{X}} = \dim \widehat{\mathcal{X}}$, because $\widetilde{\mathcal{X}}^- = \widehat{\mathcal{X}}$. Therefore, by the rank and nullity identity (Proposition 1.E), we get $\dim \mathcal{N}(\widetilde{T}) + \dim \mathcal{R}(T) = \dim \mathcal{R}(T) + \dim \mathcal{N}(\widehat{T})$. Then $\dim \mathcal{R}(T) < \infty$ implies $\dim \mathcal{N}(\widetilde{T}) = \dim \mathcal{N}(\widehat{T})$, which implies $\mathcal{N}(\widetilde{T})^- = \mathcal{N}(\widehat{T})$ since $\mathcal{N}(\widetilde{T}) \subseteq \mathcal{N}(\widehat{T})$ (the algebraic complement of $\mathcal{N}(\widetilde{T})^-$ in $\mathcal{N}(\widehat{T})$ is $\{0\}$ — Proposition 1.P). □

Recall that $\widehat{S+T} = \widehat{S} + \widehat{T}$ (cf. proof of Theorem 5.1(e)). In a similar way, $\widehat{ST} = \widehat{S}\widehat{T}$. (In fact, $\widehat{\mathcal{X}} \xrightarrow{\widehat{T}} \widehat{\mathcal{Y}} \xrightarrow{\widehat{S}} \widehat{\mathcal{Z}}$ whenever $\mathcal{X} \xrightarrow{T} \mathcal{Y} \xrightarrow{S} \mathcal{Z}$, and $\widehat{\mathcal{X}} \xrightarrow{\widehat{ST}} \widehat{\mathcal{Z}}$ with $\widehat{ST}|_{\widetilde{\mathcal{X}}} = (\widehat{S}\widehat{T})|_{\widetilde{\mathcal{X}}}$ is the unique linear continuous extension of ST.)

Remark 5.2. *Extensions of Linear Transformations.* In the above theorem $\widehat{T} \in \mathcal{B}[\widehat{\mathcal{X}}, \widehat{\mathcal{Y}}]$ was referred to as the extension over completion of $T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$. It attends to our immediate need for extension results for bounded linear transformations. It is, however, also necessary to consider other well-known extension procedures, still related to Theorem 5.1, which will often be required in the sequel. Convention: all linear spaces are over the same field $\mathbb{F}$.

(a) Linear extension. Theorem 1.1 in Chapter 1 states a purely algebraic result on extension of linear transformations between linear spaces:

> *If $\mathcal{X}$ and $\mathcal{Y}$ are linear spaces, if $\mathcal{M}$ is a linear manifold of $\mathcal{X}$, and if $L\colon \mathcal{M} \to \mathcal{Y}$ is a linear transformation, then there exists a linear extension $\widehat{L}\colon \mathcal{X} \to \mathcal{Y}$ of L defined on the whole space $\mathcal{X}$.*

Recall that the extension is not unique, as observed in Remark 1.2. (A bilinear counterpart of Theorem 1.1 was stated by Theorem 2.5 with alternate proofs given in Remarks 2.7(a) and 3.8(a).)

(b) Extension by continuity. On the other hand, extension by continuity states that *a uniformly continuous function from a dense subset of a metric space to a complete metric space has a unique continuous extension over the whole metric space, which is uniformly continuous*. Uniform continuity cannot be dismissed here: plain continuity is not enough in general (see, e.g., [52, Theorem 3.45]). For linear transformations, however, plain continuity coincides with uniform continuity. Thus, in the present normed-space context, the *extension by continuity* of bounded linear transformations reads as follows.

> *If $\mathcal{M}$ is a dense linear manifold of a normed space $\mathcal{X}$, and if $\mathcal{Y}$ is a Banach space, then every transformation $T \in \mathcal{B}[\mathcal{M}, \mathcal{Y}]$ has a unique extension $\widehat{T} \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$, and $\|\widehat{T}\| = \|T\|$.*

(See, e.g., [52, Theorem 4.35].) This played an important role in the proof of Theorem 5.1 on extension over completion for linear transformations. (Note: there is no uniform continuity for bilinear maps, and so extension by continuity does not apply to continuous bilinear maps, as we will see in Chapter 6.)

(c) Extension over completion. Therefore if a bounded linear transformation $T\colon \mathcal{X} \to \mathcal{Y}$ is defined on a linear manifold $\mathcal{X}$ of a normed space $\mathcal{Z}$ into a Banach space $\mathcal{Y}$, then it has a natural extension by continuity over the closure $\mathcal{X}^-$ of $\mathcal{X}$ in $\mathcal{Z}$. What if there is no $\mathcal{Z}$ including $\mathcal{X}$ to allow taking the closure of $\mathcal{X}$ in $\mathcal{Z}$ and, consequently, to allow extension by continuity? In this case, extension over completion is the answer, where the notion of extension by continuity applies to the isometrically isomorphic image $\widetilde{\mathcal{X}}$ of $\mathcal{X}$, which is dense in the completion $\widehat{\mathcal{X}}$ of $\mathcal{X}$. As in Theorem 5.1, this is *extension over completion*:

> *Every transformation $T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ has a unique up to isomorphism extension $\widehat{T} \in \mathcal{B}[\widehat{\mathcal{X}}, \widehat{\mathcal{Y}}]$ between the completions $\widehat{\mathcal{X}}$ and $\widehat{\mathcal{Y}}$ of the normed spaces $\mathcal{X}$ and $\mathcal{Y}$, and $\|\widehat{T}\| = \|T\|$.*

(Note: extension over completion holds for bilinear maps, but with a proof totally different from that in Theorem 5.1 — this will be discussed in Chapter 6.)

(d) Hahn–Banach extension. If a linear manifold $\mathcal{M}$ is not dense in the normed space $\mathcal{X}$, and if one wants to extend a bounded a linear transformation defined on $\mathcal{M}$ to a bounded linear transformation defined on the whole space $\mathcal{X}$, then the celebrated HAHN–BANACH THEOREM makes a proper start.

> *If $\mathcal{M}$ is a linear manifold of a normed space $\mathcal{X}$ and if $f\colon \mathcal{M} \to \mathbb{F}$ is a bounded linear functional, then there exists a bounded linear extension $\widehat{f}\colon \mathcal{X} \to \mathbb{F}$ of f over $\mathcal{X}$, and $\|\widehat{f}\| = \|f\|$.*

(See, e.g., [52, Theorem 4.62]). The extension $\widehat{f} \in \mathcal{X}^*$ of $f \in \mathcal{M}^*$ is called the *Hahn–Banach extension*. It is worth noticing that the formula (cf. Section 5.1)

$$\|x\| = \sup_{0 \neq f \in \mathcal{X}^*} \frac{|f(x)|}{\|f\|} \quad \text{for every} \quad x \in \mathcal{X}$$

is a consequence of the Hahn–Banach Theorem. In particular, if the linear manifold $\mathcal{M}$ is dense in $\mathcal{X}$, then extension by continuity as in (b) (for $\mathcal{Y} = \mathbb{F}$) ensures uniqueness for the Hahn–Banach extension (with norm preservation).

(e) Nonuniqueness of the Hahn–Banach extension. In general, however, the Hahn–Banach extension is not unique. For instance, set $\mathcal{X} = \mathbb{R}^2 = \mathbb{R} \oplus \mathbb{R}$, set $\mathcal{M} = \mathbb{R} \oplus \{0\}$, and take $f\colon \mathcal{M} \to \mathbb{R}$ given by $f(a,0) = a$ for every $a \in \mathbb{R}$, which is linear and bounded with $\|f\| = 1$ for any norm equipping $\mathbb{R}^2$. For each $\beta \in \mathbb{R}$, take $\widehat{f}_\beta\colon \mathcal{X} \to \mathbb{R}$ given by $\widehat{f}_\beta(a,b) = a + \beta b$ for every $(a,b) \in \mathbb{R}^2$, which is linear and extends f (i.e., $\widehat{f}_\beta|_{\mathcal{M}} = f$). In fact, all linear functionals on $\mathcal{X}$ that extend $f\colon \mathcal{M} \to \mathbb{R}$ are of this form. First equip $\mathbb{R}^2$ with the norm $\|\cdot\|_1$. Since

$$\|\widehat{f}_\beta\| = \sup_{\|(a,b)\|_1 = 1} |\widehat{f}_\beta(a,b)| = \sup_{|a|+|b|=1} |a + \beta b| \leq \sup_{|a|+|b|=1} |a| + |\beta|\,|b| \leq 1$$

if $|\beta| \leq 1$, and since $|\widehat{f}_\beta(1,0)| = 1$, then $\|\widehat{f}_\beta\| = 1$. Thus $\{\widehat{f}_\beta\colon |\beta| \leq 1\}$ is an uncountably infinite set of distinct Hahn–Banach extensions of f.

(f) But uniqueness may happen. If $\mathbb{R}^2$ is equipped with the Euclidean norm $\|\cdot\|_2$ (so it is a Hilbert space), then the bounded linear functional $f\colon \mathcal{M} \to \mathbb{R}$ defined above has only one Hahn–Banach extension, the trivial one: $\widehat{f}_0\colon \mathcal{X} \to \mathbb{R}$ given by $\widehat{f}_0(a,b) = a$ for every $(a,b) \in \mathbb{R}^2$. In fact, as we saw above, $\|f\| = 1$. In this case, all other bounded linear extensions $\widehat{f}_\beta\colon \mathcal{X} \to \mathbb{R}$ of f have $\|\widehat{f}_\beta\| > 1$ for $\beta \neq 0$. This can be verified as follows. For $\mathbb{R}^2$ equipped with $\|\cdot\|_2$ we get

$$\|\widehat{f}_\beta\|^2 = \sup_{\|(a,b)\|_2^2 = 1} |\widehat{f}_\beta(a,b)|^2 = \sup_{a^2+b^2=1} (a + \beta b)^2.$$

For $|\beta| > 1$ set $a = 0$ and $b = 1$ so that $(a + \beta b)^2 = |\beta|^2 > 1$. For $|\beta| = 1$ take $a^2 + b^2 = 1$ with $ab\beta > 0$ so that $(a + \beta b)^2 = a^2 + b^2 + 2ab\beta = 1 + 2ab\beta > 1$. For $0 < |\beta| < 1$ take $a^2 + b^2 = 1$ with $b > 0$, $a\beta > 0$ and $\frac{\gamma^2}{1+\gamma^2} < a^2 < 1$ where $\gamma = \frac{1-\beta^2}{2|\beta|} > 1$. This implies $(a + \beta b)^2 > 1$. Indeed,

$$a^2 > \tfrac{\gamma^2}{1+\gamma^2} \iff a^2(1+\gamma^2) > \gamma^2 \implies a^2 > \gamma^2 - a^2\gamma^2 = \gamma^2(1-a^2) = \gamma^2 b^2$$
$$\implies |a| > \gamma b \quad = \tfrac{1-\beta^2}{2|\beta|} b \implies 2|a\beta| > b(1-\beta^2) \iff 2a\beta > b(1-\beta^2) \implies$$
$$(a+\beta b)^2 = a^2 + \beta^2 b^2 + 2ab\beta = a^2 + b^2 - b^2 + \beta^2 b^2 + 2ab\beta$$
$$= 1 - b^2(1-\beta^2) + 2ab\beta = 1 + b\big[2a\beta - b(1-\beta^2)\big] > 1.$$

(g) Generalizations of the Hahn–Banach Theorem. If we could generalize the Hahn–Banach Theorem by replacing the one-dimensional Banach space $\mathbb{F}$ (the codomain of f) by an arbitrary Banach space $\mathcal{Y}$ of arbitrary dimension, then we would get a full Banach-space version of the purely algebraic result in item (a), now dealing with a bounded linear extension $\widehat{T}\colon \mathcal{X} \to \mathcal{Y}$ of a bounded linear transformation $T\colon \mathcal{M} \to \mathcal{Y}$. But this fails in general. Such an extension $\widehat{T}\colon \mathcal{X} \to \mathcal{Y}$ may not exist without extra assumptions (see, e.g., [11]). Extra assumptions may be placed on each of the four players involved, viz., the normed space $\mathcal{X}$, the linear manifold $\mathcal{M}$, the bounded linear transformation T, and the Banach space $\mathcal{Y}$ — for instance, it holds in (d) for $\dim \mathcal{Y} = 1$. (Note: there is no Hahn–Banach Theorem for bilinear forms, as will be seen in Chapter 6.)

(h) A classical approach. According to Nachbin [71], a Banach space $\mathcal{Y}$ has the *extension property* if for every Banach space $\mathcal{X}$ and every subspace $\mathcal{M}$ of $\mathcal{X}$, every bounded linear transformation $T\colon \mathcal{M} \to \mathcal{Y}$ has a bounded linear extension $\widehat{T}\colon \mathcal{X} \to \mathcal{Y}$ over $\mathcal{X}$ with the same norm. By the Hahn–Banach Theorem, a one-dimensional Banach space has the extension property, but this is not a property of every Banach space (see also [11]): *there is no Hahn–Banach Theorem for an arbitrary bounded linear transformation into an arbitrary Banach space*.

(i) The complementation approach. If we look at the normed space $\mathcal{X}$ instead, or at the lattice of all its linear manifolds $\mathcal{M}$, then a Banach-space version of the algebraic result in item (a) for bounded linear transformations can be obtained by restricting $\mathcal{M}$ to be complemented in $\mathcal{X}$. This will be considered in Theorem 5.4 below, and it shows again that *there is no Hahn–Banach Theorem for arbitrary bounded linear transformations between arbitrary Banach spaces*.

Complementation will play an important role from now on.

Remark 5.3. *Complementary Subspaces and Continuous Projection.*

(a) Recall: (1) every linear manifold of any linear space has an algebraic complement; (2) a subspace of a normed space is *complemented* if it has a subspace as an algebraic complement — in this case it is called a *complemented subspace*; (3) if two subspaces are complements of each other, then they are called *complementary subspaces*; (4) not all subspaces of an arbitrary Banach space are complemented; (5) a normed space is *complemented* if every subspace is complemented; and (6) if a Banach space is complemented, then it is isomorphic (topologically isomorphic, by the Inverse Mapping Theorem) to a Hilbert space — thus complemented Banach spaces are identified with Hilbert spaces: *only Hilbert spaces* (up to isomorphism) *are complemented* [67] (see also [45]).

(b) If $\mathcal{X}$ is a normed space and $E\colon \mathcal{X}\to\mathcal{X}$ is a continuous projection, then $\mathcal{R}(E)$ and $\mathcal{N}(E)$ are complementary subspaces of $\mathcal{X}$. This is immediate since $\mathcal{R}(E)$ and $\mathcal{N}(E)$ are complementary linear manifolds and $\mathcal{R}(E)=\mathcal{N}(I-E)$. Conversely, if $\mathcal{M}$ and $\mathcal{N}$ are complementary subspaces of a Banach space $\mathcal{X}$, then the (unique) projection $E\colon \mathcal{X}\to\mathcal{X}$ with $\mathcal{R}(E)=\mathcal{M}$ and $\mathcal{N}(E)=\mathcal{N}$ is continuous (since on a Banach space a projection with closed range and closed kernel is continuous). (See, e.g., [12, Section III.13], [70, Section 3.2], [52, Problem 435], [56, Section A.2].) Thus *if $\mathcal{X}$ is a Banach space, then the assertions below are equivalent.*

(i) *A subspace $\mathcal{M}$ of $\mathcal{X}$ is complemented.*

(ii) *There exists a continuous projection $E\colon \mathcal{X}\to\mathcal{X}$ with $\mathcal{R}(E)=\mathcal{M}$.*

Theorem 5.4. *Let $\mathcal{M}$ be a linear manifold of a Banach space $\mathcal{X}$.*

(a) *Every bounded linear transformation $T\colon \mathcal{M}\to\mathcal{Y}$ into an arbitrary Banach space $\mathcal{Y}$ has a bounded linear extension $\widehat{T}\colon \mathcal{X}\to\mathcal{Y}$ over $\mathcal{X}$ if and only if the closure $\mathcal{M}^-$ of $\mathcal{M}$ is complemented in $\mathcal{X}$.*

(b) *If $\mathcal{M}^-=\mathcal{R}(E)$ for a projection $E\in\mathcal{B}[\mathcal{X}]$ with $\|E\|=1$, then $\|\widehat{T}\|=\|T\|$.*

Proof. (a) Let $\mathcal{M}$ be a linear manifold of a Banach space $\mathcal{X}$ so that the (closed) subspace $\mathcal{M}^-$ of $\mathcal{X}$ is itself a Banach space. Take an arbitrary Banach space $\mathcal{Y}$ and an arbitrary $T\in\mathcal{B}[\mathcal{M},\mathcal{Y}]$. The following assertions are equivalent.

(a_1) For every $\mathcal{Y}$ and every $T\in\mathcal{B}[\mathcal{M},\mathcal{Y}]$ there is $\widehat{T}\in\mathcal{B}[\mathcal{X},\mathcal{Y}]$ such that $\widehat{T}|_{\mathcal{M}}=T$.

(a_2) $\mathcal{M}^-$ is complemented in $\mathcal{X}$.

(a_1)$\Rightarrow$(a_2). Consider an arbitrary linear manifold $\mathcal{M}$ of the Banach space $\mathcal{X}$ and take the Banach space $\mathcal{M}^-$. If (a_1) holds, then it holds in particular for $T\colon \mathcal{M}\to\mathcal{M}\subseteq\mathcal{M}^-$ defined by $Tu=u$ for every $u\in\mathcal{M}$. Since T acts as the identity on $\mathcal{M}$, then it is trivially linear and continuous (i.e., $T\in\mathcal{B}[\mathcal{M},\mathcal{M}^-]$). If ($a_1$) holds, then there exists a linear and continuous extension $\widehat{T}\colon \mathcal{X}\to\mathcal{M}^-$ of T over $\mathcal{X}$ (i.e., there exists $\widehat{T}\in\mathcal{B}[\mathcal{X},\mathcal{M}^-]$ such that $\widehat{T}|_{\mathcal{M}}=T$). Since $\widehat{T}|_{\mathcal{M}}$ acts as the identity on $\mathcal{M}$ and $\widehat{T}|_{\mathcal{M}}$ is linear and continuous, then $\widehat{T}|_{\mathcal{M}}$ extends by continuity (cf. Remark 5.2(b)) over $\mathcal{M}^-$ so that $\widehat{T}|_{\mathcal{M}^-}\in\mathcal{B}[\mathcal{M}^-,\mathcal{M}^-]$ is the identity operator on $\mathcal{M}^-$ and $\mathcal{R}(\widehat{T})=\mathcal{M}^-$ (as $\mathcal{M}^-=\mathcal{R}(\widehat{T}|_{\mathcal{M}^-})\subseteq\mathcal{R}(\widehat{T})\subseteq\mathcal{M}^-$). Then the bounded linear $\widehat{T}\colon \mathcal{X}\to\mathcal{M}^-\subseteq\mathcal{X}$ acts as the identity on its range, and so it is idempotent. Thus $\widehat{T}\in\mathcal{B}[\mathcal{X},\mathcal{X}]$ is a continuous projection with range $\mathcal{M}^-$. But the range of a continuous projection is a complemented subspace (cf. Remark 5.3). So $\mathcal{M}^-$ is complemented.

(a_2)$\Rightarrow$(a_1). Conversely, suppose $\mathcal{M}^-$ is complemented. Thus $\mathcal{M}^-$ is the range of a continuous projection (cf. Remark 5.3). Then let $E\in\mathcal{B}[\mathcal{X}]$ be a projection such that $\mathcal{R}(E)=\mathcal{M}^-$. Let $\mathcal{Y}$ be an arbitrary Banach space and let T be an arbitrary transformation in $\mathcal{B}[\mathcal{M},\mathcal{Y}]$. Extend T by continuity (as in Remark 5.2(b)) over $\mathcal{M}^-$ to get $\widetilde{T}\in\mathcal{B}[\mathcal{M}^-,\mathcal{Y}]$ for which $\widetilde{T}|_{\mathcal{M}}=T$. Since E acts as the

identity on $\mathcal{R}(E) = \mathcal{M}^-$, define $\widehat{T} = \widetilde{T}E \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ so that $\widehat{T}|_{\mathcal{M}} = \widetilde{T}E|_{\mathcal{M}} = \widetilde{T}|_{\mathcal{M}} = T$. Thus $\widehat{T}$ is a bounded linear extension of T over $\mathcal{X}$.

(b) Moreover, under the assumption in (a_2), $T = \widehat{T}|_{\mathcal{M}}$ and $\widehat{T} = \widetilde{T}E$. Hence $\|T\| = \|\widehat{T}|_{\mathcal{M}}\| \le \|\widehat{T}\| = \|\widetilde{T}E\| \le \|\widetilde{T}\|\|E\|$. Also $\|\widetilde{T}\| = \|T\|$ since $\widetilde{T}$ extends T by continuity. So $\|T\| \le \|\widehat{T}\| \le \|T\|\|E\|$. Thus $\|E\| = 1$ implies $\|\widehat{T}\| = \|T\|$. □

Remark 5.5. *Examples Related to Theorem 5.4.*

(a) Extensions of the identity. Set $\mathcal{Y} = \mathcal{X}$. Take $T \in \mathcal{B}[\mathcal{M}, \mathcal{Y}]$ such that T acts as the identity on $\mathcal{M}$. That is, T naturally embeds $\mathcal{M} \subseteq \mathcal{X}$ into $\mathcal{X} = \mathcal{Y}$, so that $Tu = u$ for every $u \in \mathcal{M}$ and hence $\|T\| = 1$. *If $\mathcal{M}$ is complemented in $\mathcal{X}$, then every continuous projection E in $\mathcal{B}[\mathcal{X}]$ with $\mathcal{R}(E) = \mathcal{M}$ is a continuous extension of T over $\mathcal{Y} = \mathcal{X}$* (cf. Remark 5.3). The norms, however, are not preserved for projections with norms different from 1.

(b) Norms of continuous projections. If E is a nonzero projection in $\mathcal{B}[\mathcal{X}]$ (i.e., a continuous projection on a normed space $\mathcal{X}$), then $E = E^2$ implies $\|E\| \ge 1$. *Every continuous projection on $\mathcal{X}$ has norm greater than or equal to* 1 (*and every orthogonal projection on an inner product space has norm* 1). For instance, $E(x_1, x_2) = (0, x_1 + x_2)$ and $P(x_1, x_2) = (0, x_2)$ for $(x_1, x_2) \in \mathbb{R}^2$ define continuous projections $E = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 1 \end{smallmatrix}\right)$ and $P = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ on the Euclidean space $\mathbb{R}^2$ both with the same range, $\mathcal{R}(E) = \mathcal{R}(P) = \{0\} \oplus \mathbb{R} \subseteq \mathbb{R} \oplus \mathbb{R} = \mathbb{R}^2$, such that $\|E\| = \sqrt{2}$ and $\|P\| = 1$, when $\mathbb{R}^2$ is equipped with the Euclidean norm.

(Incidentally, for bounded linear transformations A and B, if $0 \ne \|A\| = \|AB\|$, then $\|B\| \ge 1$ but not necessarily $\|B\| = 1$ — as there exists $\|E\| = \|E^2\| > 1$.)

(c) The converse of Theorem 5.4(b) may fail. Set $\mathcal{Y} = \mathcal{X}$ (same underlying set, same norm). Take a projection $E \in \mathcal{B}[\mathcal{X}]$ so that $E = E^2$ and hence $\|E\| \ge 1$. Suppose $\|E\| > 1$. Set $\mathcal{M} = \mathcal{R}(E)$ and $T = E|_{\mathcal{R}(E)} : \mathcal{R}(E) \to \mathcal{R}(E) \subseteq \mathcal{X} = \mathcal{Y}$, acting as the identity $I_{\mathcal{M}}$ on $\mathcal{M} = \mathcal{R}(E)$. Set $\widehat{T} = I_{\mathcal{X}} : \mathcal{X} \to \mathcal{X} = \mathcal{Y}$, the identity on $\mathcal{X}$, and so $\widehat{T}|_{\mathcal{M}} = I_{\mathcal{X}}|_{\mathcal{R}(E)} = E|_{\mathcal{R}(E)} = T$ with $\|T\| = \|E|_{\mathcal{R}(E)}\| = \|I_{\mathcal{M}}\| = \|I_{\mathcal{X}}\| = \|\widehat{T}\|$. Thus $\|\widehat{T}\| = \|T\|$ *and* $\mathcal{M}^- = \mathcal{R}(E)$ *for a projection* $E \in \mathcal{B}[\mathcal{X}]$ with $\|E\| \ne 1$. Does there exist another projection $P \in \mathcal{B}[\mathcal{X}]$ with $\|P\| = 1$ for which the same $\mathcal{M}$ is such that $\mathcal{M}^- = \mathcal{R}(P)$? (See Corollary 5.6 below.)

(d) Nonuniqueness of an extension. Set $\mathcal{Y} = \mathcal{X} = \mathbb{R}^2$ (the Euclidean space), consider the projection $E = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 1 \end{smallmatrix}\right) : \mathbb{R}^2 \to \mathbb{R}^2$ with $\|E\| = \sqrt{2}$ and take the subspace $\mathcal{M} = \mathcal{R}(E) = \{(x_1, x_2) : x_1 = 0\} = \{0\} \oplus \mathbb{R}$ of $\mathbb{R} \oplus \mathbb{R} = \mathbb{R}^2 = \mathcal{X}$. Set $T = E|_{\mathcal{R}(E)} : \mathcal{R}(E) \to \mathcal{R}(E) \subseteq \mathcal{X} = \mathcal{Y}$ so that $T = E|_{\mathcal{M}} \in \mathcal{B}[\mathcal{M}, \mathcal{Y}]$ acts as the identity on $\mathcal{R}(E)$. Hence $\|T\| = \|E|_{\mathcal{R}(E)}\| = 1$. Then $\widehat{T}_1 = E$ in $\mathcal{B}[\mathcal{X}, \mathcal{Y}]$ is the obvious extension of $T \in \mathcal{B}[\mathcal{M}, \mathcal{Y}]$ but $\|\widehat{T}_1\| \ne \|T\|$. Now take $I = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right) : \mathbb{R}^2 \to \mathbb{R}^2$, the identity on $\mathcal{X}$. Thus $\widehat{T}_2 = I$ in $\mathcal{B}[\mathcal{X}, \mathcal{Y}]$ is another trivial extension of $T \in \mathcal{B}[\mathcal{M}, \mathcal{Y}]$, this time with $\|\widehat{T}_2\| = \|T\|$. Next take $P = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right) : \mathbb{R}^2 \to \mathbb{R}^2$,

a nontrivial orthogonal projection in $\mathcal{B}[\mathbb{R}^2]$. Then $\|P\| = 1$ with $\mathcal{R}(P) = \mathcal{R}(E) = \mathcal{M}$ so that $P|_{\mathcal{M}} = E|_{\mathcal{M}} \in \mathcal{B}[\mathcal{M}, \mathcal{X}]$, and $\widehat{T}_3 = P$ in $\mathcal{B}[\mathcal{X}, \mathcal{Y}]$ is still another trivial extension of $T \in \mathcal{B}[\mathcal{M}, \mathcal{Y}]$, also with $\|\widehat{T}_3\| = \|T\|$. *Even with norm preservation a continuous extension is not unique.*

The following consequence of Theorem 5.4 is crucial.

Corollary 5.6. *Every bounded linear transformation $T\colon \mathcal{M} \to \mathcal{Y}$ of an arbitrary linear manifold $\mathcal{M}$ of any Hilbert space $\mathcal{X}$ into an arbitrary Banach space $\mathcal{Y}$ has a bounded linear extension $\widehat{T}\colon \mathcal{X} \to \mathcal{Y}$ over $\mathcal{X}$ such that $\|\widehat{T}\| = \|T\|$.*

Proof. (a) Let $\mathcal{X}$ be a Hilbert space. By the Projection Theorem, $\mathcal{M} + \mathcal{M}^\perp = \mathcal{X}$ *for every subspace* $\mathcal{M}$ *of* $\mathcal{X}$, where $\mathcal{M}^\perp$ is the orthogonal complement of $\mathcal{M}$, which is a subspace of $\mathcal{X}$ such that $\mathcal{M} \cap \mathcal{M}^\perp = \{0\}$ (see Proposition 5.Q in Section 5.4). So *every subspace of a Hilbert space is complemented*, and Theorem 5.4 applies to every linear manifold $\mathcal{M}$ of every Hilbert space $\mathcal{X}$. (b) If $\mathcal{X}$ is Hilbert, then we may take the orthogonal projection E onto $\mathcal{M}^-$ so that $\|E\| = 1$. Thus $\|\widehat{T}\| = \|T\|$ by Theorem 5.4(b) (cf. Proposition 5.R). □

The examples in Remark 5.5(d) show that even on a Hilbert space and under the norm identity $\|\widehat{T}\| = \|T\|$ the extension $\widehat{T}$ of T is not unique.

Now let $\mathcal{X}$ be a normed space and define the following classes of operators.

$$\Gamma_R[\mathcal{X}] = \big\{S \in \mathcal{B}[\mathcal{X}]\colon \mathcal{R}(S)^- \text{ is a complemented subspace of } \mathcal{X}\big\},$$
$$\Gamma_N[\mathcal{X}] = \big\{S \in \mathcal{B}[\mathcal{X}]\colon \mathcal{N}(S) \text{ is a complemented subspace of } \mathcal{X}\big\},$$
$$\Gamma[\mathcal{X}] = \Gamma_R[\mathcal{X}] \cap \Gamma_N[\mathcal{X}], \qquad \mathcal{S}\Gamma[\mathcal{X}] = \Gamma_R[\mathcal{X}] \cup \Gamma_N[\mathcal{X}].$$

If $\mathcal{X}$ is a Banach space, then the set $\Gamma[\mathcal{X}] = \Gamma_R[\mathcal{X}] \cap \Gamma_N[\mathcal{X}]$ is algebraically and topologically large in the sense that it includes nonempty open groups. For instance, the group of all invertible operators from $\mathcal{B}[\mathcal{X}]$ (i.e., of all operators from $\mathcal{B}[\mathcal{X}]$ with a bounded inverse) is open in $\mathcal{B}[\mathcal{X}]$ and trivially included in $\Gamma[\mathcal{X}]$. Actually, the class of all compact perturbations of invertible (more generally, of semi-Fredholm) operators lies in $\Gamma[\mathcal{X}]$ (see, e.g., [55, Remark 5.1]).

Corollary 5.7. *Let $\mathcal{X}$ be a Banach space. Let $\mathcal{M}$ be a linear manifold of $\mathcal{X}$ for which there is an operator $S \in \mathcal{S}\Gamma[\mathcal{X}]$ such that*

$$\mathcal{M}^- = \mathcal{R}(S)^- \text{ or } \mathcal{M}^- = \mathcal{N}(S),$$

according to whether S lies in $\Gamma_R[\mathcal{X}]$ or in $\Gamma_N[\mathcal{X}]$, respectively. If $T\colon \mathcal{M} \to \mathcal{Y}$ is any bounded linear transformation into an arbitrary Banach space $\mathcal{Y}$, then there exists a bounded linear extension $\widehat{T}\colon \mathcal{X} \to \mathcal{Y}$ of T over $\mathcal{X}$.

Proof. Straightforward from Theorem 5.4. If $S \in \mathcal{S}\Gamma[\mathcal{X}]$, then either $\mathcal{R}(S)^-$ or $\mathcal{N}(S)$ is complemented. If $\mathcal{M}^- = \mathcal{R}(S)^-$ or $\mathcal{M}^- = \mathcal{N}(S)$, then $T \in \mathcal{B}[\mathcal{M}, \mathcal{Y}]$ for any Banach space $\mathcal{Y}$ has an extension $\widehat{T} \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ by Theorem 5.4. □

Take $E \in \mathcal{B}[\mathcal{X}]$ and $T \in \mathcal{B}[\mathcal{M}, \mathcal{Y}]$, where $\mathcal{X}$ and $\mathcal{Y}$ are Banach spaces and $\mathcal{M}$ is a linear manifold of $\mathcal{X}$. Suppose $\mathcal{M} = \mathcal{R}(E)$. Recall from Remark 5.3 that continuous projections lie in $\Gamma[\mathcal{X}]$. Thus if E is a projection, then Corollary 5.7 says that there exists $\widehat{T} \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ such that $\widehat{T}|_{\mathcal{M}} = T$. However (cf. proof of Theorem 5.4), if $L \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ is defined by $L = TE$, then L is an extension of T. In fact, since $L|_{\mathcal{R}(E)} = T(E|_{\mathcal{R}(E)})$ and E is idempotent (so that $E|_{\mathcal{R}(E)} = I_{\mathcal{R}(E)}$, identity on $\mathcal{R}(E)$), we get $L|_{\mathcal{R}(E)} = T$. Conclusion: if $\widehat{T} \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ is an arbitrary extension of $T \in \mathcal{B}[\mathcal{M}, \mathcal{Y}]$, where $\mathcal{M} = \mathcal{R}(E)$ for some projection $E \in \mathcal{B}[\mathcal{X}]$, then $\widehat{T}|_{\mathcal{R}(E)} = T$ and therefore $\widehat{T}E = TE = L$.

5.3 Normed Quotient Space

We have agreed that by a *subspace* we mean a *closed* linear manifold of a normed space. Let $\mathcal{M}$ be a linear manifold of a normed space $\mathcal{X}$. Consider the map $\|\cdot\| : \mathcal{X}/\mathcal{M} \to \mathbb{R}$ defined for each coset $[x]$ in the quotient space $\mathcal{X}/\mathcal{M}$ by

$$\|[x]\| = \|x + \mathcal{M}\| = \inf_{u \in \mathcal{M}} \|x + u\| = d(x, \mathcal{M}),$$

the distance of x to $\mathcal{M}$, which is invariant for all representatives in $[x]$.

Theorem 5.8. *If $\mathcal{M}$ is closed, then $\|\cdot\| : \mathcal{X}/\mathcal{M} \to \mathbb{R}$ is a norm on $\mathcal{X}/\mathcal{M}$.*

Proof. Let $\mathcal{M}$ be a linear manifold of a normed space $(\mathcal{X}, \|\cdot\|)$. The map $\|\cdot\|$ defines a function from $\mathcal{X}/\mathcal{M}$ to $\mathbb{R}$ since $\|[x]\|$ depends only on the coset $[x]$ and not on a particular representative x in $[x]$. As for nonnegativeness,

$$\|[x]\| \geq 0 \quad \text{for every} \quad [x] \in \mathcal{X}/\mathcal{M}$$

trivially. Absolute homogeneity and subadditivity are easily verified. In fact,

$$\alpha[x] = [\alpha x] \quad \text{and} \quad [x] + [y] = [x + y]$$

for every $[x], [y]$ in $\mathcal{X}/\mathcal{M}$ and every scalar α. Since $\mathcal{M}$ is a linear manifold,

$$\begin{aligned} \|\alpha[x]\| = \|[\alpha x]\| &= \inf_{u \in \mathcal{M}} \|\alpha x + u\| = \inf_{u \in \mathcal{M}} \|\alpha x + \alpha u\| \\ &= |\alpha| \inf_{u \in \mathcal{M}} \|x + u\| = |\alpha| \, \|[x]\|, \end{aligned}$$

$$\begin{aligned} \|[x] + [y]\| = \|[x + y]\| &= \inf_{u \in \mathcal{M}} \|x + y + u\| = \inf_{u \in \mathcal{M}} \|x + u + y + u\| \\ &\leq \inf_{u \in \mathcal{M}} \|x + u\| + \inf_{u \in \mathcal{M}} \|y + u\| = \|[x]\| + \|[y]\|, \end{aligned}$$

for every $[x], [y]$ in $\mathcal{X}/\mathcal{M}$ and every scalar α. Thus the map $\|\cdot\| : \mathcal{X}/\mathcal{M} \to \mathbb{R}$ is a semi-norm for every linear manifold $\mathcal{M}$ of $\mathcal{X}$. Now suppose $\mathcal{M}$ is closed in $\mathcal{X}$, and take an arbitrary $x \in [x]$. If $\|[x]\| = 0$, then $d(x, \mathcal{M}) = 0$ and so

$x \in \mathcal{M}^- = \mathcal{M}$. This means $x \in [0]$ (since $\mathcal{M}$ is the origin $[0]$ of $\mathcal{X}/\mathcal{M}$ — see Section 1.3), and hence $[x] = [0]$. Equivalently,

$$\big\|[x]\big\| > 0 \quad \text{whenever} \quad [x] \neq 0. \qquad \square$$

If $\mathcal{M}$ is a subspace of a normed space $\mathcal{X}$, then equip the quotient space $\mathcal{X}/\mathcal{M}$ with this norm, which is referred to as the *quotient norm*. When we refer to the normed space $\mathcal{X}/\mathcal{M}$, it is understood that $\mathcal{M}$ is a subspace and the quotient space $\mathcal{X}/\mathcal{M}$ is equipped with the quotient norm. If $\mathcal{M}$ is a subspace so that $\mathcal{X}/\mathcal{M}$ is a normed space, then consider the *natural quotient map* $\pi\colon \mathcal{X} \to \mathcal{X}/\mathcal{M}$ of Section 1.3, which is given by $\pi(x) = [x] \in \mathcal{X}/\mathcal{M}$ for every $x \in \mathcal{X}$ (and is linear and surjective as we saw in Section 1.3). This π is a contraction. Actually, $\big\|\pi(x)\big\| = \big\|[x]\big\| = d(x, \mathcal{M}) = \inf_{u\in\mathcal{M}} \|x + u\|$ and hence

$$\big\|\pi(x)\big\| \leq \|x\| \ \text{ for every } \ x \in \mathcal{X} \quad \text{ so that } \quad \pi \in \mathcal{B}[\mathcal{X}, \mathcal{X}/\mathcal{M}].$$

More is true. First note the trivial cases.

$$\mathcal{M} = \{0\} \iff \big\|\pi(x)\big\| = \inf_{u\in\{0\}} \|x + u\| = \|x\| \ \ \forall\, x \in \mathcal{X}.$$

In fact, according to Section 1.3, $\mathcal{M} = \{0\} \Longleftrightarrow \mathcal{X}/\mathcal{M} = \mathcal{X}/\{0\} \cong \mathcal{X} \Longleftrightarrow \pi \cong I$. Also π is an isometry $\Longleftrightarrow$ π is injective (since π is injective $\Longleftrightarrow \mathcal{M} = \{0\}$, as we have also seen in Section 1.3). Moreover (compare with Section 1.3 again),

$$\mathcal{M} = \mathcal{X} \iff \big\|\pi(x)\big\| = \inf_{u\in\mathcal{X}} \|x + u\| = 0 \ \ \forall\, x \in \mathcal{X} \iff \pi = 0.$$

If $\mathcal{M}$ is a proper subspace of $\mathcal{X}$, then the contraction π has norm one:

$$\mathcal{M} \neq \mathcal{X} \iff \|\pi\| = 1.$$

Indeed, for $\mathcal{M} \neq \{0\}$ (otherwise π is an isometry), $\|\pi\| = \sup_{\|x\|=1} \big\|\pi(x)\big\| = \sup_{\|x\|=1} d(x, \mathcal{M}) = 1$ by the Riesz Lemma (see, e.g., [52, Lemma 4.33]).

Remark 5.9. *The Three-Space Property*. Recall that a linear manifold of a Banach space is itself a Banach space if and only if it is closed. If $\mathcal{M}$ is a (closed) subspace of a Banach space $\mathcal{X}$, then the quotient space $\mathcal{X}/\mathcal{M}$ is a Banach space (this an important and often used result — see, e.g., [52, Proposition 4.10], [70, Theorem 1.7.7], [12, Theorem III.4.2(b)], [87, Theorem 3.13]). Conversely, if $\mathcal{M}$ is a Banach space (i.e., $\mathcal{M}$ is complete in a normed space $\mathcal{X}$ so that $\mathcal{M}$ is closed) and the quotient space $\mathcal{X}/\mathcal{M}$ is a Banach space, then $\mathcal{X}$ is a Banach space (see, e.g., [52, Problem 4.13], [12, Exercise III.4.5]). If $\mathcal{X}/\mathcal{M}$ is a normed space equipped with the quotient norm, then $\mathcal{M}$ must be a subspace of $\mathcal{X}$; if $\mathcal{X}$ is Banach, then so is $\mathcal{M}$. Thus *if any two of* $\mathcal{M}$, $\mathcal{X}$ *and* $\mathcal{X}/\mathcal{M}$ *are Banach spaces, then so is the other* (see also [70, Theorem 1.7.9]).

Theorem 1.4 on linear transformations can be extended to bounded linear transformations when quotient spaces are defined for normed spaces, now with $\cong$ standing for topologically isomorphic rather than (algebraically) isomorphic.

Theorem 5.10. *Let $\mathcal{X}$ and $\mathcal{Z}$ be normed spaces over the same field, let $\mathcal{M}$ be a subspace of $\mathcal{X}$, consider the normed quotient space $\mathcal{X}/\mathcal{M}$, and take the natural quotient map $\pi\colon \mathcal{X} \to \mathcal{X}/\mathcal{M}$.*

(a) (Universal Property): *If $T \in \mathcal{B}[\mathcal{X}, \mathcal{Z}]$ and $\mathcal{M} \subseteq \mathcal{N}(T)$, then there exists a unique $\widehat{T} \in \mathcal{B}[\mathcal{X}/\mathcal{M}, \mathcal{Z}]$ such that*

$$T = \widehat{T} \circ \pi.$$

In other words, the diagram

$$\begin{array}{ccc} \mathcal{X} & \xrightarrow{\ T\ } & \mathcal{Z} \\ & {\scriptstyle\pi}\searrow \quad \uparrow{\scriptstyle\widehat{T}} & \\ & \mathcal{X}/\mathcal{M} & \end{array}$$

commutes. Moreover, $\mathcal{N}(\widehat{T}) = \mathcal{N}(T)/\mathcal{M}$, $\mathcal{R}(\widehat{T}) = \mathcal{R}(T)$, and $\|\widehat{T}\| = \|T\|$.

(b) (The First Isomorphism Theorem in Banach Space): *If $\mathcal{X}$ and $\mathcal{Z}$ are Banach spaces, $T \in \mathcal{B}[\mathcal{X}, \mathcal{Z}]$, and $\mathcal{R}(T)$ is closed, then $\mathcal{R}(T) \cong \mathcal{X}/\mathcal{N}(T)$.*

Proof. (a) Let $\mathcal{X}$, $\mathcal{Z}$ be normed spaces, $\mathcal{M}$ a subspace of $\mathcal{X}$, and $\pi\colon \mathcal{X} \to \mathcal{X}/\mathcal{M}$ the quotient map. Suppose the linear transformation in Theorem 1.4 is continuous. In other words, replace $L \in \mathcal{L}[\mathcal{X}, \mathcal{Z}]$ with $T \in \mathcal{B}[\mathcal{X}, \mathcal{Z}]$ in Theorem 1.4, and recall that now π is a contraction, thus continuous. We know from Theorem 1.4 that, if $\mathcal{M} \subseteq \mathcal{N}(T)$, then there is a unique linear transformation $\widehat{T}\colon \mathcal{X}/\mathcal{M} \to \mathcal{Z}$ such that $T = \widehat{T} \circ \pi$, that is, $\widehat{T}[x] = Tx$ for every $x \in \mathcal{X}$.

Claim. $\widehat{T} \in \mathcal{B}[\mathcal{X}/\mathcal{M}, \mathcal{Z}]$.

Proof of Claim. If the linear transformation $\widehat{T}\colon \mathcal{X}/\mathcal{M} \to \mathcal{Z}$ is not continuous, which means it is not continuous at the origin $[0]$, then there exists an $\mathcal{X}/\mathcal{M}$-valued sequence $\{[x_n]\}$ such that

$$[x_n] \to [0] \text{ in } \mathcal{X}/\mathcal{M} \quad \text{ while } \quad Tx_n = \widehat{T}[x_n] \not\to \widehat{T}[0] = T0 = 0 \text{ in } \mathcal{Z}.$$

Take an arbitrary $\varepsilon > 0$. Since $[x_n] \to [0] = \mathcal{M} \subseteq \mathcal{N}(T)$, then there exists a positive integer n_ε such that $\big\|[x_n] - [0]\big\| = \big\|[x_n]\big\| = \inf_{u \in \mathcal{M}} \|x_n - u\| < \varepsilon$ for every $n \geq n_\varepsilon$. Therefore there exists an $\mathcal{M}$-valued sequence $\{u_{(\varepsilon)_n}\}$ for which $\|x_n - u_{(\varepsilon)_n}\| < \varepsilon$ whenever $n \geq n_\varepsilon$. Then, since $\mathcal{M} \subseteq \mathcal{N}(T)$, for every $n \geq n_\varepsilon$

$$\|Tx_n\| = \|Tx_n - Tu_{(\varepsilon)_n}\| = \|T(x_n - u_{(\varepsilon)_n})\| \leq \|T\|\,\|x_n - u_{(\varepsilon)_n}\| < \|T\|\,\varepsilon.$$

Hence $Tx_n \to 0$, which is a contradiction. Thus $\widehat{T}$ is continuous. □

Also by Theorem 1.4 we get $\mathcal{N}(\widehat{T}) = \mathcal{N}(T)/\mathcal{M}$ and $\mathcal{R}(\widehat{T}) = \mathcal{R}(T)$. Moreover, since $\{0\} \subseteq [0] = \mathcal{M}$ so that $\mathcal{X}\backslash\mathcal{M} \subseteq \mathcal{X}\backslash\{0\}$, and since $T = \widehat{T} \circ \pi$,

$$\|\widehat{T}\| = \sup_{[x] \neq [0]} \frac{\|\widehat{T}[x]\|}{\|[x]\|} = \sup_{x \notin \mathcal{M}} \frac{\|Tx\|}{\|[x]\|} \leq \sup_{x \notin \{0\}} \frac{\|Tx\|}{\|[x]\|} = \|T\| = \|\widehat{T} \circ \pi\| \leq \|\widehat{T}\|\,\|\pi\| \leq \|\widehat{T}\|$$

because $\|\pi\| \leq 1$ (actually, $\|\pi\| = 1$). Therefore $\|\widehat{T}\| = \|T\|$.

(b) Since $T \in \mathcal{L}[\mathcal{X}, \mathcal{Z}]$ between linear spaces, then Theorem 1.4(b) ensures that $\mathcal{X}/\mathcal{N}(T)$ is algebraically isomorphic to $\mathcal{R}(T)$, and (cf. proof of Theorem 1.4(b)) $\widehat{T}\colon \mathcal{X}/\mathcal{N}(T) \to \mathcal{R}(T)$ is an isomorphism between them. However, since $\mathcal{X}$ and $\mathcal{Z}$ are normed spaces and $T \in \mathcal{B}[\mathcal{X}, \mathcal{Z}]$, the linear manifold $\mathcal{N}(T)$ is closed in $\mathcal{X}$ (i.e., it is a subspace of $\mathcal{X}$). Thus if $\mathcal{X}$ is a Banach space, then so is $\mathcal{X}/\mathcal{N}(T)$ (cf. Remark 5.9). Moreover, if $\mathcal{Z}$ is a Banach space as well and $\mathcal{R}(T)$ is closed in $\mathcal{Z}$, then $\mathcal{R}(T)$ is itself a Banach space. So $\widehat{T} \in \mathcal{B}[\mathcal{X}/\mathcal{N}(T), \mathcal{R}(T)]$ is injective and surjective between Banach spaces and therefore, by the Inverse Mapping Theorem, its inverse is continuous (cf. Proposition 5.B in Section 5.4), so that $\widehat{T}$ is a topological isomorphism of $\mathcal{X}/\mathcal{N}(T)$ onto $\mathcal{R}(T)$. □

Remark 5.11. *The Dual of the First Isomorphism Theorem.* If $\mathcal{X}$ and $\mathcal{Z}$ are Banach spaces, $T \in \mathcal{B}[\mathcal{X}, \mathcal{Z}]$, and $\mathcal{R}(T)$ is closed, then

$$\mathcal{R}(T) \;\cong\; \mathcal{X}/\mathcal{N}(T),$$

where $\cong$ means topologically isomorphic. This is the First Isomorphism Theorem in Banach Space (cf. Theorem 5.10(b)). Its dual reads as follows: if $\mathcal{X}$ and $\mathcal{Z}$ are normed spaces, $T^* \in \mathcal{B}[\mathcal{Z}^*, \mathcal{X}^*]$ is the normed-space adjoint of $T \in \mathcal{B}[\mathcal{X}, \mathcal{Z}]$, $\mathcal{R}(T)$ is closed, and $(\mathcal{Z}/\mathcal{R}(T))^*$ is the dual of $\mathcal{Z}/\mathcal{R}(T)$, then (see, e.g., [70, Theorems 1.10.17, Lemma 3.1.16])

$$\mathcal{N}(T^*) \;\cong\; \big(\mathcal{Z}/\mathcal{R}(T)\big)^*,$$

where in this case $\cong$ means isometrically isomorphic.

Definition 5.12. Let $\mathcal{X}$ and $\mathcal{Y}$ be nonzero normed spaces. A transformation $Q \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ is called a *quotient transformation* if it is surjective and

$$\|y\| = \inf_{x \in Q^{-1}(\{y\})} \|x\| \quad \text{for every} \quad y \in \mathcal{Y}.$$

Equivalently, Q is a quotient transformation if the following condition holds. Take an arbitrary $y \in \mathcal{Y}$. Then $\|y\| \le \|x\|$ for all $x \in Q^{-1}(\{y\})$, and for every $\varepsilon > 0$ there exists $x \in Q^{-1}(\{y\})$ such that $\|x\| \le \|y\| + \varepsilon$. In particular, for $y \ne 0$ and for every $\varepsilon' > 0$, set $\varepsilon = \varepsilon'\|y\|$ so that $\|x\| \le (1 + \varepsilon')\|y\|$.

If there is a quotient transformation of $\mathcal{X}$ onto $\mathcal{Y}$, then $\mathcal{Y}$ *is a quotient of* $\mathcal{X}$.

Theorem 5.13. *Let $\mathcal{X}$ and $\mathcal{Y}$ be normed spaces.*

(a) *If $Q\colon \mathcal{X} \to \mathcal{Y}$ is a linear surjective contraction and for each $y \in \mathcal{Y}$ there is an $x \in Q^{-1}(\{y\})$ such that $\|y\| = \|x\|$, then Q is a quotient transformation.*

From now on suppose $Q \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$.

(b) *Q is an injective quotient transformation if and only if Q is an isometric isomorphism.* (*So* (1) *if Q is a quotient transformation, then it is injective if and only if it is an isometric isomorphism,* (2) *every isometric isomorphism is an injective quotient transformation, and* (3) *if two normed spaces are isometrically isomorphic, then one is a quotient of the other.*)

(c) *If Q is a quotient transformation, then*

(c_1) $\|Qx\| \leq \|x\|$ *every* $x \in \mathcal{X}$ (*i.e., Q is a contraction*),

(c_2) *for every* $x \in \mathcal{X}$ *and every* $\varepsilon > 0$ *there exists an* $x_\varepsilon \in Q^{-1}(\{Qx\}) \subseteq \mathcal{X}$ *for which* $Qx = Qx_\varepsilon$ *and* $\|x_\varepsilon\| < \|Qx\| + \varepsilon$.

(d) *Q is a quotient transformation if and only if Q maps the open unit ball in $\mathcal{X}$ onto the open unit ball in $\mathcal{Y}$.* (*Thus $Q \neq O$.*)

(e) *If Q is a quotient transformation, then*

(e_1) *$Q$is an open map* (*i.e., it takes open sets into open sets*),

(e_2) $\|Q\| = 1$.

(f) *If $\mathcal{M}$ is a proper subspace of $\mathcal{X}$* (*i.e., $\mathcal{M} \neq \mathcal{X}$*), *then the quotient map $\pi\colon \mathcal{X} \to \mathcal{X}/\mathcal{M}$ is a quotient transformation.* (*So $\mathcal{X}/\mathcal{M}$ is a quotient of $\mathcal{X}$.*)

Let $\cong$ stand for isometrically isomorphic.

(g) *Let $\mathcal{X}, \widetilde{\mathcal{X}}, \mathcal{Y}, \widetilde{\mathcal{Y}}$ be normed spaces. Suppose $Q \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ and $\widetilde{Q} \in \mathcal{B}[\widetilde{\mathcal{X}}, \widetilde{\mathcal{Y}}]$ are isometrically isomorphic* (*i.e., $Q \cong \widetilde{Q}$*). *Then Q is a quotient transformation if and only if $\widetilde{Q}$ is a quotient transformation.*

(h) *If $Q \neq O$ and $\mathcal{Y} \cong \mathcal{X}/\mathcal{N}(Q)$, then Q is a quotient transformation.*

(i) *If $\mathcal{X}$ and $\mathcal{Y}$ are Banach spaces, then the converse holds: if Q is a quotient transformation then $\mathcal{Y} \cong \mathcal{X}/\mathcal{N}(Q)$.*

(j) *If $Q \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ is a quotient transformation, then so is its extension over completion $\widehat{Q} \in \mathcal{B}[\widehat{\mathcal{X}}, \widehat{\mathcal{Y}}]$.*

Proof. Let $Q : \mathcal{X} \to \mathcal{Y}$ be linear and surjective between normed spaces.

(a) Suppose Q is a contraction (i.e., $\|Qx\| \leq \|x\|$ for every $x \in \mathcal{X}$ so that $Q \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$). Since Q is surjective, being a contraction means $\|y\| \leq \|x\|$ for every $y \in \mathcal{Y}$ and every $x \in Q^{-1}(\{y\}) \subseteq \mathcal{X}$. If, in addition, for every $y \in \mathcal{Y} = \mathcal{R}(Q)$ there exists an $x_y \in Q^{-1}(\{y\}) \subseteq \mathcal{X}$ such that $\|y\| = \|x_y\|$, then Q is a quotient transformation by Definition 5.12, which proves (a). Indeed,

$$\|x_y\| = \|y\| \leq \inf_{x \in Q^{-1}(\{y\})} \|x\| \quad \Longrightarrow \quad \|y\| = \inf_{x \in Q^{-1}(\{y\})} \|x\|.$$

From now on suppose $Q \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$. According to Definition 5.12, Q is a quotient transformation if and only if Q is surjective and

$$\|Qx\| = \inf_{z \in Q^{-1}(\{Qx\})} \|z\| \quad \text{for every} \quad x \in \mathcal{X}.$$

(b) Suppose Q is a quotient transformation. Then it is surjective. If it is injective, then it is invertible and so $Q^{-1}(\{Qx\}) = \{x\}$, which implies

$\|Qx\| = \inf_{z\in\{x\}} \|z\| = \|x\|$, for every $x \in \mathcal{X}$. Thus Q is a surjective isometry, equivalently, an invertible isometry, that is, an isometric isomorphism. Conversely, every invertible isometry satisfies the above displayed identity.

(c) Let Q be a quotient transformation. (c_1) As the inclusion $A \subseteq F^{-1}(F(A))$ holds for every function $F\colon \mathcal{X} \to \mathcal{Y}$ and for every set $A \subseteq \mathcal{X}$, it follows that $\|Qx\| = \inf_{z\in Q^{-1}(Q(\{x\}))} \|z\| \le \inf_{z\in\{x\}} \|z\| = \|x\|$ for every $x \in \mathcal{X}$. (c_2) Take an arbitrary $x \in \mathcal{X}$ and an arbitrary $\varepsilon > 0$. Set $y = Qx$. Since $\|y\| = \inf_{x\in Q^{-1}(\{y\})} \|x\|$, there exists an $x_\varepsilon \in Q^{-1}(\{y\})$ such that $\|x_\varepsilon\| < \|y\| + \varepsilon = \|Qx\| + \varepsilon$. But $Q(Q^{-1}(\{y\})) = \{y\}$ because Q is surjective. So $Q(x_\varepsilon) = y = Qx$.

(d) Let $B_\mathcal{X}$ and $B_\mathcal{Y}$ be open unit balls in $\mathcal{X}$ and $\mathcal{Y}$ (centered at the origin).

Part I. Let Q be a quotient transformation. If $x \in B_\mathcal{X}$, then $\|Qx\| \le \|x\| < 1$ by (c_1). Hence $Q(B_\mathcal{X}) \subseteq B_\mathcal{Y}$. Conversely, take any $y \in B_\mathcal{Y}$. So $\|y\| < 1$. As Q is surjective, take any $x \in \mathcal{X}$ such that $Qx = y$. Take $\varepsilon \in (0, 1 - \|y\|)$. By ($c_2$) there is an $x_\varepsilon \in \mathcal{X}$ for which $Qx_\varepsilon = y = Qx$ and $\|x_\varepsilon\| < \|y\| + \varepsilon < 1$. Thus $x_\varepsilon \in B_\mathcal{X}$ and so $y \in Q(B_\mathcal{X})$. Hence $B_\mathcal{Y} \subseteq Q(B_\mathcal{X})$. Then $Q(B_\mathcal{X}) = B_\mathcal{Y}$.

Part II. Conversely, suppose $Q \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ is such that $Q(B_\mathcal{X}) = B_\mathcal{Y}$. Thus Q is surjective because $B_\mathcal{Y} \subset \mathcal{R}(Q) \subseteq \mathcal{Y}$ implies $\mathcal{R}(Q) = \mathcal{Y}$. Since $Q(B_\mathcal{X}) \subseteq B_\mathcal{Y}$,

$$\big\{x \in B_\mathcal{X} \Longrightarrow \|Qx\| \le 1\big\} \Longrightarrow \|Q\| = \sup_{x\in B_\mathcal{X}} \|Qx\| \le 1.$$

Since every $y \in B_\mathcal{Y} = Q(B_\mathcal{X})$ is such that $y = Qx$ for some $x \in B_\mathcal{X}$, then for every $\varepsilon > 0$ there exists $x_\varepsilon \in B_\mathcal{X}$ for which $\|Qx_\varepsilon\| = 1 - \varepsilon$. Hence

$$\|Q\| = \sup_{x\in B_\mathcal{X}} \|Qx\| = 1.$$

Take an arbitrary $x \in \mathcal{X}$. Recall that $x \in Q^{-1}(\{Qx\})$ (indeed, as we saw above $A \subseteq F^{-1}(F(A))$). Since $\|Q\| \le 1$ we get $\|Qx\| \le \|x\|$. Thus

$$\|Qx\| \le \inf_{x\in Q^{-1}(\{Qx\})} \|x\| \quad \text{for every} \quad x \in \mathcal{X}.$$

On the other hand, take an arbitrary $\varepsilon > 0$. Since $\|Q\| = 1$, there is an $x_\varepsilon \in B_\mathcal{X}$ such that $\|Qx_\varepsilon\| + \varepsilon > 1 = \|x_\varepsilon\|$. Again $x_\varepsilon \in Q^{-1}(Q(\{x_\varepsilon\}))$. So for every $\varepsilon > 0$

$$\|Qx_\varepsilon\| + \varepsilon > \|x_\varepsilon\| \quad \text{for some} \quad x_\varepsilon \in Q^{-1}(\{y_\varepsilon\}).$$

Therefore $\|Qx\| = \inf_{x\in Q^{-1}(\{Qx\})} \|x\|$ for every $x \in \mathcal{X}$.

(e) Let Q be a quotient transformation. (e_1) Since Q sends the open unit ball in $\mathcal{X}$ onto the open unit ball in $\mathcal{Y}$ according to (d), then it sends every open ball in $\mathcal{X}$ onto an open ball in $\mathcal{Y}$, and therefore it sends every open set in $\mathcal{X}$ into an open set in $\mathcal{Y}$, since the open balls are a topological basis for open sets, which precisely means Q is an open map. (e_2) We have already shown in the proof of (d) that a linear transformation that sends the open unit ball in $\mathcal{X}$ onto the open unit ball in $\mathcal{Y}$ has norm one.

(f) Let $\mathcal{M}$ be a subspace of the normed space $\mathcal{X}$ and consider the natural quotient map $\pi\colon \mathcal{X} \to \mathcal{X}/\mathcal{M}$ which is given by $\pi(x) = [x] = x + \mathcal{M}$. Recall from Section 1.3: $\mathcal{X}/\mathcal{M} \neq \{[0]\} \iff \mathcal{M} \neq \mathcal{X} \iff \pi \neq 0$. Observe that

$$\pi^{-1}(\{\pi(x)\}) = x + \mathcal{M} \quad \text{for every} \quad x \in \mathcal{X},$$

since $\pi^{-1}(\pi(S)) = S + \mathcal{M}$ for every subset S of $\mathcal{X}$ (cf. Section 1.3). Therefore

$$\inf_{z \in \pi^{-1}(\{\pi(x)\})} \|z\| = \inf_{z \in x + \mathcal{M}} \|z\| = \inf_{u \in \mathcal{M}} \|x + u\| = \big\|[x]\big\| = \big\|\pi(x)\big\|.$$

So the surjective bounded linear nonzero map π is a quotient transformation.

(g) Let $\cong$ stand for isometrically isomorphic. Take arbitrary transformations Q in $\mathcal{B}[\mathcal{X}, \mathcal{Y}]$ and $\widetilde{Q}$ in $\mathcal{B}[\widetilde{\mathcal{X}}, \widetilde{\mathcal{Y}}]$. To say that $\widetilde{Q} \cong Q$ means (1) $\widetilde{\mathcal{X}} = X(\mathcal{X})$ and $\widetilde{\mathcal{Y}} = Y(\mathcal{Y})$ for some invertible linear isometries (i.e., isometric isomorphisms) $X\colon \mathcal{X} \to \widetilde{\mathcal{X}}$ and $Y\colon \mathcal{Y} \to \widetilde{\mathcal{Y}}$, and also (2) $YQ = \widetilde{Q}X$. Take the open unit balls $B_{\mathcal{X}} \subseteq \mathcal{X}$ and $B_{\mathcal{Y}} \subseteq \mathcal{Y}$. Since X and Y are isometric isomorphisms, the open unit balls $B_{\widetilde{\mathcal{X}}} \subseteq \widetilde{\mathcal{X}}$ and $B_{\widetilde{\mathcal{Y}}} \subseteq \widetilde{\mathcal{Y}}$ are of the form $B_{\widetilde{\mathcal{X}}} = XB_{\mathcal{X}}$ and $B_{\widetilde{\mathcal{Y}}} = YB_{\mathcal{Y}}$. Therefore, according to (d), Q is a quotient transformation if and only if $\widetilde{Q}$ is a quotient transformation, because

$$QB_{\mathcal{X}} = B_{\mathcal{Y}} \iff Y^{-1}\widetilde{Q}XX^{-1}B_{\widetilde{\mathcal{X}}} = Y^{-1}B_{\widetilde{\mathcal{Y}}} \iff \widetilde{Q}B_{\widetilde{\mathcal{X}}} = B_{\widetilde{\mathcal{Y}}}.$$

(h) Take a nonzero $Q \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$. Since Q is nonzero, linear and continuous, then $\mathcal{N}(Q)$ is a proper subspace of $\mathcal{X}$ (i.e., $\mathcal{N}(Q) \neq \mathcal{X}$), so that the natural quotient map $\pi\colon \mathcal{X} \to \mathcal{X}/\mathcal{N}(Q)$ is a quotient transformation by (f). Thus

$$\begin{array}{ccc} \mathcal{X} & \xrightarrow{\;Q\;} & \mathcal{Y} \\ & {\scriptstyle\pi}\searrow & \\ & & \mathcal{X}/\mathcal{N}(Q)\,. \end{array}$$

Therefore according to Section 5.1 and using (g) we get

$$\mathcal{Y} \cong \mathcal{X}/\mathcal{N}(Q) \iff Q \cong \pi \implies Q \text{ is a quotient map.}$$

(i) Conversely, suppose $Q \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ is a quotient transformation (and hence $\mathcal{R}(Q) = \mathcal{Y}$ is trivially closed) between Banach spaces. By Theorem 5.10(b) and its proof we get: (1) $\mathcal{Y} \cong \mathcal{X}/\mathcal{N}(Q)$ where $\cong$ means topologically isomorphic, and (2) $\widehat{Q}\colon \mathcal{X}/\mathcal{N}(Q) \to \mathcal{Y}$ is the topological isomorphism of $\mathcal{X}/\mathcal{N}(Q)$ onto $\mathcal{Y}$ such that $Q = \widehat{Q} \circ \pi$ and $\|\widehat{Q}\| = \|Q\|$. In this case $\|Q\| = 1$ by (e_2), and so $\|\widehat{Q}\| = 1$. This ensures that $\widehat{Q}$ is an isometry, as shown in Claim 2 below. In fact, note that $\widehat{Q}[x] = \widehat{Q}(\pi(x)) = Qx$ for every $[x] \in \mathcal{X}/\mathcal{N}(Q)$ as π is surjective.

Claim 1. If $x' \in Q^{-1}(\{Qx\})$ for some $x \in \mathcal{X}$, then $[x'] = [x]$.

Proof of Claim 1. Suppose $x' \in Q^{-1}(\{Qx\})$ for some $x \in \mathcal{X}$. Since $\widehat{Q}[x] = Qx$ for every $x \in \mathcal{X}$, then $x' \in Q^{-1}(\{\widehat{Q}[x]\})$ so that $\widehat{Q}[x'] = Qx' \in \{\widehat{Q}[x]\}$ (as Q is surjective). Hence $\widehat{Q}[x'] = \widehat{Q}[x]$, which implies $[x'] = [x]$ (as $\widehat{Q}$ is invertible). □

Claim 2. $\|\widehat{Q}[x]\| = \big\|[x]\big\|$ for every $[x] \in \mathcal{X}/\mathcal{N}(Q)$.

Proof of Claim 2. Take $x \in \mathcal{X}$ and $\varepsilon > 0$ arbitrary. By (c_2), there is a vector $x_\varepsilon \in Q^{-1}(\{Qx\}) \subseteq \mathcal{X}$ such that $Qx = Qx_\varepsilon$ and $\|x_\varepsilon\| < \|Qx\| + \varepsilon$. So by Claim 1

$$\big\|[x]\big\| = \big\|[x_\varepsilon]\big\| = \|x_\varepsilon\| < \|Qx\| + \varepsilon = \|\widehat{Q}[x]\| + \varepsilon \le \|\widehat{Q}\|\,\big\|[x]\big\| + \varepsilon = \big\|[x]\big\| + \varepsilon$$

for every $[x] \in \mathcal{X}/\mathcal{N}(Q)$ because $\|\widehat{Q}\| = 1$. Hence $-\varepsilon < \|\widehat{Q}[x]\| - \big\|[x]\big\| < 0 < \varepsilon$ and so $\big|\,\|\widehat{Q}[x]\| - \big\|[x]\big\|\,\big| < \varepsilon$. Since $\varepsilon > 0$ is arbitrary (it does not depend on $[x]$), we get $\|\widehat{Q}[x]\| = \big\|[x]\big\|$ for every $[x] \in \mathcal{X}/\mathcal{N}(Q)$. □

Then the topological isomorphism $\widehat{Q}\colon \mathcal{X}/\mathcal{N}(Q) \to \mathcal{Y}$ is an isometry, thus an isometric isomorphism. So $\mathcal{X}/\mathcal{N}(Q)$ and $\mathcal{Y}$ are isometrically isomorphic.

(j) Consider the diagram in the proof of Theorem 5.1 and the isometric isomorphisms $J\colon \mathcal{X} \to \widetilde{\mathcal{X}}$ and $K\colon \mathcal{Y} \to \widetilde{\mathcal{Y}}$. Let $B_{\widehat{\mathcal{X}}}$ be an arbitrary open unit ball in $\widehat{\mathcal{X}}$. Since $\widetilde{\mathcal{X}} \subseteq \widehat{\mathcal{X}}$, $B_{\widetilde{\mathcal{X}}} = B_{\widehat{\mathcal{X}}} \cap \widetilde{\mathcal{X}}$ is an open unit ball in $\widetilde{\mathcal{X}}$. So $B_{\mathcal{X}} = J^{-1}(B_{\widetilde{\mathcal{X}}})$ is an open unit ball in $\mathcal{X}$. If $Q \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ is a quotient transformation, then $B_{\mathcal{Y}} = Q(B_{\mathcal{X}})$ is an open unit ball in $\mathcal{Y}$ by (d). Again, $B_{\widetilde{\mathcal{Y}}} = K(B_{\mathcal{Y}})$ is an open unit ball in $\widetilde{\mathcal{Y}}$, and therefore $K(B_{\mathcal{Y}}) = B_{\widehat{\mathcal{Y}}} \cap \widetilde{\mathcal{Y}}$ for some open unit ball $B_{\widehat{\mathcal{Y}}}$ in $\widehat{\mathcal{Y}}$ because $\widetilde{\mathcal{Y}} \subseteq \widehat{\mathcal{Y}}$. Hence

$$B_{\widehat{\mathcal{Y}}} \cap \widetilde{\mathcal{Y}} = K(B_{\mathcal{Y}}) = K(Q(B_{\mathcal{X}})) = K(Q(J^{-1}(B_{\widehat{\mathcal{X}}} \cap \widetilde{\mathcal{X}}))) = \widehat{Q}(B_{\widehat{\mathcal{X}}} \cap \widetilde{\mathcal{X}}).$$

So by continuity and denseness, $B_{\widehat{\mathcal{Y}}} = \widehat{Q}(B_{\widehat{\mathcal{X}}})$. As $B_{\widehat{\mathcal{X}}}$ is an arbitrary open unit ball in $\widehat{\mathcal{X}}$, it follows that $\widehat{Q} \in \mathcal{B}[\widehat{\mathcal{X}}, \widehat{\mathcal{Y}}]$ is a quotient transformation by (d). □

Remark 5.14. *Quotient Topology.* Let $\mathcal{T}_Z$ denote the topology of a topological space Z. *A topological quotient map* is a function $Q\colon X \to Y$ of a topological space X *onto* a topological space Y such that $B \in \mathcal{T}_Y \iff Q^{-1}(B) \in \mathcal{T}_X$ (i.e., a subset B of Y is open in Y if and only if its inverse image under Q is open in X). If X and Y are topological spaces and if there is a topological quotient map Q of X onto Y, then the topology of Y is called the *quotient topology induced by* Q. Thus if a surjective map $Q\colon X \to Y$ (i.e., $Q(Q^{-1}(B)) = B$ for every $B \subseteq Y$) is a continuous (i.e., $B \in \mathcal{T}_Y \Longrightarrow Q^{-1}(B) \in \mathcal{T}_X$) open map (i.e., $A \in \mathcal{T}_X \Longrightarrow Q(A) \in \mathcal{T}_Y$) of a topological space X onto a topological space Y, then Q is a topological quotient map. This definition does not involve the norm condition of Definition 5.12, and so imposes no restriction to norms (since norms play no role in general topological spaces). But a quotient transformation of a normed space $\mathcal{X}$ onto a normed space $\mathcal{Y}$ is a topological quotient map by Theorem 5.13; in particular, if $\mathcal{M}$ is a proper subspace of $\mathcal{X}$, then the natural quotient map $\pi\colon \mathcal{X} \to \mathcal{X}/\mathcal{M}$ is a topological quotient map and the topology of $\mathcal{X}/\mathcal{M}$ is the quotient topology induced by π.

5.4 Additional Propositions

It is a (true) cliché to say that the three fundamental theorems in functional analysis are the Open Mapping Theorem, the Banach–Steinhaus Theorem and the Hahn–Banach Theorem. (These are all classical and really fundamental results found in essentially all books on functional analysis). Those aspects of the Hahn–Banach Theorem required in this book have been discussed in Remark 5.2 (and will be discussed again in the next chapter). The other two (the Open Mapping and the Banach–Steinhaus Theorems, that is) will be stated in this section together with some of their immediate consequences (only those that are required in the sequel). Recall: a mapping between topological spaces is *continuous* if the inverse image of every open set is open, a mapping between topological spaces is *open* if the image of every open set is open.

Proposition 5.A. (Open Mapping Theorem). *If $\mathcal{X}$ and $\mathcal{Y}$ are Banach spaces and $T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ is surjective, then T is an open mapping.*

A crucial corollary of the Open Mapping Theorem says that *between Banach spaces, an invertible bounded linear transformation has a bounded inverse.*

Proposition 5.B. (Inverse Mapping Theorem). *If $\mathcal{X}$ and $\mathcal{Y}$ are Banach spaces and $T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ is injective and surjective, then $T^{-1} \in \mathcal{B}[\mathcal{Y}, \mathcal{X}]$.*

This is partially responsible for the next result, which plays a major role in establishing the notion of spectrum of an operator. Recall: a linear transformation T of a normed space $\mathcal{X}$ into a normed space $\mathcal{Y}$ is *bounded* if there is a constant $\beta > 0$ such that $\|Tx\| \le \beta\|x\|$ for every $x \in \mathcal{X}$, and it is *bounded below* if there is a constant $\alpha > 0$ such that $\alpha\|x\| \le \|Tx\|$ for every $x \in \mathcal{X}$.

Proposition 5.C. (Bounded Inverse Theorem). *Let $\mathcal{X}$ and $\mathcal{Y}$ be Banach spaces and take any $T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$. The following assertions are equivalent.*

(a) *T has a bounded inverse on its range.*

(b) *T is bounded below.*

(c) *T is injective and has a closed range.*

A further application of the Open Mapping Theorem that will be required later is the Closed Graph Theorem. The *graph* of a linear transformation L between linear spaces $\mathcal{X}$ and $\mathcal{Y}$ is the subset of the Cartesian product $\mathcal{X}\times\mathcal{Y}$

$$G_L = \big\{(x, y) \in \mathcal{X}\times\mathcal{Y}\colon\ y = Lx\big\} = \big\{(x, Lx) \in \mathcal{X}\times\mathcal{Y}\colon\ \text{for every } x \in \mathcal{X}\big\}.$$

The set G_L is said to be closed if it is closed in the metric space $(\mathcal{X}\times\mathcal{Y}, d)$, or in the normed space $(\mathcal{X}\oplus\mathcal{Y}, \|\cdot\|)$, which boils down to the same thing (see Section 5.1). There is, however, a significant difference: the set G_L is a linear manifold of the normed space $(\mathcal{X}\oplus\mathcal{Y}, \|\cdot\|)$ by linearity of L. If $\mathcal{X}$ and $\mathcal{Y}$ are Banach spaces, then their direct sum $\mathcal{X}\oplus\mathcal{Y}$ equipped with any of the standard norms

$\|\cdot\|$ of Section 5.1 is again a Banach space (see, e.g., [52, Example 4.E]). So if the graph G_L of a linear transformation L is regarded as a subset of the direct sum $\mathcal{X}\oplus\mathcal{Y}$ of Banach spaces $\mathcal{X}$ and $\mathcal{Y}$ equipped with any of those standard norms, and if G_L is closed, then G_L is a Banach space.

Proposition 5.D. (Closed Graph Theorem). *If $\mathcal{X}$ and $\mathcal{Y}$ are Banach spaces and $T\colon \mathcal{X}\to\mathcal{Y}$ is linear, then T is continuous if and only if G_T is closed.*

Another useful inverse theorem establishes a power series expansion for $(\lambda I - T)^{-1}$ if $\|T\| < |\lambda|$ (this condition will be weakened in Proposition 5.P).

Proposition 5.E. (Neumann Expansion). *If $T\in\mathcal{B}[\mathcal{X}]$ is an operator on a Banach space $\mathcal{X}$ and λ is any scalar such that $\|T\| < |\lambda|$, then $\lambda I - T$ has an inverse in $\mathcal{B}[\mathcal{X}]$ given by the following uniformly convergent series:*

$$(\lambda I - T)^{-1} = \tfrac{1}{\lambda}\sum\nolimits_{k=0}^{\infty}\left(\tfrac{T}{\lambda}\right)^k.$$

Proposition 5.F. (Banach–Steinhaus Theorem). *Let $\{T_\omega\}_{\omega\in\Omega}$ be an indexed family (indexed by an arbitrary index set Ω) of bounded linear transformations of a Banach space $\mathcal{X}$ into a normed space $\mathcal{Y}$. If $\sup_{\omega\in\Omega}\|T_\omega x\| < \infty$ for every $x\in\mathcal{X}$, then $\sup_{\omega\in\Omega}\|T_\omega\| < \infty$.*

A sequence of linear transformations of a normed space $\mathcal{X}$ into a normed space $\mathcal{Y}$ (i.e., an $\mathcal{L}[\mathcal{X},\mathcal{Y}]$-valued sequence) $\{L_n\}$ is *pointwise convergent* if for every $x\in\mathcal{X}$ the $\mathcal{Y}$-valued sequence $\{L_n x\}$ converges in $\mathcal{Y}$ to, say, $y_x\in\mathcal{Y}$. Since the limit is unique, this defines a transformation $L\colon\mathcal{X}\to\mathcal{Y}$ given by $L(x) = y_x$ for every $x\in\mathcal{X}$. It is readily verified that L is linear. Therefore *an $\mathcal{L}[\mathcal{X},\mathcal{Y}]$-valued sequence $\{L_n\}$ is pointwise convergent if and only if there exists $L\in\mathcal{L}[\mathcal{X},\mathcal{Y}]$ such that $\|(L_n - L)x\|\to 0$ for every $x\in\mathcal{X}$*. Now consider a sequence $\{T_n\}$ of bounded linear transformations of a normed space $\mathcal{X}$ into a normed space $\mathcal{Y}$ (i.e., a $\mathcal{B}[\mathcal{X},\mathcal{Y}]$-valued sequence). An important application of the Banach–Steinhaus Theorem ensures that if $\mathcal{X}$ is a Banach space and $\{T_n\}$ is pointwise convergent, then the limit T is bounded.

Proposition 5.G. *If $\mathcal{X}$ is a Banach space and $\mathcal{Y}$ is a normed space, then a $\mathcal{B}[\mathcal{X},\mathcal{Y}]$-valued sequence $\{T_n\}$ is pointwise convergent if and only if there is a $T\in\mathcal{B}[\mathcal{X},\mathcal{Y}]$ such that $\|(T_n - T)x\|\to 0$ for every $x\in\mathcal{X}$.*

Let $\mathcal{X}$ and $\mathcal{Y}$ be a normed spaces. If a $\mathcal{B}[\mathcal{X},\mathcal{Y}]$-valued sequence $\{T_n\}$ is such that there exists $T\in\mathcal{B}[\mathcal{X},\mathcal{Y}]$ for which $\|(T_n - T)x\|\to 0$ for every $x\in\mathcal{X}$, then the sequence $\{T_n\}$ is *strongly convergent*, or it is said to *converge strongly* to T (notation: $T_n \xrightarrow{s} T$). If $\{T_n\}$ converges to T in the induced uniform norm on $\mathcal{B}[\mathcal{X},\mathcal{Y}]$ (i.e., if $\|T_n - T\|\to 0$), then $\{T_n\}$ is *uniformly convergent*, or it is said to *converge uniformly* to $T\in\mathcal{B}[\mathcal{X},\mathcal{Y}]$ (notation: $T_n \xrightarrow{u} T$). It is clear that

$$T_n \xrightarrow{u} T \quad\Longrightarrow\quad T_n \xrightarrow{s} T,$$

and the converse fails. If $\{T_n\}$ is strongly convergent, then it is strongly bounded (i.e., $\|(T_n - T)x\|\to 0$ implies $\|T_n x\|\to\|Tx\|$, which in turn implies

$\sup_n \|T_n x\| < \infty$ for every $x \in \mathcal{X}$). If $\mathcal{X}$ is a Banach space, then the Banach–Steinhaus Theorem ensures that strong boundedness implies uniform boundedness (i.e., $\sup_n \|T_n x\| < \infty$ for every $x \in \mathcal{X}$ implies $\sup_n \|T_n\| < \infty$, and the converse holds trivially even for a plain normed space $\mathcal{X}$). Therefore if $\mathcal{X}$ is a Banach space, then the Banach–Steinhaus Theorem ensures that pointwise convergence coincides with strong convergence (Proposition 5.G), and strong boundedness coincides with uniform boundedness (Proposition 5.F). Thus

$$T_n \xrightarrow{s} T \quad \Longrightarrow \quad \sup_n \|T_n\| < \infty \ \text{ whenever } \mathcal{X} \text{ is Banach.}$$

An important special case of the above notions refers to power sequences of operators. An operator $T \in \mathcal{B}[\mathcal{X}]$ on a normed space $\mathcal{X}$ is *uniformly stable* or *strongly stable* if the power sequence $\{T^n\}$ converges uniformly or strongly to the null operator, and it is *power bounded* if the power sequence is bounded. In other words, if $T^n \xrightarrow{u} O$, $T^n \xrightarrow{s} O$, and $\sup_n \|T^n\| < \infty$, respectively.

With $\mathcal{X}$ and $\mathcal{Y}$ being normed spaces, let $\mathcal{B}_0[\mathcal{X},\mathcal{Y}]$ stand for the collection of *finite-rank* bounded linear transformations and $\mathcal{B}_\infty[\mathcal{X},\mathcal{Y}]$ for the collection of *compact* linear transformations (for the definition of compact linear transformation, see, e.g., [52, Section 4.9]). Recall: $\mathcal{B}_0[\mathcal{X},\mathcal{Y}] \subseteq \mathcal{B}_\infty[\mathcal{X},\mathcal{Y}] \subseteq \mathcal{B}[\mathcal{X},\mathcal{Y}]$. As before, set $\mathcal{B}_0[\mathcal{X}] = \mathcal{B}_0[\mathcal{X},\mathcal{X}]$, $\mathcal{B}_\infty[\mathcal{X}] = \mathcal{B}_\infty[\mathcal{X},\mathcal{X}]$, and $\mathcal{B}[\mathcal{X}] = \mathcal{B}[\mathcal{X},\mathcal{X}]$.

Proposition 5.H. *Let $\mathcal{X}$ be a normed space and $\mathcal{Y}$ be a Banach space. If a sequence of finite-rank bounded linear transformations converges uniformly, then the limit is compact (i.e., if $T_n \xrightarrow{u} T$ with $T_n \in \mathcal{B}_0[\mathcal{X},\mathcal{Y}]$, then $T \in \mathcal{B}_\infty[\mathcal{X},\mathcal{Y}]$).*

A metric space is *separable* if it has a countable dense subset.

Proposition 5.I. *The range of a compact linear transformation is separable.*

A sequence $\{x_k\}$ of vectors in a normed space $\mathcal{X}$ is a *Schauder basis* for $\mathcal{X}$ if for each x in $\mathcal{X}$ there is a *unique* sequence of scalars $\{\alpha_k(x)\}$ such that $x = \sum_k \alpha_k(x) x_k$ (i.e., $x = \lim_n \sum_{k=1}^n \alpha_k(x) x_k$). Every Schauder basis for $\mathcal{X}$ is a sequence of linearly independent vectors in $\mathcal{X}$ that spans $\mathcal{X}$ (i.e., $(\text{span}\,\{x_k\})^- = \mathcal{X}$). So if a normed space has a Schauder basis, then it is separable. The converse of Proposition 5.H holds if the Banach space $\mathcal{Y}$ has a Schauder basis.

Proposition 5.J. *If $\mathcal{Y}$ is a Banach space with a Schauder basis and $\mathcal{X}$ is any normed space, then every compact transformation $T \in \mathcal{B}_\infty[\mathcal{X},\mathcal{Y}]$ is the uniform limit of a sequence of finite-rank transformations $T_n \in \mathcal{B}_0[\mathcal{X},\mathcal{Y}]$.*

Does every separable (infinite-dimensional) Banach space have a Schauder basis? This is a famous question, raised by Banach himself in the early 1930s, that remained open for a long period. Each separable Banach space that ever came up in analysis during that period (and this includes all classical examples) had a Schauder basis. The surprising negative answer to that question was given by Enflo [25] in 1973, who constructed a separable Banach space that has no Schauder basis. In fact, he exhibited a separable (and reflexive) Banach space $\mathcal{X}$ for which $\mathcal{B}_0[\mathcal{X}]$ is not dense in $\mathcal{B}_\infty[\mathcal{X}]$, so that there exist compact

operators on $\mathcal{X}$ that are not the (uniform) limit of finite-rank operators. Hence the converse of Proposition 5.H fails in general, and by Proposition 5.J such an $\mathcal{X}$ is a separable Banach space without a Schauder basis.

Proposition 5.J induces a pair of central notions. A bounded linear transformation is *approximable* if it is the uniform limit of a sequence of finite-rank bounded linear transformations. A Banach space $\mathcal{Y}$ has the *approximation property* if every compact linear transformation of an arbitrary normed space $\mathcal{X}$ into $\mathcal{Y}$ is approximable. In other words, a Banach space $\mathcal{Y}$ has the approximation property if every $T \in \mathcal{B}_\infty[\mathcal{X},\mathcal{Y}]$ for every normed space $\mathcal{X}$ is such that there is a $\mathcal{B}_0[\mathcal{X},\mathcal{Y}]$-valued sequence $\{T_n\}$ for which $T_n \xrightarrow{u} T$. Equivalently, *a Banach space $\mathcal{Y}$ has the approximation property if $\mathcal{B}_0[\mathcal{X},\mathcal{Y}]$ is dense in $\mathcal{B}_\infty[\mathcal{X},\mathcal{Y}]$.*

By Proposition 5.J (and Proposition 5.I) we may infer the following results:

(i) *Banach spaces with a Schauder basis have the approximation property.*

(ii) *Hilbert spaces have the approximation property.*

We explore the approximation property further in Remark 8.14 (Chapter 8).

From now on up to Proposition 5.P, $\mathcal{X}$ is a *complex Banach space* and T is an *operator in the unital complex Banach algebra* $\mathcal{B}[\mathcal{X}]$ (of *bounded linear* transformations of $\mathcal{X}$ into itself). Let I stand for the identity operator in $\mathcal{B}[\mathcal{X}]$.

The *resolvent set* $\rho(T)$ of an operator T in $\mathcal{B}[\mathcal{X}]$ is the set of all scalars λ in $\mathbb{C}$ for which the operator $\lambda I - T$ in $\mathcal{B}[\mathcal{X}]$ is invertible (i.e., has a bounded inverse on $\mathcal{X}$ according to the Inverse Mapping Theorem):

$$\begin{aligned}\rho(T) &= \big\{\lambda \in \mathbb{C}\colon\ \lambda I - T \text{ has an inverse in } \mathcal{B}[\mathcal{X}]\big\}\\ &= \big\{\lambda \in \mathbb{C}\colon\ \mathcal{N}(\lambda I - T) = \{0\} \ \text{ and } \ \mathcal{R}(\lambda I - T) = \mathcal{X}\big\}.\end{aligned}$$

The *spectrum* $\sigma(T)$ of $T \in \mathcal{B}[\mathcal{X}]$ is the complement of the resolvent set,

$$\begin{aligned}\sigma(T) &= \mathbb{C} \setminus \rho(T)\\ &= \big\{\lambda \in \mathbb{C}\colon\ \lambda I - T \text{ has no inverse in } \mathcal{B}[\mathcal{X}]\big\}\\ &= \big\{\lambda \in \mathbb{C}\colon\ \mathcal{N}(\lambda I - T) \neq \{0\} \ \text{ or } \ \mathcal{R}(\lambda I - T) \neq \mathcal{X}\big\}.\end{aligned}$$

Proposition 5.K. *$\rho(T)$ is nonempty and open, and $\sigma(T)$ is compact (i.e., closed and bounded in $\mathbb{C}$). If $\mathcal{X}$ is nonzero, then $\sigma(T)$ is nonempty.*

A complex number λ for which $\lambda I - T$ is not injective (i.e., for which $\lambda I - T$ has no left inverse or, equivalently, for which $\mathcal{N}(\lambda I - T) \neq \{0\}$) is an *eigenvalue* of T. The set of all eigenvalues of T is the *point spectrum* of T,

$$\sigma_P(T) = \big\{\lambda \in \mathbb{C}\colon\ \mathcal{N}(\lambda I - T) \neq \{0\}\big\}.$$

If $\lambda I - T$ is injective, then $\lambda I - T$ has an inverse on its range which may or not be bounded. By the Bounded Inverse Theorem, it is bounded if and only if $\mathcal{R}(\lambda I - T)$ is closed. If it is not bounded but has a dense range, then set

$$\sigma_C(T) = \big\{\lambda \in \mathbb{C}\colon \mathcal{N}(\lambda I - T) = \{0\},\ \mathcal{R}(\lambda I - T)^- = \mathcal{X} \text{ and } \mathcal{R}(\lambda I - T) \neq \mathcal{X}\big\},$$

which is the *continuous spectrum* of T. The *residual spectrum* of T is the set

$$\begin{aligned}\sigma_R(T) &= \sigma(T)\backslash(\sigma_P(T) \cup \sigma_C(T)) \\ &= \big\{\lambda \in \mathbb{C}\colon \mathcal{N}(\lambda I - T) = \{0\} \text{ and } \mathcal{R}(\lambda I - T)^- \neq \mathcal{X}\big\}.\end{aligned}$$

It is clear that $\{\sigma_P(T), \sigma_C(T), \sigma_R(T)\}$ forms a partition of $\sigma(T)$, referred to as the *classical partition of the spectrum*. If Λ is a subset of $\mathbb{C}$, and $p\colon \mathbb{C} \to \mathbb{C}$ is a polynomial with complex coefficients, then set $p(\Lambda) = \{p(\lambda) \in \mathbb{C}\colon \lambda \in \Lambda\}$.

Proposition 5.L. (Spectral Mapping Theorem for Polynomials). *If p is an arbitrary polynomial with complex coefficients, then*

$$\sigma(p(T)) = p(\sigma(T)).$$

The *spectral radius* of an operator T in $\mathcal{B}[\mathcal{X}]$ is the nonnegative number $r(T) = \sup_{\lambda\in\sigma(T)} |\lambda|$. An application of the Spectral Mapping Theorem for Polynomials and the Neumann expansion yields the following expressions.

Proposition 5.M. *For every nonnegative integer n*

$$r(T)^n = r(T^n) \leq \|T^n\| \leq \|T\|^n \quad \textit{and so} \quad \sup_n \|T^n\| < \infty \Longrightarrow r(T) \leq 1.$$

Proposition 5.N. (Gelfand–Beurling Formula). *$\{\|T^n\|^{\frac{1}{n}}\}$ converges and*

$$r(T) = \lim_n \|T^n\|^{\frac{1}{n}}.$$

The Gelfand–Beurling Formula fully characterizes uniform stability.

Proposition 5.O. *The following assertions are pairwise equivalent.*

(a) $T^n \xrightarrow{u} O$.

(b) $r(T) < 1$.

(c) $\|T^n\| \leq \beta\alpha^n$ *for every* $n \geq 0$, *for some* $\beta \geq 1$ *and some* $\alpha \in (0, 1)$.

Proposition 5.P below extends the Neumann expansion. The infinite series $\sum_{k=0}^{\infty}(\frac{T}{\lambda})^k$ is said to converge uniformly or strongly if the sequence of partial sums $\big\{\sum_{k=0}^{n}(\frac{T}{\lambda})^k\big\}_{n=0}^{\infty}$ converges uniformly or strongly, respectively. In either case we denote by $\sum_{k=0}^{\infty}(\frac{T}{\lambda})^k$ the (uniform or strong) limit operator in $\mathcal{B}[\mathcal{X}]$.

Proposition 5.P. *Take any operator T and any nonzero scalar λ.*

(a) $r(T) < |\lambda|$ *if and only if* $\sum_{k=0}^{\infty}(\frac{T}{\lambda})^k$ *converges uniformly. In this case,* λ *lies in* $\rho(T)$ *and* $(\lambda I - T)^{-1} = \frac{1}{\lambda}\sum_{k=0}^{\infty}(\frac{T}{\lambda})^k$.

(b) *If* $r(T) = |\lambda|$ *and* $\sum_{k=0}^{\infty}(\frac{T}{\lambda})^k$ *converges strongly, then* λ *lies in* $\rho(T)$ *and* $(\lambda I - T)^{-1} = \frac{1}{\lambda}\sum_{k=0}^{\infty}(\frac{T}{\lambda})^k$.

(c) *If* $|\lambda| < r(T)$, *then* $\sum_{k=0}^{\infty}(\frac{T}{\lambda})^k$ *does not converge strongly.*

As in Section 2.4, an *inner product* on a linear space $\mathcal{X}$ is a Hermitian symmetric sesquilinear form $\langle \cdot\,;\cdot \rangle \colon \mathcal{X}\times\mathcal{X}\to\mathbb{F}$ generating a positive quadratic form $\|\cdot\| \colon \mathcal{X}\to\mathbb{R}$ given by $\|x\| = \langle x\,;x\rangle^{1/2}$ for every $x \in \mathcal{X}$, which is a norm on $\mathcal{X}$ (and the inner product is a continuous sesquilinear form by the *Schwartz inequality*: $|\langle x\,;y\rangle| \le \|x\|\,\|y\|$ for every $x, y \in \mathcal{X}$). The pair $(\mathcal{X}, \langle \cdot\,;\cdot\rangle)$ is an *inner product' space* — which is a special case of a normed space $(\mathcal{X}, \|\cdot\|)$. A *Hilbert space* is a complete inner product space, which is complete with respect to the norm generated by the inner product. Thus a Hilbert space is a Banach space whose norm is induced by an inner product. If $\mathbb{F} = \mathbb{C}$ or $\mathbb{F} = \mathbb{R}$, then we refer to a *complex* or *real Hilbert space*. Two vectors x, y in an inner product space $\mathcal{X}$ are *orthogonal* (notation: $x \perp y$) if their inner product $\langle x\,;y\rangle$ is zero. If $\mathcal{M}$ is a linear manifold of an inner product space $\mathcal{X}$, then its *orthogonal complement* $\mathcal{M}^\perp$ is the (closed) subspace made up of all vectors that are orthogonal to every vector in $\mathcal{M}$; that is, $\mathcal{M}^\perp = \{v \in \mathcal{X} \colon \langle v\,;u\rangle = 0 \text{ for every } u \in \mathcal{M}\}$.

The next proposition is the central result of Hilbert-space geometry.

Proposition 5.Q. (Projection Theorem). *If $\mathcal{M}$ is a subspace of a Hilbert space $\mathcal{X}$, then*

$$\mathcal{X} = \mathcal{M} + \mathcal{M}^\perp.$$

Since $\mathcal{M}$ and $\mathcal{M}^\perp$ are trivially algebraically disjoint (i.e., $\mathcal{M} \cap \mathcal{M}^\perp = \{0\}$), then Proposition 5.Q ensures that the (closed) subspace $\mathcal{M}^\perp$ is a complement (an orthogonal complement) of an arbitrary (closed) subspace $\mathcal{M}$ of $\mathcal{X}$, so that every subspace of a Hilbert space is complemented, and therefore all Hilbert spaces are complemented (and, as we saw Remark 5.3(a), only they are). Since $\mathcal{M}$ and $\mathcal{M}^\perp$ are algebraic complements, their ordinary sum is isomorphic to their direct sum, and the natural isomorphism between them is, in fact, an isometric isomorphism — where the norm of $x = u + v$ with u in $\mathcal{M}$ and v in $\mathcal{M}^\perp$ is such that $\|x\|^2 = \|u\|^2 + \|v\|^2$ (by the *Pythagorean Theorem*). Also, the inner product in a direct sum of orthogonal linear manifolds, say in $\mathcal{M} \oplus \mathcal{M}^\perp$, is given by $\langle (u_1, v_1)\,;(u_2, v_2)\rangle = \langle u_1\,;u_2\rangle + \langle v_1\,;v_2\rangle$ for every u_1, u_2 in $\mathcal{M}$ and v_1, v_2 in $\mathcal{M}^\perp$. An isometric isomorphism between inner product spaces preserves inner product, is called a *unitary transformation*, and isometrically isomorphic inner product spaces are called *unitarily equivalent*. Proposition 5.Q can be rewritten (with $\cong$ replaced by $=$ as we did before) in the form

$$\mathcal{X} = \mathcal{M} \oplus \mathcal{M}^\perp.$$

(Here $\cong$, which has been replaced by $=$, means isometrically isomorphic, i.e., unitarily equivalent.) If $\mathcal{M}$ and $\mathcal{N}$ are *orthogonal subspaces* (or *orthogonal linear manifolds*) of an inner product space (i.e., if $\langle u\,;v\rangle = 0$ for every $u \in \mathcal{M}$ and every $v \in \mathcal{N}$ — notation: $\mathcal{M} \perp \mathcal{N}$), then $\mathcal{M} \oplus \mathcal{N}$ is referred to as the *orthogonal direct sum* of $\mathcal{M}$ and $\mathcal{N}$ (which is an internal direct sum). Since $\mathcal{M} \perp \mathcal{M}^\perp$ tautologically, the above decomposition is referred to as *orthogonal decomposition*, and this leads to the notation $\mathcal{M}^\perp = \mathcal{X} \ominus \mathcal{M}$ whenever $\mathcal{M}$ is a *subspace of a Hilbert space* $\mathcal{X}$. An *orthogonal projection* E on an inner product space $\mathcal{X}$ is a projection for which $\mathcal{R}(E) \perp \mathcal{N}(E)$.

Proposition 5.R. (i) *If* $E \in \mathcal{L}[\mathcal{X}]$ *is a nonzero orthogonal projection on an inner product space* $\mathcal{X}$*, then* $E \in \mathcal{B}[\mathcal{X}]$ *and* $\|E\| = 1$*. Moreover,* (ii) *if* $E \in \mathcal{B}[\mathcal{X}]$ *is a nonzero projection on a Hilbert space* $\mathcal{X}$*, then* E *is orthogonal if and only if* $\|E\| = 1$*. Furthermore,* (iii) *if* $\mathcal{M}$ *is a subspace of a Hilbert space* $\mathcal{X}$*, then there is a unique orthogonal projection* $E \in \mathcal{B}[\mathcal{X}]$ *with* $\mathcal{R}(E) = \mathcal{M}$*.*

If $\mathcal{M}$ is a subspace of a Banach space $\mathcal{X}$, then a *minimal projection* E onto $\mathcal{M}$ (i.e., with $\mathcal{R}(E) = \mathcal{M}$) is a continuous projection $E \in \mathcal{B}[\mathcal{X}]$ such that $\|E\| \leq \|P\|$ for every projection $P \in \mathcal{B}[\mathcal{X}]$ with $\mathcal{R}(P) = \mathcal{M}$. A subspace $\mathcal{M}$ is 1-*complemented* if there is a projection $E \in \mathcal{B}[\mathcal{X}]$ with $\mathcal{R}(E) = \mathcal{M}$ and $\|E\| = 1$ (which is minimal since $\|P\| \geq 1$ for every projection $P \in \mathcal{B}[\mathcal{X}]$). So part (b) of Theorem 5.4 says: *if* $\mathcal{M}^-$ *is* 1-*complemented, then* $\|\widehat{T}\| = \|T\|$.

Proposition 5.S. *In an arbitrary Banach space, minimal projections on complemented subspaces do not necessarily exist, are not necessarily unique if they exist, and there are minimal projections of norm greater than one.*

In a Hilbert space, however, Proposition 5.R says that minimal projections exist, are orthogonal, unique, and have norm 1.

A set of mutually orthogonal vectors is an *orthogonal set*. It is an *orthonormal set* if all vectors in it have norm one. Every Hilbert space has an *orthonormal basis* (i.e., an orthonormal set whose span is dense in the space). A Hilbert space is *separable* if and only if it has a countable orthonormal basis.

Proposition 5.T. (The Fourier Series Theorem). *Let* $\{e_\gamma\}_{\gamma\in\Gamma}$ *be an orthonormal set in a Hilbert space* $\mathcal{X}$*. The assertions below are pairwise equivalent.*

(a) $\{e_\gamma\}_{\gamma\in\Gamma}$ *is an orthonormal basis for* $\mathcal{X}$*.*

(b) $x = \sum_{\gamma\in\Gamma}\langle x\,;e_\gamma\rangle e_\gamma$ *for every* $x \in \mathcal{X}$*, referred to as the* Fourier series expansion *of* x*, which is unique in terms of* $\{e_\gamma\}_{\gamma\in\Gamma}$*, and the scalars in* $\{\langle x\,;e_\gamma\rangle\}_{\gamma\in\Gamma}$ *are the* Fourier coefficients *of* x *with respect to* $\{e_\gamma\}_{\gamma\in\Gamma}$*.*

(c) $\langle x\,;y\rangle = \sum_{\gamma\in\Gamma}\langle x\,;e_\gamma\rangle\overline{\langle y\,;e_\gamma\rangle}$ *for every pair of vectors* x, y *in* $\mathcal{X}$*.*

(d) $\|x\|^2 = \sum_{\gamma\in\Gamma}|\langle x\,;e_\gamma\rangle|^2$ *for every* $x \in \mathcal{X}$*. This is the* Parseval identity*.*

(e) *Every linear manifold of* $\mathcal{X}$ *that includes* $\{e_\gamma\}_{\gamma\in\Gamma}$ *is dense in* $\mathcal{X}$*.*

It is readily verified that a bounded linear functional f on an inner product space $\mathcal{X}$ (i.e., an $f \in \mathcal{X}^*$) is naturally associated with each vector $y \in \mathcal{X}$ and $\|f\| = \|y\|$. The next result is central in the Hilbert-space theory and asserts that if $\mathcal{X}$ is complete (i.e., if $\mathcal{X}$ is a Hilbert space), then the converse holds.

Proposition 5.U. (The Riesz Representation Theorem). *If* $\mathcal{X}$ *is a Hilbert space, then for every* f *in the dual* $\mathcal{X}^*$ *of* $\mathcal{X}$ *there is a unique* y *in* $\mathcal{X}$ *such that*

$$f(x) = \langle x\,;y\rangle \quad \text{for every} \quad x \in \mathcal{X}.$$

Moreover, $\|f\| = \|y\|$*. Such a unique vector* y *in* $\mathcal{X}$ *is called the* Riesz representation *of the functional* f *in* $\mathcal{X}^* = \mathcal{B}[\mathcal{X}, \mathbb{F}]$*.*

The result below is a major corollary of the Riesz Representation Theorem.

Proposition 5.V. *For every Hilbert space $\mathcal{X}$ there exists a surjective (not necessarily linear) isometry $\Psi\colon \mathcal{X}^* \to \mathcal{X}$ of $\mathcal{X}^*$ onto $\mathcal{X}$, which is additive and conjugate homogeneous (i.e., $\Psi(\alpha f) = \overline{\alpha}\Psi(f)$ for every $f \in \mathcal{X}^*$ and $\alpha \in \mathbb{F}$).*

The Riesz Representation Theorem allows us to redefine a notion of *adjoint* for bounded linear transformations of a Hilbert space into an inner product space. If $\mathcal{X}$ is a Hilbert space and $\mathcal{Y}$ is an inner product space, then the *Hilbert-space adjoint* of $T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ is the (unique) operator $T^* \in \mathcal{B}[\mathcal{Y}, \mathcal{X}]$ such that $\langle Tx\,;y\rangle = \langle x\,;T^*y\rangle$ for every $x \in \mathcal{X}$ and $y \in \mathcal{Y}$ — the left-hand side inner product equips the inner product space $\mathcal{Y}$ and the right-hand side the Hilbert space $\mathcal{X}$. Basic properties of the adjoint, when $\mathcal{Y}$ is also a Hilbert space, are:

$$T^{**} = T \quad \text{and} \quad \|T^*\|^2 = \|T^*T\| = \|TT^*\| = \|T\|^2.$$

Remark 5.W. *On the Notions of Normed-Space and Hilbert-Space Adjoints.*

(a) The Riesz Representation Theorem and its corollary in Proposition 5.V ensure that every Hilbert space is isometrically equivalent to its dual. (In particular, every real Hilbert space is isometrically isomorphic — i.e., unitarily equivalent — to its dual). The Riesz Representation Theorem also ensures that if $\mathcal{X}$ is a Hilbert space, then for each $T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ into any inner product space $\mathcal{Y}$ there exists a unique $T^* \in \mathcal{B}[\mathcal{Y}, \mathcal{X}]$ such that $\langle Tx\,;y\rangle = \langle x\,;T^*y\rangle$ for every $x \in \mathcal{X}$ and $y \in \mathcal{Y}$, as above. This $T^* \in \mathcal{B}[\mathcal{Y}, \mathcal{X}]$ is the Hilbert-space adjoint of $T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$. In particular, for $\mathcal{Y} = \mathcal{X}$ such an inner product identity defines the Hilbert-space adjoint $T^* \in \mathcal{B}[\mathcal{X}]$ of every operator $T \in \mathcal{B}[\mathcal{X}]$ acting on a Hilbert space $\mathcal{X}$. But the Riesz Representation Theorem works only if $\mathcal{X}$ is a Hilbert space; not if $\mathcal{X}$ is an inner product space. Outcome: *there is no Hilbert-space adjoint for operators acting on inner product spaces*.

(b) Let $\mathcal{X}$ and $\mathcal{Y}$ be normed spaces. The *normed-space* adjoint $T^* \in \mathcal{B}[\mathcal{Y}^*, \mathcal{X}^*]$ of $T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ was defined in Section 5.1. It is the restriction of the algebraic adjoint $T^\sharp \in \mathcal{L}[\mathcal{Y}^\sharp, \mathcal{X}^\sharp]$ (cf. Section 1.1) to $\mathcal{Y}^* \subseteq \mathcal{Y}^\sharp$, which is the unique transformation in $\mathcal{B}[\mathcal{Y}^*, \mathcal{X}^*]$ for which $T^*g = g\,T$ for every $g \in \mathcal{Y}^*$ (i.e., for which $(T^*g)(x) = g(Tx)$ — also written as $\langle x\,;T^*g\rangle = \langle Tx\,;g\rangle$ which, of course, does not mean inner product — for every $g \in \mathcal{Y}^*$ and every $x \in \mathcal{X}$). See, e.g., [70, Definition 3.13]. In particular, for $\mathcal{Y} = \mathcal{X}$ the normed-space adjoint of an operator $T \in \mathcal{B}[\mathcal{X}]$ on a normed space $\mathcal{X}$ is the operator $T^* \in \mathcal{B}[\mathcal{X}^*]$ on the Banach space $\mathcal{X}^*$. Such a definition holds tautologically if $\mathcal{X}$ is a Banach space. If $\mathcal{X}$ is a Hilbert space, then we may compare the normed-space adjoint with the Hilbert-space adjoint (which makes sense only if $\mathcal{X}$ is a Hilbert space).

(c) This is a major difference between normed-space and Hilbert-space adjoints: normed-space adjoints are defined for transformations $T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ acting on any normed space $\mathcal{X}$, while Hilbert-space adjoints are defined only for transformations acting on a Hilbert space $\mathcal{X}$. (For the special case of $\mathcal{Y} = \mathcal{X}$, such a difference is linked to the fact that the normed-space adjoint $T^* \in \mathcal{B}[\mathcal{X}^*]$

always acts on a Banach space $\mathcal{X}^*$, while the Hilbert-space adjoint $T^* \in \mathcal{B}[\mathcal{X}]$ acts on the same Hilbert space $\mathcal{X}$ upon which T acts). A minor difference is that for the normed-space adjoint $(\alpha T)^* = \alpha T^*$, while for the Hilbert-space adjoint $(\alpha T)^* = \overline{\alpha} T^*$, for every $\alpha \in \mathbb{F}$. (By the way, the other basic properties listed in Section 5.1 coincide for both notions, viz., $\|T^*\| = \|T\|$, $(T+S)^* = T^* + S^*$, $(TS)^* = S^* T^*$, and I^* is the identity operator if $\mathcal{Y} = \mathcal{X}$.)

(d) A special instance of such a major difference is this. If $T \in \mathcal{B}[\mathcal{X}]$ acts on any normed space (in particular, on an inner product space), then its normed-space adjoint is such that $\widehat{T^*} = \widehat{T}^*$. (Actually, $\widehat{T}^* \cong T^* = \widehat{T^*}$ according to Theorem 5.1(f) and, since T^* acts on the Banach space $\mathcal{X}^*$, the identity $T^* = \widehat{T^*}$ is trivial — which holds even for the Hilbert-space adjoint when T^* acts on a Hilbert space $\mathcal{X}$.) Now suppose T acts on an incomplete inner product space $\mathcal{X}$. We can extend T to get $\widehat{T}$ on the Hilbert space $\widehat{\mathcal{X}}$, and then take its Hilbert-space adjoint $\widehat{T}^*$. But we cannot take the Hilbert-space adjoint of T (because it does not act on a Hilbert space), and hence the expression $\widehat{T^*}$ does not make sense in this case. Consequently, *the identity* $\widehat{T^*} = \widehat{T}^*$ *only makes nontrivial sense* (i.e., for $\mathcal{X}$ not complete) *when* T^* *stands for the normed-space adjoint.*

For an arbitrary subset Λ of the complex plane $\mathbb{C}$ set $\Lambda^* = \{\overline{\lambda} \in \mathbb{C} : \lambda \in \Lambda\}$, the set of all complex conjugates of elements from Λ.

Proposition 5.X. *If* $T \in \mathcal{B}[\mathcal{X}]$ *where* $\mathcal{X}$ *is a complex Hilbert space, then*

$$\sigma(T) = \sigma(T^*)^*, \quad \sigma_C(T) = \sigma_C(T^*)^*, \quad \sigma_R(T) = \sigma_P(T^*)^* \backslash \sigma_P(T).$$

Remark 5.Y. *On Point and Residual Spectra.*

(a) The point spectrum of a linear transformation $L \in \mathcal{L}[\mathcal{X}]$ acting on a linear space $\mathcal{X}$ is the set of its eigenvalues, viz., $\sigma_P(L) = \{\lambda \in \mathbb{F} : \mathcal{N}(\lambda I - L) \neq \{0\}\}$. If $T \in \mathcal{B}[\mathcal{X}]$ is a bounded linear operator acting on a normed space $\mathcal{X}$, then its point spectrum is identically defined, even if $\mathcal{X}$ is not a Banach space,

$$\sigma_P(T) = \{\lambda \in \mathbb{F} : \mathcal{N}(\lambda I - T) \neq \{0\}\}.$$

If $\widehat{\mathcal{X}}$ is the completion of a normed space $\mathcal{X}$ and $\widehat{T} \in \mathcal{B}[\widehat{\mathcal{X}}]$ is the extension over completion of $T \in \mathcal{B}[\mathcal{X}]$, then

$$\sigma_P(\widehat{T}) = \{\lambda \in \mathbb{F} : \mathcal{N}(\lambda I - \widehat{T}) \neq \{0\}\},$$

where we use the same notation λI for its extension $\widehat{\lambda I} = \lambda \widehat{I}$ on $\widehat{\mathcal{X}}$, and so write $\lambda I - \widehat{T} = \widehat{\lambda I} - \widehat{T} = \widehat{\lambda I - T}$. Recall from Theorem 5.1(i) that $\mathcal{N}(\lambda I - T) \subseteq \mathcal{N}(\lambda I - \widehat{T})$ (and the reverse inclusion of closures holds if $\mathcal{R}(\lambda I - T)$ is closed). Thus $\mathcal{N}(\lambda I - T) \neq 0$ implies $\mathcal{N}(\lambda I - \widehat{T}) \neq 0$. Hence

$$\sigma_P(T) \subseteq \sigma_P(\widehat{T}).$$

(For further results on the change of the spectrum under extensions in general — not necessarily extensions over completion — see [37, Theorem 2.16.2].)

(b) However, we can only consider the spectrum $\sigma(\widehat{T})$ of $\widehat{T}$, since spectra are defined in a unital complex Banach algebra, thus defined only for operators acting on a Banach space. (Although point spectra are defined for operators on normed spaces, spectra are not.) If $T \in \mathcal{B}[\mathcal{X}]$ acts on a Hilbert space $\mathcal{X}$, then we get the *residual spectrum formula* in Proposition 5.X,

$$\sigma_R(T) = \sigma_P(T^*)^* \backslash \sigma_P(T),$$

with respect to the Hilbert-space adjoint $T^* \in \mathcal{B}[\mathcal{X}]$. Since this expression for the residual spectrum is given entirely in terms of point spectra, and since point spectra are well-defined for operators on linear spaces, it might be tempting to try a definition of the residual spectrum on inner product spaces (not necessarily complete), just by mirroring the above residual spectrum formula which is based on point spectra only. The trouble, however, comes from another side. Hilbert-space adjoints $T^* \in \mathcal{B}[\mathcal{X}]$ are defined only on a Hilbert space, not on an inner product space (see Remark 5.W). Outcome: *there is no generalization of the residual spectrum formula for inner product spaces.*

We close this section with a simple result that will be required in Chapter 9. Let $\mathcal{X}$ and $\mathcal{Y}$ be linear spaces and equip each of them with a pair of norms, say, $\|\cdot\|_{\mathcal{X}_1}$ and $\|\cdot\|_{\mathcal{X}_2}$ on $\mathcal{X}$ and $\|\cdot\|_{\mathcal{Y}_1}$ and $\|\cdot\|_{\mathcal{Y}_2}$ on $\mathcal{Y}$. Take a nonzero linear transformation $T \in \mathcal{L}[\mathcal{X}, \mathcal{Y}]$ and suppose it is bounded when the linear spaces $\mathcal{X}$, $\mathcal{Y}$ are both equipped with the 1-norms or both equipped with the 2-norms (i.e., $T \in \mathcal{B}[(\mathcal{X}, \|\cdot\|_{\mathcal{X}_1}), (\mathcal{Y}, \|\cdot\|_{\mathcal{Y}_1})]$ and $T \in \mathcal{B}[(\mathcal{X}, \|\cdot\|_{\mathcal{X}_2}), (\mathcal{Y}, \|\cdot\|_{\mathcal{Y}_2})]$). Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be the induced uniform norms in $\mathcal{B}[(\mathcal{X}, \|\cdot\|_{\mathcal{X}_1}), (\mathcal{Y}, \|\cdot\|_{\mathcal{Y}_1})]$ and $\mathcal{B}[(\mathcal{X}, \|\cdot\|_{\mathcal{X}_2}), (\mathcal{Y}, \|\cdot\|_{\mathcal{Y}_2})]$, respectively.

Proposition 5.Z. *If* $\|\cdot\|_{\mathcal{X}_2} \le \|\cdot\|_{\mathcal{X}_1}$ *and* $\|\cdot\|_{\mathcal{Y}_2} \le \|\cdot\|_{\mathcal{Y}_1}$, *then*

$$\|T\|_1 \le \sup_{x \notin \mathcal{N}(T)} \frac{\|Tx\|_{\mathcal{Y}_1}}{\|Tx\|_{\mathcal{Y}_2}} \, \|T\|_2 \quad \textit{and} \quad \|T\|_2 \le \sup_{x \neq 0} \frac{\|x\|_{\mathcal{X}_1}}{\|x\|_{\mathcal{X}_2}} \, \|T\|_1;$$

equivalently,

$$\inf_{x \notin \mathcal{N}(T)} \frac{\|Tx\|_{\mathcal{Y}_2}}{\|Tx\|_{\mathcal{Y}_1}} \, \|T\|_1 \le \|T\|_2 \le \sup_{x \neq 0} \frac{\|x\|_{\mathcal{X}_1}}{\|x\|_{\mathcal{X}_2}} \, \|T\|_1.$$

Notes: Section 5.4 has been focused on bounded linear transformations only, taking up mainly the topics of Section 5.2, with some fundamental results in mind that are required in this and in subsequent chapters. Since all propositions listed in this section (plus the Hahn–Banach Theorem in Remark 5.2(d)) are classic, well-known, and indeed fundamental statements, they can be found in a variety of texts. The books listed below represent just a few samples among a huge collection of other possible alternatives.

For the classical results in Propositions 5.A to 5.J see, e.g., [52, Theorems 4.21, 4.22, Corollary 4.24, Theorem 4.25, Problem 4.47, Theorems 4.43, 4.44, Corollary 4.55, Proposition 4.57, and Problem 4.58], respectively. These (together with Remark 5.2) comprise an introductory basic toolbox on Banach-space operator theory, being mandatory in every functional analysis course.

Propositions 5.A to 5.E are related to bounded inverses of bounded linear transformations, while Propositions 5.F and 5.G refer mainly to sequences of bounded linear transformations acting on Banach spaces. Proposition 5.A is the Open Mapping Theorem, whose proof comes from the Baire Category Theorem on metric spaces (see, e.g., [52, Theorem 3.58]), and the inverse theorems in Propositions 5.B and 5.C (the Inverse Mapping Theorem and the Bounded Inverse Theorem) are consequences of it, as is the Closed Graph Theorem in Proposition 5.D. The Neumann Expansion (due to C.G. Neumann) in Proposition 5.E has a proof based on another basic result which says that *a normed space is complete if and only if every absolutely summable sequence is summable* (see, e.g., [52, Proposition 4.4]). The Baire Category Theorem also yields a classical proof for the Banach–Steinhaus Theorem of Proposition 5.F, which in turn leads to the notion of strong convergence in Proposition 5.G. The approximation results in Propositions 5.H and 5.J use the fact that *if* $\mathcal{Y}$ *is a Banach space, then* $\mathcal{B}_\infty[\mathcal{X},\mathcal{Y}]$ *is a* (*closed*) *subspace of* $\mathcal{B}[\mathcal{X},\mathcal{Y}]$.

Recall that if T is an operator on a normed space $\mathcal{X}$, and if $p\colon \mathbb{F} \to \mathbb{F}$ is a polynomial with coefficients α_i in $\mathbb{F}$, then the associated polynomial of T, viz., $p(T) = \sum_{i=0}^{n} \alpha_i T^i$ (as in Proposition 5.L) is again an operator on $\mathcal{X}$. Propositions 5.K to 5.P are the very first steps towards a spectral theory of operators on a complex Banach space. See, e.g., [56, Theorems 2.1, 2.2, Theorem 2.7, Corollary 2.9, Theorem 2.10, Corollaries 2.11 and 2.12]. The role played by Proposition 5.C in characterizing the spectrum of *bounded* linear operators on a *complex* Banach space is highlighted by the classical partition of the spectrum into point, continuous, and residual spectra.

The Hilbert-space geometry is centered around the Projection Theorem of Proposition 5.Q, which can be equivalently stated as in part (iii) of Proposition 5.R. This justifies the name Projection Theorem and distinguishes Hilbert spaces as the prototype of complemented Banach spaces. For Propositions 5.Q and 5.R see, e.g., [52, Theorems 5.20, 5.52 and Propositions 5.51, 5.81]. Proposition 5.S shows that such a nice scenario vanishes in an arbitrary Banach space (see, e.g., [72, Introduction, §1]). The section closes with a full version of the Fourier Series Theorem in Proposition 5.T, with the Riesz Representation Theorem in Hilbert space in Proposition 5.U and a corollary of it in Proposition 5.V, and with three fundamental results on parts of the spectrum for Hilbert-space adjoints in Proposition 5.X (see, e.g., [52, Theorems 5.48, 5.62, 5.63, 6.14]). These will often be required in Sections 7.4, 9.2 and 9.3.

Proof of Proposition 5.Z. Since $\|\cdot\|_{\mathcal{X}_2} \le \|\cdot\|_{\mathcal{X}_1}$ and $T \neq O$,

$$\begin{aligned}\|T\|_1 &= \sup_{x\neq 0} \frac{\|Tx\|_{\mathcal{Y}_1}}{\|x\|_{\mathcal{X}_1}} \le \sup_{x\neq 0} \frac{\|Tx\|_{\mathcal{Y}_1}}{\|x\|_{\mathcal{X}_2}} = \sup_{x\,\notin\mathcal{N}(T)} \frac{\|Tx\|_{\mathcal{Y}_1}}{\|x\|_{\mathcal{X}_2}} \frac{\|Tx\|_{\mathcal{Y}_2}}{\|Tx\|_{\mathcal{Y}_2}} \\ &\le \sup_{x\neq 0} \frac{\|Tx\|_{\mathcal{Y}_2}}{\|x\|_{\mathcal{X}_2}} \sup_{x\,\notin\mathcal{N}(T)} \frac{\|Tx\|_{\mathcal{Y}_1}}{\|Tx\|_{\mathcal{Y}_2}} = \|T\|_2 \sup_{x\,\notin\mathcal{N}(T)} \frac{\|Tx\|_{\mathcal{Y}_1}}{\|Tx\|_{\mathcal{Y}_2}};\end{aligned}$$

equivalently,

$$\inf_{x \notin \mathcal{N}(T)} \frac{\|Tx\|_{\mathcal{Y}_2}}{\|Tx\|_{\mathcal{Y}_1}} \|T\|_1 = \inf_{x \notin \mathcal{N}(T)} \frac{1}{\frac{\|Tx\|_{\mathcal{Y}_1}}{\|Tx\|_{\mathcal{Y}_2}}} \|T\|_1 = \frac{1}{\sup_{x \notin \mathcal{N}(T)} \frac{\|Tx\|_{\mathcal{Y}_1}}{\|Tx\|_{\mathcal{Y}_2}}} \|T\|_1 \le \|T\|_2.$$

Moreover, since $\|\cdot\|_{\mathcal{Y}_2} \le \|\cdot\|_{\mathcal{Y}_1}$ and $\|Tx\|_{\mathcal{Y}_1} \le \|T\|_1 \|x\|_{\mathcal{X}_1}$ for every $x \in \mathcal{X}$,

$$\|T\|_2 = \sup_{x \neq 0} \frac{\|Tx\|_{\mathcal{Y}_2}}{\|x\|_{\mathcal{X}_2}} \le \sup_{x \neq 0} \frac{\|Tx\|_{\mathcal{Y}_1}}{\|x\|_{\mathcal{X}_2}} \le \|T\|_1 \sup_{x \neq 0} \frac{\|x\|_{\mathcal{X}_1}}{\|x\|_{\mathcal{X}_2}}. \qquad \square$$

A final note. We use the notation ℓ^p_+, or $\ell^p(\mathbb{N})$, or $\ell^p(\mathbb{N}_0)$ with $p \ge 1$ for p-summable scalar-valued sequences $\{\xi_k\}$ (i.e., $\sum_k |\xi_k|^p < \infty$). And the notation ℓ^∞_+, or $\ell^\infty(\mathbb{N})$, or $\ell^\infty(\mathbb{N}_0)$ for bounded sequences (i.e., $\sup_k |\xi_k| < \infty$). We drop the subscript $_+$ when dealing with p-summable or bounded nets indexed by $\mathbb{Z}$, rather than sequences indexed by $\mathbb{N}$ or $\mathbb{N}_0$, so that we use ℓ^p for $\ell^p(\mathbb{Z})$ and ℓ^∞ for $\ell^\infty(\mathbb{Z})$ (where $\mathbb{N}$, $\mathbb{N}_0$ and $\mathbb{Z}$ stand for the sets of positive integers, nonnegative integers, and all integers, respectively).

Suggested Readings

Bachman and Narici [2]
Brown and Pearcy [5]
Conway [12], [13], [14]
Dunford and Schwartz [23]
Halmos [31]
Harte [34]
Kato [46]
Kubrusly [52], [56]
Megginson [70]
Reed and Simon [80]
Rudin [82]
Schechter [87]
Simon [89]
Taylor and Lay [91]
Weidmann [92]
Yosida [95]



6

Bounded Bilinear Maps

This chapter considers basic normed-space aspects of bilinear maps. The first notion is that of a bounded bilinear map, which is shown to coincide with the notion of continuous bilinear maps. The final targets are extension theorems for bounded bilinear maps with norm preservation. As in the previous chapters, all linear spaces (and therefore all normed spaces) are over the same field.

6.1 Boundedness and Continuity

Let $\mathcal{X}$, $\mathcal{Y}$, $\mathcal{Z}$ be normed spaces and let $b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$ be the linear space of all bilinear maps ϕ from $\mathcal{X}\times\mathcal{Y}$ to $\mathcal{Z}$. A *bounded bilinear map* is a map ϕ in $b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$ for which there exists a positive constant $\beta > 0$ such that

$$\|\phi(x,y)\| \le \beta\|x\|\,\|y\| \quad \text{for every} \quad (x,y) \in \mathcal{X}\times\mathcal{Y}.$$

Equivalently, ϕ is bounded if

$$\sup_{0\neq x\in\mathcal{X},\, 0\neq y\in\mathcal{Y}} \frac{\|\phi(x,y)\|}{\|x\|\,\|y\|} < \infty.$$

Let $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$ denote the collection of all bounded bilinear maps from the linear space $b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$, which is a linear manifold of it, thus being itself a linear space. The next theorem sets forth the *induced uniform norm* on $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$, thus making it into a normed space (which happens to be complete whenever $\mathcal{Z}$ is). As before, $\mathcal{B}[\mathcal{X},\mathcal{Z}]$ denotes the normed space of all bounded linear transformations from the linear space $\mathcal{L}[\mathcal{X},\mathcal{Z}]$.

Theorem 6.1. *Let $\mathcal{X},\mathcal{Y},\mathcal{Z}$ be normed spaces (over the same field $\mathbb{F}$).*

(a) *$\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$ is a linear space, the function $\|\cdot\|\colon \boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}] \to \mathbb{R}$ given by*

$$\|\phi\| = \inf\big\{\beta > 0\colon\ \|\phi(x,y)\| \le \beta\|x\|\,\|y\| \ \textit{for every}\ (x,y) \in \mathcal{X}\times\mathcal{Y}\big\},$$

or equivalently,

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C. S. Kubrusly, *Bilinear Maps and Tensor Products in Operator Theory*,
Universitext, https://doi.org/10.1007/978-3-031-34093-2_6

$$\|\phi\| = \sup_{0\neq x\in\mathcal{X},\, 0\neq y\in\mathcal{Y}} \tfrac{\|\phi(x,y)\|}{\|x\|\,\|y\|},$$

for every $\phi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$ *is a norm on* $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$*, and so*

$$\|\phi(x,y)\| \le \|\phi\|\,\|x\|\,\|y\| \quad \textit{for every} \quad (x,y)\in\mathcal{X}\times\mathcal{Y}.$$

(b) *Moreover,* $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$ *is a Banach space whenever* $\mathcal{Z}$ *is a Banach space.*

Proof. (a) Take $\phi,\phi_1,\phi_2 \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$ and $\alpha\in\mathbb{F}$ arbitrarily. Clearly, $\alpha\phi$ lies in $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$. As $b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$ is a linear space, $\phi_1+\phi_2$ is bilinear. Consider the function $\|\cdot\|\colon \boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]\to\mathbb{R}$ defined by

$$\|\phi\| = \sup_{0\neq x\in\mathcal{X},\, 0\neq y\in\mathcal{Y}} \tfrac{\|\phi(x,y)\|}{\|x\|\,\|y\|}$$

for every $\phi\in\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$. Since $\sup_{0\neq x\in\mathcal{X},\,0\neq y\in\mathcal{Y}} \frac{\|(\phi_1+\phi_2)(x,y)\|}{\|x\|\,\|y\|}\le\|\phi_1\|+\|\phi_2\|$, we get $\phi_1+\phi_2$ lies in $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$. Hence $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$ is a linear space. Now (i) $\|\phi\|\ge 0$, (ii) $\|\phi\|=0$ if and only if $\phi=0$, (iii) $\|\alpha\phi\|=\|\alpha\|\,\|\phi\|$, and (iv) $\|\phi_1+\phi_2\|\le\|\phi_1\|+\|\phi_2\|$. So $\|\cdot\|$ is a norm on $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$. This is equivalently written as $\|\phi\|=\inf\{\beta>0\colon \|\phi(x,y)\|\le\beta\|x\|\,\|y\| \text{ for every } (x,y)\in\mathcal{X}\times\mathcal{Y}\}$, and therefore $\|\phi(x,y)\|\le\|\phi\|\,\|x\|\,\|y\|$ for every $(x,y)\in\mathcal{X}\times\mathcal{Y}$.

(b) Next take any $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$-valued Cauchy sequence $\{\phi_n\}$ and an arbitrary (x,y) in $\mathcal{X}\times\mathcal{Y}$. Thus $\|\phi_m(x,y)-\phi_n(x,y)\|\le\|\phi_m-\phi_n\|\,\|x\|\,\|y\|$ and hence $\{\phi_n(x,y)\}$ is a Cauchy sequence in $\mathcal{Z}$, which converges in $\mathcal{Z}$ whenever $\mathcal{Z}$ is a Banach space. Then define a map $\phi\colon\mathcal{X}\times\mathcal{Y}\to\mathcal{Z}$ by

$$\phi(x,y) = \lim_n \phi_n(x,y) \;\in\; \mathcal{Z}.$$

This ϕ is bilinear. (Indeed, $\phi(x_1+x_2,y)=\lim_n\phi_n(x_1+x_2,y)=\phi(x_1,y)+\phi(x_2,y)$, and similarly $\phi(x,y_1+y_2)=\phi(x,y_1)+\phi(x,y_2)$, and also $\phi(\alpha x,y)=\phi(x,\alpha y)=\alpha\phi(x,y)$.) Moreover, this ϕ is bounded. In fact, take an arbitrary $\varepsilon>0$. Since $\{\phi_n\}$ is a Cauchy sequence in $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$, then $\|(\phi_m-\phi_n)(x,y)\|\le\|\phi_m-\phi_n\|\|x\|\|y\|<\varepsilon\|x\|\|y\|$ for every $(x,y)\in\mathcal{X}\times\mathcal{Y}$ for m,n large enough. So

$$\begin{aligned}\|(\phi_m-\phi)(x,y)\| &= \|\phi_m(x,y)-\phi(x,y)\| = \|\phi_m(x,y)-\lim_n\phi_n(x,y)\|\\ &= \lim_n\|(\phi_m-\phi_n)(x,y)\| < \varepsilon\|x\|\,\|y\|\end{aligned}$$

for m large enough. Thus $\phi_m-\phi$ is bounded, and so is $\phi=\phi_m-(\phi_m-\phi)$ (because $\phi_m\in\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$). Hence $\phi\in\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$. Furthermore,

$$\|\phi_m-\phi\| = \sup_{0\neq x\in\mathcal{X},\,0\neq y\in\mathcal{Y}} \tfrac{\|(\phi_m-\phi)(x,y)\|}{\|x\|\,\|y\|} < \varepsilon$$

for m large enough by the previously displayed inequality. Therefore

$$\phi_n\to\phi \;\text{ in }\; \boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}].$$

Thus every $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$-valued Cauchy sequence converges in $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$, which means $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$ is Banach. □

Boundedness means continuity for linear transformations. The same happens for bilinear maps with respect to the product topology of $\mathcal{X}\times\mathcal{Y}$, as will be verified in Theorem 6.2 below. The product topology of the Cartesian product of a pair of normed spaces $(\mathcal{X},\|\cdot\|)$ and $(\mathcal{Y},\|\cdot\|)$ — same notation for both norms — is induced by any (uniformly equivalent) product metric d on $\mathcal{X}\times\mathcal{Y}$ generated by those norms; for instance, by $d_p\colon(\mathcal{X}\times\mathcal{Y})\times(\mathcal{X}\times\mathcal{Y})\to\mathbb{R}$ for $p\geq 1$ or $d_\infty\colon(\mathcal{X}\times\mathcal{Y})\times(\mathcal{X}\times\mathcal{Y})\to\mathbb{R}$ (as in Section 5.1) given by $d_p((x,y),(x',y')) = (\|x-x'\|^p+\|y-y'\|^p)^{1/p}$ and $d_\infty((x,y),(x',y')) = \max\{\|x-x'\|,\|y-y'\|\}$ for $(x,y),(x',y')$ in $\mathcal{X}\times\mathcal{Y}$. (The same product metric is considered in Proposition 5.D.) So if d is any of the above product metrics, then an $\mathcal{X}\times\mathcal{Y}$-valued sequence $\{(x_k,y_k)\}$ converges in the metric space $(\mathcal{X}\times\mathcal{Y},d)$ to $(x,y)\in\mathcal{X}\times\mathcal{Y}$ if and only if the $\mathcal{X}$-valued and the $\mathcal{Y}$-valued sequences $\{x_k\}$ and $\{y_k\}$ both converge in $(\mathcal{X},\|\cdot\|)$ and in $(\mathcal{Y},\|\cdot\|)$ to $x\in\mathcal{X}$ and $y\in\mathcal{Y}$, respectively.

Theorem 6.2. *Let $\mathcal{X}$, $\mathcal{Y}$, $\mathcal{Z}$ be normed spaces and take $\phi\in b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$. The following three assertions are pairwise equivalent.*

(a) *ϕ is continuous.*

(b) *ϕ is continuous at $(0,0)\in\mathcal{X}\times\mathcal{Y}$.*

(c) *ϕ is bounded* (*i.e., $\phi\in\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$*).*

If any of the above assertions holds, then the linear sections are bounded,

(d) *$\phi^y\in\mathcal{B}[\mathcal{X},\mathcal{Z}]$ for every $y\in\mathcal{Y}$ and $\phi_x\in\mathcal{B}[\mathcal{Y},\mathcal{Z}]$ for every $x\in\mathcal{X}$.*

Conversely, if (d) *holds and one of $\mathcal{X}$ or $\mathcal{Y}$ is a Banach space, then any of* (a), (b), *or* (c) *holds.*

Proof. First we show that

$$\text{(a)}\Longrightarrow\text{(b)}\Longrightarrow\text{(c)}\Longrightarrow\text{(a)}.$$

The implication (a) $\Longrightarrow$ (b) is tautological. The proof of (b) $\Longrightarrow$ (c) is contrapositive. Thus suppose (c) fails. Then for an arbitrary $\alpha>0$ there is a pair $(x_\alpha,y_\alpha)\in\mathcal{X}\times\mathcal{Y}$ such that $\alpha\|x_\alpha\|\,\|y_\alpha\|<\|\phi(x_\alpha,y_\alpha)\|$. In particular, for every positive integer k there exists a pair of nonzero vectors $(x_k,y_k)\in\mathcal{X}\times\mathcal{Y}$ such that $0<\|x_k\|\,\|y_k\|<\frac{1}{k}\|\phi(x_k,y_k)\|$. Set $x'_k=\frac{x_k}{\sqrt{k}\|x_k\|}$ in $\mathcal{X}$ and $y'_k=\frac{y_k}{\sqrt{k}\|y_k\|}$ in $\mathcal{Y}$, and consider the sequences of nonzero vectors $\{x'_k\}$ and $\{y'_k\}$ which are such that $x'_k\to 0$ in $(\mathcal{X},\|\cdot\|)$ and $y'_k\to 0$ in $(\mathcal{Y},\|\cdot\|)$. Equip the Cartesian product $\mathcal{X}\times\mathcal{Y}$ with any of the equivalent product metrics d and consider the metric space $(\mathcal{X}\times\mathcal{Y},d)$, so that the sequence $\{(x'_k,y'_k)\}$ is such that

$$(x'_k,y'_k)\to(0,0)\quad\text{in}\quad(\mathcal{X}\times\mathcal{Y},d).$$

Since ϕ is bilinear, $0<\|x_k\|\,\|y_k\|<\frac{1}{k}\|\phi(x_k,y_k)\|=\|x_k\|\,\|y_k\|\|\phi(x'_k,y'_k)\|$ and hence $1<\|\phi(x'_k,y'_k)\|$ for all positive integers k. Therefore

$$\phi(x'_k, y'_k) \not\to 0 = \phi(0,0) \quad \text{in} \quad (\mathcal{Z}, \|\cdot\|).$$

Hence $\phi\colon (\mathcal{X}\times\mathcal{Y}, d) \to (\mathcal{Z}, \|\cdot\|)$ is not continuous at $(0,0)$. So (b) fails. To verify (c)$\Longrightarrow$(a), take an arbitrary $\mathcal{X}\times\mathcal{Y}$-valued convergent sequence $\{(x_k, y_k)\}$, say, $(x_k, y_k) \to (x, y)$ in $(\mathcal{X}\times\mathcal{Y}, d)$ for any equivalent product metric d. This means $x_k \to x$ in $(\mathcal{X}, \|\cdot\|)$ and $y_k \to y$ in $(\mathcal{Y}, \|\cdot\|)$. Recall: convergent sequences are bounded. If the bilinear map ϕ is bounded, then

$$\begin{aligned}\|\phi(x_k, y_k) - \phi(x, y)\| &\le \|\phi(x_k, y_k) - \phi(x_k, y)\| + \|\phi(x, y) - \phi(x_k, y)\| \\ &= \|\phi(x_k, y_k - y)\| + \|\phi(x - x_k, y)\| \\ &\le \|\phi\|\big((\sup_k \|x_k\|)\|y_k - y\| + \|x_k - x\|\,\|y\|\big),\end{aligned}$$

and so $\phi(x_k, y_k) \to \phi(x, y)$ in $(\mathcal{Z}, \|\cdot\|)$. Therefore $\phi\colon (\mathcal{X}\times\mathcal{Y}, d) \to (\mathcal{Z}, \|\cdot\|)$ preserves convergence, which means ϕ is continuous. Finally we show that

$$\text{(c)} \Longrightarrow \text{(d)} \quad \text{and} \quad \text{(d) with one of } \mathcal{X} \text{ or } \mathcal{Y} \text{ being Banach} \Longrightarrow \text{(c)}.$$

The implication (c) $\Longrightarrow$ (d) is readily verified. Indeed, fix any $0 \ne y \in \mathcal{Y}$. So

$$\sup_{0\ne x\in\mathcal{X}} \tfrac{\|\phi^y x\|}{\|x\|\,\|y\|} = \sup_{0\ne x\in\mathcal{X}} \tfrac{\|\phi(x,y)\|}{\|x\|\,\|y\|} \le \sup_{0\ne x\in\mathcal{X},\, 0\ne y\in\mathcal{Y}} \tfrac{\|\phi(x,y)\|}{\|x\|\,\|y\|} = \|\phi\|$$

whenever (c) holds. Thus the linear transformation $\phi^y\colon \mathcal{X} \to \mathcal{Z}$ is bounded. Symmetrically, (c) implies $\phi_x \in \mathcal{B}[\mathcal{Y}, \mathcal{Z}]$. So (c) implies (d), and in this case

$$\|\phi^y\| \le \|\phi\|\,\|y\| \qquad \text{and} \qquad \|\phi_x\| \le \|\phi\|\,\|x\|$$

for each $x \in \mathcal{X}$ and each $y \in \mathcal{Y}$. Conversely, suppose (d) holds. Take an arbitrary nonzero $y \in \mathcal{Y}$, set $y' = \frac{y}{\|y\|}$ in $\mathcal{Y}$, and consider the associated bounded linear transformation $\phi^{y'} \in \mathcal{B}[\mathcal{X}, \mathcal{Z}]$. Thus

$$\|\phi^{y'}x\| = \|\phi(x, y')\| = \|\phi_x y'\| \le \|\phi_x\|\,\|y'\| = \|\phi_x\| \quad \text{for every} \quad x \in \mathcal{X}.$$

So the family $\{\phi^{y'}\}_{0\ne y\in\mathcal{Y}}$ of transformations in $\mathcal{B}[\mathcal{X}, \mathcal{Z}]$ is pointwise bounded:

$$\sup_{0\ne y\in\mathcal{Y}} \|\phi^{y'}x\| < \infty \quad \text{for every} \quad x \in \mathcal{X}.$$

Suppose $\mathcal{X}$ is a Banach space to allow an application of the Banach–Steinhaus Theorem (cf. Proposition 5.F), which ensures uniform boundedness given pointwise boundedness for any family of bounded linear transformations of a Banach space into a normed space. Thus the above pointwise boundedness implies uniform boundedness by the Banach–Steinhaus Theorem:

$$\sup_{0\ne y\in\mathcal{Y}} \|\phi^{y'}\| < \infty.$$

Since $\frac{\|\phi(x,y)\|}{\|x\|\,\|y\|} = \frac{\|\phi(x,y')\|}{\|x\|} = \frac{\|\phi^{y'}x\|}{\|x\|} \le \|\phi^{y'}\|$ for every $(0,0) \ne (x, y) \in \mathcal{X}\times\mathcal{Y}$,

$$\sup_{0\neq x\in\mathcal{X},\,0\neq y\in\mathcal{Y}} \tfrac{\|\phi(x,y)\|}{\|x\|\,\|y\|} \le \sup_{0\neq y\in\mathcal{Y}} \|\phi^{y'}\| < \infty,$$

and so ϕ is bounded, that is, $\phi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$. Thus if $\mathcal{X}$ is a Banach space and $\phi^y \in \mathcal{B}[\mathcal{X}, \mathcal{Z}]$, then (c) holds. A symmetrical argument shows that if $\mathcal{Y}$ is a Banach space and $\phi_x \in \mathcal{B}[\mathcal{Y}, \mathcal{Z}]$, then (c) holds as well. □

As is readily verified, if $\phi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$, then

$$\|\phi\| = \sup_{\|x\|=\|y\|=1} \|\phi(x,y)\| = \sup_{\|x\|\le 1,\|y\|\le 1} \|\phi(x,y)\| = \sup_{\|x\|<1,\|y\|<1} \|\phi(x,y)\|.$$

6.2 Identification with Bounded Linear Transformations

The next result is a normed-space version of Theorem 2.6. We know from Theorem 2.6 that $b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] \cong \mathcal{L}[\mathcal{X}, \mathcal{L}[\mathcal{Y}, \mathcal{Z}]]$, where in that case $\cong$ meant linear-space isomorphism. Considering continuous linear transformations and continuous bilinear maps, and equipping these linear spaces with their usual norms, the program for proving Theorem 6.3 below is to verify whether the same linear-space isomorphism becomes an isometric isomorphism.

Theorem 6.3. *Take an arbitrary triple* $(\mathcal{X}, \mathcal{Y}, \mathcal{Z})$ *of normed spaces. The normed spaces* $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ *and* $\mathcal{B}[\mathcal{X}, \mathcal{B}[\mathcal{Y}, \mathcal{Z}]]$ *are isometrically isomorphic:*

$$\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] \cong \mathcal{B}[\mathcal{X}, \mathcal{B}[\mathcal{Y}, \mathcal{Z}]].$$

Proof. Take any $\phi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$. To each $x \in \mathcal{X}$ take the section ϕ_x in $\mathcal{B}[\mathcal{Y}, \mathcal{Z}]$ (recall: $\|\phi_x\| \le \|\phi\|\,\|x\|$). Associate with ϕ a map $T\colon \mathcal{X} \to \mathcal{B}[\mathcal{Y}, \mathcal{Z}]$ defined by

$$Tx = \phi_x \quad \text{for every} \quad x \in \mathcal{X}.$$

In other words, $(Tx)(\cdot) = \phi_x(\cdot) = \phi(x, \cdot) \in \mathcal{B}[\mathcal{Y}, \mathcal{Z}]$ for each $x \in \mathcal{X}$. It is clear that $(Tx)(\cdot)$ is linear and bounded ($\|(Tx)\| \le \|\phi\|\,\|x\|$ for every $x \in \mathcal{X}$). Thus

$$T \in \mathcal{B}[\mathcal{X}, \mathcal{B}[\mathcal{Y}, \mathcal{Z}]] \quad \text{for each} \quad \phi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}].$$

Now consider the map $\Im\colon \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] \to \mathcal{B}[\mathcal{X}, \mathcal{B}[\mathcal{Y}, \mathcal{Z}]]$ defined by

$$\Im(\phi) = T \quad \text{for every} \quad \phi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}].$$

It is readily verified that this is again a linear transformation. Moreover,

$$\|\Im(\phi)\| = \|T\| = \sup_{x\neq 0} \tfrac{\|Tx\|}{\|x\|} = \sup_{x\neq 0} \tfrac{\|\phi_x\|}{\|x\|} = \sup_{x\neq 0,\,y\neq 0} \tfrac{\|\phi_x(y)\|}{\|x\|\,\|y\|} = \sup_{x,y\neq 0} \tfrac{\|\phi(x,y)\|}{\|x\|\,\|y\|} = \|\phi\|.$$

Hence $\Im \in \mathcal{B}\big[\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}], \mathcal{B}[\mathcal{X}, \mathcal{B}[\mathcal{Y}, \mathcal{Z}]]\big]$ is an isometry. To show that $\Im$ is surjective, take any $T' \in \mathcal{B}[\mathcal{X}, \mathcal{B}[\mathcal{Y}, \mathcal{Z}]]$ and set $\phi'(x, y) = (T'x)(y) \in \mathcal{Z}$ for every $(x, y) \in \mathcal{X}\times\mathcal{Y}$. This defines a map $\phi'\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{Z}$ which is clearly bilinear and

bounded (i.e., $\phi' \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$). Take the image of this ϕ' under $\Im$, namely, $\Im(\phi') = T'' \in \mathcal{B}[\mathcal{X}, \mathcal{B}[\mathcal{Y}, \mathcal{Z}]]$. Therefore $(T''x)(y) = \phi'_x(y) = \phi'(x, y)$, so that $T''(x)(y) = T'(x)(y)$, for every $(x, y) \in \mathcal{X}\times\mathcal{Y}$. Then $T'' = T'$ and so $T' \in \mathcal{R}(\Im)$. Thus $\mathcal{R}(\Im) = \mathcal{B}[\mathcal{X}, \mathcal{B}[\mathcal{Y}, \mathcal{Z}]]$. Then $\Im$ is a surjective linear isometry. That is,

$$\Im\colon \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] \to \mathcal{B}[\mathcal{X}, \mathcal{B}[\mathcal{Y}, \mathcal{Z}]] \text{ is an isometric isomorphism.} \qquad \square$$

Remark 6.4. *Two Significant Consequences of Theorem 6.3.*

(a) Theorem 6.3 yields a normed-space version of Remark 2.7(b). It is clear that the isomorphism $\wp$ of Remark 2.7(b), now acting from $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ onto $\boldsymbol{b}[\mathcal{Y}\times\mathcal{X}, \mathcal{Z}]$, is an isometry (cf. norm in Theorem 6.1). Thus by Theorem 6.3

$$\mathcal{B}[\mathcal{Y}, \mathcal{B}[\mathcal{X}, \mathcal{Z}]] \cong \boldsymbol{b}[\mathcal{Y}\times\mathcal{X}, \mathcal{Z}] \cong \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] \cong \mathcal{B}[\mathcal{X}, \mathcal{B}[\mathcal{Y}, \mathcal{Z}]].$$

With $\mathcal{X}^* = \mathcal{B}[\mathcal{X}, \mathbb{F}]$ and $\mathcal{Y}^* = \mathcal{B}[\mathcal{Y}, \mathbb{F}]$ and setting $\mathcal{Z} = \mathbb{F}$ above we get

$$\mathcal{B}[\mathcal{Y}, \mathcal{X}^*] \cong \boldsymbol{b}[\mathcal{Y}\times\mathcal{X}, \mathbb{F}] \cong \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}] \cong \mathcal{B}[\mathcal{X}, \mathcal{Y}^*].$$

(b) There is no Hahn–Banach Theorem for bounded bilinear forms. In fact, suppose there is an equivalent to the Hahn–Banach Theorem for bounded bilinear forms. This implies, in particular, that

(∗) for every linear manifold $\mathcal{M}$ of every normed space $\mathcal{X}$, every bounded bilinear functional ϕ in $\boldsymbol{b}[\mathcal{M}\times\mathcal{Y}, \mathbb{F}]$ has a bounded bilinear extension $\widehat{\phi}$ in $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$ for an arbitrary normed space $\mathcal{Y}$.

Recall from Theorem 6.3 (see item (a) above) that $\boldsymbol{b}[\mathcal{M}\times\mathcal{Y}, \mathbb{F}] \cong \mathcal{B}[\mathcal{M}, \mathcal{Y}^*]$ and $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}] \cong \mathcal{B}[\mathcal{X}, \mathcal{Y}^*]$. Thus if (∗) holds, then

(∗∗) for every linear manifold $\mathcal{M}$ of every normed space $\mathcal{X}$, every bounded linear transformation T in $\mathcal{B}[\mathcal{M}, \mathcal{Z}]$ has a bounded linear extension $\widehat{T}$ in $\mathcal{B}[\mathcal{X}, \mathcal{Z}]$ for an arbitrary Banach space $\mathcal{Z}$ which is the dual of an arbitrary normed space $\mathcal{Y}$.

(Indeed, consider the isometric isomorphisms $\Im_{\mathcal{M}}\colon \boldsymbol{b}[\mathcal{M}\times\mathcal{Y}, \mathbb{F}] \to \mathcal{B}[\mathcal{M}, \mathcal{Y}^*]$ and $\Im_{\mathcal{X}}\colon \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}] \to \mathcal{B}[\mathcal{X}, \mathcal{Y}^*]$ as in the proof of Theorem 6.3. Take any T in $\mathcal{B}[\mathcal{M}, \mathcal{Y}^*]$. Then $T = \Im_{\mathcal{M}}(\phi)$ in $\mathcal{B}[\mathcal{M}, \mathcal{Y}^*]$ for some ϕ in $\boldsymbol{b}[\mathcal{M}\times\mathcal{Y}, \mathbb{F}]$. If (∗) holds, then set $\widehat{T} = \Im_{\mathcal{X}}(\widehat{\phi})$ in $\mathcal{B}[\mathcal{X}, \mathcal{Y}^*]$ for the extension $\widehat{\phi}$ in $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$ of ϕ. These are given by $Tu = \phi_u \in \mathcal{Y}^*$ and $\widehat{T}x = \widehat{\phi}_x \in \mathcal{Y}^*$ for every $u \in \mathcal{M}$ and every $x \in \mathcal{X}$ (cf. proof of Theorem 6.3 again). Thus $\widehat{T}u = \widehat{\phi}_u = \widehat{\phi}(u, \cdot) = \phi(u, \cdot) = \phi_u = Tu$ for an arbitrary $u \in \mathcal{M} \subseteq \mathcal{X}$, and hence $\widehat{T}|_{\mathcal{M}} = T$.)

But assertion (∗∗) is false if $\mathcal{M}$ is not complemented in a Banach space $\mathcal{X}$ according to Theorem 5.4. Consequently, (∗) fails. Therefore *there is no equivalent to the Hahn–Banach Theorem for bounded bilinear forms*.

Note: It is worth recalling that not every Banach space is a dual (e.g., $L^1[0, 1]$ (or c_0) is not the dual of any normed space — see, e.g., [70, Example 2.10.11]).

(c) In addition to the lack of a Hahn–Banach theorem for continuous bilinear forms, there is no uniform continuity for continuous bilinear maps either. Although this is not a consequence of Theorem 6.3, it matches the nonexistence in (b). Example: the real multiplication binary operation $\phi\colon \mathbb{R}\times\mathbb{R} = \mathbb{R}^2 \to \mathbb{R}$ given by $\phi(\alpha,\beta) = \alpha\beta$ is such that $\phi(\alpha,\alpha) = \alpha^2$, which makes the continuous bilinear form ϕ not uniformly continuous. Thus *extension by continuity does not apply to bounded bilinear maps*.

6.3 Extension of Bounded Bilinear Maps

Theorem 5.1 dealt with extensions over completion for bounded linear transformations. Its proof uses extension by continuity of bounded linear transformations, which in turn needs uniform continuity (cf. Remark 5.2(b)). However, a bilinear version of Theorem 5.1 dealing with extensions over completion for bounded bilinear maps survives the lack of uniform continuity (with a different proof avoiding extension by continuity). The proof is based on Theorem 6.3. If $\widehat{\mathcal{X}}$ is a completion of $\mathcal{X}$ and $\widehat{\mathcal{Y}}$ is a completion of $\mathcal{Y}$, then $\widehat{\phi} \in \boldsymbol{b}[\widehat{\mathcal{X}}\times\widehat{\mathcal{Y}}, \mathcal{Z}]$ is called an *extension over completion* of $\phi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ if there exist isometric isomorphisms $J\colon \mathcal{X} \to J(\mathcal{X}) = \widetilde{\mathcal{X}} \subseteq \widehat{\mathcal{X}}$ and $K\colon \mathcal{Y} \to K(\mathcal{Y}) = \widetilde{\mathcal{Y}} \subseteq \widehat{\mathcal{Y}}$ with $\widetilde{\mathcal{X}}^- = \widehat{\mathcal{X}}$ and $\widetilde{\mathcal{Y}}^- = \widehat{\mathcal{Y}}$ such that $\widehat{\phi}|_{J(\mathcal{X})\times K(\mathcal{Y})}(\cdot\,,\cdot) = \phi(J^{-1}(\cdot), K^{-1}(\cdot))$.

Theorem 6.5. *Let the Banach spaces $\widehat{\mathcal{X}}$ and $\widehat{\mathcal{Y}}$ be completions of normed spaces $\mathcal{X}$ and $\mathcal{Y}$. Let $\mathcal{Z}$ be a Banach space. Every $\phi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ has an extension $\widehat{\phi} \in \boldsymbol{b}[\widehat{\mathcal{X}}\times\widehat{\mathcal{Y}}, \mathcal{Z}]$ over the Cartesian product of the completions $\widehat{\mathcal{X}}$ and $\widehat{\mathcal{Y}}$ of $\mathcal{X}$ and $\mathcal{Y}$. Also, $\widehat{\phi}$ is unique up to isometric isomorphisms and $\|\widehat{\phi}\| = \|\phi\|$.*

Proof. There is no uniform continuity for bilinear maps. So the argument in the proof of Theorem 5.1 is not extendable *ipsis litteris* from linear transformations to bilinear maps. A way to overcome the lack of uniform continuity for bounded bilinear maps is to apply Theorem 6.3, relying on the uniform continuity of bounded linear transformations. We approach the proof in three steps.

(a) According to Theorem 6.3,

$$\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] \cong \mathcal{B}[\mathcal{X}, \mathcal{B}[\mathcal{Y}, \mathcal{Z}]].$$

Thus the proof's program is to apply extension over completion for bounded linear transformations in $\mathcal{B}[\mathcal{X}, \mathcal{B}[\mathcal{Y}, \mathcal{Z}]]$ (as in Theorem 5.1), in order to get extension over completion for bounded bilinear maps in $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ by exploring such an isometric isomorphism. Note that (i) the completion of a Banach space coincides with the Banach space itself, and that (ii) $\mathcal{Z}$ is a Banach space, and so is $\mathcal{B}[\mathcal{Y}, \mathcal{Z}]$. Thus when dealing with extensions over completion for bounded linear transformations of $\mathcal{X}$ into the Banach space $\mathcal{B}[\mathcal{Y}, \mathcal{Z}]$, only completion of the normed space $\mathcal{X}$ matters. Then for an arbitrary Banach space $\mathcal{W}$ and an arbitrary bounded linear transformation $S \in \mathcal{B}[\mathcal{X}, \mathcal{W}]$, an extension over completion $\widehat{S} \in \mathcal{B}[\widehat{\mathcal{X}}, \mathcal{W}]$ of S is constructed according to the following diagram.

$$\begin{array}{ccccc} \mathcal{W} & \xrightarrow{\ I\ } & \mathcal{W} & = & \mathcal{W} \\ {\scriptstyle S}\uparrow & & \uparrow{\scriptstyle \widehat{S}|_{\widetilde{\mathcal{X}}} = SJ^{-1}} & & \uparrow{\scriptstyle \widehat{S}} \\ \mathcal{X} & \xrightarrow{\ J\ } & \widetilde{\mathcal{X}} = J(\mathcal{X}) & \bar{\subseteq} & \widehat{\mathcal{X}}, \end{array}$$

where $J\colon \mathcal{X} \to \widetilde{\mathcal{X}} \bar{\subseteq} \widehat{\mathcal{X}}$ is an isometric isomorphism between $\mathcal{X}$ and $\widetilde{\mathcal{X}}$. (Recall: $\bar{\subseteq}$ stands for dense inclusion, i.e., $\widetilde{\mathcal{X}} \bar{\subseteq} \widehat{\mathcal{X}}$ means $\widetilde{\mathcal{X}} \subseteq \widehat{\mathcal{X}}$ and $\widetilde{\mathcal{X}}^- = \widehat{\mathcal{X}}$.)

(b) Take an arbitrary map $\phi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$. Since $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] \cong \mathcal{B}[\mathcal{X}, \mathcal{B}[\mathcal{Y}, \mathcal{Z}]]$, there are isometric isomorphisms between maps $\phi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ and transformations $T \in \mathcal{B}[\mathcal{X}, \mathcal{B}[\mathcal{Y}, \mathcal{Z}]]$. In particular (cf. proof of Theorem 6.3), there is an isometric isomorphism which is constructed as follows: $T(x) = \phi_x$ in $\mathcal{B}[\mathcal{Y}, \mathcal{Z}]$ (i.e., $T(x)(\cdot) = \phi_x(\cdot) = \phi(x, \cdot) \in \mathcal{B}[\mathcal{Y}, \mathcal{Z}]$) for each $x \in \mathcal{X}$. Thus

$$T(x)(y) = \phi(x, y) \in \mathcal{Z} \quad \text{for every} \quad (x, y) \in \mathcal{X}\times\mathcal{Y}$$

is such that $\phi = \phi(\cdot\,, \cdot) \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] \cong T = T(\cdot)(\cdot) \in \mathcal{B}[\mathcal{X}, \mathcal{B}[\mathcal{Y}, \mathcal{Z}]]$. Now apply Theorem 5.1 to extend $T \in \mathcal{B}[\mathcal{X}, \mathcal{B}[\mathcal{Y}, \mathcal{Z}]]$ to $\widehat{T} \in \mathcal{B}[\widehat{\mathcal{X}}, \mathcal{B}[\mathcal{Y}, \mathcal{Z}]]$. Take

$$\widehat{T}(\widehat{x}) = \widehat{T}(\widehat{x})(\cdot) \in \mathcal{B}[\mathcal{Y}, \mathcal{Z}] \quad \text{for each} \quad \widehat{x} \in \widehat{\mathcal{X}}$$

for which $\widehat{T}(\widetilde{x}) = T(J^{-1}\widetilde{x})$ for every $\widetilde{x} = Jx \in \widetilde{\mathcal{X}} = J(\mathcal{X})$ and $x \in \mathcal{X}$, where $J\colon \mathcal{X} \to \widetilde{\mathcal{X}} \bar{\subseteq} \widehat{\mathcal{X}}$ is an isometric isomorphism. Next apply Theorem 5.1 again to extend $\widehat{T}(\widehat{x}) \in \mathcal{B}[\mathcal{Y}, \mathcal{Z}]$ to $\widehat{T}(\widehat{x})^\wedge \in \mathcal{B}[\widehat{\mathcal{Y}}, \mathcal{Z}]$. Take

$$\widehat{T}(\widehat{x})^\wedge(\cdot) \in \mathcal{B}[\widehat{\mathcal{Y}}, \mathcal{Z}] \quad \text{for each} \quad \widehat{x} \in \widehat{\mathcal{X}}$$

for which $\widehat{T}(\widehat{x})^\wedge(\widetilde{y}) = \widehat{T}(\widehat{x})(K^{-1}\widetilde{y})$ for every $\widetilde{y} = Ky \in \widetilde{\mathcal{Y}} = K(\mathcal{Y})$ and every $y \in \mathcal{Y}$, where $K\colon \mathcal{Y} \to \widetilde{\mathcal{Y}} \bar{\subseteq} \widehat{\mathcal{Y}}$ is an isometric isomorphism. Now define the map $\widehat{\phi}\colon \widehat{\mathcal{X}}\times\widehat{\mathcal{Y}} \to \mathcal{Z}$ as follows: for every $(\widehat{x}, \widehat{y}) \in \widehat{\mathcal{X}}\times\widehat{\mathcal{Y}}$ set

$$\widehat{\phi}(\widehat{x}, \widehat{y}) = \widehat{T}(\widehat{x})^\wedge(\widehat{y}).$$

Claim. $\widehat{\phi}$ is bounded bilinear (i.e., $\widehat{\phi} \in \boldsymbol{b}[\widehat{\mathcal{X}}\times\widehat{\mathcal{Y}}, \mathcal{Z}]$).

Proof of Claim. Recall: $\widehat{T}(\widehat{x})^\wedge(\cdot) \in \mathcal{B}[\widehat{\mathcal{Y}}, \mathcal{Z}]$ for each $\widehat{x} \in \widehat{\mathcal{X}}$ (by extension). We show that $\widehat{T}(\cdot)^\wedge(\widehat{y}) \in \mathcal{B}[\widehat{\mathcal{X}}, \mathcal{Z}]$ for each $\widehat{y} \in \widehat{\mathcal{Y}}$. The proof is split into two parts.

(b_1) First take arbitrary $\widehat{x}_1, \widehat{x}_2 \in \widehat{\mathcal{X}}$. Since $\widehat{T} \in \mathcal{B}[\widehat{\mathcal{X}}, \mathcal{W}]$, we get $\widehat{T}(\widehat{x}_1 + \widehat{x}_2) = \widehat{T}(\widehat{x}_1) + \widehat{T}(\widehat{x}_2)$. By the definition of $\widehat{T}(\cdot)^\wedge(\widetilde{y})$ we get $\widehat{T}(\widehat{x}_1)^\wedge(\widetilde{y}) + \widehat{T}(\widehat{x}_2)^\wedge(\widetilde{y}) = \widehat{T}(\widehat{x}_1)(K^{-1}\widetilde{y}) + \widehat{T}(\widehat{x}_2)(K^{-1}\widetilde{y}) = \widehat{T}(\widehat{x}_1 + \widehat{x}_2)(K^{-1}\widetilde{y}) = \widehat{T}(\widehat{x}_1 + \widehat{x}_2)^\wedge(\widetilde{y})$ for every $\widetilde{y} \in \widetilde{\mathcal{Y}} \bar{\subseteq} \widehat{\mathcal{Y}}$. So $\widehat{T}(\widehat{x}_1)^\wedge(\widehat{y}) + \widehat{T}(\widehat{x}_2)^\wedge(\widehat{y}) = \widehat{T}(\widehat{x}_1 + \widehat{x}_2)^\wedge(\widehat{y})$ for each $\widehat{y} \in \widehat{\mathcal{Y}}$, proving additivity. A similar argument proves homogeneity. Hence $\widehat{T}(\cdot)^\wedge(\widehat{y}) \in \mathcal{L}[\widehat{\mathcal{X}}, \mathcal{Z}]$.

(b_2) Now take an arbitrary $\widehat{x} \in \widehat{\mathcal{X}}$. Since $\widehat{T}(\widehat{x})^\wedge(\cdot)$ extends $\widehat{T}(\widehat{x})(\cdot)$, it follows that $\|\widehat{T}(\widehat{x})^\wedge(\cdot)\| = \|\widehat{T}(\widehat{x})(\cdot)\|$ (Theorem 5.1). So $\|\widehat{T}(\widehat{x})^\wedge(\widetilde{y})\| = \|\widehat{T}(\widehat{x})(K^{-1}\widetilde{y})\|$ is (uniformly) bounded for all $\widetilde{y} = Ky \in \widetilde{\mathcal{Y}} \bar{\subseteq} \widehat{\mathcal{Y}}$. Thus for a fixed and arbitrary

$\widehat{y} \in \widehat{\mathcal{Y}}$, the vectors $\widehat{T}(\widehat{x})\widehat{\ }(\widehat{y}) \in \mathcal{Z}$ are such that $\|\widehat{T}(\widehat{x})\widehat{\ }(\widehat{y})\| < \infty$ for every $\widehat{x} \in \widehat{\mathcal{X}}$. Since $\widehat{T}(\cdot)\widehat{\ }(\widehat{y})$ is linear and $\widehat{\mathcal{X}}$ is a Banach space, the Banach–Steinhaus Theorem (cf. Proposition 5.F) ensures $\|\widehat{T}(\cdot)\widehat{\ }(\widehat{y})\| < \infty$. Hence $\widehat{T}(\cdot)\widehat{\ }(\widehat{y}) \in \mathcal{B}[\widehat{\mathcal{X}}, \mathcal{Z}]$.

Then $\widehat{\phi}(\widehat{x}, \cdot) = \widehat{T}(\widehat{x})\widehat{\ }(\cdot)$ and $\widehat{\phi}(\cdot, \widehat{y}) = \widehat{T}(\cdot)\widehat{\ }(\widehat{y})$ are both linear and bounded. As $\widehat{\mathcal{X}}$ and $\widehat{\mathcal{Y}}$ are Banach spaces, Theorem 6.2 ensures $\widehat{\phi} \in \boldsymbol{b}[\widehat{\mathcal{X}} \times \widehat{\mathcal{Y}}, \mathcal{Z}]$. □

Since $\widehat{\phi}(\widetilde{x}, \widetilde{y}) = \widehat{T}(\widetilde{x})\widehat{\ }(\widetilde{y}) = \widehat{T}(\widetilde{x})(K^{-1}y) = T(J^{-1}\widetilde{x})(K^{-1}\widetilde{y}) = \phi(J^{-1}\widetilde{x}, K^{-1}\widetilde{y})$ for every $(\widetilde{x}, \widetilde{y}) \in \widetilde{\mathcal{X}} \times \widetilde{\mathcal{Y}} = J(\mathcal{X}) \times K(\mathcal{Y})$,

$$\widehat{\phi}|_{J(\mathcal{X}) \times K(\mathcal{Y})}(\cdot, \cdot) = \phi(J^{-1}(\cdot), K^{-1}(\cdot))$$

and so $\widehat{\phi}$ in $\boldsymbol{b}[\widehat{\mathcal{X}} \times \widehat{\mathcal{Y}}, \mathcal{Z}]$ is an extension over completion of ϕ in $\boldsymbol{b}[\mathcal{X} \times \mathcal{Y}, \mathcal{Z}]$.

(c) Extensions over completion of bounded linear transformations are unique up to isometric isomorphisms and preserve norms (Theorem 5.1), and since $\phi \cong T$, then $\|\widehat{\phi}\| = \|\widehat{T}(\cdot)\widehat{\ }(\cdot)\| = \|T(\cdot)(\cdot)\| = \|T\| = \|\phi\|$ and extensions over completion of bilinear maps are also unique up to isometric isomorphisms. □

Remark 6.6. *Extension over Completion for Incomplete $\mathcal{Z}$.* According to Theorem 6.5, the map $\widehat{\phi} \in \boldsymbol{b}[\widehat{\mathcal{X}} \times \widehat{\mathcal{Y}}, \mathcal{Z}]$ is the extension over completion of the map $\phi \in \boldsymbol{b}[\mathcal{X} \times \mathcal{Y}, \mathcal{Z}]$. It considers completions $\widehat{\mathcal{X}}$ of $\mathcal{X}$ and $\widehat{\mathcal{Y}}$ of $\mathcal{Y}$, but assumes $\mathcal{Z}$ is already complete so that there is no need to take a completion of $\mathcal{B}[\mathcal{Y}, \mathcal{Z}]$. This is the standard form of extension over completion for bounded bilinear maps (see, e.g., [15, Section 1.6, p. 11]). A generalization where $\mathcal{Z}$ is not complete but requires a completion $\widehat{\mathcal{Z}}$ can obtained as follows. If the normed space $\mathcal{Z}$ is not complete, then let $\widehat{\mathcal{Z}}$ be its completion so that $\widehat{\mathcal{Z}} = \widetilde{\mathcal{Z}}^{-} = L(\mathcal{Z})^{-}$ for some isometric isomorphism

$$L \colon \mathcal{Z} \to L(\mathcal{Z}) = \widetilde{\mathcal{Z}} \subseteq \widehat{\mathcal{Z}}.$$

Since the argument in the proof of Theorem 6.5 assumes completeness for $\mathcal{Z}$, then completing $\mathcal{Z}$ must be the first step. Define $\phi' \colon \mathcal{X} \times \mathcal{Y} \to \widehat{\mathcal{Z}}$ by setting

$$\phi'(x, y) = L\phi(x, y) \in \widetilde{\mathcal{Z}} \subseteq \widehat{\mathcal{Z}} \quad \text{for every} \quad (x, y) \in \mathcal{X} \times \mathcal{Y}$$

so that $\phi' = L\phi \colon \mathcal{X} \times \mathcal{Y} \to \widehat{\mathcal{Z}}$ is bilinear and bounded (as L is linear and bounded) where $\widehat{\mathcal{Z}}$ is complete and $\|\phi'\| = \|\phi\|$ (as L is an isometric isomorphism). Replace ϕ with ϕ' and replace $\mathcal{Z}$ with $\widehat{\mathcal{Z}}$ in the proof of Theorem 6.5 to get the extension $\widehat{\phi} \in \boldsymbol{b}[\widehat{\mathcal{X}} \times \widehat{\mathcal{Y}}, \widehat{\mathcal{Z}}]$ of $\phi' = L\phi \in \boldsymbol{b}[\mathcal{X} \times \mathcal{Y}, \widehat{\mathcal{Z}}]$ such that

$$\widehat{\phi}|_{\widetilde{\mathcal{X}} \times \widetilde{\mathcal{Y}}}(\widetilde{x}, \widetilde{y}) = \widehat{\phi'}(\widetilde{x}, \widetilde{y}) = \phi'(J^{-1}\widetilde{x}, K^{-1}\widetilde{y}) = L\phi(J^{-1}\widetilde{x}, K^{-1}\widetilde{y})$$

with $(\widetilde{x}, \widetilde{y}) = (Jx, Ky) \in \widetilde{\mathcal{X}} \times \widetilde{\mathcal{Y}}$ for every $(x, y) \in \mathcal{X} \times \mathcal{Y}$. This means the map $\widehat{\phi} : \widehat{\mathcal{X}} \times \widehat{\mathcal{Y}} \to \widehat{\mathcal{Z}}$ is the extension over completion of $L\phi \colon \mathcal{X} \times \mathcal{Y} \to \widehat{\mathcal{Z}}$, where

$$\widehat{\phi}|_{\widetilde{\mathcal{X}} \times \widetilde{\mathcal{Y}}}(\cdot, \cdot) = L\phi(J^{-1}(\cdot), K^{-1}(\cdot)) \quad \text{and} \quad \|\widehat{\phi}\| = \|\phi'\| = \|\phi\|,$$

and in this sense $\widehat{\phi}: \widehat{\mathcal{X}}\times\widehat{\mathcal{Y}} \to \widehat{\mathcal{Z}}$ is referred to as an extension over completion of $\phi: \mathcal{X}\times\mathcal{Y} \to \mathcal{Z}$, where $\widehat{\mathcal{X}}$, $\widehat{\mathcal{Y}}$, and $\widehat{\mathcal{Z}}$ are the completions of $\mathcal{X}$, $\mathcal{Y}$, and $\mathcal{Z}$ with $J(\mathcal{X}) = \widetilde{\mathcal{X}} \subseteq \widehat{\mathcal{X}}$, $K(\mathcal{Y}) = \widetilde{\mathcal{Y}} \subseteq \widehat{\mathcal{Y}}$, and $L(\mathcal{Z}) = \widetilde{\mathcal{Z}} \subseteq \widehat{\mathcal{Z}}$ for each of the isomorphisms J, K, and L. The diagram below illustrates the above argument.

$$\begin{array}{ccc}
\mathcal{X} \times \mathcal{Y} & \xrightarrow{\ \phi\ } & \mathcal{Z} \\
{\scriptstyle J}\downarrow \quad \downarrow{\scriptstyle K} & & \downarrow{\scriptstyle L} \\
\widetilde{\mathcal{X}} \times \widetilde{\mathcal{Y}} & \xrightarrow{\ \widehat{\phi}|_{\widetilde{\mathcal{X}}\times\widetilde{\mathcal{Y}}}\ } & \widetilde{\mathcal{Z}} \\
\subseteq \quad \subseteq & & \subseteq \\
\widehat{\mathcal{X}} \times \widehat{\mathcal{Y}} & \xrightarrow{\ \widehat{\phi}\ } & \widehat{\mathcal{Z}}\,.
\end{array}$$

Then a new statement for Theorem 6.5 reads as follows. *Let the Banach spaces $\widehat{\mathcal{X}}$, $\widehat{\mathcal{Y}}$, $\widehat{\mathcal{Z}}$ be completions of normed spaces $\mathcal{X}$, $\mathcal{Y}$, $\mathcal{Z}$. Every bilinear map $\phi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ has an extension $\widehat{\phi} \in \boldsymbol{b}[\widehat{\mathcal{X}}\times\widehat{\mathcal{Y}}, \widehat{\mathcal{Z}}]$ over the Cartesian product of the completions $\widehat{\mathcal{X}}$ and $\widehat{\mathcal{Y}}$ of $\mathcal{X}$ and $\mathcal{Y}$ to the completion $\widehat{\mathcal{Z}}$ of $\mathcal{Z}$. Moreover, $\widehat{\phi}$ is unique up to isometric isomorphisms and $\|\widehat{\phi}\| = \|\phi\|$.*

The bilinear counterpart of Remark 5.2 reads as follows.

Remark 6.7. *Extensions of Bilinear Maps*. As before, the previous theorem is again referred to as extension over completion, and it will likewise be enough for our present needs regarding extention results for bounded bilinear maps. As in Remark 5.2, we consider here some further extension procedures, now for bilinear maps (same convention: all linear spaces are over the same field).

(a) An algebraic result on extensions of bilinear maps was considered in Theorem 2.5, which says that *if $\mathcal{M}$ and $\mathcal{N}$ are nonzero linear manifolds of linear spaces $\mathcal{X}$ and $\mathcal{Y}$, and if $\phi: \mathcal{M}\times\mathcal{N} \to \mathcal{Z}$ is a bilinear map into a linear space $\mathcal{Z}$, then there exists a bilinear extension $\widehat{\phi}: \mathcal{X}\times\mathcal{Y} \to \mathcal{Z}$ of ϕ defined on $\mathcal{X}\times\mathcal{Y}$* (alternate proofs appeared in Remarks 2.7(a) and 3.8(a)).

(b) Continuity and boundedness coincide for bilinear maps (Theorem 6.2), as they do for linear transformations. Bilinear, however, is definitely nonlinear (bilinear forms are closer to quadratic forms than to linear forms), and a sharp distinction between bilinear and linear emerges when, on the one hand, every continuous linear transformation is uniformly continuous, while on the other hand, uniform continuity fails to hold for continuous bilinear maps (cf. Remark 6.4(c)). Thus, since uniform continuity is required for the notion of extension by continuity (in any metric-space setting), *there is no extension by continuity for* (*continuous*) *bilinear maps* (cf. Remark 6.4(c)).

(c) Extension over completion for continuous bilinear maps, however, survives the lack of extension by continuity, as we saw in Theorem 6.5, which extends Theorem 5.1 towards bilinear maps. To bypass the lack of extension by continuity for continuous bilinear maps, Theorem 6.3 was used to transfer the bilinear problem to a linear one, where extension by continuity holds.

(d) Unlike continuous linear functionals on a linear manifold $\mathcal{M}$ of a normed space $\mathcal{X}$, there is no Hahn–Banach Theorem for continuous bilinear functionals on $\mathcal{M}\times\mathcal{N}$ where $\mathcal{M}$ and $\mathcal{N}$ are linear manifolds of normed spaces $\mathcal{X}$ and $\mathcal{Y}$. This was shown in Remark 6.4(b) (with some help from Theorem 6.3), and the failure persists even when $\mathcal{X}$ and $\mathcal{Y}$ are Banach spaces. (For bilinear versions of Hahn–Banach-like results for bilinear forms, see [36].)

(e) Although there is no Hahn–Banach Theorem for bilinear functionals, variants of it may hold if extra assumptions are imposed (even generalizations for bilinear maps taking values in a Banach space $\mathcal{Z}$ rather than just in $\mathbb{F}$). This may lead to a Banach-space version of the purely algebraic result in item (a), now dealing with a bounded bilinear extension $\widehat{\phi}\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{Z}$ of a bounded bilinear map $\phi\colon \mathcal{M}\times\mathcal{N} \to \mathcal{Z}$. As in the linear case (cf. Remark 5.2(g,h,i)), extra assumptions may be placed on each of the four players involved, namely, the normed (or Banach) spaces $\mathcal{X}$ and $\mathcal{Y}$, the linear manifolds $\mathcal{M}$ and $\mathcal{N}$, the bounded bilinear map ϕ, and the Banach space $\mathcal{Z}$ (see, e.g., [36]).

(f) There are several approaches to adding extra assumptions onto some of the above mentioned players, either by considering concrete Banach spaces $\mathcal{X}$ and $\mathcal{Y}$, or by imposing conditions on the linear manifolds $\mathcal{M}$ and $\mathcal{N}$. We will consider a generalization of a result from [36] in Theorem 6.8. Other ways to face the bounded bilinear extension problem have included assuming $\mathcal{X}=\mathcal{Y}$ and $\mathcal{Z}=\mathbb{F}$ (i.e., bilinear or multilinear forms rather than maps). Many authors also assume $\mathcal{M}=\mathcal{N}$. Along this line, the role played by the embedding of $\mathcal{M}$ into $\mathcal{X}$ was considered in [9]; the case of $\mathcal{X}=L^1$ was considered in [10] and [16]; properties of the bilinear (multilinear) form itself were investigated in [43]; the restriction operator taking bounded bilinear forms on $\mathcal{X}\times\mathcal{X}$ into bounded bilinear forms into $\mathcal{M}\times\mathcal{M}$ was considered in [8]; and a characterization of the sequence space c_0 in terms of extensible bilinear forms was given in [7]. All these have been done for bilinear (multilinear) forms (rather than for maps), having in mind the quest for conditions (or particularizations) ensuring Hahn–Banach-type results for continuous bilinear functionals.

(g) In particular, if we impose a rather standard condition on the linear manifolds $\mathcal{M}$ and $\mathcal{N}$, then a Banach-space version of the algebraic result in item (a) for bounded bilinear maps can also be obtained by restricting $\mathcal{M}$ and $\mathcal{N}$ to be complemented in $\mathcal{X}$ and $\mathcal{Y}$. (More particularly, leading to a version of a Hahn–Banach-like result for bilinear forms by setting $\mathcal{Z}=\mathbb{F}$). This extends Theorem 5.4 on bounded linear transformations to bounded bilinear maps.

We borrow Theorem 6.8 below from [59]. It is the bilinear counterpart of Theorem 5.4. This extends the result in [36] by showing that the norm-one condition is required only for the norm identity (not for the extension itself), and it also holds for bilinear maps in general.

Theorem 6.8. *Let $\mathcal{M}$ and $\mathcal{N}$ be linear manifolds of Banach spaces $\mathcal{X}$ and $\mathcal{Y}$.*

(a) *Every bounded bilinear map* $\phi\colon \mathcal{M}\times\mathcal{N}\to\mathcal{Z}$ *into an arbitrary Banach space* $\mathcal{Z}$ *has a bounded bilinear extension* $\widehat{\phi}\colon \mathcal{X}\times\mathcal{Y}\to\mathcal{Z}$ *over* $\mathcal{X}\times\mathcal{Y}$ *if and only if the closures* $\mathcal{M}^-$ *and* $\mathcal{N}^-$ *of* $\mathcal{M}$ *and* $\mathcal{N}$ *are complemented in* $\mathcal{X}$ *and* $\mathcal{Y}$.

(b) *Moreover, if* $\mathcal{M}^- = \mathcal{R}(E)$ *and* $\mathcal{N}^- = \mathcal{R}(P)$ *for some projections* $E \in \mathcal{B}[\mathcal{X}]$ *and* $P \in \mathcal{B}[\mathcal{Y}]$ *with* $\|E\| = \|P\| = 1$, *then* $\|\widehat{\phi}\| = \|\phi\|$.

Proof. (a) Let $\mathcal{M}$ and $\mathcal{N}$ be nonzero linear manifolds of Banach spaces $\mathcal{X}$ and $\mathcal{Y}$. Let $\mathcal{M}^-$ and $\mathcal{N}^-$ be their closures. Take an arbitrary Banach space $\mathcal{Z}$ and an arbitrary $\phi \in \boldsymbol{b}[\mathcal{M}\times\mathcal{N}, \mathcal{Z}]$. The next two assertions are equivalent.

(a_1) For every $\mathcal{Z}$ and every $\phi \in \boldsymbol{b}[\mathcal{M}\times\mathcal{N}, \mathcal{Z}]$, there exists $\widehat{\phi} \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ such that $\widehat{\phi}|_{\mathcal{M}\times\mathcal{N}} = \phi$.

(a_2) $\mathcal{M}^-$ and $\mathcal{N}^-$ are complemented subspaces of $\mathcal{X}$ and $\mathcal{Y}$.

(a_1)$\Rightarrow$(a_2). Let $g\colon \mathcal{N}\to\mathbb{F}$ be an arbitrary nonzero bounded linear functional on the linear manifold $\mathcal{N}$ and let $T\colon \mathcal{M}\to\mathcal{Z}$ be an arbitrary nonzero bounded linear transformation on the linear manifold $\mathcal{M}$. (Existence of each is ensured by the Hahn–Banach Theorem.) Consider the map $\phi\colon \mathcal{M}\times\mathcal{N}\to\mathcal{Z}$ defined by

$$\phi(u, v) = g(v)Tu \quad \text{for every} \quad (u, v) \in \mathcal{M}\times\mathcal{N}.$$

Since g and T are both linear and bounded, it is immediate to verify that ϕ is bilinear and bounded.

(Indeed, $\phi(u, v_1 + v_2) = g(v_1 + v_2)Tu = (g(v_1) + g(v_2))Tu = g(v_1)Tu + g(v_2)Tu = \phi(u, v_1) + \phi(u, v_2)$, $\phi(u_1 + u_2, v) = g(v)T(u_1 + u_2) = g(v)(Tu_1 + Tu_2) = g(v)Tu_1 + g(v)Tu_2 = \phi(u_1, v) + \phi(u_2, v)$, $\phi(\alpha u, v) = g(v)(\alpha Tu) = \alpha g(v)Tu = \alpha\phi(u, v) = g(\alpha v)Tu = \phi(u, \alpha v)$, $\|\phi(u, v)\| \le \|g\|\|T\|\|v\|\|u\|$ for every $u, u_1, u_2 \in \mathcal{M}$, every $v, v_1, v_2 \in \mathcal{N}$, and every $\alpha \in \mathbb{F}$.)

So $\phi \in \boldsymbol{b}[\mathcal{M}\times\mathcal{N}, \mathcal{Z}]$. If ($\mathrm{a}_1$) holds, then consider its extension $\widehat{\phi} \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$, and let $\phi^v \in \mathcal{B}[\mathcal{M}, \mathcal{Z}]$ and $\widehat{\phi}^v \in \mathcal{B}[\mathcal{X}, \mathcal{Z}]$ be their v-sections for each $v \in \mathcal{N} \subseteq \mathcal{Y}$. Then for every $u \in \mathcal{M}$ and every $v \in \mathcal{N}$,

$$\widehat{\phi}^v|_{\mathcal{M}}(u) = \widehat{\phi}(u, v) = \widehat{\phi}|_{\mathcal{M}\times\mathcal{N}}(u, v) = \phi(u, v) = \phi^v(u).$$

Fix an arbitrary $v \in \mathcal{N}$ for which $g(v) \ne 0$ and set $0 \ne \alpha = g(v) \in \mathbb{F}$. Thus

$$\widehat{\phi}^v|_{\mathcal{M}}(\cdot) = \phi^v(\cdot) = \alpha T(\cdot) \in \mathcal{B}[\mathcal{M}, \mathcal{Z}].$$

Hence $\alpha T = \phi^v \in \mathcal{B}[\mathcal{M}, \mathcal{Z}]$ has an extension $\widehat{\alpha T} = \widehat{\phi}^v \in \mathcal{B}[\mathcal{X}, \mathcal{Z}]$. Then there exists $\widehat{T} \in \mathcal{B}[\mathcal{X}, \mathcal{Z}]$ for which $\widehat{T}|_{\mathcal{M}} = T$. Since $\mathcal{X}$ is a Banach space and since this holds for every $T \in \mathcal{B}[\mathcal{M}, \mathcal{Z}]$ and every Banach space $\mathcal{Z}$, then Theorem 5.4 says that $\mathcal{M}^-$ is complemented. Similarly, by taking an arbitrary nonzero bounded linear functional $f\colon \mathcal{M}\to\mathbb{F}$ and an arbitrary nonzero bounded linear transformation $S\colon \mathcal{N}\to\mathcal{Z}$, and defining $\psi \in \boldsymbol{b}[\mathcal{M}\times\mathcal{N}, \mathcal{Z}]$ by

$$\psi(u,v) = f(u)Sv \quad \text{for every} \quad (u,v) \in \mathcal{M}\times\mathcal{N},$$

the same argument ensures that $\mathcal{N}^-$ is complemented. So (a_2) holds.

$(\mathrm{a}_2) \Rightarrow (\mathrm{a}_1)$. Consider the normed spaces $\mathcal{M}$ and $\mathcal{N}$. By Theorem 6.3,

$$\boldsymbol{b}[\mathcal{M}\times\mathcal{N}, \mathcal{Z}] \cong \mathcal{B}[\mathcal{M}, \mathcal{B}[\mathcal{N}, \mathcal{Z}]]$$

for any normed space $\mathcal{Z}$. Let $\Im\colon \boldsymbol{b}[\mathcal{M}\times\mathcal{N}, \mathcal{Z}] \to \mathcal{B}[\mathcal{M}, \mathcal{B}[\mathcal{N}, \mathcal{Z}]]$ be the natural isometric isomorphism sending an arbitrary $\phi \in \boldsymbol{b}[\mathcal{M}\times\mathcal{N}, \mathcal{Z}]$ to $T = \Im(\phi)$ in $\mathcal{B}[\mathcal{M}, \mathcal{B}[\mathcal{N}, \mathcal{Z}]]$ which is given by $T(u) = \phi_u \in \mathcal{B}[\mathcal{N}, \mathcal{Z}]$ for each $u \in \mathcal{M}$. Thus

$$\phi(u,v) = T(u)(v) \quad \text{for every} \quad (u,v) \in \mathcal{M}\times\mathcal{N}.$$

First suppose $\mathcal{Z}$ is a Banach space (and so is $\mathcal{B}[\mathcal{N}, \mathcal{Z}]$) in order to allow extension by continuity of uniformly continuous functions on dense sets. Consider the extension by continuity $\widetilde{T} \in \mathcal{B}[\mathcal{M}^-, \mathcal{B}[\mathcal{N}, \mathcal{Z}]]$ of $T \in \mathcal{B}[\mathcal{M}, \mathcal{B}[\mathcal{N}, \mathcal{Z}]]$. Now for each $\widetilde{u} \in \mathcal{M}^-$ consider the extension by continuity $\widetilde{T}^{\sim}(\widetilde{u}) \in \mathcal{B}[\mathcal{N}^-, \mathcal{Z}]$ of $\widetilde{T}(\widetilde{u}) \in \mathcal{B}[\mathcal{N}, \mathcal{Z}]$, defining a transformation $\widetilde{T}^{\sim} \in \mathcal{B}[\mathcal{M}^-, \mathcal{B}[\mathcal{N}^-, \mathcal{Z}]]$ for which

$$\widetilde{T}^{\sim}(u)(v) = T(u)(v) \quad \text{for every} \quad (u,v) \in \mathcal{M}\times\mathcal{N}.$$

Applying Theorem 6.3 again,

$$\boldsymbol{b}[\mathcal{M}^-\times\mathcal{N}^-, \mathcal{Z}] \cong \mathcal{B}[\mathcal{M}^-, \mathcal{B}[\mathcal{N}^-, \mathcal{Z}]].$$

Let $\widetilde{\Im}\colon \boldsymbol{b}[\mathcal{M}^-\times\mathcal{N}^-, \mathcal{Z}] \to \mathcal{B}[\mathcal{M}^-, \mathcal{B}[\mathcal{N}^-, \mathcal{Z}]]$ be the natural isometric isomorphism. Set $\widetilde{\phi} = \widetilde{\Im}^{-1}(\widetilde{T}^{\sim})$ in $\boldsymbol{b}[\mathcal{M}^-\times\mathcal{N}^-, \mathcal{Z}]$ for each $\widetilde{T}^{\sim}$ in $\mathcal{B}[\mathcal{M}^-, \mathcal{B}[\mathcal{N}^-, \mathcal{Z}]]$ so that $\widetilde{\phi}_{\widetilde{u}} = \widetilde{T}^{\sim}(\widetilde{u}) \in \mathcal{B}[\mathcal{N}^-, \mathcal{Z}]$ for each $\widetilde{u} \in \mathcal{M}^-$, and hence

$$\widetilde{\phi}(\widetilde{u}, \widetilde{v}) = \widetilde{T}^{\sim}(\widetilde{u})(\widetilde{v}) \quad \text{for every} \quad (\widetilde{u}, \widetilde{v}) \in \mathcal{M}^-\times\mathcal{N}^-.$$

Thus $\widetilde{\phi}(u,v) = \widetilde{T}^{\sim}(u)(v) = T(u)(v) = \phi(u,v)$ for every $(u,v) \in \mathcal{M}\times\mathcal{N}$. So

$$\widetilde{\phi}|_{\mathcal{M}\times\mathcal{N}} = \phi.$$

Next suppose (a_2) holds. That is, there are projections $E \in \mathcal{B}[\mathcal{X}]$ and $P \in \mathcal{B}[\mathcal{Y}]$ with $\mathcal{R}(E) = \mathcal{M}^-$ and $\mathcal{R}(P) = \mathcal{N}^-$ (Remark 5.3). Set

$$\widehat{\phi}(x,y) = \widetilde{\phi}(Ex, Py) \quad \text{for every} \quad (x,y) \in \mathcal{X}\times\mathcal{Y}.$$

Again, since $E \in \mathcal{B}[\mathcal{X}]$ and $P \in \mathcal{B}[\mathcal{Y}]$ are both linear and bounded, and since $\widetilde{\phi} \in \boldsymbol{b}[\mathcal{M}^-\times\mathcal{N}^-, \mathcal{Z}]$ is bilinear and bounded, it is readily verified that $\widehat{\phi}$ is again bilinear and bounded (i.e., $\widehat{\phi} \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$). Moreover, since E and P act as the identity on $\mathcal{M} \subseteq \mathcal{R}(E) = \mathcal{M}^-$ and $\mathcal{N} \subseteq \mathcal{R}(P) = \mathcal{N}^-$, we get (a_1):

$$\widehat{\phi}|_{\mathcal{M}\times\mathcal{N}} = \widetilde{\phi}(E|_{\mathcal{M}}, P|_{\mathcal{N}}) = \widetilde{\phi}|_{\mathcal{M}\times\mathcal{N}} = \phi.$$

(b) Hence $\|\phi\| = \|\widehat{\phi}|_{\mathcal{M}\times\mathcal{N}}\| \le \|\widehat{\phi}\| = \|\widetilde{\phi}(E(\cdot), P(\cdot))\| \le \|\widetilde{\phi}\|\|E\|\|P\|$. Also,

$$\begin{aligned}\|\widetilde{\phi}\| &= \sup_{0\ne\widetilde{u}\in\mathcal{M}^-,\,0\ne\widetilde{v}\in\mathcal{N}^-} \tfrac{\|\phi(\widetilde{u},\widetilde{v})\|}{\|\widetilde{u}\|\|\widetilde{v}\|} = \sup_{0\ne\widetilde{u}\in\mathcal{M}^-,\,0\ne\widetilde{v}\in\mathcal{N}^-} \tfrac{\|\widetilde{T}^\sim(\widetilde{u})(\widetilde{v})\|}{\|\widetilde{u}\|\|\widetilde{v}\|}\\ &= \sup_{0\ne u\in\mathcal{M},\,0\ne v\in\mathcal{N}} \tfrac{\|\widetilde{T}^\sim(u)(v)\|}{\|u\|\|v\|} = \sup_{0\ne u\in\mathcal{M},\,0\ne v\in\mathcal{N}} \tfrac{\|T(u)(v)\|}{\|u\|\|v\|}\\ &= \sup_{0\ne u\in\mathcal{M},\,0\ne v\in\mathcal{N}} \tfrac{\|\phi(u,v)\|}{\|u\|\|v\|} = \|\phi\|.\end{aligned}$$

Thus $\|\phi\| \le \|\widehat{\phi}\| \le \|\phi\|\|E\|\|P\|$. Then $\|E\| = \|P\| = 1$ implies $\|\widehat{\phi}\| = \|\phi\|$. □

Remark 6.9. *The Converse of Theorem 6.8(b) Fails.*

(a) $\|\widehat{\phi}\| = \|\phi\| \not\Longrightarrow \|E\| = \|P\| = 1$. Set $F = \left(\begin{smallmatrix}0&0\\0&1\end{smallmatrix}\right) \in \mathcal{B}[\mathbb{R}^2,\mathbb{R}^2]$, so that $\|F\| = 1$. Next take $E\colon \mathbb{R}^2 \to \mathbb{R}^2$ defined by $E(\alpha,\beta) = (0, \alpha+\beta)$ for $x = (\alpha,\beta) \in \mathbb{R}^2$. This also is a continuous projection $E = \left(\begin{smallmatrix}0&0\\1&1\end{smallmatrix}\right) \in \mathcal{B}[\mathbb{R}^2,\mathbb{R}^2]$ on the Euclidean space $\mathcal{X} = \mathbb{R}^2$, now with $\|E\| = \sqrt{2}$. Set $\mathcal{M} = \mathcal{R}(E) = \mathcal{R}(F) = \{0\}\oplus\mathbb{R}$, a subspace of $\mathcal{X} = \mathbb{R}^2 = \mathbb{R}\oplus\mathbb{R}$. Take a bounded linear form $f \in \mathcal{B}[\mathbb{R}^2,\mathbb{R}] = \mathbb{R}^{2*}$ given by $f((\alpha,\beta)) = \beta$ for every $x = (\alpha,\beta) \in \mathbb{R}^2$, so that $\|f\| = 1$. Construct a bounded bilinear form $\psi \in \boldsymbol{b}[\mathbb{R}^2\times\mathbb{R}^2,\mathbb{R}]$ as $\psi(x,y) = f(x)f(y)$ for every $(x,y) \in \mathbb{R}^2\times\mathbb{R}^2$, so that $\|\psi\| = \|f\|^2$. The restriction $\phi = \psi|_{\mathcal{M}\times\mathcal{M}} = f|_{\mathcal{M}} f|_{\mathcal{M}} = f|_{\{0\}\oplus\mathbb{R}} f|_{\{0\}\oplus\mathbb{R}}$ of ψ to $\mathcal{M}\times\mathcal{M}$ is a bounded bilinear form given by $\phi((0,\beta_1),(0,\beta_2)) = \beta_1\beta_2$ for every $(0,\beta_1),(0,\beta_2) \in \mathcal{M} = \{0\}\oplus\mathbb{R}$, with $\|\phi\| = 1$. Hence $\phi \in \boldsymbol{b}[\mathcal{M}\times\mathcal{M},\mathbb{R}]$ has an extension $\widehat{\phi} = \psi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{X},\mathbb{R}]$ such that $\|\phi\| = \|\widehat{\phi}\|$. However, we can have either $\|E\| = \|P\| > 1$ or $\|E\| > \|P\| = 1$, by taking $P = E$ or $P = F$.

(b) Another example with a similar argument. Let $\mathcal{X}$ be a Hilbert space and consider the Banach algebras $\mathcal{B}[\mathcal{X}]$ and $\mathcal{Z} = \mathcal{B}[\mathcal{X}]\oplus\mathcal{B}[\mathcal{X}]$, where $\mathcal{Z}$ consists of all orthogonal direct sums $A\oplus B = \left(\begin{smallmatrix}A&O\\O&B\end{smallmatrix}\right)$ of A, B in $\mathcal{B}[\mathcal{X}]$. Take the projection $\mathcal{E} \in \mathcal{B}[\mathcal{Z}]$ given by $\mathcal{E}(A\oplus B) = O\oplus(A+B)$ for $A, B \in \mathcal{B}[\mathcal{X}]$, so that $\|\mathcal{E}\| > 1$. Set $\mathcal{M} = \mathcal{R}(\mathcal{E}) = \{0\}\oplus\mathcal{B}[\mathcal{X}] \subseteq \mathcal{B}[\mathcal{X}]\oplus\mathcal{B}[\mathcal{X}] = \mathcal{Z}$. Consider the restriction of $\mathcal{E}$ to $\mathcal{M}$, $\mathcal{E}|_{\mathcal{M}} = \mathcal{I}_{\mathcal{M}}\colon \mathcal{M} \to \mathcal{Z}$, which acts as the identity $\mathcal{I}_{\mathcal{M}}$ on $\mathcal{M}$. Now take $\phi \in \boldsymbol{b}[\mathcal{M}\times\mathcal{M},\mathcal{Z}]$ defined by $\phi = \mathcal{I}_{\mathcal{M}}\cdot\mathcal{I}_{\mathcal{M}}$ and let $\widehat{\phi} \in \boldsymbol{b}[\mathcal{Z}\times\mathcal{Z},\mathcal{Z}]$ be defined by $\widehat{\phi} = \mathcal{I}\cdot\mathcal{I}$ where $\mathcal{I} \in \mathcal{B}[\mathcal{Z}]$ denotes the identity on $\mathcal{Z}$. In other words, ϕ is given by $\phi(O\oplus B_1, O\oplus B_2) = (O\oplus B_1)\cdot(O\oplus B_2) = O\oplus(B_1\cdot B_2)$ for every $B_1, B_2 \in \mathcal{B}[\mathcal{X}]$, and $\widehat{\phi}$ is given by $\widehat{\phi}(A_1\oplus B_1, A_2\oplus B_2) = (A_1\oplus B_1)\cdot(A_2\oplus B_2) = (A_1\cdot A_2)\oplus(B_1\cdot B_2)$ for A_1, A_2, B_1, B_2 in $\mathcal{B}[\mathcal{X}]$. Thus $\widehat{\phi}|_{\mathcal{M}\times\mathcal{M}} = \mathcal{I}_{\mathcal{M}}\cdot\mathcal{I}_{\mathcal{M}} = \phi$ and $\|\phi\| = \|\mathcal{I}_{\mathcal{M}}\cdot\mathcal{I}_{\mathcal{M}}\| = 1 = \|\mathcal{I}\cdot\mathcal{I}\| = \|\widehat{\phi}\|$. Hence $\|\widehat{\phi}\| = \|\phi\| \not\Longrightarrow \|\mathcal{E}\| = \|\mathcal{P}\| = 1$ (with $\mathcal{P} \in \mathcal{B}[\mathcal{Z}]$ taken in a similar fashion as P was taken in item (a) above).

As expected, if $\mathcal{X}$ and $\mathcal{Y}$ are Hilbert spaces, then we get a full extension result for bounded bilinear maps with no restrictions on the linear manifolds $\mathcal{M}$ and $\mathcal{N}$, which is stated in Corollary 6.10 below. In particular, with $\mathcal{Z} = \mathbb{F}$ we get a Hahn–Banach-like theorem for bounded bilinear functionals on the Cartesian product of linear manifolds of Hilbert spaces.

Corollary 6.10. *Every bounded bilinear map $\phi\colon \mathcal{M}\times\mathcal{N}\to\mathcal{Z}$ defined on the Cartesian product of arbitrary linear manifolds $\mathcal{M}$ and $\mathcal{N}$ of arbitrary Hilbert spaces $\mathcal{X}$ and $\mathcal{Y}$ into an arbitrary Banach space $\mathcal{Z}$ has a bounded bilinear extension $\widehat{\phi}\colon \mathcal{X}\times\mathcal{Y}\to\mathcal{Z}$ over $\mathcal{X}\times\mathcal{Y}$ such that $\|\widehat{\phi}\| = \|\phi\|$.*

Proof. (a) If $\mathcal{X}$ is a Hilbert space, then $\mathcal{M}^- + \mathcal{M}^\perp = \mathcal{X}$ for every linear manifold $\mathcal{M}$ of $\mathcal{X}$ by the Projection Theorem (cf. Proposition 5.Q), where $\mathcal{M}^- \cap \mathcal{M}^\perp = \{0\}$. So every subspace of a Hilbert space is complemented (cf. Remark 5.3). Thus Theorem 6.8(a) applies to every linear manifold $\mathcal{M}$ of a Hilbert space $\mathcal{X}$ and to every linear manifold $\mathcal{N}$ of a Hilbert space $\mathcal{Y}$.

(b) If $\mathcal{X}$ and $\mathcal{Y}$ are Hilbert spaces, then take the orthogonal projections E in $\mathcal{B}[\mathcal{X}]$ with $\mathcal{R}(E) = \mathcal{M}^-$ and P in $\mathcal{B}[\mathcal{Y}]$ with $\mathcal{R}(P) = \mathcal{N}^-$, which are such that $\|E\| = \|P\| = 1$ (cf. Proposition 5.R), and so $\|\widehat{\phi}\| = \|\phi\|$ by Theorem 6.8(b). □

Extensions of linear transformations are not unique (cf. example in Remark 1.2). Thus extensions of bilinear maps are not unique by Theorem 2.4.

If $\mathcal{M}$ and $\mathcal{N}$ are ranges or kernels of operators in $\mathcal{SI}[\mathcal{X}] \subseteq \mathcal{B}[\mathcal{X}]$ and in $\mathcal{SI}[\mathcal{Y}] \subseteq \mathcal{B}[\mathcal{Y}]$ (as defined in the paragraph before Corollary 5.7) respectively, then we get the following extension result for bounded bilinear maps.

Corollary 6.11. *Let $\mathcal{X}$ and $\mathcal{Y}$ be Banach spaces. Let $\mathcal{M}$ and $\mathcal{N}$ be linear manifolds of $\mathcal{X}$ and $\mathcal{Y}$ for which there are operators $T \in \mathcal{SI}[\mathcal{X}] \subseteq \mathcal{B}[\mathcal{X}]$ and $S \in \mathcal{SI}[\mathcal{Y}] \subseteq \mathcal{B}[\mathcal{Y}]$ such that*

$$\mathcal{M}^- = \mathcal{R}(T)^- \text{ or } \mathcal{M}^- = \mathcal{N}(T) \quad \text{and} \quad \mathcal{N}^- = \mathcal{R}(S)^- \text{ or } \mathcal{N}^- = \mathcal{N}(S),$$

according to whether T lies in $\mathit{\Gamma}_R[\mathcal{X}]$ or $\mathit{\Gamma}_N[\mathcal{X}]$ and S lies in $\mathit{\Gamma}_R[\mathcal{Y}]$ or $\mathit{\Gamma}_N[\mathcal{Y}]$. If $\phi\colon \mathcal{M}\times\mathcal{N}\to\mathcal{Z}$ is a bounded bilinear map into an arbitrary Banach space $\mathcal{Z}$, then there exists a bounded bilinear extension $\widehat{\phi}\colon \mathcal{X}\times\mathcal{Y}\to\mathcal{Z}$ of ϕ over $\mathcal{X}\times\mathcal{Y}$.

Proof. As in the proof of Corollary 5.7, this follows immediately from Theorem 6.8. If $T \in \mathcal{SI}[\mathcal{X}]$ and $S \in \mathcal{SI}[\mathcal{Y}]$, then either $\mathcal{R}(T)^-$ or $\mathcal{N}(T)$ is complemented in $\mathcal{X}$ and either $\mathcal{R}(S)^-$ or $\mathcal{N}(S)$ is complemented in $\mathcal{Y}$. If $\mathcal{M}^- = \mathcal{R}(T)^-$ or $\mathcal{M}^- = \mathcal{N}(T)$ and if $\mathcal{N}^- = \mathcal{R}(S)^-$ or $\mathcal{N}^- = \mathcal{N}(S)$, then $\phi \in \boldsymbol{b}[\mathcal{M}\times\mathcal{N}, \mathcal{Z}]$ for any Banach space $\mathcal{Z}$ has an extension $\widehat{\phi} \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ by Theorem 6.8. □

Take arbitrary nonzero operators $T \in \mathcal{B}[\mathcal{X}]$ and $S \in \mathcal{B}[\mathcal{Y}]$, and take an arbitrary bounded bilinear map $\phi \in \boldsymbol{b}[\mathcal{R}(T)\times\mathcal{R}(S), \mathcal{Z}]$. Now define a map $\psi\colon \mathcal{X}\times\mathcal{Y}\to\mathcal{Z}$ by $\psi(x,y) = \phi(Tx, Sy)$ for every $(x,y) \in \mathcal{X}\times\mathcal{Y}$. According to Proposition 6.B below, $\psi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$. *If T and S are projections, then ψ is an extension of ϕ.* (Indeed, since $\psi|_{\mathcal{R}(T)\times\mathcal{R}(S)} = \phi(T|_{\mathcal{R}(T)}, S|_{\mathcal{R}(S)})$, if T and S are projections, or equivalently, if $T|_{\mathcal{R}(T)} = I_{\mathcal{R}(T)}$, identity on $\mathcal{R}(T)$, and $S|_{\mathcal{R}(S)} = I_{\mathcal{R}(S)}$, identity on $\mathcal{R}(S)$, then $\psi|_{\mathcal{R}(T)\times\mathcal{R}(S)} = \phi$.) However, if there is a $\widehat{\phi} \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ such that $\widehat{\phi}|_{\mathcal{R}(T)\times\mathcal{R}(S)} = \phi$ (e.g., under the assumptions of Corollary 6.10), then $\widehat{\phi}(Tx, Sy) = \phi(Tx, Sy) = \psi(x,y)$ for every $(x,y) \in \mathcal{X}\times\mathcal{Y}$.

6.4 Additional Propositions

Let $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ be nonzero normed spaces over the same field $\mathbb{F}$, let $b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ and $\mathcal{L}[\mathcal{X}, \mathcal{Z}]$ denote the linear space of all bilinear maps from $\mathcal{X}\times\mathcal{Y}$ to $\mathcal{Z}$ and of all linear transformations of $\mathcal{X}$ into $\mathcal{Z}$, and let $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ and $\mathcal{B}[\mathcal{X}, \mathcal{Z}]$ denote the normed spaces of all bounded (i.e., continuous) bilinear maps from $\mathcal{X}\times\mathcal{Y}$ to $\mathcal{Z}$ and of all bounded linear transformations of $\mathcal{X}$ into $\mathcal{Z}$.

Proposition 6.A. *If $\mathcal{X}$ and $\mathcal{Y}$ are finite-dimensional normed spaces, then*

$$\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] = b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}].$$

Proposition 6.B. *Let $\mathcal{X}, \mathcal{X}', \mathcal{Y}, \mathcal{Y}', \mathcal{Z}, \mathcal{Z}'$ be normed spaces (over the same field). Take bounded linear transformations $A \in \mathcal{B}[\mathcal{X}', \mathcal{X}]$, $B \in \mathcal{B}[\mathcal{Y}', \mathcal{Y}]$ and $C \in \mathcal{B}[\mathcal{Z}, \mathcal{Z}']$, and take a bounded bilinear map $\phi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$. The mapping $\psi\colon \mathcal{X}'\times\mathcal{Y}' \to \mathcal{Z}'$ given by $\psi(x', y') = C\big(\phi(Ax', By')\big)$ for every $(x', y') \in \mathcal{X}'\times\mathcal{Y}'$ is a bounded bilinear map. Actually,*

$$\psi = C \circ \phi(A, B) \in \boldsymbol{b}[\mathcal{X}'\times\mathcal{Y}', \mathcal{Z}'] \quad \text{with} \quad \|\psi\| \le \|\phi\|\|A\|\|B\|\|C\|.$$

Proposition 6.C. *The* evaluation *map $\phi\colon \mathcal{B}[\mathcal{X}, \mathcal{Y}]\times\mathcal{X} \to \mathcal{Y}$ given by*

$$\phi(T, x) = Tx \quad \text{for every} \quad (T, x) \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]\times\mathcal{X}$$

is a surjective bounded bilinear map with norm one.

Recall: all nonzero bilinear forms (i.e., for $\mathcal{Z} = \mathbb{F}$) are trivially surjective.

Proposition 6.D. *Let $\mathcal{Z}$ be a normed algebra. Take any bounded linear transformations $S \in \mathcal{B}[\mathcal{X}, \mathcal{Z}]$ and $T \in \mathcal{B}[\mathcal{Y}, \mathcal{Z}]$. The map $\phi\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{Z}$ defined by*

$$\phi(x, y) = Sx \cdot Ty \quad \text{for every} \quad (x, y) \in \mathcal{X}\times\mathcal{Y}$$

is a bounded bilinear map with $\|\phi\| \le \|S\|\|T\|$. In particular, for fixed bounded linear functionals $f \in \mathcal{X}^*$ and $g \in \mathcal{Y}^*$ the bilinear form $\phi\colon \mathcal{X}\times\mathcal{Y} \to \mathbb{F}$ given by $\phi(x, y) = f(x) \cdot g(y)$ for every $(x, y) \in \mathcal{X}\times\mathcal{Y}$ is bounded with $\|\phi\| = \|f\|\|g\|$.

There is no Hahn–Banach Theorem and no uniform continuity for bounded bilinear maps (cf. Remark 6.4(b,c)) but there is extension over completion (cf. Theorem 6.5). The next proposition shows that there is no Open Mapping Theorem for surjective bounded bilinear maps of the Cartesian product of Banach spaces onto a Banach space.

Proposition 6.E. *Let $\mathcal{Z}$ be a finite-dimensional normed space over a field $\mathbb{F}$ of dimension greater than one. The* scalar multiplication *$\mathbb{F}\times\mathcal{Z} \to \mathcal{Z}$ (given by $(\alpha, z) \mapsto \alpha z$) is a surjective bilinear map but it is not an open mapping.*

The Closed Graph Theorem, however, travels well from linear transformations to bilinear maps. The *graph* G_ϕ of a bilinear map $\phi\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{Z}$,

$$G_\phi = \big\{((x,y),z) \in (\mathcal{X}\times\mathcal{Y})\times\mathcal{Z}\colon\ z = \phi(x,y)\big\}$$
$$= \big\{(x,y,\phi(x,y)) \in \mathcal{X}\times\mathcal{Y}\times\mathcal{Z}\colon \text{for every } (x,y) \in \mathcal{X}\times\mathcal{Y}\big\},$$

is a subset of the Cartesian product $(\mathcal{X}\times\mathcal{Y})\times\mathcal{Z} = \mathcal{X}\times\mathcal{Y}\times\mathcal{Z}$. If $\mathcal{X},\mathcal{Y},\mathcal{Z}$ are normed spaces, then equip the Cartesian product with a product metric d generated by their norms (cf. Section 5.1 — as in Proposition 5.D in Section 5.4). The Closed Graph Theorem for bilinear maps is given below.

Proposition 6.F. *If $\mathcal{X},\mathcal{Y},\mathcal{Z}$ are Banach spaces and $\phi\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{Z}$ is bilinear, then ϕ is continuous if and only if G_ϕ is closed.*

As for the Banach–Steinhaus Theorem, its consequence ensuring a bounded limit for pointwise convergence of sequences of bounded mappings, which holds for sequences of bounded linear transformations (as in Proposition 5.G), also holds for sequences of bounded bilinear maps (as in Proposition 6.H below).

Proposition 6.G. *Let $\{\phi_{(\omega)}\}_{\omega\in\Omega}$ be an indexed family of bounded bilinear maps $\phi_{(\omega)}$ of the Cartesian product $\mathcal{X}\times\mathcal{Y}$ of Banach spaces $\mathcal{X}$ and $\mathcal{Y}$ into a normed space $\mathcal{Z}$. If $\sup_{\omega\in\Omega}\|\phi_{(\omega)}(x,y)\| < \infty$ for every $(x,y) \in \mathcal{X}\times\mathcal{Y}$, then $\sup_{\omega\in\Omega}\|\phi_{x(\omega)}\| < \infty$ for each $x \in \mathcal{X}$ and $\sup_{\omega\in\Omega}\|\phi_{y(\omega)}\| < \infty$ for each $y \in \mathcal{Y}$.*

A sequence of bilinear maps of the Cartesian product $\mathcal{X}\times\mathcal{Y}$ of normed spaces $\mathcal{X}$ and $\mathcal{Y}$ into a normed space $\mathcal{Z}$ (i.e., a $b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$-valued sequence) $\{\phi_{(n)}\}$ is *pointwise convergent* if for every pair (x,y) in $\mathcal{X}\times\mathcal{Y}$ the $\mathcal{Z}$-valued sequence $\{\phi_{(n)}(x,y)\}$ converges in $\mathcal{Z}$ to, say, $z_{(x,y)}$ in $\mathcal{Z}$. Since the limit is unique, this defines a map $\phi : \mathcal{X}\times\mathcal{Y} \to \mathcal{Z}$ given by $\phi(x,y) = z_{(x,y)}$ for every (x,y) in $\mathcal{X}\times\mathcal{Y}$. It is readily verified that ϕ is bilinear. Thus *a $b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$-valued sequence $\{\phi_{(n)}\}$ is pointwise convergent if and only if there exists a map ϕ in $b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$ such that $\|(\phi_{(n)} - \phi)(x,y)\| \to 0$ for every (x,y) in $\mathcal{X}\times\mathcal{Y}$.* Now consider a sequence $\{\phi_{(n)}\}$ of bounded bilinear maps of $\mathcal{X}\times\mathcal{Y}$ into $\mathcal{Z}$ (i.e., a $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$-valued sequence). An application of the Banach–Steinhaus Theorem for bounded linear transformations ensures that if $\mathcal{X}$ and $\mathcal{Y}$ are Banach spaces and $\{\phi_{(n)}\}$ is pointwise convergent, then the limit ϕ is bounded.

The next result is the bilinear counterpart of Proposition 5.G. Since Proposition 5.G is a consequence of Proposition 5.F, the proposition below is regarded as the Banach–Steinhaus Theorem for bilinear maps.

Proposition 6.H. *If $\mathcal{X}$ and $\mathcal{Y}$ are Banach spaces and $\mathcal{Z}$ is a normed space, then a $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$-valued sequence $\{\phi_{(n)}\}$ is pointwise convergent if and only if there is a map $\phi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$ such that $\|(\phi_{(n)} - \phi)(x,y)\| \to 0$ for every pair $(x,y) \in \mathcal{X}\times\mathcal{Y}$ (i.e., pointwise convergence implies a bounded limit).*

Recall that a function between topological spaces is continuous if and only if it preserves convergent nets (i.e., if and only if $f(x_\alpha) \to f(x)$ whenever $\{x_\alpha\}$ is a net converging to x). For metric spaces, nets can be replaced by sequences. A function between topological spaces is sequentially continuous if it preserves convergent sequences. Thus continuity implies sequential continuity, and the

concepts coincide in a metric space. The weak topology of (infinite-dimensional) normed spaces is not metrizable. Along these lines, also recall that a linear functional on a normed space $\mathcal{X}$ is weakly continuous (i.e., continuous with respect to the weak topology of $\mathcal{X}$) if and only if it is strongly continuous (i.e., continuous with respect to the norm topology of $\mathcal{X}$). This also does not survive when transferred from linear functionals to bilinear functionals.

Proposition 6.I. *Continuous bilinear functionals on the Cartesian product of Banach spaces may not be weakly continuous.*

Let $\mathcal{Z}^{**} = (\mathcal{Z}^*)^*$ denote the *second dual* (or *bidual*) of a normed space $\mathcal{Z}$. The next proposition states a significant fact that will be discussed below.

Proposition 6.J. *If $\mathcal{X}$ and $\mathcal{Y}$ are Banach spaces, then every bounded bilinear form $\phi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$ has an extension $\widehat{\phi} \in \boldsymbol{b}[\mathcal{X}^{**}\times\mathcal{Y}^{**}, \mathbb{F}]$ with $\|\widehat{\phi}\| = \|\phi\|$.*

Sesquilinear forms $\sigma : \mathcal{X}\times\mathcal{Y} \to \mathbb{F}$ were defined in Section 2.4 for linear spaces $\mathcal{X}$ and $\mathcal{Y}$. Inner products on a linear space $\mathcal{X}$ were defined in Section 5.4 as sesquilinear forms $\langle \cdot\,;\cdot \rangle : \mathcal{X}\times\mathcal{X} \to \mathbb{F}$ generating a quadratic form $\|\cdot\| : \mathcal{X} \to \mathbb{R}$. Inner product spaces and Hilbert spaces were defined in Section 5.4. The notion of completion of inner product spaces is similar to that of normed spaces.

> *If the image of a linear isometry on an inner product space $\mathcal{X}$ is dense in a Hilbert space $\widehat{\mathcal{X}}$, then $\widehat{\mathcal{X}}$ is a completion of $\mathcal{X}$. Equivalently, if an inner product space $\mathcal{X}$ is unitarily equivalent to a dense linear manifold of a Hilbert space $\widehat{\mathcal{X}}$, then $\widehat{\mathcal{X}}$ is a completion of $\mathcal{X}$.*

Inner product spaces, being normed spaces, satisfy in particular the assumptions of Theorem 5.1. The remaining question is whether the Banach spaces obtained by completion of inner product spaces are Hilbert spaces. Yes, they are. So completion of inner product spaces agrees with the above definition. The counterpart of Theorem 5.1 for inner product spaces reads as follows.

> *Every inner product space has a completion. Two completions of an inner product space are unitarily equivalent. If $\widehat{\mathcal{X}}$ and $\widehat{\mathcal{Y}}$ are completions of inner product spaces $\mathcal{X}$ and $\mathcal{Y}$, then every $T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ has an extension $\widehat{T} \in \mathcal{B}[\widehat{\mathcal{X}}, \widehat{\mathcal{Y}}]$ over the completion $\widehat{\mathcal{X}}$ of $\mathcal{X}$ into the completion $\widehat{\mathcal{Y}}$ of $\mathcal{Y}$. Also, $\widehat{T}$ is unique up to unitary transformations, and $\|\widehat{T}\| = \|T\|$.*

(See, e.g., [52, Definition 5.22 and Theorem 5.23].) Moreover, the inner product equipping the Hilbert space $\widehat{\mathcal{X}}$ (completion of $\mathcal{X}$) is precisely the extension over completion of the inner product space equipping $\mathcal{X}$. This is the sesquilinear version of the extension over completion for bilinear forms in Theorem 6.5.

Proposition 6.K. *Let $\widehat{\mathcal{X}}$ be the completion of an inner product space $\mathcal{X}$. The inner product $\langle \cdot\,;\cdot \rangle_{\mathcal{X}} : \mathcal{X}\times\mathcal{X} \to \mathbb{F}$ has an extension over completion on $\widehat{\mathcal{X}}\times\widehat{\mathcal{X}}$, $\langle \cdot\,;\cdot \rangle_{\widehat{\mathcal{X}}} : \widehat{\mathcal{X}}\times\widehat{\mathcal{X}} \to \mathbb{F}$ which is unique up to unitary equivalence. If $\widehat{x}, \widehat{y} \in \widehat{\mathcal{X}}$ are extensions of $x, y \in \mathcal{X}$, then $\langle \widehat{x}\,;\widehat{y} \rangle_{\widehat{\mathcal{X}}} = \langle x\,;y \rangle_{\mathcal{X}}$.*

Notes: Bilinear maps in a linear-space setting, considered in Chapter 2, played a central role in defining tensor products of linear spaces as elaborated in Chapter 3. The next chapter will consider tensor products of normed spaces, where the tensor product space will be equipped with a suitable norm, and this naturally requires the notion of bounded bilinear maps in a normed-space setting. Proposition 6.A is an immediate consequence of Theorem 6.2 since the identities $\mathcal{B}[\mathcal{X},\mathcal{Z}] = \mathcal{L}[\mathcal{X},\mathcal{Z}]$ and $\mathcal{B}[\mathcal{Y},\mathcal{Z}] = \mathcal{L}[\mathcal{Y},\mathcal{Z}]$ apply to the linear transformations ϕ^y and ϕ_x whenever $\mathcal{X}$ and $\mathcal{Y}$ are finite-dimensional, since every finite-dimensional normed space is a Banach space, and since linear transformations on finite-dimensional normed spaces are continuous (see, e.g., [52, Corollaries 4.28 and 4.30]). The bilinearity in Proposition 6.B follows from Remark 2.2 and boundedness (as well as the norm inequalities) from Theorem 6.1. The evaluation map ϕ of Proposition 6.C is bilinear (cf. Proposition 2.H), surjective (indeed, if $y \in \mathcal{Y}$ and $f \in \mathcal{X}^*$, then $Tx = f(x)y$ defines $T \in \mathcal{B}[\mathcal{X},\mathcal{Y}]$ for which $y \in \mathcal{R}(T)$), clearly contractive (i.e., $\|\phi\| \le 1$), and by setting $T(\cdot) = f(\cdot)y$ with $\|f\| = \|y\| = 1$ it follows that $\|\phi\| = 1$. Proposition 6.D is straightforward in light of the pertinent definitions, where $\|Sx{\cdot}Ty\| \le \|Sx\|\|Ty\|$ since $\mathcal{Z}$ is a normed algebra, and the norm identity for the scalar case holds because $|f(x){\cdot}g(y)| = |f(x)|\,|g(y)|$. For Proposition 6.E see, e.g., [15, Section 1.7].

Proof of Proposition 6.F. Consider the normed space obtained by the direct sum $(\mathcal{X} \oplus \mathcal{Y} \oplus \mathcal{Z}, \|\cdot\|) = (\mathcal{X}{\times}\mathcal{Y}{\times}\mathcal{Z}, \oplus, \centerdot, \mathbb{F}, \|\cdot\|)$ where the norm $\|\cdot\|$ on the linear space $\mathcal{X} \oplus \mathcal{Y} \oplus \mathcal{Z}$ is the one generating the product metric d on the Cartesian product $\mathcal{X}{\times}\mathcal{Y}{\times}\mathcal{Z}$. Suppose the graph G_ϕ of ϕ is a closed subset of $(\mathcal{X} \oplus \mathcal{Y} \oplus \mathcal{Z}, \|\cdot\|)$. Take the (closed) subspace $\mathcal{M} = \{(0,y,z) \in \mathcal{X} \oplus \mathcal{Y} \oplus \mathcal{Z} : (y,z) \in \mathcal{Y} \oplus \mathcal{Z}\}$ of $(\mathcal{X} \oplus \mathcal{Y} \oplus \mathcal{Z}, \|\cdot\|)$. Identify each x in $\mathcal{X}$ with $(x,0,0)$ in $\mathcal{X} \oplus \mathcal{Y} \oplus \mathcal{Z}$. Consider the translation $[x] = \mathcal{M} \oplus x$ for each $x \in \mathcal{X}$, again a closed subset of $(\mathcal{X} \oplus \mathcal{Y} \oplus \mathcal{Z}, \|\cdot\|)$. Thus the intersection $G_\phi \cap [x]$ is closed. The graph of each linear transformation $\phi_x \in \mathcal{L}[\mathcal{Y},\mathcal{Z}]$,

$$G_{\phi_x} = \big\{(y,z) \in \mathcal{Y} \oplus \mathcal{Z} \colon z = \phi_x(y)\big\} = (G_\phi \cap [x]) \oplus (-x) \;\subseteq\; \mathcal{Y} \oplus \mathcal{Z},$$

regarded as a subset of $\mathcal{X} \oplus \mathcal{Y} \oplus \mathcal{Z}$, is the translation of a closed set, thus closed in $(\mathcal{X} \oplus \mathcal{Y} \oplus \mathcal{Z}, \|\cdot\|)$, and naturally (and isometrically) embedded as a closed set in the subspace $(\mathcal{Y} \oplus \mathcal{Z}, \|\cdot\|) \cong (\mathcal{M}, \|\cdot\|)$. Then by the Closed Graph Theorem for linear transformations (Proposition 5.D), each ϕ_x is continuous. Symmetrically, each ϕ^y is continuous. Therefore ϕ is continuous by Theorem 6.2. The converse is readily verified by the Closed Set Theorem. □

(Indeed, suppose ϕ is continuous. Take any G_ϕ-valued convergent sequence $\{(x_n,y_n,\phi(x_n,y_n))\}$, converging in $(\mathcal{X}{\times}\mathcal{Y}{\times}\mathcal{Z}, d)$ to $(x,y,z) \in \mathcal{X}{\times}\mathcal{Y}{\times}\mathcal{Z}$. Since d is a product metric, the $\mathcal{X}$-valued, the $\mathcal{Y}$-valued, and the $\mathcal{Z}$-valued sequences $\{x_k\}$, $\{y_k\}$, $\{\phi(x_n,y_n)\}$ converge in $(\mathcal{X},\|\cdot\|)$, $(\mathcal{Y},\|\cdot\|)$, $(\mathcal{Z},\|\cdot\|)$ to $x \in \mathcal{X}$, $y \in \mathcal{Y}$, and $z \in \mathcal{Z}$, respectively. Since ϕ is continuous, it preserves convergence. By uniqueness of the limit, $\phi(x_n,y_n) \to \phi(\lim_n x_n, \lim_n y_n) = \phi(x,y) = z$ in $(\mathcal{Z},\|\cdot\|)$. Thus (x,y,z) lies in G_ϕ and hence G_ϕ is closed in the metric space $(\mathcal{X}{\times}\mathcal{Y}{\times}\mathcal{Z}, d)$ by the Closed Set Theorem (see, e.g., [52, Theorem 3.30]).)

Proof of Proposition 6.G. Since $\phi(\omega)(x,y) = \phi_x(\omega)(y) = \phi^y(\omega)(x)$ for every $(\omega, x, y) \in \Omega \times \mathcal{X} \times \mathcal{Y}$, it follows that if $\sup_{\omega\in\Omega} \|\phi(\omega)(x,y)\| < \infty$ for every $(x,y) \in \mathcal{X}\times\mathcal{Y}$, then for each $x \in \mathcal{X}$ we get $\sup_{\omega\in\Omega} \|\phi_x(\omega)(y)\| < \infty$ for every $y \in \mathcal{Y}$, and for each $y \in \mathcal{Y}$ we get $\sup_{\omega\in\Omega} \|\phi^y(\omega)(x)\| < \infty$ for every $x \in \mathcal{X}$. Now apply the Banach–Steinhaus Theorem (Proposition 5.F) to the bounded linear transformations $\phi_x(\omega) \in \mathcal{B}[\mathcal{Y}, \mathcal{Z}]$ and $\phi^y(\omega) \in \mathcal{B}[\mathcal{X}, \mathcal{Z}]$. □

Proof of Proposition 6.H. Suppose a $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$-valued sequence $\{\phi(n)\}$ converges pointwise to $\phi \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$. This means $\|(\phi(n) - \phi)(x,y)\| \to 0$ for every $(x,y) \in \mathcal{X}\times\mathcal{Y}$, which implies $\|\phi(n)(x,y)\| \to \|\phi(x,y)\|$, which in turn implies $\sup_n \|\phi(n)(x,y)\| < \infty$ for every $(x,y) \in \mathcal{X}\times\mathcal{Y}$. Thus $\sup_n \|\phi_x(n)\| < \infty$ for every $x \in \mathcal{X}$ by Proposition 6.G. Moreover, for every $(x,y) \in \mathcal{X}\times\mathcal{Y}$,

$$\|\phi_x(y)\| = \|\phi(x,y)\| = \lim_n \|\phi(n)(x,y)\| = \lim_n \|\phi_x(n)(y)\| \le \sup_n \|\phi_x(n)\|\,\|y\|.$$

So $\|\phi_x\| = \sup_{y\neq 0} \frac{\|\phi_x(y)\|}{\|y\|} \le \sup_n \|\phi_x(n)\| < \infty$ for each $x \in \mathcal{X}$. Symmetrically, $\|\phi^y\| < \infty$ for each $y \in \mathcal{Y}$. Thus $\phi_x \in \mathcal{B}[\mathcal{Y}, \mathcal{Z}]$ and $\phi^y \in \mathcal{B}[\mathcal{X}, \mathcal{Z}]$ for every x and y in the Banach spaces $\mathcal{X}$ and $\mathcal{Y}$. Then $\phi \in \mathcal{B}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ by Theorem 6.2. □

Proof of Proposition 6.I. The inner product on the real Hilbert space ℓ^2_+ (of all square-summable sequences of real numbers) is a bilinear form which is trivially continuous (by the Schwartz inequality) but it is not weakly sequentially continuous. In fact, two ℓ^2_+-valued sequences may individually converge weakly on ℓ^2_+ but the scalar sequence of their inner products may not converge. For instance, the sequence of unit sequences $\{e_n\}$ converges weakly to zero (since $\langle e_n\,; x\rangle \to 0$ for every $x \in \ell^2_+$) and $\langle e_n\,; e_n\rangle = 1$ for all positive integers n. □

Proposition 6.J is a particularly important result on bilinear extension to biduals. It is based on the following classical result (which is another consequence of the Hahn–Banach Theorem).

Every normed space $\mathcal{Z}$ is isometrically embedded in its second dual $\mathcal{Z}^{**}$.

Indeed, the *natural embedding* $\Phi\colon \mathcal{Z} \hookrightarrow \mathcal{Z}^{**}$ sets an isometric isomorphism of $\mathcal{Z}$ onto its range $\Phi(\mathcal{Z}) \subseteq \mathcal{Z}^{**}$. The isometric embedding Φ is defined by $\Phi(z) = \zeta_z \in \mathcal{Z}^{**}$ for every $z \in \mathcal{Z}$, with ζ_z given by $\zeta_z(h) = h(z)$ for every $h \in \mathcal{Z}^*$, for each $z \in \mathcal{Z}$ (see, e.g., [52, Theorem 4.66]). We drop the subscript and denote $\zeta_z \in \Phi(\mathcal{Z})$ by ζ for simplicity. Hence

$$\langle h\,; \zeta\rangle = \zeta(h) = h(z) = \langle z\,; h\rangle \quad \text{for every} \quad h \in \mathcal{Z}^* \qquad (*)$$

establishes the identification $z \leftrightarrow \zeta$ between $z \in \mathcal{Z}$ and $\zeta \in \Phi(\mathcal{Z}) \subseteq \mathcal{Z}^{**}$. So $\mathcal{Z}$ can be regarded as a linear manifold of $\mathcal{Z}^{**}$ where $\mathcal{Z}$ is identified with the subspace $\Phi(\mathcal{Z})$ of $\mathcal{Z}^{**}$ (i.e., $\mathcal{Z}$ is identified with the range $\Phi(\mathcal{Z})$ of the natural embedding Φ of $\mathcal{Z}$ into $\mathcal{Z}^{**}$).

Proof of Proposition 6.J. Let $\mathcal{X}$ and $\mathcal{Y}$ be arbitrary nonzero Banach spaces. Identify $\mathcal{X}$ and $\mathcal{Y}$ as subspaces of $\mathcal{X}^{**}$ and $\mathcal{Y}^{**}$. Take an arbitrary bounded bilinear form $\phi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$. According to Theorem 6.3, $\phi \cong T$ for $T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}^*]$ given by $(Tx)(\cdot) = \phi_x(\cdot)\colon \mathcal{Y} \to \mathbb{F}$ so that

$$\phi(x,y) = (Tx)(y) = \langle y\,;Tx\rangle$$

for every $(x,y)\in\mathcal{X}\times\mathcal{Y}$. Consider the (normed-space) adjoint $T^*\in\mathcal{B}[\mathcal{Y}^{**},\mathcal{X}^*]$ of $T\in\mathcal{B}[\mathcal{X},\mathcal{Y}^*]$. Take a form $\widehat{\phi}\colon \mathcal{X}^{**}\times\mathcal{Y}^{**}\to\mathbb{F}$ defined by

$$\widehat{\phi}(\varphi,\eta) = \varphi(T^*\eta) = \langle T^*\eta\,;\varphi\rangle$$

for every $(\varphi,\eta)\in\mathcal{X}^{**}\times\mathcal{Y}^{**}$. It is clear that the form $\widehat{\phi}$ is bilinear and bounded. Consider the restriction $\widehat{\phi}|_{\Phi(\mathcal{X})\times\Phi(\mathcal{Y})}$ of $\widehat{\phi}$ to the Cartesian product $\Phi(\mathcal{X})\times\Phi(\mathcal{Y})$ of subspaces $\Phi(\mathcal{X})$ of $\mathcal{X}^{**}$ and $\Phi(\mathcal{Y})$ of $\mathcal{Y}^{**}$, which are identified with $\mathcal{X}$ and $\mathcal{Y}$, respectively. Therefore when acting on $\Phi(\mathcal{X})\times\Phi(\mathcal{Y})$, and identifying each pair $(\varphi,\eta)\in\Phi(\mathcal{X})\times\Phi(\mathcal{Y})$ with the unique pair $(x,y)\in\mathcal{X}\times\mathcal{Y}$, we write (by using $(*)$ twice, and according to the definition of adjoint)

$$\widehat{\phi}(x,y) = \widehat{\phi}(\varphi,\eta) = \langle T^*\eta\,;\varphi\rangle = \langle x\,;T^*\eta\rangle = \langle Tx\,;\eta\rangle = \langle y\,;Tx\rangle = \phi(x,y)$$

for every $(x,y)\in\mathcal{X}\times\mathcal{Y}$. Thus, identifying $\Phi(\mathcal{X})$ with $\mathcal{X}$ and $\Phi(\mathcal{Y})$ with $\mathcal{Y}$,

$$\phi = \widehat{\phi}|_{\mathcal{X}\times\mathcal{Y}}.$$

Moreover, since $\phi\cong T$,

$$\|\widehat{\phi}\| = \sup_{\|\varphi\|=1,\|\eta\|=1}\|\phi(T^*\eta)\| \le \|T^*\| = \|T\| = \|\phi\| = \|\widehat{\phi}|_{\Phi(\mathcal{X})\times\Phi(\mathcal{Y})}\| \le \|\widehat{\phi}\|. \quad \square$$

(It is worth noticing *en passant* that for an arbitrary $\phi\in\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$ there is an *adjoint* $\phi^*\in\boldsymbol{b}[\mathcal{Z}^*\times\mathcal{X},\mathcal{Y}^*]$ of ϕ, given by $\phi^*(f,x) = f(\phi(x,y))$ for every $f\in\mathcal{Z}^*$, $x\in\mathcal{X}$ and $y\in\mathcal{Y}$, whose third iteration $\phi^{***}\in\boldsymbol{b}[\mathcal{X}^{**}\times\mathcal{Y}^{**},\mathcal{Z}^{**}]$ yields a norm-preserving extension of ϕ [1, p. 839] — as $\widehat{\phi}\in\boldsymbol{b}[\mathcal{X}^{**}\times\mathcal{Y}^{**},\mathbb{F}]$ in Proposition 6.J yields a norm-preserving extension of $\phi\in\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathbb{F}]$.)

Proof of Proposition 6.K. Let $\|\cdot\|_{\mathcal{X}}$ be the norm on $\mathcal{X}$ induced by the inner product $\langle\cdot\,;\cdot\rangle_{\mathcal{X}}$ on $\mathcal{X}$. It can be shown (see, e.g., [52, proof of Theorem 5.23]) that the norm $\|\cdot\|_{\widehat{\mathcal{X}}}$ on the completion $\widehat{\mathcal{X}}$ of $\mathcal{X}$ satisfies the parallelogram law (see, e.g., [52, Proposition 5.4]). This is equivalent to saying that the norm $\|\cdot\|_{\widehat{\mathcal{X}}}$ is generated by an inner product $\langle\cdot\,;\cdot\rangle_{\widehat{\mathcal{X}}}$ (see, e.g., [52, Theorem 5.5]). Also, uniqueness comes naturally from the definition of completion. Moreover, according to the completion procedure (see Section 5.1), let $\widehat{x}\in\widehat{\mathcal{X}}$ be the correspondent (i.e., the extension) of $x\in\mathcal{X}$ so that $\|\widehat{x}\|_{\widehat{\mathcal{X}}} = \|x\|_{\mathcal{X}}$. Using the polarization identity (see, e.g., [52, Proposition 5.4]), and recalling that $\widehat{x+y} = \widehat{x}+\widehat{y}$, we get the inner product identity from the norm identity:

$$\langle\widehat{x}\,;\widehat{y}\rangle_{\widehat{\mathcal{X}}} = \langle x\,;y\rangle_{\mathcal{X}}. \qquad \square$$

Suggested Readings

Brown and Pearcy [5]
Defant and Floret [15]
Greub [27]
Ryan [83]



7

Norms on Tensor Products

This chapter contains a detailed account of basic results on tensor products of Banach spaces. It focuses on suitable norms equipping a tensor product space. Essentially, the attributes belonging to a norm on a tensor product space (of a pair of normed spaces over the same field) that make it suitable (the proper term will be "reasonable") are the properties (a) and (b) in Theorem 7.2.

7.1 Reasonable Crossnorms

Usually we adopt a generic notation $\|\cdot\|$ for norms on normed spaces, indexed if necessary as we agreed in Section 5.1. Norms on tensor product spaces, however, will always be indexed. If the linear space $\mathcal{X} \otimes \mathcal{Y}$ is equipped with a norm $\|\cdot\|_\alpha$, then we write $\mathcal{X} \otimes_\alpha \mathcal{Y}$ for the normed space $(\mathcal{X} \otimes \mathcal{Y}, \|\cdot\|_\alpha)$. Let $\|\cdot\|_{*\alpha}$ be the norm on the dual $(\mathcal{X} \otimes_\alpha \mathcal{Y})^*$ when $\mathcal{X} \otimes \mathcal{Y}$ is equipped with the norm $\|\cdot\|_\alpha$ (i.e., $\|\cdot\|_{*\alpha}$ is the induced uniform norm on $\mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathbb{F}]$). Summing up:

$$\mathcal{X} \otimes_\alpha \mathcal{Y} = (\mathcal{X} \otimes \mathcal{Y}, \|\cdot\|_\alpha) \quad \text{and} \quad (\mathcal{X} \otimes_\alpha \mathcal{Y})^* = (\mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathbb{F}], \|\cdot\|_{*\alpha}).$$

When necessary we write $\|\cdot\|_{\mathcal{X} \otimes_\alpha \mathcal{Y}}$ for $\|\cdot\|_\alpha$ and $\|\cdot\|_{(\mathcal{X} \otimes_\alpha \mathcal{Y})^*}$ for $\|\cdot\|_{*\alpha}$.

Definition 7.1. Let $\mathcal{X}$ and $\mathcal{Y}$ be normed spaces. A norm $\|\cdot\|_\alpha$ on the linear space $\mathcal{X} \otimes \mathcal{Y}$ is a *reasonable crossnorm* (sometimes denoted by $\alpha(\cdot)$) if, for every $x \in \mathcal{X}$, $y \in \mathcal{Y}$, $f \in \mathcal{X}^*$, $g \in \mathcal{Y}^*$,

(a) $\|x \otimes y\|_\alpha \le \|x\| \, \|y\|$,

(b) $f \otimes g$ lies in $(\mathcal{X} \otimes_\alpha \mathcal{Y})^*$, and $\|f \otimes g\|_{*\alpha} \le \|f\| \, \|g\|$.

Definition 7.1 imposes conditions on the norms of single tensors $x \otimes y$ and $f \otimes g$. The first basic observations about it are: (i) the above inequalities are in fact identities (as we will see next), and (ii) a reasonable crossnorm is such that the inclusion involving algebraic duals in Remark 3.18(b), viz., $\mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp \subseteq (\mathcal{X} \otimes \mathcal{Y})^\sharp$, is sharpened to $\mathcal{X}^* \otimes \mathcal{Y}^* \subseteq (\mathcal{X} \otimes_\alpha \mathcal{Y})^*$.

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Theorem 7.2. *Let $\mathcal{X}$ and $\mathcal{Y}$ be normed spaces. A norm $\|\cdot\|_\alpha$ on $\mathcal{X}\otimes\mathcal{Y}$ is a reasonable crossnorm if and only if, for every $x\in\mathcal{X}$, $y\in\mathcal{Y}$, $f\in\mathcal{X}^*$, $g\in\mathcal{Y}^*$,*

(a) $\|x\otimes y\|_\alpha = \|x\|\,\|y\|$,

(b) $f\otimes g$ *lies in* $(\mathcal{X}\otimes_\alpha\mathcal{Y})^*$, *and* $\|f\otimes g\|_{*\alpha} = \|f\|\,\|g\|$.

Proof. Consider the setup of Definition 7.1 where $\|\cdot\|_{*\alpha}$ denotes the (induced uniform) norm on $(\mathcal{X}\otimes\mathcal{Y})^*$ when $\mathcal{X}\otimes\mathcal{Y}$ is equipped with the norm $\|\cdot\|_\alpha$. If the identities in (a) and (b) hold, then the respective inequalities in Definition 7.1 hold tautologically. The converse goes as follows. Suppose $\|\cdot\|_\alpha$ is a reasonable crossnorm on $\mathcal{X}\otimes\mathcal{Y}$ as in Definition 7.1.

(a) Take arbitrary $x\in\mathcal{X}$ and $y\in\mathcal{Y}$. The Hahn–Banach Theorem ensures the existence of $f\in\mathcal{X}^*$ and $g\in\mathcal{Y}^*$ with $\|f\|=\|g\|=1$ for which $|f(x)|=\|x\|$ and $|g(y)|=\|y\|$ (see, e.g., [52, Corollary 4.64]). Thus $\|f\otimes g\|_{*\alpha}\le 1$ by Definition 7.1(b), and so (since $(f\otimes g)(x\otimes y)=f(x)\,g(y)$ — cf. Remark 3.18(b))

$$\|x\|\,\|y\| = |f(x)\,g(y)| = |(f\otimes g)(x\otimes y)| \le \|f\otimes g\|_{*\alpha}\|x\otimes y\|_\alpha \le \|x\otimes y\|_\alpha.$$

Hence $\|x\otimes y\|_\alpha = \|x\|\,\|y\|$ by Definition 7.1(a).

(b) Take any $f\in\mathcal{X}^*$ and $g\in\mathcal{Y}^*$. As $\|x\otimes y\|_\alpha\le\|x\|\,\|y\|$ by Definition 7.1(a),

$$\begin{aligned}\|f\|\,\|g\| &= \sup_{x\ne0}\frac{|f(x)|}{\|x\|}\sup_{y\ne0}\frac{|g(y)|}{\|y\|} = \sup_{x,y\ne0}\frac{|f(x)\,g(y)|}{\|x\|\|y\|} = \sup_{x,y\ne0}\frac{|(f\otimes g)\,(x\otimes y)|}{\|x\|\|y\|}\\ &\le \sup_{x\otimes y\ne0}\frac{|(f\otimes g)\,(x\otimes y)|}{\|x\otimes y\|_\alpha} \le \sup_{0\ne F\in\mathcal{X}\otimes\mathcal{Y}}\frac{|(f\otimes g)F|}{\|F\|_\alpha} = \|f\otimes g\|_{*\alpha}.\end{aligned}$$

Hence $\|f\otimes g\|_{*\alpha}=\|f\|\,\|g\|$ by Definition 7.1(b). □

Thus a tensor product space $\mathcal{X}\otimes\mathcal{Y}$ equipped with a reasonable crossnorm $\|\cdot\|_\alpha$ yields the normed space $\mathcal{X}\otimes_\alpha\mathcal{Y}=(\mathcal{X}\otimes\mathcal{Y},\|\cdot\|_\alpha)$, and its dual

$$(\mathcal{X}\otimes_\alpha\mathcal{Y})^* = (\mathcal{X}\otimes\mathcal{Y},\|\cdot\|_\alpha)^* = \mathcal{B}[(\mathcal{X}\otimes\mathcal{Y},\|\cdot\|_\alpha),\mathbb{F}] = \mathcal{B}[\mathcal{X}\otimes_\alpha\mathcal{Y},\mathbb{F}]$$

is equipped with $\|\cdot\|_{*\alpha}$, which stands for the induced uniform norm on the Banach space $\mathcal{B}[\mathcal{X}\otimes_\alpha\mathcal{Y},\mathbb{F}]$. By Definition 7.1, every single tensor $f\otimes g$ in the tensor product space $\mathcal{X}^*\otimes\mathcal{Y}^*$ lies in $(\mathcal{X}\otimes_\alpha\mathcal{Y})^*$. Then every element $\Omega=\sum_i f_i\otimes g_i$ in $\mathcal{X}^*\otimes\mathcal{Y}^*$ lies in $(\mathcal{X}\otimes_\alpha\mathcal{Y})^*$. In fact, the linear space $\mathcal{X}^*\otimes\mathcal{Y}^*$ is an algebraic linear manifold of the linear space $(\mathcal{X}\otimes_\alpha\mathcal{Y})^*$,

$$\mathcal{X}^*\otimes\mathcal{Y}^*\subseteq(\mathcal{X}\otimes_\alpha\mathcal{Y})^*,$$

whenever the tensor product space $\mathcal{X}\otimes\mathcal{Y}$ is equipped with any reasonable crossnorm $\|\cdot\|_\alpha$. The above linear-space inclusion follows the pattern

$$\mathcal{X}^\sharp\otimes\mathcal{Y}^\sharp\subseteq(\mathcal{X}\otimes\mathcal{Y})^\sharp$$

of Remark 3.18(b), and the next linear-space inclusions are clear:

$$\mathcal{X}^*\otimes\mathcal{Y}^*\subseteq\mathcal{X}^\sharp\otimes\mathcal{Y}^\sharp \quad\text{and}\quad (\mathcal{X}\otimes_\alpha\mathcal{Y})^*\subseteq(\mathcal{X}\otimes\mathcal{Y})^\sharp.$$

Thus what is required in Definition 7.1(b) is that every single tensor

$$f \otimes g \quad \text{in} \quad \mathcal{X}^* \otimes \mathcal{Y}^* \subseteq \mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp \subseteq (\mathcal{X} \otimes \mathcal{Y})^\sharp$$

lies in $(\mathcal{X} \otimes_\alpha \mathcal{Y})^*$. So besides lying in $(\mathcal{X} \otimes \mathcal{Y})^\sharp$ it is continuous as a functional from the normed space $\mathcal{X} \otimes_\alpha \mathcal{Y} = (\mathcal{X} \otimes \mathcal{Y}, \|\cdot\|_\alpha)$ (equipped with any reasonable crossnorm $\|\cdot\|_\alpha$) to $\mathbb{F}$ (equipped with its usual norm $|\cdot|$). Since $\|\cdot\|_{*\alpha}$ is the norm on $(\mathcal{X} \otimes_\alpha \mathcal{Y})^* = \mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathbb{F}]$, and $\mathcal{X}^* \otimes \mathcal{Y}^* \subseteq (\mathcal{X} \otimes_\alpha \mathcal{Y})^*$, we get

$$\|f \otimes g\|_{*\alpha} = \|f \otimes g\|_{(\mathcal{X} \otimes_\alpha \mathcal{Y})^*} = \|f \otimes g\|_{\mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathbb{F}]} = \|f\| \, \|g\|$$

according to Theorem 7.2. In general, for every $\Lambda \in (\mathcal{X} \otimes_\alpha \mathcal{Y})^*$,

$$\|\Lambda\|_{*\alpha} = \|\Lambda\|_{(\mathcal{X} \otimes_\alpha \mathcal{Y})^*} = \|\Lambda\|_{\mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathbb{F}]} = \sup_{F \in \mathcal{X} \otimes \mathcal{Y},\, \|F\|_\alpha \le 1} |\Lambda(F)|$$

$$= \sup_{0 \neq F \in \mathcal{X} \otimes \mathcal{Y}} \frac{|\Lambda(F)|}{\|F\|_\alpha} = \sup_{0 \neq \Sigma_i x_i \otimes y_i \in \mathcal{X} \otimes \mathcal{Y}} \frac{\left|\Lambda\left(\sum_i x_i \otimes y_i\right)\right|}{\left\|\sum_i x_i \otimes y_i\right\|_\alpha}.$$

In particular, for every $\Omega = \sum_j f_j \otimes g_j \in \mathcal{X}^* \otimes \mathcal{Y}^* \subseteq (\mathcal{X} \otimes_\alpha \mathcal{Y})^*$,

$$\|\Omega\|_{*\alpha} = \sup_{0 \neq F \in \mathcal{X} \otimes \mathcal{Y}} \frac{|\Omega(F)|}{\|F\|_\alpha} = \sup_{0 \neq F \in \mathcal{X} \otimes \mathcal{Y}} \frac{\left|\sum_j f_j \otimes g_j(F)\right|}{\|F\|_\alpha} = \sup_{0 \neq \Sigma_i x_i \otimes y_i \in \mathcal{X} \otimes \mathcal{Y}} \frac{\left|\left(\sum_{i,j} f_j(x_i)\, g_j(y_i)\right)\right|}{\left\|\sum_i x_i \otimes y_i\right\|_\alpha}.$$

More particularly, for every single tensor $f \otimes g \in \mathcal{X}^* \otimes \mathcal{Y}^* \subseteq (\mathcal{X} \otimes_\alpha \mathcal{Y})^*$,

$$\|f \otimes g\|_{*\alpha} = \sup_{0 \neq \Sigma_i x_i \otimes y_i \in \mathcal{X} \otimes \mathcal{Y}} \frac{\left|f \otimes g\left(\sum_i x_i \otimes y_i\right)\right|}{\left\|\sum_i x_i \otimes y_i\right\|_\alpha} = \sup_{0 \neq \Sigma_i x_i \otimes y_i \in \mathcal{X} \otimes \mathcal{Y}} \frac{\left|\left(\sum_i f(x_i)\, g(y_i)\right|\right.}{\left\|\sum_i x_i \otimes y_i\right\|_\alpha}.$$

However, by Theorem 7.2 again,

$$\|f \otimes g\|_{*\alpha} = \|f \otimes g\|_\beta = \|f\| \, \|g\|$$

for every single tensor $f \otimes g \in \mathcal{X}^* \otimes \mathcal{Y}^*$, for every reasonable crossnorm $\|\cdot\|_\alpha$ on $\mathcal{X} \otimes \mathcal{Y}$, and for every reasonable crossnorm $\|\cdot\|_\beta$ on $\mathcal{X}^* \otimes \mathcal{Y}^*$.

Theorem 7.3. *Let $\|\cdot\|_\alpha$ be a reasonable crossnorm on a tensor product space $\mathcal{X} \otimes \mathcal{Y}$ of normed spaces $\mathcal{X}$ and $\mathcal{Y}$. Take the dual $(\mathcal{X} \otimes_\alpha \mathcal{Y})^*$ of $\mathcal{X} \otimes_\alpha \mathcal{Y} = (\mathcal{X} \otimes \mathcal{Y}, \|\cdot\|_\alpha)$. When restricted to $\mathcal{X}^* \otimes \mathcal{Y}^*$, the norm $\|\cdot\|_{*\alpha}$ on $(\mathcal{X} \otimes_\alpha \mathcal{Y})^*$ is a reasonable crossnorm on $\mathcal{X}^* \otimes \mathcal{Y}^*$.*

In other words, since $\mathcal{X}^* \otimes \mathcal{Y}^* \subseteq (\mathcal{X} \otimes_\alpha \mathcal{Y})^*$, when restricted to the tensor product space $\mathcal{X}^* \otimes \mathcal{Y}^*$ the norm $\|\cdot\|_{*\alpha}$ on the Banach space $(\mathcal{X} \otimes_\alpha \mathcal{Y})^*$ is such that for arbitrary single tensors $f \otimes g$ in $\mathcal{X}^* \otimes \mathcal{Y}^*$ and $\varphi \otimes \eta$ in $\mathcal{X}^{**} \otimes \mathcal{Y}^{**}$,

(a) $\|f \otimes g\|_{*\alpha} \le \|f\| \, \|g\|$,

(b) *$\varphi \otimes \eta$ lies in $(\mathcal{X}^* \otimes_{*\alpha} \mathcal{Y}^*)^*$ so that*

$$\mathcal{X}^{**} \otimes \mathcal{Y}^{**} \subseteq (\mathcal{X}^* \otimes_{*\alpha} \mathcal{Y}^*)^*,$$

and $\|\varphi \otimes \eta\|_{**\alpha} \le \|\varphi\| \|\eta\|$ (where $\|\cdot\|_{**\alpha}$ is the norm on $(\mathcal{X}^* \otimes_{*\alpha} \mathcal{Y}^*)^*$ when $\mathcal{X}^* \otimes \mathcal{Y}^*$ is equipped with the norm $\|\cdot\|_{*\alpha}$),

and the inequalities in (a) *and* (b) *are in fact identities (by Theorem 7.2).*

Proof. Let $\|\cdot\|_\alpha$ be a reasonable crossnorm on $\mathcal{X}\otimes\mathcal{Y}$. Take the normed space $\mathcal{X}\otimes_\alpha\mathcal{Y}$. Let $\|\cdot\|_{*\alpha}$ be the norm on the dual of $\mathcal{X}\otimes_\alpha\mathcal{Y}$; that is, the induced uniform norm on $(\mathcal{X}\otimes_\alpha\mathcal{Y})^* = \mathcal{B}[\mathcal{X}\otimes_\alpha\mathcal{Y},\mathbb{F}]$. Since $\mathcal{X}^*\otimes\mathcal{Y}^*\subseteq(\mathcal{X}\otimes_\alpha\mathcal{Y})^*$, equip $\mathcal{X}^*\otimes\mathcal{Y}^*$ with the norm $\|\cdot\|_{*\alpha}$ inherited from $(\mathcal{X}\otimes_\alpha\mathcal{Y})^*$. So $\mathcal{X}^*\otimes_{*\alpha}\mathcal{Y}^*$ is a linear manifold of $(\mathcal{X}\otimes_\alpha\mathcal{Y})^*$. Let $\|\cdot\|_{**\alpha}$ be the norm on the dual of $\mathcal{X}^*\otimes_{*\alpha}\mathcal{Y}^*$; that is, the induced uniform norm on $(\mathcal{X}^*\otimes_{*\alpha}\mathcal{Y}^*)^* = \mathcal{B}[\mathcal{X}^*\otimes_{*\alpha}\mathcal{Y}^*,\mathbb{F}]$.

The inequality in (a) holds according to Definition 7.1(b). Take an arbitrary single tensor $\varphi\otimes\eta$ in $\mathcal{X}^{**}\otimes\mathcal{Y}^{**}$, so that $\varphi\in\mathcal{X}^{**}=\mathcal{B}[\mathcal{X}^*,\mathbb{F}]$ and $\eta\in\mathcal{Y}^{**}=\mathcal{B}[\mathcal{Y}^*,\mathbb{F}]$. Since (cf. Remark 3.18(b) again)

$$\mathcal{X}^{**}\otimes\mathcal{Y}^{**}\subseteq\mathcal{X}^{*\sharp}\otimes\mathcal{Y}^{*\sharp}\subseteq(\mathcal{X}^*\otimes\mathcal{Y}^*)^\sharp,$$

then $\varphi\otimes\eta$ lies in $(\mathcal{X}^*\otimes\mathcal{Y}^*)^\sharp$. Thus $\varphi\otimes\eta\colon\mathcal{X}^*\otimes\mathcal{Y}^*\to\mathbb{F}$ is a linear functional. We apply the next claim to verify that the assertions in (b) hold as well.

Claim. Let $\mathcal{X}$ and $\mathcal{Y}$ be normed spaces. Take arbitrary nonzero $\varphi\in\mathcal{X}^{**}$ and $\eta\in\mathcal{Y}^{**}$. There are nets $\{x_\gamma\}_{\gamma\in\Gamma}$ and $\{y_{\gamma'}\}_{\gamma'\in\Gamma'}$ of elements in $\mathcal{X}$ and $\mathcal{Y}$ with

$$\|x_\gamma\|\le\|\varphi\| \quad\text{and}\quad \|y_{\gamma'}\|\le\|\eta\|,$$

such that, for every $f\in\mathcal{X}^*$ and $g\in\mathcal{Y}^*$,

$$\lim\nolimits_\gamma f(x_\gamma)=\varphi(f) \quad\text{and}\quad \lim\nolimits_{\gamma'} g(y_{\gamma'})=\eta(g).$$

Proof of Claim. For each $x\in\mathcal{X}$ consider the functional $\varphi_x\colon\mathcal{X}^*\to\mathbb{F}$ given by $\varphi_x(f)=f(x)$ for every $f\in\mathcal{X}^*$, which is linear and bounded (i.e., $\varphi_x\in\mathcal{X}^{**}$) with $\|\varphi_x\|=\|x\|$. Let $\Phi\colon\mathcal{X}\to\mathcal{X}^{**}$ be defined by $\Phi(x)=\varphi_x$, so that

$$\Phi(x)(f)=\varphi_x(f)=f(x)$$

for every $x\in\mathcal{X}$ and every $f\in\mathcal{X}^*$. This Φ is an isometric isomorphism of $\mathcal{X}$ onto $\mathcal{R}(\Phi)=\Phi(\mathcal{X})\subseteq\mathcal{X}^{**}$, which is the natural embedding of $\mathcal{X}$ into $\mathcal{X}^{**}$ (cf. proof of Proposition 6.J in Section 6.4). The Goldstine Theorem (see, e.g., [70, Theorem 2.6.26]) says that $\Phi(\boldsymbol{B}_\mathcal{X})\subset\mathcal{X}^{**}$ is weakly* dense in $\boldsymbol{B}_{\mathcal{X}^{**}}\subset\mathcal{X}^{**}$ (where $\boldsymbol{B}_\mathcal{X}$ and $\boldsymbol{B}_{\mathcal{X}^{**}}$ are the closed unit balls in $\mathcal{X}$ and in $\mathcal{X}^{**}$).

(Recall that a set S is dense in a topological space X if and only if for every point in X there is an S-valued net converging to it (see, e.g., [12, Proposition A.2.2]). Thus every φ in $\boldsymbol{B}_{\mathcal{X}^{**}}\subset\mathcal{X}^{**}$ is the weak* limit of a $\Phi(\boldsymbol{B}_\mathcal{X})$-valued net $\{\Phi(x_\gamma)\}_{\gamma\in\Gamma}$ in the dual $\mathcal{X}^{**}$ of $\mathcal{X}^*$, where $\{x_\gamma\}_{\gamma\in\Gamma}$ is a $\boldsymbol{B}_\mathcal{X}$-valued net of elements from $\mathcal{X}$ (i.e., $\Phi(x_\gamma)\xrightarrow{w*}\varphi$, which means $\Phi(x_\gamma)(f)\to\varphi(f)$ for every $f\in\mathcal{X}^*$). In other words, every functional φ in $\boldsymbol{B}_{\mathcal{X}^{**}}\subset\mathcal{X}^{**}$ is such that $\varphi(f)=\lim_\gamma\Phi(x_\gamma)(f)$ for every functional $f\in\mathcal{X}^*$.)

Therefore, for every $0\ne\varphi\in\mathcal{X}^{**}$, set $\frac{\varphi}{\|\varphi\|}\in\boldsymbol{B}_{\mathcal{X}^{**}}\subset\mathcal{X}^{**}$ (with $\|\frac{\varphi}{\|\varphi\|}\|=1$) so that there exists an $\mathcal{X}$-valued net $\{x_\gamma\}_{\gamma\in\Gamma}$ with $\|x_\gamma\|\le\|\varphi\|$ (which implies $\frac{\|x_\gamma\|}{\|\varphi\|}\le 1$ and hence $\frac{\|x_\gamma\|}{\|\varphi\|}\in\boldsymbol{B}_\mathcal{X}\subset\mathcal{X}$) such that for every $f\in\mathcal{X}^*$

$$\tfrac{\varphi}{\|\varphi\|}(f) = \lim_\gamma \Phi(\tfrac{x_\gamma}{\|\varphi\|})(f), \quad \text{which means} \quad \tfrac{\varphi}{\|\varphi\|}(f)(x) = \lim_\gamma f(\tfrac{x_\gamma}{\|\varphi\|}),$$

and so $\lim_\gamma f(x_\gamma) = \varphi(f)$. Similarly, $\|y'_\gamma\| \le \|\eta\|$ and $\lim_{\gamma'} g(y_{\gamma'}) = \eta(g)$. □

Take an arbitrary $\Omega = \sum_i f_i \otimes g_i$ from $\mathcal{X}^* \otimes \mathcal{Y}^*$. Thus by the above claim, and since the norm $\|\cdot\|_\alpha$ on $\mathcal{X} \otimes \mathcal{Y}$ is a reasonable crossnorm,

$$\begin{aligned}
|(\varphi \otimes \eta)\Omega| &= \Big|(\varphi \otimes \eta)\sum\nolimits_i f_i \otimes g_i\Big| = \Big|\sum\nolimits_i \varphi(f_i)\,\eta(g_i)\Big| \\
&= \Big|\sum\nolimits_i \lim\nolimits_\gamma f_i(x_\gamma) \lim\nolimits_{\gamma'} g_i(y_{\gamma'})\Big| = \lim\nolimits_{\gamma,\gamma'} \Big|\sum\nolimits_i f_i(x_\gamma)\, g_i(y_{\gamma'})\Big| \\
&= \lim\nolimits_{\gamma,\gamma'} \Big|\sum\nolimits_i (f_i \otimes g_i)(x_\gamma \otimes y_{\gamma'})\Big| = \lim\nolimits_{\gamma,\gamma'} \Big|\Omega(x_\gamma \otimes y_{\gamma'})\Big| \\
&\le \limsup\nolimits_{\gamma,\gamma'} \|\Omega\|_{*\alpha} \|x_\gamma \otimes y_{\gamma'}\|_\alpha = \limsup\nolimits_{\gamma,\gamma'} \|\Omega\|_{*\alpha} \|x_\gamma\|\, \|y_{\gamma'}\| \\
&\le \|\varphi\|\, \|\eta\|\, \|\Omega\|_{*\alpha}.
\end{aligned}$$

Thus $\varphi \otimes \eta$ is bounded, so that $\mathcal{X}^{**} \otimes \mathcal{Y}^{**} \subseteq (\mathcal{X}^* \otimes_{*\alpha} \mathcal{Y}^*)^*$, and

$$\|\varphi \otimes \eta\|_{**\alpha} = \sup_{\Omega \ne 0} \frac{\big|(\varphi \otimes \eta)\Omega\big|}{\|\Omega\|_{*\alpha}} \le \|\varphi\|\, \|\eta\|,$$

concluding the proof of (b). □

Let $\mathcal{X}$ and $\mathcal{Y}$ be normed spaces and consider a function $\|\cdot\|_\vee \colon \mathcal{X} \otimes \mathcal{Y} \to \mathbb{R}$ (sometimes denoted by $\varepsilon(\cdot)$) defined for every $F = \sum_i x_i \otimes y_i \in \mathcal{X} \otimes \mathcal{Y}$ as

$$\|F\|_\vee = \sup_{\|f\| \le 1,\, \|g\| \le 1,\, f \in \mathcal{X}^*,\, g \in \mathcal{Y}^*} \Big|\sum\nolimits_i f(x_i)\, g(y_i)\Big|,$$

where the supremum is taken over all functionals $f \in \mathcal{X}^*$ with $\|f\| \le 1$ and $g \in \mathcal{Y}^*$ with $\|g\| \le 1$ for any (finite) representation $\sum_i x_i \otimes y_i$ of $F \in \mathcal{X} \otimes \mathcal{Y}$. Note: (i) $\sum_i f(x_i)\, g(y_i) = \sum_i (f \otimes g)(x_i \otimes y_i) = (f \otimes g)F$ does not depend on the representation $\sum_i x_i \otimes y_i$ of F, and $\|F\|_\vee \le \sum_i \|x_i\|\, \|y_i\|$. Now consider another function $\|\cdot\|_\wedge \colon \mathcal{X} \otimes \mathcal{Y} \to \mathbb{R}$ (sometimes denoted by $\pi(\cdot)$) defined by

$$\|F\|_\wedge = \inf_{\{x_i\},\, \{y_i\},\, F = \sum_i x_i \otimes y_i} \sum\nolimits_i \|x_i\|\, \|y_i\|$$

for every $F \in \mathcal{X} \otimes \mathcal{Y}$, where the infimum is taken over all (finite) representations of $F \in \mathcal{X} \otimes \mathcal{Y}$ of the form $F = \sum_i x_i \otimes y_i$.

Theorem 7.4. *Let $\mathcal{X}$ and $\mathcal{Y}$ be normed spaces. The functions $\|\cdot\|_\vee$ and $\|\cdot\|_\wedge$ are reasonable crossnorms on $\mathcal{X} \otimes \mathcal{Y}$. Moreover, a norm $\|\cdot\|_\alpha$ is a reasonable crossnorm on $\mathcal{X} \otimes \mathcal{Y}$ if and only if*

$$\|F\|_\vee \le \|F\|_\alpha \le \|F\|_\wedge \quad \textit{for every} \quad F \in \mathcal{X} \otimes \mathcal{Y}.$$

So $\|\cdot\|_\vee$ is the least and $\|\cdot\|_\wedge$ is the greatest reasonable crossnorm on $\mathcal{X} \otimes \mathcal{Y}$.

Proof. Consider the functions $\|\cdot\|_\vee \colon \mathcal{X} \otimes \mathcal{Y} \to \mathbb{R}$ and $\|\cdot\|_\wedge \colon \mathcal{X} \otimes \mathcal{Y} \to \mathbb{R}$.

(a) *$\|\cdot\|_\vee$ and $\|\cdot\|_\wedge$ are norms on $\mathcal{X} \otimes \mathcal{Y}$.*

If $F \neq 0$, then we may represent it as $\sum_i x_i \otimes y_i$ where each of $\{x_i\}$ and $\{y_i\}$ is linearly independent (cf. proof of Claim in the proof of Corollary 4.10). Thus $\|0\|_\vee = \|0\|_\wedge = 0$. By the above mentioned claim, $\sum_i f(x_i)g(y_i) = 0$ for $0 < \|f\|$ and $0 < \|g\|$ implies $\sum_i x_i \otimes y_i = 0$, and so $\|F\|_\vee = 0$ implies $F = 0$. The proof that $\|F\|_\wedge = 0$ implies $F = 0$ is postponed until item (c), where it is shown that $\|F\|_\vee \leq \|F\|_\wedge$. Moreover, it is clear that $\|\alpha F\|_\vee = |\alpha|\,\|F\|_\vee$ and $\|\alpha F\|_\wedge = |\alpha|\,\|F\|_\wedge$. Finally, we check the triangle inequality.

$$\begin{aligned}\|F + F'\|_\vee &= \Big\|\sum\nolimits_i x_i \otimes y_i + \sum\nolimits_j x'_j \otimes y'_j\Big\|_\vee \\ &= \sup_{\|f\|\leq 1,\, \|g\|\leq 1} \Big|\sum\nolimits_i f(x_i)\, g(y_i) + \sum\nolimits_j f(x'_j)\, g(y'_j)\Big| \leq \|F\|_\vee + \|F'\|_\vee.\end{aligned}$$

For every $\varepsilon > 0$ there are representations $\sum_i x_i \otimes y_i$ and $\sum_j x'_j \otimes y'_j$ of F and F' such that $\sum_i \|x_i\|\,\|y_i\| \leq \|F\|_\wedge + \varepsilon$ and $\sum_j \|x'_j\|\,\|y'_j\| \leq \|F'\|_\wedge + \varepsilon$, and so

$$\begin{aligned}\|F + F'\|_\wedge &= \Big\|\sum\nolimits_i x_i \otimes y_i + \sum\nolimits_j x'_i \otimes y'_i\Big\|_\wedge \\ &\leq \sum\nolimits_i \|x_i\|\,\|y_i\| + \sum\nolimits_j \|x'_j\|\,\|y'_j\| \leq \|F\|_\wedge + \|F'\|_\wedge + 2\varepsilon.\end{aligned}$$

(b) $\|\cdot\|_\vee$ and $\|\cdot\|_\wedge$ *are reasonable crossnorms on* $\mathcal{X} \otimes \mathcal{Y}$.

Take arbitrary $0 \neq x \otimes y \in \mathcal{X} \otimes \mathcal{Y}$ and $0 \neq f \otimes g \in \mathcal{X}^* \otimes \mathcal{Y}^*$. First note that

$$\|x \otimes y\|_\vee = \sup_{\|f\|\leq 1,\, \|g\|\leq 1} |f(x)\, g(y)| = \sup_{\|f\|\leq 1} |f(x)| \sup_{\|g\|\leq 1} |g(y)| = \|x\|\,\|y\| = \|x \otimes y\|_\wedge.$$

Now let $\|\cdot\|_{*\vee}$ be the norm on $(\mathcal{X} \otimes_\vee \mathcal{Y})^*$ when $\mathcal{X} \otimes \mathcal{Y}$ is equipped with $\|\cdot\|_\vee$. Take an arbitrary $F = \sum_i x_i \otimes y_i \in \mathcal{X} \otimes \mathcal{Y}$. If $\|f\| \neq 0$ and $\|g\| \neq 0$, then

$$|(f \otimes g)(F)| = \|f\|\,\|g\| \left|\sum\nolimits_i \left(\tfrac{f}{\|f\|}(x_i)\, \tfrac{g}{\|g\|}(y_i)\right)\right| \leq \|f\|\,\|g\|\,\|F\|_\vee.$$

Thus $f \otimes g \in (\mathcal{X} \otimes_\vee \mathcal{Y})^*$ and $\|f \otimes g\|_{*\vee} \leq \|f\|\,\|g\|$. So $\|\cdot\|_\vee$ is a reasonable crossnorm according to Definition 7.1.

Remark. Therefore if $\|\cdot\|_\gamma$ is any norm on $\mathcal{X} \otimes \mathcal{Y}$ (not necessarily a reasonable crossnorm), then $\|\cdot\|_\vee \leq \|\cdot\|_\gamma \Longrightarrow \mathcal{X}^* \otimes \mathcal{Y}^* \subseteq (\mathcal{X} \otimes_\gamma \mathcal{Y})^*$.

Next let $\|\cdot\|_{*\wedge}$ be the norm on $(\mathcal{X} \otimes_\wedge \mathcal{Y})^*$ when $\mathcal{X} \otimes \mathcal{Y}$ is equipped with $\|\cdot\|_\wedge$. Take an arbitrary $\varepsilon > 0$, and let $\sum_i x_i \otimes y_i$ be a representation of an arbitrary F in $\mathcal{X} \otimes \mathcal{Y}$ for which $\sum_i \|x_i\|\,\|y_i\| \leq \|F\|_\wedge + \varepsilon$. Then

$$|(f \otimes g)(F)| = \Big|\sum\nolimits_i f(x_i)\, g(y_i)\Big| \leq \|f\|\,\|g\| \sum\nolimits_i \|x_i\|\,\|y_i\| \leq \|f\|\,\|g\|(\|F\|_\wedge + \varepsilon).$$

Thus, again, $f \otimes g \in (\mathcal{X} \otimes_\wedge \mathcal{Y})^*$ and $\|f \otimes g\|_{*\wedge} \leq \|f\|\,\|g\|$ since the above inequality holds for every $\varepsilon > 0$. Hence $\|\cdot\|_\wedge$ is a reasonable crossnorm as well.

(c) *If* $\|\cdot\|_\alpha$ *is a reasonable crossnorm on* $\mathcal{X} \otimes \mathcal{Y}$, *then* $\|\cdot\|_\vee \leq \|\cdot\|_\alpha \leq \|\cdot\|_\wedge$.

Let $\|\cdot\|_\alpha$ be a reasonable crossnorm on $\mathcal{X} \otimes \mathcal{Y}$ according to Definition 7.1. Let $\sum_i x_i \otimes y_i$ be any representation of an arbitrary $F \in \mathcal{X} \otimes \mathcal{Y}$. Then

$$\|F\|_\alpha = \Big\|\sum\nolimits_i x_i \otimes y_i\Big\|_\alpha \le \sum\nolimits_i \|x_i \otimes y_i\|_\alpha \le \sum\nolimits_i \|x_i\|\,\|y_i\|$$

for all representations of F. So $\|F\|_\alpha \le \inf_{\{x_i\},\{y_i\}} \sum_i \|x_i\|\,\|y_i\| = \|F\|_\wedge$. Also, letting $\|\cdot\|_{*\alpha}$ be the norm on the dual $(\mathcal{X}\otimes\mathcal{Y})^*$ of $\mathcal{X}\otimes\mathcal{Y}$ when $\mathcal{X}\otimes\mathcal{Y}$ is equipped with the reasonable crossnorm $\|\cdot\|_\alpha$,

$$\begin{aligned}\Big|\sum\nolimits_i f(x_i)g(y_i)\Big| &= \Big|\sum\nolimits_i (f\otimes g)(x_i\otimes y_i)\Big| = \Big|(f\otimes g)\sum\nolimits_i x_i\otimes y_i\Big| \\ &\le \|f\otimes g\|_{*\alpha}\Big\|\sum\nolimits_i x_i\otimes y_i\Big\|_\alpha = \|f\otimes g\|_{*\alpha}\|F\|_\alpha = \|f\|\,\|g\|\,\|F\|_\alpha.\end{aligned}$$

So $\|F\|_\vee = \sup_{\|f\|\le 1,\|g\|\le 1}\big|\sum_i f(x_i)g(y_i)\big| \le \|F\|_\alpha$. Thus $\|\cdot\|_\vee \le \|\cdot\|_\alpha \le \|\cdot\|_\wedge$.

(d) *If a norm* $\|\cdot\|_\alpha$ *on* $\mathcal{X}\otimes\mathcal{Y}$ *is such that* $\|\cdot\|_\vee \le \|\cdot\|_\alpha \le \|\cdot\|_\wedge$, *then it is a reasonable crossnorm.*

Consider Definition 7.1. For an arbitrary single tensor $x\otimes y$ in $\mathcal{X}\otimes\mathcal{Y}$ we get

$$\|\cdot\|_\alpha \le \|\cdot\|_\wedge \quad\Longrightarrow\quad \|x\otimes y\|_\alpha \le \|x\otimes y\|_\wedge = \|x\|\,\|y\|.$$

On the other hand, take an arbitrary single tensor $f\otimes g$ in $\mathcal{X}^*\otimes\mathcal{Y}^*$. As we saw in item (b), $|(f\otimes g)(F)| \le \|f\|\,\|g\|\,\|F\|_\vee$ for every $F\in\mathcal{X}\otimes\mathcal{Y}$. Therefore

$$\|\cdot\|_\vee \le \|\cdot\|_\alpha \quad\Longrightarrow\quad \|f\otimes g\|_{*\alpha} = \sup_{F\ne 0}\tfrac{|(f\otimes g)(F)|}{\|F\|_\alpha} \le \sup_{F\ne 0}\tfrac{\|f\|\|g\|\|F\|_\alpha}{\|F\|_\alpha} \le \|f\|\,\|g\|,$$

where $\|\cdot\|_{*\alpha}$ is the norm on the dual $(\mathcal{X}\otimes\mathcal{Y},\|\cdot\|_\alpha)^*$ of $(\mathcal{X}\otimes\mathcal{Y},\|\cdot\|_\alpha)$. Thus $\|x\otimes y\|_\alpha \le \|x\|\,\|y\|$ and $f\otimes g\in(\mathcal{X}\otimes\mathcal{Y})^*$ with $\|f\otimes g\|_{*\alpha}\le\|f\|\,\|g\|$. □

The reasonable crossnorms $\|\cdot\|_\vee$ and $\|\cdot\|_\wedge$ on $\mathcal{X}\otimes\mathcal{Y}$ are referred to as the *injective norm* and the *projective norm*, respectively. (Terminology will be justified in due course.) When equipped with a reasonable crossnorm $\|\cdot\|_\alpha$, a tensor product linear space $\mathcal{X}\otimes\mathcal{Y}$ yields different normed spaces: $(\mathcal{X}\otimes\mathcal{Y},\|\cdot\|_\vee)$, $(\mathcal{X}\otimes\mathcal{Y},\|\cdot\|_\alpha)$, and $(\mathcal{X}\otimes\mathcal{Y},\|\cdot\|_\wedge)$. These will be denoted by $\mathcal{X}\otimes_\vee\mathcal{Y}$, $\mathcal{X}\otimes_\alpha\mathcal{Y}$, and $\mathcal{X}\otimes_\wedge\mathcal{Y}$. Since $\|\cdot\|_\vee\le\|\cdot\|_\alpha\le\|\cdot\|_\wedge$, the reverse inclusion $\boldsymbol{B}_\wedge\subseteq\boldsymbol{B}_\alpha\subseteq\boldsymbol{B}_\vee$ holds, where $\boldsymbol{B}_\vee=\{F\in\mathcal{X}\otimes\mathcal{Y}\colon\|F\|_\vee\le 1\}$, $\boldsymbol{B}_\alpha=\{F\in\mathcal{X}\otimes\mathcal{Y}\colon\|F\|_\alpha\le 1\}$, $\boldsymbol{B}_\wedge=\{F\in\mathcal{X}\otimes\mathcal{Y}\colon\|F\|_\wedge\le 1\}$ are closed unit balls in $\mathcal{X}\otimes_\vee\mathcal{Y}$, $\mathcal{X}\otimes_\alpha\mathcal{Y}$, $\mathcal{X}\otimes_\wedge\mathcal{Y}$.

Even if $\mathcal{X}$ and $\mathcal{Y}$ are Banach spaces, a tensor product space $\mathcal{X}\otimes\mathcal{Y}$ equipped with an arbitrary reasonable crossnorm $\|\cdot\|_\alpha$, say $\mathcal{X}\otimes_\alpha\mathcal{Y}=(\mathcal{X}\otimes\mathcal{Y},\|\cdot\|_\alpha)$, may not be complete. The completion of $\mathcal{X}\otimes_\alpha\mathcal{Y}$ is denoted by $\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$. The same notation $\|\cdot\|_\alpha$ is used for the norms on both normed spaces $\mathcal{X}\otimes_\alpha\mathcal{Y}$ and its completion $\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$, both referred to as reasonable crossnorms. Saying that $\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$ is the completion of $\mathcal{X}\otimes_\alpha\mathcal{Y}$ means $\mathcal{X}\otimes_\alpha\mathcal{Y}$ is densely embedded in the Banach space $\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$. So $\mathcal{X}\otimes_\alpha\mathcal{Y}$ is identified with the range $\mathcal{R}(J)=J(\mathcal{X}\otimes_\alpha\mathcal{Y})$ of an isometric isomorphism $J\colon\mathcal{X}\otimes_\alpha\mathcal{Y}\to\mathcal{R}(J)\subseteq\!\!\!\!\!\!\;\;\;\;\;\;\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$ onto $\mathcal{R}(J)$ for which $\mathcal{R}(J)$ is dense in $\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$; that is, $\mathcal{R}(J)^-=\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$ (recall: the upper bar $^-$ stands for closure and $\subseteq\!\!\!\!\!\!\;\;\;\;\;\;$ for densely included). If $T\in\mathcal{B}[\mathcal{X}\otimes_\alpha\mathcal{Y},\mathcal{Z}]$ for some Banach space $\mathcal{Z}$, then let $\widehat{T}\in\mathcal{B}[\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y},\mathcal{Z}]$ be its extension over completion as in Theorem 5.1, which is represented by the diagram

$$\begin{array}{ccccc} \mathcal{Z} & \xrightarrow{\ I\ } & \mathcal{Z} & = & \mathcal{Z} \\ {\scriptstyle T}\uparrow & & \uparrow{\scriptstyle \widehat{T}|_{\mathcal{X}\widetilde{\otimes}_\alpha\mathcal{Y}} = TJ^{-1}} & & \uparrow{\scriptstyle \widehat{T}} \\ \mathcal{X}\otimes_\alpha\mathcal{Y} & \xrightarrow{\ J\ } & \mathcal{X}\,\widetilde{\otimes}_\alpha\mathcal{Y} = J(\mathcal{X}\otimes_\alpha\mathcal{Y}) \subseteq & & \mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}. \end{array}$$

Since $\mathcal{X}\otimes_\alpha\mathcal{Y}$ is identified with $\mathcal{R}(J) = J(\mathcal{X}\otimes_\alpha\mathcal{Y})$, write $\mathcal{X}\otimes_\alpha\mathcal{Y}$ for $\mathcal{R}(J)$, then write $F = \sum_{i=1}^n x_i \otimes y_i$ for elements in $\mathcal{R}(J)$, and finally, regarding $\mathcal{X}\otimes_\alpha\mathcal{Y}$ as being dense in $\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$, and by an abuse of notation, write $(\mathcal{X}\otimes_\alpha\mathcal{Y})^- = \mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$. Note that the linear space $\mathcal{X}\otimes\mathcal{Y}$ is already the result of an identification with some tensor product space $\mathcal{T}$ according to Corollary 3.6, and the further identification with $\mathcal{R}(J)$ remains (isometrically) identified with the original $\mathcal{T}$.

For normed spaces (in particular, for Banach spaces) $\mathcal{X}$ and $\mathcal{Y}$, the completions $\mathcal{X}\,\widehat{\otimes}_\vee\mathcal{Y}$ and $\mathcal{X}\,\widehat{\otimes}_\wedge\mathcal{Y}$ (of $\mathcal{X}\otimes_\vee\mathcal{Y}$ and $\mathcal{X}\otimes_\wedge\mathcal{Y}$) are referred to as the *injective tensor product* and the *projective tensor product* of $\mathcal{X}$ and $\mathcal{Y}$, respectively.

Remark 7.5. *Regular Subspaces of Crossnormed Tensor Products.*

(a) Let $\mathcal{S}$ be any (algebraic) linear manifold of a linear space $\mathcal{Z}$. Equip $\mathcal{Z}$ with a norm $\|\cdot\|_\mathcal{Z}$ and consider the normed space $(\mathcal{Z}, \|\cdot\|_\mathcal{Z})$. Definition: $\mathcal{S}$ *is a linear manifold of the normed space* $\mathcal{Z}$ if, besides being an algebraic linear manifold of the linear space $\mathcal{Z}$, it is equipped with the restriction to it of the norm on $\mathcal{Z}$ (same notation $\|\cdot\|_\mathcal{Z}$), so that $(\mathcal{S}, \|\cdot\|_\mathcal{Z})$ is a normed space. In other words, if $\mathcal{S}$ *inherits the norm from* $\mathcal{Z}$. (Recall: a norm defined on $\mathcal{Z}$, when restricted to $\mathcal{S}$, is a norm on the linear space $\mathcal{S}$). Another way to say exactly the same thing (which will be useful in the sequel) is this. The identity I on $\mathcal{S}$ defines the trivial embedding $\mathcal{S} \hookrightarrow \mathcal{Z}$ (i.e., it defines the trivial linear-space isomorphism $I\colon \mathcal{S} \to I(\mathcal{S}) = \mathcal{S} \subseteq \mathcal{Z}$). Suppose $\mathcal{S}$ is equipped with a norm $\|\cdot\|_\mathcal{S}$. We say that $(\mathcal{S}, \|\cdot\|_\mathcal{S})$ is a *linear manifold* of $(\mathcal{Z}, \|\cdot\|_\mathcal{Z})$ if and only if $\|I(s)\|_\mathcal{Z} = \|s\|_\mathcal{S}$ for every $s \in \mathcal{S}$. That is,

$$I\colon (\mathcal{S}, \|\cdot\|_\mathcal{S}) \to (\mathcal{Z}, \|\cdot\|_\mathcal{Z}) \text{ is an isometry} \iff \mathcal{S} \text{ is a linear manifold of } \mathcal{Z}.$$

(b) Let $\mathcal{M}$ and $\mathcal{N}$ be linear manifolds of normed spaces $\mathcal{X}$ and $\mathcal{Y}$. Consider the regular linear manifold $\mathcal{M}\otimes\mathcal{N}$ of the linear space $\mathcal{X}\otimes\mathcal{Y}$. Equip the linear spaces $\mathcal{M}\otimes\mathcal{N}$ and $\mathcal{X}\otimes\mathcal{Y}$ with a reasonable crossnorm $\|\cdot\|_\alpha$.

Claim 1. $\mathcal{M}\otimes_\alpha\mathcal{N}$ may not be a linear manifold of $\mathcal{X}\otimes_\alpha\mathcal{Y}$.

Indeed, if a linear space $\mathcal{X}\otimes\mathcal{Y}$ is equipped with a reasonable crossnorm $\|\cdot\|_\alpha$, then it may happen that the restriction of $\|\cdot\|_\alpha$ to a regular linear manifold $\mathcal{M}\otimes\mathcal{N}$ does not act on $\mathcal{M}\otimes\mathcal{N}$ as a norm inherited from $\mathcal{X}\otimes\mathcal{Y}$, in the sense that $\|F\|_{\mathcal{M}\otimes_\alpha\mathcal{N}} \neq \|F\|_{\mathcal{X}\otimes_\alpha\mathcal{Y}}$ for some $F \in \mathcal{M}\otimes\mathcal{N} \subseteq \mathcal{X}\otimes\mathcal{Y}$, as discussed in (a).

Example 1. $\mathcal{M}\otimes_\wedge\mathcal{N}$ is not always a linear manifold of $\mathcal{X}\otimes_\wedge\mathcal{Y}$.

For instance, if a linear manifold $\mathcal{M}$ of a linear space $\mathcal{X}$ is properly included in $\mathcal{X}$ and if $F = \sum_i u_i \otimes y_i \in \mathcal{M}\otimes\mathcal{Y} \subset \mathcal{X}\otimes\mathcal{Y}$, then it may occur that $\|F\|_{\mathcal{X}\otimes_\wedge\mathcal{Y}}$ becomes less than $\|F\|_{\mathcal{M}\otimes_\wedge\mathcal{Y}}$. In fact,

$$\begin{aligned}\|F\|_{\mathcal{M}\otimes_\wedge\mathcal{Y}} &= \inf_{u_i\in\mathcal{M},\ y_i\in\mathcal{Y},\ F=\Sigma_i u_i\otimes y_i} \sum\nolimits_i \|u_i\otimes y_i\|_\wedge \\ &\geq \inf_{x_i\in\mathcal{X},\ x_i\otimes y_i = u_i\otimes y_i} \sum\nolimits_i \|x_i\otimes y_i\|_\wedge = \|F\|_{\mathcal{X}\otimes_\wedge\mathcal{Y}},\end{aligned}$$

with each infimum taken over all representations of $F=\sum_i u_i\otimes y_i\in\mathcal{M}\otimes\mathcal{Y}$, while the infimum taken in the second expression runs over a larger set (including all $u_i\in\mathcal{M}$ and also some possible $x_i\in\mathcal{X}\setminus\mathcal{M}$). Then in general

$$\|F\|_{\mathcal{X}\otimes_\wedge\mathcal{Y}} \leq \|F\|_{\mathcal{M}\otimes_\wedge\mathcal{N}},$$

and the inequality may be strict (i.e., it may be that $\|F\|_{\mathcal{X}\otimes_\wedge\mathcal{Y}} < \|F\|_{\mathcal{M}\otimes_\wedge\mathcal{N}}$ — e.g., by applying Theorem 7.9 below). Thus when the above inequality is strict for some $F\in\mathcal{M}\otimes\mathcal{Y}$ under the projective norm $\|\cdot\|_\wedge$, then $\mathcal{M}\otimes_\wedge\mathcal{Y}$ is not a linear manifold of $\mathcal{X}\otimes_\wedge\mathcal{Y}$ according to (a).

Example 2. $\mathcal{M}\otimes_\vee\mathcal{N}$ is a linear manifold of $\mathcal{X}\otimes_\vee\mathcal{Y}$.

This will be proved later in Theorem 7.19 (a little more familiarity with the injective norm $\|\cdot\|_\vee$, as will be summarized in Remark 7.18, is needed). By the way, this is the reason why the injective norm is called *injective*.

(c) Now let $\|\cdot\|_\mathcal{Z}$ be the norm of a Banach space $(\mathcal{Z},\|\cdot\|_\mathcal{Z})$ and let $\mathcal{S}$ be any linear manifold of the linear space $\mathcal{Z}$. The identity I on $\mathcal{S}$ defines the trivial embedding $\mathcal{S}\hookrightarrow\mathcal{Z}$. Instead of equipping $\mathcal{S}$ with the norm inherited from $\mathcal{Z}$, equip it with an arbitrary norm $\|\cdot\|_\mathcal{S}$. Suppose $(\mathcal{S},\|\cdot\|_\mathcal{S})$ is a Banach space and consider the trivial linear-space isomorphism $I\colon\mathcal{S}\to I(\mathcal{S})=\mathcal{S}\subseteq\mathcal{Z}$.

Claim 2. $I\colon(\mathcal{S},\|\cdot\|_\mathcal{S})\to(\mathcal{Z},\|\cdot\|_\mathcal{Z})$ is an isometry $\Longleftrightarrow\mathcal{S}$ is a subspace of $\mathcal{Z}$.

Indeed, suppose $I\colon(\mathcal{S},\|\cdot\|_\mathcal{S})\to(I(\mathcal{S}),\|\cdot\|_\mathcal{Z})$ is an isometric isomorphism. Since $(\mathcal{S},\|\cdot\|_\mathcal{S})$ is complete and $I(\mathcal{S})=\mathcal{S}\subseteq\mathcal{Z}$, then $(\mathcal{S},\|\cdot\|_\mathcal{Z})$ is complete (see, e.g., [52, Theorem 3.44]), and therefore the linear manifold $\mathcal{S}$ is closed in $(\mathcal{Z},\|\cdot\|_\mathcal{Z})$, that is, $\mathcal{S}$ is a subspace of $\mathcal{Z}$ (see, e.g., [52, Corollary 3.41]). The converse comes from (a) since a subspace is a linear manifold.

(d) Let $\mathcal{X}$ and $\mathcal{Y}$ be Banach spaces and suppose $\mathcal{M}$ and $\mathcal{N}$ are subspaces of $\mathcal{X}$ and $\mathcal{Y}$ (i.e., they are closed linear manifolds of $\mathcal{X}$ and $\mathcal{Y}$ inheriting the norms of $\mathcal{X}$ and $\mathcal{Y}$), so that $\mathcal{M}$ and $\mathcal{N}$ are Banach spaces themselves. Consider the regular linear manifold $\mathcal{M}\otimes\mathcal{N}$ of the linear space $\mathcal{X}\otimes\mathcal{Y}$. Equip them with an arbitrary reasonable crossnorm $\|\cdot\|_\alpha$. As we saw in (b), the crossnormed tensor product space $\mathcal{M}\otimes_\alpha\mathcal{N}$ is not necessarily a linear manifold of the crossnormed tensor product space $\mathcal{X}\otimes_\alpha\mathcal{Y}$. Take the completions $\mathcal{M}\,\widehat{\otimes}_\alpha\mathcal{N}$ and $\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$ of $\mathcal{M}\otimes_\alpha\mathcal{N}$ and $\mathcal{X}\otimes_\alpha\mathcal{Y}$. If $\mathcal{M}\otimes_\alpha\mathcal{N}$ is a linear manifold of $\mathcal{X}\otimes_\alpha\mathcal{Y}$, then $\mathcal{M}\,\widehat{\otimes}_\alpha\mathcal{N}$ is a subspace of $\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$ by Theorem 5.1(g).

(e) According to (a), (c), (d), a way to show that $\mathcal{M}\otimes_\alpha\mathcal{N}$ is a linear manifold of $\mathcal{X}\otimes_\alpha\mathcal{Y}$, and so $\mathcal{M}\,\widehat{\otimes}_\alpha\mathcal{N}$ is a subspace of $\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$, is to verify whether the trivial embeddings are isometries: for every F in $\mathcal{M}\otimes_\alpha\mathcal{N}$,

$$\|F\|_{\mathcal{M}\otimes_\alpha\mathcal{N}} = \|F\|_{\mathcal{X}\otimes_\alpha\mathcal{Y}} \quad \Longrightarrow \quad \mathcal{M}\otimes_\alpha\mathcal{N} \text{ is a linear manifold of } \mathcal{X}\otimes_\alpha\mathcal{Y}$$
$$\Longrightarrow \quad \mathcal{M}\,\widehat{\otimes}_\alpha\mathcal{N} \text{ is a subspace of } \mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}.$$

The converses hold trivially. (If $\mathcal{S}$ is any linear manifold of an arbitrary normed space $\mathcal{Z}$ and if $\mathcal{Z}\xrightarrow{J}J(\mathcal{Z})\subseteq\widehat{\mathcal{Z}}$ and $\mathcal{S}\xrightarrow{J}J(\mathcal{S})\subseteq\widehat{\mathcal{S}}$ for an arbitrary isometric isomorphism J on $\mathcal{Z}$, then $\|s\|_{\mathcal{Z}} = \|s\|_{\mathcal{S}} = \|\widehat{s}\|_{\widehat{\mathcal{S}}} = \|\widehat{s}\|_{\widehat{\mathcal{Z}}}$ for every $s\in\mathcal{S}$.)

We close the section with the notion of *transpose norm* $\|\cdot\|_{t_\alpha}$.

Let $\mathcal{X}$ and $\mathcal{Y}$ be linear spaces and consider the natural permutation isomorphism $\Pi\colon\mathcal{X}\otimes\mathcal{Y}\to\mathcal{Y}\otimes\mathcal{X}$ between linear spaces $\mathcal{X}\otimes\mathcal{Y}$ and $\mathcal{Y}\otimes\mathcal{X}$ given by $\Pi(\sum_i x_i\otimes y_i)=\sum_i y_i\otimes x_i$. For each $F\in\mathcal{X}\otimes\mathcal{Y}$ and each $\mathrm{G}\in\mathcal{Y}\otimes\mathcal{X}$ set

$$^tF = \Pi F \in \mathcal{Y}\otimes\mathcal{X} \quad \text{and} \quad {}^t\mathrm{G} = \Pi^{-1}\mathrm{G} \in \mathcal{X}\otimes\mathcal{Y}.$$

This ${}^tF$ is the *transpose of* F and, symmetrically, ${}^t\mathrm{G}$ is the *transpose of* G. So

$${}^{tt}F = {}^t({}^tF) = \Pi^{-1}({}^tF) = \Pi^{-1}(\Pi F) = F \quad \text{for every} \quad F\in\mathcal{X}\otimes\mathcal{Y},$$

$${}^{tt}\mathrm{G} = {}^t({}^t\mathrm{G}) = \Pi({}^t\mathrm{G}) = \Pi(\Pi^{-1}\mathrm{G}) = \mathrm{G} \quad \text{for every} \quad \mathrm{G}\in\mathcal{Y}\otimes\mathcal{X}.$$

Now suppose $\mathcal{X}$ and $\mathcal{Y}$ are normed spaces. Let $\|\cdot\|_\alpha$ be a reasonable crossnorm on $\mathcal{Y}\otimes\mathcal{X}$. Associate with $\|\cdot\|_\alpha$ a norm $\|\cdot\|_{t_\alpha}$ on $\mathcal{X}\otimes\mathcal{Y}$ defined as follows:

$$\|F\|_{t_\alpha} = \|{}^tF\|_\alpha \quad \text{for every} \quad F\in\mathcal{X}\otimes\mathcal{Y} \text{ for which } {}^tF\in\mathcal{Y}\otimes_\alpha\mathcal{X}.$$

Consider the iterated norm $\|\cdot\|_{t t_\alpha} = \|\cdot\|_{t(t_\alpha)}$ on $\mathcal{Y}\otimes\mathcal{X}$ defined by

$$\|\mathrm{G}\|_{t t_\alpha} = \|{}^t\mathrm{G}\|_{t_\alpha} \quad \text{for every} \quad \mathrm{G}\in\mathcal{Y}\otimes\mathcal{X} \text{ for which } {}^t\mathrm{G}\in\mathcal{X}\otimes\mathcal{Y}.$$

Since ${}^{tt}\mathrm{G}=\mathrm{G}$, and by the above two identities, we get

$$\|\mathrm{G}\|_\alpha = \|{}^{tt}\mathrm{G}\|_\alpha = \|{}^t\mathrm{G}\|_{t_\alpha} = \|\mathrm{G}\|_{t t_\alpha} \quad \text{for every} \quad \mathrm{G}\in\mathcal{Y}\otimes\mathcal{X},$$

and hence $\mathcal{X}\otimes_{t t_\alpha}\mathcal{Y} = \mathcal{X}\otimes_\alpha\mathcal{Y}$ because

$$\|\cdot\|_{t t_\alpha} = \|\cdot\|_\alpha.$$

Theorem 7.6. *If $\|\cdot\|_\alpha$ is a reasonable crossnorm on $\mathcal{Y}\otimes\mathcal{X}$, then*

(a) *$\|\cdot\|_{t_\alpha}$ is a reasonable crossnorm on $\mathcal{X}\otimes\mathcal{Y}$, and*

(b) *the natural permutation $\Pi\colon\mathcal{X}\otimes_{t_\alpha}\mathcal{Y}\to\mathcal{Y}\otimes_\alpha\mathcal{X}$ is an isometric isomorphism so that $\mathcal{X}\otimes_{t_\alpha}\mathcal{Y}\cong\mathcal{Y}\otimes_\alpha\mathcal{X}$.*

If $\|\cdot\|_\beta$ is another reasonable crossnorm on $\mathcal{Y}\otimes\mathcal{X}$, then

(c) *$\|\cdot\|_\alpha\le\|\cdot\|_\beta$ if and only if $\|\cdot\|_{t_\alpha}\le\|\cdot\|_{t_\beta}$.*

Proof. (a) Let $\|\cdot\|_\alpha$ be a reasonable crossnorm acting on the tensor product space of an arbitrary pair of normed spaces, say, $\|\cdot\|_\alpha\colon\mathcal{Y}\otimes\mathcal{X}\to\mathbb{R}$. It is clear that $\|\cdot\|_{t_\alpha}\colon\mathcal{X}\otimes\mathcal{Y}\to\mathbb{R}$ defines a norm acting on $\mathcal{X}\otimes\mathcal{Y}$ for which

$$\|x\otimes y\|_{t_\alpha} = \|\Pi(x\otimes y)\|_\alpha = \|y\otimes x\|_\alpha = \|y\|\,\|x\|$$

for every single tensor $x \otimes y$ in $\mathcal{X} \otimes \mathcal{Y}$. Moreover, take an arbitrary single tensor $f \otimes g$ in $\mathcal{X}^* \otimes \mathcal{Y}^* \subseteq \mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp \subseteq (\mathcal{X} \otimes \mathcal{Y})^\sharp$. Recall that $(\mathcal{X} \otimes_{t\alpha} \mathcal{Y})^* = \{\Lambda \in (\mathcal{X} \otimes \mathcal{Y})^\sharp : \sup_{\|F\|_{t\alpha} \le 1,\, F \in \mathcal{X} \otimes \mathcal{Y}} |\Lambda(F)| < \infty\}$, and also that for any $F = \sum_i x_i \otimes y_i \in \mathcal{X} \otimes \mathcal{Y}$ we get $(f \otimes g)F = (f \otimes g) \sum_i x_i \otimes y_i = \sum_i f(x_i) g(y_i) = \sum_i g(y_i) f(x_i) = (g \otimes f) \sum_i y_i \otimes x_i = (g \otimes f)\, {}^tF$. Therefore

$$\begin{aligned} \|f \otimes g\|_{(\mathcal{X} \otimes_{t\alpha} \mathcal{Y})^*} &= \sup_{\|F\|_{t\alpha} \le 1,\, F \in \mathcal{X} \otimes \mathcal{Y}} |(f \otimes g)F| = \sup_{\|{}^tF\|_\alpha \le 1,\, {}^tF \in \mathcal{Y} \otimes \mathcal{X}} |(g \otimes f)\, {}^tF| \\ &= \sup_{\|\mathrm{G}\|_\alpha \le 1,\, \mathrm{G} \in \mathcal{Y} \otimes \mathcal{X}} |(g \otimes f)\mathrm{G}| = \|g \otimes f\|_{(\mathcal{Y} \otimes_\alpha \mathcal{X})^*} = \|g\|\,\|f\|, \end{aligned}$$

so that $f \otimes g$ lies in $(\mathcal{X} \otimes_{t\alpha} \mathcal{Y})^*$ and $\|f \otimes g\|_{*^t\alpha} = \|f \otimes g\|_{(\mathcal{X} \otimes_{t\alpha} \mathcal{Y})^*} = \|f\|\,\|g\|$, and hence $\|\cdot\|_{t\alpha}$ is a reasonable crossnorm on $\mathcal{X} \otimes \mathcal{Y}$ by Theorem 7.2.

(b) Since $\|\Pi F\|_{\mathcal{Y} \otimes_\alpha \mathcal{X}} = \|{}^tF\|_{\mathcal{Y} \otimes_\alpha \mathcal{X}} = \|F\|_{\mathcal{X} \otimes_{t\alpha} \mathcal{Y}}$ for every $F \in \mathcal{X} \otimes_{t\alpha} \mathcal{Y}$,

$$\Pi : \mathcal{X} \otimes_{t\alpha} \mathcal{Y} \to \mathcal{Y} \otimes_\alpha \mathcal{X} \text{ is an isometric isomorphism.}$$

(c) Thus $\|F\|_{t\alpha} = \|\Pi F\|_\alpha \le \|\Pi F\|_\beta = \|F\|_{t\beta}$ for every $F \in \mathcal{X} \otimes \mathcal{Y}$ whenever $\|\cdot\|_\alpha \le \|\cdot\|_\beta$. For the converse, set $\|\cdot\|_\alpha = \|\cdot\|_{t^t\alpha}$ and $\|\cdot\|_\beta = \|\cdot\|_{t^t\beta}$. □

This norm $\|\cdot\|_{t\alpha} : \mathcal{X} \otimes \mathcal{Y} \to \mathbb{R}$ is the *transpose norm of* $\|\cdot\|_\alpha : \mathcal{Y} \otimes \mathcal{X} \to \mathbb{R}$.

Remark 7.7. *Symmetric Reasonable Crossnorms.* A reasonable crossnorm is defined for a given pair of normed spaces $\mathcal{X}$ and $\mathcal{Y}$. However, suppose for a moment that a reasonable crossnorm $\|\cdot\|_\alpha$ may equip both linear spaces $\mathcal{X} \otimes \mathcal{Y}$ and $\mathcal{Y} \otimes \mathcal{X}$ (as will be the case for the uniform crossnorms of Chapter 8). So its transpose $\|\cdot\|_{t\alpha}$ also equips both linear spaces $\mathcal{Y} \otimes \mathcal{X}$ and $\mathcal{X} \otimes \mathcal{Y}$. Define:

(a) $\|\cdot\|_\alpha$ is a *symmetric reasonable crossnorm* if $\|\cdot\|_{t\alpha} = \|\cdot\|_\alpha$.

By Theorem 7.6(b), if $\|\cdot\|_{t\alpha} = \|\cdot\|_\alpha$, then $\mathcal{X} \otimes_\alpha \mathcal{Y} \cong \mathcal{Y} \otimes_\alpha \mathcal{X}$, which implies $\Pi : \mathcal{X} \otimes_\alpha \mathcal{Y} \to \mathcal{Y} \otimes_\alpha \mathcal{X}$ is an isometric isomorphism. Therefore

(b) $\|\cdot\|_\alpha$ *is a symmetric reasonable crossnorm* $\iff$ $\mathcal{X} \otimes_\alpha \mathcal{Y} \cong \mathcal{Y} \otimes_\alpha \mathcal{X}$ $\iff$ $\Pi : \mathcal{X} \otimes_\alpha \mathcal{Y} \to \mathcal{Y} \otimes_\alpha \mathcal{X}$ *is an isometric isomorphism.*

Moreover, let $\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}$ and $\mathcal{Y} \mathbin{\widehat{\otimes}_\alpha} \mathcal{X}$ be completions of $\mathcal{X} \otimes_\alpha \mathcal{Y}$ and $\mathcal{Y} \otimes_\alpha \mathcal{X}$, and let $\widehat{\Pi}$ in $\mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}, \mathcal{Y} \mathbin{\widehat{\otimes}_\alpha} \mathcal{X}]$ be the extension over completion of Π in $\mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{Y} \otimes_\alpha \mathcal{X}]$, which are isometric isomorphisms together (see Theorem 5.1(d)). Hence $\Pi : \mathcal{X} \otimes_\alpha \mathcal{Y} \to \mathcal{Y} \otimes_\alpha \mathcal{X}$ is an isometric isomorphism if and only if $\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y} \cong \mathcal{Y} \mathbin{\widehat{\otimes}_\alpha} \mathcal{X}$, which means the norms $\|\cdot\|_\alpha$ (same notation) on $\mathcal{X} \otimes_\alpha \mathcal{Y}$ and on $\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}$ are symmetric reasonable crossnorms together. Summing up,

(c) $\|\cdot\|_\alpha$ *is a symmetric reasonable crossnorm* $\iff$ $\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y} \cong \mathcal{Y} \mathbin{\widehat{\otimes}_\alpha} \mathcal{X}$.

(d) Furthermore, *the projective and the injective norms are symmetric.*

Indeed, $\|\cdot\|_\wedge$ and $\|\cdot\|_\vee$ can be assigned to $\mathcal{X} \otimes \mathcal{Y}$ and to $\mathcal{Y} \otimes \mathcal{X}$. Take $F = \sum_i x_i \otimes y_i$ in $\mathcal{X} \otimes \mathcal{Y}$. (i) If F lies in $\mathcal{X} \otimes_\wedge \mathcal{Y}$, then $\|F\|_\wedge = \inf \sum_i \|x_i\|\,\|y_i\| = \inf \sum_i \|y_i\|\|x_i\| = \|{}^tF\|_\wedge = \|F\|_{t\wedge}$. Also, (ii) if F lies in $\mathcal{X} \otimes_\vee \mathcal{Y}$, then $\|F\|_\vee = \sup_{\|f\|=\|g\|=1} |\sum_i f(x_i)\, g(y_i)| = \sup_{\|g\|=\|f\|=1} |\sum_i g(y_i)\, f(x_i)| = \|{}^tF\|_\vee = \|F\|_{t\vee}$.

7.2 Projective Tensor Product

Take a tensor product space $\mathcal{X} \otimes \mathcal{Y}$ of a pair of normed spaces $\mathcal{X}$ and $\mathcal{Y}$. Consider the characterization of a reasonable crossnorm $\|\cdot\|_\alpha$ as in Theorem 7.2, which shows that $\|x \otimes y\|_\alpha = \|x\|\,\|y\|$ for every single tensor $x \otimes y \in \mathcal{X} \otimes \mathcal{Y}$. Let the finite sum $\sum_i x_i \otimes y_i$ be any representation of an arbitrary $f \in \mathcal{X} \otimes \mathcal{Y}$. If $\|\cdot\|_\alpha$ is a reasonable crossnorm on the linear space $\mathcal{X} \otimes \mathcal{Y}$, then

$$\|f\|_\alpha \le \sum\nolimits_i \|x_i \otimes y_i\|_\alpha = \sum\nolimits_i \|x_i\|\,\|y_i\|$$

for every representation of f. The projective norm $\|f\|_\wedge$ is precisely the infimum of the above sum taken over all possible representations of f.

Let $\mathcal{M}$ and $\mathcal{N}$ be linear manifolds of normed spaces $\mathcal{X}$ and $\mathcal{Y}$, respectively. Take the regular linear manifold $\mathcal{M} \otimes \mathcal{N}$ of the linear space $\mathcal{X} \otimes \mathcal{Y}$. The projective norm can be assigned to both spaces $\mathcal{M} \otimes \mathcal{N}$ and $\mathcal{X} \otimes \mathcal{Y}$. Thus consider the normed spaces $\mathcal{M} \otimes_\wedge \mathcal{N} = (\mathcal{M} \otimes \mathcal{N}, \|\cdot\|_\wedge)$ and $\mathcal{X} \otimes_\wedge \mathcal{Y} = (\mathcal{X} \otimes \mathcal{Y}, \|\cdot\|_\wedge)$. If $\mathcal{X}$ and $\mathcal{Y}$ are Banach spaces and $\mathcal{M}$ and $\mathcal{N}$ are subspaces of them, then take their completions $\mathcal{M} \mathbin{\widehat{\otimes}_\wedge} \mathcal{N}$ and $\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}$. In light of the previous Remark 7.5(b) on linear manifolds and subspaces of tensor products, we begin by establishing when $\mathcal{M} \otimes_\wedge \mathcal{N}$ is a linear manifold of the normed space $\mathcal{X} \otimes_\wedge \mathcal{Y}$ and, consequently, when $\mathcal{M} \mathbin{\widehat{\otimes}_\wedge} \mathcal{N}$ is a subspace of the Banach space $\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}$.

Theorem 7.8. *Suppose $\mathcal{M}$ and $\mathcal{N}$ are complemented subspaces of Banach spaces $\mathcal{X}$ and $\mathcal{Y}$, respectively.*

(a) *Then the projective norm on $\mathcal{M} \otimes_\wedge \mathcal{N}$ and the norm on $\mathcal{M} \otimes \mathcal{N}$ inherited from $\mathcal{X} \otimes_\wedge \mathcal{Y}$ are equivalent.*

(b) *If $\mathcal{M} \otimes_\wedge \mathcal{N}$ is a linear manifold of the normed space $\mathcal{X} \otimes_\wedge \mathcal{Y}$, then it is closed in $\mathcal{X} \otimes_\wedge \mathcal{Y}$ (thus a subspace of $\mathcal{X} \otimes_\wedge \mathcal{Y}$).*

(c) *If $\mathcal{M} \otimes_\wedge \mathcal{N}$ is a linear manifold of the normed space $\mathcal{X} \otimes_\wedge \mathcal{Y}$, then $\mathcal{M} \otimes_\wedge \mathcal{N}$ is a complemented subspace of the normed space $\mathcal{X} \otimes_\wedge \mathcal{Y}$, and $\mathcal{M} \mathbin{\widehat{\otimes}_\wedge} \mathcal{N}$ is a complemented subspace of the Banach space $\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}$.*

(d) *If $\mathcal{M} = \mathcal{R}(E)$ and $\mathcal{N} = \mathcal{R}(P)$ for projections E in $\mathcal{B}[\mathcal{X}]$ and P in $\mathcal{B}[\mathcal{Y}]$ with $\|E\| = \|P\| = 1$, then $\mathcal{M} \otimes_\wedge \mathcal{N}$ is a linear manifold of $\mathcal{X} \otimes_\wedge \mathcal{Y}$.*

Proof. Suppose $\mathcal{M}$ and $\mathcal{N}$ are complemented in $\mathcal{X}$ and $\mathcal{Y}$. Let E in $\mathcal{B}[\mathcal{X}]$ and P in $\mathcal{B}[\mathcal{Y}]$ be projections such that $\mathcal{M} = \mathcal{R}(E)$ and $\mathcal{N} = \mathcal{R}(P)$ (Remark 5.3). Take the tensor product transformation $E \otimes P$ in $\mathcal{L}[\mathcal{X} \otimes \mathcal{Y}]$ (Theorem 3.17), which is idempotent, thus a projection on the linear space $\mathcal{X} \otimes \mathcal{Y}$. Indeed,

$$(E \otimes P)(E \otimes P)f = (E \otimes P)\sum\nolimits_i Ex_i \otimes Py_i = \sum\nolimits_i E^2 x_i \otimes P^2 y_i = (E \otimes P)f$$

for every $f = \sum_i x_i \otimes y_i \in \mathcal{X} \otimes \mathcal{Y}$. Also (recall Theorem 3.19(e))

$$\mathcal{R}(E \otimes P) = \mathcal{M} \otimes \mathcal{N},$$

so that $F = (E \otimes P)F$ for every F in the regular linear manifold $\mathcal{M} \otimes \mathcal{N}$ of the linear space $\mathcal{X} \otimes \mathcal{Y}$. Equip $\mathcal{X} \otimes \mathcal{Y}$ with the projective norm and take $F \in \mathcal{X} \otimes_\wedge \mathcal{Y} = (\mathcal{X} \otimes \mathcal{Y}, \|\cdot\|_\wedge) = (\mathcal{X} \otimes \mathcal{Y}, \|\cdot\|_{\mathcal{X} \otimes_\wedge \mathcal{Y}})$. Since E and P are bounded,

$$\sup_{0 \neq F \in \mathcal{X} \otimes_\wedge \mathcal{Y}} \frac{\|(E \otimes P)F\|_{\mathcal{X} \otimes_\wedge \mathcal{Y}}}{\|F\|_{\mathcal{X} \otimes_\wedge \mathcal{Y}}} = \sup_{0 \neq F \in \mathcal{X} \otimes_\wedge \mathcal{Y}} \frac{\inf_{\{x_i\},\,\{y_i\}} \sum_i \|Ex_i\| \|Py_i\|}{\inf_{\{x_i\},\,\{y_i\}} \sum_i \|x_i\| \|y_i\|} \leq \|E\| \, \|P\|.$$

The above shows that the linear transformation $E \otimes P$ as considered in Theorem 3.17 is bounded when acting from the normed space $\mathcal{X} \otimes_\wedge \mathcal{Y}$ to itself. (We will consider *bounded* linear tensor product transformations in detail in due course.) Thus $E \otimes P$ is a continuous projection, $E \otimes P \in \mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}]$, with

$$\|E \otimes P\|_{\mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}]} \leq \|E\| \, \|P\|.$$

On the other hand, since $\|\cdot\|_{\mathcal{X} \otimes_\wedge \mathcal{Y}}$ is a reasonable crossnorm,

$$\|Ex\| \|Py\| = \|Ex \otimes Py\|_{\mathcal{X} \otimes_\wedge \mathcal{Y}} = \|(E \otimes P)(x \otimes y)\|_{\mathcal{X} \otimes_\wedge \mathcal{Y}} \leq \|E \otimes P\|_{\mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}]} \|x\| \|y\|$$

for every $x \in \mathcal{X}$ and $y \in \mathcal{Y}$. Hence

$$\|E\| \, \|P\| = \sup_{x \neq 0} \frac{\|Ex\|}{\|x\|} \sup_{y \neq 0} \frac{\|Py\|}{\|y\|} \leq \|E \otimes P\|_{\mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}]}.$$

Since E and P are continuous projections, $1 \leq \|E\|$ and $1 \leq \|P\|$, and so

$$1 \leq \|E \otimes P\|_{\mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}]} = \|E\| \, \|P\|.$$

(a) Take an arbitrary $F \in \mathcal{M} \otimes \mathcal{N}$. Consider a representation of it in $\mathcal{X} \otimes \mathcal{Y}$, say $F = \sum_i x_i \otimes y_i$. Recall from the argument in Remark 7.5(b) that this representation may occur for $x_i \in \mathcal{X} \backslash \mathcal{M}$ and $y_i \in \mathcal{Y} \backslash \mathcal{N}$. Since $F = (E \otimes P)F$, $(E \otimes P) \sum_i x_i \otimes y_i = \sum_i Ex_i \otimes Py_i$ is a representation of F in $\mathcal{M} \otimes \mathcal{N}$ with $Ex_i \in \mathcal{M}$ and $Px_i \in \mathcal{N}$. Since $\|F\|_{\mathcal{M} \otimes_\wedge \mathcal{N}} = \inf_{\{u_i\},\{v_i\},\, F = \Sigma_i u_i \otimes v_i} \sum_i \|u_i\| \, \|v_i\|$,

$$\|F\|_{\mathcal{M} \otimes_\wedge \mathcal{N}} \leq \sum\nolimits_i \|Ex_i\| \, \|Py_i\| \leq \|E\| \, \|P\| \sum\nolimits_i \|x_i\| \, \|y_i\|,$$

and hence

$$\|F\|_{\mathcal{M} \otimes_\wedge \mathcal{N}} \leq \inf_{\{x_i\},\,\{y_i\},\, F = \Sigma_i x_i \otimes y_i} \|E\| \, \|P\| \sum\nolimits_i \|x_i\| \, \|y_i\| = \|E\| \, \|P\| \, \|F\|_{\mathcal{X} \otimes_\wedge \mathcal{Y}}.$$

Therefore, with the help of Remark 7.5(b),

$$\|F\|_{\mathcal{X} \otimes_\wedge \mathcal{Y}} \leq \|F\|_{\mathcal{M} \otimes_\wedge \mathcal{N}} \leq \|E\| \, \|P\| \, \|F\|_{\mathcal{X} \otimes_\wedge \mathcal{Y}}, \qquad (*)$$

and so the normed spaces $\mathcal{M} \otimes_\wedge \mathcal{N} = (\mathcal{M} \otimes \mathcal{N}, \|\cdot\|_\wedge) = (\mathcal{M} \otimes \mathcal{N}, \|\cdot\|_{\mathcal{M} \otimes_\wedge \mathcal{N}})$ and $(\mathcal{M} \otimes \mathcal{N}, \|\cdot\|_{\mathcal{X} \otimes_\wedge \mathcal{Y}})$ with a common underlying linear space $\mathcal{M} \otimes \mathcal{N}$ are equipped with equivalent norms, which proves (a).

(b) The range of the above continuous projection $E \otimes P$ in $\mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}]$ is given by $\mathcal{R}(E \otimes P) = (\mathcal{M} \otimes \mathcal{N}, \|\cdot\|_{\mathcal{X} \otimes_\wedge \mathcal{Y}})$, which is a linear manifold of the normed

space $\mathcal{X} \otimes_\wedge \mathcal{Y}$ inheriting the norm of $\mathcal{X} \otimes_\wedge \mathcal{Y}$. But ranges of continuous projections are closed. Thus this is a closed linear manifold of $\mathcal{X} \otimes_\wedge \mathcal{Y}$, and hence a subspace of $\mathcal{X} \otimes_\wedge \mathcal{Y}$. Now if $\mathcal{M} \otimes_\wedge \mathcal{N}$ is a linear manifold of $\mathcal{X} \otimes_\wedge \mathcal{Y}$, then it coincides with $(\mathcal{M} \otimes \mathcal{N}, \|\cdot\|_{\mathcal{X} \otimes_\wedge \mathcal{Y}})$ and so it is a subspace of $\mathcal{X} \otimes_\wedge \mathcal{Y}$.

(c) Suppose $\mathcal{M} \otimes_\wedge \mathcal{N}$ is a linear manifold of the normed space $\mathcal{X} \otimes_\wedge \mathcal{Y}$, so that it is a subspace of $\mathcal{X} \otimes_\wedge \mathcal{Y}$ according to (b). Let $E \otimes P$ in $\mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}]$ be the projection with $\mathcal{R}(E \otimes P) = \mathcal{M} \otimes_\wedge \mathcal{N}$, as in the proof of (b) above. Thus $\mathcal{M} \otimes_\wedge \mathcal{N}$ is complemented by Remark 5.3. Now let $E \mathbin{\widehat{\otimes}} P$ in $\mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}}_\wedge \mathcal{Y}]$ be the extension over completion of $E \otimes P$ in $\mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}]$. This is the extension by continuity of $J(E \otimes P)J^{-1}$ for some isometric isomorphism J, so that $J(E \otimes P)J^{-1}$ is a continuous projection with $\mathcal{R}(J(E \otimes P)J^{-1}) = J(\mathcal{R}(E \otimes P)) = J(\mathcal{M} \otimes_\wedge \mathcal{N})$, and so is its extension by continuity $E \mathbin{\widehat{\otimes}} P$ into the closure $J(\mathcal{X} \otimes_\wedge \mathcal{Y})^- = \mathcal{X} \mathbin{\widehat{\otimes}}_\wedge \mathcal{Y}$ of $J(\mathcal{X} \otimes_\wedge \mathcal{Y})$. Thus $E \mathbin{\widehat{\otimes}} P$ is a continuous projection with the same closed range $\mathcal{R}(E \mathbin{\widehat{\otimes}}_\wedge P) = \mathcal{R}(J(E \otimes P)J^{-1}) = J(\mathcal{R}(E \otimes P)) = J(\mathcal{M} \otimes_\wedge \mathcal{N}) = \mathcal{M} \mathbin{\widehat{\otimes}}_\wedge \mathcal{N}$. (See Theorem 5.1(a,f) and the diagram in the proof of Theorem 5.1.) So $\mathcal{M} \mathbin{\widehat{\otimes}}_\wedge \mathcal{N}$ is a complemented subspace of $\mathcal{X} \mathbin{\widehat{\otimes}}_\wedge \mathcal{Y}$ (see Remark 5.3 once again).

(d) If the complemented $\mathcal{M}$ and $\mathcal{N}$ are such that $\mathcal{M} = \mathcal{R}(E)$ and $\mathcal{N} = \mathcal{R}(P)$ for projections $E \in \mathcal{B}[\mathcal{X}]$ and $P \in \mathcal{B}[\mathcal{Y}]$ with $\|E\| = \|P\| = 1$, then by $(*)$

$$\|F\|_{\mathcal{M} \otimes_\wedge \mathcal{N}} = \|F\|_{\mathcal{X} \otimes_\wedge \mathcal{Y}}.$$

Therefore, according to Remark 7.5(e), this means $\mathcal{M} \otimes_\wedge \mathcal{N}$ is a linear manifold of the normed space $\mathcal{X} \otimes_\wedge \mathcal{Y}$, which proves (d). □

Since a projective tensor product $\mathcal{X} \mathbin{\widehat{\otimes}}_\wedge \mathcal{Y}$ is the completion of the normed space $\mathcal{X} \otimes_\wedge \mathcal{Y} = (\mathcal{X} \otimes \mathcal{Y}, \|\cdot\|_\wedge)$, this $\mathcal{X} \otimes_\wedge \mathcal{Y}$ is naturally identified with the range $J(\mathcal{X} \otimes_\wedge \mathcal{Y})$ of an isometric isomorphism $J : \mathcal{X} \otimes_\wedge \mathcal{Y} \to J(\mathcal{X} \otimes_\wedge \mathcal{Y}) \subseteq \mathcal{X} \mathbin{\widehat{\otimes}}_\wedge \mathcal{Y}$, as we saw above. Write $F = \sum_{i=1}^n x_i \otimes y_i$ with $\|F\|_\wedge = \inf \sum_{i=1}^n \|x_i\| \, \|y_i\|$ for the elements in $J(\mathcal{X} \otimes_\wedge \mathcal{Y})$ identified with $\mathcal{X} \otimes_\wedge \mathcal{Y}$, regard $\mathcal{X} \otimes_\wedge \mathcal{Y}$ as being dense in $\mathcal{X} \mathbin{\widehat{\otimes}}_\wedge \mathcal{Y}$, and write $\mathcal{X} \mathbin{\widehat{\otimes}}_\wedge \mathcal{Y} \setminus \mathcal{X} \otimes_\wedge \mathcal{Y}$ for $\mathcal{X} \mathbin{\widehat{\otimes}}_\wedge \mathcal{Y} \setminus J(\mathcal{X} \otimes_\wedge \mathcal{Y})$.

The next result is sometimes referred to as the Grothendieck Theorem. In fact, most of the results in this chapter are Grothendieck's (see, e.g., [17]).

Theorem 7.9. (Grothendieck). *If $\mathcal{X}$ and $\mathcal{Y}$ are Banach spaces, then for every $F \in \mathcal{X} \mathbin{\widehat{\otimes}}_\wedge \mathcal{Y}$ there exist $\mathcal{X}$-valued and $\mathcal{Y}$-valued sequences $\{x_k\}$ and $\{y_k\}$ for which the real sequence $\{\|x_k\| \, \|y_k\|\}$ is summable and*

$$F = \sum\nolimits_k x_k \otimes y_k$$

(i.e., every $F \in \mathcal{X} \mathbin{\widehat{\otimes}}_\wedge \mathcal{Y}$ is representable in the form of a countable sum $F = \sum_k x_k \otimes y_k$ in the sense that it is either a finite or a countably infinite representation). Moreover, the norm $\|\cdot\|_\wedge$ on $\mathcal{X} \mathbin{\widehat{\otimes}}_\wedge \mathcal{Y}$ is given by

$$\|F\|_\wedge = \inf \sum\nolimits_k \|x_k\| \, \|y_k\|,$$

where the infimum is taken over all representations $\sum_k x_k \otimes y_k$ of $F \in \mathcal{X} \mathbin{\widehat{\otimes}}_\wedge \mathcal{Y}$.

Proof. Let $\mathcal{X}\widehat{\otimes}_\wedge\mathcal{Y}$ be the completion of $\mathcal{X}\otimes_\wedge\mathcal{Y}$. Take any $F\in\mathcal{X}\widehat{\otimes}_\wedge\mathcal{Y}\setminus\mathcal{X}\otimes_\wedge\mathcal{Y}$ (otherwise the resulting finite sum is trivially obtained). Thus F is arbitrarily close to elements in $\mathcal{X}\otimes_\wedge\mathcal{Y}$. Take an arbitrary $\varepsilon>0$. For each positive integer k take $F_k=\sum_{i=1}^{n_k}x_i\otimes y_i$ in $\mathcal{X}\otimes_\wedge\mathcal{Y}$ such that

$$\|F-F_k\|_\wedge<\tfrac{\varepsilon}{2}\tfrac{1}{2^k}.$$

In particular, $|\,\|F_1\|_\wedge-\|F\|_\wedge|\le\|F_1-F\|_\wedge\le\frac{\varepsilon}{2}\frac{1}{2}$ so $\|F_1\|_\wedge\le\|F\|_\wedge+\frac{\varepsilon}{4}$. Take a representation $\sum_{i=1}^{n_1}x_i\otimes y_i$ for F_1 for which $\sum_{i=1}^{n_1}\|x_i\|\,\|y_i\|$ is close enough to $\|F_1\|_\wedge=\inf_{\{x_i\}_{i=1}^{n_1},\,\{y_i\}_{i=1}^{n_1}}\sum_{i=1}^{n_1}\|x_i\|\,\|y_i\|$, say

$$\sum\nolimits_{i=1}^{n_1}\|x_i\|\,\|y_i\|\le\|F_1\|_\wedge+\tfrac{3\varepsilon}{4}\le\|F\|_\wedge+\varepsilon.$$

Computing the norm of $F_{k+1}-F_k\in\mathcal{X}\otimes_\wedge\mathcal{Y}$ we get

$$\|F_{k+1}-F_k\|_\wedge\le\|F_{k+1}-F\|_\wedge+\|F-F_k\|_\wedge<\tfrac{\varepsilon}{2}\tfrac{1}{2^{k+1}}+\tfrac{\varepsilon}{2}\tfrac{1}{2^k}=\tfrac{\varepsilon}{2}\tfrac{1}{2^k}\tfrac{3}{2}<\varepsilon\tfrac{1}{2^k},$$

so we can take a representation $\sum_{i=n_k+1}^{n_{k+1}}x_i\otimes y_i$ for $F_{k+1}-F_k\in\mathcal{X}\otimes_\wedge\mathcal{Y}$ for which $\sum_{i=n_k+1}^{n_{k+1}}\|x_i\|\,\|y_i\|$ is close enough to $\|F_{k+1}-F_k\|_\wedge$, say

$$\sum\nolimits_{i=n_k+1}^{n_{k+1}}\|x_i\|\,\|y_i\|\le\|F_{k+1}-F_k\|_\wedge+\tfrac{\varepsilon}{4}\tfrac{1}{2^k}<\tfrac{\varepsilon}{2}\tfrac{1}{2^k}\tfrac{3}{2}+\tfrac{\varepsilon}{2}\tfrac{1}{2^k}\tfrac{1}{2}=\varepsilon\tfrac{1}{2^k}.$$

Since $\|\cdot\|_\wedge$ is a reasonable crossnorm, we get for each $k>1$

$$\begin{aligned}\sum\nolimits_{i=1}^{n_{k+1}}\|x_i\otimes y_i\|_\wedge&=\sum\nolimits_{i=1}^{n_1}\|x_i\|\,\|y_i\|+\sum\nolimits_{j=1}^{k}\sum\nolimits_{i=n_j+1}^{n_{j+1}}\|x_i\|\,\|y_i\|\\&<\|F\|_\wedge+\varepsilon+\varepsilon\sum\nolimits_{j=1}^{k}\tfrac{1}{2^j}<\|F\|_\wedge+2\varepsilon.\end{aligned}$$

Hence the sequence $\{x_i\otimes y_i\}$ is absolutely summable with $\sum_i\|x_i\otimes y_i\|_\wedge<\|F\|_\wedge+2\varepsilon$. Then the sequence is summable because $\mathcal{X}\widehat{\otimes}_\wedge\mathcal{Y}$ is a Banach space. This means the sequence of partial sums $\{\sum_{i=1}^{n_k}x_i\otimes y_i\}$ converges in $\mathcal{X}\widehat{\otimes}_\wedge\mathcal{Y}$. Thus the limit $F\in\mathcal{X}\widehat{\otimes}_\wedge\mathcal{Y}$ of $F_k=\sum_{i=1}^{n_k}x_i\otimes y_i\in\mathcal{X}\otimes_\wedge\mathcal{Y}$ is represented as

$$F=\sum\nolimits_i x_i\otimes y_i,$$

and

$$\|F\|_\wedge\le\sum\nolimits_i\|x_i\otimes y_i\|_\wedge=\sum\nolimits_i\|x_i\|\,\|y_i\|\le\|F\|_\wedge+2\varepsilon,$$

so that the projective norm $\|F\|_\wedge$ of F is the infimum of $\sum_i\|x_i\|\,\|y_i\|$ over all representations of F of the form $\sum_i x_i\otimes y_i$. □

This is the classical statement of Theorem 7.9. Do we need that $\mathcal{X}$ and $\mathcal{Y}$ be complete? And $\mathcal{X}$ and $\mathcal{Z}$ in the next theorem? And $\mathcal{X}$ and $\mathcal{Y}$ in its corollary?

As for the next result, recall that (i) by a subspace of a normed space we mean a *closed* linear manifold of it, (ii) a normed space $\mathcal{Y}$ is a quotient of a normed space $\mathcal{X}$ if there is a quotient transformation Q of $\mathcal{X}$ onto $\mathcal{Y}$, (iii) the natural quotient map π of $\mathcal{X}$ onto the quotient space $\mathcal{X}/\mathcal{M}$ is a quotient transformation, and so $\mathcal{X}/\mathcal{M}$ is a quotient of $\mathcal{X}$, for every proper subspace $\mathcal{M}$ of $\mathcal{X}$.

The next theorem justifies the terminology *projective* for the norm $\|\cdot\|_\wedge$. It shows that when tensor products are equipped with the projective norm, then the fact that $\mathcal{X}/\mathcal{M}$ is a quotient of $\mathcal{X}$ is transferred (projected) to tensor products in the sense that $\mathcal{Z}\,\widehat{\otimes}_\wedge(\mathcal{X}/\mathcal{M})$ is a quotient of $\mathcal{Z}\,\widehat{\otimes}_\wedge\mathcal{X}$ for every $\mathcal{Z}$.

Theorem 7.10. *If $\mathcal{M}$ is a subspace of a Banach space $\mathcal{X}$, then $\mathcal{Z}\,\widehat{\otimes}_\wedge(\mathcal{X}/\mathcal{M})$ is a quotient of $\mathcal{Z}\,\widehat{\otimes}_\wedge\mathcal{X}$ for every Banach space $\mathcal{Z}$.*

Proof. Suppose all linear spaces are nonzero. Let $\mathcal{M}$ be a subspace of a Banach space $\mathcal{X}$ and take an arbitrary Banach space $\mathcal{Z}$. Consider the tensor product spaces $\mathcal{Z}\,\widehat{\otimes}_\wedge\mathcal{X}$ and $\mathcal{Z}\,\widehat{\otimes}_\wedge(\mathcal{X}/\mathcal{M})$. Take an arbitrary $\mathrm{G}\in\mathcal{Z}\,\widehat{\otimes}_\wedge(\mathcal{X}/\mathcal{M})$. Since $\mathcal{Z}$ and $\mathcal{X}/\mathcal{M}$ are Banach spaces, we may apply Theorem 7.9, and so there are $\mathcal{Z}$-valued and $\mathcal{X}/\mathcal{M}$-valued sequences $\{z_k\}$ and $\{[x_k]\}$ such that

$$\mathrm{G}=\sum\nolimits_k z_k\otimes[x_k]\quad\text{ with }\quad\|\mathrm{G}\|_{\mathcal{Z}\,\widehat{\otimes}_\wedge(\mathcal{X}/\mathcal{M})}=\inf\sum\nolimits_k\|z_k\|_{\mathcal{Z}}\big\|[x_k]\big\|_{\mathcal{X}/\mathcal{M}}.$$

Let $\pi\colon\mathcal{X}\to\mathcal{X}/\mathcal{M}$ be the natural quotient map, which is a linear surjective contraction (see Sections 1.3 and 5.3), and consider the transformation $Q=I\otimes\pi\colon\mathcal{Z}\,\widehat{\otimes}_\wedge\mathcal{X}\to\mathcal{Z}\,\widehat{\otimes}_\wedge(\mathcal{X}/\mathcal{M})$ where I is the identity on $\mathcal{Z}$, which is linear (see Section 3.3). It is surjective. In fact, if $F'=\sum_k z_k\otimes x'_k\in\mathcal{Z}\,\widehat{\otimes}_\wedge\mathcal{X}$ for any $\mathcal{X}$-valued sequence $\{x'_k\}$ with $x'_k\in[x_k]$, then $Q(F')=\mathrm{G}$ (Definition 3.16). Also, Q is a contraction. Indeed, an arbitrary $F\in\mathcal{Z}\,\widehat{\otimes}_\wedge\mathcal{X}$ can be represented as $F=\sum_k y_k\otimes v_k$ for some $\mathcal{Z}$-valued sequence $\{y_k\}$ and some $\mathcal{X}$-valued sequence $\{v_k\}$ by Theorem 7.9, so that (as $\big\|[v]\big\|_{\mathcal{X}/\mathcal{M}}\le\|v'\|_{\mathcal{X}}$ for every $v'\in[v]\subseteq\mathcal{X}$)

$$\|Q(F)\|_{\mathcal{Z}\,\widehat{\otimes}_\wedge(\mathcal{X}/\mathcal{M})}=\inf\sum\nolimits_k\|y_k\|_{\mathcal{Z}}\big\|[v_k]\big\|_{\mathcal{X}/\mathcal{M}}\le\inf\sum\nolimits_k\|y_k\|_{\mathcal{Z}}\|v_k\|_{\mathcal{X}}=\|F\|_{\mathcal{Z}\,\widehat{\otimes}_\wedge\mathcal{X}}.$$

Now take an arbitrary $\varepsilon>0$. Associated with ε there is a $\mathcal{Z}$-valued sequence $\{z_k\}$ with $\|z_k\|_{\mathcal{Z}}\le1$ and an $\mathcal{X}/\mathcal{M}$-valued sequence $\{[x_k]\}$ such that $\mathrm{G}=\sum_k z_k\otimes[x_k]$ with $\|\mathrm{G}\|_{\mathcal{Z}\,\widehat{\otimes}_\wedge(\mathcal{X}/\mathcal{M})}=\inf\sum_k\|z_k\|_{\mathcal{Z}}\big\|[x_k]\big\|_{\mathcal{X}/\mathcal{M}}$ and

$$\|\mathrm{G}\|_{\mathcal{Z}\,\widehat{\otimes}_\wedge(\mathcal{X}/\mathcal{M})}\le\sum\nolimits_k\|z_k\|_{\mathcal{Z}}\big\|[x_k]\big\|_{\mathcal{X}/\mathcal{M}}\le\|\mathrm{G}\|_{\mathcal{Z}\,\widehat{\otimes}_\wedge(\mathcal{X}/\mathcal{M})}+\varepsilon\tfrac{1}{2}.\qquad(*)$$

Since $[x_k]=x_k+\mathcal{M}$ and $\big\|[x_k]\big\|_{\mathcal{X}/\mathcal{M}}=\inf_{u\in\mathcal{M}}\|x_k+u\|_{\mathcal{X}}\le\|x_k\|_{\mathcal{X}}$, then for each k there exists $x''_k\in[x_k]$ (so that $[x_k]=[x''_k]$) for which

$$\big\|[x_k]\big\|_{\mathcal{X}/\mathcal{M}}\le\|x''_k\|_{\mathcal{X}}\le\big\|[x_k]\big\|_{\mathcal{X}/\mathcal{M}}+\varepsilon\tfrac{1}{2^{k+1}}.\qquad(**)$$

Thus with $F''=\sum_k z_k\otimes x''_k\in\mathcal{Z}\,\widehat{\otimes}_\wedge\mathcal{X}$ we get $\mathrm{G}=\sum_k z_k\otimes[x_k]=\sum_k z_k\otimes[x''_k]=Q(F'')$. Hence by $(*)$ and $(**)$, and recalling that $\|z_k\|_{\mathcal{Z}}\le1$,

$$\begin{aligned}\|\mathrm{G}\|_{\mathcal{Z}\,\widehat{\otimes}_\wedge(\mathcal{X}/\mathcal{M})}=\|Q(F'')\|_{\mathcal{Z}\,\widehat{\otimes}_\wedge(\mathcal{X}/\mathcal{M})}&\le\|F''\|_{\mathcal{Z}\,\widehat{\otimes}_\wedge\mathcal{X}}=\inf\sum\nolimits_k\|z_k\|_{\mathcal{Z}}\|x''_k\|_{\mathcal{X}}\\&\le\sum\nolimits_k\|z_k\|_{\mathcal{Z}}\|x''_k\|_{\mathcal{X}}\le\sum\nolimits_k\|z_k\|_{\mathcal{Z}}\big(\big\|[x_k]\big\|_{\mathcal{X}/\mathcal{M}}+\varepsilon\tfrac{1}{2^{k+1}}\big)\\&\le\sum\nolimits_k\|z_k\|_{\mathcal{Z}}\big\|[x_k]\big\|_{\mathcal{X}/\mathcal{M}}+\varepsilon\sum\nolimits_k\tfrac{1}{2^{k+1}}\\&\le\|\mathrm{G}\|_{\mathcal{Z}\,\widehat{\otimes}_\wedge(\mathcal{X}/\mathcal{M})}+\varepsilon\tfrac{1}{2}+\varepsilon\tfrac{1}{2}=\|\mathrm{G}\|_{\mathcal{Z}\,\widehat{\otimes}_\wedge(\mathcal{X}/\mathcal{M})}+\varepsilon.\end{aligned}$$

As this holds for every $\varepsilon > 0$, then for every $\mathrm{G} \in \mathcal{Z}\,\widehat{\otimes}_{\wedge}(\mathcal{X}/\mathcal{M})$ there is an $F = Q^{-1}(\mathrm{G}) \in \mathcal{Z}\,\widehat{\otimes}_{\wedge}\mathcal{X}$ such that $\|\mathrm{G}\|_{\mathcal{Z}\,\widehat{\otimes}_{\wedge}(\mathcal{X}/\mathcal{M})} = \|F\|_{\mathcal{Z}\,\widehat{\otimes}_{\wedge}\mathcal{X}}$. So the linear surjective contraction $Q\colon \mathcal{Z}\,\widehat{\otimes}_{\wedge}\mathcal{X} \to \mathcal{Z}\,\widehat{\otimes}_{\wedge}(\mathcal{X}/\mathcal{M})$ is a quotient transformation (Theorem 5.13(a)). Thus $\mathcal{Z}\,\widehat{\otimes}_{\wedge}(\mathcal{X}/\mathcal{M})$ is a quotient of $\mathcal{Z}\,\widehat{\otimes}_{\wedge}\mathcal{X}$ (Definition 5.12). □

Corollary 7.11. *If $\mathcal{M}$ and $\mathcal{N}$ are subspaces of Banach spaces $\mathcal{X}$ and $\mathcal{Y}$, then $(\mathcal{X}/\mathcal{M})\,\widehat{\otimes}_{\wedge}(\mathcal{Y}/\mathcal{N})$ is a quotient of $\mathcal{X}\,\widehat{\otimes}_{\wedge}\mathcal{Y}$.*

Proof. Symmetrically (by Theorem 7.10), $(\mathcal{X}/\mathcal{M})\,\widehat{\otimes}_{\wedge}\mathcal{Z}$ is a quotient of $\mathcal{X}\,\widehat{\otimes}_{\wedge}\mathcal{Z}$ for every Banach space $\mathcal{Z}$. Since a quotient space of a Banach space modulo a subspace is a Banach space (cf. Remark 5.9), and since composition of quotient transformations is again a quotient transformation (according to Theorem 5.13(d)), then by transitivity $(\mathcal{X}/\mathcal{M})\,\widehat{\otimes}_{\wedge}(\mathcal{Y}/\mathcal{N})$ is a quotient of $\mathcal{X}\,\widehat{\otimes}_{\wedge}\mathcal{Y}$. □

Theorem 3.7 played a central role in the theory of tensor products of linear spaces. Theorem 7.12 below is a counterpart of it, which holds when the tensor product space is equipped with a suitable reasonable crossnorm. This plays an equally crucial role in the theory of tensor products of Banach spaces, and the suitable reasonable crossnorm is precisely the projective norm.

The next result is often referred to as the *universal mapping property* of the projective norm, or the *universal mapping principle* of the projective tensor product. It says that a bilinear map $\phi\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{Z}$ is continuous if and only if its linearization $\Phi\colon \mathcal{X}\otimes_{\wedge}\mathcal{Y} \to \mathcal{Z}$ is. In its classical form it reads as follows.

Theorem 7.12. *Take an arbitrary triple $(\mathcal{X}, \mathcal{Y}, \mathcal{Z})$ of Banach spaces. The Banach spaces $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ and $\mathcal{B}[\mathcal{X}\,\widehat{\otimes}_{\wedge}\mathcal{Y}, \mathcal{Z}]$ are isometrically isomorphic:*

$$\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] \cong \mathcal{B}[\mathcal{X}\,\widehat{\otimes}_{\wedge}\mathcal{Y}, \mathcal{Z}].$$

Proof. Take the tensor product space $\mathcal{X}\otimes\mathcal{Y}$ and the natural bilinear map $\theta \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{X}\otimes\mathcal{Y}]$ associated with it. Theorem 3.7 ensures the existence of a linear-space isomorphism $C_\theta\colon \mathcal{L}[\mathcal{X}\otimes\mathcal{Y}, \mathcal{Z}] \to b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ between the linear spaces $\mathcal{L}[\mathcal{X}\otimes\mathcal{Y}, \mathcal{Z}]$ and $b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ defined as the composition with θ:

$$C_\theta(\Phi) = \Phi\circ\theta \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] \quad \text{for every} \quad \Phi \in \mathcal{L}[\mathcal{X}\otimes\mathcal{Y}, \mathcal{Z}].$$

Let $\mathcal{X}$ and $\mathcal{Y}$ be Banach spaces and equip the linear space $\mathcal{X}\otimes\mathcal{Y}$ with the projective norm to get the normed space $\mathcal{X}\otimes_{\wedge}\mathcal{Y}$. Suppose $\mathcal{Z}$ is a Banach space and consider the Banach space $\mathcal{B}[\mathcal{X}\otimes_{\wedge}\mathcal{Y}, \mathcal{Z}]$, which is a linear manifold of $\mathcal{L}[\mathcal{X}\otimes_{\wedge}\mathcal{Y}, \mathcal{Z}]$. Let $\mathcal{J}$ be the restriction to $\mathcal{B}[\mathcal{X}\otimes_{\wedge}\mathcal{Y}, \mathcal{Z}]$ of the linear-space isomorphism C_θ on $\mathcal{L}[\mathcal{X}\otimes_{\wedge}\mathcal{Y}, \mathcal{Z}]$, which remains linear and injective,

$$\mathcal{J} = C_\theta|_{\mathcal{B}[\mathcal{X}\otimes_{\wedge}\mathcal{Y}, \mathcal{Z}]}\colon \mathcal{B}[\mathcal{X}\otimes_{\wedge}\mathcal{Y}, \mathcal{Z}] \to \mathcal{R}(\mathcal{J}) \subseteq b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}].$$

We show that (a) $\mathcal{R}(\mathcal{J}) = \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ and (b) $\mathcal{J}$ is an isometry onto $\mathcal{R}(\mathcal{J})$. Let $\|\cdot\|_{\mathcal{B}}$ and $\|\cdot\|_{\boldsymbol{b}}$ stand for the norms in $\mathcal{B}[\mathcal{X}\otimes_{\wedge}\mathcal{Y}, \mathcal{Z}]$ and $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$.

(a_1) If $\Phi \in \mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}]$, then $\mathcal{J}(\Phi) = C_\theta(\Phi) = \Phi \circ \theta \in b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ is such that

$$\phi(x,y) = \mathcal{J}(\Phi)(x,y) = (\Phi \circ \theta)(x,y) = \Phi(x \otimes y) \quad \text{for every} \quad (x,y) \in \mathcal{X}\times\mathcal{Y},$$

and hence

$$\|\mathcal{J}(\Phi)(x,y)\|_{\mathcal{Z}} = \|\Phi(x \otimes y)\|_{\mathcal{Z}} \le \|\Phi\|_{\mathcal{B}} \|x \otimes y\|_\wedge = \|\Phi\|_{\mathcal{B}} \|x\|_{\mathcal{X}} \|y\|_{\mathcal{Y}}. \qquad (*)$$

Then the bilinear map $\mathcal{J}(\Phi)$ is bounded. Thus $\mathcal{J}(\Phi) \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ for every Φ in $\mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}]$, and so $\mathcal{R}(\mathcal{J}) \subseteq \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$.

(a_2) Conversely, if $\phi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$, then there is a unique $\Phi \in \mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}]$ such that $C_\theta(\Phi) = \Phi \circ \theta = \phi$. So

$$\|\Phi(x \otimes y)\|_{\mathcal{Z}} = \|\Phi \circ \theta(x,y)\|_{\mathcal{Z}} = \|\phi(x,y)\|_{\mathcal{Z}} \le \|\phi\|_{\boldsymbol{b}} \|x\|_{\mathcal{X}} \|y\|_{\mathcal{Y}}$$

for every $(x,y) \in \mathcal{X}\times\mathcal{Y}$. Hence for an arbitrary $F = \sum_i x_i \otimes y_i \in \mathcal{X} \otimes_\wedge \mathcal{Y}$,

$$\begin{aligned} \|\Phi(F)\|_{\mathcal{Z}} &= \Big\|\Phi\sum\nolimits_i x_i \otimes y_i\Big\|_{\mathcal{Z}} = \Big\|\sum\nolimits_i \Phi(x_i \otimes y_i)\Big\|_{\mathcal{Z}} \\ &\le \sum\nolimits_i \|\phi\|_{\boldsymbol{b}} \|x_i\|_{\mathcal{X}} \|y_i\|_{\mathcal{Y}} = \|\phi\|_{\boldsymbol{b}} \sum\nolimits_i \|x_i\|_{\mathcal{X}} \|y_i\|_{\mathcal{Y}}. \end{aligned}$$

As this holds for every (finite) representation of $F = \sum_i x_i \otimes y_i$, and since $\|F\|_\wedge = \inf \sum_i \|x_i\|_{\mathcal{X}} \|y_i\|_{\mathcal{Y}}$ over all representations, then for every $F \in \mathcal{X} \otimes_\wedge \mathcal{Y}$

$$\|\Phi(F)\|_{\mathcal{Z}} \le \|\phi\|_{\boldsymbol{b}} \|F\|_\wedge, \qquad (**)$$

and the linear Φ is bounded: $\Phi \in \mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}]$. So $\phi = C_\theta(\Phi) = \mathcal{J}(\Phi) \in \mathcal{R}(\mathcal{J})$. Since this holds for every $\phi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$, we get $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] \subseteq \mathcal{R}(\mathcal{J})$.

(a) By (a_1) and (a_2) we get $\mathcal{R}(\mathcal{J}) = \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$.

(b) Take an arbitrary $\Phi \in \mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}]$. As we saw above in $(*)$,

$$\|\mathcal{J}(\Phi)(x,y)\|_{\mathcal{Z}} \le \|\Phi\|_{\mathcal{B}} \|x\|_{\mathcal{X}} \|y\|_{\mathcal{Y}}$$

for every $(x,y) \in \mathcal{X}\times\mathcal{Y}$. Then the bilinear map $\mathcal{J}(\Phi)$ is bounded with norm

$$\|\mathcal{J}(\Phi)\|_{\boldsymbol{b}} \le \|\Phi\|_{\mathcal{B}}.$$

Conversely, the unique $\phi = C_\theta(\Phi) = \mathcal{J}(\Phi)$ is such that $\phi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ because $\mathcal{R}(\mathcal{J}) \subseteq \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ by (a_1) and hence, as we saw above in $(**)$,

$$\|\Phi(F)\|_{\mathcal{Z}} \le \|\phi\|_{\boldsymbol{b}} \|F\|_\wedge = \|\mathcal{J}(\Phi)\|_{\boldsymbol{b}} \|F\|_\wedge$$

for every $F \in \mathcal{X} \otimes_\wedge \mathcal{Y}$. Therefore,

$$\|\Phi\|_{\mathcal{B}} \le \|\phi\|_{\boldsymbol{b}} = \|\mathcal{J}(\Phi)\|_{\boldsymbol{b}}.$$

Thus $\|\mathcal{J}(\Phi)\|_{\boldsymbol{b}} = \|\Phi\|_{\mathcal{B}}$ for every $\Phi \in \mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}]$. That is, $\mathcal{J}$ is an isometry.

(c) Then the linear transformation $\mathcal{J}\colon \mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}] \to \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ is a surjective isometry by (a) and (b), which means $\mathcal{J}$ is an isometric isomorphism. So

$$\mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}] \cong \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}].$$

Consider the completion $\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}$ of $\mathcal{X} \otimes_\wedge \mathcal{Y}$. According to Theorem 5.1(e),

$$\mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}, \mathcal{Z}] \cong \mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}]$$

because $\mathcal{Z}$ is a Banach space. So the stated result follows by transitivity. □

Do we need that $\mathcal{X}$ and $\mathcal{Y}$ in Theorem 7.12 be Banach spaces?

Remark 7.13. *Dualities and the Projective Universal Mapping Principle.*

(a) Let $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ be arbitrary linear spaces (over the same field $\mathbb{F}$) and let $\cong$ denote linear-space isomorphism. From a purely algebraic point of view, Theorem 2.6, Remark 2.7(b) and Theorem 3.7 ensure that

$$\mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{Z}] \cong b[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] \cong \mathcal{L}[\mathcal{X}, \mathcal{L}[\mathcal{Y}, \mathcal{Z}]] \cong \mathcal{L}[\mathcal{Y}, \mathcal{L}[\mathcal{X}, \mathcal{Z}]].$$

(b) In particular, if $\mathcal{Z} = \mathbb{F}$ and $^\sharp$ stands for algebraic dual, and recalling Remarks 3.8(b) and 2.7(b) again,

$$\mathcal{X}^\sharp \otimes \mathcal{Y}^\sharp \subseteq (\mathcal{X} \otimes \mathcal{Y})^\sharp \cong b[\mathcal{X}\times\mathcal{Y}, \mathbb{F}] \cong \mathcal{L}[\mathcal{X}, \mathcal{Y}^\sharp] \cong \mathcal{L}[\mathcal{Y}, \mathcal{X}^\sharp].$$

(c) If $\mathcal{X}$, $\mathcal{Y}$, $\mathcal{Z}$ are Banach spaces and $\mathcal{X} \otimes \mathcal{Y}$ is equipped with the projective norm, then $\mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}, \mathcal{Z}]$ is naturally isometrically isomorphic to $\mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}]$, which in turn is isometrically isomorphic to $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ by Theorem 7.12. Thus using Theorem 6.3 (with the help of Remark 6.4(a)), where $\cong$ now means isometric isomorphism, we also get

$$\mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}, \mathcal{Z}] \cong \mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}] \cong \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}] \cong \mathcal{B}[\mathcal{X}, \mathcal{B}[\mathcal{Y}, \mathcal{Z}]] \cong \mathcal{B}[\mathcal{Y}, \mathcal{B}[\mathcal{X}, \mathcal{Z}]].$$

(d) Now set $\mathcal{Z} = \mathbb{F}$ and replace $^\sharp$ by the normed-space dual *. By Definition 7.1, $\mathcal{X}^* \otimes \mathcal{Y}^* \subseteq (\mathcal{X} \otimes_\alpha \mathcal{Y})^*$ for any reasonable crossnorm $\|\cdot\|_\alpha$ on $\mathcal{X} \otimes \mathcal{Y}$. By Theorem 7.3, the induced uniform norm $\|\cdot\|_{*\alpha}$ on $(\mathcal{X} \otimes_\alpha \mathcal{Y})^*$ is a reasonable crossnorm on $\mathcal{X}^* \otimes \mathcal{Y}^*$ when restricted to it. Equip it with this norm and get $\mathcal{X}^* \otimes_{*\alpha} \mathcal{Y}^*$. Thus $\mathcal{X}^* \otimes_{*\alpha} \mathcal{Y}^* \subseteq (\mathcal{X} \otimes_\alpha \mathcal{Y})^*$. (Note: $\mathcal{X}^* \otimes_{*\alpha} \mathcal{Y}^*$ is a linear manifold of the normed space $(\mathcal{X} \otimes_\alpha \mathcal{Y})^*$ — its norm is inherited from $(\mathcal{X} \otimes_\alpha \mathcal{Y})^*$.) Thus Theorem 5.1(g) ensures that $\mathcal{X}^* \mathbin{\widehat{\otimes}_{*\alpha}} \mathcal{Y}^* \subseteq (\mathcal{X} \otimes_\alpha \mathcal{Y})^*$, which is its own natural completion since dual spaces are complete. Moreover, Theorem 5.1(f) says that $(\mathcal{X} \otimes_\alpha \mathcal{Y})^* \cong (\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y})^*$ and $\mathcal{X}^* \otimes_{*\alpha} \mathcal{Y}^* \cong \mathcal{X}^* \mathbin{\widehat{\otimes}_{*\alpha}} \mathcal{Y}^*$. Hence

$$\mathcal{X}^* \mathbin{\widehat{\otimes}_{*\alpha}} \mathcal{Y}^* \cong \mathcal{X}^* \otimes_{*\alpha} \mathcal{Y}^* \subseteq (\mathcal{X} \otimes_\alpha \mathcal{Y})^* \cong (\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y})^*$$

for every reasonable crossnorm $\|\cdot\|_\alpha$ on $\mathcal{X} \otimes \mathcal{Y}$. Also, by Remark 6.4(a),

$$\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}] \cong \mathcal{B}[\mathcal{X}, \mathcal{Y}^*] \cong \mathcal{B}[\mathcal{Y}, \mathcal{X}^*].$$

(e) By setting $\mathcal{Z} = \mathbb{F}$ in Theorem 7.12 we get $(\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y})^* \cong \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$. Thus

$$\mathcal{X}^* \mathbin{\widehat{\otimes}_{*\wedge}} \mathcal{Y}^* \cong \mathcal{X}^* \otimes_{*\wedge} \mathcal{Y}^* \subseteq (\mathcal{X} \otimes_\wedge \mathcal{Y})^* \cong (\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y})^* \cong \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}] \cong \mathcal{B}[\mathcal{X}, \mathcal{Y}^*] \cong \mathcal{B}[\mathcal{Y}, \mathcal{X}^*]$$

for the *projective* norm $\|\cdot\|_\wedge$ (compare with Remark 4.9(e,f,g,h)). It is worth noticing that although the above equivalences (i.e., isometric isomorphisms) give interpretations for the dual $(\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y})^*$ of the projective tensor product, the above inclusion (embedding) does not supply any interpretation for the projective tensor product $\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}$ itself (but it gives interpretations for the tensor product $\mathcal{X}^* \mathbin{\widehat{\otimes}_{*\wedge}} \mathcal{Y}^*$ under the reasonable crossnorm $\|\cdot\|_{*\wedge}$).

(f) The proof of Theorem 7.12 for $\mathcal{Z} = \mathbb{F}$ shows that $(\mathcal{X} \otimes_\wedge \mathcal{Y})^* \cong (\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y})^* \cong \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$, where an arbitrary isometrically isomorphic pair $\Lambda \in (\mathcal{X} \otimes_\wedge \mathcal{Y})^* \cong \phi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$ is related by $\Lambda(x \otimes y) = \phi(x, y)$ for every $(x, y) \in \mathcal{X}\times\mathcal{Y}$. Then

$$\Lambda(F) = \sum\nolimits_i \Lambda(x_i \otimes y_i) = \sum\nolimits_i \phi(x_i, y_i) \quad \text{for every} \quad F = \sum\nolimits_i x_i \otimes y_i \in \mathcal{X} \otimes_\wedge \mathcal{Y},$$

as Λ is linear. (Note: the value $\Lambda(F)$ of the functional Λ at a point F does not depend on the representation of F — cf. Remark 3.18(a).) As Λ is bounded and $\cong$ stands for isometric isomorphism, $\|\Lambda\|_{*\wedge} = \|\phi\|$. This yields another expression for the projective norm of $F = \sum_i x_i \otimes y_i \in \mathcal{X} \otimes_\wedge \mathcal{Y}$ (cf. Remark 5.2(d)),

$$\|F\|_\wedge = \sup_{\|\Lambda\|_{*\wedge} \le 1,\, \Lambda \in (\mathcal{X} \otimes_\wedge \mathcal{Y})^*} |\Lambda(F)| = \sup_{\|\phi\| \le 1,\, \phi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]} \Big|\sum\nolimits_i \phi(x_i, y_i)\Big|.$$

(g) On the other hand, by Definition 7.1 the linear space $\mathcal{X}^* \otimes \mathcal{Y}^*$ is a linear manifold of the linear space $(\mathcal{X} \otimes_\alpha \mathcal{Y})^* = \mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathbb{F}]$, which is isometrically isomorphic to $\mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}, \mathbb{F}] = (\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y})^*$ (by Theorem 5.1(e)) for any reasonable crossnorm. Then the linear space $\mathcal{X}^* \otimes_\wedge \mathcal{Y}^*$ (now equipped with projective norm) remains algebraically embedded in the Banach space $(\mathcal{X} \otimes_\wedge \mathcal{Y})^*$,

$$\mathcal{X}^* \otimes_\wedge \mathcal{Y}^* \subseteq (\mathcal{X} \otimes_\wedge \mathcal{Y})^*,$$

and so its completion $\mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}^*$ remains algebraically embedded in $(\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y})^*$,

$$\mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}^* \subseteq (\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y})^*,$$

where, differently from items (d) and (e) above, $\subseteq$ now means linear-space embedding but not normed-space embedding as the norms are different: $\|\cdot\|_\wedge$ equips $\mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}^*$ and $\|\cdot\|_{*\wedge}$ equips $(\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y})^* = \mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}, \mathbb{F}]$. Next take an arbitrary $\Omega \in \mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}^* \subseteq (\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y})^*$. Theorem 7.9 ensures that there are $\mathcal{X}^*$-valued and $\mathcal{Y}^*$-valued sequences $\{f_j\}$ and $\{g_j\}$ with $\sum_j \|f_j\|\,\|g_j\| < \infty$ such that

$$\Omega = \sum\nolimits_j f_j \otimes g_j \quad \text{and} \quad \|\Omega\|_\wedge = \inf \sum\nolimits_j \|f_j\|\,\|g_j\|,$$

where the infimum is taken over all representations $\sum_j f_j \otimes g_j$ of Ω. Hence

$$\begin{aligned} |\Omega(F)| &= \Big|\sum\nolimits_j f_j \otimes g_j \Big(\sum\nolimits_i x_i \otimes y_i\Big)\Big| = \Big|\sum\nolimits_j \sum\nolimits_i f_j(x_i)\, g_j(y_i)\Big| \\ &\le \sum\nolimits_j \sum\nolimits_i \|f_j\|\,\|g_j\|\,\|x_i\|\,\|y_i\| \le \Big(\sum\nolimits_j \|f_j\|\,\|g_j\|\Big)\Big(\sum\nolimits_i \|x_i\|\,\|y_i\|\Big) \end{aligned}$$

for every $F = \sum_i x_i \otimes y_i \in \mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}$ with $\|F\|_\wedge = \inf \sum_i \|x_i\| \, \|y_i\|$ (Theorem 7.9). So $|\Omega(F)| \le \|\Omega\|_\wedge \|F\|_\wedge$. Then $\|\Omega\|_{*\wedge} = \sup_{\|F\|_\wedge \le 1} |\Omega(F)| \le \|\Omega\|_\wedge$. Thus

$$\|\Omega\|_{*\wedge} \le \|\Omega\|_\wedge \quad \text{for every} \quad \Omega \in \mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}^* \subseteq (\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y})^*.$$

(h) Next take the isometric isomorphism $\mathcal{J} \colon (\mathcal{X} \otimes_\wedge \mathcal{Y})^* \to \boldsymbol{b}[\mathcal{X} \times \mathcal{Y}, \mathbb{F}]$ in the proof of Theorem 7.12 with $\mathcal{Z} = \mathbb{F}$, where the value of $\mathcal{J}$ at an arbitrary single tensor $f \otimes g \in \mathcal{X}^* \otimes_\wedge \mathcal{Y}^* \subseteq (\mathcal{X} \otimes_\wedge \mathcal{Y})^*$ is such that

$$\mathcal{J}(f \otimes g) = \phi \in \boldsymbol{b}[\mathcal{X} \times \mathcal{Y}, \mathbb{F}] \quad \text{if and only if} \quad \phi(\cdot\,, \cdot) = f(\cdot)\, g(\cdot).$$

Indeed, if $f \otimes g$ lies in $\mathcal{X}^* \otimes \mathcal{Y}^* \subseteq (\mathcal{X} \otimes_\wedge \mathcal{Y})^* = \mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathbb{F}]$, then

$$\mathcal{J}(f \otimes g)(x, y) = \big((f \otimes g) \circ \theta\big)(x, y) = (f \otimes g)\theta(x, y) = (f \otimes g)(x \otimes y) = f(x) g(y)$$

for every $(x, y) \in \mathcal{X} \times \mathcal{Y}$ (cf. proof of Theorem 3.7 with $C_\theta = \mathcal{J}$ and $\Phi = f \otimes g$, and see also Definition 3.16). Conversely, if $\phi(\cdot\,, \cdot) = f(\cdot)\, g(\cdot)$ lies in $\boldsymbol{b}[\mathcal{X} \times \mathcal{Y}, \mathbb{F}]$, then $\mathcal{J}^{-1}(\phi) = f \otimes g$. Therefore, for an arbitrary $\Omega = \sum_j f_j \otimes g_j \in \mathcal{X}^* \otimes \mathcal{Y}^*$,

$$\mathcal{J}(\Omega) = \phi \in \boldsymbol{b}[\mathcal{X} \times \mathcal{Y}, \mathbb{F}] \quad \text{if and only if} \quad \phi(\cdot\,, \cdot) = \sum\nolimits_j f_j(\cdot)\, g_j(\cdot),$$

since $\mathcal{J}(\Omega) = \sum_j \mathcal{J}(f_j \otimes g_j)$ because $\mathcal{J}$ is linear. This characterizes the image of $\mathcal{X}^* \otimes_\wedge \mathcal{Y}^*$ under the isometric isomorphism $\mathcal{J} \colon (\mathcal{X} \otimes_\wedge \mathcal{Y})^* \to \boldsymbol{b}[\mathcal{X} \times \mathcal{Y}, \mathbb{F}]$:

$$\mathcal{J}(\mathcal{X}^* \otimes_\wedge \mathcal{Y}^*) = \big\{\phi \in \boldsymbol{b}[\mathcal{X} \times \mathcal{Y}, \mathbb{F}] \colon \phi(\cdot, \cdot) = \textstyle\sum_j f_j(\cdot)\, g_j(\cdot) \text{ for every finite sum with each } f_j \text{ in } \mathcal{X}^* \text{ and each } g_j \text{ in } \mathcal{Y}^*\big\}.$$

So (using the same notation twice) with $\Omega \in \mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}^* \subseteq (\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y})^*$ and for the isometric isomorphism $\mathcal{J} \colon (\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y})^* \to \boldsymbol{b}[\mathcal{X} \times \mathcal{Y}, \mathbb{F}]$ we get, by Theorem 7.9,

$$\mathcal{J}(\mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}^*) = \big\{\phi \in \boldsymbol{b}[\mathcal{X} \times \mathcal{Y}, \mathbb{F}] \colon \phi(\cdot, \cdot) = \textstyle\sum_j f_j(\cdot)\, g_j(\cdot) \text{ for every countable sum with } f_j \text{ in } \mathcal{X}^* \text{ and } g_j \text{ in } \mathcal{Y}^* \text{ such that } \sum_j \|f_j\| \, \|g_j\| < \infty\big\}.$$

Corollary 7.14. *Let $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ be arbitrary Banach spaces, let $\mathcal{M}$ and $\mathcal{N}$ be subspaces of $\mathcal{X}$ and $\mathcal{Y}$, respectively, and consider the following assertions.*

(o) *$\mathcal{M}$ and $\mathcal{N}$ are complemented in $\mathcal{X}$ and $\mathcal{Y}$ with $\mathcal{M} = \mathcal{R}(E)$ and $\mathcal{N} = \mathcal{R}(P)$ for projections $E \in \mathcal{B}[\mathcal{X}]$ and $P \in \mathcal{B}[\mathcal{Y}]$ such that $\|E\| = \|P\| = 1$.*

(a) *$\mathcal{M} \otimes_\wedge \mathcal{N}$ is a linear manifold of $\mathcal{X} \otimes_\wedge \mathcal{Y}$.*

(b) *Every bounded bilinear map $\phi \colon \mathcal{M} \times \mathcal{N} \to \mathcal{Z}$ has a bounded bilinear extension $\widehat{\phi} \colon \mathcal{X} \times \mathcal{Y} \to \mathcal{Z}$ with $\|\widehat{\phi}\| = \|\phi\|$.*

(c) *Every bounded linear transformation $T \colon \mathcal{M} \otimes_\wedge \mathcal{N} \to \mathcal{Z}$ has a bounded linear extension $\widetilde{T} \colon \mathcal{X} \otimes_\wedge \mathcal{Y} \to \mathcal{Z}$ with $\|\widetilde{T}\| = \|T\|$.*

(d) *Every bounded linear transformation $\overline{T} \colon \mathcal{M} \mathbin{\widehat{\otimes}_\wedge} \mathcal{N} \to \mathcal{Z}$ has a bounded linear extension $\widehat{T} \colon \mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y} \to \mathcal{Z}$ with $\|\widehat{T}\| = \|\overline{T}\|$.*

The above assertions are related as follows:

$$\text{(o)} \Longrightarrow \text{(a, b)}, \qquad \text{(a, b)} \iff \text{(c)} \iff \text{(d)}.$$

Proof. Let $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ be arbitrary Banach spaces.

(o) $\Longrightarrow$ (a,b). If (o) holds, then according to Theorems 6.8 and 7.8(d)

(a) $\mathcal{M} \otimes_\wedge \mathcal{N}$ is a linear manifold of $\mathcal{X} \otimes_\wedge \mathcal{Y}$, and

(b) every $\phi \in \boldsymbol{b}[\mathcal{M}\times\mathcal{N}, \mathcal{Z}]$ into an arbitrary Banach space $\mathcal{Z}$ has an extension $\widehat{\phi} \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ with $\|\widehat{\phi}\| = \|\phi\|$.

Now we proceed to show that (a,b), (c), and (d) are pairwise equivalent. Note that each extension in (c) or (d) only makes sense if (a) holds (since the linear transformation $\widetilde{T}|_{\mathcal{M} \otimes_\wedge \mathcal{N}}$ is a restriction to linear manifold, and so is $\widehat{T}|_{\mathcal{M} \,\widehat{\otimes}_\wedge \mathcal{N}}$ which acts on the completion of a linear manifold of $\mathcal{X} \otimes_\wedge \mathcal{Y}$). Let $\cong$ stand for isometrically isomorphic. As we saw in the proof of Theorem 7.12,

$$\mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}] \cong \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}].$$

The natural isometric isomorphism between them,

$$\mathcal{J} : \mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}] \to \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$$

such that $\psi = \mathcal{J}(S)$ for $S \in \mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}]$ and $\psi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$, is given by

$$\psi(x, y) = \mathcal{J}(S)(x, y) = S(x \otimes y) \quad \text{and so} \quad S(x \otimes y) = \mathcal{J}^{-1}(\psi)(x \otimes y) = \psi(x, y)$$

for every $(x, y) \in \mathcal{X}\times\mathcal{Y}$. By Theorem 6.8(a), $\mathcal{M}$ and $\mathcal{N}$ are complemented subspaces of $\mathcal{X}$ and $\mathcal{Y}$ if and only if every ϕ in $\boldsymbol{b}[\mathcal{M}\times\mathcal{N}, \mathcal{Z}]$ into an arbitrary Banach space $\mathcal{Z}$ has an extension $\widehat{\phi}$ in $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$. So in this case, with T in $\mathcal{B}[\mathcal{M} \otimes_\wedge \mathcal{N}, \mathcal{Z}]$ and $\widetilde{T}$ in $\mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}]$ being the isometrically isomorphic images of ϕ in $\boldsymbol{b}[\mathcal{M}\times\mathcal{N}, \mathcal{Z}]$ and $\widehat{\phi}$ in $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$, respectively, we get

$$T \cong \phi = \widehat{\phi}|_{\mathcal{M}\times\mathcal{N}} \quad \text{and} \quad \widehat{\phi} \cong \widetilde{T}.$$

Since the restriction $\widetilde{T}|_{\mathcal{M} \otimes_\wedge \mathcal{N}} \in \mathcal{B}[\mathcal{M} \otimes_\wedge \mathcal{N}, \mathcal{Z}]$ of $\widetilde{T} \in \mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}]$ to $\mathcal{M} \otimes_\wedge \mathcal{N}$ only makes sense if (a) holds, then in this case with $\phi = \mathcal{J}(T)$ and $\widehat{\phi} = \mathcal{J}(\widetilde{T})$,

$$\begin{aligned} T(u \otimes v) &= \mathcal{J}^{-1}(\phi)(u \otimes v) = \phi(u, v) = \widehat{\phi}|_{\mathcal{M}\times\mathcal{N}}(u, v) = \widehat{\phi}(u, v) = \mathcal{J}(\widetilde{T})(u, v) \\ &= \widetilde{T}(u \otimes v) = \widetilde{T}|_{\mathcal{M} \otimes_\wedge \mathcal{N}}(u \otimes v) \quad \text{for every} \quad (u \otimes v) \in \mathcal{M} \otimes_\wedge \mathcal{N}. \end{aligned}$$

Hence $T(F) = \widetilde{T}|_{\mathcal{M} \otimes_\wedge \mathcal{N}}(F)$ for every $F = \sum_i u_i \otimes v_i \in \mathcal{M} \otimes_\wedge \mathcal{N}$. Therefore

$$\widehat{\phi}|_{\mathcal{M}\times\mathcal{N}} = \phi \quad \text{implies} \quad \widetilde{T}|_{\mathcal{M} \otimes_\wedge \mathcal{N}} = T.$$

Thus *if every* $\phi \in \boldsymbol{b}[\mathcal{M}\times\mathcal{N}, \mathcal{Z}]$ *has an extension* $\widehat{\phi} \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathcal{Z}]$ *and* $\mathcal{M} \otimes_\wedge \mathcal{N}$ *is a linear manifold of* $\mathcal{X} \otimes_\wedge \mathcal{Y}$, then every $T \in \mathcal{B}[\mathcal{M} \otimes_\wedge \mathcal{N}, \mathcal{Z}]$ *has an extension* $\widetilde{T} \in \mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}]$.

A symmetric argument shows the converse. Suppose every $T \in \mathcal{B}[\mathcal{M} \otimes_\wedge \mathcal{N}, \mathcal{Z}]$ has an extension $\widetilde{T} \in \mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}]$. Then by the above isometric isomorphisms

$$\phi \cong T = \widetilde{T}|_{\mathcal{M} \otimes_\wedge \mathcal{N}} \quad \text{and} \quad \widetilde{T} \cong \widehat{\phi}$$

whenever $\mathcal{M} \otimes_\wedge \mathcal{N}$ is a linear manifold of $\mathcal{X} \otimes_\wedge \mathcal{Y}$. Hence

$$\begin{aligned}\phi(u,v) &= \mathcal{J}(T)(u,v) = T(u \otimes v) = \widetilde{T}|_{\mathcal{M} \otimes_\wedge \mathcal{N}}(u \otimes v) \\ &= \widetilde{T}(u \otimes v) = \mathcal{J}^{-1}(\widehat{\phi})(u \otimes v) = \widehat{\phi}(u,v) = \widehat{\phi}|_{\mathcal{M} \times \mathcal{N}}(u,v)\end{aligned}$$

for every $(u,v) \in \mathcal{M} \times \mathcal{N}$. Therefore

$$\widetilde{T}|_{\mathcal{M} \otimes_\wedge \mathcal{N}} = T \quad \text{implies} \quad \widehat{\phi}|_{\mathcal{M} \times \mathcal{N}} = \phi.$$

Thus *if every* $T \in \mathcal{B}[\mathcal{M} \otimes_\wedge \mathcal{N}, \mathcal{Z}]$ *has an extension* $\widetilde{T} \in \mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}]$, *then every* $\phi \in \boldsymbol{b}[\mathcal{M} \times \mathcal{N}, \mathcal{Z}]$ *has an extension* $\widehat{\phi} \in \boldsymbol{b}[\mathcal{X} \times \mathcal{Y}, \mathcal{Z}]$ *and also* $\mathcal{M} \otimes_\wedge \mathcal{N}$ *is a linear manifold of* $\mathcal{X} \otimes_\wedge \mathcal{Y}$.

Next assume one of the extensions in either (c) or (d) holds, which implies (a). Take the completions $\mathcal{M} \widehat{\otimes}_\wedge \mathcal{N}$ of $\mathcal{M} \otimes_\wedge \mathcal{N}$ and $\mathcal{X} \widehat{\otimes}_\wedge \mathcal{Y}$ of $\mathcal{X} \otimes_\wedge \mathcal{Y}$. By Theorem 5.1(g), $\mathcal{M} \widehat{\otimes}_\wedge \mathcal{N}$ is a subspace of $\mathcal{X} \widehat{\otimes}_\wedge \mathcal{Y}$. By Theorem 5.1(e), as $\mathcal{Z}$ is a Banach space, $\mathcal{B}[\mathcal{M} \otimes_\wedge \mathcal{N}, \mathcal{Z}]$ is isometrically isomorphic to $\mathcal{B}[\mathcal{M} \widehat{\otimes}_\wedge \mathcal{N}, \mathcal{Z}]$ and $\mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}]$ is isometrically isomorphic to $\mathcal{B}[\mathcal{X} \widehat{\otimes}_\wedge \mathcal{Y}, \mathcal{Z}]$:

$$\mathcal{B}[\mathcal{M} \widehat{\otimes}_\wedge \mathcal{N}, \mathcal{Z}] \cong \mathcal{B}[\mathcal{M} \otimes_\wedge \mathcal{N}, \mathcal{Z}] \quad \text{and} \quad \mathcal{B}[\mathcal{X} \widehat{\otimes}_\wedge \mathcal{Y}, \mathcal{Z}] \cong \mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}].$$

Proceeding as before, let $\overline{T}$ in $\mathcal{B}[\mathcal{M} \widehat{\otimes}_\wedge \mathcal{N}, \mathcal{Z}]$ be the extension over completion of T in $\mathcal{B}[\mathcal{M} \otimes_\wedge \mathcal{N}, \mathcal{Z}]$ so that $\overline{T} \cong T$, and let $\widehat{T}$ in $\mathcal{B}[\mathcal{X} \widehat{\otimes}_\wedge \mathcal{Y}, \mathcal{Z}]$ be the extension over completion of $\widetilde{T}$ in $\mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}]$ so that $\widehat{T} \cong \widetilde{T}$. Extensions over completion are unique up to isometric isomorphism (cf. Theorem 5.1(c)). Hence $\widehat{T}$ in $\mathcal{B}[\mathcal{X} \widehat{\otimes}_\wedge \mathcal{Y}, \mathcal{Z}]$ extends $\overline{T}$ in $\mathcal{B}[\mathcal{M} \widehat{\otimes}_\wedge \mathcal{N}, \mathcal{Z}]$ if and only if $\widetilde{T}$ in $\mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}]$ extends T in $\mathcal{B}[\mathcal{M} \otimes_\wedge \mathcal{N}, \mathcal{Z}]$:

$$\widetilde{T}|_{\mathcal{M} \otimes_\wedge \mathcal{N}} \cong \widehat{T}|_{\mathcal{M} \widehat{\otimes}_\wedge \mathcal{N}} = \overline{T} \cong T \quad \text{and} \quad \widehat{T}|_{\mathcal{M} \widehat{\otimes}_\wedge \mathcal{N}} \cong \widetilde{T}|_{\mathcal{M} \otimes_\wedge \mathcal{N}} = T \cong \overline{T}.$$

Thus *every* $\overline{T} \in \mathcal{B}[\mathcal{M} \widehat{\otimes}_\wedge \mathcal{N}, \mathcal{Z}]$ *has an extension* $\widehat{T} \in \mathcal{B}[\mathcal{X} \widehat{\otimes}_\wedge \mathcal{Y}, \mathcal{Z}]$ *if and only if every* $T \in \mathcal{B}[\mathcal{M} \otimes_\wedge \mathcal{N}, \mathcal{Z}]$ *has an extension* $\widetilde{T} \in \mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{Z}]$.

Finally, since $\widehat{\phi} \cong \widetilde{T}$ and $\phi \cong T$, then $\|\widehat{\phi}\| = \|\phi\|$ if and only if $\|\widetilde{T}\| = \|T\|$. Similarly, since $\widetilde{T} \cong \widehat{T}$ and $T \cong \overline{T}$, then $\|T\| = \|\widetilde{T}\|$ if and only if $\|\widehat{T}\| = \|\overline{T}\|$. Since the equivalence between the extensions in (b) under assumption (a) and (c), and the equivalence between the extensions in (c) and (d) had been verified before, then (a,b) $\Longleftrightarrow$ (c) $\Longleftrightarrow$ (d). □

Corollary 7.14 is an application of Theorems 7.8 and 7.12 and, especially, an application of Theorem 6.8 on extension of bilinear maps. A very special particular case of Corollary 7.14 with a rather simplified statement is obtained by fixing $\mathcal{Z} = \mathbb{F}$. What is special in this particular case is this: it allows us to use the Hahn–Banach Theorem for the bounded linear functional in (c) whenever $\mathcal{M} \otimes_\wedge \mathcal{N}$ is a linear manifold of $\mathcal{X} \otimes_\wedge \mathcal{Y}$.

Corollary 7.15. *If $\mathcal{M}$ and $\mathcal{N}$ are subspaces of Banach spaces $\mathcal{X}$ and $\mathcal{Y}$, then the following assertions are pairwise equivalent.*

(a) *$\mathcal{M} \otimes_\wedge \mathcal{N}$ is a linear manifold of $\mathcal{X} \otimes_\wedge \mathcal{Y}$.*

(b) *Every bounded bilinear form $\phi\colon \mathcal{M}\times\mathcal{N} \to \mathbb{F}$ has a bounded bilinear extension $\widehat{\phi}\colon \mathcal{X}\times\mathcal{Y} \to \mathbb{F}$ with $\|\widehat{\phi}\| = \|\phi\|$.*

(c) *Every bounded linear form $f\colon \mathcal{M} \otimes_\wedge \mathcal{N} \to \mathbb{F}$ has a bounded linear extension $\widetilde{f} \in \mathcal{X} \otimes_\wedge \mathcal{Y}$ with $\|\widetilde{f}\| = \|f\|$.*

(d) *Every bounded linear form $\overline{f}\colon \mathcal{M} \mathbin{\widehat{\otimes}_\wedge} \mathcal{N} \to \mathbb{F}$ has a bounded linear extension $\widehat{f}\colon \mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y} \to \mathbb{F}$ with $\|\widehat{f}\| = \|\overline{f}\|$.*

Proof. Let $\mathcal{M}$ and $\mathcal{N}$ be subspaces of Banach spaces $\mathcal{X}$ and $\mathcal{Y}$. Equip tensor products with the projective norm, set $\mathcal{Z} = \mathbb{F}$, and recall from Remark 7.13(e):

$$(\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y})^* \cong (\mathcal{X} \otimes_\wedge \mathcal{Y})^* = \mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathbb{F}] \cong \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}].$$

Take an arbitrary $F = \sum_i u_i \otimes v_i \in \mathcal{M} \otimes_\wedge \mathcal{N}$. If assertion (b) holds, then every ϕ in $\boldsymbol{b}[\mathcal{M}\times\mathcal{N}, \mathbb{F}]$ has an extension $\widehat{\phi}$ in $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$ with the same norm. So the set of all bilinear forms in $\boldsymbol{b}[\mathcal{M}\times\mathcal{N}, \mathbb{F}]$ with norm less than 1 is included in the set of all restrictions to $\mathcal{M}\times\mathcal{N}$ of all bilinear forms in $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$ with norm less than 1. Then by Remark 7.13(f)

$$\begin{aligned} \|F\|_{\mathcal{M} \otimes_\wedge \mathcal{N}} &= \sup_{\|\psi\|\le 1,\, \psi\in \boldsymbol{b}[\mathcal{M}\times\mathcal{N}, \mathbb{F}]} \left|\sum\nolimits_i \psi(u_i, v_i)\right| \\ &\le \sup_{\|\psi\|\le 1,\, \psi\in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]} \left|\sum\nolimits_i \psi(u_i, v_i)\right| = \|F\|_{\mathcal{X} \otimes_\wedge \mathcal{Y}}. \end{aligned}$$

Since $\|F\|_{\mathcal{X} \otimes_\wedge \mathcal{Y}} \le \|F\|_{\mathcal{M} \otimes_\wedge \mathcal{N}}$ as we saw in Remark 7.5(b), then

$$\|F\|_{\mathcal{M} \otimes_\wedge \mathcal{N}} = \|F\|_{\mathcal{X} \otimes_\wedge \mathcal{Y}}.$$

Equivalently, $\mathcal{M} \otimes_\wedge \mathcal{N}$ is a linear manifold of $\mathcal{X} \otimes_\wedge \mathcal{Y}$ by Remark 7.5(e). Thus (b) implies (a). The converse goes through (c). Actually, (a) implies (c) by the Hahn–Banach Theorem (and (c) only makes sense if (a) holds). Also, (c) implies (a,b) by setting $\mathcal{Z} = \mathbb{F}$ in Corollary 7.14(c), and (a,b) in turn is equivalent to (b) (since (b) implies (a)), and so (c) implies (b). Finally (c) is equivalent to (d) by Corollary 7.14(c,d) with $\mathcal{Z} = \mathbb{F}$. Summing up,

$$\text{(b)} \Longrightarrow \text{(a)} \Longrightarrow \text{(c)} \Longrightarrow \text{(b)} \quad \text{and} \quad \text{(c)} \iff \text{(d)}. \qquad \square$$

By Remark 5.3, in a Hilbert-space setting each assertion in Corollaries 7.14 and 7.15 holds true (see also Proposition 5.R and Corollaries 5.6, 6.10).

Remark 7.16. *Immediate Consequences of Corollaries 7.14 and 7.15.*

(i) All norms in the proof of Corollary 7.14 are not only pairwise preserved but they coincide due to the isometric isomorphisms:

$$\|\widehat{\phi}\| = \|\phi\| = \|\widetilde{T}\| = \|T\| = \|\widehat{T}\| = \|\overline{T}\|.$$

It is also worth noticing that extension over completion preserves norms so that $\|\overline{T}\| = \|T\|$ and $\|\widehat{T}\| = \|\widetilde{T}\|$ independently of Corollary 7.14. In particular, all norms in Corollary 7.15 coincide as well:

$$\|\widehat{\phi}\| = \|\phi\| = \|\widetilde{f}\| = \|f\| = \|\widehat{f}\| = \|\overline{f}\|.$$

(ii) By Theorem 5.13(a) and since Corollary 7.14 holds for every Banach space $\mathcal{Z}$, any of the assertions (b), (c), (d) in Corollary 7.14 implies

(o′) $\mathcal{M}$ and $\mathcal{N}$ are complemented.

Moreover, according to Theorem 7.8(b) and Remark 7.5(e), if (o′) holds, then assertion (a) in both Corollaries 7.14 and 7.15 is equivalent to each of

(a′) $\mathcal{M} \otimes_\wedge \mathcal{N}$ *is a subspace of* $\mathcal{X} \otimes_\wedge \mathcal{Y}$,

(a″) $\mathcal{M} \widehat{\otimes}_\wedge \mathcal{N}$ *is a subspace of* $\mathcal{X} \widehat{\otimes}_\wedge \mathcal{Y}$.

(iii) The implication (o) $\Longrightarrow$ (b) in Corollary 7.14 has nothing to do with reasonable crossnorms. The equivalence (b) $\Longleftrightarrow$ (c) in both corollaries holds for the projective norm only (by Theorem 7.12). The equivalence (c) $\Longleftrightarrow$ (d) in both corollaries does not depend on the particular reasonable crossnorms.

(iv) As we saw in Remark 7.5(b), $\mathcal{M} \otimes_\wedge \mathcal{N}$ may not be a linear manifold of $\mathcal{X} \otimes_\wedge \mathcal{Y}$. Corollaries 7.14 and 7.15 gave equivalent conditions to ensure that (a) holds. However, as we also saw in Remark 7.5(e), *if* $\mathcal{M} \otimes_\wedge \mathcal{N}$ *is a linear manifold of* $\mathcal{X} \otimes_\wedge \mathcal{Y}$, *then* $\mathcal{M} \widehat{\otimes}_\wedge \mathcal{N}$ *is a subspace of* $\mathcal{X} \widehat{\otimes}_\wedge \mathcal{Y}$. Regarding this particular issue, things get simpler for the injective tensor product: no extra condition is needed for the injective tensor product $\mathcal{M} \widehat{\otimes}_\vee \mathcal{N}$ to be a subspace of the injective tensor product $\mathcal{X} \widehat{\otimes}_\vee \mathcal{Y}$, as we will see in Theorem 7.19 below.

7.3 Injective Tensor Product

Unlike the projective tensor product, there is no representation of elements in the injective tensor product $\mathcal{X} \widehat{\otimes}_\vee \mathcal{Y}$: there is no injective counterpart of Theorem 7.9. There are, however, several interpretations of injective tensor products $\mathcal{X} \widehat{\otimes}_\vee \mathcal{Y}$ of normed spaces $\mathcal{X}$ and $\mathcal{Y}$ by embedding them into Banach spaces of bounded linear transformations or bounded bilinear forms following the patterns in Remark 4.9. Next we describe three such embeddings.

Theorem 7.17. *Let* $\mathcal{X}$ *and* $\mathcal{Y}$ *be normed spaces.*

(a_1) *If* $F \in \mathcal{X} \otimes_\vee \mathcal{Y}$, *then* $\|F\|_\vee = \|\psi_F\|_{\boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]}$ *where* $\psi_F \in \boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]$ *is a bounded bilinear form associated with* $F \in \mathcal{X} \widehat{\otimes}_\vee \mathcal{Y}$ *defined by*

$$\psi_F(f, g) = \sum\nolimits_i f(x_i)\, g(y_i) \quad \text{for every} \quad (f, g) \in \mathcal{X}^* \times \mathcal{Y}^*,$$

which does not depend on the (finite) representation $\sum_i x_i \otimes y_i$ *of* F.

(a$_2$) *The injective tensor product* $\mathcal{X}\,\widehat{\otimes}_\vee\mathcal{Y}$ *is isometrically embedded in the Banach space* $\boldsymbol{b}[\mathcal{X}^*\times\mathcal{Y}^*,\mathbb{F}]$*, and so is viewed as a subspace of* $\boldsymbol{b}[\mathcal{X}^*\times\mathcal{Y}^*,\mathbb{F}]$*:*

$$\mathcal{X}\otimes_\vee\mathcal{Y}\subseteq\boldsymbol{b}[\mathcal{X}^*\times\mathcal{Y}^*,\mathbb{F}],\quad\textit{which implies}\quad\mathcal{X}\,\widehat{\otimes}_\vee\mathcal{Y}\subseteq\boldsymbol{b}[\mathcal{X}^*\times\mathcal{Y}^*,\mathbb{F}].$$

(b$_1$) *If* $F'\in\mathcal{X}^*\otimes_\vee\mathcal{Y}$*, then* $\|F'\|_\vee=\|\Psi_{F'}\|_{\mathcal{B}[\mathcal{X},\mathcal{Y}]}$*, where* $\Psi_{F'}\in\mathcal{B}[\mathcal{X},\mathcal{Y}]$ *is a bounded linear transformation associated with* $F'\in\mathcal{X}^*\,\widehat{\otimes}_\vee\mathcal{Y}$ *defined by*

$$\Psi_{F'}x=\sum\nolimits_i f_i(x)\,y_i\quad\textit{for every}\quad x\in\mathcal{X},$$

which does not depend on the (finite) representation $\sum_i f_i\otimes y_i$ *of* F'*.*

(b$_2$) *The injective tensor product* $\mathcal{X}^*\,\widehat{\otimes}_\vee\mathcal{Y}$ *is isometrically embedded in the Banach space* $\mathcal{B}[\mathcal{X},\mathcal{Y}]$*, and so it is viewed as a subspace of* $\mathcal{B}[\mathcal{X},\mathcal{Y}]$*:*

$$\mathcal{X}^*\otimes_\vee\mathcal{Y}\subseteq\mathcal{B}[\mathcal{X},\mathcal{Y}],\quad\textit{which implies}\quad\mathcal{X}^*\,\widehat{\otimes}_\vee\mathcal{Y}\subseteq\mathcal{B}[\mathcal{X},\mathcal{Y}]\ \textit{if}\ \mathcal{Y}\ \textit{is Banach.}$$

(c$_1$) *If* $F''\in\mathcal{X}\otimes_\vee\mathcal{Y}^*$*, then* $\|F''\|_\vee=\|\Psi_{F''}\|_{\mathcal{B}[\mathcal{Y},\mathcal{X}]}$*, where* $\Psi_{F''}\in\mathcal{B}[\mathcal{Y},\mathcal{X}]$ *is a bounded linear transformation associated with* $F''\in\mathcal{X}\,\widehat{\otimes}_\vee\mathcal{Y}^*$ *defined by*

$$\Psi_{F''}y=\sum\nolimits_i g_i(y)\,x_i\quad\textit{for every}\quad y\in\mathcal{Y},$$

which does not depend on the (finite) representation $\sum_i x_i\otimes g_i$ *of* F''*.*

(c$_2$) *The injective tensor product* $\mathcal{X}\,\widehat{\otimes}_\vee\mathcal{Y}^*$ *is isometrically embedded in the Banach space* $\mathcal{B}[\mathcal{Y},\mathcal{X}]$*, and so it is viewed as a subspace of* $\mathcal{B}[\mathcal{Y},\mathcal{X}]$*:*

$$\mathcal{X}\otimes_\vee\mathcal{Y}^*\subseteq\mathcal{B}[\mathcal{Y},\mathcal{X}],\quad\textit{which implies}\quad\mathcal{X}\,\widehat{\otimes}_\vee\mathcal{Y}^*\subseteq\mathcal{B}[\mathcal{Y},\mathcal{X}]\ \textit{if}\ \mathcal{X}\ \textit{is Banach.}$$

Proof. (a) If $\mathcal{X}$ and $\mathcal{Y}$ are linear spaces, then $\mathcal{X}\otimes\mathcal{Y}\subseteq b[\mathcal{X}^\sharp\times\mathcal{Y}^\sharp,\mathbb{F}]$ by Remark 4.9(b). We will show that if $\mathcal{X}$ and $\mathcal{Y}$ are normed spaces and the tensor product $\mathcal{X}\otimes\mathcal{Y}$ is equipped with the injective norm $\|\cdot\|_\vee$, then the above embedding is lifted to $\mathcal{X}\otimes_\vee\mathcal{Y}\subseteq\boldsymbol{b}[\mathcal{X}^*\times\mathcal{Y}^*,\mathbb{F}]$. This is somewhat related to Corollary 4.10.

Let $\mathcal{X}$ and $\mathcal{Y}$ be normed spaces. Fix an arbitrary $F=\sum_i x_i\otimes y_i$ in $\mathcal{X}\otimes\mathcal{Y}$. Consider the map $\psi_F:\mathcal{X}^*\otimes\mathcal{Y}^*\to\mathbb{F}$ associated with F defined by

$$\psi_F(f,g)=(f\otimes g)(F)=\sum\nolimits_i f(x_i)\otimes g(y_i)=\sum\nolimits_i f(x_i)\,g(y_i)$$

for each $(f,g)\in\mathcal{X}^*\times\mathcal{Y}^*$. This does not depend on the representation of F, by Theorem 3.17(a). As is readily verified, ψ_F is bilinear: $\psi_F\in b[\mathcal{X}^*\times\mathcal{Y}^*,\mathbb{F}]$. Take the transformation $\mathcal{K}\colon\mathcal{X}\otimes\mathcal{Y}\to b[\mathcal{X}^*\times\mathcal{Y}^*,\mathbb{F}]$ given by $\mathcal{K}(F)=\psi_F$ for every F in $\mathcal{X}\otimes\mathcal{Y}$. This is clearly linear, and injective. (In fact, if $\mathcal{K}(F)=0$, then $\psi_F=0$ so that $\psi_F(f,g)=0$ for every $(f,g)\in\mathcal{X}^*\times\mathcal{Y}^*$ and hence $F=0$ by the Claim in the proof of Corollary 4.10.) Thus $\mathcal{K}$ is a linear-space embedding:

$$\mathcal{K}\colon\mathcal{X}\otimes\mathcal{Y}\hookrightarrow b[\mathcal{X}^*\times\mathcal{Y}^*,\mathbb{F}],\quad\text{which means}\quad\mathcal{X}\otimes\mathcal{Y}\subseteq b[\mathcal{X}^*\times\mathcal{Y}^*,\mathbb{F}].$$

Now equip $\mathcal{X} \otimes \mathcal{Y}$ with the injective norm $\|\cdot\|_\vee$. In this case ψ_F is bounded, that is, $\psi_F \in \boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]$ and, moreover (cf. Theorem 6.1),

$$\|F\|_\vee = \sup_{\|f\|\le 1,\, \|g\|\le 1} \left|\sum\nolimits_i f(x_i)\, g(y_i)\right| = \sup_{\|f\|\le 1, \|g\|\le 1} |\psi_F(f,g)| = \|\psi_F\|_{\boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]}.$$

As $\|\mathcal{K}(F)\|_{\boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]} = \|F\|_\vee$, the injective linear transformation $\mathcal{K}$ is an isometry and so $\mathcal{K}\colon \mathcal{X} \otimes_\vee \mathcal{Y} \to \mathcal{R}(\mathcal{K}) \subseteq \boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]$ is an isometric isomorphism onto its range $\mathcal{R}(\mathcal{K})$, which means $\mathcal{K}$ is an isometric embedding of $\mathcal{X} \otimes_\vee \mathcal{Y}$ into $\boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]$, and so $\mathcal{X} \otimes_\vee \mathcal{Y}$ is interpreted as a linear manifold of $\boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]$,

$$\mathcal{X} \otimes_\vee \mathcal{Y} \subseteq \boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}].$$

The isometric isomorphism $\mathcal{K} \in \mathcal{B}[\mathcal{X} \otimes_\vee \mathcal{Y}, \mathcal{R}(\mathcal{K})]$ extends over the completion $\mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y}$ of $\mathcal{X} \otimes_\vee \mathcal{Y}$ to an isometric isomorphism $\widehat{\mathcal{K}} \in \mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y}, \widehat{\mathcal{R}(\mathcal{K})}]$ by Theorem 5.1(d). Note: as $\boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]$ is a Banach space, it is its own completion, and the completion of $\mathcal{R}(\mathcal{K})$ is its closure in $\boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]$ (i.e., $\widehat{\mathcal{R}(\mathcal{K})} = \mathcal{R}(\mathcal{K})^-$), and so $\widehat{\mathcal{R}(\mathcal{K})}$ is a subspace of $\boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]$ (i.e., $\widehat{\mathcal{R}(\mathcal{K})} \subseteq \boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]$). Then $\widehat{\mathcal{K}}$ is an isometric embedding of $\mathcal{X} \otimes_\vee \mathcal{Y}$ into $\boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]$), and therefore

$$\mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y} \subseteq \boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}].$$

In other words, the injective tensor product $\mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y}^*$ is included/isometrically embedded in the Banach space $\boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]$, and in this sense it is regarded as a subspace (i.e., a closed linear manifold) of $\boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]$.

(b) This follows a different but similar path from item (a). By Remark 4.9(i), $\mathcal{X}^\sharp \otimes \mathcal{Y} \subseteq \mathcal{L}[\mathcal{X}, \mathcal{Y}]$ for linear spaces $\mathcal{X}$ and $\mathcal{Y}$. Equip $\mathcal{X}^* \otimes \mathcal{Y}$ with the injective norm and the inclusion is lifted to $\mathcal{X}^* \otimes_\vee \mathcal{Y} \subseteq \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ (so $\mathcal{X}^* \mathbin{\widehat{\otimes}_\vee} \mathcal{Y} \subseteq \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ if $\mathcal{Y}$ is Banach). Explanations have been given in (a) so we get straight to the point.

Equip $\mathcal{X}^* \otimes \mathcal{Y}$ with the injective norm $\|\cdot\|_\vee$. Fix an arbitrary $F' = \sum_i f_i \otimes y_i$ in $\mathcal{X}^* \otimes_\vee \mathcal{Y}$, and consider the linear transformation $\Psi_{F'}\colon \mathcal{X} \to \mathcal{Y}$ given by

$$\Psi_{F'} x = \sum\nolimits_i f_i(x)\, y_i \quad \text{for every} \quad x \in \mathcal{X}.$$

(This also does not depend on the representation of F' since for each $x \in \mathcal{X}$, $\Psi_{(\cdot)}(x)\colon \mathcal{X}^* \otimes \mathcal{Y} \to \mathcal{Y}$ is linear — same argument as in Theorem 3.17(a)'s proof.)

Claim. $\Psi_{F'} \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ and $\|F'\|_\vee = \|\Psi_{F'}\|_{\mathcal{B}[\mathcal{X}, \mathcal{Y}]}$.

Proof of Claim. Let $\varphi \in \mathcal{X}^{**}$, $g \in \mathcal{Y}^*$ and $x \in \mathcal{X}$ be elements in $\mathcal{X}^{**}$, $\mathcal{Y}^*$ and $\mathcal{X}$.

$$\|F'\|_\vee = \sup_{\|\varphi\|\le 1, \|g\|\le 1} \left|\sum\nolimits_i \varphi(f_i)\, g(y_i)\right| = \sup_{\|\varphi\|\le 1, \|g\|\le 1} \left|\varphi\Big(\sum\nolimits_i g(y_i)\, f_i\Big)\right|.$$

By the Claim in Theorem 7.3's proof (a corollary of Goldstine's Theorem),

$$\varphi(f) = \lim\nolimits_\gamma f(x_\gamma) \quad \text{with} \quad \|x_\gamma\| \le \|\varphi\|$$

for every $\varphi \in \mathcal{X}^{**}$, every $f \in \mathcal{X}^*$, and some $\mathcal{X}$-valued net $\{x_\gamma\}$. Therefore

$$\varphi\Big(\sum\nolimits_i g(y_i)\, f_i\Big) = \lim\nolimits_\gamma \Big(\sum\nolimits_i g(y_i) f_i\Big)(x_\gamma) = \lim\nolimits_\gamma \Big(\sum\nolimits_i g(y_i) f_i(x_\gamma)\Big)$$
$$= \lim\nolimits_\gamma g\Big(\sum\nolimits_i y_i f_i(x_\gamma)\Big) = \lim\nolimits_\gamma g(\Psi_{F'} x_\gamma) = g(\Psi_{F'} \lim\nolimits_\gamma x_\gamma).$$

Since $\|\varphi\| = \sup_{\|f\|\le 1} |\varphi(f)| = \sup_{\|f\|\le 1} |f(\lim_\gamma x_\gamma)| = \|\lim_\gamma x_\gamma\|$,

$$\sup_{\|\varphi\|\le 1} \Big|\varphi\Big(\sum\nolimits_i g(y_i)\, f_i\Big)\Big| = \sup_{\|x\|\le 1} |g(\Psi_{F'} x)|.$$

Moreover, since $\sup_{\|g\|\le 1} |g(y)| = \|y\|$ for every $y \in \mathcal{Y}$ and every $g \in \mathcal{Y}^*$ (by the Hahn–Banach Theorem — cf. Remark 5.2(d)),

$$\sup_{\|\varphi\|\le 1, \|g\|\le 1} \Big|\varphi\Big(\sum\nolimits_i g(y_i)\, f_i\Big)\Big| = \sup_{\|x\|\le 1,\, \|g\|\le 1} |g(\Psi_{F'} x)| = \sup_{\|x\|\le 1} \|\Psi_{F'} x\| = \|\Psi_{F'}\|. \quad \square$$

The map $F' \mapsto \Psi_{F'}$ of $\mathcal{X}^* \otimes_\vee \mathcal{Y}$ into $\mathcal{B}[\mathcal{X}, \mathcal{Y}]$ is clearly linear. It is an isometry (as seen above, thus injective). Then it is an isometric isomorphism onto its range, so an isometric embedding of $\mathcal{X}^* \otimes_\vee \mathcal{Y}$ into the normed space $\mathcal{B}[\mathcal{X}, \mathcal{Y}]$, which extends over the completion $\mathcal{X}^* \,\widehat{\otimes}_\vee \mathcal{Y}$ of $\mathcal{X}^* \otimes_\vee \mathcal{Y}$ (Theorem 5.1(d)).

(c) This is a symmetric version of (b). Fix an arbitrary $F'' = \sum_i x_i \otimes g_i$ in $\mathcal{X} \otimes_\vee \mathcal{Y}^*$ and consider the linear transformation $\Psi_{F''} \colon \mathcal{Y} \to \mathcal{X}$ given by

$$\Psi_{F''} y = \sum\nolimits_i g_i(y)\, x_i \quad \text{for every} \quad y \in \mathcal{Y}.$$

Using the same argument as in the proof of item (b) we get

$$\Psi_{F''} \in \mathcal{B}[\mathcal{Y}, \mathcal{X}] \quad \text{and} \quad \|F''\|_\vee = \|\Psi_{F''}\|_{\mathcal{B}[\mathcal{Y},\mathcal{X}]}.$$

Thus the map $F'' \mapsto \Psi_{F''}$ is a linear isometry of $\mathcal{X} \otimes_\vee \mathcal{Y}^*$ into $\mathcal{B}[\mathcal{Y}, \mathcal{X}]$, which implies that $\mathcal{X} \,\widehat{\otimes}_\vee \mathcal{Y}^*$ is isometrically embedded in $\mathcal{B}[\mathcal{Y}, \mathcal{X}]$ if $\mathcal{X}$ is Banach. $\square$

Before proceeding, it is convenient to remark on a few additional points.

Remark 7.18. *Further Properties of the Injective Norm.*

(a) Let $\mathcal{X}$ and $\mathcal{Y}$ be normed spaces. Take an arbitrary $F = \sum_i x_i \otimes y_i \in \mathcal{X} \otimes \mathcal{Y}$. If $F = \sum_i x_i \otimes y_i \in \mathcal{X} \otimes_\vee \mathcal{Y}$, then as we saw in proof of Theorem 7.17(a_1)

$$\|F\|_\vee = \sup_{\|f\|\le 1,\, \|g\|\le 1} \Big|\sum\nolimits_i f(x_i) g(y_i)\Big| = \sup_{\|f\|\le 1, \|g\|\le 1} |\psi_F(f, g)| = \|\psi_F\|_{\boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]},$$

the injective norm $\|F\|_\vee$ *of each* $F \in \mathcal{X} \otimes_\vee \mathcal{Y}$ *is the induced uniform norm* $\|\psi_F\|_{\boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]}$ *of the bounded bilinear map* $\psi_F \in \boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]$,

given by $\psi_F(f, g) = \sum_i f(x_i)\, g(y_i)$ for every $(f, g) \in \mathcal{X}^* \times \mathcal{Y}^*$. The above expression implies another two expressions for $\|\cdot\|_\vee$ on $\mathcal{X} \otimes \mathcal{Y}$ as in (b) and (c) below.

(b) Since $\|y\| = \sup_{\|g\|\le 1} |g(y)|$ for every $y \in \mathcal{Y}$ and $g \in \mathcal{Y}^*$ (Remark 5.2(d)),

$$\|F\|_\vee = \sup_{\|f\|\le 1,\,\|g\|\le 1} \left|g\Big(\sum\nolimits_i f(x_i)y_i\Big)\right| = \sup_{\|f\|\le 1} \left\|\sum\nolimits_i f(x_i)y_i\right\|_{\mathcal{Y}} = \|\Psi'_F\|_{\mathcal{B}[\mathcal{X}^*,\mathcal{Y}]},$$

the injective norm $\|F\|_\vee$ *of each* $F \in \mathcal{X}\otimes_\vee\mathcal{Y}$ *is the induced uniform norm* $\|\Psi'_F\|_{\mathcal{B}[\mathcal{X}^*,\mathcal{Y}]}$ *of the bounded linear transformation* $\Psi'_F \in \mathcal{B}[\mathcal{X}^*,\mathcal{Y}]$,

given by $\Psi'_F f = \sum_i f(x_i)y_i$ for every $f \in \mathcal{X}^*$.

(c) Since $\|x\| = \sup_{\|f\|\le 1} |f(x)|$ for every $x \in \mathcal{X}$ and $f \in \mathcal{X}^*$ (Remark 5.2(d)),

$$\|F\|_\vee = \sup_{\|f\|\le 1,\,\|g\|\le 1} \left|f\Big(\sum\nolimits_i g(y_i)x_i\Big)\right| = \sup_{\|g\|\le 1} \left\|\sum\nolimits_i g(y_i)x_i\right\|_{\mathcal{X}} = \|\Psi''_F\|_{\mathcal{B}[\mathcal{Y}^*,\mathcal{X}]},$$

the injective norm $\|F\|_\vee$ *of each* $F \in \mathcal{X}\otimes_\vee\mathcal{Y}$ *is the induced uniform norm* $\|\Psi''_F\|_{\mathcal{B}[\mathcal{Y}^*,\mathcal{X}]}$ *of the bounded linear transformation* $\Psi''_F \in \mathcal{B}[\mathcal{Y}^*,\mathcal{X}]$,

which is given by $\Psi''_F g = \sum_i g(y_i)x_i$ for every $g \in \mathcal{Y}^*$.

The next result justifies the terminology *injective* for the norm $\|\cdot\|_\vee$. It shows that if $\mathcal{M}$ and $\mathcal{N}$ are linear manifolds of $\mathcal{X}$ and $\mathcal{Y}$, respectively, then $\mathcal{M}\widehat{\otimes}_\vee\mathcal{N}$ is a subspace of $\mathcal{X}\widehat{\otimes}_\vee\mathcal{Y}$, thus embedded (injected) in $\mathcal{X}\widehat{\otimes}_\vee\mathcal{Y}$. In other words, take a regular linear manifold $\mathcal{M}\otimes\mathcal{N}$ of the linear space $\mathcal{X}\otimes\mathcal{Y}$. Equip each of them individually with the injective norm. Then $\mathcal{M}\otimes_\vee\mathcal{N}$ is a linear manifold of $\mathcal{X}\otimes_\vee\mathcal{Y}$ (see Remark 7.5). Consequently, the completion $\mathcal{M}\widehat{\otimes}_\vee\mathcal{N}$ is a subspace (i.e., a closed linear manifold) of the completion $\mathcal{X}\widehat{\otimes}_\vee\mathcal{Y}$.

Theorem 7.19. *If* $\mathcal{M}$ *and* $\mathcal{N}$ *are linear manifolds of normed spaces* $\mathcal{X}$ *and and* $\mathcal{Y}$*, respectively, then* $\mathcal{M}\otimes_\vee\mathcal{N}$ *is a linear manifold of* $\mathcal{X}\otimes_\vee\mathcal{Y}$.

Proof. Let $\mathcal{M}$ and $\mathcal{N}$ be linear manifolds of normed spaces $\mathcal{X}$ and $\mathcal{Y}$, respectively. Consider the (algebraic) linear manifold $\mathcal{M}\otimes\mathcal{N}$ of $\mathcal{X}\otimes\mathcal{Y}$. Equip these tensor products of normed spaces with the injective norm. We show that the norm identity in Remark 7.5(e) holds for the injective norm $\|\cdot\|_\vee$. Therefore $\mathcal{M}\otimes_\vee\mathcal{N}$ is a linear manifold of the normed space $\mathcal{X}\otimes_\vee\mathcal{Y}$ (and consequently $\mathcal{M}\widehat{\otimes}_\vee\mathcal{N}$ is a subspace of $\mathcal{X}\widehat{\otimes}_\vee\mathcal{Y}$ as we saw in Remark 7.5(d)). In fact, if $\mathcal{X}$ and $\mathcal{Y}$ are normed spaces and $\mathcal{M}$ and $\mathcal{N}$ are linear manifolds of $\mathcal{X}$ and $\mathcal{Y}$ (and so $\|u\|_{\mathcal{M}} = \|u\|_{\mathcal{X}}$ for every $u \in \mathcal{M}$ and $\|v\|_{\mathcal{N}} = \|v\|_{\mathcal{Y}}$ for every $v \in \mathcal{N}$), then for an arbitrary $F = \sum_i u_i \otimes v_i \in \mathcal{M}\otimes_\vee\mathcal{N}$ we get, by Remark 7.18(b,c) (which was obtained by means of the Hahn–Banach Theorem as in Remark 5.2(d)),

$$\|F\|_{\mathcal{M}\otimes_\vee\mathcal{N}} = \sup_{\|f\|\le 1} \left\|\sum\nolimits_i f(u_i)v_i\right\|_{\mathcal{N}} = \sup_{\|f\|\le 1} \left\|\sum\nolimits_i f(u_i)v_i\right\|_{\mathcal{Y}} = \|F\|_{\mathcal{M}\otimes_\vee\mathcal{Y}},$$

$$\|F\|_{\mathcal{M}\otimes_\vee\mathcal{Y}} = \sup_{\|g\|\le 1} \left\|\sum\nolimits_i g(v_i)u_i\right\|_{\mathcal{M}} = \sup_{\|g\|\le 1} \left\|\sum\nolimits_i g(v_i)u_i\right\|_{\mathcal{X}} = \|F\|_{\mathcal{X}\otimes_\vee\mathcal{Y}}.$$

So $\|F\|_{\mathcal{M}\otimes_\vee\mathcal{N}} = \|F\|_{\mathcal{X}\otimes_\vee\mathcal{Y}}$. The claimed result follows by Remark 7.5(e). □

Remark 7.20. *Embeddings Related to Injective and Projective Norms.*

(a) According to Theorem 7.17,

$$\mathcal{X} \otimes_\vee \mathcal{Y} \subseteq \boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}], \quad \mathcal{X}^* \otimes_\vee \mathcal{Y} \subseteq \mathcal{B}[\mathcal{X}, \mathcal{Y}], \quad \mathcal{X} \otimes_\vee \mathcal{Y}^* \subseteq \mathcal{B}[\mathcal{Y}, \mathcal{X}].$$

By the last two inclusions, $\mathcal{X}^* \mathbin{\widehat{\otimes}_\vee} \mathcal{Y}^* \subseteq \mathcal{B}[\mathcal{X}, \mathcal{Y}^*] \cap \mathcal{B}[\mathcal{Y}, \mathcal{X}^*]$ (Theorem 5.1(g), as $\mathcal{B}[\mathcal{X}, \mathcal{Y}^*]$ and $\mathcal{B}[\mathcal{Y}, \mathcal{X}^*]$ are Banach spaces) for the *injective* norm. By Remark 6.4(a), $\mathcal{B}[\mathcal{X}, \mathcal{Y}^*] \cong \mathcal{B}[\mathcal{Y}, \mathcal{X}^*] \cong \boldsymbol{b}[\mathcal{X} \times \mathcal{Y}, \mathbb{F}]$. Also, $\boldsymbol{b}[\mathcal{X} \times \mathcal{Y}, \mathbb{F}] \cong (\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y})^*$ for the *projective* norm (Theorem 7.12 for normed spaces $\mathcal{X}$ and $\mathcal{Y}$). Thus

$$\mathcal{X}^* \mathbin{\widehat{\otimes}_\vee} \mathcal{Y}^* \subseteq \mathcal{B}[\mathcal{X}, \mathcal{Y}^*] \cong \mathcal{B}[\mathcal{Y}, \mathcal{X}^*] \cong \boldsymbol{b}[\mathcal{X} \times \mathcal{Y}, \mathbb{F}] \cong (\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y})^*.$$

(Compare the embeddings involving the injective tensor product of the duals $\mathcal{X}^* \mathbin{\widehat{\otimes}_\vee} \mathcal{Y}^*$ with those algebraic embeddings in Remark 4.9(b,f,g,h,i,ℓ).) Summarizing the relation between the *injective* tensor product of the duals $\mathcal{X}^* \mathbin{\widehat{\otimes}_\vee} \mathcal{Y}^*$ and the dual $(\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y})^*$ of the *projective* tensor product $\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}$:

$$\mathcal{X}^* \mathbin{\widehat{\otimes}_\vee} \mathcal{Y}^* \subseteq (\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y})^* \cong \boldsymbol{b}[\mathcal{X} \times \mathcal{Y}, \mathbb{F}].$$

(b) We show that a similar relation holds if the injective tensor product of the duals $\mathcal{X}^* \mathbin{\widehat{\otimes}_\vee} \mathcal{Y}^*$ is replaced by the dual $(\mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y})^*$ of the injective tensor product $\mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y}$. In fact, for any normed space $\mathcal{Z}$, let $\boldsymbol{B}_\mathcal{Z} = \{z \in \mathcal{Z} \colon \|z\|_\mathcal{Z} \leq 1\}$ be the closed unit ball in $\mathcal{Z}$ (centered at the origin). Equip the tensor product $\mathcal{X} \otimes \mathcal{Y}$ with both injective and projective norms. That is, take the normed spaces $\mathcal{X} \otimes_\vee \mathcal{Y} = (\mathcal{X} \otimes \mathcal{Y}, \|\cdot\|_\vee)$ and $\mathcal{X} \otimes_\wedge \mathcal{Y} = (\mathcal{X} \otimes \mathcal{Y}, \|\cdot\|_\wedge)$. By Theorem 7.4

$$\|\cdot\|_\vee \leq \|\cdot\|_\wedge \quad \text{and so} \quad \mathcal{X} \otimes_\wedge \mathcal{Y} \subseteq \mathcal{X} \otimes_\vee \mathcal{Y},$$

which is just a set inclusion: different norms on the same underlying linear space. Then $\|F\|_\wedge \leq 1$ implies $\|F\|_\vee \leq 1$ for every $F \in \mathcal{X} \otimes_\wedge \mathcal{Y} \subseteq \mathcal{X} \otimes_\vee \mathcal{Y}$. Therefore $\boldsymbol{B}_{\mathcal{X} \otimes_\wedge \mathcal{Y}} \subseteq \boldsymbol{B}_{\mathcal{X} \otimes_\vee \mathcal{Y}}$, which in turn implies

$$\begin{aligned}(\mathcal{X} \otimes_\vee \mathcal{Y})^* &= \big\{f \in \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathbb{F}] \colon \sup_{F \in \boldsymbol{B}_{\mathcal{X} \otimes_\vee \mathcal{Y}}} |f(F)| < \infty\big\} \\ &\subseteq \big\{f \in \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathbb{F}] \colon \sup_{F \in \boldsymbol{B}_{\mathcal{X} \otimes_\wedge \mathcal{Y}}} |f(F)| < \infty\big\} \\ &= (\mathcal{X} \otimes_\wedge \mathcal{Y})^*.\end{aligned}$$

Since $(\mathcal{X} \otimes_\vee \mathcal{Y})^* \cong (\mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y})^*$ and $(\mathcal{X} \otimes_\wedge \mathcal{Y})^* \cong (\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y})^*$ by Theorem 5.1(f),

$$(\mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y})^* \subseteq (\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y})^* \cong \boldsymbol{b}[\mathcal{X} \times \mathcal{Y}, \mathbb{F}].$$

Thus *the dual of* $\mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y}$ *is embedded* (*algebraically*) *in the dual of* $\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}$, *which in turn is identified* (*isometrically*) *with* $\boldsymbol{b}[\mathcal{X} \times \mathcal{Y}, \mathbb{F}]$.

Hence $(\mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y})^*$ is (algebraically) embedded in $\boldsymbol{b}[\mathcal{X} \times \mathcal{Y}, \mathbb{F}]$. Then each element Λ in $(\mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y})^*$ is identified with a (unique) bilinear form ϕ in $\boldsymbol{b}[\mathcal{X} \times \mathcal{Y}, \mathbb{F}]$. The next result shows which class of bounded bilinear forms from $\boldsymbol{b}[\mathcal{X} \times \mathcal{Y}, \mathbb{F}]$ represents all elements Λ in the dual $(\mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y})^*$ of the injective tensor product.

Theorem 7.21. *A bounded bilinear form* $\phi \in \boldsymbol{b}[\mathcal{X} \times \mathcal{Y}, \mathbb{F}]$ *on the Cartesian product of normed spaces* $\mathcal{X}$ *and* $\mathcal{Y}$ *is the identification of an element* Λ *from* $(\mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y})^*$ *if and only if there is an* $\mathbb{F}$*-valued measure* μ *on the Borel* σ*-algebra of subsets of the compact space* $\boldsymbol{B}_{\mathcal{X}^*} \times \boldsymbol{B}_{\mathcal{Y}^*}$ (*where* $\boldsymbol{B}_{\mathcal{X}^*}$ *and* $\boldsymbol{B}_{\mathcal{Y}^*}$ *are the closed unit balls in* $\mathcal{X}^*$ *and* $\mathcal{Y}^*$ *equipped with their weak* topology*) *such that*

$$\phi(x,y) = \int_{\boldsymbol{B}_{\mathcal{X}^*}\times\boldsymbol{B}_{\mathcal{Y}^*}} (f'\otimes g')(x,y)\,d\mu(f',g') \quad \textit{for every} \quad (x,y)\in\mathcal{X}\times\mathcal{Y}.$$

Before proving it we need four fundamental results as summarized below.

(1) For any normed space $\mathcal{Z}$ let $\boldsymbol{B}_{\mathcal{Z}^*} = \{f\in\mathcal{Z}^*\colon \|f\|_{\mathcal{Z}}* \le 1\}$ be the closed unit ball in $\mathcal{Z}^*$, where $\|f\|_{\mathcal{Z}}* = \sup_{\|z\|_{\mathcal{Z}}\le 1}|f(z)|$. The Banach–Alaoglu Theorem says that $\boldsymbol{B}_{\mathcal{Z}^*}$ *is weak* compact* (see, e.g., [70, Theorem 2.6.18]).

Note that $\boldsymbol{B}_{\mathcal{Z}^*}$ is a Hausdorff space. Reason: every topological vector space (in particular, $\mathcal{Z}^*$ equipped with its weak* topology) is a Hausdorff space (see, e.g., [82, Theorem 1.12]), and a subspace of a Hausdorff space is Hausdorff (see, e.g., [19, Theorem 1.3]) — in this context, subspace means topological subspace (equipped with the induced topology).

(2) Equip sets X and Y each with any topology. So their Cartesian product $X\times Y$ (equipped with the product topology) *is compact if and only if* X *and* Y *are compact* (Tychonoff Theorem — see, e.g., [19, Theorem XI.1.4(4)]).

(3) Let Z be a topological space. With respect to the topology in Z and the usual (metric) topology in $\mathbb{F}$ ($\mathbb{F}$ is either $\mathbb{R}$ or $\mathbb{C}$) consider the linear space $C[Z,\mathbb{F}]\subseteq\mathbb{F}^Z$ of all $\mathbb{F}$-valued continuous functions defined on Z. *If* Z *is compact, then every* $c\in C[Z,\mathbb{F}]$ *is bounded* in the sense that $\sup_{z\in Z}|c(z)|<\infty$ (Weierstrass (little) Theorem — see, e.g., [19, Theorem XI.2.3]).

(4) This is one of the general forms of the Riesz Representation Theorem. *Let* X *be a compact Hausdorff space. If* $\Gamma\colon C[X,\mathbb{F}]\to\mathbb{F}$ *is an* $\mathbb{F}$*-valued bounded linear functional on* $C[X,\mathbb{F}]$*, then there is a unique* $\mathbb{F}$*-valued measure* μ *on the Borel* σ*-algebra* $\mathcal{X}_{\mathcal{T}}$ *of subsets of* X *such that*

$$\Gamma(c) = \int c\,d\mu \quad \textit{for every} \quad c\in C[X,\mathbb{F}].$$

Moreover, $\|\Gamma\|_{C[X,\mathbb{F}]^*} = |\mu|(X)$ (see, e.g., [54, Theorems 12.6, 12.7]).

Here $|\mu|$ is the total variation of μ, a positive finite (thus Borel) measure on $\mathcal{X}_{\mathcal{T}}$ — hence, unlike positive measures, $\mathbb{F}$-valued measures (i.e., either real-valued measures, the so-called signed measures, or complex-valued measures) are naturally assumed to be Borel measures in the sense that their total variations are assumed to be finite and so Borel; and such a unique measure happens to be regular. Precisely, $|\mu|$ is the least positive finite measure dominating the $\mathbb{F}$-valued measure μ (i.e., $|\mu| = \min\lambda$ for all positive finite measures λ on $\mathcal{X}_{\mathcal{T}}$ such that $|\mu(E)|\le\lambda(E)$ for every $E\in\mathcal{X}_{\mathcal{T}}$) which also satisfies $|\mu|(X) = \sup_{|f|\le 1}|\int f\,d\mu|$ where the supremum is taken over all $\mathbb{F}$-valued integrable functions f on X for which $|f|\le 1$ (see, e.g., [54, Chapter 10]). If $\boldsymbol{X}$ denotes the linear space of all $\mathbb{F}$-valued measures on $\mathcal{X}_{\mathcal{T}}$, then the function $\|\cdot\|_{\boldsymbol{X}}\colon \boldsymbol{X}\to\mathbb{R}$ given by $\|\mu\|_{\boldsymbol{X}} = |\mu|(X)$ is a norm on $\boldsymbol{X}$, and so $\|\Gamma\|_{C[X,\mathbb{F}]^*} = \|\mu\|_{\boldsymbol{X}}$.

Proof. Let $C[\boldsymbol{B}_{\mathcal{X}^*}\times\boldsymbol{B}_{\mathcal{Y}^*},\mathbb{F}]$ denote the linear space of all $\mathbb{F}$-valued continuous functions defined on the Cartesian product of the closed unit balls $\boldsymbol{B}_{\mathcal{X}^*}\times\boldsymbol{B}_{\mathcal{Y}^*}$.

Take the transformation $\psi \in \boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}] \mapsto c \in C[\boldsymbol{B}_{\mathcal{X}^*} \times \boldsymbol{B}_{\mathcal{Y}^*}, \mathbb{F}]$ such that $c(f', g') = \psi\big(\frac{f}{\|f\|_{\mathcal{X}^*}}, \frac{g}{\|g\|_{\mathcal{Y}^*}}\big)$ with $f' = \frac{f}{\|f\|_{\mathcal{X}^*}} \in \boldsymbol{B}_{\mathcal{X}^*}$ and $g' = \frac{g}{\|g\|_{\mathcal{Y}^*}} \in \boldsymbol{B}_{\mathcal{Y}^*}$ for $(f, g) \neq 0$ in $\mathcal{X}^* \times \mathcal{Y}^*$. The transformation $\psi \mapsto c$ is clearly linear, and injective (since $c = 0$ only if $\psi = 0$). Equip $\boldsymbol{B}_{\mathcal{X}^*}$ and $\boldsymbol{B}_{\mathcal{Y}^*}$ with their weak* topologies. As they are compact, then so is $\boldsymbol{B}_{\mathcal{X}^*} \times \boldsymbol{B}_{\mathcal{Y}^*}$. Hence the $\mathbb{F}$-valued continuous functions on the compact set $\boldsymbol{B}_{\mathcal{X}^*} \times \boldsymbol{B}_{\mathcal{Y}^*}$ in $C[\boldsymbol{B}_{\mathcal{X}^*} \times \boldsymbol{B}_{\mathcal{Y}^*}, \mathbb{F}]$ are all bounded. Thus equip the linear space $C[\boldsymbol{B}_{\mathcal{X}^*} \times \boldsymbol{B}_{\mathcal{Y}^*}, \mathbb{F}]$ with the sup-norm $\|\cdot\|_\infty$ so that

$$\|\psi\|_{\boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]} = \sup_{(f', g') \in \boldsymbol{B}_{\mathcal{X}^*} \times \boldsymbol{B}_{\mathcal{Y}^*}} |c(f', g')| = \|c\|_\infty.$$

Then the linear injective transformation $\psi \mapsto c$ is an isometry, and so it is an isometric isomorphism onto its range, which means $\boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]$ is isometrically embedded in $C[\boldsymbol{B}_{\mathcal{X}^*} \times \boldsymbol{B}_{\mathcal{Y}^*}, \mathbb{F}]$. Let $\mathcal{I}$ be such an isometric embedding taking $\psi \in \boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]$ to $c = \mathcal{I}(\psi) \in \mathcal{R}(\mathcal{I}) \subseteq C[\boldsymbol{B}_{\mathcal{X}^*} \times \boldsymbol{B}_{\mathcal{Y}^*}, \mathbb{F}]$; that is,

$$\mathcal{I} : \boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}] \hookrightarrow C[\boldsymbol{B}_{\mathcal{X}^*} \times \boldsymbol{B}_{\mathcal{Y}^*}, \mathbb{F}], \text{ meaning } \boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}] \subseteq C[\boldsymbol{B}_{\mathcal{X}^*} \times \boldsymbol{B}_{\mathcal{Y}^*}, \mathbb{F}].$$

The injective tensor product $\mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y}$ is isometrically embedded in $\boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]$ according to Theorem 7.17(a_2). Thus let $\mathcal{K}$ be that isometric embedding which takes $F \in \mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y}$ to $\psi = \mathcal{K}(F) \in \mathcal{R}(\mathcal{K}) \subseteq \boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]$ so that

$$\mathcal{K} \colon \mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y} \hookrightarrow \boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}], \text{ which means } \mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y} \subseteq \boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}].$$

Therefore the composition $\mathcal{I}\mathcal{K}$ is an isometric embedding taking $F \in \mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y}$ to $c = \mathcal{I}\mathcal{K}(F) \in \mathcal{R}(\mathcal{I}\mathcal{K}) \subseteq C[\boldsymbol{B}_{\mathcal{X}^*} \times \boldsymbol{B}_{\mathcal{Y}^*}, \mathbb{F}]$ and so write

$$\mathcal{I}\mathcal{K} \colon \mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y} \hookrightarrow C[\boldsymbol{B}_{\mathcal{X}^*} \times \boldsymbol{B}_{\mathcal{Y}^*}, \mathbb{F}], \text{ meaning } \mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y} \subseteq C[\boldsymbol{B}_{\mathcal{X}^*} \times \boldsymbol{B}_{\mathcal{Y}^*}, \mathbb{F}].$$

Take an arbitrary $\Lambda \in (\mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y})^* = \mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y}, \mathbb{F}]$, and consider the above inclusion/embedding. By the Hahn–Banach Theorem (cf. Remark 5.2(d)), we can extend Λ to $\widehat{\Lambda} \in \mathcal{B}[C[\boldsymbol{B}_{\mathcal{X}^*} \times \boldsymbol{B}_{\mathcal{Y}^*}, \mathbb{F}], \mathbb{F}] = C[\boldsymbol{B}_{\mathcal{X}^*} \times \boldsymbol{B}_{\mathcal{Y}^*}, \mathbb{F}]^*$ with norm preservation. By the Riesz Representation Theorem, there is an $\mathbb{F}$-valued measure μ on the Borel σ-algebra of subsets of the compact set $\boldsymbol{B}_{\mathcal{X}^*} \times \boldsymbol{B}_{\mathcal{Y}^*}$ such that

$$\widehat{\Lambda}(c) = \int c \, d\mu = \int_{\boldsymbol{B}_{\mathcal{X}^*} \times \boldsymbol{B}_{\mathcal{Y}^*}} c(f', g') \, d\mu(f', g') \quad \text{for every } c \in C[\boldsymbol{B}_{\mathcal{X}^*} \times \boldsymbol{B}_{\mathcal{Y}^*}, \mathbb{F}].$$

In particular, for $F \in \mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y} \subseteq C[\boldsymbol{B}_{\mathcal{X}^*} \times \boldsymbol{B}_{\mathcal{Y}^*}, \mathbb{F}]$,

$$\Lambda(F) = \widehat{\Lambda}(F) = \int F \, d\mu \quad \text{for every} \quad F \in \mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y}.$$

More particularly, for an arbitrary single tensor $x \otimes y \subseteq \mathcal{X} \mathbin{\otimes_\vee} \mathcal{Y} \subseteq \mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y}$,

$$\Lambda(x \otimes y) = \int x \otimes y \, d\mu \quad \text{for every} \quad (x, y) \in \mathcal{X} \times \mathcal{Y}.$$

Consider again the isometric embedding $\mathcal{K}$ of Theorem 7.17(a_1,a_2). Thus each single tensor $x \otimes y$ in $\mathcal{X} \mathbin{\otimes_\vee} \mathcal{Y}$ is identified with a bounded bilinear form $\psi_{x \otimes y}$ in $\boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}]$ given by $\psi_{x \otimes y}(f, g) = f(x)\, g(y)$ for every (f, g) in $\mathcal{X}^* \times \mathcal{Y}^*$. So

$$\int x \otimes y \, d\mu = \int \psi_{x\otimes y} \, d\mu = \int_{B_{\mathcal{X}^*}\times B_{\mathcal{Y}^*}} \psi_{x\otimes y}(f', g') \, d\mu(f', g')$$

$$= \int_{B_{\mathcal{X}^*}\times B_{\mathcal{Y}^*}} f'(x)g'(y) \, d\mu(f', g') = \phi(x, y) \quad \text{for every} \quad (x, y) \in \mathcal{X}\times\mathcal{Y},$$

defining a bounded bilinear form $\phi\colon \mathcal{X}\times\mathcal{Y} \to \mathbb{F}$ (i.e., a form $\phi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$). As $(\mathcal{X}\,\widehat{\otimes}_\vee\mathcal{Y})^*$ is algebraically embedded in $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$ (cf. Remark 7.20(b)), $(\mathcal{X}\,\widehat{\otimes}_\vee\mathcal{Y})^*$ is regarded as an algebraic linear manifold of $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$, and

$$\Lambda(x \otimes y) = \phi(x, y) \quad \text{for every} \quad x \in \mathcal{X} \text{ and } y \in \mathcal{Y}$$

establishes the claimed identification: $\Lambda \in (\mathcal{X}\,\widehat{\otimes}_\vee\mathcal{Y})^* \hookrightarrow \phi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$ for

$$\phi(x, y) = \int_{B_{\mathcal{X}^*}\times B_{\mathcal{Y}^*}} (f' \otimes g')(x, y) \, d\mu(f', g') \quad \text{for every} \quad (x, y) \in \mathcal{X}\times\mathcal{Y}. \qquad \square$$

Note that, since norms in the above proof are preserved by the Riesz representation and by the Hahn–Banach extension, and since ϕ has been identified with the restriction of Λ to single tensors, we get

$$\|\mu\|_{\boldsymbol{X}} = \|\widehat{\Lambda}\|_{C[B_{\mathcal{X}^*}\times B_{\mathcal{Y}^*}, \mathbb{F}]^*} = \|\Lambda\|_{(\mathcal{X}\,\widehat{\otimes}_\vee\mathcal{Y})^*} \quad \text{and} \quad \|\phi\|_{\boldsymbol{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]} = \sup_{x\otimes y\neq 0} \frac{|\Lambda(x\otimes y)|}{\|x\otimes y\|_\vee}.$$

7.4 Additional Propositions

Throughout this section $\mathcal{X}$, $\mathcal{Y}$, $\mathcal{Z}$ are nonzero Banach spaces over a field $\mathbb{F}$. Equip the tensor product space $\mathcal{X} \otimes \mathcal{Y}$ with the injective and projective norms $\|\cdot\|_\vee$ and $\|\cdot\|_\wedge$. Set $\mathcal{X} \otimes_\vee \mathcal{Y} = (\mathcal{X} \otimes \mathcal{Y}, \|\cdot\|_\vee)$ and $\mathcal{X} \otimes_\wedge \mathcal{Y} = (\mathcal{X} \otimes \mathcal{Y}, \|\cdot\|_\wedge)$. Their completions $\mathcal{X}\,\widehat{\otimes}_\vee\mathcal{Y}$ and $\mathcal{X}\,\widehat{\otimes}_\wedge\mathcal{Y}$ are the injective and the projective tensor products. As before, $\cong$ means isometrically isomorphic. By a scalar sequence (function) we mean a scalar-valued sequence (function).

Proposition 7.A. *$\mathcal{X} \otimes_\wedge \mathcal{Y}$ is complete $\iff$ $\mathcal{X}$ or $\mathcal{Y}$ is finite-dimensional.*

The next two results exhibit a Banach-space projective version of the algebraic results in Corollary 3.11(a) and Proposition 3.D(c).

Proposition 7.B. $\mathcal{X}\,\widehat{\otimes}_\wedge\mathcal{Y} \cong \mathcal{Y}\,\widehat{\otimes}_\wedge\mathcal{X}$.

Proposition 7.C. $(\mathcal{X}\,\widehat{\otimes}_\wedge\mathcal{Y})\,\widehat{\otimes}_\wedge\mathcal{Z} \cong \mathcal{X}\,\widehat{\otimes}_\wedge(\mathcal{Y}\,\widehat{\otimes}_\wedge\mathcal{Z})$.

By a closed unit ball we mean one centered at the origin of a normed space. Also, in light of Theorem 3.13(b), if $\mathcal{X}$ and $\mathcal{Y}$ are linear spaces and if $M \subseteq \mathcal{X}$ and $N \subseteq \mathcal{Y}$ are subsets (but not linear manifolds) of $\mathcal{X}$ and $\mathcal{Y}$, respectively, then by $M \otimes N$ we mean the range of $\theta|_{M\times N}$, where $\theta\colon \mathcal{X}\times\mathcal{Y} \to \mathcal{X} \otimes \mathcal{Y}$ is the natural bilinear map associated with the tensor product space $\mathcal{X} \otimes \mathcal{Y}$.

Proposition 7.D. *Let $\mathcal{X}$ and $\mathcal{Y}$ be Banach spaces and let $\boldsymbol{B}_\mathcal{X}$ and $\boldsymbol{B}_\mathcal{Y}$ denote the closed unit balls in $\mathcal{X}$ and $\mathcal{Y}$. The closed unit ball in $\mathcal{X}\,\widehat{\otimes}_\wedge\mathcal{Y}$ is (isometrically isomorphic to) the closed convex hull of $\boldsymbol{B}_\mathcal{X} \otimes \boldsymbol{B}_\mathcal{Y}$.*

Set $\mathbb{F}^m(\mathcal{Z}) = \mathcal{Z}^m = \bigoplus_{i=1}^m \mathcal{Z} = \{(z_1, \ldots, z_m)\colon z_i \in \mathcal{Z},\ i = 1, \ldots, m\}$ for any linear space $\mathcal{Z}$ (cf. Sections 1.1 and 4.4). Proposition 3.F says that $\mathbb{F}^m \otimes \mathcal{Z} = \mathbb{F}^m(\mathcal{Z})$ for every linear space $\mathcal{Z}$. The next result goes along this line extending from $\mathbb{F}^m$ to ℓ_+^1. If $\mathcal{Z}$ is a Banach space, then let $\ell_+^1(\mathcal{Z})$ denote the Banach space of all absolutely summable $\mathcal{Z}$-valued sequences $\{z_k\}$ where the norm on $\ell_+^1(\mathcal{Z})$ is given by $\|\{z_k\}\|_1 = \sum_k \|z_k\|$. For $\mathcal{Z} = \mathbb{F}$ we get the standard Banach space $\ell_+^1 = \ell_+^1(\mathbb{F})$ of all absolutely summable scalar sequences.

Proposition 7.E. *$\ell_+^1 \mathbin{\widehat{\otimes}_\wedge} \mathcal{Z} \cong \ell_+^1(\mathcal{Z})$.*

Proposition 7.F. *$\ell_+^1 \mathbin{\widehat{\otimes}_\wedge} \ell_+^1 \cong \ell_+^1(\ell_+^1) \cong \ell_+^1$.*

If $\mathcal{M}$ and $\mathcal{N}$ are subspaces of $\mathcal{X}$ and $\mathcal{Y}$, then $\mathcal{M} \otimes_\wedge \mathcal{N}$ may not be a linear manifold of $\mathcal{X} \otimes_\wedge \mathcal{Y}$ under the projective norm (cf. Remark 7.5). Here are two examples where the projective tensor product of subspaces yields a subspace. The next result is a relevant application of Proposition 6.J from Section 6.4.

Proposition 7.G. *$\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}$ is a subspace of $\mathcal{X}^{**} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}^{**}$.*

Proposition 7.H. *$\ell_+^1 \mathbin{\widehat{\otimes}_\wedge} \mathcal{M}$ is a subspace of $\ell_+^1 \mathbin{\widehat{\otimes}_\wedge} \mathcal{Z}$ if $\mathcal{M}$ is a subspace of $\mathcal{Z}$.*

Proposition 7.E is a particular case of Proposition 7.I below for the counting measure. Let $L^1(\mathcal{Z}) = L^1(\mu, \mathcal{Z}) = L^1(\Omega, \mathcal{A}, \mu; \mathcal{Z})$ stand for the Banach space of all (equivalence class of) $\mathcal{Z}$-valued integrable functions with respect to a measure space $(\Omega, \mathcal{A}, \mu)$ (where μ is a measure on a σ-algebra $\mathcal{A}$ of subsets of a nonempty set Ω), and by a $\mathcal{Z}$-valued integrable function $F\colon \Omega \to \mathcal{Z}$ we mean it is integrable in the Bochner sense with $\|F\|_1 = \int \|F\|\, d\mu$ (see, e.g., [54, Proposition 10.H]). Again, for $\mathcal{Z} = \mathbb{F}$ we get the standard Banach space $L^1 = L^1(\mu)$ of all integrable scalar functions.

Proposition 7.I. *$L^1 \mathbin{\widehat{\otimes}_\wedge} \mathcal{Z} \cong L^1(\mathcal{Z})$.*

The graph G_L of a linear transformation $L\colon \mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y} \to \mathcal{Z}$,

$$G_L = \big\{(F, z) \in (\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}) \times \mathcal{Z}\colon\ z = L(F)\big\},$$

is a subset of the Cartesian product $(\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}) \times \mathcal{Z}$. Equip the Cartesian product with a product metric d generated by the norms in $\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}$ and in $\mathcal{Z}$ (as in Propositions 5.D and 6.F in Sections 5.4 and 6.4). The Closed Graph Theorem still holds for linear transformations on projective tensor products as follows.

Proposition 7.J. *If $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ are Banach spaces and $T\colon \mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y} \to \mathcal{Z}$ is linear, then T is continuous if and only if its graph G_T is closed.*

Up to this point all propositions in this section dealt with the projective norm. The next four propositions deal with the injective norm. For a Banach space $\mathcal{Z}$ let $c_0(\mathcal{Z})$ be the Banach space of all $\mathcal{Z}$-valued sequences converging to zero under the sup-norm. For $\mathcal{Z} = \mathbb{F}$ we get the standard Banach space $c_0 = c_0(\mathbb{F})$ of all scalar sequences converging to zero.

Proposition 7.K. $c_0 \mathbin{\widehat{\otimes}_\vee} \mathcal{Z} \cong c_0(\mathcal{Z})$.

Proposition 7.L. $c_0 \mathbin{\widehat{\otimes}_\vee} c_0 \cong c_0(c_0) \cong c_0$.

Let K be a compact set in a Hausdorff space, let $C(K)$ denote the Banach space of all $\mathbb{F}$-valued continuous functionals on K equipped with the sup-norm $\|\cdot\|_\infty$, and let $C(K,\mathcal{Z})$ be the Banach space of all $\mathcal{Z}$-valued continuous mappings $F\colon K \to \mathcal{Z}$ on K equipped with the sup-norm ($\|F\|_\infty = \sup_{x\in K}\|F(x)\|_\mathcal{Z}$).

Proposition 7.M. $C(K) \mathbin{\widehat{\otimes}_\vee} \mathcal{Z} \cong C(K,\mathcal{Z})$.

Proposition 7.N. $\mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y} \cong \mathcal{Y} \mathbin{\widehat{\otimes}_\vee} \mathcal{X}$.

Back to the projective norm, consider the dual of the projective tensor product. Theorem 7.12 ensures the existence of an isometric isomorphism

$$\mathcal{J}\colon (\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y})^* \to \boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathbb{F}].$$

Since $\mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}^* \subseteq (\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y})^* \cong \boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathbb{F}]$ (cf. Remark 7.13(g)), take the restriction of $\mathcal{J}$ to the linear space $\mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}^*$. As we saw in Remark 7.13(g), the above inclusion $\subseteq$ means linear-space embedding but not normed-space embedding, and so $\mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}^*$ is not regarded as a linear manifold of the normed space $(\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y})^*$ (they are equipped with different norms). Thus set

$$J = \mathcal{J}|_{\mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}^*}\colon \mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}^* \to \boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathbb{F}].$$

Proposition 7.O. *$J\colon \mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}^* \to \boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathbb{F}]$ has the following properties.*

(a) *J is linear.*

(b) *J is a contraction with $\|J\| = 1$.*

(c) *$\phi \in \boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathbb{F}]$ lies in the range $\mathcal{R}(J)$ of J if and only if there are $\mathcal{X}^*$-valued and $\mathcal{Y}^*$-valued sequences $\{f_k\}$ and $\{g_k\}$ for which the real sequence $\{\|f_k\|\,\|g_k\|\}$ is summable and*

$$\phi(x,y) = \sum\nolimits_k f_k(x)\,g_k(y) \quad \text{for every} \quad (x,y) \in \mathcal{X}\times\mathcal{Y}.$$

(d) *For each $\phi \in \mathcal{R}(J)$,*

$$\|\phi\|_N = \inf_{\{f_k\},\{g_k\},\ \phi=\Sigma_k f_k(\cdot)\,g_k(\cdot)} \sum\nolimits_k \|f_k\|\,\|g_k\|$$

defines a norm on the linear space $\mathcal{R}(J)$ for which $\|\phi\| \le \|\phi\|_N$.

The range $\mathcal{R}(J)$ of J is referred to as *the linear space of nuclear bilinear forms*. Thus set $\boldsymbol{b}_N[\mathcal{X}\times\mathcal{Y},\mathbb{F}] = \mathcal{R}(J) \subseteq \boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathbb{F}]$.

A bilinear form ϕ is nuclear *(i.e., $\phi \in \boldsymbol{b}_N[\mathcal{X}\times\mathcal{Y},\mathbb{F}]$) if and only if there are $\mathcal{X}^*$-valued and $\mathcal{Y}^*$-valued sequences $\{f_k\}$ and $\{g_k\}$ such that $\sum_k \|f_k\|\|g_k\| < \infty$ and $\phi(x,y) = \sum_k f_k(x)g_k(y)$ for every $(x,y) \in \mathcal{X}\times\mathcal{Y}$.*

The expression $\phi(x,y) = \sum_k f_k(x)g_k(y)$ is a *nuclear representation* of ϕ, and

$$\|\phi\|_N = \inf \sum\nolimits_k \|f_k\|\,\|g_k\| < \infty \quad \text{for every} \quad \phi \in \boldsymbol{b}_N[\mathcal{X}\times\mathcal{Y},\mathbb{F}]$$

(where the infimum is taken over all nuclear representations of ϕ) defines a norm on the linear space $\boldsymbol{b}_N[\mathcal{X}\times\mathcal{Y},\mathbb{F}]$. This norm dominates the usual norm on $\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathbb{F}]$ (i.e., $\|\phi\| \le \|\phi\|_N$), and $(\boldsymbol{b}_N[\mathcal{X}\times\mathcal{Y},\mathbb{F}], \|\cdot\|_N)$ is a Banach space. The linear contraction J of norm one in Proposition 7.O is not necessarily injective (i.e., $\mathcal{N}(J)$ is not necessarily equal to $\{0\}$), so J is not necessarily isometric, nor is it necessarily an isomorphism of $\mathcal{X}^* \widehat{\otimes}_\wedge \mathcal{Y}^*$ onto $\mathcal{R}(J) = \boldsymbol{b}_N[\mathcal{X}\times\mathcal{Y},\mathbb{F}]$. In fact, $\mathcal{R}(J)$ is identified with the quotient space of $\mathcal{X}^* \widehat{\otimes}_\wedge \mathcal{Y}^*$ modulo $\mathcal{N}(J)$, that is, $\boldsymbol{b}_N[\mathcal{X}\times\mathcal{Y},\mathbb{F}] \cong (\mathcal{X}^* \widehat{\otimes}_\wedge \mathcal{Y}^*)/\mathcal{N}(J)$ (see, e.g., [83, p. 40]).

From Theorem 7.17 we know that there are several interpretations of injective tensor products by embedding them into Banach spaces of bounded linear transformations following the patterns in Remark 4.9. In particular, we saw in Theorem 7.17(b_2) that $\mathcal{X}^* \widehat{\otimes}_\vee \mathcal{Y} \subseteq \mathcal{B}[\mathcal{X},\mathcal{Y}]$. This isometric embedding does not hold in general for the projective norm. The result below shows how far one can get along this line for the projective norm in the general case.

Proposition 7.P. *There is a transformation $K : \mathcal{X}^* \widehat{\otimes}_\wedge \mathcal{Y} \to \mathcal{B}[\mathcal{X},\mathcal{Y}]$ such that*

(a) *K is a linear contraction with $\|K\| = 1$,*

(b) *whose range $\mathcal{R}(K)$ is characterized as follows: $T \in \mathcal{B}[\mathcal{X},\mathcal{Y}]$ lies in $\mathcal{R}(K)$ if and only if there are $\mathcal{X}^*$-valued and $\mathcal{Y}$-valued sequences $\{f_k\}$ and $\{y_k\}$ for which the real sequence $\{\|f_k\|\,\|y_k\|\}$ is summable and*

$$Tx = \sum\nolimits_k f_k(x)\, y_k \qquad \text{for every} \quad x \in \mathcal{X},$$

(c) *and for each $T \in \mathcal{R}(K)$,*

$$\|T\|_N = \inf_{\{f_k\},\{y_k\},\, T=\Sigma_k f_k(\cdot)\, y_k} \sum\nolimits_k \|f_k\|\,\|y_k\|$$

defines a norm on the linear space $\mathcal{R}(K)$ for which $\|T\| \le \|T\|_N$.

A transformation $T \in \mathcal{B}[\mathcal{X},\mathcal{Y}]$ is *nuclear* if it lies in the range $\mathcal{R}(K)$ of the contraction $K \in \mathcal{B}[\mathcal{X}^* \widehat{\otimes}_\wedge \mathcal{Y}, \mathcal{B}[\mathcal{X},\mathcal{Y}]]$ defined in Proposition 7.P, and the range $\mathcal{R}(K)$ itself is called *the linear space of nuclear transformations*. Thus set

$$\mathcal{B}_N[\mathcal{X},\mathcal{Y}] = \mathcal{R}(K) \subseteq \mathcal{B}[\mathcal{X},\mathcal{Y}]. \quad \text{In particular,} \quad \mathcal{B}_N[\mathcal{X}] = \mathcal{B}_N[\mathcal{X},\mathcal{X}] \subseteq \mathcal{B}[\mathcal{X}].$$

A linear transformation T is nuclear *(i.e., $T \in \mathcal{B}_N[\mathcal{X},\mathcal{Y}]$) if and only if there exist $\mathcal{X}^*$-valued and $\mathcal{Y}$-valued sequences $\{f_k\}$ and $\{y_k\}$ such that $\sum_k \|f_k\|\,\|y_k\| < \infty$ and $Tx = \sum_k f_k(x)\, y_k$ for every $x \in \mathcal{X}$.*

The expression $Tx = \sum_k f_k(x)\, y_k$ is a *nuclear representation* of T, and

$$\|T\|_N = \inf \sum\nolimits_k \|f_k\|\,\|y_k\| \quad \text{for every} \quad T \in \mathcal{B}_N[\mathcal{X},\mathcal{Y}]$$

(where the infimum is taken over all nuclear representations of T) defines a norm on the linear space $\mathcal{B}_N[\mathcal{X},\mathcal{Y}]$, the *nuclear norm*, such that $\|T\| \le \|T\|_N$. As before, $(\mathcal{B}_N[\mathcal{X},\mathcal{Y}], \|\cdot\|_N)$ is a Banach space, the linear contraction K of norm one in Proposition 7.P is not necessarily injective — thus K may not be an isometry — and $\mathcal{R}(K)$ is identified with the quotient space of $\mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}$ modulo $\mathcal{N}(K)$, that is, $\mathcal{B}_N[\mathcal{X},\mathcal{Y}] \cong (\mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y})/\mathcal{N}(K)$ (see, e.g., [83, p. 41]).

It is worth noting the following special case. Suppose $\mathcal{Y}$ is such that $\mathcal{Y} \cong \mathcal{V}^*$ for some Banach space $\mathcal{V}$ (for instance, if $\mathcal{Y}$ is reflexive). By Theorems 7.12 and 6.3 we get $(\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{V})^* \cong \mathcal{b}[\mathcal{X}\times\mathcal{V}, \mathbb{F}] \cong \mathcal{B}[\mathcal{X},\mathcal{V}^*]$. Moreover, by Definition 7.1 the linear space $\mathcal{X}^* \otimes \mathcal{V}^*$ is a linear manifold of the linear space $(\mathcal{X} \mathbin{\otimes_\wedge} \mathcal{V})^*$, which is isometrically isomorphic to $(\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{V})^*$ by Theorem 5.1(f). If equipped with the projective norm, the linear space $\mathcal{X}^* \mathbin{\otimes_\wedge} \mathcal{V}^*$ remains algebraically embedded in the Banach space $(\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{V})^*$ (with different norms, see Remark 7.13(g)). So

$$\mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y} \cong \mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{V}^* \subseteq (\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{V})^* \cong \mathcal{B}[\mathcal{X},\mathcal{V}^*] \cong \mathcal{B}[\mathcal{X},\mathcal{Y}],$$

where $\subseteq$ means algebraic embedding and $\cong$ isometric isomorphism. Therefore

$$\mathcal{Y} \cong \mathcal{V}^* \text{ for some } \mathcal{V} \quad \Longrightarrow \quad \mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y} \subseteq \mathcal{B}[\mathcal{X},\mathcal{Y}].$$

Since there is an isometric isomorphism $\mathcal{K}\colon (\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{V})^* \to \mathcal{B}[\mathcal{X},\mathcal{Y}]$, the linear contraction $K\colon \mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y} \cong \mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{V}^* \to \mathcal{B}[\mathcal{X},\mathcal{Y}]$ of Proposition 7.P can in this case be taken as the restriction of the isometric isomorphism $\mathcal{K}$ to $\mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}$:

$$K = \mathcal{K}|_{\mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}}\colon \mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y} \to \mathcal{B}[\mathcal{X},\mathcal{Y}].$$

Thus if $\mathcal{Y} \cong \mathcal{V}^*$ (i.e., if the Banach space $\mathcal{Y}$ is the dual of a Banach space $\mathcal{V}$), then Proposition 7.P has a proof following the argument of Proposition 7.O.

Proposition 7.Q. *Let $\mathcal{X},\mathcal{Y},\mathcal{V},\mathcal{W}$ be Banach spaces, Suppose $T \in \mathcal{B}_N[\mathcal{X},\mathcal{Y}]$, Then for every $R \in \mathcal{B}[\mathcal{V},\mathcal{X}]$ and $L \in \mathcal{B}[\mathcal{Y},\mathcal{W}]$ it follows that $LTR \in \mathcal{B}_N[\mathcal{V},\mathcal{W}]$ and $\|LTR\|_N \le \|L\|\,\|T\|_N\|R\|$.*

This shows that $\mathcal{B}_N[\mathcal{X}]$ is an ideal of $\mathcal{B}[\mathcal{X}]$. Also, $\mathcal{B}_N[\mathcal{X},\mathcal{Y}] \subseteq \mathcal{B}_\infty[\mathcal{X},\mathcal{Y}]$ (nuclear transformations are compact, as a consequence of Proposition 5.H).

From now on let $\mathcal{X}$ be a Hilbert space, denote its inner product by $\langle\cdot\,;\cdot\rangle$, and take the Banach algebra $\mathcal{B}[\mathcal{X}]$ of all operators on $\mathcal{X}$. Consider the Fourier Series and the Riesz Representation Theorems (Propositions 5.T and 5.U). A functional f lies in $\mathcal{X}^*$ if and only if there exists a unique z in $\mathcal{X}$ (which does not depend on x) such that $f(x) = \langle x\,; z\rangle$ for every $x \in \mathcal{X}$ and $\|z\| = \|f\|$. This unique z is the Riesz representation of f. Thus a *nuclear operator* acting on a Hilbert space is defined as follows.

> *On a Hilbert space $\mathcal{X}$ an operator $T \in \mathcal{B}[\mathcal{X}]$ is* nuclear *(i.e., $T \in \mathcal{B}_N[\mathcal{X}]$) if and only if there are $\mathcal{X}$-valued sequences $\{z_k\}$ and $\{y_k\}$ such that $\sum_k \|z_k\|\,\|y_k\| < \infty$ and $Tx = \sum_k \langle x\,; z_k\rangle\, y_k$ for every $x \in \mathcal{X}$.*

For each $T \in \mathcal{B}[\mathcal{X}]$ let $T^* \in \mathcal{B}[\mathcal{X}]$ stand for its Hilbert-space adjoint.

Proposition 7.R. *Let $T \in \mathcal{B}[\mathcal{X}]$ be an operator on a Hilbert space $\mathcal{X}$.*

(a) *If there exist $\mathcal{X}$-valued sequences $\{z_k\}$ and $\{y_k\}$ such that*

$$Tx = \sum\nolimits_k \langle x\,; z_k\rangle y_k \qquad \textit{for every} \quad x \in \mathcal{X},$$

then

$$T^*x = \sum\nolimits_k \langle x\,; y_k\rangle z_k \qquad \textit{for every} \quad x \in \mathcal{X}.$$

(b) *T is nuclear if and only if its adjoint T^* is nuclear and $\|T^*\|_N = \|T\|_N$.*

For each operator $T \in \mathcal{B}[\mathcal{X}]$ on a Hilbert space $\mathcal{X}$, set $|T| = (T^*T)^{1/2} \in \mathcal{B}[\mathcal{X}]$. Take an arbitrary orthonormal basis $\{e_\gamma\}$ for $\mathcal{X}$. Consider the family of nonnegative numbers $\{\langle |T|e_\gamma\,; e_\gamma\rangle\}$. If $\sum_\gamma \langle |T|e_\gamma\,; e_\gamma\rangle < \infty$, then its value in $\mathbb{R}$ does not depend on the choice of the orthonormal basis $\{e_\gamma\}$. An operator $T \in \mathcal{B}[\mathcal{X}]$ is called *trace-class* if $\sum_\gamma \langle |T|e_\gamma\,; e_\gamma\rangle < \infty$ for any orthonormal basis $\{e_\gamma\}$. This implies that $\sum_\gamma |\langle Te_\gamma\,; e_\gamma\rangle| < \infty$ for every orthonormal basis $\{e_\gamma\}$. In this case $\{\langle Te_\gamma\,; e_\gamma\rangle\}$ is a summable family of scalars, and its sum is denoted by $\mathrm{tr}(T) = \sum_\gamma \langle Te_\gamma\,; e_\gamma\rangle$ in $\mathbb{F}$ and referred to as the *trace* of T (which again does not depend on the choice of the orthonormal basis). Let $\mathcal{B}_1[\mathcal{X}]$ denote the collection of all trace-class operators from $\mathcal{B}[\mathcal{X}]$, which is a linear manifold of the linear space $\mathcal{B}[\mathcal{X}]$. Actually, $\mathcal{B}_1[\mathcal{X}]$ is an ideal of the algebra $\mathcal{B}[\mathcal{X}]$, and every operator in $\mathcal{B}_1[\mathcal{X}]$ is compact (i.e., $\mathcal{B}_1[\mathcal{X}] \subseteq \mathcal{B}_\infty[\mathcal{X}]$). For each $T \in \mathcal{B}_1[\mathcal{X}]$, set

$$\|T\|_1 = \mathrm{tr}(|T|) = \sum\nolimits_\gamma \langle |T|e_\gamma\,; e_\gamma\rangle = \sum\nolimits_k \||T|^{1/2}e_\gamma\|^2.$$

The function $\|\cdot\|_1 \colon \mathcal{B}_1[\mathcal{X}] \to \mathbb{R}$ is a norm on the linear space $\mathcal{B}_1[\mathcal{X}]$, the *trace-norm*, and $\|T\| \le \|T\|_1$. $(\mathcal{B}_1[\mathcal{X}], \|\cdot\|_1)$ is a Banach space. Also, $T \in \mathcal{B}_1[\mathcal{X}]$ if and only if $|T| \in \mathcal{B}_1[\mathcal{X}]$ and $\||T|\|_1 = \|T\|_1$. Moreover, $T \in \mathcal{B}_1[\mathcal{X}]$ if and only if $T^* \in \mathcal{B}_1[\mathcal{X}]$ and $\|T^*\|_1 = \|T\|_1$. (See, e.g., [52, Problems 5.64 to 5.69].)

Proposition 7.S. *Suppose $\mathcal{X}$ is a Hilbert space.*

(a) *$\mathcal{B}_1[\mathcal{X}] = \mathcal{B}_N[\mathcal{X}]$ and $\|\cdot\|_1 = \|\cdot\|_N$.*

Take an arbitrary operator $T \in \mathcal{B}[\mathcal{X}]$.

(b) *$T \in \mathcal{B}_1[\mathcal{X}] \iff \sum_\gamma \|Te_\gamma\| < \infty$ for some orthonormal basis $\{e_\gamma\}$ for $\mathcal{X}$.*

(c) *$T \in \mathcal{B}_1[\mathcal{X}] \Longrightarrow \sum_\gamma |\langle Te_\gamma\,; e_\gamma\rangle| < \infty$ for every orthonormal basis $\{e_\gamma\}$ for $\mathcal{X}$.*

(d) *The converse of* (c) *fails.*

(*As $T \in \mathcal{B}_1[\mathcal{X}] \iff T^* \in \mathcal{B}_1[\mathcal{X}]$, we may replace $\sum_\gamma \|Te_\gamma\| < \infty$ in* (b) *by $\sum_\gamma \|T^*e_\gamma\| < \infty$ and $\sum_\gamma |\langle Te_\gamma\,; e_\gamma\rangle| < \infty$ in* (c) *by $\sum_\gamma |\langle T^*e_\gamma\,; e_\gamma\rangle| < \infty$.*)

Notes: The first two paragraphs of the Notes in Sections 3.4 and 4.4 summarized a little of the tensor product saga since Schatten's [85] monograph in 1950, which resulted from his thesis as of 1942. Thus such a tensor product theory is indeed a very young child. In 1953 the theory was further developed

by Grothendieck [28, 29] in his celebrated *Résumé*. This chapter gave a brief introduction to the notion of reasonable crossnorms, highlighting the elementary properties of their extremes, namely, the projective and injective norms.

A question which goes further, much further, than the commutation properties in Propositions 7.B and 7.N asks whether commutativity survives with mixed norms. It is clear by Corollary 3.10 that if $\mathcal{X}$ and $\mathcal{Y}$ are finite-dimensional normed spaces (where all norms are equivalent and normed spaces are complete), then $\mathcal{X} \otimes_\vee \mathcal{Y} \cong \mathcal{Y} \otimes_\wedge \mathcal{X}$. Grothendieck left as an open question whether $\mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y} \cong \mathcal{Y} \mathbin{\widehat{\otimes}_\wedge} \mathcal{X}$ holds for some pair of infinite-dimensional Banach spaces $\mathcal{X}$ and $\mathcal{Y}$. A solution was given by Pisier [73, 74], who exhibited in 1981 a separable infinite-dimensional Banach space $\mathcal{P}$, now called *Pisier space*, such that

$$\mathcal{P} \mathbin{\widehat{\otimes}_\vee} \mathcal{P} \cong \mathcal{P} \mathbin{\widehat{\otimes}_\wedge} \mathcal{P}.$$

This is a seminal result in itself, which shows in addition that all reasonable crossnorms on $\mathcal{P} \otimes \mathcal{P}$ are equivalent.

Proposition 7.A explains why completions are required, and Propositions 7.B and 7.C show that some elementary algebraic properties survive when the tensor product spaces are equipped with the projective norm (see, e.g., [83, Exercises 2.2–2.5] — Proposition 7.B also follows from Remark 7.7(d)). For Proposition 7.D see, e.g., [83, Proposition 2.3]. As we saw in Section 4.4, tensor products of finite-dimensional linear spaces are identified with Kronecker products (the identification there is attained by algebraic isomorphism $\cong$), and so $\mathbb{F}^m \otimes \mathbb{F}^n \cong \mathbb{F}^m(\mathbb{F}^n)$. This is generalized in Proposition 3.F to $\mathbb{F}^m \otimes \mathcal{Z} \cong \mathbb{F}^m(\mathcal{Z})$ for any linear space $\mathcal{Z}$, and it survives isometrically the extension from $\mathbb{F}^m$ to the infinite-dimensional space $\ell^1_+ = (\ell^1_+, \|\cdot\|_1)$ when tensor products are normed with the projective norm $\|\cdot\|_\wedge$, viz., $\ell^1_+ \mathbin{\widehat{\otimes}_\wedge} \mathcal{Z} \cong \ell^1_+(\mathcal{Z})$ for every Banach space $\mathcal{Z}$ as in Proposition 7.E (see, e.g., [83, Example 2.6]). Proposition 7.F is a consequence of Proposition 7.E (see also [83, Exercise 2.6]).

Proof of Proposition 7.G. Apply Proposition 6.J (Section 6.4) and Corollary 7.15 to verify that $\mathcal{X} \otimes_\wedge \mathcal{Y}$ is a linear manifold of $\mathcal{X}^{**} \otimes_\wedge \mathcal{Y}^{**}$, and therefore $\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}$ is a subspace of $\mathcal{X}^{**} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}^{**}$ by Theorem 5.1(g). □

Proof of Proposition 7.H. As $\mathcal{M}$ and $\mathcal{Z}$ are Banach spaces, $\ell^1_+ \mathbin{\widehat{\otimes}_\wedge} \mathcal{M} \cong \ell^1_+(\mathcal{M})$ and $\ell^1_+ \mathbin{\widehat{\otimes}_\wedge} \mathcal{Z} \cong \ell^1_+(\mathcal{Z})$ by Proposition 7.E. If $\mathcal{M}$ is a subspace of $\mathcal{Z}$, then $\ell^1_+(\mathcal{M})$ is a subspace of $\ell^1_+(\mathcal{Z})$, and so $\ell^1_+ \mathbin{\widehat{\otimes}_\wedge} \mathcal{M}$ is isometrically embedded in $\ell^1_+ \mathbin{\widehat{\otimes}_\wedge} \mathcal{Z}$. □

For Proposition 7.I see, e.g., [17, Theorem 1.1.10]. Proposition 7.J follows from the Closed Graph Theorem for bounded bilinear maps in Proposition 6.F by using the universal mapping property of Theorem 7.12 (see, e.g., [15, Exercise 3.10]). Proposition 7.K goes along the line of Proposition 7.E, now with respect to the injective norm $\|\cdot\|_\vee$ (see, e.g., [17, Theorem 1.1.11]). Proposition 7.L follows partially from Proposition 7.K (see, e.g., [83, Exercise 3.1]). Proposition 7.M can be viewed as the injective counterpart of Proposition 7.I (see, e.g., [17, Theorem 1.1.10]). Remark 7.7(d) implies Proposition 7.N.

Proof of Proposition 7.O. (a) Since a restriction of a linear transformation to a linear manifold of its domain is again linear, $J = \mathcal{J}|_{\mathcal{X}^* \widehat{\otimes}_\wedge \mathcal{Y}^*}$ is linear.

(b) Take an arbitrary $\Omega \in \mathcal{X}^* \widehat{\otimes}_\wedge \mathcal{Y}^* \subseteq (\mathcal{X} \widehat{\otimes}_\wedge \mathcal{Y})^*$. Consider the isometric isomorphism $\mathcal{J}\colon (\mathcal{X} \widehat{\otimes}_\wedge \mathcal{Y})^* \to \mathcal{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$. Since $\|\Omega\|_{*\wedge} \le \|\Omega\|_\wedge$ (Remark 7.13(g)),

$$\|J\| = \|\mathcal{J}|_{\mathcal{X}^* \widehat{\otimes}_\wedge \mathcal{Y}^*}\| = \sup_{\|\Omega\|_\wedge \le 1} \|\mathcal{J}(\Omega)\| = \sup_{\|\Omega\|_\wedge \le 1} \|\Omega\|_{*\wedge} \le \sup_{\|\Omega\|_\wedge \le 1} \|\Omega\|_\wedge = 1.$$

Take $(f, g) \in \mathcal{X}^* \times \mathcal{Y}^*$ with $\|f\| = \|g\| = 1$ so that $f \otimes g \in \mathcal{X}^* \widehat{\otimes}_\wedge \mathcal{Y}^* \subseteq (\mathcal{X} \widehat{\otimes}_\wedge \mathcal{Y})^*$ is such that $\|f \otimes g\|_\wedge = 1$. As we saw in Remark 7.13(h), $\phi = J(f \otimes g) = \mathcal{J}(f \otimes g)$ in $\mathcal{b}[\mathcal{X}\times\mathcal{Y}, \mathbb{F}]$ is such that $\phi(x, y) = f(x)\, g(y)$ for every $(x, y) \in \mathcal{X}\times\mathcal{Y}$. So $\|J(f \otimes g)\| = \|\phi\| = \|f\|\,\|g\| = 1$. Hence there exists $\Omega = f \otimes g \in \mathcal{X}^* \widehat{\otimes}_\wedge \mathcal{Y}^*$ with $\|\Omega\|_\wedge = 1$ such that $\|J(\Omega)\| = 1$. Then $\|J\| \ge 1$. Therefore $\|J\| = 1$.

(c) This was proved in Remark 7.13(h) since $\mathcal{R}(J) = \mathcal{J}(\mathcal{X}^* \widehat{\otimes}_\wedge \mathcal{Y}^*)$.

(d) It is readily verified that $\|\cdot\|_N$ is a norm on $\mathcal{R}(J)$. We show the triangle inequality. Take $\phi, \phi' \in \mathcal{R}(J)$ so that $\phi + \phi' = \sum_k f_k(\cdot) g_k(\cdot) + \sum_k f'_k(\cdot) g'_k(\cdot) = \sum_k f''_k(\cdot) g''_k(\cdot)$. Thus $\|\phi + \phi'\|_N = \inf_{f_k, g_k, f'_k, g'_k} (\sum_k \|f_k\|\,\|g_k\| + \|f'_k\|\,\|g'_k\|) = \inf_{f''_k, g''_k} \sum_k \|f''_k\|\,\|g''_k\| \le \inf_{f_k, g_k} \sum_k \|f_k\|\,\|g_k\| + \inf_{f'_k, g'_k} \sum_k \|f'_k\|\,\|g'_k\| = \|\phi\|_N + \|\phi'\|_N$. As for the norm inequality, if $\phi \in \mathcal{R}(J) = \mathcal{J}(\mathcal{X}^* \widehat{\otimes}_\wedge \mathcal{Y}^*)$, then there is $\Omega \in \mathcal{X}^* \widehat{\otimes}_\wedge \mathcal{Y}^* \subseteq (\mathcal{X} \widehat{\otimes}_\wedge \mathcal{Y})^*$ such that $\phi = \mathcal{J}(\Omega)$. So (Remark 7.13(g)),

$$\|\phi\| = \|\mathcal{J}(\Omega)\| = \|\Omega\|_{*\wedge} \le \|\Omega\|_\wedge = \inf \sum\nolimits_k \|f_k\|\,\|g_k\| = \|\phi\|_N. \qquad \square$$

Proof of Proposition 7.P. Let $\mathcal{X}^*$ be the dual of $\mathcal{X}$. By Theorem 7.9, take an arbitrary $F = \sum_k f_k \otimes y_k$ in $\mathcal{X}^* \widehat{\otimes}_\wedge \mathcal{Y}$ so that $\|F\|_\wedge = \inf \sum_k \|f_k\|\,\|y_k\|$. Associated with F, consider the transformation $\Psi_F : \mathcal{X} \to \mathcal{Y}$ given by

$$\Psi_F x = \sum\nolimits_k f_k(x)\, y_k \quad \text{for every} \quad x \in \mathcal{X},$$

which is clearly linear and does not depend on the representation $\sum_k f_k \otimes y_k$ of F (same argument as in Theorem 3.17(a)'s proof). Also, Ψ_F is bounded. In fact, $\|\Psi_F x\| \le \sum_k \|f_k\|\,\|y_k\|\,\|x\|$, so that $\|\Psi_F x\| \le \|F\|_\wedge \|x\|$, for every $x \in \mathcal{X}$. Thus Ψ_F lies in $\mathcal{B}[\mathcal{X}, \mathcal{Y}]$ for each $F \in \mathcal{X}^* \widehat{\otimes}_\wedge \mathcal{Y}$. This defines a transformation

$$K\colon \mathcal{X}^* \widehat{\otimes}_\wedge \mathcal{Y} \to \mathcal{B}[\mathcal{X}, \mathcal{Y}] \quad \text{such that} \quad K(F) = \Psi_F \ \text{ for every } \ F \in \mathcal{X}^* \widehat{\otimes}_\wedge \mathcal{Y}.$$

(a_1) It is clear that this K is linear. Indeed, $F + F' = \sum_k f_k \otimes y_k + \sum_k f'_k \otimes y'_k$, so that $\Psi_{F+F'} x = \sum_k f_k(x) \otimes y_k + \sum_k f'_k(x) \otimes y'_k = \Psi_F x + \Psi_{F'} x$ for each x, and hence $K(F + F') = \Psi_{F+F'} = \Psi_F + \Psi_{F'} = K(F) + \mathcal{K}(F')$. Similarly, $K(\alpha F) = \alpha F$.

(a_2) $\mathcal{K}$ is a contraction. In fact, as $\|\Psi_F x\| \le \|F\|_\wedge \|x\|$ for every $x \in \mathcal{X}$, $\|K(F)\| = \|\Psi_F\| = \sup_{\|x\|=1} \|\Psi_F x\| \le \sup_{\|x\|=1} \|F\|_\wedge \|x\| = \|F\|_\wedge$ for every $F \in \mathcal{X}^* \widehat{\otimes}_\wedge \mathcal{Y}$. So

$$K \in \mathcal{B}\big[\mathcal{X}^* \widehat{\otimes}_\wedge \mathcal{Y}, \mathcal{B}[\mathcal{X}, \mathcal{Y}]\big]$$

with $\|K\| = \sup_{\|F\|_\wedge = 1} \|K(F)\| \le 1$. Reversely, for $f \otimes y$ in $\mathcal{X}^* \widehat{\otimes}_\wedge \mathcal{Y}$ with $\|f\| = \|y\| = 1$ we get $\|\Psi_{f\otimes y} x\| = |f(x)|\,\|y\| = |f(x)|$. Hence $\|K(f \otimes y)\| = \|\Psi_{f\otimes y}\| =$

$\sup_{\|x\|=1} |f(x)| = \|f\| = 1 = \|f\|\,\|y\| = \|f \otimes y\|_\wedge$. Then there exists $F = f \otimes y$ in $\mathcal{X}^* \otimes_\wedge \mathcal{Y}$ with $\|F\|_\wedge = 1$ for which $\|K(F)\| = \|F\|_\wedge$. So $\|K\| \geq 1$. Thus $\|K\| = 1$.

(b) By definition $\mathcal{K}(f \otimes y) = T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ if and only if $Tx = \Psi_{f \otimes y}(x) = f(x)y$ for every $x \in \mathcal{X}$. Then for an arbitrary $F = \sum_k f_k \otimes y_k \in \mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}$,

$$\mathcal{K}(F) = T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}] \quad \text{ if and only if } \quad Tx = \Psi_F(x) = \sum\nolimits_k f_k(x)\, y_k$$

for every $x \in \mathcal{X}$, since $K(F) = \sum_k K(f_k \otimes y_k)$ because K is linear and bounded. This characterizes the range $\mathcal{R}(K)$ of K.

(c) It is easy to see that $\|\cdot\|_N$ is a norm on $\mathcal{R}(K)$. We show the triangle inequality only. Take $T, T' \in \mathcal{R}(K)$ so that $T + T' = \sum_k f_k(\cdot)y_k + \sum_k f'_k(\cdot)y'_k = \sum_k f''_k(\cdot)y''_k$. Thus $\|T + T'\|_N = \inf_{f_k, y_k, f'_k, y'_k} (\sum_k \|f_k\|\,\|y_k\| + \|f'_k\|\,\|y'_k\|) = \inf_{f''_k, y''_k} \sum_k \|f''_k\|\,\|y''_k\| \leq \inf_{f_k, y_k} \sum_k \|f_k\|\,\|y_k\| + \inf_{f'_k, y'_k} \sum_k \|f'_k\|\,\|y'_k\| = \|T\|_N + \|T'\|_N$. To verify the norm inequality, if $T \in \mathcal{R}(K)$, then by (b) there are $\{f_k\}$ and $\{y_k\}$ with $\{\|f_k\|\,\|y_k\|\}$ summable such that $T(x) = \sum_k f_k(x) y_k$, and hence $\|Tx\| = \|\sum_k f_k(x) y_k\|$, for every $x \in \mathcal{X}$. Therefore

$$\|T\| = \sup_{\|x\|=1} \|Tx\| \leq \sup_{\|x\|=1} \sum\nolimits_k \|f_k\|\,\|y_k\|\,\|x\| = \sum\nolimits_k \|f_k\|\,\|y_k\|$$

so that $\|T\| \leq \inf \sum_k \|f_k\|\,\|y_k\| = \|T\|_N$. □

Proof of Proposition 7.Q. This is immediate by definition, and it shows that $\mathcal{B}_N[\mathcal{X}]$ is a two-sided ideal of the algebra $\mathcal{B}[\mathcal{X}]$. Since $\sum_k \|f_k\|\,\|y_k\| < \infty$ and $Tx = \sum_k f_k(x)\, y_k$, set $g_k = f_k R \in \mathcal{V}^*$ and $w_k = L y_k \in \mathcal{W}$. So $\sum_k \|g_k\|\,\|y_k\| \leq \sum_k \|f_k\|\,\|y_k\|\,\|R\| < \infty$ and $TRx = \sum_k g_k(x)\, y_k$, as well as $\sum_k \|f_k\|\,\|w_k\| \leq \sum_k \|f_k\|\,\|y_k\|\,\|L\| < \infty$ and $LTx = \sum_k f_k(x)\, w_k$. □

Proof of Proposition 7.R. Take an operator $T \in \mathcal{B}[\mathcal{X}]$ on a Hilbert space $\mathcal{X}$.

(a) Suppose there are $\mathcal{X}$-valued sequences $\{z_k\}$ and $\{y_k\}$ such that

$$Tx = \sum\nolimits_k \langle x\,; z_k \rangle\, y_k \quad \text{ for every } \quad x \in \mathcal{X}.$$

Thus $\langle Tx\,; y \rangle = \big\langle \sum_k \langle x\,; z_k \rangle\, y_k ; y \big\rangle = \sum_k \langle x\,; z_k \rangle\, \langle y_k\,; y \rangle = \sum_k \big\langle x\,; \langle y\,; y_k \rangle\, z_k \big\rangle = \big\langle x\,; \sum_k \langle y\,; y_k \rangle\, z_k \big\rangle$ for every $x, y \in \mathcal{X}$, and hence (by uniqueness of the adjoint)

$$T^* y = \sum\nolimits_k \langle y\,; y_k \rangle\, z_k \quad \text{ for every } \quad y \in \mathcal{X}.$$

(b) An operator T lies in $\mathcal{B}_N[\mathcal{X}]$ if and only if there exist $\mathcal{X}$-valued sequences $\{z_k\}$ and $\{y_k\}$ such that $\sum_k \|z_k\|\,\|y_k\| < \infty$ and $Tx = \sum_k \langle x\,; z_k \rangle\, y_k$ for every $x \in \mathcal{X}$. Therefore if T lies in $\mathcal{B}_N[\mathcal{X}]$, then T^* lies in $\mathcal{B}_N[\mathcal{X}]$ by item (a). The converse holds since $T^{**} = T$. Moreover, with the infimum taken over all sequences $\{y_k\}$ and $\{z_k\}$ such that $\sum_k \|z_k\|\,\|y_k\| < \infty$,

$$\|T^*\|_N = \inf \sum\nolimits_k \|y_k\|\,\|z_k\| = \|T\|_N.$$ □

Proof of Proposition 7.S. Take any operator $T \in \mathcal{B}[\mathcal{X}]$ on a Hilbert space $\mathcal{X}$. First we need the following four auxiliary results.

Claim 1. $\||T|x\| = \|Tx\|$ for every $x \in \mathcal{X}$.

Proof of Claim 1. $\langle |T|^2 x\,;x\rangle = \langle T^*Tx\,;x\rangle = \langle Tx\,;Tx\rangle$ for every $x \in \mathcal{X}$. □

Claim 2. If T is compact, then there exist an orthonormal basis $\{e_\gamma\}$ for $\mathcal{X}$ and a family of nonnegative numbers $\{\mu_\gamma\}$ such that

$$|T|x = \sum\nolimits_\gamma \mu_\gamma \langle x\,;e_\gamma\rangle e_\gamma \quad \text{for every} \quad x \in \mathcal{X}.$$

Proof of Claim 2. The operator $|T| \in \mathcal{B}[\mathcal{X}]$ is nonnegative (thus self-adjoint and so normal) and compact. (In fact, as the class of compact operators from $\mathcal{B}[\mathcal{X}]$ is an ideal of $\mathcal{B}[\mathcal{X}]$, the nonnegative square root $|T|$ of the nonnegative compact $|T|^2$ is compact because $|T|^2 = T^*T$ is compact — see, e.g., [52, Problem 5.62].) Observe from Claim 1 that $\mathcal{N}(|T|) = \mathcal{N}(T)$. Then by the Spectral Theorem there is an orthonormal basis $\{e_k\}$ for the separable Hilbert space $\mathcal{H} = \mathcal{N}(T)^\perp$ consisting of eigenvectors of $|T|$ associated with positive eigenvalues $\{\mu_k\}$ of $|T|$ such that $|T|u = \sum_k \mu_k \langle u\,;e_k\rangle e_k$ for every $u \in \mathcal{H}$ (see, e.g., [52, Corollary 6.44]). Since $\mathcal{X} = \mathcal{H} \oplus \mathcal{N}$ with $\mathcal{N} = \mathcal{N}(T)$, there is an orthonormal basis $\{e_\gamma\} = \{e_k\} \cup \{e_\delta\}$ for $\mathcal{X}$, where $\{e_\delta\}$ is any orthonormal basis for the (not necessarily separable) Hilbert space $\mathcal{N}$ where $|T|v = 0$ for every $v \in \mathcal{N}$, so that the above expansion on $\mathcal{H}$ actually describes $|T|x$ for all $x = u \oplus v$ in $\mathcal{X} = \mathcal{H} \oplus \mathcal{N}$ with $|T|e_\delta = 0$, $Te_k = \mu_k e_k$, and $\mu_\gamma = 0$ if $\gamma \neq k$. □

Claim 3. An operator T lies in $\mathcal{B}_1[\mathcal{X}]$ if and only if there are unit $\mathcal{X}$-valued sequences $\{z_k\}$ and $\{y_k\}$ (i.e., $\|z_k\| = \|y_k\| = 1$) and a summable scalar sequence $\{\alpha_k\}$ (i.e., $\sum_k |\alpha_k| < \infty$) such that $Tx = \sum_k \alpha_k \langle x\,;z_k\rangle y_k$ for every x in $\mathcal{X}$. Moreover, $\|T\|_1 = \inf \sum_k |\alpha_k|$, where the infimum is taken over all summable scalar sequences for which the above expression for Tx holds.

Proof of Claim 3. If T lies in $\mathcal{B}_1[\mathcal{X}]$, then T is compact. Consider Claim 2 and the setup in its proof. Take the direct sums $T = T|_\mathcal{H} \oplus O$ and $|T| = |T||_\mathcal{H} \oplus O$ on $\mathcal{X} = \mathcal{H} \oplus \mathcal{N}$, where $T|_\mathcal{H} \in \mathcal{B}[\mathcal{H}]$ is injective. Thus $T|_\mathcal{H}$ has a polar decomposition $T|_\mathcal{H} = V|T||_\mathcal{H}$ where $V \in \mathcal{B}[\mathcal{H}]$ is an isometry (see, e.g., [52, Corollary 5.90]). Set $h_k = Ve_k$ so that $\{h_k\}$ is an $\mathcal{H}$-valued orthonormal sequence. Then

$$|T|x = \sum\nolimits_k \mu_k \langle x\,;e_k\rangle e_k \quad \text{and so} \quad Tx = \sum\nolimits_k \mu_k \langle x\,;e_k\rangle h_k \quad \text{for every} \quad x \in \mathcal{X},$$

since $\langle v\,;e_k\rangle = 0$ and hence $\langle x\,;e_k\rangle = \langle u\,;e_k\rangle$ for all k and every $x = u + v$ in $\mathcal{X}$ with $u \in \mathcal{H}$ and $v \in \mathcal{N}$. Moreover, with $\mu_\gamma = 0$ if $\gamma \neq k$,

$$\sum\nolimits_\gamma \mu_\gamma = \sum\nolimits_\gamma \sum\nolimits_\delta \mu_\gamma \langle e_\gamma\,;e_\delta\rangle e_\delta = \sum\nolimits_\gamma \langle |T|e_\gamma\,;e_\gamma\rangle = \|T\|_1 < \infty.$$

Therefore $T \in \mathcal{B}_1[\mathcal{X}]$ implies $Tx = \sum_k \mu_k \langle x\,;e_k\rangle h_k$ for every $x \in \mathcal{X}$, where $\{e_k\}$ and $\{h_k\}$ are unit $\mathcal{X}$-valued sequences and $\{\mu_k\}$ is a summable scalar sequence. Conversely, as $\mathcal{X} = \mathcal{N}(T)^\perp \oplus \mathcal{N}(T)$ we may regard the action of $T = T|_\mathcal{H} \oplus O$ and $|T| = |T||_\mathcal{H} \oplus O$ only on $\mathcal{H} = \mathcal{N}(T)^\perp$. So for notational simplicity, write T and $|T|$ for the injective operators $T|_\mathcal{H}$ and $|T||_\mathcal{H}$. Suppose

$$Tx = \sum\nolimits_k \alpha_k \langle x\,;z_k\rangle y_k \quad \text{for every} \quad x \in \mathcal{H},$$

for some $\mathcal{H}$-valued sequences $\{z_k\}$ and $\{y_k\}$ and for some scalar sequence $\{\alpha_k\}$ with $\|z_k\| = \|y_k\| = 1$ and $\sum_k |\alpha_k| < \infty$. Thus (by polar decomposition)

$$|T|x = V^*Tx = \sum\nolimits_k \alpha_k \langle x\,; z_k\rangle w_k \quad \text{for every} \quad x \in \mathcal{H},$$

where $w_k = V^*y_k$ so that $\|w_k\| \le 1$. For any orthonormal basis $\{e_j\}$ for $\mathcal{H}$,

$$\sum\nolimits_j \langle |T|e_j\,; e_j\rangle = \sum\nolimits_j \Big\langle \sum\nolimits_k \alpha_k \langle e_j\,; z_k\rangle w_k\,; e_j \Big\rangle = \sum\nolimits_j \sum\nolimits_k \alpha_k \langle e_j\,; z_k\rangle \langle w_k\,; e_j\rangle$$
$$\le \sum\nolimits_k |\alpha_k| \left| \sum\nolimits_j \langle e_j\,; z_k\rangle \langle w_k\,; e_j\rangle \right| = \sum\nolimits_k |\alpha_k|\, |\langle w_k\,; z_k\rangle| \le \sum\nolimits_k |\alpha_k| < \infty.$$

So T lies in $\mathcal{B}_1[\mathcal{X}]$. Moreover, since $Tx = \sum_k \alpha_k \langle x\,; z_k\rangle y_k$ for every $x \in \mathcal{H}$,

$$\sum\nolimits_k \mu_k = \|T\|_1 = \sum\nolimits_k \langle |T|e_k\,; e_k\rangle \le \sum\nolimits_k \| \, |T|e_k\| = \sum\nolimits_k \|Te_k\| \le \sum\nolimits_k |\alpha_k|,$$

for every orthonormal basis $\{e_k\}$ for $\mathcal{H}$. Therefore $\|T\|_1 = \min \sum_k |\alpha_k|$ where the minimum is taken over all scalar sequences $\{\alpha_k\}$ satisfying a (nuclear) representation of T as above. □

Claim 4. Take an arbitrary orthonormal basis $\{e_\gamma\}$ for $\mathcal{X}$. If $T \in \mathcal{B}[\mathcal{X}]$ is such that $\sum_\gamma \|Te_\gamma\|^2 < \infty$, then for every orthonormal basis $\{h_\gamma\}$ for $\mathcal{X}$,

$$\sum\nolimits_\gamma \|Te_\gamma\|^2 = \sum\nolimits_\gamma \|T^*h_\gamma\|^2.$$

Proof of Claim 4. For arbitrary orthonormal bases $\{e_\gamma\}$ and $\{h_\gamma\}$,

$$\|Te_\gamma\|^2 = \sum\nolimits_\delta |\langle Te_\gamma\,; h_\delta\rangle|^2 = \sum\nolimits_\delta |\langle h_\delta\,; Te_\gamma\rangle|^2 = \sum\nolimits_\delta |\langle T^*h_\delta\,; e_\gamma\rangle|^2$$

so that

$$\sum\nolimits_\gamma \|Te_\gamma\|^2 = \sum\nolimits_\gamma \sum\nolimits_\delta |\langle Te_\gamma\,; h_\delta\rangle|^2 = \sum\nolimits_\delta \sum\nolimits_\gamma |\langle T^*h_\delta\,; e_\gamma\rangle|^2 = \sum\nolimits_\delta \|T^*h_\delta\|^2$$

whenever any of the families $\{\|Te_\gamma\|^2\}$ or $\{\|T^*h_\delta\|^2\}$ is summable. □

(a) By the definition of nuclear operator on a Hilbert space and nuclear norm,

$$\mathcal{B}_1[\mathcal{X}] = \mathcal{B}_N[\mathcal{X}] \quad \text{and} \quad \|\cdot\|_1 = \|\cdot\|_N.$$

Indeed, the expression for Tx in Claim 3 is precisely a nuclear representation for T. So with the infimum taken over all nuclear representations of $T \in \mathcal{B}_N[\mathcal{X}]$, we get (for arbitrary unit sequences $\{z_k\}$, $\{y_k\}$ and summable scalar sequence $\{\alpha_k\}$) $\|T\|_N = \inf \sum_k |\alpha_k|\, \|z_k\|\, \|y_k\| = \inf \sum_k |\alpha_k| = \|T\|_1$, thus proving (a).

(b) For any orthonormal basis $\{e_\gamma\}$ for $\mathcal{X}$ we get, by Claim 1,

$$\sum\nolimits_\gamma \langle |T|e_\gamma\,; e_\gamma\rangle \le \sum\nolimits_\gamma \| \, |T|e_\gamma\| = \sum\nolimits_\gamma \|Te_\gamma\|.$$

So if $\sum_\gamma \|Te_\gamma\| < \infty$, then $T \in \mathcal{B}_1[\mathcal{X}]$. Conversely, since trace-class operators are compact, Claim 2 ensures the existence of an orthonormal basis $\{e_\gamma\}$ for $\mathcal{X}$ and a sequence of nonnegative numbers $\{\mu_\gamma\}$ such that

$$|T|x = \sum\nolimits_\gamma \mu_\gamma \langle x\,; e_\gamma\rangle e_\gamma \quad \text{for every} \quad x \in \mathcal{X},$$

where $|T|e_\gamma = \mu_\gamma e_\gamma$ (cf. proof of Claim 2). Then

$$\sum\nolimits_\gamma \|Te_\gamma\| = \sum\nolimits_\gamma \||T|e_\gamma\| = \sum\nolimits_\gamma \mu_\gamma$$

by Claim 1 again. If T lies in $\mathcal{B}_1[\mathcal{X}]$, then

$$\|T\|_1 = \sum\nolimits_\gamma \langle |T|e_\gamma; e_\gamma\rangle = \sum\nolimits_\gamma \mu_\gamma < \infty.$$

Therefore if $T \in \mathcal{B}_1[\mathcal{X}]$, then $\sum_\gamma \|Te_\gamma\| < \infty$. This concludes the proof of (b).

(c) Every $T \in \mathcal{B}_1[\mathcal{X}]$ can be factored as $T = AB$ where $A, B \in \mathcal{B}[\mathcal{X}]$ are such that $|A|^2$ and $|B|^2$ lie in $\mathcal{B}_1[\mathcal{X}]$ (see, e.g., [52, Problem 5.66(e)]). Equivalently, $A, B \in \mathcal{B}[\mathcal{X}]$ are such that $\sum_\gamma \||A|e_\gamma\|^2 < \infty$ and $\sum_\gamma \||B|e_\gamma\|^2 < \infty$ for an arbitrary orthonormal basis $\{e_\gamma\}$ for $\mathcal{X}$. Since $2|\langle Te_\gamma; e_\gamma\rangle| = 2|\langle ABe_\gamma; e_\gamma\rangle| = 2|\langle Be_\gamma; A^*e_\gamma\rangle| \le 2\|Be_\gamma\|\,\|A^*e_\gamma\| \le \|Be_\gamma\|^2 + \|A^*e_\gamma\|^2$, then by Claims 1 and 4

$$\begin{aligned} 2\sum\nolimits_\gamma |\langle Te_\gamma; e_\gamma\rangle| &\le \sum\nolimits_\gamma \|Be_\gamma\|^2 + \sum\nolimits_\gamma \|A^*e_\gamma\|^2 = \sum\nolimits_\gamma \|Be_\gamma\|^2 + \sum\nolimits_\gamma \|Ae_\gamma\|^2 \\ &= \sum\nolimits_\gamma \||B|e_\gamma\|^2 + \sum\nolimits_\gamma \||A|e_\gamma\|^2 < \infty. \end{aligned}$$

Thus if T lies in $\mathcal{B}_1[\mathcal{X}]$, then $\sum_\gamma |\langle Te_\gamma; e_\gamma\rangle| < \infty$, concluding the proof of (c).

(d) If $S \in \mathcal{B}[\mathcal{X}]$ is a unilateral shift of multiplicity one on a separable Hilbert space $\mathcal{X}$ (where orthonormal bases are countable), then (by definition) S shifts an orthonormal basis for $\mathcal{X}$, which means $Se_k = e_{k+1}$ for each positive integer k for some orthonormal basis $\{e_k\}$ for $\mathcal{X}$. Observe that

$$\langle Sh_k; h_k\rangle = 0 \text{ for every orthonormal basis } \{h_k\} \text{ for } \mathcal{X}.$$

Indeed, take any orthonormal basis $\{h_k\}$ for $\mathcal{X}$ and the Fourier expansion of h_k in terms of $\{e_k\}$, say $h_k = \sum_j \langle h_k; e_j\rangle e_j$. So $Sh_k = \sum_j \langle h_k; e_j\rangle Se_j$. Hence

$$\begin{aligned} \langle Sh_k; h_k\rangle &= \Big\langle \sum\nolimits_j \langle h_k; e_j\rangle e_{j+1}; \sum\nolimits_i \langle h_k; e_i\rangle e_i \Big\rangle = \sum\nolimits_{i,j} \langle h_k; e_j\rangle \overline{\langle h_k; e_i\rangle} \langle e_{j+1}; e_i\rangle \\ &= \sum\nolimits_j \langle h_k; e_j\rangle \overline{\langle h_k; e_{j+1}\rangle} = \sum\nolimits_j \langle e_{j+1}; h_k\rangle \overline{\langle e_j; h_k\rangle} = \langle e_{j+1}; e_j\rangle = 0, \end{aligned}$$

by considering the Fourier expansion of each e_k in terms of $\{h_k\}$. Therefore $\sum_k |\langle Sh_k; h_k\rangle| = 0$ for every orthonormal basis $\{h_k\}$. However, as S is an isometry, $S^*S = I$ and so $|S| = I$, the identity on $\mathcal{X}$. Thus $S \notin \mathcal{B}_1[\mathcal{X}]$ since $\sum_{k=1}^n \langle |S|h_k; h_k\rangle = n$ for every orthonormal basis $\{h_k\}$ for $\mathcal{X}$, thus proving (d).

This completes the proof of Proposition 7.S. (More along this line in [58].) □

For a notion of trace beyond Hilbert spaces see, e.g., [26, 44, 75].

Suggested Readings

Defant and Floret [15]
Diestel, Fourie and Swart [17]
Ryan [83]
Schatten [85]



8

Operator Norms

If $\mathcal{X}$, $\mathcal{Y}$, and $\mathcal{Z}$ are normed spaces over the same scalar field, and if T lies in $\mathcal{B}[\mathcal{X},\mathcal{Y}]$ and S lies in $\mathcal{B}[\mathcal{Y},\mathcal{Z}]$, then ST lies in $\mathcal{B}[\mathcal{X},\mathcal{Z}]$ and

$$\|ST\| \leq \|S\|\,\|T\|.$$

The above inequality is a crucial property shared by the induced uniform norm of bounded linear transformations, referred to as the *operator norm property*. For $\mathcal{X} = \mathcal{Y} = \mathcal{Z}$ this property ensures that $\mathcal{B}[\mathcal{X}]$ is a normed algebra, where the composition $S \circ T$ stands for the product ST of S and T in the algebra. This chapter deals with a counterpart of that inequality for the case of tensor products rather than ordinary products. When is it the case that

$$\|A \otimes B\| \leq \|A\|\,\|B\|\,?$$

8.1 Uniform Crossnorms

Each reasonable crossnorm in Chapter 7 has been defined for a given pair of normed spaces. From now on we consider a class of reasonable crossnorms that, among other attributes, are defined for every pair of normed spaces.

Throughout the chapter, $\mathcal{X},\mathcal{Y},\mathcal{V},\mathcal{W}$ stand for normed spaces (obviously, over the same field). Take the tensor product spaces $\mathcal{X} \otimes \mathcal{Y}$ and $\mathcal{V} \otimes \mathcal{W}$, let $A \in \mathcal{B}[\mathcal{X},\mathcal{V}]$ and $B \in \mathcal{B}[\mathcal{Y},\mathcal{W}]$ be bounded linear transformations, and consider the linear tensor product transformation $A \otimes B \in \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{V} \otimes \mathcal{W}]$ as in Definition 3.16 and Theorem 3.17.

Remark 8.1. *Bounded Tensor Product Transformation.* Let the tensor product spaces $\mathcal{X} \otimes \mathcal{Y}$ and $\mathcal{V} \otimes \mathcal{W}$ be equipped with reasonable crossnorms $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$, respectively, so that $\mathcal{X} \otimes_\alpha \mathcal{Y} = (\mathcal{X} \otimes \mathcal{Y}, \|\cdot\|_\alpha)$ and $\mathcal{V} \otimes_\beta \mathcal{W} = (\mathcal{V} \otimes \mathcal{W}, \|\cdot\|_\beta)$. For arbitrary bounded linear transformations $A \in \mathcal{B}[\mathcal{X},\mathcal{V}]$ and $B \in \mathcal{B}[\mathcal{Y},\mathcal{W}]$, consider their tensor product $A \otimes B \in \mathcal{L}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{V} \otimes_\beta \mathcal{W}]$.

If $A \otimes B$ is bounded (i.e., if $A \otimes B \in \mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{V} \otimes_\beta \mathcal{W}]$), then

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Universitext, https://doi.org/10.1007/978-3-031-34093-2_8

(a) $\|(A\otimes B)F\|_{\mathcal{V}\otimes_\beta\mathcal{W}} \le \|A\otimes B\|\,\|F\|_{\mathcal{X}\otimes_\alpha\mathcal{Y}}$ *for every* $F\in\mathcal{X}\otimes\mathcal{Y}$,

(b) $\|A\|\,\|B\| \le \|A\otimes B\|$.

Indeed, as happens with bounded linear operators between normed spaces,
(a) $\|A\otimes B\| = \inf\{c>0\colon \|(A\otimes B)F\|_{\mathcal{V}\otimes_\beta\mathcal{W}} \le c\|F\|_{\mathcal{X}\otimes_\alpha\mathcal{Y}},\ F\in\mathcal{X}\otimes_\alpha\mathcal{Y}\}$, and
(b) $\|A\otimes B\| = \sup_{\|F\|_\alpha=1}\|(A\otimes B)F\|_\beta \ge \sup_{\|x\otimes y\|_\alpha=1}\|(A\otimes B)(x\otimes y)\|_\beta = \sup_{\|x\|\|y\|=1}\|Ax\|\,\|By\| \ge \sup_{\|x\|=1}\|Ax\|\,\sup_{\|y\|=1}\|By\| = \|A\|\,\|B\|$.
(We have already used this argument in the proof of Theorem 7.8.)

Let $\|\cdot\|_\alpha$ be a reasonable crossnorm acting on the tensor product space $\mathcal{X}\otimes\mathcal{Y}$ for every pair of normed spaces $\mathcal{X}$ and $\mathcal{Y}$. So for arbitrary normed spaces $\mathcal{X},\mathcal{Y},\mathcal{V},\mathcal{W}$, it is a reasonable crossnorm on both tensor product spaces $\mathcal{X}\otimes\mathcal{Y}$ and $\mathcal{V}\otimes\mathcal{W}$. Take bounded linear transformations $A\in\mathcal{B}[\mathcal{X},\mathcal{V}]$ and $B\in\mathcal{B}[\mathcal{Y},\mathcal{W}]$ and consider their tensor product $A\otimes B\colon \mathcal{X}\otimes_\alpha\mathcal{Y}\to\mathcal{V}\otimes_\alpha\mathcal{W}$, which is linear, acting between the crossnormed spaces $\mathcal{X}\otimes_\alpha\mathcal{Y}$ and $\mathcal{V}\otimes_\alpha\mathcal{W}$.

Definition 8.2. (a) A *uniform crossnorm* (*plain uniform crossnorm*) $\|\cdot\|_\alpha$ is

(a_1) a reasonable crossnorm acting on every tensor product space such that

(a_2) for arbitrary normed spaces $\mathcal{X},\mathcal{Y},\mathcal{V},\mathcal{W}$, and for every pair of bounded linear transformations $A\in\mathcal{B}[\mathcal{X},\mathcal{V}]$ and $B\in\mathcal{B}[\mathcal{Y},\mathcal{W}]$, the tensor product $A\otimes B$ is bounded — that is, $A\otimes B\in\mathcal{B}[\mathcal{X}\otimes_\alpha\mathcal{Y},\mathcal{V}\otimes_\alpha\mathcal{W}]$ — and

$$\|A\otimes B\| \le \|A\|\,\|B\|.$$

This is the standard definition. It will be helpful to define two particular cases.

(b) A *finite-dimensional uniform crossnorm* (or a uniform crossnorm on tensor product spaces of finite-dimensional normed spaces) is a reasonable crossnorm on the class of all tensor product spaces of finite-dimensional normed spaces, which acts on all such spaces as a uniform crossnorm. In other words, such that (a_1) and (a_2) above hold for all finite-dimensional normed spaces $\mathcal{X},\mathcal{Y},\mathcal{V},\mathcal{W}$.

Tautologically, (a) implies (b). A useful intermediate notion reads as follows.

(c) A *uniform crossnorm on finite-dimensional spaces* is a reasonable crossnorm on every tensor product space (of arbitrary normed spaces) acting as a uniform crossnorm for bounded linear transformations defined on finite-dimensional normed spaces. In other words, $\|\cdot\|_\alpha$ is defined as a reasonable crossnorm on every tensor product space (of arbitrary normed spaces), but (a_2) above holds for finite-dimensional normed spaces $\mathcal{X},\mathcal{Y}$ and for arbitrary normed spaces $\mathcal{V},\mathcal{W}$ (i.e., (a_2) holds for bounded linear transformations with arbitrary finite-dimensional domains, with no restriction on codomains).

The difference between (b) and (c) above is this. In (c) it is assumed that (1) the reasonable crossnorm $\|\cdot\|_\alpha$ is defined on every tensor product space (of arbitrary normed spaces), (2) the tensor product transformation $A\otimes B$ lies in $\mathcal{B}[\mathcal{X}\otimes_\alpha\mathcal{Y},\mathcal{V}\otimes_\alpha\mathcal{W}]$ for every $A\in\mathcal{B}[\mathcal{X},\mathcal{V}]$ and $B\in\mathcal{B}[\mathcal{Y},\mathcal{W}]$ if $\mathcal{X}$ and $\mathcal{Y}$ are finite-dimensional and, in this case, (3) the norm inequality in (a_2) holds true.

Note that $\mathcal{R} = \mathcal{R}(A) \subseteq \mathcal{V}$ and $\mathcal{S} = \mathcal{R}(B) \subseteq \mathcal{W}$ are finite-dimensional spaces (by the rank and nullity identity in Proposition 1.E), and so $\mathcal{R} \otimes_\alpha \mathcal{S}$ is a finite-dimensional (algebraic) linear manifold of (the possibly infinite-dimensional) $\mathcal{V} \otimes_\alpha \mathcal{W}$. Thus we may regard the bounded linear transformations $A \in \mathcal{B}[\mathcal{X}, \mathcal{R}]$, $B \in \mathcal{B}[\mathcal{Y}, \mathcal{S}]$, and $A \otimes B \in \mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{R} \otimes_\alpha \mathcal{S}]$, as operating in a finite-dimensional setup. In (b) a similar scheme is assumed, but it is restricted to a further condition where $\mathcal{V}$ and $\mathcal{W}$ are required to be finite-dimensional, and $\|\cdot\|_\alpha$ is not assumed to act on infinite-dimensional tensor product spaces.

Theorem 8.3. *Let $\|\cdot\|_\alpha$ be an assignment of a reasonable crossnorm to every tensor product space. Let $\mathcal{X}, \mathcal{Y}, \mathcal{V}, \mathcal{W}$ be arbitrary normed spaces.*

(a) *The norm $\|\cdot\|_\alpha$ is a uniform crossnorm of one of the types defined in Definition 8.2 if and only if*

$$\|A \otimes B\| = \|A\| \, \|B\|$$

whenever $A \otimes B \in \mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{V} \otimes_\alpha \mathcal{W}]$ for $A \in \mathcal{B}[\mathcal{X}, \mathcal{V}]$ and $B \in \mathcal{B}[\mathcal{Y}, \mathcal{W}]$.

(b) *If $\mathcal{M}$ and $\mathcal{N}$ are linear manifolds of arbitrary normed spaces $\mathcal{X}$ and $\mathcal{Y}$, respectively, and if $\|\cdot\|_\alpha$ is a uniform crossnorm (or a uniform crossnorm on finite-dimensional spaces), then*

$$\|F\|_{\mathcal{X} \otimes_\alpha \mathcal{Y}} \leq \|F\|_{\mathcal{M} \otimes_\alpha \mathcal{N}} \quad \text{for every} \quad F \in \mathcal{M} \otimes \mathcal{N}.$$

Note: in case of a uniform crossnorm on finite-dimensional spaces as in Definition 8.2(c), $\mathcal{M}$ and $\mathcal{N}$ are assumed to be finite-dimensional.

(c) *If $\|\cdot\|_\alpha$ is a uniform crossnorm, then*

$$\mathcal{B}[\mathcal{X}, \mathcal{V}] \otimes \mathcal{B}[\mathcal{Y}, \mathcal{W}] \subseteq \mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{V} \otimes_\alpha \mathcal{W}].$$

Note: this extends a condition satisfied for every reasonable crossnorm,

$$\mathcal{X}^* \otimes \mathcal{Y}^* \subseteq \mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathbb{F} \otimes_\alpha \mathbb{F}] \cong (\mathcal{X} \otimes_\alpha \mathcal{Y})^*.$$

Proof. (a) The identity in item (a) is straightforward from Remark 8.1(b) and Definition 8.2 (regarding the nature — finite-dimensional or not — of $\mathcal{X}, \mathcal{Y}, \mathcal{V}, \mathcal{W}$ according to each of the options (a), (b) or (c) in Definition 8.2).

(b) Suppose $\|\cdot\|_\alpha$ is a uniform crossnorm or a uniform crossnorm on finite-dimensional spaces as in Definition 8.2(a,c). Let $\mathcal{M}$ and $\mathcal{N}$ be (finite-dimensional, in the case of Definition 8.2(c)) linear manifolds of $\mathcal{X}$ and $\mathcal{Y}$. Take the natural isometric embeddings $I_\mathcal{M} \colon \mathcal{M} \hookrightarrow \mathcal{X}$ and $I_\mathcal{N} \colon \mathcal{N} \hookrightarrow \mathcal{Y}$. Since $\|\cdot\|_\alpha$ is uniform, $I_\mathcal{M} \in \mathcal{B}[\mathcal{M}, \mathcal{X}]$, $I_\mathcal{N} \in \mathcal{B}[\mathcal{N}, \mathcal{Y}]$, $\|I_\mathcal{M}\| = \|I_\mathcal{N}\| = 1$, it follows that $I_\mathcal{M} \otimes I_\mathcal{N}$ lies in $\mathcal{B}[\mathcal{M} \otimes_\alpha \mathcal{N}, \mathcal{X} \otimes_\alpha \mathcal{Y}]$ with $\|I_\mathcal{M} \otimes I_\mathcal{N}\| = \|I_\mathcal{M}\| \, \|I_\mathcal{N}\| = 1$. So for $F \in \mathcal{M} \otimes \mathcal{N}$,

$$\|F\|_{\mathcal{X} \otimes_\alpha \mathcal{Y}} = \|(I_\mathcal{M} \otimes I_\mathcal{N})F\|_{\mathcal{X} \otimes_\alpha \mathcal{Y}} \leq \|I_\mathcal{M} \otimes I_\mathcal{N}\| \, \|F\|_{\mathcal{M} \otimes_\alpha \mathcal{N}} = \|F\|_{\mathcal{M} \otimes_\alpha \mathcal{N}}.$$

(c) An element in the tensor product space $\mathcal{B}[\mathcal{X}, \mathcal{V}] \otimes \mathcal{B}[\mathcal{Y}, \mathcal{W}]$ is a finite sum $\sum_i A_i \otimes B_i$ of single tensors with $A_i \in \mathcal{B}[\mathcal{X}, \mathcal{V}]$ and $B_i \in \mathcal{B}[\mathcal{Y}, \mathcal{W}]$. Since $\|\cdot\|_\alpha$

is a uniform crossnorm, each single tensor $A_i \otimes B_i$ lies in $\mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{V} \otimes_\alpha \mathcal{W}]$ by Definition 8.2(a). So $\sum_i A_i \otimes B_i$ lies in the linear space $\mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{V} \otimes_\alpha \mathcal{W}]$, leading to the first claimed inclusion (compare with Theorem 3.17 (d)). For the particular case, set $\mathcal{V} = \mathcal{W} = \mathbb{F}$ and note that $\mathbb{F} \otimes_\alpha \mathbb{F} \cong \mathbb{F}$ for every reasonable crossnorm $\|\cdot\|_\alpha$ on the one-dimensional tensor product space $\mathbb{F} \otimes \mathbb{F}$ (i.e., such that $\|a \otimes b\|_\alpha = \|a \cdot b\|_\alpha \le |a|\,|b|$ for every $a, b \in \mathbb{F}$). Thus $\mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathbb{F} \otimes_\alpha \mathbb{F}] \cong \mathcal{B}[(\mathcal{X} \otimes \mathcal{Y}, \|\cdot\|_\alpha), (\mathbb{F}, |\cdot|)] = (\mathcal{X} \otimes_\alpha \mathcal{Y})^*$ (cf. Remark 3.18). □

Let $\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}$ and $\mathcal{V} \mathbin{\widehat{\otimes}_\alpha} \mathcal{W}$ be the completions of $\mathcal{X} \otimes_\alpha \mathcal{Y}$ and $\mathcal{V} \otimes_\alpha \mathcal{W}$ (where $\|\cdot\|_\alpha$ is a reasonable crossnorm on $\mathcal{X} \otimes \mathcal{Y}$ and on $\mathcal{V} \otimes \mathcal{W}$). Suppose $A \otimes B$ lies in $\mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{V} \otimes_\alpha \mathcal{W}]$ for some $A \in \mathcal{B}[\mathcal{X}, \mathcal{V}]$ and $B \in \mathcal{B}[\mathcal{Y}, \mathcal{W}]$. Let $A \mathbin{\widehat{\otimes}} B$ in $\mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}, \mathcal{V} \mathbin{\widehat{\otimes}_\alpha} \mathcal{W}]$ be the extension over completion of $A \otimes B$. Recall that $\|A \mathbin{\widehat{\otimes}} B\| = \|A \otimes B\|$, and hence $\|\cdot\|_\alpha$ is a uniform crossnorm if and only if $\|A \mathbin{\widehat{\otimes}} B\| = \|A\|\,\|B\|$ for every A and B by Theorem 8.3(a).

Theorem 8.4. *Projective and injective norms are uniform crossnorms.*

Proof. Let $\mathcal{X}, \mathcal{Y}, \mathcal{V}, \mathcal{W}$ be normed spaces. Suppose a reasonable crossnorm $\|\cdot\|_\alpha$ acts on both tensor product spaces $\mathcal{X} \otimes \mathcal{Y}$ and $\mathcal{V} \otimes \mathcal{W}$. Take $A \in \mathcal{B}[\mathcal{X}, \mathcal{V}]$ and $B \in \mathcal{B}[\mathcal{Y}, \mathcal{W}]$. Consider their tensor product $A \otimes B$ in $\mathcal{L}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{V} \otimes_\alpha \mathcal{W}]$. Let $F = \sum_i x_i \otimes y_i$ be any (finite) representation of an arbitrary $F \in \mathcal{X} \otimes_\alpha \mathcal{Y}$.

(a) Thus for every representation $\sum_i x_i \otimes y_i$ of $F \in \mathcal{X} \otimes \mathcal{Y}$,

$$\begin{aligned}\|(A \otimes B)F\|_\alpha &= \Big\|\sum\nolimits_i Ax_i \otimes By_i\Big\|_\alpha \le \sum\nolimits_i \|Ax_i \otimes By_i\|_\alpha \\ &= \sum\nolimits_i \|Ax_i\|\,\|By_i\| \le \|A\|\,\|B\| \sum\nolimits_i \|x_i\|\,\|y_i\|.\end{aligned}$$

By definition, the injective $\|\cdot\|_\vee$ and projective $\|\cdot\|_\wedge$ norms can be assigned to every tensor product of normed spaces. Then, to begin with, set $\|\cdot\|_\alpha = \|\cdot\|_\wedge$. In this case $\|F\|_\wedge = \inf \sum_i \|x_i\|\,\|y_i\|$, and so by the above inequality we get

$$\|(A \otimes B)F\|_\wedge \le \|A\|\,\|B\| \inf \sum\nolimits_i \|x_i\|\,\|y_i\| = \|A\|\,\|B\|\,\|F\|_\wedge.$$

Therefore $A \otimes B \in \mathcal{L}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{V} \otimes_\wedge \mathcal{W}]$ is bounded and $\|A \otimes B\| \le \|A\|\,\|B\|$. Thus $\|\cdot\|_\wedge$ is a uniform crossnorm by Definition 8.2.

(b) Set $\|\cdot\|_\alpha = \|\cdot\|_\vee$. Consider the adjoints $A^* \in \mathcal{B}[\mathcal{V}^*, \mathcal{X}^*]$ and $B^* \in \mathcal{B}[\mathcal{W}^*, \mathcal{Y}^*]$. Take the balls $\boldsymbol{B}_{\mathcal{V}^*} = \{\widetilde{f} \in \mathcal{V}^* \colon \|\widetilde{f}\| \le 1\}$ and $\boldsymbol{B}_{\mathcal{W}^*} = \{\widetilde{g} \in \mathcal{W}^* \colon \|\widetilde{g}\| \le 1\}$. Thus

$$\begin{aligned}\|(A \otimes B)F\|_\vee &= \Big\|\sum\nolimits_i Ax_i \otimes By_i\Big\|_\vee = \sup_{\widetilde{f} \in \boldsymbol{B}_{\mathcal{V}^*},\, \widetilde{g} \in \boldsymbol{B}_{\mathcal{W}^*}} \Big|\sum\nolimits_i \widetilde{f}(Ax_i)\,\widetilde{g}(By_i)\Big| \\ &= \sup_{\widetilde{f} \in \boldsymbol{B}_{\mathcal{V}^*},\, \widetilde{g} \in \boldsymbol{B}_{\mathcal{W}^*}} \Big|\sum\nolimits_i (A^*\widetilde{f})(x_i)\,(B^*\widetilde{g})(y_i)\Big| \\ &= \sup_{\widetilde{f} \in \boldsymbol{B}_{\mathcal{V}^*},\, \widetilde{g} \in \boldsymbol{B}_{\mathcal{W}^*}} \|A^*\|\,\|B^*\| \Big|\sum\nolimits_i \Big(\tfrac{A^*}{\|A^*\|}\widetilde{f}\Big)(x_i)\,\Big(\tfrac{B^*}{\|B^*\|}\widetilde{g}\Big)(y_i)\Big|.\end{aligned}$$

For each $\widetilde{f} \in \mathcal{V}^*$ and $\widetilde{g} \in \mathcal{W}^*$ set $f' = \frac{A^*\widetilde{f}}{\|A^*\|} \in \mathcal{X}^*$ and $g' = \frac{B^*\widetilde{g}}{\|B^*\|} \in \mathcal{Y}^*$, and take the normalized images of the unit balls $\boldsymbol{B}_{\mathcal{V}^*}$ and $\boldsymbol{B}_{\mathcal{W}^*}$ under A^* and B^*,

$$\boldsymbol{B}'_{\mathcal{X}^*} = \tfrac{1}{\|A^*\|} A^*(\boldsymbol{B}_{\mathcal{V}^*}) = \Big\{ f' = \tfrac{A^* \widetilde{f}}{\|A^*\|} \in \mathcal{X}^* \colon \widetilde{f} \in \mathcal{V}^*,\ \|\widetilde{f}\| \le 1 \Big\},$$

$$\boldsymbol{B}'_{\mathcal{Y}^*} = \tfrac{1}{\|B^*\|} B^*(\boldsymbol{B}_{\mathcal{W}^*}) = \Big\{ g' = \tfrac{B^* \widetilde{g}}{\|B^*\|} \in \mathcal{Y}^* \colon \widetilde{g} \in \mathcal{W}^*,\ \|\widetilde{g}\| \le 1 \Big\}.$$

Then (as $\widetilde{f} \in \boldsymbol{B}_{\mathcal{V}^*} \Leftrightarrow f' \in \boldsymbol{B}'_{\mathcal{X}^*}$ and $\widetilde{g} \in \boldsymbol{B}_{\mathcal{W}^*} \Leftrightarrow g' \in \boldsymbol{B}'_{\mathcal{Y}^*}$)

$$\|(A \otimes B)F\|_\vee = \|A^*\|\,\|B^*\| \sup_{f' \in \boldsymbol{B}'_{\mathcal{X}^*},\, g' \in \boldsymbol{B}'_{\mathcal{Y}^*}} \Big| \sum\nolimits_i f'(x_i)\, g'(y_i) \Big|.$$

Take the balls $\boldsymbol{B}_{\mathcal{X}^*} = \{f \in \mathcal{X}^* \colon \|f\| \le 1\}$ and $\boldsymbol{B}_{\mathcal{Y}^*} = \{g \in \mathcal{Y}^* \colon \|g\| \le 1\}$ so that $\boldsymbol{B}'_{\mathcal{X}^*} \subseteq \boldsymbol{B}_{\mathcal{X}^*}$ and $\boldsymbol{B}'_{\mathcal{Y}^*} \subseteq \boldsymbol{B}_{\mathcal{Y}^*}$ (as $\|\widetilde{f}\| \le 1 \Leftrightarrow \|f'\| \le 1$ and $\|\widetilde{g}\| \le 1 \Leftrightarrow \|g'\| \le 1$). Hence (since $\|A^*\| = \|A\|$ and $\|B^*\| = \|B\|$) we get

$$\|(A \otimes B)F\|_\vee \le \|A\|\,\|B\| \sup_{f \in \boldsymbol{B}_{\mathcal{X}^*},\, g \in \boldsymbol{B}_{\mathcal{Y}^*}} \Big| \sum\nolimits_i f(x_i)\, g(y_i) \Big| = \|A\|\,\|B\|\,\|F\|_\vee.$$

Therefore $A \otimes B \in \mathcal{L}[\mathcal{X} \otimes_\vee \mathcal{Y}, \mathcal{V} \otimes_\vee \mathcal{W}]$ is bounded and $\|A \otimes B\| \le \|A\|\,\|B\|$. Thus $\|\cdot\|_\vee$ is a uniform crossnorm by Definition 8.2. □

Corollary 8.5. *If $A \in \mathcal{B}[\mathcal{X}, \mathcal{V}]$ and $B \in \mathcal{B}[\mathcal{Y}, \mathcal{W}]$ are quotient transformations, then $A \mathbin{\widehat{\otimes}} B \in \mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}}_\wedge \mathcal{Y}, \mathcal{V} \mathbin{\widehat{\otimes}}_\wedge \mathcal{W}]$ is a quotient transformation.*

Proof. Let $\mathcal{X}, \mathcal{Y}, \mathcal{V}, \mathcal{W}$ be normed spaces. Take $A \in \mathcal{B}[\mathcal{X}, \mathcal{V}]$ and $B \in \mathcal{B}[\mathcal{X}, \mathcal{W}]$. Equip the tensor product spaces $\mathcal{X} \otimes \mathcal{Y}$ and $\mathcal{V} \otimes \mathcal{W}$ with the projective norm. Thus $A \otimes B \in \mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{V} \otimes_\wedge \mathcal{W}]$ and $\|A \otimes B\| = \|A\|\,\|B\|$ by Theorems 8.3 and 8.4. Suppose $A \colon \mathcal{X} \to \mathcal{V}$ and $B \colon \mathcal{Y} \to \mathcal{W}$ are quotient transformations, and take an arbitrary $\varepsilon > 0$. Then, according to Definition 5.12,

(i) A and B are surjective and hence $A \otimes B$ is surjective by Theorem 3.19(e),

(ii) for every $0 \ne v \in \mathcal{V}$ and every $0 \ne w \in \mathcal{W}$ there are $x \in A^{-1}(\{v\}) \subseteq \mathcal{X}$ and $y \in B^{-1}(\{w\}) \subseteq \mathcal{Y}$ such that $\|x\|^2 < (1+\varepsilon)\|v\|^2$ and $\|y\|^2 < (1+\varepsilon)\|w\|^2$,

(iii) $\|A\| = \|B\| = 1$ and so $\|A \otimes B\| = 1$ by Theorems 5.13(e), 8.3 and 8.4.

Take an arbitrary $0 \ne \mathrm{G} \in \mathcal{V} \otimes_\wedge \mathcal{W}$. Consider the definition of $\|\mathrm{G}\|_\wedge$ and let $\sum_i v_i \otimes w_i \in \mathcal{V} \otimes_\wedge \mathcal{W}$ be a representation of G with nonzero single tensors such that $\sum_i \|v_i\|\,\|w_i\| < \|\mathrm{G}\|_\wedge + \varepsilon$. By (i) and (ii), for each $0 \ne v_i \in \mathcal{V}$ and $0 \ne w_i \in \mathcal{W}$ there are $x_i \in A^{-1}(\{v_i\}) \subseteq \mathcal{X}$ and $y_i \in B^{-1}(\{w_i\}) \subseteq \mathcal{Y}$ such that

(1) $\|x_i\|^2 \le (1+\varepsilon)\|v_i\|^2$ and $\|y_i\|^2 \le (1+\varepsilon)\|w_i\|^2$.

Associated with such a $\mathrm{G} \in \mathcal{V} \otimes_\wedge \mathcal{W}$, set $F = \sum_i x_i \otimes y_i \in \mathcal{X} \otimes \mathcal{Y}$ so that $(A \otimes B)F = \sum_i Ax_i \otimes By_i = \sum_i v_i \otimes w_i = \mathrm{G}$ by Definition 3.16. Then

(2) $F = \sum_i x_i \otimes y_i \in (A \otimes B)^{-1}(\{\mathrm{G}\}) \subseteq (\mathcal{X} \otimes_\wedge \mathcal{Y})$.

Moreover, by property (iii),

(3) $\|\mathrm{G}\|_\wedge = \|(A \otimes B)F\|_\wedge \le \|F\|_\wedge$.

Also, by property (1),

$$\|F\|_\wedge = \Big\|\sum_i x_i \otimes y_i\Big\|_\wedge \le \sum_i \|x_i\|\|y_i\| \le (1+\varepsilon)\sum_i \|v_i\|\|w_i\| < (1+\varepsilon)(\|G\|_\wedge + \varepsilon).$$

Thus properties (1), (2) and (3) say that for an arbitrary nonzero $G \in \mathcal{V} \otimes_\wedge \mathcal{W}$ and an arbitrary $\varepsilon > 0$ there exists $F = F_{G,\varepsilon} \in (A \otimes B)^{-1}(\{G\})$ for which

$$\|G\|_\wedge \le \|F\|_\wedge \le (1+\varepsilon)\,(\|G\|_\wedge + \varepsilon).$$

Hence

$$\|G\|_\wedge = \inf_{F \in (A \otimes B)^{-1}(\{G\})} \|F\|_\wedge.$$

That is, $A \otimes B$ in $\mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{V} \otimes_\wedge \mathcal{W}]$ is a quotient transformation (Definition 5.12). Thus Theorem 5.13(j) says that its extension $A \,\widehat{\otimes}\, B$ in $\mathcal{B}[\mathcal{X} \,\widehat{\otimes}_\wedge\, \mathcal{Y}, \mathcal{V} \,\widehat{\otimes}_\wedge\, \mathcal{W}]$ over the completions $\mathcal{X} \,\widehat{\otimes}_\wedge\, \mathcal{Y}$ and $\mathcal{V} \,\widehat{\otimes}_\wedge\, \mathcal{W}$ of the tensor products $\mathcal{X} \otimes_\wedge \mathcal{Y}$ and $\mathcal{V} \otimes_\wedge \mathcal{W}$ is a quotient transformation as well. □

Remark 8.6. *Characterizing Projective and Injective Uniform Crossnorms.* Suppose $\mathcal{M}$ and $\mathcal{N}$ are subspaces of $\mathcal{X}$ and $\mathcal{Y}$, respectively.

(a) By Theorem 5.8, $\mathcal{M}$ and $\mathcal{N}$ must be closed so that $\mathcal{X}/\mathcal{M}$ and $\mathcal{Y}/\mathcal{N}$ are naturally normed. By Theorem 5.13(f), the quotient maps $\pi_X \in \mathcal{B}[\mathcal{X}, \mathcal{X}/\mathcal{M}]$ and $\pi_Y \in \mathcal{B}[\mathcal{Y}, \mathcal{Y}/\mathcal{N}]$ (which are contractions with norm one) are quotient transformations for every pair of proper subspaces $\mathcal{M}$ and $\mathcal{N}$ of $\mathcal{X}$ and $\mathcal{Y}$. Then, by setting $\mathcal{V} = \mathcal{X}/\mathcal{M}$ and $\mathcal{W} = \mathcal{Y}/\mathcal{N}$ in Corollary 8.5, it follows that $\pi_X \otimes_\wedge \pi_Y$ in $\mathcal{B}[\mathcal{X} \otimes_\wedge \mathcal{Y}, \mathcal{X}/\mathcal{M} \otimes_\wedge \mathcal{Y}/\mathcal{N}]$ is a quotient transformation, and thus is $\pi_X \,\widehat{\otimes}_\wedge\, \pi_Y$ in $\mathcal{B}[\mathcal{X} \,\widehat{\otimes}_\wedge\, \mathcal{Y}, \mathcal{X}/\mathcal{M} \,\widehat{\otimes}_\wedge\, \mathcal{Y}/\mathcal{N}]$. So Corollary 7.11 and Theorem 7.10 (whose proofs are based on Theorem 7.9) can be viewed as consequences of Corollary 8.5.

> A *projective uniform crossnorm* is a uniform crossnorm for which Corollary 8.5 holds (and so Corollary 7.11 and Theorem 7.10 hold). That is, a *projective uniform crossnorm* is a uniform crossnorm $\|\cdot\|_\alpha$ for which $A \otimes B \in \mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{V} \otimes_\alpha \mathcal{W}]$ is a quotient transformation whenever $A \in \mathcal{B}[\mathcal{X}, \mathcal{V}]$ and $B \in \mathcal{B}[\mathcal{Y}, \mathcal{W}]$ are.

(b) Let $\mathcal{M}$ and $\mathcal{N}$ be subspaces of $\mathcal{X}$ and $\mathcal{Y}$ and take an arbitrary F in $\mathcal{M} \otimes \mathcal{N}$. By Theorem 8.3(b), $\|F\|_{\mathcal{M} \otimes_\alpha \mathcal{N}} \le \|F\|_{\mathcal{X} \otimes_\alpha \mathcal{Y}}$ for every uniform crossnorm $\|\cdot\|_\alpha$. By Theorem 8.4, the projective norm $\|\cdot\|_\wedge$ is uniform, and so $\|F\|_{\mathcal{M} \otimes_\wedge \mathcal{N}} \le \|F\|_{\mathcal{X} \otimes_\wedge \mathcal{Y}}$ but, as observed in Remark 7.5, the inequality may be strict. In other words, although $\mathcal{M} \otimes \mathcal{N}$ is a regular manifold of the linear space $\mathcal{X} \otimes \mathcal{Y}$, when both are equipped with the projective norm $\|\cdot\|_\wedge$ it may happen that $\|F\|_{\mathcal{M} \otimes_\wedge \mathcal{N}} < \|F\|_{\mathcal{X} \otimes_\wedge \mathcal{Y}}$ for some F in $\mathcal{M} \otimes_\wedge \mathcal{N}$ and, consequently, $\mathcal{M} \otimes_\wedge \mathcal{N}$ may not be a linear manifold of the normed space $\mathcal{X} \otimes_\wedge \mathcal{Y}$. For the injective norm $\|\cdot\|_\vee$, however, Theorem 7.19 says that $\|F\|_{\mathcal{M} \otimes_\vee \mathcal{N}} = \|F\|_{\mathcal{X} \otimes_\vee \mathcal{Y}}$ for every F in $\mathcal{M} \otimes_\vee \mathcal{N}$, and so $\mathcal{M} \otimes_\vee \mathcal{N}$ is a linear manifold of the normed space $\mathcal{X} \otimes_\vee \mathcal{Y}$.

> An *injective uniform crossnorm* is a uniform crossnorm for which Theorem 7.19 holds. That is, an *injective uniform crossnorm* is a uniform crossnorm $\|\cdot\|_\alpha$ for which $\mathcal{M} \otimes_\alpha \mathcal{N}$ is a linear manifold of the normed space $\mathcal{X} \otimes_\alpha \mathcal{Y}$ whenever $\mathcal{M}$ and $\mathcal{N}$ are subspaces of $\mathcal{X}$ and $\mathcal{Y}$. Equivalently, for which $\|F\|_{\mathcal{M} \otimes_\alpha \mathcal{N}} = \|F\|_{\mathcal{X} \otimes_\alpha \mathcal{Y}}$ for every F in $\mathcal{M} \otimes_\alpha \mathcal{N}$.

8.2 Tensor Norms

The notion of uniform crossnorms allows us to proceed towards a theory of normed tensor products of bounded linear operators. Indeed, uniform crossnorms apply to tensor product spaces of every pair of normed spaces. Moreover, a tensor product of bounded linear operators acting between tensor product spaces equipped with a uniform crossnorm is bounded and, in addition, its norm satisfies the operator norm property (regarding tensor products). There is, however, an additional property that is still desirable: *finite generation*.

Definition 8.7. Let $\|\cdot\|_\alpha$ be a reasonable crossnorm defined on every tensor product space $\mathcal{X}\otimes\mathcal{Y}$ for arbitrary normed spaces $\mathcal{X}$ and $\mathcal{Y}$. Let $\mathcal{X}\otimes_\alpha\mathcal{Y}$ stand for the normed space $(\mathcal{X}\otimes\mathcal{Y},\|\cdot\|_\alpha)$, as usual.

(a) The reasonable crossnorm $\|\cdot\|_\alpha$ is said to be *finitely generated* if, for every pair $\{\mathcal{X},\mathcal{Y}\}$ of normed spaces and every $F\in\mathcal{X}\otimes\mathcal{Y}$,

$$\|F\|_{\mathcal{X}\otimes_\alpha\mathcal{Y}} = \inf_{\dim\mathcal{M}<\infty,\,\dim\mathcal{N}<\infty,\,F\in\mathcal{M}\otimes\mathcal{N}} \|F\|_{\mathcal{M}\otimes_\alpha\mathcal{N}},$$

where the infimum is taken over all finite-dimensional subspaces $\mathcal{M}$ and $\mathcal{N}$ of $\mathcal{X}$ and $\mathcal{Y}$ such that $F\in\mathcal{M}\otimes\mathcal{N}$.

(b) A *tensor norm* is a finitely generated uniform crossnorm.

Remark 8.8. *Two Notes on Finite Generation.*

(a) Consider the completion $\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$ of $\mathcal{X}\otimes_\alpha\mathcal{Y}$ equipped with any reasonable crossnorm $\|\cdot\|_\alpha$ for any pair of normed spaces $\mathcal{X}$ and $\mathcal{Y}$. Since $\|F\|_\alpha = \|F\|_{\mathcal{X}\otimes_\alpha\mathcal{Y}} = \|\widehat{F}\|_{\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}} = \|\widehat{F}\|_\alpha$ for every $F\in\mathcal{X}\otimes_\alpha\mathcal{Y}$ and its correspondent $\widehat{F}\in\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$, then the condition for a reasonable crossnorm $\|\cdot\|_\alpha$ to be finitely generated can be equivalently written with $\|F\|_{\mathcal{X}\otimes_\alpha\mathcal{Y}}$ replaced by $\|\widehat{F}\|_{\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}}$. Recall: if $\mathcal{M}$ and $\mathcal{N}$ are finite-dimensional, then $\mathcal{M}\otimes\mathcal{N}$ is finite-dimensional by Corollary 3.10, and therefore $\mathcal{M}\otimes_\alpha\mathcal{N} = \mathcal{M}\,\widehat{\otimes}_\alpha\mathcal{N}$.

(b) Let $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ be finitely generated reasonable crossnorms on $\mathcal{X}\otimes\mathcal{Y}$. If $\|\cdot\|_{\mathcal{M}\otimes_\alpha\mathcal{N}} \le \|\cdot\|_{\mathcal{M}\otimes_\beta\mathcal{N}}$ for any finite-dimensional subspaces $\mathcal{M}$ and $\mathcal{N}$ of $\mathcal{X}$ and $\mathcal{Y}$, then $\|\cdot\|_{\mathcal{X}\otimes_\alpha\mathcal{Y}} \le \|\cdot\|_{\mathcal{X}\otimes_\beta\mathcal{Y}}$ (Definition 8.7(a)). The converse holds as well, since a norm of F does not depend on the (finite) representation of it:

$$\|\cdot\|_{\mathcal{X}\otimes_\alpha\mathcal{Y}} \le \|\cdot\|_{\mathcal{X}\otimes_\beta\mathcal{Y}} \iff \|\cdot\|_{\mathcal{M}\otimes_\alpha\mathcal{N}} \le \|\cdot\|_{\mathcal{M}\otimes_\beta\mathcal{N}}$$

for every pair of finite-dimensional subspaces $\mathcal{M}$ and $\mathcal{N}$ of $\mathcal{X}$ and $\mathcal{Y}$.

Theorem 8.9. *Projective and injective norms are finitely generated.*

Proof. Take an arbitrary pair $\{\mathcal{X},\mathcal{Y}\}$ of normed spaces. A representation of any element F in the tensor product space $\mathcal{X}\otimes\mathcal{Y}$ is given by a finite sum $F = \sum_i x_i\otimes y_i$ of single tensors with $x_i\in\mathcal{X}$ and $y_i\in\mathcal{Y}$. A representation is not

unique, but every representation is a finite sum, and for each representation, F lies in a regular linear manifold $\mathcal{M}_F \otimes \mathcal{N}_F$ of the linear space $\mathcal{X} \otimes \mathcal{Y}$ with $\mathcal{M}_F = \operatorname{span}\{x_i\}$ and $\mathcal{N}_F = \operatorname{span}\{y_i\}$, which are finite-dimensional linear manifolds (equivalently, finite-dimensional subspaces) of $\mathcal{X}$ and $\mathcal{Y}$.

(a) The injective reasonable crossnorm $\|\cdot\|_\vee$ respects regular subspaces as in Theorem 7.19 and Remark 7.5(e). Therefore, for every pair $\{\mathcal{M},\mathcal{N}\}$ of finite-dimensional subspaces of $\mathcal{X}$ and $\mathcal{Y}$ for which $F \in \mathcal{M} \otimes \mathcal{N}$,

$$\|F\|_{\mathcal{X}\otimes_\vee\mathcal{Y}} = \|F\|_{\mathcal{M}\otimes_\vee\mathcal{N}}.$$

Thus, in particular, $\|F\|_{\mathcal{X}\otimes_\vee\mathcal{Y}} = \inf_{\dim\mathcal{M}<\infty,\,\dim\mathcal{N}<\infty,\,F\in\mathcal{M}\otimes\mathcal{N}} \|F\|_{\mathcal{M}\otimes_\vee\mathcal{N}}$.

(b) The projective reasonable crossnorm $\|\cdot\|_\wedge$ on $\mathcal{X}\otimes\mathcal{Y}$ is given by $\|F\|_{\mathcal{X}\otimes_\wedge\mathcal{Y}} = \inf_{x_i\in\mathcal{X},\,y_i\in\mathcal{Y},\,F=\Sigma_i x_i\otimes y_i} \sum_i \|x_i\|\,\|y_i\|$ for every F in $\mathcal{X}\otimes\mathcal{Y}$, where the infimum is taken over all finite representations $\sum_i x_i \otimes y_i$ of F. Fix an element F in $\mathcal{X}\otimes\mathcal{Y}$. Take $\varepsilon > 0$ arbitrary. Thus there is a finite representation $\sum_i x_i' \otimes y_i'$ of F such that $\|F\|_{\mathcal{X}\otimes_\wedge\mathcal{Y}} \le \sum_i \|x_i'\|\,\|y_i'\| \le \|F\|_{\mathcal{X}\otimes_\wedge\mathcal{Y}} + \varepsilon$. Set $\mathcal{M}_\varepsilon = \operatorname{span}\{x_i'\}$ and $\mathcal{N}_\varepsilon = \operatorname{span}\{y_i'\}$, which are finite-dimensional subspaces of $\mathcal{X}$ and $\mathcal{Y}$. Then F lies in $\mathcal{M}_\varepsilon \otimes \mathcal{N}_\varepsilon$ and $\|F\|_{\mathcal{M}_\varepsilon\otimes_\wedge\mathcal{N}_\varepsilon} = \inf_{u_i\in\mathcal{M}_\varepsilon,\,v_i\in\mathcal{N}_\varepsilon,\,F=\Sigma_i u_i\otimes v_i} \sum_i \|u_i\|\,\|v_i\| \le \sum_i \|x_i'\|\,\|y_i'\|$. Therefore, according to Theorems 8.3(b) and 8.4,

$$\|F\|_{\mathcal{X}\otimes_\wedge\mathcal{Y}} \le \|F\|_{\mathcal{M}_\varepsilon\otimes_\wedge\mathcal{N}_\varepsilon} \le \|F\|_{\mathcal{X}\otimes_\wedge\mathcal{Y}} + \varepsilon.$$

Taking the infimum over all finite-dimensional subspaces $\mathcal{M}$ and $\mathcal{N}$ for which $F \in \mathcal{M}\otimes\mathcal{N}$, we get $\|F\|_{\mathcal{X}\otimes_\wedge\mathcal{Y}} = \inf_{\dim\mathcal{M}<\infty,\,\dim\mathcal{N}<\infty,\,F\in\mathcal{M}\otimes\mathcal{N}} \|F\|_{\mathcal{M}\otimes_\wedge\mathcal{N}}$. □

Corollary 8.10. *Projective and injective norms are tensor norms.*

Proof. By Theorems 8.4 and 8.9, according to Definition 8.7. □

Observe that $\|\cdot\|_\vee$ is the least and $\|\cdot\|_\wedge$ is the greatest uniform crossnorm/tensor norm (since $\|\cdot\|_\alpha$ is a reasonable crossnorm if and only if $\|\cdot\|_\vee \le \|\cdot\|_\alpha \le \|\cdot\|_\vee$ — Theorem 7.4).

Take a reasonable crossnorm $\|\cdot\|_\alpha$ acting on a tensor product space $\mathcal{X}\otimes\mathcal{Y}$ of an arbitrary pair of normed spaces and consider its transpose $\|\cdot\|_{t\alpha}$, which was defined in Section 7.1 in terms of the natural permutation isomorphism $\Pi : \mathcal{X}\otimes_{t\alpha}\mathcal{Y} \to \mathcal{Y}\otimes_\alpha\mathcal{X}$ given by $\Pi(\sum_i x_i\otimes y_i) = \sum_i y_i \otimes x_i$, which in turn is an isometric isomorphism according to Theorem 7.6(b).

Theorem 8.11. $\|\cdot\|_{t\alpha}$ *is a tensor norm whenever* $\|\cdot\|_\alpha$ *is a tensor norm.*

Proof. Let $\|\cdot\|_\alpha$ be a uniform crossnorm on the tensor product space of an arbitrary pair of normed spaces. We split the proof into three parts.

(1) $\|\cdot\|_{t\alpha}$ *is a reasonable crossnorm whenever* $\|\cdot\|_\alpha$ *is*. This is Theorem 7.6(a).

(2) $\|\cdot\|_{t\alpha}$ *is finitely generated whenever* $\|\cdot\|_\alpha$ *is*. This is clear by the definition of transpose in Section 7.1. In fact, take $F \in \mathcal{X}\otimes\mathcal{Y}$ and consider its transpose ${}^tF \in \mathcal{Y}\otimes\mathcal{X}$. If $\|\cdot\|_\alpha$ on $\mathcal{Y}\otimes\mathcal{X}$ is finitely generated, then (cf. Definition 8.7)

$$\|{}^tF\|_{\mathcal{Y}\otimes_\alpha\mathcal{X}} = \inf_{\dim\mathcal{N}<\infty,\ \dim\mathcal{M}<\infty,\ {}^tF\in\mathcal{N}\otimes\mathcal{M}} \|{}^tF\|_{\mathcal{N}\otimes_\alpha\mathcal{M}}.$$

Since $\|F\|_{\mathcal{X}\otimes_{t_\alpha}\mathcal{Y}} = \|{}^tF\|_{\mathcal{Y}\otimes_\alpha\mathcal{X}}$ and $\|F\|_{\mathcal{M}\otimes_{t_\alpha}\mathcal{N}} = \|{}^tF\|_{\mathcal{N}\otimes_\alpha\mathcal{M}}$, and since F lies in $\mathcal{M}\otimes_\alpha\mathcal{N}$ if and only if ${}^tF$ lies in $\mathcal{N}\otimes_{t_\alpha}\mathcal{M}$, we get

$$\|F\|_{\mathcal{X}\otimes_{t_\alpha}\mathcal{Y}} = \inf_{\dim\mathcal{M}<\infty,\ \dim\mathcal{N}<\infty,\ F\in\mathcal{M}\otimes\mathcal{N}} \|F\|_{\mathcal{M}\otimes_{t_\alpha}\mathcal{N}}.$$

(3) $\|\cdot\|_{t_\alpha}$ *is a uniform crossnorm whenever* $\|\cdot\|_\alpha$ *is*. Indeed, let $\mathcal{X}$, $\mathcal{Y}$, $\mathcal{V}$, $\mathcal{W}$ be normed spaces and take arbitrary bounded linear transformations

$$A\colon \mathcal{X}\to\mathcal{V} \quad \text{and} \quad B\colon \mathcal{Y}\to\mathcal{W}.$$

Suppose $\|\cdot\|_\alpha$ is a reasonable crossnorm equipping both linear spaces $\mathcal{Y}\otimes\mathcal{X}$ and $\mathcal{W}\otimes\mathcal{V}$ so that its transpose $\|\cdot\|_{t_\alpha}$ equips both linear spaces $\mathcal{X}\otimes\mathcal{Y}$ and $\mathcal{V}\otimes\mathcal{W}$. Take the tensor product linear transformations

$$A\otimes B\colon \mathcal{X}\otimes_{t_\alpha}\mathcal{Y}\to\mathcal{V}\otimes_{t_\alpha}\mathcal{W} \quad \text{and} \quad B\otimes A\colon \mathcal{Y}\otimes_\alpha\mathcal{X}\to\mathcal{W}\otimes_\alpha\mathcal{V}.$$

Consider the natural permutations,

$$\Pi_1\colon \mathcal{X}\otimes_{t_\alpha}\mathcal{Y}\to\mathcal{Y}\otimes_\alpha\mathcal{X} \quad \text{and} \quad \Pi_2\colon \mathcal{V}\otimes_{t_\alpha}\mathcal{W}\to\mathcal{W}\otimes_\alpha\mathcal{V},$$

which are isometric isomorphisms (cf. proof of Theorem 7.6(b)). Also, by their own definition, and by the definition of tensor product transformation (cf. Definition 3.16), we get $\Pi_2(A\otimes B) = (B\otimes A)\Pi_1$. Equivalently, the diagram

$$\begin{array}{ccc} \mathcal{X}\otimes_{t_\alpha}\mathcal{Y} & \xrightarrow{A\otimes B} & \mathcal{V}\otimes_{t_\alpha}\mathcal{W} \\ {\scriptstyle\Pi_1}\downarrow & & \downarrow{\scriptstyle\Pi_2} \\ \mathcal{Y}\otimes_\alpha\mathcal{X} & \xrightarrow{B\otimes A} & \mathcal{W}\otimes_\alpha\mathcal{V} \end{array}$$

commutes. Therefore boundedness is preserved,

$$A\otimes B\in\mathcal{B}[\mathcal{X}\otimes_{t_\alpha}\mathcal{Y},\mathcal{V}\otimes_{t_\alpha}\mathcal{W}] \iff B\otimes A\in\mathcal{B}[\mathcal{Y}\otimes_\alpha\mathcal{X},\mathcal{W}\otimes_\alpha\mathcal{V}],$$

and in this case $A\otimes B\cong B\otimes A$ (isometrically equivalent bounded linear transformations), and so their norms coincide (cf. Section 5.1). Thus if $\|\cdot\|_\alpha$ is uniform (and recalling that $A\otimes B$ and $B\otimes A$ are bounded together), then

$$\|A\otimes B\|_{\mathcal{B}[\mathcal{X}\otimes_{t_\alpha}\mathcal{Y},\,\mathcal{V}\otimes_{t_\alpha}\mathcal{W}]} = \|B\otimes A\|_{\mathcal{B}[\mathcal{Y}\otimes_\alpha\mathcal{X},\,\mathcal{W}\otimes_\alpha\mathcal{V}]} \le \|A\|\,\|B\|,$$

which implies $\|\cdot\|_{t_\alpha}$ is uniform. □

The next result is a sort of reverse of Definition 8.7. There, we had a fixed reasonable crossnorm which was supposed to satisfy a given condition. Here, we define a tensor norm out of a given uniform crossnorm defined on arbitrary tensor products of finite-dimensional normed spaces by means of a certain condition. As it happens, the conditions in both cases coincide.

Theorem 8.12. *Let $\|\cdot\|_\alpha$ be a finite-dimensional uniform crossnorm (i.e., a reasonable crossnorm on tensor product spaces of finite-dimensional normed spaces which is uniform there). Let $\mathcal{X}$ and $\mathcal{Y}$ be arbitrary normed spaces. Consider the functional $\|\cdot\|_{\widehat{\alpha}}\colon \mathcal{X}\otimes\mathcal{Y}\to\mathbb{R}$ defined for every $F\in\mathcal{X}\otimes\mathcal{Y}$ by*

$$\|F\|_{\widehat{\alpha}} = \|F\|_{\mathcal{X}\otimes_{\widehat{\alpha}}\mathcal{Y}} = \inf_{\dim\mathcal{M}<\infty,\ \dim\mathcal{N}<\infty,\ F\in\mathcal{M}\otimes\mathcal{N}} \|F\|_{\mathcal{M}\otimes_\alpha\mathcal{N}}, \qquad (*)$$

where the infimum is taken over all finite-dimensional subspaces $\mathcal{M}$ and $\mathcal{N}$ of $\mathcal{X}$ and $\mathcal{Y}$ for which $F\in\mathcal{M}\otimes\mathcal{N}$. This $\|\cdot\|_{\widehat{\alpha}}$ is a tensor norm on $\mathcal{X}\otimes\mathcal{Y}$. The norm $\|\cdot\|_{\widehat{\alpha}}$ coincides with $\|\cdot\|_\alpha$ if restricted to tensor products of finite-dimensional normed spaces.

Proof. Let $\mathcal{X}$ and $\mathcal{Y}$ be arbitrary normed spaces. First observe that $\mathcal{X}\otimes\mathcal{Y} = \bigcup_{\mathcal{M},\mathcal{N};\,\dim\mathcal{M}<\infty,\,\dim\mathcal{N}<\infty} \mathcal{M}\otimes\mathcal{N}$, union over all pairs of finite-dimensional subspaces $\mathcal{M}$ and $\mathcal{N}$ of $\mathcal{X}$ and $\mathcal{Y}$. Take the partial inclusion ordering $\subseteq$ on $\mathcal{X}$ and $\mathcal{Y}$. For each pair of pairs of finite-dimensional subspaces of $\mathcal{X}$ and $\mathcal{Y}$, set

$$(\mathcal{M},\mathcal{N}) \le (\mathcal{M}',\mathcal{N}') \quad \text{whenever} \quad \mathcal{M}\subseteq\mathcal{M}' \ \text{ and } \ \mathcal{N}\subseteq\mathcal{N}'.$$

It is readily verified that the relation $\le$ makes a partial ordering on the collection of all pairs of finite-dimensional subspaces of $\mathcal{X}$ and $\mathcal{Y}$. Since every pair in this collection is bounded above, it becomes a *directed set*, and so every family indexed by this collection is a *net*. Let $I_{\mathcal{M}}\in\mathcal{B}[\mathcal{M},\mathcal{M}']$ and $I_{\mathcal{N}}\in\mathcal{B}[\mathcal{N},\mathcal{N}']$ be the natural isometric embeddings $I_{\mathcal{M}}\hookrightarrow\mathcal{M}'$ and $I_{\mathcal{N}}\hookrightarrow\mathcal{N}'$. Suppose $\|\cdot\|_\alpha$ is a uniform crossnorm defined on every tensor product space of finite-dimensional normed spaces. Take an arbitrary $F\in\mathcal{M}\otimes\mathcal{N}$ where $\mathcal{M},\mathcal{M}'$ and $\mathcal{N},\mathcal{N}'$ are finite-dimensional subspaces of $\mathcal{X}$ and $\mathcal{Y}$ such that $\mathcal{M}\subseteq\mathcal{M}'$ and $\mathcal{N}\subseteq\mathcal{N}'$, and therefore $\mathcal{M}\otimes\mathcal{N}\subseteq\mathcal{M}'\otimes\mathcal{N}'$. Since $\|\cdot\|_\alpha$ is a uniform crossnorm, and since $\|I_{\mathcal{M}}\| = \|I_{\mathcal{N}}\| = 1$, it follows that $I_{\mathcal{M}}\otimes I_{\mathcal{N}}\in\mathcal{B}[\mathcal{M}\otimes_\alpha\mathcal{N},\mathcal{M}'\otimes_\alpha\mathcal{N}']$ with $\|I_{\mathcal{M}}\otimes I_{\mathcal{N}}\| = \|I_{\mathcal{M}}\|\,\|I_{\mathcal{N}}\| = 1$. Hence

$$\|F\|_{\mathcal{M}'\otimes_\alpha\mathcal{N}'} = \|(I_{\mathcal{M}}\otimes I_{\mathcal{N}})F\|_{\mathcal{M}'\otimes_\alpha\mathcal{N}'} \le \|I_{\mathcal{M}}\|\,\|I_{\mathcal{N}}\|\,\|F\|_{\mathcal{M}\otimes_\alpha\mathcal{N}} = \|F\|_{\mathcal{M}\otimes_\alpha\mathcal{N}}$$

(compare with Theorem 8.3(b)). Thus $\{\|F\|_{\mathcal{M}\otimes_\alpha\mathcal{N}}\}_{(\mathcal{M},\mathcal{N})}$ is a nonincreasing net. This ensures that the functional $\|\cdot\|_{\widehat{\alpha}}\colon\mathcal{X}\otimes\mathcal{Y}\to\mathbb{R}$ given by $(*)$ is well-defined (it exists and is unique). Note that this in fact is the limit (over the same directed set) of the nonincreasing net $\{\|F\|_{\mathcal{M}\otimes_\alpha\mathcal{N}}\}_{(\mathcal{M},\mathcal{N})}$,

$$\|F\|_{\mathcal{X}\otimes_{\widehat{\alpha}}\mathcal{Y}} = \lim_{\dim\mathcal{M}<\infty,\ \dim\mathcal{N}<\infty,\ F\in\mathcal{M}\otimes\mathcal{N}} \|F\|_{\mathcal{M}\otimes_\alpha\mathcal{N}} \quad \text{for every} \quad F\in\mathcal{X}\otimes\mathcal{Y}.$$

The above limit of a nonincreasing net shows at once that for every $F\in\mathcal{X}\otimes\mathcal{Y}$,

$$\dim\mathcal{X}<\infty \text{ and } \dim\mathcal{Y}<\infty \implies \|F\|_{\mathcal{X}\otimes_{\widehat{\alpha}}\mathcal{Y}} = \|F\|_{\mathcal{X}\otimes_\alpha\mathcal{Y}}.$$

(1) *$\|\cdot\|_{\widehat{\alpha}}$ is a norm.*

Nonnegativeness ($\|F\|_{\widehat{\alpha}}\ge 0$) and absolute homogeneity ($\|\lambda F\|_{\widehat{\alpha}} = |\lambda|\,\|F\|_{\widehat{\alpha}}$) are readily verified by $(*)$ since $\|\cdot\|_\alpha$ is a norm, and positiveness ($\|F\|_{\widehat{\alpha}} = 0 \implies$

$F = 0$) also follows from $(*)$ and the fact that $\|F\|_\vee \le \|F\|_\alpha \le \|F\|_\wedge$ since $\|\cdot\|_\alpha$ is a reasonable crossnorm (see the argument in the proof of Theorem 7.4). For the triangle inequality (subadditivity), proceed as follows. Take $F, F' \in \mathcal{X} \otimes \mathcal{Y}$ arbitrarily, and take any (finite) representation, say $F = \sum_i x_i \otimes y_i$ and $F' = \sum_i x'_i \otimes y'_i$ in $\mathcal{M} \otimes \mathcal{N} \subseteq \mathcal{X} \otimes \mathcal{Y}$, and so $F + F'$ lies in $\mathcal{M} \otimes \mathcal{N}$, where $\mathcal{M} = \operatorname{span}(\{x_i\} \cup \{x'_i\})$ and $\mathcal{N} = \operatorname{span}(\{y_i\} \cup \{y'_i\})$ are finite-dimensional subspaces of $\mathcal{X}$ and $\mathcal{Y}$. Since $\|\cdot\|_\alpha$ is a norm on finite-dimensional tensor product spaces, $\|F + F'\|_\alpha \le \|F\|_\alpha + \|F'\|_\alpha$, and so $(*)$ ensures $\|F + F'\|_{\widehat{\alpha}} \le \|F\|_{\widehat{\alpha}} + \|F'\|_{\widehat{\alpha}}$.

(2) $\|\cdot\|_{\widehat{\alpha}}$ *is a reasonable crossnorm.*

Since $\|\cdot\|_\alpha$ is a reasonable crossnorm, Definition 7.1(a) and $(*)$ ensure that $\|x \otimes y\|_{\widehat{\alpha}} = \|x \otimes y\|_\alpha \le \|x\|\,\|y\|$ for every single tensor $x \otimes y \in \mathcal{X} \otimes \mathcal{Y}$. Moreover, as we saw above (since $\|\cdot\|_\alpha$ is a reasonable crossnorm), $\|F\|_\vee \le \|F\|_\alpha$ for every F in an arbitrary finite-dimensional regular (algebraic) linear manifold of the linear space $\mathcal{X} \otimes \mathcal{Y}$. Thus $\|F\|_\vee \le \|F\|_{\widehat{\alpha}}$ for every $F \in \mathcal{X} \otimes \mathcal{Y}$ according to $(*)$, and hence $|(f \otimes g)F| \le \|f\|\,\|g\|\,\|F\|_\vee \le \|f\|\,\|g\|\,\|F\|_{\widehat{\alpha}}$ (cf. proof of Theorem 7.4 part (b)). Therefore $\|f \otimes g\|_{*\widehat{\alpha}} \le \|f\|\,\|g\|$ for every single tensor $f \otimes g \in \mathcal{X}^* \otimes \mathcal{Y}^*$, thus satisfying Definition 7.1(b) as well.

(3) $\|\cdot\|_{\widehat{\alpha}}$ *is a uniform crossnorm.*

Let $\mathcal{X}, \mathcal{Y}, \mathcal{V}, \mathcal{W}$ be normed spaces. Take $A \in \mathcal{B}[\mathcal{X}, \mathcal{V}]$ and $B \in \mathcal{B}[\mathcal{Y}, \mathcal{W}]$. Consider their restrictions to finite-dimensional subspaces $\mathcal{M}, \mathcal{N}, \mathcal{R}, \mathcal{S}$ of $\mathcal{X}, \mathcal{Y}, \mathcal{V}, \mathcal{W}$, respectively, so that by using the same notation for restrictions we write $A = A|_{\mathcal{M}} \in \mathcal{B}[\mathcal{M}, \mathcal{R}]$ and $B = B|_{\mathcal{N}} \in \mathcal{B}[\mathcal{N}, \mathcal{S}]$, where $A(\mathcal{M}) = \mathcal{R}$ and $B(\mathcal{N}) = \mathcal{S}$. Since $\|\cdot\|_\alpha$ is a finite-dimensional uniform crossnorm, the linear transformation $A \otimes B \colon \mathcal{M} \otimes_\alpha \mathcal{N} \to \mathcal{R} \otimes_\alpha \mathcal{S}$ is bounded, $A \otimes B \in \mathcal{B}[\mathcal{M} \otimes_\alpha \mathcal{N}, \mathcal{R} \otimes_\alpha \mathcal{S}]$, and $\|A \otimes B\| \le \|A\|\,\|B\|$. Take an arbitrary $F \in \mathcal{X} \otimes \mathcal{Y}$ and an arbitrary $\varepsilon > 0$. According to $(*)$, there are finite-dimensional subspaces $\mathcal{M}_\varepsilon$ and $\mathcal{N}_\varepsilon$ of $\mathcal{X}$ and $\mathcal{Y}$ such that if $(\mathcal{M}, \mathcal{N}) \ge (\mathcal{M}_\varepsilon, \mathcal{N}_\varepsilon)$, then

$$\|F\|_{\mathcal{M} \otimes_\alpha \mathcal{N}} \le \|F\|_{\mathcal{X} \otimes_{\widehat{\alpha}} \mathcal{Y}} + \varepsilon\,\|F\|_{\mathcal{X} \otimes_{\widehat{\alpha}} \mathcal{Y}} = (1 + \varepsilon)\|F\|_{\mathcal{X} \otimes_{\widehat{\alpha}} \mathcal{Y}}.$$

So for every $(\mathcal{M}, \mathcal{N}) \ge (\mathcal{M}_\varepsilon, \mathcal{N}_\varepsilon)$, with $\mathcal{R} = A(\mathcal{M})$ and $\mathcal{S} = B(\mathcal{N})$ being finite-dimensional subspaces of $\mathcal{V}$ and $\mathcal{W}$, and since $(*)$ holds concerning every pair of finite-dimensional subspaces of every pair of normed spaces, we get by the expression in $(*)$ and by the above inequality that

$$\begin{aligned}\|(A \otimes B)F\|_{\mathcal{V} \otimes_{\widehat{\alpha}} \mathcal{W}} &\le \|(A \otimes B)F\|_{\mathcal{R} \otimes_\alpha \mathcal{S}} \le \|A\|\,\|B\|\,\|F\|_{\mathcal{M} \otimes_\alpha \mathcal{N}} \\ &\le (1 + \varepsilon)\|A\|\,\|B\|\,\|F\|_{\mathcal{X} \otimes_{\widehat{\alpha}} \mathcal{Y}}.\end{aligned}$$

As this holds for every $F \in \mathcal{X} \otimes \mathcal{Y}$ and every $\varepsilon > 0$, $\|A \otimes B\| \le \|A\|\,\|B\|$.

(4) $\|\cdot\|_{\widehat{\alpha}}$ *is a tensor norm.*

$\|\cdot\|_{\widehat{\alpha}}$ is uniform by (3) and naturally finitely generated by $(*)$. □

Corollary 8.13. *A finitely generated uniform crossnorm on finite-dimensional spaces (i.e., a finitely generated reasonable crossnorm on every tensor product space which is uniform on finite-dimensional spaces) is a tensor norm.*

Proof. If $\|\cdot\|_\alpha$ is finitely generated, then the expression in Definition 8.7(a) holds true. If it is a uniform crossnorm on tensor product spaces of finite-dimensional normed spaces, then the expression $(*)$ in Theorem 8.12 also holds true. By the uniqueness of $(*)$ (since limits of nets are unique — cf. proof of Theorem 8.12), $\|\cdot\|_\alpha = \|\cdot\|_{\widetilde{\alpha}}$ and $\|\cdot\|_\alpha$ is a tensor norm by Theorem 8.12. □

Remark 8.14. *Approximation Properties*. Let $\mathcal{B}_0[\mathcal{X},\mathcal{Y}]$ denote the collection of all finite-rank bounded linear transformations, and let $\mathcal{B}_\infty[\mathcal{X},\mathcal{Y}]$ denote the collection of all compact linear transformations.

> A Banach space $\mathcal{Y}$ has the *approximation property* if $\mathcal{B}_0[\mathcal{X},\mathcal{Y}]$ is dense in $\mathcal{B}_\infty[\mathcal{X},\mathcal{Y}]$ for every normed space $\mathcal{X}$,

as defined in Section 5.4 after Proposition 5.J, which says that a Banach space with a Schauder basis (in particular, a separable Hilbert space — actually, any Hilbert space, as in (g) below) has the approximation property. An equivalent definition (due to Grothendieck [29]) gives an alternative and useful characterization of the approximation property (see, e.g., [70, Theorem 3.4.32]):

(a) *A Banach space $\mathcal{Y}$ has the approximation property if and only if, for every compact subset K of $\mathcal{Y}$ and every $\varepsilon > 0$, there is a finite-rank operator $T = T_{K,\varepsilon} \in \mathcal{B}_0[\mathcal{Y}]$ such that $\|Ty - y\| < \varepsilon$ for every $y \in K$.*

A stronger notion of approximation is given when the above finite-rank operators are such that $\sup_{K,\varepsilon} \|T\| < \infty$ (see, e.g., [68, Definition 1.e.11]):

(b) A Banach space $\mathcal{Y}$ has the *bounded approximation property* if there is a real constant $c \geq 1$ such that, for every compact subset K of $\mathcal{Y}$ and every $\varepsilon > 0$, there is a finite-rank operator $T = T_{K,\varepsilon} \in \mathcal{B}_0[\mathcal{Y}]$ for which $\|Ty - y\| < \varepsilon$ for every $y \in K$, with $\|T\| \leq c$.

An even stronger notion is obtained if the above operators are contractions.

(c) A Banach space $\mathcal{Y}$ has the *metric approximation property* if it has the bounded approximation property for $c = 1$.

Tautologically,

$$\begin{aligned}\text{metric approximation property} &\Longrightarrow \text{bounded approximation property}\\ &\Longrightarrow \text{approximation property.}\end{aligned}$$

Recalling Proposition 5.J of Section 5.4 again,

(d) existence of Schauder basis $\Longrightarrow$ approximation property.

A Schauder basis $\{y_k\}$ for $\mathcal{Y}$ is *monotone* if the projections $E_n \in \mathcal{B}[\mathcal{Y}]$ for which $E_n\big(\sum_{k=1}^\infty \alpha_k(y)y_k\big) = \sum_{k=1}^n \alpha_k(y)y_k$ are such that $\|E_n\| = 1$ for all n. A stronger version of (d) reads as follows (cf. [68, p. 38 and Proposition 1.a.2]):

(e) existence of monotone Schauder basis $\Longrightarrow$ metric approximation property.

Thus (by Fourier expansion since orthogonal projections have norm 1),

(f) being a separable Hilbert space $\Longrightarrow$ existence of monotone Schauder basis,

and so (cf. Proposition 5.I — see, e.g., [52, Problem 5.42]),

(g) Hilbert spaces have the metric approximation property.

As we see next, on Hilbert spaces uniform crossnorms are finitely generated.

Theorem 8.15. *Consider a reasonable crossnorm on a tensor product space of Banach spaces.*

(a) *If it is a uniform crossnorm and if the Banach spaces have the metric approximation property, then it is finitely generated.*

Consequently, (b) *every uniform crossnorm on a tensor product space of Banach spaces having the metric approximation property is a tensor norm.*

Proof. Let $\mathcal{M}$ and $\mathcal{N}$ be arbitrary finite-dimensional subspaces of an arbitrary pair of normed spaces $\mathcal{X}$ and $\mathcal{Y}$, respectively. If $\|\cdot\|_\alpha$ is a uniform crossnorm on finite-dimensional spaces as in Definition 8.2(c) (i.e., if $\|\cdot\|_\alpha$ is defined on every tensor product space and acts as uniform crossnorm on tensor product spaces of finite-dimensional normed spaces), then Theorem 8.3(b) says that $\|F\|_{\mathcal{X}\otimes_\alpha\mathcal{Y}} \le \|F\|_{\mathcal{M}\otimes_\alpha\mathcal{N}}$ for every $\mathcal{M}$ and $\mathcal{N}$ for which $F \in \mathcal{M}\otimes\mathcal{N}$. Therefore

$$\|F\|_{\mathcal{X}\otimes_\alpha\mathcal{Y}} \le \inf_{\dim\mathcal{M}<\infty,\, \dim\mathcal{N}<\infty,\, F\in\mathcal{M}\otimes\mathcal{N}} \|F\|_{\mathcal{M}\otimes_\alpha\mathcal{N}}.$$

To prove the reverse inequality, take an arbitrary $F \in \mathcal{X}\otimes\mathcal{Y}$ and consider a fixed (finite) representation of it, say $F = \sum_{i=1}^n x_i \otimes y_i$ for some $n \in \mathbb{N}$. Now take compact subsets $K_\mathcal{X} = K_{\mathcal{X},F}$ and $K_\mathcal{Y} = K_{\mathcal{Y},F}$ of $\mathcal{X}$ and $\mathcal{Y}$ including the finite sets $\{x_i\}_{i=1}^n$ and $\{y_i\}_{i=1}^n$, say $\{x_i\}_{i=1}^n \subset K_\mathcal{X} \subset \mathcal{X}$ and $\{y_i\}_{i=1}^n \subset K_\mathcal{Y} \subset \mathcal{Y}$. Take an arbitrary $\varepsilon > 0$. Suppose $\mathcal{X}$ and $\mathcal{Y}$ are Banach spaces having the metric approximation property. This implies that there are finite-rank contractions $X = X_{K_\mathcal{X},\varepsilon}$ in $\mathcal{B}_0[\mathcal{X}] \subset \mathcal{B}[\mathcal{X}]$ and $Y = Y_{K_\mathcal{Y},\varepsilon}$ in $\mathcal{B}_0[\mathcal{Y}] \subset \mathcal{B}[\mathcal{Y}]$ (so that $\|X\| \le 1$ and $\|Y\| \le 1$) for which (cf. Remark 8.14(a,b,c))

$$\|Xx_i - x_i\| < \varepsilon \quad \text{and} \quad \|Yy_i - y_i\| < \varepsilon \quad \text{for all} \quad i = 1, \dots, n.$$

Next consider the subspaces $\mathcal{M} = \operatorname{span}\big\{\operatorname{span}\{x_i\}_{i=1}^n \cup \mathcal{R}(X)\big\} \subset \mathcal{X}$ and $\mathcal{N} = \operatorname{span}\big\{\operatorname{span}\{y_i\}_{i=1}^n \cup \mathcal{R}(Y)\big\} \subset \mathcal{Y}$ of $\mathcal{X}$ and $\mathcal{Y}$. Since $\{x_i\}_{i=1}^n$ and $\{y_i\}_{i=1}^n$ are finite sets, and since the ranges $\mathcal{R}(X)$ and $\mathcal{R}(Y)$ are finite-dimensional, it follows that $\mathcal{M}$ and $\mathcal{N}$ are finite-dimensional spaces. Since $F = \sum_{i=1}^n x_i \otimes y_i$ and $(X\otimes Y)F$ lie in $\mathcal{M}\otimes\mathcal{N}$ (cf. Theorem 3.19(e)), we get

$$\begin{aligned}\|(X\otimes Y)F - F\|_{\mathcal{M}\otimes_\alpha\mathcal{N}} &= \Big\|\sum\nolimits_{i=1}^n (Xx_i \otimes Yy_i) - (x_i \otimes y_i)\Big\|_{\mathcal{M}\otimes_\alpha\mathcal{N}} \\ &= \Big\|\sum\nolimits_{i=1}^n \big((Xx_i - x_i) \otimes Yy_i\big) + \big(x_i \otimes (Yy_i - y_i)\big)\Big\|_{\mathcal{M}\otimes_\alpha\mathcal{N}}\end{aligned}$$

$$\leq \sum_{i=1}^{n} \|Xx_i - x_i\| \, \|Y\| \, \|y_i\| + \|x_i\| \, \|Yy_i - y_i\|$$
$$\leq \Big(\sum_{i=1}^{n} \|y_i\| + \|x_i\| \Big) \varepsilon = c\,\varepsilon,$$

where $c = c(F) = \sum_{i=1}^{n} \|y_i\| + \|x_i\|$ is a nonnegative constant. Moreover, since $\mathcal{R}(X) \subseteq \mathcal{M}$ and $\mathcal{R}(Y) \subseteq \mathcal{N}$, we may regard X and Y as bounded linear transformations into $\mathcal{M}$ and $\mathcal{N}$ so that $X \in \mathcal{B}[\mathcal{X}, \mathcal{M}]$ and $Y \in \mathcal{B}[\mathcal{Y}, \mathcal{N}]$. Take $X \otimes Y : \mathcal{X} \otimes \mathcal{Y} \to \mathcal{M} \otimes \mathcal{N}$. Thus (cf. Remark 8.1(a))

$$\|(X \otimes Y)F\|_{\mathcal{M}\otimes_\alpha \mathcal{N}} \leq \|X \otimes Y\| \, \|F\|_{\mathcal{X}\otimes_\alpha \mathcal{Y}}$$

whenever $X \otimes Y$ lies in $\mathcal{B}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{M} \otimes \mathcal{N}]$, which happens whenever $\|\cdot\|_\alpha$ is a uniform crossnorm and, in this case (Definition 8.2(a)),

$$\|X \otimes Y\| \leq \|X\| \, \|Y\| \leq 1.$$

Then, by the above two inequalities,

$$\|(X \otimes Y)F\|_{\mathcal{M}\otimes_\alpha \mathcal{N}} \leq \|F\|_{\mathcal{X}\otimes_\alpha \mathcal{Y}}.$$

Therefore, since there is a constant c such that $\|(X \otimes Y)F - F\|_{\mathcal{M}\otimes_\alpha \mathcal{N}} \leq c\,\varepsilon$ for every $\varepsilon > 0$, we get (by using the triangle inequality of $\|\cdot\|_{\mathcal{M}\otimes_\alpha \mathcal{N}}$)

$$\|F\|_{\mathcal{M}\otimes_\alpha \mathcal{N}} \leq \|(X \otimes Y)F - F\|_{\mathcal{M}\otimes_\alpha \mathcal{N}} + \|(X \otimes Y)F\|_{\mathcal{M}\otimes_\alpha \mathcal{N}} \leq \|F\|_{\mathcal{X}\otimes_\alpha \mathcal{Y}} + c\,\varepsilon.$$

As these subspaces $\mathcal{M}$ and $\mathcal{N}$ are finite-dimensional, the inequality ensures

$$\inf_{\dim \mathcal{M}<\infty,\, \dim \mathcal{N}<\infty,\, F \in \mathcal{M}\otimes\mathcal{N}} \|F\|_{\mathcal{M}\otimes_\alpha \mathcal{N}} \leq \|F\|_{\mathcal{X}\otimes_\alpha \mathcal{Y}}.$$

Hence $\|F\|_{\mathcal{X}\otimes_\alpha \mathcal{Y}} = \inf_{\dim \mathcal{M}<\infty,\, \dim \mathcal{N}<\infty,\, F \in \mathcal{M}\otimes\mathcal{N}} \|F\|_{\mathcal{M}\otimes_\alpha \mathcal{N}}$, which proves (a), and (a) implies (b) via Definition 8.7. □

Corollary 8.16. *If the Banach spaces $\mathcal{X}$ and $\mathcal{Y}$ have a monotone Schauder basis, then every uniform crossnorm on $\mathcal{X} \otimes \mathcal{Y}$ is a tensor norm.*

Proof. By Remark 8.14(e) and Theorem 8.15. □

Corollary 8.17. *If $\mathcal{X}$ and $\mathcal{Y}$ are Hilbert spaces, then every uniform crossnorm on $\mathcal{X} \otimes \mathcal{Y}$ is a tensor norm.*

Proof. Corollary 8.16 and Remark 8.14(g). □

8.3 Dual Norms

We close the chapter with the notion of *dual norm* $\|\cdot\|_{\alpha^*}$.

Let $\mathcal{X}$ and $\mathcal{Y}$ be normed spaces and take their duals $\mathcal{X}^*$ and $\mathcal{Y}^*$. Let the tensor product space $\mathcal{X} \otimes \mathcal{Y}$ be equipped with a reasonable crossnorm $\|\cdot\|_\alpha$ on

it, so that $\mathcal{X} \otimes_\alpha \mathcal{Y} = (\mathcal{X} \otimes \mathcal{Y}, \|\cdot\|_\alpha)$. Recall from Section 7.1 that $\|\cdot\|_{*\alpha}$ stands for the induced uniform norm on $\mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathbb{F}]$ when $\mathcal{X} \otimes \mathcal{Y}$ is equipped with the reasonable crossnorm $\|\cdot\|_\alpha$, so that $(\mathcal{X} \otimes_\alpha \mathcal{Y})^* = (\mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathbb{F}], \|\cdot\|_{*\alpha})$. According to Definition 7.1, $\mathcal{X}^* \otimes \mathcal{Y}^* \subseteq (\mathcal{X} \otimes_\alpha \mathcal{Y})^* = \mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathbb{F}]$ and

$$\|\cdot\|_{*\alpha} \text{ is a reasonable crossnorm on } \mathcal{X}^* \otimes \mathcal{Y}^*$$

by Theorem 7.3, so that $\mathcal{X}^* \otimes_{*\alpha} \mathcal{Y}^* = (\mathcal{X}^* \otimes \mathcal{Y}^*, \|\cdot\|_{*\alpha})$.

Now equip the tensor product space $\mathcal{X}^* \otimes \mathcal{Y}^*$ with an arbitrary reasonable crossnorm $\|\cdot\|_\alpha$, so that $\mathcal{X}^* \otimes_\alpha \mathcal{Y}^* = (\mathcal{X}^* \otimes \mathcal{Y}^*, \|\cdot\|_\alpha)$. Denote by $\|\cdot\|_{\alpha^\sharp}$ the induced uniform norm on $\mathcal{B}[\mathcal{X}^* \otimes_\alpha \mathcal{Y}^*, \mathbb{F}]$ when $\mathcal{X}^* \otimes \mathcal{Y}^*$ is equipped with the reasonable crossnorm $\|\cdot\|_\alpha$, so that $(\mathcal{X}^* \otimes_\alpha \mathcal{Y}^*)^* = (\mathcal{B}[\mathcal{X}^* \otimes_\alpha \mathcal{Y}^*, \mathbb{F}], \|\cdot\|_{\alpha^\sharp})$. Accordingly, $\mathcal{X}^{**} \otimes \mathcal{Y}^{**} \subseteq (\mathcal{X}^* \otimes_\alpha \mathcal{Y}^*)^* = \mathcal{B}[\mathcal{X}^* \otimes_\alpha \mathcal{Y}^*, \mathbb{F}]$ and

$$\|\cdot\|_{\alpha^\sharp} \text{ is a reasonable crossnorm on } \mathcal{X}^{**} \otimes \mathcal{Y}^{**}$$

by Theorem 7.3, so that $\mathcal{X}^{**} \otimes_{\alpha^\sharp} \mathcal{Y}^{**} = (\mathcal{X}^{**} \otimes \mathcal{Y}^{**}, \|\cdot\|_{\alpha^\sharp})$.

For a while we restrict our attention to finite-dimensional spaces. First, recall that in finite dimensions all normed spaces are complete, all norms are equivalent, and all linear transformations between normed spaces are continuous. Thus, in this case, the notation * standing for dual and adjoint agrees with the notation $^\sharp$ of Sections 1.1, 3.2 and 3.3 standing for algebraic dual and algebraic adjoint. So if $\mathcal{M}$ and $\mathcal{N}$ are finite-dimensional normed spaces, then let $\mathcal{M}^\sharp$ and $\mathcal{N}^\sharp$ stand for the duals of the normed spaces $\mathcal{M}$ and $\mathcal{N}$ (which are again finite-dimensional — cf. Proposition 1.V), and let $A^\sharp$ in $\mathcal{L}[\mathcal{N}^\sharp, \mathcal{M}^\sharp] = \mathcal{B}[\mathcal{N}^\sharp, \mathcal{M}^\sharp]$ stand for the normed-space adjoint of A in $\mathcal{L}[\mathcal{M}, \mathcal{N}] = \mathcal{B}[\mathcal{M}, \mathcal{N}]$.

Then take finite-dimensional normed spaces $\mathcal{M}$ and $\mathcal{N}$ (e.g., they might be finite-dimensional subspaces of $\mathcal{X}$ and $\mathcal{Y}$). Suppose both linear (tensor product) spaces $\mathcal{M} \otimes \mathcal{N}$ and $\mathcal{M}^\sharp \otimes \mathcal{N}^\sharp$ are equipped with the same reasonable crossnorm $\|\cdot\|_\alpha$. First, as we saw above for the general case,

$$\mathcal{M}^\sharp \otimes \mathcal{N}^\sharp \subseteq (\mathcal{M} \otimes_\alpha \mathcal{N})^\sharp = (\mathcal{B}[\mathcal{M} \otimes_\alpha \mathcal{N}, \mathbb{F}], \|\cdot\|_{\sharp\alpha}),$$

with $\|\cdot\|_{\sharp\alpha}$ standing for the induced uniform norm on $\mathcal{B}[\mathcal{M} \otimes_\alpha \mathcal{N}, \mathbb{F}]$ when $\mathcal{M} \otimes \mathcal{N}$ is equipped with $\|\cdot\|_\alpha$, and so $\|\cdot\|_{\sharp\alpha}$ becomes a reasonable crossnorm on $\mathcal{M}^\sharp \otimes \mathcal{N}^\sharp$. Next, besides the reasonable crossnorm $\|\cdot\|_{\sharp\alpha}$, equip $\mathcal{M}^\sharp \otimes \mathcal{N}^\sharp$ with the reasonable crossnorm $\|\cdot\|_\alpha$ and consider the tensor normed space

$$\mathcal{M}^\sharp \otimes_\alpha \mathcal{N}^\sharp = (\mathcal{M}^\sharp \otimes \mathcal{N}^\sharp, \|\cdot\|_\alpha).$$

Let $\|\cdot\|_{\alpha^\sharp}$ be the induced uniform norm on $(\mathcal{M}^\sharp \otimes_\alpha \mathcal{N}^\sharp)^\sharp = \mathcal{B}[\mathcal{M}^\sharp \otimes_\alpha \mathcal{N}^\sharp, \mathbb{F}]$ when $\mathcal{M}^\sharp \otimes \mathcal{N}^\sharp$ is equipped with $\|\cdot\|_\alpha$ so that

$$\mathcal{M}^{\sharp\sharp} \otimes \mathcal{N}^{\sharp\sharp} \subseteq (\mathcal{M}^\sharp \otimes_\alpha \mathcal{N}^\sharp)^\sharp = (\mathcal{B}[\mathcal{M}^\sharp \otimes_\alpha \mathcal{N}^\sharp, \mathbb{F}], \|\cdot\|_{\alpha^\sharp}).$$

As we saw above, $\|\cdot\|_{\alpha^\sharp}$ is a reasonable crossnorm on the linear space $\mathcal{M}^{\sharp\sharp} \otimes \mathcal{N}^{\sharp\sharp}$. In such a finite-dimensional setting, the linear spaces $\mathcal{M} \otimes \mathcal{N}$,

$\mathcal{M}^{\sharp\sharp} \otimes \mathcal{N}^{\sharp\sharp}$ and $(\mathcal{M}^\sharp \otimes_\alpha \mathcal{N}^\sharp)^\sharp$ are pairwise (algebraically) isomorphic to each other — reason: all linear spaces involved are finite-dimensional, and so the dimensions of $\mathcal{M} \otimes \mathcal{N}$, $\mathcal{M}^{\sharp\sharp} \otimes \mathcal{N}^{\sharp\sharp}$ and $(\mathcal{M}^\sharp \otimes_\alpha \mathcal{N}^\sharp)^\sharp$ coincide (cf. Propositions 1.C and 1.V). Equip the linear space $\mathcal{M} \otimes \mathcal{N}$ with the induced uniform norm $\|\cdot\|_{\alpha^\sharp}$ on $(\mathcal{M}^\sharp \otimes_\alpha \mathcal{N}^\sharp)^\sharp$, which is a reasonable crossnorm on $\mathcal{M}^{\sharp\sharp} \otimes \mathcal{N}^{\sharp\sharp}$, thus a reasonable crossnorm on $\mathcal{M} \otimes \mathcal{N}$. We summarize the argument as follows.

Lemma 8.18. *Let $\mathcal{M}$ and $\mathcal{N}$ be finite-dimensional normed spaces. Let $\|\cdot\|_\alpha$ be a reasonable crossnorm on $\mathcal{M} \otimes \mathcal{N}$. Define another reasonable crossnorm*

$$\|\cdot\|_{\alpha^\sharp} : \mathcal{M} \otimes \mathcal{N} \to \mathbb{R}$$

on $\mathcal{M} \otimes \mathcal{N}$ as follows: $\|\cdot\|_{\alpha^\sharp}$ is the induced uniform norm on $(\mathcal{M}^\sharp \otimes_\alpha \mathcal{N}^\sharp)^\sharp$, which is a reasonable crossnorm on $\mathcal{M}^{\sharp\sharp} \otimes \mathcal{N}^{\sharp\sharp}$, which in turn is identified with $\mathcal{M} \otimes \mathcal{N}$, so that

(a) $\mathcal{M}^\sharp \otimes \mathcal{N}^\sharp \subseteq (\mathcal{M} \otimes_\alpha \mathcal{N})^\sharp$ (*algebraic embedding*; *isomorphism, actually*),

(b) $\mathcal{M} \otimes_{\alpha^\sharp} \mathcal{N} \cong (\mathcal{M}^\sharp \otimes_\alpha \mathcal{N}^\sharp)^\sharp$ (*isometric isomorphism*).

Moreover, with $F = \sum_i x_i \otimes y_i \in \mathcal{M} \otimes_{\alpha^\sharp} \mathcal{N} \cong (\mathcal{M}^\sharp \otimes_\alpha \mathcal{N}^\sharp)^\sharp = \mathcal{B}[\mathcal{M}^\sharp \otimes_\alpha \mathcal{N}^\sharp, \mathbb{F}]$ and $\Omega = \sum_i f_i \otimes g_i \in \mathcal{M}^\sharp \otimes \mathcal{N}^\sharp \subseteq (\mathcal{M} \otimes_{\alpha^\sharp} \mathcal{N})^\sharp = \mathcal{B}[\mathcal{M} \otimes_{\alpha^\sharp} \mathcal{N}, \mathbb{F}]$, we get

$$\|\Omega\|_{\mathcal{B}[\mathcal{M} \otimes_{\alpha^\sharp} \mathcal{N}, \mathbb{F}]} = \sup_{F \in \mathcal{M} \otimes \mathcal{N},\, \|F\|_{\alpha^\sharp} \le 1} |\Omega(F)|,$$

and dually,

$$\|F\|_{\alpha^\sharp} = \sup_{\Omega \in \mathcal{M}^\sharp \otimes \mathcal{N}^\sharp,\, \|\Omega\|_{\mathcal{B}[\mathcal{M} \otimes_{\alpha^\sharp} \mathcal{N}, \mathbb{F}]} \le 1} |\Omega(F)|,$$

or equivalently,

$$\|F\|_{\alpha^\sharp} = \|F\|_{\mathcal{B}[\mathcal{M}^\sharp \otimes_\alpha \mathcal{N}^\sharp, \mathbb{F}]} = \sup_{\Omega \in \mathcal{M}^\sharp \otimes \mathcal{N}^\sharp,\, \|\Omega\|_\alpha \le 1} |F(\Omega)|, \qquad (**)$$

and dually,

$$\|\Omega\|_\alpha = \sup_{F \in (\mathcal{M}^\sharp \otimes_\alpha \mathcal{N}^\sharp)^\sharp,\, \|F\|_{\alpha^\sharp} \le 1} |F(\Omega)|.$$

Lemma 8.19. *If $\|\cdot\|_\alpha$ is a reasonable crossnorm associated with finite-dimensional normed spaces, then*

(a) $\|\cdot\|_{(\alpha^\sharp)^\sharp} = \|\cdot\|_\alpha$ (*dropping the parenthesis*: $\|\cdot\|_{\alpha^{\sharp\sharp}} = \|\cdot\|_{(\alpha^\sharp)^\sharp}$),

(b) $\|\cdot\|_{({}^t\alpha)^\sharp} = \|\cdot\|_{{}^t(\alpha^\sharp)}$ (*this is the* adjoint *or* contragradient *norm*),

(c) $\|\cdot\|_{\vee^\sharp} = \|\cdot\|_\wedge$ *and* $\|\cdot\|_{\wedge^\sharp} = \|\cdot\|_\vee$,

(d) $\|\cdot\|_{\alpha^\sharp}$ *is a uniform crossnorm whenever* $\|\cdot\|_\alpha$ *is a uniform crossnorm*,

(e) $\|\cdot\|_{({}^t\alpha)^\sharp}$ *is a uniform crossnorm whenever* $\|\cdot\|_\alpha$ *is a uniform crossnorm.*

If $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ are reasonable crossnorms associated with finite-dimensional normed spaces, then

(f) $\|\cdot\|_\alpha \le \|\cdot\|_\beta$ *if and only if* $\|\cdot\|_{\beta^\sharp} \le \|\cdot\|_{\alpha^\sharp}$.

Proof. Let $\mathcal{M},\mathcal{N},\mathcal{R},\mathcal{S}$ be finite-dimensional normed spaces and let $\|\cdot\|_\alpha$ be a reasonable crossnorm. Equip $\mathcal{M}^\sharp \otimes \mathcal{N}^\sharp$ and $\mathcal{R}^\sharp \otimes \mathcal{S}^\sharp$ with $\|\cdot\|_\alpha$. Recall:

$$\mathcal{M} \otimes_{\alpha^\sharp} \mathcal{N} \cong (\mathcal{M}^\sharp \otimes_\alpha \mathcal{N}^\sharp)^\sharp \quad \text{and} \quad \mathcal{R} \otimes_{\alpha^\sharp} \mathcal{S} \cong (\mathcal{R}^\sharp \otimes_\alpha \mathcal{S}^\sharp)^\sharp.$$

(a) By finite dimensionality, identify dual with primal (cf. Propositions 1.C and 1.V and Remark 3.12) so that

$$\mathcal{M} \otimes_{(\alpha^\sharp)^\sharp} \mathcal{N} \cong (\mathcal{M}^\sharp \otimes_{\alpha^\sharp} \mathcal{N}^\sharp)^\sharp \cong (\mathcal{M}^{\sharp\sharp} \otimes_\alpha \mathcal{N}^{\sharp\sharp})^{\sharp\sharp} \cong \mathcal{M} \otimes_\alpha \mathcal{N}.$$

(b) According to Theorem 7.6(a,b), $\|\cdot\|_{t_\alpha}$ is a reasonable crossnorm whenever $\|\cdot\|_\alpha$ is, and $\mathcal{X} \otimes_{t_\alpha} \mathcal{Y} \cong \mathcal{Y} \otimes_\alpha \mathcal{X}$. Therefore

$$\mathcal{M} \otimes_{t_{(\alpha^\sharp)}} \mathcal{N} \cong \mathcal{N} \otimes_{\alpha^\sharp} \mathcal{M} \cong (\mathcal{N}^\sharp \otimes_\alpha \mathcal{M}^\sharp)^\sharp \cong (\mathcal{M}^\sharp \otimes_{t_\alpha} \mathcal{N}^\sharp)^\sharp \cong \mathcal{M} \otimes_{(t_\alpha)^\sharp} \mathcal{N}.$$

(c) From Theorems 7.17(a) and 7.12, $\mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y} \subseteq \boldsymbol{b}[\mathcal{X}^* \times \mathcal{Y}^*, \mathbb{F}] \cong (\mathcal{X}^* \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}^*)^*$. In the finite-dimensional case, the above tensor product spaces have the same finite dimension. Thus the isometric embedding $\subseteq$ becomes an isometric isomorphism $\cong$ and so, since there is no need of completion in finite dimension,

$$\mathcal{M} \otimes_\vee \mathcal{N} \cong (\mathcal{M}^\sharp \otimes_\wedge \mathcal{Y}^\sharp)^\sharp \cong \mathcal{M} \otimes_{\wedge^\sharp} \mathcal{N}.$$

Then $\|\cdot\|_\vee = \|\cdot\|_{\wedge^\sharp}$, and so by (a), $\|\cdot\|_{\vee^\sharp} = \|\cdot\|_{\wedge^{\sharp\sharp}} = \|\cdot\|_\wedge$.

(d) Take bounded linear transformations

$$A\colon \mathcal{M} \to \mathcal{R}, \quad A^\sharp\colon \mathcal{R}^\sharp \to \mathcal{M}^\sharp \quad \text{and} \quad B\colon \mathcal{N} \to \mathcal{S}, \quad B^\sharp\colon \mathcal{S}^\sharp \to \mathcal{N}^\sharp,$$

and consider the linear tensor product transformations

$$A \otimes B\colon \mathcal{M} \otimes \mathcal{N} \to \mathcal{R} \otimes \mathcal{S} \quad \text{and} \quad A^\sharp \otimes B^\sharp\colon \mathcal{R}^\sharp \otimes \mathcal{S}^\sharp \to \mathcal{M}^\sharp \otimes \mathcal{N}^\sharp.$$

Equip the tensor product spaces with reasonable crossnorms as follows.

$$A \otimes B\colon \mathcal{M} \otimes_{\alpha^\sharp} \mathcal{N} \to \mathcal{R} \otimes_{\alpha^\sharp} \mathcal{S} \quad \text{and} \quad A^\sharp \otimes B^\sharp\colon \mathcal{R}^\sharp \otimes_\alpha \mathcal{S}^\sharp \to \mathcal{M}^\sharp \otimes_\alpha \mathcal{N}^\sharp.$$

Hence

$$(A^\sharp \otimes B^\sharp)^\sharp\colon (\mathcal{M}^\sharp \otimes_\alpha \mathcal{N}^\sharp)^\sharp \to (\mathcal{R}^\sharp \otimes_\alpha \mathcal{S}^\sharp)^\sharp.$$

Finite dimensionality allowed us to conclude the isometrically isomorphic equivalence $\mathcal{M} \otimes_{\alpha^\sharp} \mathcal{N} \cong (\mathcal{M}^\sharp \otimes_\alpha \mathcal{N}^\sharp)^\sharp$ and $\mathcal{R} \otimes_{\alpha^\sharp} \mathcal{S} \cong (\mathcal{R}^\sharp \otimes_\alpha \mathcal{S}^\sharp)^\sharp$. In the realm of finite dimensions (where linear transformations are bounded), we can infer from Theorem 3.19(j) that $A \otimes B \cong (A^\sharp \otimes B^\sharp)^\sharp$. Thus there exist isometric isomorphisms $J\colon \mathcal{M} \otimes_{\alpha^\sharp} \mathcal{N} \to (\mathcal{M}^\sharp \otimes_\alpha \mathcal{N}^\sharp)^\sharp$ and $K\colon \mathcal{R} \otimes_{\alpha^\sharp} \mathcal{S} \to (\mathcal{R}^\sharp \otimes_\alpha \mathcal{S}^\sharp)^\sharp$ such that $K(A \otimes B) = (A^\sharp \otimes B^\sharp)^\sharp J$. Equivalently, the diagram

$$\begin{array}{ccc} \mathcal{M} \otimes_{\alpha^\sharp} \mathcal{N} & \xrightarrow{A \otimes B} & \mathcal{R} \otimes_{\alpha^\sharp} \mathcal{S} \\ {\scriptstyle J}\big\downarrow & & \big\downarrow{\scriptstyle K} \\ (\mathcal{M}^\sharp \otimes_\alpha \mathcal{N}^\sharp)^\sharp & \xrightarrow{(A^\sharp \otimes B^\sharp)^\sharp} & (\mathcal{R}^\sharp \otimes_\alpha \mathcal{S}^\sharp)^\sharp \end{array}$$

commutes. Hence $A \otimes B$ and $(A^\sharp \otimes B^\sharp)^\sharp$ are isometrically equivalent bounded linear transformations, and therefore their norms coincide — cf. Section 5.1. So if $\|\cdot\|_\alpha$ is uniform, then $A^\sharp \otimes B^\sharp$ is bounded (and so is $(A^\sharp \otimes B^\sharp)^\sharp$), and

$$\begin{aligned}\|A \otimes B\|_{\mathcal{B}[\mathcal{M}\otimes_{\alpha^\sharp}\mathcal{N},\, \mathcal{R}\otimes_{\alpha^\sharp}\mathcal{S}]} &= \|(A^\sharp \otimes B^\sharp)^\sharp\|_{\mathcal{B}[(\mathcal{M}^\sharp\otimes_\alpha\mathcal{N}^\sharp)^\sharp,\, (\mathcal{R}^\sharp\otimes_\alpha\mathcal{S}^\sharp)^\sharp]} \\ &= \|A^\sharp \otimes B^\sharp\|_{\mathcal{B}[\mathcal{R}^\sharp\otimes_\alpha\mathcal{S}^\sharp,\, \mathcal{M}^\sharp\otimes_\alpha\mathcal{N}^\sharp]} \\ &\le \|A^\sharp\|\,\|B^\sharp\| = \|A\|\,\|B\|,\end{aligned}$$

which implies $\|\cdot\|_{\alpha^\sharp}$ is uniform.

(e) If $\|\cdot\|_\alpha$ is a uniform crossnorm, then $\|\cdot\|_{t\alpha}$ is a uniform crossnorm by Theorem 8.11, and so $\|\cdot\|_{(t\alpha)^\sharp}$ is a uniform crossnorm by (d).

(f) Let $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ be reasonable crossnorms such that $\|\cdot\|_\alpha \le \|\cdot\|_\beta$. Thus $\|\Omega\|_\beta \le 1$ implies $\|\Omega\|_\alpha \le 1$, and so the unit ball equipped with $\|\cdot\|_\beta$ is included in the unit ball equipped $\|\cdot\|_\alpha$. Hence, for an arbitrary $F \in \mathcal{M}\otimes\mathcal{N}$,

$$\begin{aligned}\|F\|_{\beta^\sharp} &= \|F\|_{(\mathcal{M}^\sharp\otimes_\beta\mathcal{N}^\sharp)^\sharp} = \|F\|_{\mathcal{B}[\mathcal{M}^\sharp\otimes_\beta\mathcal{Y}^\sharp,\mathbb{F}]} \\ &= \sup_{\Omega\in\mathcal{M}^\sharp\otimes\mathcal{N}^\sharp,\, \|\Omega\|_\beta\le 1} |F(\Omega)| \\ &\le \sup_{\Omega\in\mathcal{M}^\sharp\otimes\mathcal{N}^\sharp,\, \|\Omega\|_\alpha\le 1} |F(\Omega)| = \|F\|_{\alpha^\sharp}.\end{aligned}$$

So $\|\cdot\|_\alpha \le \|\cdot\|_\beta$ implies $\|\cdot\|_{\beta^\sharp} \le \|\cdot\|_{\alpha^\sharp}$. The converse follows by (a). □

Remark 8.20. *Lattice of Tensor Norms.* Take a family $\{\|\cdot\|_{\alpha_\gamma}\}_{\gamma\in\Gamma}$ of reasonable crossnorms, each of them assigned to every finite-dimensional tensor product space. For an arbitrary pair of finite-dimensional normed spaces $\mathcal{M}$ and $\mathcal{N}$, consider the functional $\|\cdot\|_{\vee\alpha_\gamma} : \mathcal{M}\otimes\mathcal{N} \to \mathbb{R}$ defined by

$$\|F\|_{\vee\alpha_\gamma} = \sup_{\gamma\in\Gamma} \|F\|_{\alpha_\gamma} \quad \text{for each} \quad F \in \mathcal{M}\otimes\mathcal{N}.$$

This is well-defined since it is naturally bounded by the projective norm (i.e., $\|\cdot\|_{\vee\alpha_\gamma} \le \|\cdot\|_\wedge$). It is readily verified that $\|\cdot\|_{\vee\alpha_\gamma}$ is a norm on $\mathcal{M}\otimes\mathcal{N}$, and also that $\|x\otimes y\|_{\vee\alpha_\gamma} = \|x\|\,\|y\|$ for every single tensor $x\otimes y$ in $\mathcal{M}\otimes\mathcal{N}$. Moreover, it is also easy to verify that $\|f\otimes g\|_{\mathcal{B}[\mathcal{M}\otimes_{\vee\alpha_\gamma}\mathcal{N},\,\mathbb{F}]} \le \|f\|\,\|g\|$ for every single tensor $f\otimes g$ in $\mathcal{M}^*\otimes\mathcal{N}^*$. Hence

$$\|\cdot\|_{\vee\alpha_\gamma} \text{ is a reasonable crossnorm on } \mathcal{M}\otimes\mathcal{N}.$$

Now, if each reasonable crossnorm $\|\cdot\|_{\alpha_\gamma}$ is uniform, then so is $\|\cdot\|_{\vee\alpha_\gamma}$. In fact, let $\mathcal{M},\mathcal{N},\mathcal{R},\mathcal{S}$ be finite-dimensional normed spaces, take $A\in\mathcal{B}[\mathcal{M},\mathcal{R}]$ and $B\in\mathcal{B}[\mathcal{N},\mathcal{S}]$, and take $A\otimes B \in \mathcal{B}[\mathcal{M}\otimes\mathcal{N},\mathcal{R}\otimes\mathcal{S}]$. If $\|\cdot\|_{\alpha_\gamma}$ is uniform, then $\|A\otimes B\|_{\mathcal{B}[\mathcal{M}\otimes_{\alpha_\gamma}\mathcal{N},\mathcal{R}\otimes_{\alpha_\gamma}\mathcal{S}]} \le \|A\|\,\|B\|$ for each $\gamma\in\Gamma$, and so for $F\in\mathcal{M}\otimes\mathcal{N}$,

$$\|(A\otimes B)F\|_{\vee\alpha_\gamma} = \sup_{\gamma\in\Gamma}\|(A\otimes B)F\|_{\alpha_\gamma} \le \sup_{\gamma\in\Gamma}\|A\|\,\|B\|\,\|F\|_{\alpha_\gamma} = \|A\|\,\|B\|\,\|F\|_{\vee\alpha_\gamma}.$$

Then $\|A\otimes B\|_{\mathcal{B}[\mathcal{M}\otimes_{\vee\alpha_\gamma}\mathcal{N},\mathcal{R}\otimes_{\vee\alpha_\gamma}\mathcal{S}]} \le \|A\|\,\|B\|$. Hence $\|\cdot\|_{\vee\alpha_\gamma}$ is uniform. Thus

$\|\cdot\|_{\vee\alpha_\gamma}$ is a uniform crossnorm whenever every $\|\cdot\|_{\alpha_\gamma}$ is.

As happens with any real-valued function, (i) if $\|\cdot\|_{\alpha_\gamma} \le \|\cdot\|_{\alpha_\delta}$ and $\|\cdot\|_{\alpha_\delta} \le \|\cdot\|_{\alpha_\gamma}$, then $\|\cdot\|_{\alpha_\gamma} = \|\cdot\|_{\alpha_\delta}$ (i.e., the relation $\le$ is antisymmetric), (ii) $\|\cdot\|_{\alpha_\gamma} \le \|\cdot\|_{\alpha_\gamma}$ (i.e., it is reflexive), and (iii) $\|\cdot\|_{\alpha_\gamma} \le \|\cdot\|_{\alpha_\delta}$ and $\|\cdot\|_{\alpha_\delta} \le \|\cdot\|_{\alpha_\epsilon}$ imply $\|\cdot\|_{\alpha_\gamma} \le \|\cdot\|_{\alpha_\epsilon}$ (i.e., it is transitive). Then the relation $\le$ establishes a partial ordering on the collection of all reasonable (uniform) crossnorms associated with finite-dimensional normed spaces. Finally, recall that a *complete lattice* is a partially ordered set such that every subset of it has a supremum and an infimum in the set (see, e.g., [52, Section 1.6]). Consequently,

> *the collection of all reasonable crossnorms (uniform crossnorms) on tensor products of finite-dimensional normed spaces is a complete lattice.*

Now return to arbitrary normed spaces (of arbitrary dimension) $\mathcal{X}$ and $\mathcal{Y}$. Let $\|\cdot\|_\alpha$ be a uniform crossnorm. Thus it acts, in particular, on $\mathcal{X}\otimes\mathcal{Y}$. Let $\mathcal{FD}(\mathcal{X}\otimes\mathcal{Y})$ be the collection of all tensor product spaces $\mathcal{M}\otimes\mathcal{N}$ of pairs of finite-dimensional subspaces $\mathcal{M}$ and $\mathcal{N}$ of $\mathcal{X}$ and $\mathcal{Y}$. Write $\|\cdot\|_\alpha|_{\mathcal{FD}(\mathcal{X}\otimes\mathcal{Y})}$ for a generic restriction $\|\cdot\|_\alpha|_{\mathcal{M}\otimes\mathcal{N}}$ of the norm $\|\cdot\|_\alpha$ to an arbitrary $\mathcal{M}\otimes\mathcal{N}$ in $\mathcal{FD}(\mathcal{X}\otimes\mathcal{Y})$. For each restriction of $\|\cdot\|_\alpha$ to any $\mathcal{M}\otimes\mathcal{N}$ in $\mathcal{FD}(\mathcal{X}\otimes\mathcal{Y})$, take the reasonable crossnorm $\|\cdot\|_{\alpha^\sharp}$ associated with it as in Lemma 8.18, where

$$\|\cdot\|_{\alpha^\sharp} \text{ is the induced uniform norm on } \mathcal{B}[\mathcal{M}^\sharp\otimes_\alpha\mathcal{N}^\sharp,\mathbb{F}] = \mathcal{B}[\mathcal{M}^*\otimes_\alpha\mathcal{N}^*,\mathbb{F}]$$

for every $\mathcal{M}\otimes\mathcal{N}\in\mathcal{FD}(\mathcal{X}\otimes\mathcal{Y})$. Since $\|\cdot\|_\alpha$ is a uniform crossnorm, then so is $\|\cdot\|_\alpha|_{\mathcal{FD}(\mathcal{X}\otimes\mathcal{Y})}$, which implies that $\|\cdot\|_{\alpha^\sharp}$ is a uniform crossnorm by Lemma 8.19(d), independently of the pair of finite-dimensional subspaces $\mathcal{M}$ and $\mathcal{N}$.

Theorem 8.21. *Let $\mathcal{X}$ and $\mathcal{Y}$ be arbitrary normed spaces. Take a uniform crossnorm $\|\cdot\|_\alpha$. Let $\|\cdot\|_{\alpha^\sharp}$ be the uniform crossnorm associated with $\|\cdot\|_\alpha$ acting on finite-dimensional tensor products from $\mathcal{FD}(\mathcal{X}\otimes\mathcal{Y})$. The functional $\|\cdot\|_{\alpha^*}\colon\mathcal{X}\otimes\mathcal{Y}\to\mathbb{R}$ defined for every $f\in\mathcal{X}\otimes\mathcal{Y}$ by*

$$\|f\|_{\alpha^*} = \|f\|_{\mathcal{X}\otimes_{\alpha^*}\mathcal{Y}} = \inf_{\dim\mathcal{M}<\infty,\,\dim\mathcal{N}<\infty,\,f\in\mathcal{M}\otimes\mathcal{N}} \|f\|_{\mathcal{M}\otimes_{\alpha^\sharp}\mathcal{N}} \qquad (*)'$$

is a tensor norm that coincides with $\|\cdot\|_{\alpha^\sharp}$ when acting on a finite-dimensional tensor product space (i.e., $\|\cdot\|_{\mathcal{M}\otimes_{\alpha^*}\mathcal{N}} = \|\cdot\|_{\mathcal{M}\otimes_{\alpha^\sharp}\mathcal{N}}$ on $\mathcal{M}\otimes\mathcal{N}$ in $\mathcal{FD}(\mathcal{X}\otimes\mathcal{Y})$).

Proof. As $\|\cdot\|_{\alpha^\sharp}$ is a finite-dimensional uniform crossnorm, Theorem 8.12 via $(*)$ says that $\|\cdot\|_{\alpha^*}$ in $(*)'$ is a tensor norm that agrees with $\|\cdot\|_{\alpha^\sharp}$ when restricted to tensor products of finite-dimensional subspaces of $\mathcal{X}$ and $\mathcal{Y}$. □

This norm $\|\cdot\|_{\alpha^*}\colon\mathcal{X}\otimes\mathcal{Y}\to\mathbb{R}$ is the *dual norm* of $\|\cdot\|_\alpha\colon\mathcal{X}\otimes\mathcal{Y}\to\mathbb{R}$. Theorem 8.21 leads to a restatement of Lemma 8.19 in terms of $\|\cdot\|_{\alpha^*}$. In particular, $\|\cdot\|_{\wedge^*} = \|\cdot\|_\vee$ and $\|\cdot\|_{\vee^*} = \|\cdot\|_\wedge$ as in Proposition 8.H.

Remark 8.22. *An Expression for the Dual Norm.* Let $\mathcal{X}$ and $\mathcal{Y}$ be arbitrary normed spaces and let $\|\cdot\|_\alpha$ be a reasonable crossnorm on $\mathcal{X}^*\otimes\mathcal{Y}^*$. The norm

$\|\cdot\|_{\alpha^\sharp}$ in Theorem 8.21 was restricted to finite-dimensional tensor products. However, $\|\cdot\|_{\alpha^\sharp}$ has been originally defined on arbitrary tensor products as the induced uniform norm on $\mathcal{B}[\mathcal{X}^* \otimes_\alpha \mathcal{Y}^*, \mathbb{F}]$, that is,

$$\|\cdot\|_{\alpha^\sharp} \text{ is the induced uniform norm } (\mathcal{X}^* \otimes_\alpha \mathcal{Y}^*)^*.$$

Since $\mathcal{X}^{**} \otimes \mathcal{Y}^{**} \subseteq (\mathcal{X}^* \otimes_\alpha \mathcal{Y}^*)^*$ and since $\|\cdot\|_{\alpha^\sharp}$ is a reasonable crossnorm on $\mathcal{X}^{**} \otimes \mathcal{Y}^{**}$ by Theorem 7.3, we consider the crossnormed space $\mathcal{X}^{**} \otimes_{\alpha^\sharp} \mathcal{Y}^{**}$:

$$\|\cdot\|_{\alpha^\sharp} \text{ is a reasonable crossnorm on } \mathcal{X}^{**} \otimes \mathcal{Y}^{**}.$$

Now recall the natural isometric embeddings $\mathcal{X} \hookrightarrow \mathcal{X}^{**}$ and $\mathcal{Y} \hookrightarrow \mathcal{Y}^{**}$ (every normed space is isometrically isomorphic to a linear manifold of its second dual). Thus $\mathcal{X} \otimes \mathcal{Y}$ is a regular (algebraic) linear manifold of $\mathcal{X}^{**} \otimes \mathcal{Y}^{**}$. Hence the reasonable crossnorm $\|\cdot\|_{\alpha^\sharp}$ on $\mathcal{X}^{**} \otimes \mathcal{Y}^{**}$ induces a norm on the linear space $\mathcal{X} \otimes \mathcal{Y}$. So equip it with this norm, and consider the following setup:

$$(\mathcal{X} \otimes \mathcal{Y}, \|\cdot\|_{\alpha^\sharp}) \subseteq (\mathcal{X}^* \otimes_\alpha \mathcal{Y}^*)^* = \mathcal{B}[\mathcal{X}^* \otimes_\alpha \mathcal{Y}^*, \mathbb{F}].$$

Then $F \in (\mathcal{X} \otimes \mathcal{Y}, \|\cdot\|_{\alpha^\sharp})$ is identified with a functional from $(\mathcal{X}^* \otimes_\alpha \mathcal{Y}^*)^* = \mathcal{B}[\mathcal{X}^* \otimes_\alpha \mathcal{Y}^*, \mathbb{F}] = (\mathcal{B}[\mathcal{X}^* \otimes_\alpha \mathcal{Y}^*, \mathbb{F}], \|\cdot\|_{\alpha^\sharp})$, and so we set

$$\|F\|_{\alpha^\sharp} = \|F\|_{(\mathcal{X}\otimes\mathcal{Y}, \|\cdot\|_{\alpha^\sharp})} = \|F\|_{\mathcal{B}[\mathcal{X}^* \otimes_\alpha \mathcal{Y}^*, \mathbb{F}]} = \sup_{\Omega \in \mathcal{X}^* \otimes \mathcal{Y}^*, \|\Omega\|_\alpha \le 1} |F(\Omega)| \quad (**)'$$

(compare with $(**)$ in Lemma 8.18). Thus if $\mathcal{X}$ and $\mathcal{Y}$ are normed spaces, and if $\|\cdot\|_\alpha$ is a reasonable crossnorm on $\mathcal{X}^* \otimes \mathcal{Y}^*$, then let $\|\cdot\|_{\alpha^\sharp} : \mathcal{X} \otimes \mathcal{Y} \to \mathbb{R}$ be defined as in $(**)'$ above. Suppose $\|\cdot\|_\alpha$ is a uniform crossnorm and consider its action on $\mathcal{X} \otimes \mathcal{Y}$ as well. Let $\|\cdot\|_{\alpha^*}$ be the tensor norm on $\mathcal{X} \otimes \mathcal{Y}$ as in $(*)'$ of Theorem 8.21 (i.e., the dual norm of $\|\cdot\|_\alpha : \mathcal{X} \otimes \mathcal{Y} \to \mathbb{R}$).

If $\mathcal{X}$ and $\mathcal{Y}$ are Banach spaces with the metric approximation property,

$$\|F\|_{\alpha^*} = \|F\|_{\alpha^\sharp}$$

for every $F \in \mathcal{X} \otimes \mathcal{Y}$, and so $\|\cdot\|_{\alpha^\sharp}$ becomes a tensor norm on $\mathcal{X} \otimes \mathcal{Y}$.

Indeed, let $\|\cdot\|_\alpha$ be a uniform crossnorm acting on both $\mathcal{X} \otimes \mathcal{Y}$ and $\mathcal{X}^* \otimes \mathcal{Y}^*$. When acting on $\mathcal{X}^* \otimes \mathcal{Y}^*$, it yields $\|\cdot\|_{\alpha^\sharp}$ on $\mathcal{X} \otimes \mathcal{Y}$ as defined in $(**)'$ above. When acting on $\mathcal{X} \otimes \mathcal{Y}$, its restriction to tensor products of finite-dimensional subspaces yields $\|\cdot\|_\alpha|_{\mathcal{FD}(\mathcal{X}\otimes\mathcal{Y})}$, which in turn yields $\|\cdot\|_{\alpha^\sharp}|_{\mathcal{FD}(\mathcal{X}\otimes\mathcal{Y})}$ as in $(**)$ of Lemma 8.18. The following six steps lead to the norms identity (and therefore Theorem 8.21 ensures that the norms are tensor norms).

(1) If $\|\cdot\|_\alpha$ on $\mathcal{X} \otimes \mathcal{Y}$ is uniform, then so are their restrictions by Definition 8.2.

(2) As $\|\cdot\|_\alpha|_{\mathcal{FD}(\mathcal{X}\otimes\mathcal{Y})}$ is uniform, then so is $\|\cdot\|_{\alpha^\sharp}|_{\mathcal{FD}(\mathcal{X}\otimes\mathcal{Y})}$ by Lemma 8.19(d).

(3) Consider $\|\cdot\|_{\alpha^\sharp}$ defined in $(**)'$ above. Its restriction $\|\cdot\|_{\alpha^\sharp}|_{\mathcal{FD}(\mathcal{X}\otimes\mathcal{Y})}$ coincides with $\|\cdot\|_{\alpha^\sharp}$ in $(**)$ of Lemma 8.18, which is uniform according to (2).

(4) So $\|\cdot\|_{\alpha^\sharp}$ in $(**)'$ is uniform on $\mathcal{FD}(\mathcal{X}\otimes\mathcal{Y})$, which can be extended to a uniform crossnorm (as it is the induced uniform norm on $\mathcal{B}[\mathcal{X}^*\otimes_\alpha\mathcal{Y}^*,\mathbb{F}]$).

(5) Since $\|\cdot\|_{\alpha^\sharp}$ in $(**)'$ is uniform, and since $\mathcal{X}$ and $\mathcal{Y}$ are Banach spaces with the approximation property, Theorem 8.15 and Definition 8.7(a) ensure that

$$\|F\|_{\alpha^\sharp} = \inf_{\dim\mathcal{M}<\infty,\,\dim\mathcal{N}<\infty,\,F\in\mathcal{M}\otimes\mathcal{N}} \|F\|_{\mathcal{M}\otimes_{\alpha^\sharp}\mathcal{N}} \quad \text{for every} \quad F\in\mathcal{X}\otimes\mathcal{Y}.$$

(6) Hence $\|\cdot\|_{\alpha^*} = \|\cdot\|_{\alpha^\sharp}$ by Theorem 8.21 (and so $\|\cdot\|_{\alpha^\sharp}$ is a tensor norm).

8.4 Additional Propositions

Proposition 5.H, and its converse for Banach spaces with a Schauder basis in Proposition 5.J, motivated the notion of the approximation property for a Banach space $\mathcal{Y}$, namely, $\mathcal{B}_0[\mathcal{X},\mathcal{Y}]$ dense in $\mathcal{B}_\infty[\mathcal{X},\mathcal{Y}]$ for every normed space $\mathcal{X}$, which was the subject of Remark 8.14. The approximation property has played a central role in subsequent results in this chapter, mainly through the stronger notion of the metric approximation property. Propositions 8.A to 8.F below compile further applications of the approximation property.

Proposition 8.A. *If a Banach space $\mathcal{X}$ is such that $\mathcal{X}^*$ has the approximation property, then so has $\mathcal{X}$.*

Proposition 8.B. *If a Banach space $\mathcal{X}$ is such that $\mathcal{X}^*$ has the metric approximation property, then so has $\mathcal{X}$.*

Proposition 8.C. *Let $\mathcal{X}$ and $\mathcal{Y}$ be Banach spaces. Take the projective norm.*

(a) *If $\mathcal{X}^*$ or $\mathcal{Y}^*$ has the approximation property, then $\mathcal{X}^*\,\widehat{\otimes}_\wedge\,\mathcal{Y}^* \cong \boldsymbol{b}_N[\mathcal{X}\times\mathcal{Y},\mathbb{F}]$,*

(b) *If $\mathcal{X}^*$ or $\mathcal{Y}$ has the approximation property, then $\mathcal{X}^*\,\widehat{\otimes}_\wedge\,\mathcal{Y} \cong \mathcal{B}_N[\mathcal{X},\mathcal{Y}]$.*

Compare Proposition 8.C with Propositions 7.O and 7.P. (The spaces of nuclear bilinear forms and nuclear transformations, $\boldsymbol{b}_N[\mathcal{X}\times\mathcal{Y},\mathbb{F}]$ and $\mathcal{B}_N[\mathcal{X},\mathcal{Y}]$, were defined in the paragraphs following those propositions.) As noted there, (a) $\boldsymbol{b}_N[\mathcal{X}\times\mathcal{Y},\mathbb{F}] \cong (\mathcal{X}^*\,\widehat{\otimes}_\wedge\,\mathcal{Y}^*)/\mathcal{N}(J)$ and (b) $\mathcal{B}_N[\mathcal{X},\mathcal{Y}] \cong (\mathcal{X}^*\,\widehat{\otimes}_\wedge\,\mathcal{Y})/\mathcal{N}(K)$. In fact, if $\mathcal{X}^*$ or $\mathcal{Y}^*$ in case (a), or $\mathcal{X}^*$ or $\mathcal{Y}$ in case (b), has the approximation property, then those kernels are null (see, e.g., [83, Proposition 4.6]). Also compare the result below with Proposition 7.R. (Recall from Remark 8.14(g): Hilbert spaces have the metric approximation property.)

Proposition 8.D. *Let $\mathcal{X}$ be a Banach space. Suppose $\mathcal{X}^*$ has the approximation property. Take the normed-space adjoint $T^*\in\mathcal{B}[\mathcal{X}^*]$ of $T\in\mathcal{B}[\mathcal{X}]$. If T^* is nuclear then so is T (i.e., $T^*\in\mathcal{B}_N[\mathcal{X}^*] \Longrightarrow T\in\mathcal{B}_N[\mathcal{X}]$) and $\|T\|_N = \|T^*\|_N$.*

Proposition 8.E. *Let $\mathcal{X}$ and $\mathcal{Y}$ be Banach spaces. Take the injective norm $\|\cdot\|_\vee$. If $\mathcal{X}^*$ or $\mathcal{Y}$ has the approximation property, then $\mathcal{X}^*\,\widehat{\otimes}_\vee\,\mathcal{Y} \cong \mathcal{B}_\infty[\mathcal{X},\mathcal{Y}]$.*

Theorem 3.9 ensures that if $\{e_i\}$ and $\{d_j\}$ are countable Hamel bases for linear spaces $\mathcal{X}$ and $\mathcal{Y}$, then $\{e_i \otimes d_j\}$ is a countable Hamel basis for the tensor product space $\mathcal{X} \otimes \mathcal{Y}$. A Banach-space version holds for Schauder bases. Recall from Section 5.4: a sequence $\{z_k\}$ with values in a Banach space $\mathcal{Z}$ is a *Schauder basis* for $\mathcal{Z}$ if for each $z \in \mathcal{Z}$ there is a unique scalar sequence $\{\alpha_k(z)\}$ such that $z = \lim_n \sum_{k=1}^{n} \alpha_k(z) z_k$. A condition (among others) for getting a Schauder basis for some reasonably crossnormed tensor product $\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}$ out of Schauder bases $\{e_i\}$ and $\{d_j\}$ for $\mathcal{X}$ and $\mathcal{Y}$ is this: endow the double-indexed countable family of single tensors $\{e_i \otimes d_j\}$ with a suitable order as follows. The *square ordering* in $\mathbb{N} \times \mathbb{N}$ complies with the pattern in the array below:

$$
\begin{array}{cccccccc}
i \uparrow & & & & & & & \\
 & & & & & & & \\
4 & & & & \leftarrow & 13 & & \\
 & & & & & \uparrow & & \\
3 & 9 & \leftarrow 8 & \leftarrow & 7 & 12 & & \\
 & & & & \uparrow & \uparrow & & \\
2 & 4 & \leftarrow 3 & & 6 & 11 & & \\
 & & \uparrow & & \uparrow & \uparrow & & \\
1 & 1 & \rightarrow 2 & & 5 & 10 & & \\
 & & & & & & & j \\
 & 1 & 2 & 3 & 4 & 5 & \xrightarrow{} & .
\end{array}
$$

Let $\{e_i\}$ and $\{d_j\}$ be $\mathcal{X}$-valued and $\mathcal{Y}$-valued sequences, take the $\mathcal{X} \otimes \mathcal{Y}$-valued countable family of single tensors $\{e_i \otimes d_j\}$ and square-order it so that

$$\{e_i \otimes d_j\} = \{e_1 \otimes d_1,\ e_1 \otimes d_2,\ e_2 \otimes d_2,\ e_2 \otimes d_1,\ e_1 \otimes d_3,\ e_2 \otimes d_3,\ \ldots\}.$$

Proposition 8.F. *If $\{e_i\}$ and $\{d_j\}$ are Schauder bases for Banach spaces $\mathcal{X}$ and $\mathcal{Y}$ and if the family of single tensors $\{e_i \otimes d_j\}$ is square-ordered, then $\{e_i \otimes d_j\}$ is a Schauder basis for both $\mathcal{X} \mathbin{\widehat{\otimes}_\vee} \mathcal{Y}$ and $\mathcal{X} \mathbin{\widehat{\otimes}_\wedge} \mathcal{Y}$.*

Proposition 8.G. *If $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ are tensor norms on $\mathcal{X} \otimes \mathcal{Y}$ for Banach spaces $\mathcal{X}$ and $\mathcal{Y}$, then $\|\cdot\|_\alpha \le \|\cdot\|_\beta$ if and only if $\|\cdot\|_{\mathcal{M} \otimes_\alpha \mathcal{N}} \le \|\cdot\|_{\mathcal{M} \otimes_\beta \mathcal{N}}$ for every pair of finite-dimensional subspaces $\mathcal{M}$ and $\mathcal{N}$ of $\mathcal{X}$ and $\mathcal{Y}$, respectively.*

Projective and injective norms (which are tensor norms by Theorem 8.9) are duals of each other in the sense of Theorem 8.21, as stated in Proposition 8.H below. This extends Lemma 8.19(c). Actually, as we said before, Theorem 8.21 leads to a natural restatement of the entire Lemma 8.19.

Proposition 8.H. $\|\cdot\|_{\wedge^*} = \|\cdot\|_\vee$ *and* $\|\cdot\|_{\vee^*} = \|\cdot\|_\wedge$.

Besides its intrinsic importance, the next proposition can also be viewed as an extension of Proposition 7.G (since the projective norm is a tensor norm by Corollary 8.10), or as a sort of counterpart of the canonical isometric embedding $\mathcal{X} \otimes_\alpha \mathcal{Y} \hookrightarrow (\mathcal{X} \otimes_\alpha \mathcal{Y})^{**}$ of a normed space into its second dual (as in the paragraph preceding the proof of Proposition 6.J).

Proposition 8.I. *If $\|\cdot\|_\alpha$ is a tensor norm on the tensor product of Banach spaces $\mathcal{X}$ and $\mathcal{Y}$, then $\mathcal{X} \otimes_\alpha \mathcal{Y}$ is a linear manifold of $\mathcal{X}^{**} \otimes_\alpha \mathcal{Y}^{**}$.*

As we did in Chapter 5, we close this chapter with a simple set of results (brought together in Remark 8.J below) that will be required in Chapter 9,

Remark 8.J. *On Extensions over Completion under Uniform Crossnorms.*
(a) Let $\|\cdot\|_\alpha$ be a uniform crossnorm, and let $\mathcal{X}\otimes\mathcal{Y}$ be the tensor product of arbitrary normed spaces $\mathcal{X}$ and $\mathcal{Y}$. As before, use that same notation $\|\cdot\|_\alpha$ for the uniform crossnorm in $\mathcal{X}\otimes_\alpha\mathcal{Y}$ and its extension in $\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$. Take an arbitrary pair $x\in\mathcal{X}$ and $y\in\mathcal{Y}$, and consider the single tensor $x\otimes y\in\mathcal{X}\otimes_\alpha\mathcal{Y}$. Let $x\,\widetilde{\otimes}\,y$ be its isometrically isomorphic image in $\mathcal{X}\,\widetilde{\otimes}_\alpha\mathcal{Y}\subseteq\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$, and let $x\,\widehat{\otimes}\,y$ be its correspondent (i.e., its extension) in $\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$. Thus

$$x\,\widehat{\otimes}\,y = x\,\widetilde{\otimes}\,y \cong x\otimes y \quad\text{and}\quad \|x\,\widehat{\otimes}\,y\|_\alpha = \|x\|\,\|y\|.$$

(b) Now take any pair of transformations $A\in\mathcal{B}[\mathcal{X},\mathcal{V}]$ and $B\in\mathcal{B}[\mathcal{Y},\mathcal{W}]$, for arbitrary normed spaces $\mathcal{V}$ and $\mathcal{W}$. According to item (a) above, we get

$$(A\,\widehat{\otimes}\,B)(x\,\widehat{\otimes}\,y) = (A\,\widetilde{\otimes}\,B)(x\,\widetilde{\otimes}\,y) \cong (A\otimes B)(x\otimes\ y) = Ax\otimes By$$
$$\text{and}\qquad \|(A\,\widehat{\otimes}\,B)(x\,\widehat{\otimes}\,y)\|_\alpha = \|Ax\|\,\|Bx\|.$$

(c) Observe that if $x_n\to x$ in $\mathcal{X}$ and $y_n\to y$ in $\mathcal{Y}$, then $x_n\otimes y_n\to x\otimes y$ in $\mathcal{X}\otimes_\alpha\mathcal{Y}$. (Indeed, $\|x_n\otimes y_n - x\otimes y\|_\alpha = \|x_n\otimes y_n - x\otimes y_n + x\otimes y_n - x\otimes y\|_\alpha \le \|x_n\otimes y_n - x\otimes y_n\|_\alpha + \|x\otimes y_n - x\otimes y\|_\alpha = \|(x_n - x)\otimes y_n\|_\alpha + \|x\otimes(y_n - y)\|_\alpha = \|x_n - x\|\,\|y_n\| + \|x\|\,\|y_n - y\| \to 0$.) Thus the isometrically isomorphic image $\{x_n\,\widetilde{\otimes}\,y_n\}$ of $\{x_n\otimes y_n\}$ in $\mathcal{X}\,\widetilde{\otimes}_\alpha\mathcal{Y}\subseteq\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$ converges in $\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$:

$$x_n\to x \;\text{ and }\; y_n\to y \quad\Longrightarrow\quad x_n\,\widehat{\otimes}\,y_n = x_n\,\widetilde{\otimes}\,y_n \to x\,\widetilde{\otimes}\,y = x\,\widehat{\otimes}\,y.$$

(d) Again, take any $A\in\mathcal{B}[\mathcal{X},\mathcal{V}]$ and $B\in\mathcal{B}[\mathcal{Y},\mathcal{W}]$ for arbitrary normed spaces $\mathcal{X},\mathcal{Y},\mathcal{V},\mathcal{W}$. If $x_n\to x$ in $\mathcal{X}$ and $y_n\to y$ in $\mathcal{Y}$, then $Ax_n\to Ax$ in $\mathcal{V}$ and $By_n\to By$ in $\mathcal{W}$ and so, according to the above results in (a), (b) and (c), and with convergence in $\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$ (since $\|\cdot\|_\alpha$ is a uniform crossnorm), we get

$$x_n\to x \;\text{ and }\; y_n\to y \quad\Longrightarrow\quad (A\,\widehat{\otimes}\,B)(x_n\,\widehat{\otimes}\,y_n)\to(A\,\widehat{\otimes}\,B)(x\,\widehat{\otimes}\,y)\cong Ax\otimes By.$$

Notes: Propositions 8.A and 8.B show that the approximation properties travel well from dual to primal (see, e.g., [83, Corollaries 4.7 and 4.15]). For the isometric isomorphisms in Proposition 8.C see, for instance, [83, Corollary 4.8], which shows that in this particular case there are invertible isometries playing the role of J and K in Propositions 7.O and 7.P in Section 5.4. Observe that, since Hilbert spaces have the metric approximation property (cf. Remark 8.14(g)), the equivalence in Proposition 8.C(b) holds for Hilbert space. In particular, recalling Propositions 7.B and 7.S(a),

$$\mathcal{X}\,\widehat{\otimes}_\wedge\mathcal{X}^* \cong \mathcal{X}^*\,\widehat{\otimes}_\wedge\mathcal{X} \cong \mathcal{B}_N[\mathcal{X}] = \mathcal{B}_1[\mathcal{X}]$$

if $\mathcal{X}$ is a Hilbert space, and this take us back to the original Schatten's definition of Hilbert-space trace-class operators: "The trace-class may be also interpreted as $\mathcal{X}\,\widehat{\otimes}_\wedge\mathcal{X}^*$" [85, Theorem 5.12]).

Proposition 8.D is a Banach-space version of Proposition 7.R in Section 7.4, and Proposition 8.E is a version of Proposition 8.C(b) for the injective norm (extended to compact operators — see, e.g., [83, Corollaries 4.10 and 4.13]). Proposition 8.F shows one of the possible ways of building a Schauder basis for injective and projective tensor products out of Schauder bases for the respective Banach spaces (see [83, Proposition 4.25]). (We will be back to Schauder — actually, orthonormal — bases for tensor products of Hilbert spaces in Remark 9.5(a) in the next chapter.) Proposition 8.G shows order preservation for tensor norms generated by Theorem 8.12 (see, e.g., [83, Proposition 6.3(a)]). As mentioned before, Proposition 8.H is an application of Lemma 8.19(c) and Theorem 8.21. For Proposition 8.I, see, e.g., [83, Proposition 6.4]).

Proposition 8.I is a significant result. To see the argument behind its proof, first recall that every normed space is isometrically embedded in its second dual (see proof of Proposition 6.J). So we may regard $\mathcal{X}$ as a subspace of $\mathcal{X}^{**}$ and $\mathcal{Y}$ as a subspace of $\mathcal{Y}^{**}$. Thus, according to Theorem 7.19, Proposition 8.I shows that a tensor norm behaves somewhat similar to the injective norm. (Moreover, as we have commented above, since the projective norm is a tensor norm, Proposition 8.I may also be regarded as extending the projective norm result in Proposition 7.G.) Indeed, it is shown in [83, proof of Proposition 6.4] that $\|F\|_{\mathcal{X}\otimes_\alpha\mathcal{Y}} \le \|F\|_{\mathcal{X}^{**}\otimes_\alpha\mathcal{Y}^{**}}$ whenever $\|\cdot\|_\alpha$ is a tensor norm. On the other hand, regarding $\mathcal{X}$ and $\mathcal{Y}$ as subspaces of $\mathcal{X}^{**}$ and $\mathcal{Y}^{**}$, we get $\|F\|_{\mathcal{X}^{**}\otimes_\alpha\mathcal{Y}^{**}} \le \|F\|_{\mathcal{X}\otimes_\alpha\mathcal{Y}}$ whenever $\|\cdot\|_\alpha$ is a uniform crossnorm by Theorem 8.3(b). Thus Proposition 8.I follows from Remark 7.5(e). Note: the completion version of it, viz., $\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$ is a linear manifold of $\mathcal{X}^{**}\,\widehat{\otimes}_\alpha\mathcal{Y}^{**}$, comes from Theorem 5.1(g).

Suggested Readings

Defant and Floret [15]
Diestel, Fourie and Swart [17]
Ryan [83]
Schatten [85]



9

Tensor Product Operators

As everywhere in this book, all linear spaces are over the same field $\mathbb{F}$, which is either $\mathbb{C}$ or $\mathbb{R}$. If $\mathcal{V}, \mathcal{W}, \mathcal{X}, \mathcal{Y}$ are nonzero linear spaces and $A \in \mathcal{L}[\mathcal{X}, \mathcal{V}]$ and $B \in \mathcal{L}[\mathcal{Y}, \mathcal{W}]$ are linear transformations, then take the tensor product transformation $A \otimes B \in \mathcal{L}[\mathcal{X}, \mathcal{V}] \otimes \mathcal{L}[\mathcal{Y}, \mathcal{W}] \subseteq \mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{V} \otimes \mathcal{W}]$ defined in Chapter 3, and the collection of its properties presented in Theorem 3.19. Now replace those linear spaces with normed spaces, and consider bounded linear transformations A and B instead. The aim of this chapter is to extend Section 3.3 to tensor products of bounded linear transformations acting on tensor products of normed spaces equipped with a uniform crossnorm. This includes the rather important particular case where $\mathcal{V}, \mathcal{W}, \mathcal{X}, \mathcal{Y}$ are Hilbert spaces and their tensor product spaces are equipped with a tensor norm naturally induced by inner products, following an idea whose seed was planted in Remark 4.6.

9.1 Tensor Product of Bounded Linear Transformations

Let $\mathcal{X}, \mathcal{Y}, \mathcal{V}, \mathcal{W}$ be normed spaces. Consider the tensor products $\mathcal{X} \otimes_\alpha \mathcal{Y}$ and $\mathcal{V} \otimes_\beta \mathcal{W}$ where $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ are reasonable crossnorms on $\mathcal{X} \otimes \mathcal{Y}$ and $\mathcal{V} \otimes \mathcal{W}$, respectively. Take bounded linear transformations A in $\mathcal{B}[\mathcal{X}, \mathcal{V}]$ and B in $\mathcal{B}[\mathcal{Y}, \mathcal{W}]$, and consider the linear transformation $A \otimes B$ in $\mathcal{L}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{V} \otimes_\beta \mathcal{W}]$ as in Definition 3.16 and Theorem 3.17. If we expect $A \otimes B$ to be bounded for every A, B on every $\mathcal{X}, \mathcal{Y}, \mathcal{V}, \mathcal{W}$ and, in addition, to satisfy the norm inequality in Definition 8.2 leading to the norm identity in Theorem 8.3, then we need a redefinition: suppose $\|\cdot\|_\alpha$ is a uniform crossnorm and take the bounded linear transformation $A \otimes B$ in $\mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{V} \otimes_\alpha \mathcal{W}]$ so that $\|A \otimes B\| = \|A\| \, \|B\|$ (Theorem 8.3(a)), where $\|A \otimes B\|$ stands for the induced uniform norm of the bounded linear transformation $A \otimes B$ in $\mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{V} \otimes_\alpha \mathcal{W}]$.

Let $\|\cdot\|_{[\alpha,\alpha]}$ denote such induced uniform norm on the normed space of bounded linear transformations $\mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{V} \otimes_\alpha \mathcal{W}]$ (i.e., $\|\cdot\|_{[\alpha,\alpha]}$ stands for the induced uniform norm $\|\cdot\|$ on $\mathcal{B}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{V} \otimes \mathcal{W}]$ when $\mathcal{X} \otimes \mathcal{Y}$ and $\mathcal{V} \otimes \mathcal{W}$

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are equipped with the uniform crossnorm $\|\cdot\|_\alpha$). The notation $\|\cdot\|_{[\alpha,\alpha]}$ highlights the dependence of $\|\cdot\|$ on the uniform crossnorm $\|\cdot\|_\alpha$ equipping both $\mathcal{X}\otimes\mathcal{Y}$ and $\mathcal{V}\otimes\mathcal{W}$. If $\|\cdot\|_\alpha$ is a uniform crossnorm, then Theorem 8.3(c) says that $\mathcal{B}[\mathcal{X},\mathcal{V}]\otimes\mathcal{B}[\mathcal{Y},\mathcal{W}]\subseteq\mathcal{B}[\mathcal{X}\otimes_\alpha\mathcal{Y},\mathcal{V}\otimes_\alpha\mathcal{W}]$. Equip $\mathcal{B}[\mathcal{X},\mathcal{V}]\otimes\mathcal{B}[\mathcal{Y},\mathcal{W}]$ with the induced uniform norm on $\mathcal{B}[\mathcal{X}\otimes_\alpha\mathcal{Y},\mathcal{V}\otimes_\alpha\mathcal{W}]$ and set $\mathcal{B}[\mathcal{X},\mathcal{V}]\otimes_{[\alpha,\alpha]}\mathcal{B}[\mathcal{Y},\mathcal{W}]$ $=(\mathcal{B}[\mathcal{X},\mathcal{V}]\otimes\mathcal{B}[\mathcal{Y},\mathcal{W}],\ \|\cdot\|_{[\alpha,\alpha]})$ so that

$$\mathcal{B}[\mathcal{X},\mathcal{V}]\otimes_{[\alpha,\alpha]}\mathcal{B}[\mathcal{Y},\mathcal{W}]\subseteq\mathcal{B}[\mathcal{X}\otimes_\alpha\mathcal{Y},\mathcal{V}\otimes_\alpha\mathcal{W}].$$

Theorem 9.1 below extends Theorem 7.3, where $\mathcal{X}^*\otimes\mathcal{Y}^*$ and $(\mathcal{X}\otimes_\alpha\mathcal{Y})^*$ are replaced by $\mathcal{B}[\mathcal{X},\mathcal{V}]\otimes\mathcal{B}[\mathcal{Y},\mathcal{W}]$ and $\mathcal{B}[\mathcal{X}\otimes_\alpha\mathcal{Y},\mathcal{V}\otimes_\alpha\mathcal{W}]$. We give two proofs for Theorem 9.1. The first is a straightforward consequence of Proposition 9.A. The second shows how $\|\cdot\|_{[\alpha,\alpha]}$ lies between the injective and projective norms, leading to a natural question involving the induced uniform norms $\|\cdot\|_{[\vee,\vee]}$ and $\|\cdot\|_{[\wedge,\wedge]}$. A first entry upon that question is given in Proposition 9.B.

Theorem 9.1. *If $\|\cdot\|_\alpha$ is a uniform crossnorm, then the restriction of $\|\cdot\|_{[\alpha,\alpha]}$ to $\mathcal{B}[\mathcal{X},\mathcal{V}]\otimes\mathcal{B}[\mathcal{Y},\mathcal{W}]$ is a reasonable crossnorm on $\mathcal{B}[\mathcal{X},\mathcal{V}]\otimes\mathcal{B}[\mathcal{Y},\mathcal{W}]$.*

First Proof. Take an arbitrary quadruple $(\mathcal{X},\mathcal{Y},\mathcal{V},\mathcal{W})$ of normed spaces. Let $\|\cdot\|_\alpha$ is a uniform crossnorm, and take the induced uniform norm $\|\cdot\|_{[\alpha,\alpha]}$ on $\mathcal{B}[\mathcal{X}\otimes_\alpha\mathcal{Y},\mathcal{V}\otimes_\alpha\mathcal{W}]$. By Definition 8.2(a),

$$\|A\otimes B\|_{[\alpha,\alpha]}\le\|A\|\,\|B\| \qquad (*)$$

for every single tensor $A\otimes B\in\mathcal{B}[\mathcal{X},\mathcal{V}]\otimes\mathcal{B}[\mathcal{Y},\mathcal{W}]$. Also,

$$\begin{aligned}\mathcal{B}[\mathcal{X},\mathcal{V}]^*\otimes\mathcal{B}[\mathcal{Y},\mathcal{W}]^* &= \mathcal{B}\big[\mathcal{B}[\mathcal{X},\mathcal{V}],\mathbb{F}\big]\otimes\mathcal{B}\big[\mathcal{B}[\mathcal{Y},\mathcal{W}],\mathbb{F}\big]\big]\\ &\subseteq \mathcal{L}\big[\mathcal{B}[\mathcal{X},\mathcal{V}],\mathbb{F}\big]\otimes\mathcal{L}\big[\mathcal{B}[\mathcal{Y},\mathcal{W}],\mathbb{F}\big]\big]\\ &\subseteq \mathcal{L}\big[\mathcal{B}[\mathcal{X},\mathcal{V}]\otimes\mathcal{B}[\mathcal{Y},\mathcal{W}],\mathbb{F}\otimes\mathbb{F}\big]\end{aligned}$$

(cf. Theorem 3.17(d)). As $\mathcal{B}[\mathcal{X},\mathcal{V}]\otimes\mathcal{B}[\mathcal{Y},\mathcal{W}]\subseteq\mathcal{B}[\mathcal{X}\otimes_\alpha\mathcal{Y},\mathcal{V}\otimes_\alpha\mathcal{W}]$ by Theorem 8.3(c), equip $\mathcal{B}[\mathcal{X},\mathcal{V}]\otimes\mathcal{B}[\mathcal{Y},\mathcal{W}]$ with the norm $\|\cdot\|_{[\alpha,\alpha]}$ on $\mathcal{B}[\mathcal{X}\otimes_\alpha\mathcal{Y},\mathcal{V}\otimes_\alpha\mathcal{W}]$, so that (recall: we write $\mathbb{F}\otimes\mathbb{F}=\mathbb{F}$, as discussed in Remark 3.18(b))

$$\mathcal{B}[\mathcal{X},\mathcal{V}]^*\otimes\mathcal{B}[\mathcal{Y},\mathcal{W}]^*\subseteq\mathcal{L}\big[\mathcal{B}[\mathcal{X},\mathcal{V}]\otimes_{[\alpha,\alpha]}\mathcal{B}[\mathcal{Y},\mathcal{W}],\mathbb{F}\big].$$

Take an arbitrary single tensor $\varphi\otimes\eta\in\mathcal{B}[\mathcal{X},\mathcal{V}]^*\otimes\mathcal{B}[\mathcal{Y},\mathcal{W}]^*$. By the above inclusion, $\varphi\otimes\eta\in\mathcal{L}\big[\mathcal{B}[\mathcal{X},\mathcal{V}]\otimes_{[\alpha,\alpha]}\mathcal{B}[\mathcal{Y},\mathcal{W}],\mathbb{F}\big]$. Let $\|\cdot\|_{*[\alpha,\alpha]}$ be the induced uniform norm on $\mathcal{B}\big[\mathcal{B}[\mathcal{X},\mathcal{V}]\otimes_{[\alpha,\alpha]}\mathcal{B}[\mathcal{Y},\mathcal{W}],\mathbb{F}\big]$, and compute

$$\|\varphi\otimes\eta\|_{*[\alpha,\alpha]}=\sup_{\mathbb{L}\in\mathcal{B}[\mathcal{X},\mathcal{V}]\otimes\mathcal{B}[\mathcal{Y},\mathcal{W}],\ \|\mathbb{L}\|_{[\alpha,\alpha]}\le1}|(\varphi\otimes\eta)\mathbb{L}|.$$

Proposition 9.A ensures that the above supremum is finite. Now take an arbitrary element $\Bbbk$ in $\mathcal{B}[\mathcal{X},\mathcal{V}]^*\otimes\mathcal{B}[\mathcal{Y},\mathcal{W}]^*\subseteq\mathcal{L}\big[\mathcal{B}[\mathcal{X},\mathcal{V}]\otimes_{[\alpha,\alpha]}\mathcal{B}[\mathcal{Y},\mathcal{W}],\mathbb{F}\big]$, which is represented by a finite sum of single tensors, $\Bbbk=\sum_k\varphi_k\otimes\eta_k$, with φ_k in $\mathcal{B}[\mathcal{X},\mathcal{V}]^*$ and η_k in $\mathcal{B}[\mathcal{Y},\mathcal{W}]^*$. Therefore

$$\|\Bbbk\|_{*[\alpha,\alpha]} = \sup_{\mathbb{L}\in\mathcal{B}[\mathcal{X},\mathcal{V}]\otimes\mathcal{B}[\mathcal{Y},\mathcal{W}],\ \|\mathbb{L}\|_{[\alpha,\alpha]}\le 1} \left|\left(\sum\nolimits_k \varphi_k\otimes\eta_k\right)\mathbb{L}\right|$$
$$\le \sup_{\|\mathbb{L}\|_{[\alpha,\alpha]}\le 1} \sum\nolimits_k \|\varphi_k\otimes\eta_k\|_{*[\alpha,\alpha]}\|\mathbb{L}\|_{[\alpha,\alpha]} < \infty,$$

because the sum is finite. Hence $\Bbbk \in \mathcal{B}\big[\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes_{[\alpha,\alpha]} \mathcal{B}[\mathcal{Y},\mathcal{W}],\mathbb{F}\big]$. Then

$$\mathcal{B}[\mathcal{X},\mathcal{V}]^* \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}]^* = (\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes_{[\alpha,\alpha]} \mathcal{B}[\mathcal{Y},\mathcal{W}])^*. \qquad (**)$$

Finally, applying Proposition 9.A again,

$$\|\varphi\otimes\eta\|_{*[\alpha,\alpha]} \le \|\varphi\|\,\|\eta\|, \qquad (***)$$

for every $\varphi\otimes\eta \in \mathcal{B}[\mathcal{X},\mathcal{V}]^* \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}]^*$. Thus, by $(*)$, $(**)$, and $(**)$, $\|\cdot\|_{[\alpha,\alpha]}$ is a reasonable crossnorm on $\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}]$ (cf. Definition 7.1). □

Note. If $\|\cdot\|_{[\alpha,\alpha]}$ is a reasonable crossnorm on $\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}]$, then

$$\|\mathbb{L}\|_\vee \le \|\mathbb{L}\|_{[\alpha,\alpha]} \le \|\mathbb{L}\|_\wedge,$$

for every $\mathbb{L}$ in $\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}]$, where $\|\mathbb{L}\|_\vee$ and $\|\mathbb{L}\|_\wedge$ stand for the injective and projective norms on $\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}]$, respectively, according to Theorem 7.4. The second proof below goes the other way around: it proves the above inequalities in order to conclude that $\|\cdot\|_{[\alpha,\alpha]}$ is a reasonable crossnorm.

Second Proof. If $\|\cdot\|_\alpha$ is a uniform crossnorm, then

$$\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}] \subseteq \mathcal{B}[\mathcal{X}\otimes_\alpha\mathcal{Y},\mathcal{V}\otimes_\alpha\mathcal{W}].$$

by Theorem 8.3(c), as we have seen above. Take an arbitrary element $\mathbb{L}$ in $\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}]$ and let $\sum_j A_j\otimes B_j$ be a finite-sum representation of it in terms of single tensors in $\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}]$. By the above inclusion, regard $\mathbb{L}$ as a transformation in $\mathcal{B}[\mathcal{X}\otimes_\alpha\mathcal{Y},\mathcal{V}\otimes_\alpha\mathcal{W}]$. Set $\|\mathbb{L}(F)\|_{\mathcal{V}\otimes_\alpha\mathcal{W}} = \|\mathbb{L}(F)\|_\alpha$ and $\|F\|_{\mathcal{X}\otimes_\alpha\mathcal{Y}} = \|F\|_\alpha$. If $\|\cdot\|_{[\alpha,\alpha]}$ is the norm on $\mathcal{B}[\mathcal{X}\otimes_\alpha\mathcal{Y},\mathcal{V}\otimes_\alpha\mathcal{W}]$, then (cf. Remark 8.1(a) and Theorem 8.3(a))

$$\|\mathbb{L}(F)\|_\alpha = \left\|\left(\sum\nolimits_j A_j\otimes B_j\right)F\right\|_\alpha \le \sum\nolimits_j \|(A_j\otimes B_j)F\|_\alpha$$
$$\le \sum\nolimits_j \|A_j\otimes B_j\|_{[\alpha,\alpha]}\|F\|_\alpha = \left(\sum\nolimits_j \|A_j\|\,\|B_j\|\right)\|F\|_\alpha$$

for every $F\in\mathcal{X}\otimes\mathcal{Y}$. Since $\|\mathbb{L}\|_{[\alpha,\alpha]} = \sup_{\|F\|_\alpha\le 1}\|\mathbb{L}(F)\|_\alpha$, we get

$$\|\mathbb{L}\|_{[\alpha,\alpha]} \le \sum\nolimits_j \|A_j\|\,\|B_j\|.$$

(Actually, since $\|\cdot\|_{[\alpha,\alpha]}$ is a norm on the linear space $\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}]$ — the restriction of the induced uniform norm on $\mathcal{B}[\mathcal{X}\otimes_\alpha\mathcal{Y},\mathcal{V}\otimes_\alpha\mathcal{W}]$ — the above inequality follows at once: $\|\mathbb{L}\|_{[\alpha,\alpha]} = \|\sum_j A_j\otimes B_j\|_{[\alpha,\alpha]} \le \sum_j \|A_j\otimes B_j\|_{[\alpha,\alpha]}$.) Since the above inequality holds for every representation $\sum_j A_j\otimes B_j$ of $\mathbb{L}$,

$$\|\mathbb{L}\|_{[\alpha,\alpha]} \le \inf_{\mathbb{L}=\sum_j A_j\otimes B_j} \sum\nolimits_j \|A_j\|\,\|B_j\| = \|\mathbb{L}\|_\wedge,$$

where $\|\cdot\|_\wedge$ is the projective norm on $\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}]$. On the other hand, $\mathcal{B}[\mathcal{X},\mathcal{V}]^* \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}]^* \subseteq (\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes_{[\alpha,\alpha]} \mathcal{B}[\mathcal{Y},\mathcal{W}])^*$ by Proposition 9.A, and so

$$\mathcal{B}[\mathcal{X},\mathcal{V}]^* \otimes_{*[\alpha,\alpha]} \mathcal{B}[\mathcal{Y},\mathcal{W}]^* \subseteq (\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes_{[\alpha,\alpha]} \mathcal{B}[\mathcal{Y},\mathcal{W}])^*$$

as in (∗∗) for the induced uniform norm $\|\cdot\|_{*[\alpha,\alpha]}$ on $(\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes_{[\alpha,\alpha]} \mathcal{B}[\mathcal{Y},\mathcal{W}])^*$. Take $\mathbb{L} = \sum_j A_j \otimes B_j$ in $\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes_{[\alpha,\alpha]} \mathcal{B}[\mathcal{Y},\mathcal{W}] \subseteq \mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{V} \otimes_\alpha \mathcal{W}]$ and take $\Bbbk = \sum_k \varphi_k \otimes \eta_k$ in $\mathcal{B}[\mathcal{X},\mathcal{V}]^* \otimes_{*[\alpha,\alpha]} \mathcal{B}[\mathcal{Y},\mathcal{W}]^* \subseteq (\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes_{[\alpha,\alpha]} \mathcal{B}[\mathcal{Y},\mathcal{W}])^*$ arbitrarily. Regard $\Bbbk$ as a functional in $(\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes_{[\alpha,\alpha]} \mathcal{B}[\mathcal{Y},\mathcal{W}])^*$ so that $\Bbbk(\mathbb{L}) = \big(\sum_k \varphi_k \otimes \eta_k\big)\sum_j A_j \otimes B_j = \sum_{k,j} (\varphi_k \otimes \eta_k)(A_j \otimes B_j) = \sum_{k,j} \varphi_k(A_j)\, \eta_k(B_j)$. Since $\|\cdot\|_{*[\alpha,\alpha]}$ is the induced uniform norm in $\mathcal{B}\big[\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes_{[\alpha,\alpha]} \mathcal{B}[\mathcal{Y},\mathcal{W}], \mathbb{F}\big]$, we get $\|\Bbbk\|_{*[\alpha,\alpha]} = \sup_{\|\mathbb{L}\|_{[\alpha,\alpha]} \le 1} |\Bbbk(\mathbb{L})|$. Dually (cf. Remark 5.2(d) again),

$$\begin{aligned} \|\mathbb{L}\|_{[\alpha,\alpha]} &= \sup_{\|\Bbbk\|_{*[\alpha,\alpha]} \le 1} |\Bbbk(\mathbb{L})| = \sup_{\|\Sigma_k \varphi_k \otimes \eta_k\|_{*[\alpha,\alpha]} \le 1} \left| \sum\nolimits_{k,j} \varphi_k(A_j)\, \eta_k(B_j) \right| \\ &\ge \sup_{\|\varphi \otimes \eta\|_{*[\alpha,\alpha]} \le 1} \left| \sum\nolimits_j \varphi(A_j)\, \eta(B_j) \right| \ge \sup_{\|\varphi\| \|\eta\| \le 1} \left| \sum\nolimits_j \varphi(A_j)\, \eta(B_j) \right|, \end{aligned}$$

as $\|\varphi \otimes \eta\|_{*[\alpha,\alpha]} \le \|\varphi\|\,\|\eta\|$ by Proposition 9.A (so that $\|\varphi\|\,\|\eta\| \le 1$ implies $\|\varphi \otimes \eta\|_{*[\alpha,\alpha]} \le 1$ and hence $\{\varphi,\eta\colon \|\varphi\|\|\eta\| \le 1\} \subseteq \{\varphi,\eta\colon \|\varphi \otimes \eta\|_{*[\alpha,\alpha]} \le 1\}$). Thus the above inequality ensures that

$$\|\mathbb{L}\|_{[\alpha,\alpha]} \ge \sup_{\|\varphi\|\|\eta\| \le 1} \left| \sum\nolimits_j \varphi(A_j)\, \eta(B_j) \right| \ge \sup_{\|\varphi\|=1, \|\eta\|=1} \left| \sum\nolimits_j \varphi(A_j)\, \eta(B_j) \right| = \|\mathbb{L}\|_\vee,$$

where $\|\cdot\|_\vee$ stands for the injective norm on $\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}]$. Consequently,

$$\|\mathbb{L}\|_\vee \le \|\mathbb{L}\|_{[\alpha,\alpha]} \le \|\mathbb{L}\|_\wedge$$

for every $\mathbb{L} \in \mathcal{B}[\mathcal{X},\mathcal{V}] \otimes_{[\alpha,\alpha]} \mathcal{B}[\mathcal{Y},\mathcal{W}]$, and so $\|\cdot\|_{[\alpha,\alpha]}$ is a reasonable crossnorm on $\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}]$ according to Theorem 7.4. □

The injective and projective norms $\|\cdot\|_\vee$ and $\|\cdot\|_\wedge$ are the least and the greatest uniform crossnorms (Theorem 8.4) acting on every tensor product space; in particular, on $\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}]$ (as in the above inequalities). Also, $\|\cdot\|_{[\vee,\vee]}$ and $\|\cdot\|_{[\wedge,\wedge]}$ are reasonable crossnorms on $\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}]$ by the above theorem since $\|\cdot\|_\vee$ and $\|\cdot\|_\wedge$ are uniform crossnorms. Therefore

$$\|\mathbb{L}\|_\vee \le \|\mathbb{L}\|_{[\vee,\vee]} \quad \text{and} \quad \|\mathbb{L}\|_{[\wedge,\wedge]} \le \|\mathbb{L}\|_\wedge.$$

Question: does $\|\mathbb{L}\|_{[\alpha,\alpha]}$ lie in between, so that $\|\mathbb{L}\|_{[\vee,\vee]}$ is the least and $\|\mathbb{L}\|_{[\wedge,\wedge]}$ is the greatest reasonable crossnorm on $\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}]$ that are inherited from $\mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{V} \otimes_\alpha \mathcal{W}]$? (Note that there are reasonable crossnorms on $\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}]$ that are not restrictions of the induced uniform norm on $\mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{V} \otimes_\alpha \mathcal{W}]$, as is the case of $\|\mathbb{L}\|_\vee$ and $\|\mathbb{L}\|_\wedge$.) Proposition 9.B gives a first (and natural) entry upon such a question. (We borrowed the second proof of Theorem 9.1 from [60].)

Theorem 3.19 listed standard algebraic properties of tensor products of linear transformations acting between tensor product spaces of linear spaces. As

$\mathcal{B}[\mathcal{X},\mathcal{Y}]$ is a linear manifold of $\mathcal{L}[\mathcal{X},\mathcal{Y}]$, all hold for tensor products of bounded linear transformations acting between crossnormed tensor product spaces of normed spaces (when the crossnorm is uniform). Among the properties in Theorem 3.19, just the last two need a reformulation for the normed-space case.

Theorem 9.2. *Let $\mathcal{V},\mathcal{W},\mathcal{X},\mathcal{Y}$ be nonzero normed spaces. Take $A \in \mathcal{B}[\mathcal{X},\mathcal{V}]$ and $B \in \mathcal{B}[\mathcal{Y},\mathcal{W}]$. If $\|\cdot\|_\alpha$ is a uniform crossnorm, then*

(a) $A \otimes B \cong B \otimes A$ *with* $\|A \otimes B\| = \|B \otimes A\| = \|A\|\,\|B\|$,

(b) $(A \otimes B)^* = A^* \otimes B^*$ *with* $\|(A \otimes B)^*\| = \|A \otimes B\|$
(*in* $\mathcal{B}[(\mathcal{V} \otimes_\alpha \mathcal{W})^*, (\mathcal{X} \otimes_\alpha \mathcal{Y})^*]$, *where* * *stands for normed-space adjoint*).

Proof. Let $\mathcal{V}$,$\mathcal{W}$,$\mathcal{X}$,$\mathcal{Y}$ be nonzero normed spaces and $\|\cdot\|_\alpha$ a uniform crossnorm.

(a) The *tour de force* proof. Let $\|\cdot\|_{t_\alpha}$ be the transpose norm, again a uniform crossnorm by Theorem 8.11 (part 3). Take $A \in \mathcal{B}[\mathcal{X},\mathcal{V}]$ and $B \in \mathcal{B}[\mathcal{Y},\mathcal{W}]$ and consider their tensor products $A \otimes B$ in $\mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{V} \otimes_\alpha \mathcal{W}]$ and $B \otimes A$ in $\mathcal{B}[\mathcal{Y} \otimes_{t_\alpha} \mathcal{X}, \mathcal{W} \otimes_{t_\alpha} \mathcal{V}]$. Take the natural permutations $\Pi_1 : \mathcal{X} \otimes_\alpha \mathcal{Y} \to \mathcal{Y} \otimes_{t_\alpha} \mathcal{X}$ and $\Pi_2 : \mathcal{V} \otimes_\alpha \mathcal{W} \to \mathcal{W} \otimes_{t_\alpha} \mathcal{V}$ such that $\Pi_2(A \otimes B) = (B \otimes A)\Pi_1$, which are linear-space isomorphisms as in the proof of Theorem 3.19(i). So the diagram

$$\begin{array}{ccc} \mathcal{X} \otimes_\alpha \mathcal{Y} & \xrightarrow{A \otimes B} & \mathcal{V} \otimes_\alpha \mathcal{W} \\ {\scriptstyle \Pi_1}\downarrow & & \downarrow{\scriptstyle \Pi_2} \\ \mathcal{Y} \otimes_{t_\alpha} \mathcal{X} & \xrightarrow{B \otimes A} & \mathcal{W} \otimes_{t_\alpha} \mathcal{V} \end{array}$$

commutes. Recalling that $\|\cdot\|_{t t_\alpha} = \|\cdot\|_\alpha$, we get from Theorem 7.6(b) that the natural permutations $\Pi_1 : \mathcal{X} \otimes_\alpha \mathcal{Y} \to \mathcal{Y} \otimes_{t_\alpha} \mathcal{X}$ and $\Pi_2 : \mathcal{V} \otimes_\alpha \mathcal{W} \to \mathcal{W} \otimes_{t_\alpha} \mathcal{V}$ are isometric isomorphisms. Hence $A \otimes B \cong B \otimes A$ (where $\cong$ stands for isometric equivalence; i.e., $A \otimes B$ and $B \otimes A$ are isometrically equivalent transformations as in Section 5.1). The second norm identity comes from Theorem 8.3(a).

(a′) A clever proof. Theorem 8.3(a): $\|A \otimes B\| = \|B \otimes A\|$. Use Theorem 3.19(i).

(b) First recall that, for every T in $\mathcal{B}[\mathcal{X},\mathcal{Y}]$, its normed-space adjoint T^* in $\mathcal{B}[\mathcal{Y}^*,\mathcal{X}^*]$ is the restriction of its algebraic adjoint $T^\sharp$ in $\mathcal{L}[\mathcal{Y}^\sharp,\mathcal{X}^\sharp]$ to $\mathcal{Y}^* \subseteq \mathcal{Y}^\sharp$ (see, e.g., [70, Proposition 3.1.2]). Then replace $^\sharp$ with * and $\mathcal{L}$ with $\mathcal{B}$ in the proof of Theorem 3.19(j). This shows that $A^* \otimes B^* = (A \otimes B)^*$ as transformations in $\mathcal{B}[(\mathcal{V} \otimes_\alpha \mathcal{W})^*, (\mathcal{X} \otimes_\alpha \mathcal{Y})^*]$ since (cf. Theorem 8.3(c) and Definition 7.1)

$$\mathcal{B}[\mathcal{V}^*,\mathcal{X}^*] \otimes \mathcal{B}[\mathcal{W}^*,\mathcal{Y}^*] \subseteq \mathcal{B}[\mathcal{V}^* \otimes_{*\alpha} \mathcal{W}^*,\ \mathcal{X}^* \otimes_{*\alpha} \mathcal{Y}^*] \subseteq \mathcal{B}[(\mathcal{V} \otimes_\alpha \mathcal{W})^*,\ (\mathcal{X} \otimes_\alpha \mathcal{Y})^*].$$

For the norm identity, recall that $\|T^*\| = \|T\|$ for every bounded linear T. □

Let $\mathcal{V},\mathcal{W},\mathcal{X},\mathcal{Y}$ be normed spaces, and take $A \in \mathcal{B}[\mathcal{X},\mathcal{V}]$ and $B \in \mathcal{B}[\mathcal{Y},\mathcal{W}]$. Let $\|\cdot\|_\alpha$ be a uniform crossnorm, and take the bounded linear transformation $A \otimes B$ in $\mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{V} \otimes_\alpha \mathcal{W}]$ so that $\|A \otimes B\| = \|A\|\,\|B\|$, where $\|A \otimes B\| = \|A \otimes B\|_{[\alpha,\alpha]}$ is the induced uniform norm of $A \otimes \mathcal{B}$ in $\mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{V} \otimes_\alpha \mathcal{W}]$. Let $\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}$ and $\mathcal{V} \mathbin{\widehat{\otimes}_\alpha} \mathcal{W}$ be the completions of $\mathcal{X} \otimes_\alpha \mathcal{Y}$ and $\mathcal{V} \otimes_\alpha \mathcal{W}$, and let

$A \mathbin{\widehat{\otimes}} B$ in $\mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}, \mathcal{V} \mathbin{\widehat{\otimes}_\alpha} \mathcal{W}]$ be the extension over completion of $A \otimes B$ so that $\|A \mathbin{\widehat{\otimes}} B\| = \|A \otimes B\|$ according to Theorem 5.1.

Theorem 9.3. *Let $\mathcal{V}, \mathcal{W}, \mathcal{X}, \mathcal{Y}, \mathcal{X}', \mathcal{Y}'$ be nonzero normed spaces, equip their tensor product spaces with a uniform crossnorm $\|\cdot\|_\alpha$, and let λ and μ be arbitrary scalars in their common field. Take arbitrary bounded linear transformations $A, A_1, A_2 \in \mathcal{B}[\mathcal{X}, \mathcal{V}]$, $B, B_1, B_2 \in \mathcal{B}[\mathcal{Y}, \mathcal{W}]$, $C \in \mathcal{B}[\mathcal{X}', \mathcal{X}]$ and $D \in \mathcal{B}[\mathcal{Y}', \mathcal{Y}]$. Then the following identities hold true.*

(a) $\lambda\mu\,(A \mathbin{\widehat{\otimes}} B) = \lambda A \mathbin{\widehat{\otimes}} \mu B = (\lambda\mu\, A) \mathbin{\widehat{\otimes}} B = A \mathbin{\widehat{\otimes}} (\lambda\mu\, B)$.

(b) $(A_1 + A_2) \mathbin{\widehat{\otimes}} (B_1 + B_2) = A_1 \mathbin{\widehat{\otimes}} B_1 + A_2 \mathbin{\widehat{\otimes}} B_1 + A_1 \mathbin{\widehat{\otimes}} B_2 + A_2 \mathbin{\widehat{\otimes}} B_2$.

(c) $AC \mathbin{\widehat{\otimes}} BD = (A \mathbin{\widehat{\otimes}} B)\,(C \mathbin{\widehat{\otimes}} D)$.

(d) *If A and B are invertible, then so is $A \mathbin{\widehat{\otimes}} B$ and* $(A \mathbin{\widehat{\otimes}} B)^{-1} = A^{-1} \mathbin{\widehat{\otimes}} B^{-1}$.

(e) $\mathcal{R}(A) \mathbin{\widehat{\otimes}_\alpha} \mathcal{R}(B) = \mathcal{R}(A \mathbin{\widehat{\otimes}} B)^-$.

(f) *$A \mathbin{\widehat{\otimes}} B$ has dense range if and only if A and B are surjective:* $\mathcal{R}(A \mathbin{\widehat{\otimes}} B)^- = \mathcal{V} \mathbin{\widehat{\otimes}_\alpha} \mathcal{W} \iff \mathcal{R}(A) = \mathcal{V}$ *and* $\mathcal{R}(B) = \mathcal{W}$.

(g) $\mathcal{N}(A) \mathbin{\widehat{\otimes}_\alpha} \mathcal{N}(B) \subseteq \mathcal{N}(A \mathbin{\widehat{\otimes}} B) \subseteq \big(\mathcal{N}(A) \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}\big) + \big(\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{N}(B)\big)$. ($\subseteq$ *stands for set inclusion*).

(h) *$A \mathbin{\widehat{\otimes}} B$ is injective if and only if A and B are injective:* $\mathcal{N}(A \mathbin{\widehat{\otimes}} B) = \{0\} \iff \mathcal{N}(A) = \{0\}$ *and* $\mathcal{N}(B) = \{0\}$.

(i) $A \mathbin{\widehat{\otimes}} B \cong B \mathbin{\widehat{\otimes}} A$ *with* $\|A \mathbin{\widehat{\otimes}} B\| = \|B \mathbin{\widehat{\otimes}} A\| = \|A\|\,\|B\|$.

(j) $(A \mathbin{\widehat{\otimes}} B)^* = A^* \mathbin{\widehat{\otimes}} B^*$ *with* $\|(A \mathbin{\widehat{\otimes}} B)^*\| = \|A \mathbin{\widehat{\otimes}} B\|$ (*in* $\mathcal{B}[(\mathcal{V} \mathbin{\widehat{\otimes}_\alpha} \mathcal{W})^*, (\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y})^*]$, *where * stands for normed-space adjoint*).

Proof. Consider the setup of Section 7.1 (before Remark 7.5) with $\mathcal{Z}$ replaced by $\mathcal{V} \otimes_\alpha \mathcal{W}$, or $\mathcal{V} \mathbin{\widetilde{\otimes}_\alpha} \mathcal{W}$, or $\mathcal{V} \mathbin{\widehat{\otimes}_\alpha} \mathcal{W}$, T replaced by $A \otimes B$, and I replaced by the appropriate isometric isomorphism K, which is represented by the diagram

$$\begin{array}{ccccc} \mathcal{V} \otimes_\alpha \mathcal{W} & \xrightarrow{\ K\ } & \mathcal{V} \mathbin{\widetilde{\otimes}_\alpha} \mathcal{W} = K(\mathcal{V} \otimes_\alpha \mathcal{W}) & \subseteq & \mathcal{V} \mathbin{\widehat{\otimes}_\alpha} \mathcal{W} \\ {\scriptstyle A\otimes B}\uparrow & & \uparrow{\scriptstyle (A \mathbin{\widehat{\otimes}} B)\big|_{\mathcal{X} \mathbin{\widetilde{\otimes}_\alpha} \mathcal{Y}} = K(A\otimes B)J^{-1}} & & \uparrow{\scriptstyle A \mathbin{\widehat{\otimes}} B} \\ \mathcal{X} \otimes_\alpha \mathcal{Y} & \xrightarrow{\ J\ } & \mathcal{X} \mathbin{\widetilde{\otimes}_\alpha} \mathcal{Y} = J(\mathcal{X} \otimes_\alpha \mathcal{Y}) & \subseteq & \mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y} \end{array}$$

(cf. diagram in the proof of Theorem 5.1). The proof comes from Theorems 3.19 and 5.1 for items (a) to (h), and also with the help of Theorem 9.2 for items (i) and (j). For items (a) to (d) the argument is essentially this: (1) use Theorem 3.19 to ensure the unextended identities since $A \otimes B$ is a single tensor in $\mathcal{B}[\mathcal{X}, \mathcal{V}] \otimes \mathcal{B}[\mathcal{Y}, \mathcal{W}]$, then (2) recall that these identities are preserved under isometric equivalence, and so (3) apply extension by continuity (for the bounded linear transformations, thus uniformly continuous functions) on a dense linear manifold. Computations are verified with the help of the above diagram, of Theorems 3.19 and 5.1, and are synthesized as follows.

(a) $\big(\lambda\,\mu\,(A\,\widehat{\otimes}\,B)\big)|_{\mathcal{X}\,\widetilde{\otimes}_\alpha\mathcal{Y}} = K\big(\lambda\,\mu\,(A\otimes B)\big)J^{-1} = K(\lambda\,A\otimes\,\mu\,B)J^{-1}$
$= (\lambda\,A\,\widehat{\otimes}\,\mu\,B)|_{\mathcal{X}\,\widetilde{\otimes}_\alpha\mathcal{Y}}.$

(b) $\big(A\,\widehat{\otimes}\,(B_1+B_2)\big)|_{\mathcal{X}\,\widetilde{\otimes}_\alpha\mathcal{Y}} = K(A\otimes(B_1+B_2))J^{-1} = K(A\otimes B_1+A\otimes B_2)J^{-1}$
$= K(A\otimes B_1)J^{-1}+K(A\otimes B_2)J^{-1} = A\,\widehat{\otimes}\,B_1|_{\mathcal{X}\,\widetilde{\otimes}_\alpha\mathcal{Y}} + A\,\widehat{\otimes}\,B_2|_{\mathcal{X}\,\widetilde{\otimes}_\alpha\mathcal{Y}}.$

(c) $\big(AC\,\widehat{\otimes}\,BD\big)|_{\mathcal{X}\,\widetilde{\otimes}_\alpha\mathcal{Y}} = K(AC\otimes BD)J^{-1} = K(A\otimes B)(C\otimes D)J^{-1}$
$= \widehat{(A\,\widehat{\otimes}\,B)(C\,\widehat{\otimes}\,D)}|_{\mathcal{X}\,\widetilde{\otimes}_\alpha\mathcal{Y}} = (A\,\widehat{\otimes}\,B)(C\,\widehat{\otimes}\,D)|_{\mathcal{X}\,\widetilde{\otimes}_\alpha\mathcal{Y}}.$

(d) $(A\,\widehat{\otimes}\,B)(A^{-1}\,\widehat{\otimes}\,B^{-1}) = A\,A^{-1}\,\widehat{\otimes}\,BB^{-1} = I_{\mathcal{V}}\,\widehat{\otimes}\,I_{\mathcal{W}} = I_{\mathcal{V}\,\widehat{\otimes}\,\mathcal{W}}$,
$(A^{-1}\,\widehat{\otimes}\,B^{-1})(A\,\widehat{\otimes}\,B) = A^{-1}A\,\widehat{\otimes}\,B^{-1}B = I_{\mathcal{X}}\,\widehat{\otimes}\,I_{\mathcal{Y}} = I_{\mathcal{X}\,\widehat{\otimes}\,\mathcal{Y}}.$

(e) As $\mathcal{X}\otimes_\alpha\mathcal{Y}\cong\mathcal{X}\,\widetilde{\otimes}_\alpha\mathcal{Y}$ and $(\mathcal{X}\,\widetilde{\otimes}_\alpha\mathcal{Y})^- = \mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$ for arbitrary normed spaces,

$$\mathcal{R}(A)\,\widehat{\otimes}_\alpha\mathcal{R}(B) = \big(\mathcal{R}(A)\,\widetilde{\otimes}_\alpha\mathcal{R}(B)\big)^- \cong \big(\mathcal{R}(A)\otimes_\alpha\mathcal{R}(B)\big)^-.$$

Moreover, according to Theorems 3.19(e) and 5.1(h),

$$\big(\mathcal{R}(A)\otimes_\alpha\mathcal{R}(B)\big)^- = \mathcal{R}(A\otimes B)^- \cong \mathcal{R}(A\,\widehat{\otimes}\,B)^-.$$

(f) $\mathcal{R}(A)\,\widehat{\otimes}_\alpha\mathcal{R}(B) = \mathcal{V}\,\widehat{\otimes}_\alpha\mathcal{W} \iff \mathcal{R}(A)\cong\mathcal{V}$ and $\mathcal{R}(B)\cong\mathcal{W}$ by Theorem 5.1(c). Now apply (e).

(g) $\mathcal{N}(A)\,\widehat{\otimes}_\alpha\mathcal{N}(B)\subseteq\mathcal{N}(A\,\widehat{\otimes}\,B)\subseteq\big(\mathcal{N}(A)\,\widehat{\otimes}_\alpha\mathcal{Y}\big)+\big(\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{N}(B)\big)$ by Theorem 3.19(h).

(h) $\mathcal{N}(A)=\{0\}$ and $\mathcal{N}(B)=\{0\}$ implies $\mathcal{N}(A\,\widehat{\otimes}\,B)=\{0\}$ by (g), which implies $\mathcal{N}(A\otimes B)=\{0\}$ by Theorem 5.1(i), which implies $\mathcal{N}(A)=\{0\}$ and $\mathcal{N}(B)=\{0\}$ by Theorem 3.19(g).

(i) $A\,\widehat{\otimes}\,B\cong B\,\widehat{\otimes}\,A$ and $\|A\,\widehat{\otimes}\,B\| = \|A\otimes B\|$ by Theorems 5.1(b) and 9.2(a).

(j) $A^*\,\widehat{\otimes}\,B^* = \big[\widehat{A^*\otimes B^*}\big] = \big[\widehat{(A\otimes B)^*}\big] = (A\otimes B)^*\cong\big[\widehat{A\otimes B}\big]^* = (A\,\widehat{\otimes}\,B)^*$ by Theorems 5.1(b,f) and 9.2(b) (with $=$ for $\cong$ in the statement of (j)). □

Note: $\mathcal{R}(A\,\widehat{\otimes}\,B)^- = \mathcal{V}\,\widehat{\otimes}_\alpha\mathcal{W} \iff \mathcal{R}(A\otimes B)=\mathcal{V}\otimes\mathcal{W}$ and $\mathcal{N}(A\,\widehat{\otimes}\,B)=\{0\}$ $\iff \mathcal{N}(A\otimes B)=\{0\}$ by Theorems 3.19(f,g) and 9.3(f,h). Also, in Theorems 9.2 and 9.3, equality and inclusion of tensor product spaces are interpreted as isometric isomorphisms or embeddings, and equality between tensor products of transformations as isometric equivalence (unless otherwise stated).

9.2 A Hilbert-Space Setting

Let $\mathcal{X}$ and $\mathcal{Y}$ be normed spaces. Consider the collection

$$\big\{\psi\in\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathbb{F}]\colon \psi(x,y)=f(x)\,g(y)\text{ for }(f,g)\in\mathcal{X}^*\times\mathcal{Y}^*\big\}$$

of all bounded bilinear forms ψ on $\mathcal{X}\times\mathcal{Y}$ that are products of bounded linear forms f and g on $\mathcal{X}$ and $\mathcal{Y}$. For each pair $(x,y)\in\mathcal{X}\times\mathcal{Y}$, define a bilinear form

$$\theta'(x,y)\colon \mathcal{X}^*\times\mathcal{Y}^* \to \mathbb{F}$$

in $\boldsymbol{b}[\mathcal{X}^*\times\mathcal{Y}^*, \mathbb{F}]$ given for each $(x,y)\in\mathcal{X}\times\mathcal{Y}$ by

$$\theta'(x,y)(f,g) = f(x)\,g(y) \;\in \mathbb{F} \quad \text{for every} \quad (f,g)\in\mathcal{X}^*\times\mathcal{Y}^*.$$

Take the set of all those bounded bilinear forms $\theta'(x,y)$, namely

$$\top'_{\mathcal{X},\mathcal{Y}} = \big\{\theta'(x,y)\in \boldsymbol{b}[\mathcal{X}^*\times\mathcal{Y}^*, \mathbb{F}]\colon\ x\in\mathcal{X} \text{ and } y\in\mathcal{Y}\big\},$$

and consider the linear space consisting of the linear span of it, say,

$$\mathcal{X}\otimes\mathcal{Y} = \operatorname{span}\top'_{\mathcal{X},\mathcal{Y}} \subseteq \boldsymbol{b}[\mathcal{X}^*\times\mathcal{Y}^*, \mathbb{F}] = \boldsymbol{b}[\mathcal{B}[\mathcal{X},\mathbb{F}]\times\mathcal{B}[\mathcal{Y},\mathbb{F}], \mathbb{F}].$$

According to Corollary 4.5, the linear space $\mathcal{X}\otimes\mathcal{Y} = \operatorname{span}\top'_{\mathcal{X},\mathcal{Y}}$ is a tensor product space, where the map $\theta'\colon \mathcal{X}\times\mathcal{Y}\to\top'_{\mathcal{X},\mathcal{Y}}$ is bilinear — the natural bilinear map associated to $\operatorname{span}\top'_{\mathcal{X},\mathcal{Y}}$ — so that the pair $(\operatorname{span}\top'_{\mathcal{X},\mathcal{Y}}, \theta')$ satisfies the axioms of Definition 3.1, leading to the commutative diagram

$$\begin{array}{ccc} \mathcal{X}\times\mathcal{Y} & \xrightarrow{\ \phi\ } & \mathcal{Z} \\ & {\scriptstyle\theta'}\searrow \quad \uparrow{\scriptstyle\Phi} & \\ & \mathcal{X}\otimes\mathcal{Y} = \operatorname{span}\top'_{\mathcal{X},\mathcal{Y}}\,. & \end{array}$$

Let $x\otimes y$ stand for the value of $\theta'\colon \mathcal{X}\times\mathcal{Y}\to\top'_{\mathcal{X},\mathcal{Y}}$ at $(x,y)\in\mathcal{X}\times\mathcal{Y}$,

$$x\otimes y = \theta'(x,y)\in \top'_{\mathcal{X},\mathcal{Y}} \subseteq \operatorname{span}\top'_{\mathcal{X},\mathcal{Y}} \subseteq \boldsymbol{b}[\mathcal{X}^*\times\mathcal{Y}^*, \mathbb{F}].$$

This characterizes the single tensors $x\otimes y$ in $\mathcal{X}\otimes\mathcal{Y}$ as bounded bilinear forms $x\otimes y = \theta'(x,y)\colon \mathcal{X}^*\otimes\mathcal{Y}^*\to\mathbb{F}$ defined for every $(f,g)\in\mathcal{X}^*\times\mathcal{Y}^*$ by

$$(x\otimes y)(f,g) = \theta'(x,y)(f,g) = f(x)\,g(y).$$

Now suppose $\mathcal{X}$ and $\mathcal{Y}$ are nonzero Hilbert spaces with $\langle\cdot\,;\cdot\rangle_{\mathcal{X}}$ and $\langle\cdot\,;\cdot\rangle_{\mathcal{Y}}$ denoting their respective inner products (defined to be linear in the first argument). Sometimes we will drop the subscript and use the same notation $\langle\cdot\,;\cdot\rangle$ for both inner products. As we did in the second half of Section 7.4, we will often apply the Fourier Series and the Riesz Representation Theorems (Propositions 5.T and 5.U in Section 5.4), which are pertinent to Hilbert spaces. Recall from Remark 4.6 that the Riesz Representation Theorem in Hilbert spaces allows us to identify $f\in\mathcal{X}^*$ and $g\in\mathcal{Y}^*$ with $u\in\mathcal{X}$ and $v\in\mathcal{Y}$ such that the pair of vectors $(u,v)\in\mathcal{X}\times\mathcal{Y}$ is identified with the pair of bounded linear functionals $(f,g)\in\mathcal{X}^*\times\mathcal{Y}^*$ by means of the identities $f(\cdot) = \langle\cdot\,;u\rangle_{\mathcal{X}}$ and $g(\cdot) = \langle\cdot\,;v\rangle_{\mathcal{Y}}$. So, along the lines of Sections 4.2 and 4.3 (in particular, according to Corollary 4.5), a single tensor $x\otimes y$ in $\mathcal{X}\otimes\mathcal{Y}$ associated with (x,y) in $\mathcal{X}\times\mathcal{Y}$ is a bilinear form $x\otimes y\colon \mathcal{X}^*\times\mathcal{Y}^*\to\mathbb{F}$ defined by

$$(x\otimes y)(f,g) = f(x)\,g(y) \quad \text{for every} \quad (f,g)\in\mathcal{X}^*\times\mathcal{Y}^*,$$

which is equivalently written (via the identification $f \cong u$ and $g \cong v$) as

$$(x \otimes y)(u, v) = \langle x\,; u\rangle_{\mathcal{X}}\, \langle y\,; v\rangle_{\mathcal{Y}} \quad \text{for every} \quad (u, v) \in \mathcal{X}\times\mathcal{Y}$$

(still a bilinear form $x \otimes y$ on $\mathcal{X}^*\times\mathcal{Y}^*$ with $\langle \cdot\,; u\rangle_{\mathcal{X}} \in \mathcal{X}^*$ and $\langle \cdot\,; v\rangle_{\mathcal{Y}} \in \mathcal{Y}^*$).

Next consider the functional

$$\langle \cdot\,;\cdot\rangle_\eta \colon (\mathcal{X} \otimes \mathcal{Y}) \times (\mathcal{X} \otimes \mathcal{Y}) \to \mathbb{F}$$

defined on the Cartesian product $(\mathcal{X} \otimes \mathcal{Y}) \times (\mathcal{X} \otimes \mathcal{Y})$ of a tensor product space $\mathcal{X} \otimes \mathcal{Y}$ of Hilbert spaces $\mathcal{X}$ and $\mathcal{Y}$ with itself as follows:

$$\langle F\,; F'\rangle_\eta = \sum\nolimits_{i=1}^{n}\sum\nolimits_{j=1}^{m} \langle x_i\,; x'_j\rangle_{\mathcal{X}}\, \langle y_i\,; y'_j\rangle_{\mathcal{Y}} \tag{IP}$$

for every $F = \sum_{i=1}^{n} x_i \otimes y_i$ and $F' = \sum_{j=1}^{m} x'_j \otimes y'_j$ in $\mathcal{X} \otimes \mathcal{Y}$. In particular,

$$\langle x \otimes y\,; u \otimes v\rangle_\eta = \langle x\,; u\rangle_{\mathcal{X}}\, \langle y\,; v\rangle_{\mathcal{Y}} \tag{IP}$$

for every $x, u \in \mathcal{X}$ and $y, v \in \mathcal{Y}$. We will drop the subscripts and write $\langle \cdot\,;\cdot\rangle$ for both $\langle \cdot\,;\cdot\rangle_{\mathcal{X}}$ and $\langle \cdot\,;\cdot\rangle_{\mathcal{Y}}$ (and also $\|\cdot\|$ for $\|\cdot\|_{\mathcal{X}}$ and $\|\cdot\|_{\mathcal{Y}}$) from now on.

Theorem 9.4. *Let $\mathcal{X}$ and $\mathcal{Y}$ be arbitrary Hilbert spaces. Consider the functional $\langle \cdot\,;\cdot\rangle_\eta \colon (\mathcal{X} \otimes \mathcal{Y})\times(\mathcal{X} \otimes \mathcal{Y}) \to \mathbb{F}$ defined in* (IP) *above.*

(a) *$\langle \cdot\,;\cdot\rangle_\eta$ does not depend on the representation of each element in $\mathcal{X} \otimes \mathcal{Y}$.*

(b) *$\langle \cdot\,;\cdot\rangle_\eta$ is an inner product on $\mathcal{X} \otimes \mathcal{Y}$. (Set $\mathcal{X} \otimes_\eta \mathcal{Y} = (\mathcal{X} \otimes \mathcal{Y}, \langle \cdot\,;\cdot\rangle_\eta)$).*

(c) *The norm $\|\cdot\|_\eta$ induced by $\langle \cdot\,;\cdot\rangle_\eta$ is a reasonable crossnorm on $\mathcal{X} \otimes \mathcal{Y}$.*

Proof. (a) Since single tensors are bilinear functionals, it is indeed readily verified that the above functional $\langle \cdot\,;\cdot\rangle_\eta$ is a Hermitian symmetric sesquilinear form on $(\mathcal{X} \otimes \mathcal{Y}) \times (\mathcal{X} \otimes \mathcal{Y})$ (i.e., $\langle F + F'; F''\rangle_\eta = \langle F; F''\rangle_\eta + \langle F'; F''\rangle_\eta$, $\langle aF; F'\rangle_\eta = a\langle F; F'\rangle_\eta$ and $\langle F; F'\rangle_\eta = \overline{\langle F'; F\rangle_\eta}$ for every $F, F', F'' \in \mathcal{X} \otimes \mathcal{Y}$ and $a \in \mathbb{F}$). Moreover, since $\langle F - F'; F''\rangle_\eta = \langle F; F''\rangle_\eta - \langle F'; F''\rangle_\eta$ for every F'', it follows that if F and F' are two representations of the same element in $\mathcal{X} \otimes \mathcal{Y}$, then $\langle F; F''\rangle_\eta = \langle F'; F''\rangle_\eta$ for every $F'' \in \mathcal{X} \otimes \mathcal{Y}$. Similarly, $\langle F''; F\rangle_\eta = \langle F''; F'\rangle_\eta$ for every $F'' \in \mathcal{X} \otimes \mathcal{Y}$. Therefore the functional $\langle \cdot\,;\cdot\rangle_\eta$ does not depend on the representation of $F \in \mathcal{X} \otimes \mathcal{Y}$. (This is a consequence of additivity and real homogeneity that holds on both entries of $\langle \cdot\,;\cdot\rangle_\eta$ — we have used a similar argument in the proof of Theorem 3.17(a).)

(b) To conclude that $\langle \cdot\,;\cdot\rangle_\eta$ is an inner product on the linear space $\mathcal{X} \otimes \mathcal{Y}$, we need to show that it induces a positive quadratic form. Thus let $\mathcal{X}$ and $\mathcal{Y}$ be Hilbert spaces. Take an arbitrary (but fixed) $F = \sum_i x_i \otimes y_i$ in $\mathcal{X} \otimes \mathcal{Y}$. Since this sum is finite, consider the finite sets $\{x_i\}$ and $\{y_i\}$, and take the finite-dimensional subspaces $\mathcal{M} = \operatorname{span}\{x_i\}$ and $\mathcal{N} = \operatorname{span}\{y_i\}$ of $\mathcal{X}$ and $\mathcal{Y}$, respectively. Since $\mathcal{M}$ and $\mathcal{N}$ are finite-dimensional subspaces of $\mathcal{X}$ and $\mathcal{Y}$, let

$\{e_k\}$ and $\{d_\ell\}$ be *finite* orthonormal bases for $\mathcal{M}$ and $\mathcal{N}$, which are subsets of orthonormal bases for the Hilbert spaces $\mathcal{X}$ and $\mathcal{Y}$, respectively. Since each x_i lies in $\mathcal{M}$ and each y_i lies in $\mathcal{N}$, consider their *finite* Fourier series expansions $x_i = \sum_k \langle x_i\,; e_k\rangle\, e_k$ and $y_i = \sum_\ell \langle y_i\,; d_\ell\rangle\, d_\ell$ (Proposition 5.T). Since each single tensor $x \otimes y = \theta'(x,y)$ is the image of a bilinear map $\theta' \colon \mathcal{X}\times\mathcal{Y} \to \top'_{\mathcal{X},\mathcal{Y}}$ on the linear space $\top'_{\mathcal{X},\mathcal{Y}}$ (cf. algebraic manipulations in Section 3.1),

$$x_i \otimes y_i = \Big(\sum\nolimits_k \langle x_i\,; e_k\rangle\, e_k\Big) \otimes \Big(\sum\nolimits_\ell \langle y_i\,; d_\ell\rangle\, d_\ell\Big) = \sum\nolimits_{k,\ell} \langle x_i\,; e_k\rangle\, \langle y_i\,; d_\ell\rangle\, (e_k \otimes d_\ell).$$

For such an F set $\gamma_{i,k,\ell} = \langle x_i\,; e_k\rangle\, \langle y_i\,; d_\ell\rangle$ and

$$\sigma_{k,\ell} = \sum\nolimits_i \gamma_{i,k,\ell} = \sum\nolimits_i \langle x_i\,; e_k\rangle\, \langle y_i\,; d_\ell\rangle.$$

Thus $x_i \otimes y_i = \sum_{k,\ell} \gamma_{i,k,\ell}\, (e_k \otimes d_\ell)$ and

$$\begin{aligned} F = \sum\nolimits_i x_i \otimes y_i &= \sum\nolimits_i \Big(\sum\nolimits_{k,\ell} \gamma_{i,k,\ell}\, (e_k \otimes d_\ell)\Big) \\ &= \sum\nolimits_{k,\ell} \Big(\sum\nolimits_i \gamma_{i,k,\ell}\Big) (e_k \otimes d_\ell) = \sum\nolimits_{k,\ell} \sigma_{k,\ell}\, (e_k \otimes d_\ell). \end{aligned}$$

Recall that all sums above and below are finite. Since $\{e_k\}$ and $\{d_\ell\}$ are (finite) subsets of orthonormal bases for $\mathcal{X}$ and $\mathcal{Y}$, the set $\{e_k \otimes d_\ell\}$ is orthonormal in the sense that $\langle e_k \otimes d_\ell\,; e_{k'} \otimes d_{\ell'}\rangle_\eta = \langle e_k\,; e_{k'}\rangle\, \langle d_\ell\,; d_{\ell'}\rangle = \delta_{k,k'}\, \delta_{\ell,\ell'}$ according to $(\text{IP})_1$, where the deltas stand for Kronecker functions ($\delta_{k,k'} = 0$ if $k \neq k'$ and $\delta_{k,k} = 1$). Hence the above orthogonal expansion yields

$$\begin{aligned} \langle F\,; F\rangle_\eta &= \Big\langle \sum\nolimits_{k,\ell} \sigma_{k,\ell}\, (e_k \otimes d_\ell)\,;\, \sum\nolimits_{k',\ell'} \sigma_{k',\ell'}\, (e_{k'} \otimes d_{\ell'}) \Big\rangle_\eta \\ &= \sum\nolimits_{k,\ell,k',\ell'} \sigma_{k,\ell}\, \overline{\sigma_{k',\ell'}}\, \langle e_k \otimes d_\ell\,; e_{k'} \otimes d_{\ell'}\rangle_\eta = \sum\nolimits_{k,\ell} |\sigma_{k,\ell}|^2. \qquad (\text{IP})_2 \end{aligned}$$

So $\langle F\,; F\rangle_\eta \geq 0$, and $\langle F\,; F\rangle_\eta = 0$ only if $\sigma_{k,\ell} = 0$ for all k, ℓ, which implies $F = 0$ (by the above expression for F). As F was taken arbitrarily in $\mathcal{X} \otimes \mathcal{Y}$, the functional $\langle \cdot\,; \cdot\rangle_\eta \colon (\mathcal{X} \otimes \mathcal{Y}) \times (\mathcal{X} \otimes \mathcal{Y}) \to \mathbb{F}$ is an inner product on $\mathcal{X} \otimes \mathcal{Y}$. Thus let $\|\cdot\|_\eta \colon \mathcal{X} \otimes \mathcal{Y} \to \mathbb{R}$ be the norm it induces, and set $\mathcal{X} \otimes_\eta \mathcal{Y} = (\mathcal{X} \otimes \mathcal{Y}, \langle \cdot\,; \cdot\rangle_\eta)$.

(c) Take any $F = \sum_i x_i \otimes y_i \in \mathcal{X} \otimes_\eta \mathcal{Y}$. Computing the value of $\|F\|_\eta^2 = \langle F\,; F\rangle_\eta$,

$$\begin{aligned} \Big\|\sum\nolimits_i x_i \otimes y_i\Big\|_\eta^2 &= \Big\langle \sum\nolimits_i x_i \otimes y_i\,;\, \sum\nolimits_j x_j \otimes y_j \Big\rangle_\eta \\ &= \sum\nolimits_{i,j} \langle x_i \otimes y_i\,; x_j \otimes y_j\rangle_\eta = \sum\nolimits_{i,j} \langle x_i\,; x_j\rangle \langle y_i\,; y_j\rangle. \qquad (\text{N}) \end{aligned}$$

This gives the expression for the norm induced by the inner product defined in (IP). In particular, $\|x \otimes y\|_\eta^2 = \langle x\,; x\rangle\, \langle y\,; y\rangle$ so that

$$\|x \otimes y\|_\eta = \|x\|\, \|y\|. \qquad (\text{N})_1$$

According to $(\text{N})_1$,

$$\|F\|_\eta = \left\|\sum\nolimits_i x_i \otimes y_i\right\|_\eta \le \sum\nolimits_i \|x_i \otimes y_i\|_\eta = \sum\nolimits_i \|x_i\|\,\|y_i\|.$$

As $\|F\|_\eta$ does not depend on the representation of F (since $\langle\cdot\,;\cdot\rangle_\eta$ does not),

$$\|F\|_\eta \le \inf_{\{x_i\},\,\{y_i\},\,F=\Sigma_i x_i\otimes y_i} \sum\nolimits_i \|x_i\|\,\|y_i\| = \|F\|_\wedge.$$

On the other hand, recall that our arbitrary (but fixed) $F \in \mathcal{X}\otimes\mathcal{Y}$ actually acts on the regular linear manifold $\mathcal{M}\otimes\mathcal{N}$ of $\mathcal{X}\otimes\mathcal{Y}$, which is a finite-dimensional linear space, and hence when equipped with any inner product (e.g., with the restriction of $\langle\cdot\,;\cdot\rangle_\eta$ to $\mathcal{M}\otimes\mathcal{N}$) becomes a Hilbert space and we can apply the Riesz Representation Theorem (Proposition 5.U) to it. Thus take an arbitrary $f\otimes g \in \mathcal{M}^*\otimes\mathcal{N}^* = \mathcal{M}^\sharp\otimes\mathcal{N}^\sharp \subseteq (\mathcal{M}\otimes_\eta\mathcal{N})^\sharp = (\mathcal{M}\otimes_\eta\mathcal{N})^*$ (all spaces are finite-dimensional, so linear functionals are continuous) acting on the domain $\mathcal{M}\otimes_\eta\mathcal{N} = (\mathcal{M}\otimes\mathcal{N}, \langle\cdot\,;\cdot\rangle_\eta)$ of F. Let $u\otimes v \in \mathcal{M}\otimes_\eta\mathcal{N}$ be the Riesz representation of $f\otimes g \in (\mathcal{M}\otimes_\eta\mathcal{N})^*$. That is, $(f\otimes g)F = \langle u\otimes v\,;F\rangle_\eta$ for every $F = \sum_i x_i\otimes y_i \in \mathcal{M}\otimes_\eta\mathcal{N}$. Observe that

> *the Riesz representation $u\otimes v \in \mathcal{M}\otimes_\eta\mathcal{N}$ of the functional $f\otimes g$ in $(\mathcal{M}\otimes_\eta\mathcal{N})^*$ coincides with the single tensor of the Riesz representations $u\in\mathcal{M}$ and $v\in\mathcal{N}$ of the functionals $f\in\mathcal{M}^*$ and $g\in\mathcal{N}^*$.*

Indeed, if u and v are Riesz representations of f and g so that $f(x)=\langle x\,;u\rangle$ and $g(x)=\langle y\,;u\rangle$ for every $x\in\mathcal{M}$ and $y\in\mathcal{N}$, then $(f\otimes g)F = \sum_i f(x_i)\,g(y_i) = \sum_i\langle x_i\,;u\rangle\,\langle y_i\,;v\rangle = \sum_i\langle x_i\otimes y_i\,;u\otimes v\rangle_\eta = \langle F\,;u\otimes v\rangle_\eta$ for every $F\in\mathcal{M}\otimes_\eta\mathcal{N}$, and so $u\otimes v$ is the Riesz representation of $f\otimes g$. Since these representations are unique, the converse holds as well. Moreover, $\|f\| = \|u\|$, $\|g\| = \|v\|$, and $\|f\otimes g\|_{(\mathcal{M}\otimes\mathcal{N})^*} = \|u\otimes v\|_\eta = \|u\|\,\|v\| = \|f\|\,\|g\|$ according to $(\text{N})_1$. Then

$$\left|\sum\nolimits_i f(x_i)\,g(y_i)\right| = |(f\otimes g)F| = |\langle F\,;u\otimes v\rangle_\eta| \le \|F\|_\eta\|u\otimes v\|_\eta = \|F\|_\eta\|u\|\,\|v\|$$

by the Schwartz inequality, so that

$$\|F\|_\vee = \sup_{\|f\|\le1,\|g\|\le1}\left|\sum\nolimits_i f(x_i)\,g(y_i)\right| \le \sup_{\|u\|\le1,\|v\|\le1}\|F\|_\eta\|u\|\,\|v\| = \|F\|_\eta.$$

Therefore $\|\cdot\|_\eta$ is a reasonable crossnorm by Theorem 7.4. □

Remark 9.5. *Orthonormal Bases for $\mathcal{X}\otimes\mathcal{Y}$ and Single Tensors in $\mathcal{X}^*\otimes\mathcal{Y}^*$.*

(a) If $\{e_k\}$ and $\{d_\ell\}$ are arbitrary finite subsets of orthonormal bases $\{e_\epsilon\}$ and $\{d_\delta\}$ for Hilbert spaces $\mathcal{X}$ and $\mathcal{Y}$, respectively, then the set $\{e_k\otimes d_\ell\}$ is linearly independent (Theorem 3.9(b)) and orthonormal (Proposition 9.D) in the inner product space $(\mathcal{X}\otimes\mathcal{Y}, \langle\cdot\,;\cdot\rangle_\eta)$ (Theorem 9.4), and so is $\{e_\epsilon\otimes d_\delta\}$. Then one might ask whether $\{e_\epsilon\otimes d_\delta\}$ would be an orthonormal basis for $(\mathcal{X}\otimes\mathcal{Y}, \langle\cdot\,;\cdot\rangle_\eta)$ and the orthogonal expansion $F = \sum_{k,\ell}\sigma_{k,\ell}\,(e_k\otimes d_\ell)$ in the proof of Theorem 9.4(b) would be a Fourier series expansion of F. But $(\mathcal{X}\otimes\mathcal{Y}, \langle\cdot\,;\cdot\rangle_\eta)$ may not be complete (compare with Proposition 7.A), thus it may not be Hilbert, and so the Fourier Series Theorem of Proposition 5.T may not apply to it yet. However (see, e.g., [92, Theorem 3.12(b)]),

if $\{e_\epsilon\}$ *and* $\{d_\delta\}$ *are orthonormal bases for Hilbert spaces* $\mathcal{X}$ *and* $\mathcal{Y}$*, then* $\{e_\epsilon \otimes d_\delta\}$ *is an orthonormal basis for the completion of* $(\mathcal{X} \otimes \mathcal{Y}, \langle \cdot\,;\cdot \rangle_\eta)$.

(b) Because $\|\cdot\|_\eta$ is a reasonable crossnorm on $\mathcal{X} \otimes \mathcal{Y}$, Definition 7.1 says $\mathcal{X}^* \otimes \mathcal{Y}^* \subseteq (\mathcal{X} \otimes_\eta \mathcal{Y})^*$ and, in addition to $(\text{N})_1$, we get by Theorem 7.2

$$\|f \otimes g\|_{*\eta} = \|f\|\,\|g\| \qquad (\text{N})_2$$

for every $f \in \mathcal{X}^*$ and $g \in \mathcal{Y}^*$. Again, drop the subscripts and write $\|\cdot\|$ for both $\|\cdot\|_{\mathcal{X}^*}$ and $\|\cdot\|_{\mathcal{Y}^*}$. Here $\|\cdot\|_{*\eta}$ is defined as in Section 7.1: the induced uniform norm on $(\mathcal{X} \otimes_\eta \mathcal{Y})^* = \mathcal{B}[\mathcal{X} \otimes_\eta \mathcal{Y}, \mathbb{F}]$. By Theorem 7.3, the norm $\|\cdot\|_{*\eta}$ on $(\mathcal{X} \otimes_\eta \mathcal{Y})^*$ is a reasonable crossnorm on $\mathcal{X}^* \otimes \mathcal{Y}^*$, and so we write

$$\mathcal{X}^* \otimes_{*\eta} \mathcal{Y}^* = (\mathcal{X}^* \otimes \mathcal{Y}^*, \|\cdot\|_{*\eta}) \quad \big(\text{as well as } \mathcal{X} \otimes_\eta \mathcal{Y} = (\mathcal{X} \otimes \mathcal{Y}, \|\cdot\|_\eta)\big).$$

Theorem 9.6. *The reasonable crossnorm* $\|\cdot\|_\eta$ *is a symmetric tensor norm.*

Proof. We split the proof into three parts. The reasonable crossnorm $\|\cdot\|_\eta$

(a) is symmetric,

(b) is a uniform crossnorm (on tensor products of arbitrary Hilbert spaces),

(c) and is a tensor norm (i.e., a finitely generated uniform crossnorm).

(a) Let $\mathcal{X}$ and $\mathcal{Y}$ be Hilbert spaces and consider the inner product $\langle \cdot\,;\cdot \rangle_\eta$ defined in (IP). Take the norm $\|\cdot\|_\eta$ on $\mathcal{X} \otimes \mathcal{Y}$ induced by this inner product and set $\mathcal{X} \otimes_\eta \mathcal{Y} = (\mathcal{X} \otimes \mathcal{Y}, \|\cdot\|_\eta)$ as usual. According to Theorem 9.4, the inner product $\langle \cdot\,;\cdot \rangle_\eta$ is well defined for every pair of Hilbert spaces, and so is the norm it generates. Thus consider the norm $\|\cdot\|_\eta$ acting on $\mathcal{Y} \otimes \mathcal{X}$ as well and set $\mathcal{Y} \otimes_\eta \mathcal{X} = (\mathcal{Y} \otimes \mathcal{X}, \|\cdot\|_\eta)$. By the norm expression in (N) we get

$$\|F\|_\eta = \Big\|\sum\nolimits_i x_i \otimes y_i\Big\|_{\mathcal{X} \otimes_\eta \mathcal{Y}} = \Big\|\sum\nolimits_i y_i \otimes x_i\Big\|_{\mathcal{Y} \otimes_\eta \mathcal{X}} = \|{}^tF\|_\eta = \|F\|_{t\eta}$$

for every $F = \sum_i x_i \otimes y_i \in \mathcal{X} \otimes \mathcal{Y}$, where ${}^tF = \sum_i y_i \otimes x_i \in \mathcal{Y} \otimes \mathcal{X}$ is the transpose of F and $\|\cdot\|_{t\eta} \colon \mathcal{X} \otimes \mathcal{Y} \to \mathbb{R}$ is the transpose norm of $\|\cdot\|_\eta \colon \mathcal{Y} \otimes \mathcal{X} \to \mathbb{R}$ given by $\|F\|_{t\eta} = \|{}^tF\|_\eta$ for every $F \in \mathcal{X} \otimes \mathcal{Y}$ so that ${}^tF \in \mathcal{Y} \otimes \mathcal{X}$. Theorem 9.4 ensures that the norm $\|\cdot\|_\eta$ given in (N) acts as a reasonable crossnorm on both $\mathcal{X} \otimes \mathcal{Y}$ and $\mathcal{Y} \otimes \mathcal{X}$. Then Theorem 7.6 says that its transpose $\|\cdot\|_{t\eta}$ acts as a reasonable crossnorm on both $\mathcal{X} \otimes \mathcal{Y}$ and $\mathcal{Y} \otimes \mathcal{X}$. The above identity shows that they actually are symmetric according to Remark 7.7.

(b) Let $\mathcal{X}, \mathcal{Y}, \mathcal{V}, \mathcal{W}$ be Hilbert spaces. Equip tensor product spaces of any combination of them with the inner product $\langle \cdot\,;\cdot \rangle_\eta$ defined in (IP), which acts on the Cartesian product of tensor products of every pair of Hilbert spaces, and so the reasonable crossnorm $\|\cdot\|_\eta$ induced by it acts on tensor products of every pair of Hilbert spaces (Theorem 9.4). Take A in $\mathcal{B}[\mathcal{X}, \mathcal{V}]$, B in $\mathcal{B}[\mathcal{Y}, \mathcal{W}]$, and the tensor product $A \otimes B$ in $\mathcal{L}[\mathcal{X} \otimes_\eta \mathcal{Y}, \mathcal{V} \otimes_\eta \mathcal{W}]$. Let $I_\mathcal{X}$ and $I_\mathcal{Y}$ be the identity operators on $\mathcal{X}$ and $\mathcal{Y}$. Consider the expansion of an arbitrary $F = \sum_i x_i \otimes y_i$ in $\mathcal{X} \otimes_\eta \mathcal{Y}$ and its norm $\|F\|_\eta$ as in $(\text{IP})_2$, namely, $F = \sum_{k,\ell} \sigma_{k,\ell}(e_k \otimes d_\ell)$ and $\langle F\,;F \rangle_\eta = \sum_{k,\ell} |\sigma_{k,\ell}|^2$. Reorder and reindex these finite sums to get

$$F = \sum\nolimits_i x_i \otimes y_i = \sum\nolimits_j \sigma_j (e_j \otimes d_j) \quad \text{with} \quad \|F\|_\eta^2 = \langle F; F\rangle_\eta = \sum\nolimits_j |\sigma_j|^2.$$

Since $\{e_j \otimes d_j\}$ is an orthonormal set (cf. Proposition 9.D), and using the Pythagorean Theorem, it follows that $\|\sum_j \sigma_j (e_j \otimes d_j)\|_\eta^2 = \sum_j \|\sigma_j (e_j \otimes d_j)\|_\eta^2$. Also, $\{Te_j \otimes d_j\}$ remains orthogonal for every $T \in \mathcal{B}[\mathcal{X}]$. Indeed, by $(\text{IP})_1$

$$\langle Te_j \otimes d_j ; Te_{j'} \otimes d_{j'}\rangle_\eta = \langle Te_j ; Te_{j'}\rangle \langle d_j ; d_{j'}\rangle = \|Te_j\|^2 \delta_{j,j'},$$

as $\{d_j\}$ is an orthonormal set. By the Pythagorean Theorem again and $(\text{N})_1$,

$$\begin{aligned} \|(A \otimes I_\mathcal{Y})F\|_\eta^2 &= \Big\|(A \otimes I_\mathcal{Y}) \sum\nolimits_j \sigma_j \, (e_j \otimes d_j)\Big\|_\eta^2 = \Big\|\sum\nolimits_j \sigma_j \, (Ae_j \otimes d_j)\Big\|_\eta^2 \\ &= \sum\nolimits_j |\sigma_j|^2 \, \|Ae_j \otimes d_j\|_\eta^2 = \sum\nolimits_j |\sigma_j|^2 \, \|Ae_j\|^2 \\ &\le \|A\|^2 \sum\nolimits_j |\sigma_j|^2 = \|A\|^2 \|F\|_\eta^2. \end{aligned}$$

Similarly, $\|(I_\mathcal{X} \otimes B)F\|_\eta^2 \le \|B\|^2 \|F\|_\eta^2$. Therefore

$$\|A \otimes I_\mathcal{Y}\| \le \|A\| \quad \text{and} \quad \|I_\mathcal{X} \otimes B\| \le \|B\|,$$

where the left-hand side norms are the induced uniform norms of elements in $\mathcal{B}[\mathcal{X} \otimes_\eta \mathcal{Y}, \mathcal{V} \otimes_\eta \mathcal{W}]$. Moreover, by Theorem 3.19(c) (see also Remark 3.20),

$$A \otimes B = (A \otimes I_\mathcal{Y})(I_\mathcal{X} \otimes B).$$

Consequently, taking the induced uniform norm,

$$\|A \otimes B\| = \|(A \otimes I_\mathcal{Y})(I_\mathcal{X} \otimes B)\| \le \|A \otimes I_\mathcal{Y}\| \, \|I_\mathcal{X} \otimes B\| \le \|A\| \, \|B\|.$$

Thus $\|\cdot\|_\eta$ is a uniform crossnorm by Definition 8.2(a).

(c) Take an arbitrary (but fixed) F in $\mathcal{X} \otimes \mathcal{Y}$, and let the finite sum $\sum_i x_i \otimes y_i$ be a representation of it. Building from the finite sets $\{x_i\}$ and $\{y_i\}$, consider the finite-dimensional subspaces $\mathcal{M} = \text{span}\,\{x_i\}$ and $\mathcal{N} = \text{span}\,\{y_i\}$ of $\mathcal{X}$ and $\mathcal{Y}$ (as in the proof of Theorem 9.4(b)). The very construction of $\|F\|_\eta$ yields

$$\|F\|_{\mathcal{X} \otimes_\eta \mathcal{Y}} = \|F\|_\eta = \|F\|_{\mathcal{M} \otimes_\eta \mathcal{N}}.$$

Since $\mathcal{M}$ and $\mathcal{N}$ are finite-dimensional subspaces of $\mathcal{X}$ and $\mathcal{Y}$, and since (by its own construction) the norm $\|F\|_\eta$ does not depend on the finite-dimensional subspaces $\mathcal{M}$ and $\mathcal{N}$ spanned by the finite sets of any representation, $\|F\|_{\mathcal{X} \otimes_\eta \mathcal{Y}}$ satisfies Definition 8.7(a). (Actually, Corollary 8.17 says that for Hilbert spaces every uniform crossnorm — not only this one — is finitely generated). □

Remark 9.7. *The Norms* $\|f \otimes g\|_{*\eta}$ *and* $\|f \otimes g\|_\eta$ *Coincide on* $\mathcal{X}^* \otimes \mathcal{Y}^*$. Proposition 5.V stated a crucial corollary of the Riesz Representation Theorem which says that for every Hilbert space $\mathcal{X}$ there exists an additive and conjugate homogeneous surjective isometry $\Psi \colon \mathcal{X}^* \to \mathcal{X}$. Thus the functional

$\langle\cdot\,;\cdot\rangle_*\colon \mathcal{X}^*\times\mathcal{X}^*\to\mathbb{F}$ given by $\langle f\,;g\rangle_* = \langle \Psi(g)\,;\Psi(f)\rangle_{\mathcal{X}}$ for every $f,g\in\mathcal{X}^*$ is an inner product on $\mathcal{X}^*$, and so it induces a norm $\|\cdot\|_*\colon \mathcal{X}^*\to\mathbb{R}$ such that $\|f\|_* = \|\Psi(f)\|_{\mathcal{X}} = \|f\|_{\mathcal{X}^*} = \|f\|_{\mathcal{B}[\mathcal{X},\mathbb{F}]}$ for every $f\in\mathcal{X}^*$: *the norm induced on* $\mathcal{X}^*$ *by the inner product* $\langle\cdot\,;\cdot\rangle_*$ *coincides with the induced uniform norm on* $\mathcal{X}^* = \mathcal{B}[\mathcal{X},\mathbb{F}]$ (i.e., the induced uniform norm on $\mathcal{X}^* = (\mathcal{B}[\mathcal{X},\mathbb{F}],\|\cdot\|_{\mathcal{B}[\mathcal{X},\mathbb{F}]})$ when $\mathcal{X}$ is equipped with $\|\cdot\|_{\mathcal{X}}$ makes $\mathcal{X}^*$ into an inner product space, and so a Hilbert space). Then consider the same argument used to define the inner product $\langle\cdot\,;\cdot\rangle_\eta$ on $(\mathcal{X}\otimes\mathcal{Y})\times(\mathcal{X}\otimes\mathcal{Y})$ for Hilbert spaces $\mathcal{X}$ and $\mathcal{Y}$ to redefine it on $(\mathcal{X}^*\otimes\mathcal{Y}^*)\times(\mathcal{X}^*\otimes\mathcal{Y}^*)$ for Hilbert spaces $\mathcal{X}^*$ and $\mathcal{Y}^*$ as follows. Take

$$\langle\cdot\,;\cdot\rangle_\eta\colon (\mathcal{X}^*\otimes\mathcal{Y}^*)\times(\mathcal{X}^*\otimes\mathcal{Y}^*)\to\mathbb{F}$$

defined on the Cartesian product $(\mathcal{X}^*\otimes\mathcal{Y}^*)\times(\mathcal{X}^*\otimes\mathcal{Y}^*)$ of a tensor product space $\mathcal{X}^*\otimes\mathcal{Y}^*$ of Hilbert spaces $\mathcal{X}^*$ and $\mathcal{Y}^*$ with itself (as in (IP)) by

$$\langle\Omega\,;\Omega'\rangle_\eta = \sum\nolimits_{i=1}^{n}\sum\nolimits_{j=1}^{m}\langle f_i\,;f'_j\rangle_{\mathcal{X}^*}\langle g_i\,;g'_j\rangle_{\mathcal{Y}^*}$$

for every $\Omega = \sum_{i=1}^n f_i\otimes g_i$ and $\Omega' = \sum_{j=1}^m f'_j\otimes g'_j$ in $\mathcal{X}^*\otimes\mathcal{Y}^*$. As before, drop the subscripts and write $\langle\cdot\,;\cdot\rangle$ for both $\langle\cdot\,;\cdot\rangle_{\mathcal{X}^*}$ and $\langle\cdot\,;\cdot\rangle_{\mathcal{Y}^*}$ (and also $\|\cdot\|$ for $\|\cdot\|_{\mathcal{X}^*}$ and $\|\cdot\|_{\mathcal{Y}^*}$). Now apply Theorem 9.4(a) to verify that this is an inner product, and hence $\langle\cdot\,;\cdot\rangle_\eta$ induces the norm (as $\|F\|_\eta$ in (N)), viz.,

$$\|\Omega\|_\eta^2 = \langle\Omega\,;\Omega\rangle_\eta = \Big\|\sum\nolimits_i f_i\otimes g_i\Big\|_\eta^2 = \sum\nolimits_{i,j}\langle f_i\,;f_j\rangle\langle g_i\,;g_j\rangle,$$

which is a reasonable crossnorm on $\mathcal{X}^*\otimes\mathcal{Y}^*$ by Theorem 9.4(c); in fact, it is a symmetric tensor norm according Theorem 9.6. In particular,

$$\|f\otimes g\|_\eta = \|f\|\|g\|,$$

a reasonable crossnorm on $\mathcal{X}^*\otimes\mathcal{Y}^*$ coinciding with $(\text{N})_2$ in Remark 9.5(b).

Theorems 3.19, 9.2, and 9.3 hold in particular for bounded linear transformations between inner product spaces. The adjoints in Theorems 9.2(b) and 9.3(j) are normed-space ones, applicable to tensor products of bounded linear transformations between any pair of normed spaces, and so are applicable to tensor products of bounded linear transformations between inner product spaces. But the Hilbert-space adjoint is the natural notion of adjoint for operators acting on a Hilbert space. Recall from Remark 5.W that it does not make sense to speak of the Hilbert-space adjoint of $A\otimes B\in\mathcal{B}[\mathcal{X}\otimes_\eta\mathcal{Y},\mathcal{V}\otimes_\eta\mathcal{W}]$ because $\mathcal{X}\otimes_\eta\mathcal{Y}$ may not be a Hilbert space (even if $\mathcal{V},\mathcal{W},\mathcal{X},\mathcal{Y}$ are).

For Hilbert spaces $\mathcal{X}$ and $\mathcal{Y}$, take the inner product space $\mathcal{X}\otimes_\eta\mathcal{Y}$ equipped with the inner product $\langle\cdot\,;\cdot\rangle_\eta$. Let $\mathcal{X}\,\widehat{\otimes}_\eta\mathcal{Y}$ be the completion of $\mathcal{X}\otimes_\eta\mathcal{Y}$, which is a Hilbert space whose inner product is the extension over completion of the inner product on $\mathcal{X}\otimes_\eta\mathcal{Y}$ (cf. Proposition 6.K— using the same notation $\langle\cdot\,;\cdot\rangle_\eta$ for the extension over completion of the inner product, and the same notation $\|\cdot\|_\eta$ for the extended norm induced by it). The Hilbert space $\mathcal{X}\,\widehat{\otimes}_\eta\mathcal{Y}$ and the norm $\|\cdot\|_\eta$ are sometimes referred to as *Hilbert–Schmidt tensor product* and *Hilbert–Schmidt norm* (see, e.g., [15, Section 26.7] — cf. Remark 9.13 below).

Theorem 9.8. *Let $\mathcal{V},\mathcal{W},\mathcal{X},\mathcal{Y}$ be Hilbert spaces, and equip their tensor product spaces with the inner product $\langle\cdot\,;\cdot\rangle_\eta$ inducing the tensor norm $\|\cdot\|_\eta$. Take bounded linear transformations $A\in\mathcal{B}[\mathcal{X},\mathcal{V}]$ and $B\in\mathcal{B}[\mathcal{Y},\mathcal{W}]$. Consider their tensor product $A\otimes B\in\mathcal{B}[\mathcal{X}\otimes_\eta\mathcal{Y},\mathcal{V}\otimes_\eta\mathcal{W}]$ and its extension over completion $A\mathbin{\widehat{\otimes}}B\in\mathcal{B}[\mathcal{X}\mathbin{\widehat{\otimes}}_\eta\mathcal{Y},\mathcal{V}\mathbin{\widehat{\otimes}}_\eta\mathcal{W}]$. The Hilbert-space adjoint of $A\mathbin{\widehat{\otimes}}B$ is such that*

$$(A\mathbin{\widehat{\otimes}}B)^* = A^*\mathbin{\widehat{\otimes}}B^* \quad \text{and} \quad \|(A\mathbin{\widehat{\otimes}}B)^*\| = \|A\mathbin{\widehat{\otimes}}B\| = \|A\|\,\|B\|,$$

where $A^*\in\mathcal{B}[\mathcal{V},\mathcal{X}]$, $B^*\in\mathcal{B}[\mathcal{W},\mathcal{Y}]$ and $(A\mathbin{\widehat{\otimes}}B)^*$ are Hilbert-space adjoints of A, B and $A\mathbin{\widehat{\otimes}}B$, with $A^*\mathbin{\widehat{\otimes}}B^*$ and $(A\mathbin{\widehat{\otimes}}B)^*$ lying in $\mathcal{B}[\mathcal{V}\mathbin{\widehat{\otimes}}_\eta\mathcal{W},\mathcal{X}\mathbin{\widehat{\otimes}}_\eta\mathcal{Y}]$.

Proof. Let $\mathcal{V},\mathcal{W},\mathcal{X},\mathcal{Y}$ be Hilbert spaces. Take $A\in\mathcal{B}[\mathcal{X},\mathcal{V}]$, $B\in\mathcal{B}[\mathcal{Y},\mathcal{W}]$, $\mathrm{F}=\sum_i x_i\otimes y_i\in\mathcal{X}\otimes_\eta\mathcal{Y}$ and $\mathrm{G}=\sum_i v_i\otimes w_i\in\mathcal{V}\otimes_\eta\mathcal{W}$, arbitrarily. Also take the Hilbert-space adjoints $A^*\in\mathcal{B}[\mathcal{V},\mathcal{X}]$ and $B^*\in\mathcal{B}[\mathcal{W},\mathcal{Y}]$ and the tensor products $A\otimes B\in\mathcal{B}[\mathcal{X}\otimes_\eta\mathcal{Y},\mathcal{V}\otimes_\eta\mathcal{W}]$ and $A^*\otimes B^*\in\mathcal{B}[\mathcal{V}\otimes_\eta\mathcal{W},\mathcal{X}\otimes_\eta\mathcal{Y}]$. By (IP) we get

$$\begin{aligned}\langle(A\otimes B)\mathrm{F}\,;\mathrm{G}\rangle_\eta &= \Big\langle\sum\nolimits_i Ax_i\otimes By_i\,;\sum\nolimits_j v_j\otimes w_j\Big\rangle_\eta\\ &= \sum\nolimits_{i,j}\langle Ax_i\,;v_i\rangle\,\langle By_j\,;w_j\rangle = \sum\nolimits_{i,j}\langle x_i\,;A^*v_i\rangle\,\langle y_j\,;B^*w_j\rangle\\ &= \Big\langle\sum\nolimits_i x_i\otimes y_i\,;\sum\nolimits_j A^*v_j\otimes B^*w_j\Big\rangle_\eta = \langle\mathrm{F}\,;(A^*\otimes B^*)\mathrm{G}\rangle_\eta.\end{aligned}$$

Consider the Hilbert spaces $\mathcal{X}\mathbin{\widehat{\otimes}}_\eta\mathcal{Y}$ and $\mathcal{V}\mathbin{\widehat{\otimes}}_\eta\mathcal{X}$, completions of the inner product spaces $\mathcal{X}\otimes_\eta\mathcal{Y}$ and $\mathcal{V}\otimes_\eta\mathcal{W}$. Thus if $\widehat{\mathrm{F}},\widehat{\mathrm{F}}'\in\mathcal{X}\mathbin{\widehat{\otimes}}_\eta\mathcal{Y}$ are the extensions of $\mathrm{F},\mathrm{F}'\in\mathcal{X}\otimes_\eta\mathcal{Y}$ and $\widehat{\mathrm{G}},\widehat{\mathrm{G}}'\in\mathcal{V}\mathbin{\widehat{\otimes}}_\eta\mathcal{W}$ are the extensions of $\mathrm{G},\mathrm{G}'\in\mathcal{V}\otimes_\eta\mathcal{W}$, then

$$\|\widehat{\mathrm{F}}\|_\eta=\|\mathrm{F}\|_\eta\,,\quad \|\widehat{\mathrm{G}}\|_\eta=\|\mathrm{G}\|_\eta\,,\quad \langle\widehat{\mathrm{F}}\,;\widehat{\mathrm{F}}'\rangle_\eta=\langle\mathrm{F}\,;\mathrm{F}'\rangle_\eta\,,\quad \langle\widehat{\mathrm{G}}\,;\widehat{\mathrm{G}}'\rangle_\eta=\langle\mathrm{G}\,;\mathrm{G}'\rangle_\eta\,.$$

(See Proposition 6.K.) Take $A^*\mathbin{\widehat{\otimes}}B^*$ in $\mathcal{B}[\mathcal{V}\mathbin{\widehat{\otimes}}_\eta\mathcal{W}\,;\mathcal{X}\mathbin{\widehat{\otimes}}_\eta\mathcal{Y}]$, the extension over completion of $A^*\otimes B^*$ in $\mathcal{B}[\mathcal{V},\mathcal{X}]\otimes\mathcal{B}[\mathcal{W},\mathcal{Y}]\subseteq\mathcal{B}[\mathcal{V}\otimes_\eta\mathcal{W}\,;\mathcal{X}\otimes_\eta\mathcal{Y}]$ as in Theorem 5.1. Extend the above identity $\langle(A\otimes B)\mathrm{F}\,;\mathrm{G}\rangle_\eta=\langle\mathrm{F}\,;(A^*\otimes B^*)\mathrm{G}\rangle_\eta$ from $\mathcal{X}\otimes_\eta\mathcal{Y}$ and $\mathcal{V}\otimes_\eta\mathcal{W}$ to $\mathcal{X}\mathbin{\widehat{\otimes}}_\eta\mathcal{Y}$ and $\mathcal{V}\mathbin{\widehat{\otimes}}_\eta\mathcal{W}$ (cf. Proposition 6.K again), so that

$$\big\langle(A\mathbin{\widehat{\otimes}}B)\widehat{\mathrm{F}}\,;\widehat{\mathrm{G}}\big\rangle_\eta=\big\langle\widehat{(A\otimes B)}\mathrm{F}\,;\widehat{\mathrm{G}}\big\rangle_\eta=\big\langle\widehat{\mathrm{F}}\,;\widehat{(A^*\otimes B^*)}\mathrm{G}\big\rangle_\eta=\big\langle\widehat{\mathrm{F}}\,;(A^*\mathbin{\widehat{\otimes}}B^*)\widehat{\mathrm{G}}\big\rangle_\eta$$

for every $\widehat{\mathrm{F}}\in\mathcal{X}\mathbin{\widehat{\otimes}}_\eta\mathcal{Y}$ and $\widehat{\mathrm{G}}\in\mathcal{V}\mathbin{\widehat{\otimes}}_\eta\mathcal{W}$, which determines the Hilbert-space adjoint $(A\mathbin{\widehat{\otimes}}B)^*\in\mathcal{B}[\mathcal{V}\mathbin{\widehat{\otimes}}_\eta\mathcal{W},\mathcal{X}\mathbin{\widehat{\otimes}}_\eta\mathcal{Y}]$ of $A\mathbin{\widehat{\otimes}}B\in\mathcal{B}[\mathcal{X}\mathbin{\widehat{\otimes}}_\eta\mathcal{Y},\mathcal{V}\mathbin{\widehat{\otimes}}_\eta\mathcal{W}]$ as

$$(A\mathbin{\widehat{\otimes}}B)^*=A^*\mathbin{\widehat{\otimes}}B^*.$$

Since $\|T^*\|=\|T\|=\|\widehat{T}\|$ in general, and $\|\cdot\|_\eta$ is a uniform crossnorm,

$$\|(A\mathbin{\widehat{\otimes}}B)^*\|=\|A\mathbin{\widehat{\otimes}}B\|=\|A\otimes B\|=\|A\|\,\|B\|.$$ □

9.3 Some Spectral Properties

Let $\Delta\cdot\Lambda=\{\nu\in\mathbb{C}\colon\ \nu=\delta\lambda$ for every $\delta\in\Delta$ and $\lambda\in\Lambda\}$ stand for the product of two subsets Δ and Λ of the complex plane $\mathbb{C}$. We use the same notation

I for the identity operator everywhere: on $\mathcal{X}$, on $\mathcal{Y}$, on $\mathcal{X} \otimes \mathcal{Y}$, on $\mathcal{X} \otimes_\alpha \mathcal{Y}$, and on $\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}$ (i.e., $I = I \otimes I$ on $\mathcal{X} \otimes \mathcal{Y}$ and on $\mathcal{X} \otimes_\alpha \mathcal{Y}$, and $I = \widehat{I} = I \mathbin{\widehat{\otimes}} I$ on $\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}$). Recall from Theorems 3.19(c) and 9.3(c): $A \otimes B = (A \otimes I)(I \otimes B) = (I \otimes B)(A \otimes I)$, and $A \mathbin{\widehat{\otimes}} B = (A \mathbin{\widehat{\otimes}} I)(I \mathbin{\widehat{\otimes}} B) = (I \mathbin{\widehat{\otimes}} B)(A \mathbin{\widehat{\otimes}} I)$.

From now on all normed spaces are complex (and nonzero), and so are all Banach and Hilbert spaces. Let $\mathcal{X}$ and $\mathcal{Y}$ be Banach spaces and take $A \in \mathcal{B}[\mathcal{X}]$ and $B \in \mathcal{B}[\mathcal{Y}]$. Let the Banach space $\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}$ be the completion of the normed space $\mathcal{X} \otimes_\alpha \mathcal{Y}$ consisting of a linear space $\mathcal{X} \otimes \mathcal{Y}$ equipped with a uniform crossnorm $\|\cdot\|_\alpha$. Take $A \mathbin{\widehat{\otimes}} B \in \mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}]$ on the Banach space $\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}$. According to the classical partition $\{\sigma_P(\cdot), \sigma_C(\cdot), \sigma_R(\cdot)\}$ of the spectrum $\sigma(\cdot)$ for bounded linear operators on a complex Banach space as in Section 5.4, we get

$$\sigma(A \mathbin{\widehat{\otimes}} B) = \big\{\lambda \in \mathbb{C} \colon \mathcal{N}(\lambda I - A \mathbin{\widehat{\otimes}} B) \neq \{0\} \ \text{ or } \ \mathcal{R}(\lambda I - A \mathbin{\widehat{\otimes}} B) \neq \mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}\big\};$$

$$\sigma_P(A \mathbin{\widehat{\otimes}} B) = \big\{\lambda \in \mathbb{C} \colon \mathcal{N}(\lambda I - A \mathbin{\widehat{\otimes}} B) \neq \{0\}\big\};$$

$$\sigma_C(A \mathbin{\widehat{\otimes}} B) = \big\{\lambda \in \mathbb{C} \colon \mathcal{N}(\lambda I - A \mathbin{\widehat{\otimes}} B) = \{0\}, \ \mathcal{R}(\lambda I - A \mathbin{\widehat{\otimes}} B)^- = \mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y} \ \text{ and } \ \mathcal{R}(\lambda I - A \mathbin{\widehat{\otimes}} B) \neq \mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}\big\};$$

$$\sigma_R(A \mathbin{\widehat{\otimes}} B) = \big\{\lambda \in \mathbb{C} \colon \mathcal{N}(\lambda I - A \mathbin{\widehat{\otimes}} B) = \{0\} \ \text{ and } \ \mathcal{R}(\lambda I - A \mathbin{\widehat{\otimes}} B)^- \neq \mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}\big\}.$$

As the point spectrum is a purely algebraic notion, we may speak of $\sigma_P(A \otimes B)$ for operators acting on $\mathcal{X} \otimes_\alpha \mathcal{Y}$, but we can speak of $\sigma(A \mathbin{\widehat{\otimes}} B)$ only for operators acting on $\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}$, as the spectrum is defined in a unital complex Banach algebra, thus defined only for operators on a Banach space (recall: in the finite-dimensional case, the spectrum coincides with the point spectrum).

Theorem 9.9. *Let $\mathcal{X}$ and $\mathcal{Y}$ be Banach spaces. Consider the tensor product space $\mathcal{X} \otimes_\alpha \mathcal{Y}$ equipped with a uniform crossnorm and its completion $\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}$. Take $A \in \mathcal{B}[\mathcal{X}]$, $B \in \mathcal{B}[\mathcal{Y}]$, $A \otimes B \in \mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}]$ and $A \mathbin{\widehat{\otimes}} B \in \mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}]$. Then*

(a) $\sigma_P(A \otimes B) \subseteq \sigma_P(A \mathbin{\widehat{\otimes}} B)$,

(b) $\sigma_P(A) \cdot \sigma_P(B) \subseteq \sigma_P(A \otimes B)$.

Moreover, the above inclusion may be proper and $0 \in \mathbb{C}$ plays a special role:

(c) *there exist A and B such that $\sigma_P(A \otimes B) \not\subseteq \sigma_P(A) \cdot \sigma_P(B)$;*

(d) *if $0 \in \sigma_P(A) \cup \sigma_P(B)$ and $\sigma_P(A)$ or $\sigma_P(B)$ is empty, then $0 \in \sigma_P(A \otimes B)$;*

(e) *if $0 \neq \nu \in \sigma_P(A \otimes B)$, and if ν has a single tensor among its eigenvectors, then $\nu \in \sigma_P(A) \cdot \sigma_P(B)$.*

Proof. (a) The inclusion in (a) comes from Remark 5.Y in Section 5.4.

(b) If $\lambda \in \sigma_P(A)$ and $\mu \in \sigma_P(B)$, then there exist $x \neq 0$ in $\mathcal{X}$ and $y \neq 0$ in $\mathcal{Y}$ such that $Ax = \lambda x$ and $By = \mu y$, and hence $(A \otimes B)(x \otimes y) = Ax \otimes By = \lambda x \otimes \mu y = \lambda \mu (x \otimes y)$ with $x \otimes y \neq 0$ in $\mathcal{X} \otimes \mathcal{Y}$ (cf. Proposition 3.C(a)). Thus $\lambda\mu \in \sigma_P(A \otimes B)$, and hence $\sigma_P(A) \cdot \sigma_P(B) \subseteq \sigma_P(A \otimes B)$.

(c) The reverse inclusion may fail. Let S be a unilateral shift (e.g., of multiplicity 1 on the Hilbert space ℓ^2_+) so that $\sigma_P(S)$ is empty and $\sigma_P(S^*)$ is the open unit disc (see, e.g., [52, Example 6.D]), and so $\sigma_P(S^*)\cdot\sigma_P(S)=\varnothing$. However, $\sigma_P(S^*\otimes S)=\{0\}$ (cf. [52, Example H]). Thus $\sigma_P(S^*\otimes S)\not\subseteq\sigma_P(S^*)\cdot\sigma_P(S)$.

(d) If $0\in\sigma_P(A)\cup\sigma_P(B)$, say $0\in\sigma_P(A)$, then there is $x\neq 0$ such that $Ax=0$. Therefore $(A\otimes B)(x\otimes y)=Ax\otimes By=0$ and $x\otimes y\neq 0$ for every $y\neq 0$, so that 0 is an eigenvalue of $A\otimes B$ (i.e., $0\in\sigma_P(A\otimes B)$), even if $\sigma_P(B)=\varnothing$.

(e) If there exists $0\neq x\otimes y\in\mathcal{N}(\nu I-A\otimes B)$ for some $\nu\neq 0$, then we get $0\neq Ax\otimes By=(A\otimes B)(x\otimes y)=\nu(x\otimes y)=(\nu x)\otimes y$. So Proposition 3.C(b) ensures the existence of a nonzero $\kappa\in\mathbb{F}$ for which $\kappa\nu x=Ax$ and $\kappa^{-1}y=By$. Hence $\kappa\nu\in\sigma_P(A)$ and $\kappa^{-1}\in\sigma_P(B)$, and therefore $\nu\in\sigma_P(A)\cdot\sigma_P(B)$. □

Corollary 9.10. *If $\mathcal{X}$ and $\mathcal{Y}$ are Hilbert spaces, then*

(a) $\sigma_R(A\mathbin{\widehat{\otimes}}B)=\sigma_P(A^*\mathbin{\widehat{\otimes}}B^*)^*\backslash\sigma_P(A\mathbin{\widehat{\otimes}}B)$,

(b) $\sigma_R(A)\cdot\sigma_R(B)\subseteq\big[\sigma_P\big((A\mathbin{\widehat{\otimes}}B)^*\big)\big]^*\subseteq\sigma(A\mathbin{\widehat{\otimes}}B)$,

(c) $\sigma_P(A^*)^*\cdot\sigma_P(B^*)^*\backslash\sigma_P(A\mathbin{\widehat{\otimes}}B)\subseteq\sigma_R(A\mathbin{\widehat{\otimes}}B)$.

Proof. (a) Apply Proposition 5.X and Theorem 9.8:

$$\sigma_R(A\mathbin{\widehat{\otimes}}B)=\sigma_P\big((A\mathbin{\widehat{\otimes}}B)^*\big)^*\backslash\sigma_P(A\mathbin{\widehat{\otimes}}B)=\sigma_P\big(A^*\mathbin{\widehat{\otimes}}B^*\big)^*\backslash\sigma_P(A\mathbin{\widehat{\otimes}}B).$$

(b) Again, applying Proposition 5.X (twice) and Theorems 9.9(a,b) and 9.8,

$$\begin{aligned}\sigma_R(A)\cdot\sigma_R(B)&=\sigma_P(A^*)^*\backslash\sigma_P(A)\cdot\sigma_P(B^*)^*\backslash\sigma_P(B)\subseteq\sigma_P(A^*)^*\cdot\sigma_P(B^*)^*\\&=\big[\sigma_P(A^*)\cdot\sigma_P(B^*)\big]^*\subseteq\big[\sigma_P(A^*\mathbin{\widehat{\otimes}}B^*)\big]^*=\big[\sigma_P\big((A\mathbin{\widehat{\otimes}}B)^*\big)\big]^*\\&\subseteq\big[\sigma\big((A\mathbin{\widehat{\otimes}}B)^*\big)\big]^*=\sigma(A\mathbin{\widehat{\otimes}}B).\end{aligned}$$

(c) As before, compute using Theorem 9.9(a,b) and item (a) above:

$$\begin{aligned}\sigma_P(A^*)^*\cdot\sigma_P(B^*)^*\backslash\sigma_P(A\mathbin{\widehat{\otimes}}B)&=\big[\sigma_P(A^*)\cdot\sigma_P(B^*)\big]^*\backslash\sigma_P(A\mathbin{\widehat{\otimes}}B)\\&\subseteq\sigma_P(A^*\mathbin{\widehat{\otimes}}B^*)^*\backslash\sigma_P(A\mathbin{\widehat{\otimes}}B).\end{aligned}$$ □

Theorem 9.9's central assertion is the inclusion,

$$\sigma_P(A)\cdot\sigma_P(B)\subseteq\sigma_P(A\otimes B),$$

which may be proper, and its item (e) shows a particular case where the reverse inclusion holds. Exploring further along this line, consider the subset $\sigma_{PF}(L)$ of the point spectrum $\sigma_P(L)$ for any $L\in\mathcal{L}[\mathcal{Z}]$ on any linear space $\mathcal{Z}$, consisting of all eigenvalues of finite multiplicity:

$$\sigma_{PF}(L)=\big\{\lambda\in\sigma_P(L)\colon\dim\mathcal{N}(\lambda I-L)<\infty\big\}=\big\{\lambda\in\mathbb{C}\colon 0<\dim\mathcal{N}(\lambda I-L)<\infty\big\}.$$

(This is meaningful only if $\dim\mathcal{Z}=\infty$; otherwise, $\sigma_{PF}(L)=\sigma_P(L)=\sigma(L)$.)

Corollary 9.11. *Consider the setup in Theorem 9.9 ($\mathcal{X}, \mathcal{Y}$ are Banach spaces).*

(a) $\sigma_{PF}(A \otimes B) \subseteq \sigma_{PF}(A \mathbin{\widehat{\otimes}} B)$.

(b) $0 \notin \sigma_{PF}(A \mathbin{\widehat{\otimes}} B)$ *whenever* $\dim \mathcal{X} = \infty$ *or* $\dim \mathcal{Y} = \infty$.

(c) $\big(\sigma_P(A) \cdot \sigma_P(B)\big) \cap \sigma_{PF}(A \mathbin{\widehat{\otimes}} B) \subseteq \sigma_{PF}(A) \cdot \sigma_{PF}(B)$.

Moreover,

(d) $\sigma_{PF}(A) \cdot \sigma_{PF}(B) \subseteq \sigma_{PF}(A \otimes B)$ *if and only if* $\big(\sigma_P(A) \cdot \sigma_P(B)\big) \cap \sigma_{PF}(A \otimes B) = \sigma_{PF}(A) \cdot \sigma_{PF}(B)$,

(e) $\big(\sigma_P(A) \backslash \sigma_{PF}(A) \cdot \sigma_P(B)\big) \cup \big(\sigma_P(A) \cdot \sigma_P(B) \backslash \sigma_{PF}(B)\big) \subseteq \sigma_P(A \otimes B) \backslash \sigma_{PF}(A \otimes B)$.

Proof. (a) This is straightforward by Theorem 9.9(a).

(b) If $0 \in \sigma_{PF}(A \mathbin{\widehat{\otimes}} B)$, then $0 \in \sigma_P(A \mathbin{\widehat{\otimes}} B)$. Thus $\mathcal{N}(A \mathbin{\widehat{\otimes}} B) \neq \{0\}$, and hence $\mathcal{N}(A) \neq \{0\}$ or $\mathcal{N}(B) \neq \{0\}$ by Theorem 9.3(h). Also, if $0 \in \sigma_{PF}(A \mathbin{\widehat{\otimes}} B)$, then $\dim \mathcal{N}(A \mathbin{\widehat{\otimes}} B) < \infty$ so that $\dim \mathcal{N}(A \otimes B) < \infty$. By Proposition 3.M(b), this implies that $\big\{\dim \mathcal{X} < \infty$ or $\mathcal{N}(A) = \{0\}\big\}$ and $\big\{\dim \mathcal{Y} < \infty$ or $\mathcal{N}(B) = \{0\}\big\}$. But as we saw above, $\dim \mathcal{N}(A) \neq \{0\}$ and $\dim \mathcal{N}(B) \neq \{0\}$. Therefore

$$0 \in \sigma_{PF}(A \mathbin{\widehat{\otimes}} B) \implies \dim \mathcal{Y} < \infty \text{ and } \dim \mathcal{X} < \infty.$$

(c) According to Proposition 3.M(c) we get,

$$\mathcal{N}(\lambda I - A) \otimes \mathcal{N}(\mu I - B) \subseteq \mathcal{N}(\lambda\mu I - A \otimes B).$$

If $0 < \dim \mathcal{N}(\lambda I - A)$, $0 < \dim \mathcal{N}(\mu I - B)$, and $\dim \mathcal{N}(\lambda\mu I - A \mathbin{\widehat{\otimes}} B) < \infty$, then all kernels above are nonzero and finite-dimensional (cf. Corollary 3.10). Thus if $\lambda \in \sigma_P(A)$, $\mu \in \sigma_P(B)$ (so that $\lambda\mu \in \sigma_P(A \mathbin{\widehat{\otimes}} B)$ by Theorem 9.9(a,b)), and $\lambda\mu \in \sigma_{PF}(A \mathbin{\widehat{\otimes}} B)$, then $\lambda \in \sigma_{PF}(A)$ and $\mu \in \sigma_{PF}(B)$, which proves the inclusion in (c): $\big(\sigma_P(A) \cdot \sigma_P(B)\big) \cap \sigma_{PF}(A \mathbin{\widehat{\otimes}} B) \subseteq \sigma_{PF}(A) \cdot \sigma_{PF}(B)$.

(d) This is almost tautological. Just for a while, simplify notation as follows:

$$\mathcal{A} = \sigma_P(A), \quad \mathcal{B} = \sigma_P(B), \quad \mathcal{A}' = \sigma_{PF}(A), \quad \mathcal{B}' = \sigma_{PF}(B), \quad \mathcal{C}' = \sigma_{PF}(A \otimes B).$$

Since $\mathcal{A}' \cdot \mathcal{B}' \subseteq \mathcal{A} \cdot \mathcal{B}$ trivially, (c) yields $(\mathcal{A} \cdot \mathcal{B}) \cap \mathcal{C}' \subseteq \mathcal{A}' \cdot \mathcal{B}' \subseteq \mathcal{A} \cdot \mathcal{B}$. Then

$$\mathcal{A}' \cdot \mathcal{B}' \subseteq \mathcal{C}' \implies (\mathcal{A} \cdot \mathcal{B}) \cap \mathcal{C}' \subseteq \mathcal{A}' \cdot \mathcal{B}' \subseteq (\mathcal{A} \cdot \mathcal{B}) \cap \mathcal{C}' \implies (\mathcal{A} \cdot \mathcal{B}) \cap \mathcal{C}' = \mathcal{A}' \cdot \mathcal{B}'.$$

The converse is trivial, viz., $(\mathcal{A} \cdot \mathcal{B}) \cap \mathcal{C}' = \mathcal{A}' \cdot \mathcal{B}' \implies \mathcal{A}' \cdot \mathcal{B}' \subseteq \mathcal{C}'$. Conclusion: $\mathcal{A}' \cdot \mathcal{B}' \subseteq \mathcal{C}' \iff (\mathcal{A} \cdot \mathcal{B}) \cap \mathcal{C}' = \mathcal{A}' \cdot \mathcal{B}'$, proving (d).

(e) The inclusion in (e) also follows by Proposition 3.M(c) and Corollary 3.10. Indeed, $\lambda \in \sigma_P(A) \backslash \sigma_{PF}(A)$ if and only if $\dim \mathcal{N}(\lambda I - A) = \infty$, and $\mu \in \sigma_P(B)$ if and only if $0 < \dim \mathcal{N}(\mu I - B)$. Therefore by Corollary 3.10 and Proposition 3.M(c), $\dim \mathcal{N}(\lambda\mu I - A \otimes B) = \infty$, that is, $\lambda\mu \in \sigma_P(A \otimes B) \backslash \sigma_{PF}(A \otimes B)$. Similarly, if $\mu \in \sigma_P(B) \backslash \sigma_{PF}(B)$ and $\lambda \in \sigma_P(A)$, then $\lambda\mu \in \sigma_P(A \otimes B) \backslash \sigma_{PF}(A \otimes B)$.

To verify that the inclusions may all be proper, let one of A or B (or both) have zero as an eigenvalue of finite multiplicity, and apply item (b). □

Let $\mathcal{X}$ and $\mathcal{Y}$ be (complex nonzero) Hilbert spaces, and let $\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$ be the completion of the normed space $\mathcal{X}\otimes_\alpha\mathcal{Y}$ with respect to any uniform crossnorm $\|\cdot\|_\alpha$ (as agreed before, same notation for norm on the normed space $\mathcal{X}\otimes_\alpha\mathcal{Y}$ and on the Banach space $\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}$). In particular, we can take the completion $\mathcal{X}\,\widehat{\otimes}_\eta\mathcal{Y}$ of the inner product space $\mathcal{X}\otimes_\eta\mathcal{Y}$ with the inner product $\langle\cdot\,;\cdot\rangle_\eta$ inducing the uniform crossnorm $\|\cdot\|_\eta$, which is a Hilbert space. A central spectral result is the following *spectral formula* (compare with Proposition 4.I(a)).

Theorem 9.12. *If $\mathcal{X}$ and $\mathcal{Y}$ are complex Hilbert spaces, then*

$$\sigma(A\,\widehat{\otimes}\,B)=\sigma(A)\cdot\sigma(B), \tag{SF}$$

when $\mathcal{X}\otimes\mathcal{Y}$ is equipped with any uniform crossnorm $\|\cdot\|_\alpha$; in particular, if it is equipped with the tensor norm $\|\cdot\|_\eta$ induced by the inner product $\langle\cdot\,;\cdot\rangle_\eta$.

Proof. To begin with, consider the following auxiliary result.

Claim 1. For $A\in\mathcal{B}[\mathcal{X}]$ and $B\in\mathcal{B}[\mathcal{Y}]$ on normed spaces $\mathcal{X}$ and $\mathcal{Y}$ and $\lambda,\mu\in\mathbb{F}$,

$$\begin{aligned}\text{(a)}\quad \lambda\mu I-A\,\widehat{\otimes}\,B&=\big((\lambda I-A)\,\widehat{\otimes}\,B\big)+\big(\lambda I\,\widehat{\otimes}\,(\mu I-B)\big)\\&=\big(A\,\widehat{\otimes}\,(\mu I-B)\big)+\big((\lambda I-A)\,\widehat{\otimes}\,\mu I\big).\end{aligned}$$

Consequently, for every $x\in\mathcal{X}$ and $y\in\mathcal{Y}$,

$$\begin{aligned}\text{(b)}\quad (\lambda\mu I-A\,\widehat{\otimes}\,B)(x\,\widehat{\otimes}\,y)&\cong\big((\lambda I-A)x\otimes By\big)+\big(\lambda x\otimes(\mu I-B)y\big)\\&=\big(Ax\otimes(\mu I-B)y\big)+\big((\lambda I-A)\otimes\mu y\big).\end{aligned}$$

Proof of Claim 1. Theorem 9.3(a,b) ensures item (a):

$$\begin{aligned}(\lambda\mu I-A\,\widehat{\otimes}\,B)&=\lambda I\,\widehat{\otimes}\,B-A\,\widehat{\otimes}\,B+\lambda I\,\widehat{\otimes}\,\mu I-\lambda I\,\widehat{\otimes}\,B\\&=\big((\lambda I-A)\,\widehat{\otimes}\,B\big)+\big(\lambda I\,\widehat{\otimes}\,(\mu I-B)\big).\end{aligned}$$

Symmetrically,

$$\begin{aligned}(\lambda\mu I-A\,\widehat{\otimes}\,B)&=A\,\widehat{\otimes}\,\mu I-A\,\widehat{\otimes}\,B+\lambda I\,\widehat{\otimes}\,\mu I-A\,\widehat{\otimes}\,\mu I\\&=\big(A\,\widehat{\otimes}\,(\mu I-B)\big)+\big((\lambda I-A)\,\widehat{\otimes}\,\mu I\big).\end{aligned}$$

Item (b) is a consequence of (a), Remark 8.J, and Definition 3.16. □

From now on suppose $\mathcal{X}$ and $\mathcal{Y}$ are Banach spaces. Recall from Section 5.4 that $\lambda\in\rho(A)$ if and only if there exists $(\lambda I-A)^{-1}$ in $\mathcal{B}[\mathcal{X}]$. Theorem 9.3(d) ensures that if there exists $(\lambda I-A)^{-1}$ in $\mathcal{B}[\mathcal{X}]$, then there exists $((\lambda I-A)\,\widehat{\otimes}\,I)^{-1}$ in $\mathcal{B}[\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}]$. By setting $B=I$ and $\mu=1$ in Claim 1, there is $((\lambda I-A)\,\widehat{\otimes}\,I)^{-1}$ in $\mathcal{B}[\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}]$ if and only if there is $(\lambda I-A\,\widehat{\otimes}\,I)^{-1}$ in $\mathcal{B}[\mathcal{X}\,\widehat{\otimes}_\alpha\mathcal{Y}]$. This means $\lambda\in\rho(A\,\widehat{\otimes}\,I)$. Outcome: $\rho(A)\subseteq\rho(A\,\widehat{\otimes}\,I)$. Equivalently, $\sigma(A\,\widehat{\otimes}\,I)\subseteq\sigma(A)$. Moreover, since $(A\,\widehat{\otimes}\,I)\cong(I\,\widehat{\otimes}\,A)$ (Theorem 9.3(i)), $\sigma(A\,\widehat{\otimes}\,I)=\sigma(I\,\widehat{\otimes}\,A)$ (spectra are preserved under isometric isomorphisms). So $\sigma(I\,\widehat{\otimes}\,A)=\sigma(A\,\widehat{\otimes}\,I)\subseteq\sigma(A)$. Recall from Theorem 9.3(c) that $(A\,\widehat{\otimes}\,B)=(A\,\widehat{\otimes}\,I)\,(I\,\widehat{\otimes}\,B)=(I\,\widehat{\otimes}\,B)\,(A\,\widehat{\otimes}\,I)$. Also recall that the spectrum of the product of commuting operators is included in the product of their spectra. (This property actually holds in every unital

complex Banach algebra: *if $\mathcal{A}$ is a unital complex Banach algebra, and if S, T in $\mathcal{A}$ commute, then $\sigma(S+T) \subseteq \sigma(S) + \sigma(T)$ and $\sigma(ST) \subseteq \sigma(S) \cdot \sigma(T)$* — see, e.g., [82, Theorem 11.22].) Therefore

$$\sigma(A \mathbin{\widehat{\otimes}} B) = \sigma\big((A \mathbin{\widehat{\otimes}} I)\,(I \mathbin{\widehat{\otimes}} B)\big) \subseteq \sigma(A \mathbin{\widehat{\otimes}} I) \cdot \sigma(I \mathbin{\widehat{\otimes}} B) \subseteq \sigma(A) \cdot \sigma(B).$$

For the reverse inclusion, consider the classical partition of the spectrum as in Section 5.4: for an arbitrary $T \in \mathcal{B}[\mathcal{X}]$ on a complex nonzero Banach space $\mathcal{X}$, $\sigma(T) = \{\sigma_P(T), \sigma_C(T), \sigma_R(T)\}$. Refine the spectrum partition by considering further partitions of the point spectrum and the residual spectrum, as follows: $\sigma_P(T) = \bigcup_{i=1}^4 \sigma_{P_i}(T)$ and $\sigma_R(T) = \sigma_{R_1}(T) \cup \sigma_{R_2}(T)$, where

$$\begin{aligned}
\sigma_{P_1}(T) &= \{\lambda \in \sigma_P(T) \colon \mathcal{R}(\lambda I - T) = \mathcal{R}(\lambda I - T)^- = \mathcal{X}\},\\
\sigma_{P_2}(T) &= \{\lambda \in \sigma_P(T) \colon \mathcal{R}(\lambda I - T) \neq \mathcal{R}(\lambda I - T)^- = \mathcal{X}\},\\
\sigma_{P_3}(T) &= \{\lambda \in \sigma_P(T) \colon \mathcal{R}(\lambda I - T) \neq \mathcal{R}(\lambda I - T)^- \neq \mathcal{X}\},\\
\sigma_{P_4}(T) &= \{\lambda \in \sigma_P(T) \colon \mathcal{R}(\lambda I - T) = \mathcal{R}(\lambda I - T)^- \neq \mathcal{X}\},\\
\sigma_{R_1}(T) &= \{\lambda \in \sigma_R(T) \colon \mathcal{R}(\lambda I - T) = \mathcal{R}(\lambda I - T)^-\},\\
\sigma_{R_2}(T) &= \{\lambda \in \sigma_R(T) \colon \mathcal{R}(\lambda I - T) \neq \mathcal{R}(\lambda I - T)^-\}.
\end{aligned}$$

Consider the *approximate point spectrum*,

$$\sigma_{AP}(T) = \sigma(T) \backslash \sigma_{R_1}(T),$$

where $\sigma_{R_1}(T)$ is an open subset of the compact set $\sigma(T)$, thus open in $\mathbb{C}$ (since $\sigma_{AP}(T)$ is a nonempty closed subset of $\mathbb{C}$). If $\mathcal{X}$ is a Hilbert space, then $\sigma_{R_1}(T) = \sigma_{P_1}(T^*)^*$ with respect to the Hilbert-space adjoint (see, e.g., [56, Fig. §2.2, Theorems 2.5 and 2.6]). Thus suppose $\mathcal{X}$ and $\mathcal{Y}$ are Hilbert spaces, and equip the tensor product space $\mathcal{X} \otimes \mathcal{Y}$ with any uniform crossnorm $\|\cdot\|_\alpha$.

Claim 2. (a) $\sigma_{AP}(A) \cdot \sigma_{AP}(B) \subseteq \sigma_{AP}(A \mathbin{\widehat{\otimes}} B)$.
(b) $0 \in \sigma_{AP}(A) \cup \sigma_{AP}(B) \implies 0 \in \sigma_{AP}(A \mathbin{\widehat{\otimes}} B)$.

Proof of Claim 2. (a) Take $\lambda \in \sigma_{AP}(A)$ and $\mu \in \sigma_{AP}(B)$. That is, there are sequences $\{x_k\}$ and $\{y_k\}$ of unit vectors in $\mathcal{X}$ and $\mathcal{Y}$ such that $\|(\lambda I - A)x_k\| \to 0$ and $\|(\mu I - B)y_k\| \to 0$ (see, e.g., [56, Theorem 2.4]). By Claim 1,

$$\|(\lambda\mu\, I - A \mathbin{\widehat{\otimes}} B)(x_k \mathbin{\widehat{\otimes}} y_k)\|_\alpha \leq \|(\lambda I - A)x_k\|\,\|B\| + |\lambda|\,\|(\mu I - B)y_k\| \to 0.$$

Since $\|x_k \mathbin{\widehat{\otimes}} y_k\|_\alpha = \|x_k\|\,\|y_k\| = 1$ (cf. Remark 8.J), we get $\lambda\mu \in \sigma_{AP}(A \mathbin{\widehat{\otimes}} B)$.

(b) If $\lambda = 0 \in \sigma_{AP}(A)$, then $\|Ax_k\| \to 0$ for some sequence $\{x_k\}$ of unit vectors in $\mathcal{X}$, and the above inequality with $\lambda = 0$ shows that $\|(A \mathbin{\widehat{\otimes}} B)(x_k \mathbin{\widehat{\otimes}} y_k)\|_\alpha \to 0$ for every sequence $\{y_k\}$ of unit vectors in $\mathcal{Y}$. Thus $0 \in \sigma_{AP}(A \mathbin{\widehat{\otimes}} B)$. A symmetric argument shows that if $\mu = 0 \in \sigma_{AP}(B)$, then $0 \in \sigma_{AP}(A \mathbin{\widehat{\otimes}} B)$. □

Claim 3. $\sigma_{R_1}(A) \cdot \sigma_{R_1}(B) \subseteq \sigma(A \mathbin{\widehat{\otimes}} B)$.

Proof of Claim 3. This is a particular case of Corollary 9.10(b). □

Claim 4. $\big(\sigma_{AP}(A)\cdot\sigma_{R_1}(B)\big)\cup\big(\sigma_{R_1}(A)\cdot\sigma_{AP}(B)\big)\subseteq\sigma(A\,\widehat{\otimes}\,B)$.

Proof of Claim 4. We split the proof of Claim 4 into three parts, and will use Proposition 5.X and Theorem 9.8 freely. Take

$$\lambda\in\sigma_{AP}(A)\quad\text{ and }\quad\mu\in\sigma_{R_1}(B).$$

Part I. If $\lambda=0$, then $\lambda\mu\in\sigma_{AP}(A\,\widehat{\otimes}\,B)\subseteq\sigma(A\,\widehat{\otimes}\,B)$ by Claim 2(b). Since $\sigma_{R_1}(B)=\sigma_{P_1}(B^*)^*\subseteq\sigma_{AP}(B^*)^*$, if $\mu=0$, then $\lambda\mu=0\in\sigma_{AP}(B^*)$, and hence $\lambda\mu=0\in\sigma_{AP}(A^*\,\widehat{\otimes}\,B^*)$ by Claim 2(b), so that $\lambda\mu=0\in\sigma_{AP}(A^*\,\widehat{\otimes}\,B^*)^*\subseteq\sigma(A^*\,\widehat{\otimes}\,B^*)^*=\sigma\big((A\,\widehat{\otimes}\,B)^*\big)^*=\sigma(A\,\widehat{\otimes}\,B)$. Thus suppose

$$\lambda\neq0\quad\text{ and }\quad\mu\neq0.$$

Part II. If $\lambda\in\sigma_{AP}(A^*)^*$, then $\overline{\lambda}\in\sigma_{AP}(A^*)$. Since $\mu\in\sigma_{R_1}(B)$, it follows that $\overline{\mu}\in\sigma_{R_1}(B)^*=\sigma_{P_1}(B^*)$. So $\overline{\mu}\in\sigma_{AP}(B^*)$ because $\sigma_{AP}(B^*)=\sigma(B^*)\backslash\sigma_{R_1}(B^*)$. Then by Claim 2(a), $\overline{\lambda\mu}\in\sigma_{AP}(A^*)\cdot\sigma_{AP}(B^*)\subseteq\sigma_{AP}(A^*\,\widehat{\otimes}\,B^*)\subseteq\sigma(A^*\,\widehat{\otimes}\,B^*)=\sigma\big((A\,\widehat{\otimes}\,B)^*\big)=\sigma(A\,\widehat{\otimes}\,B)^*$ so that $\lambda\mu\in\sigma(A\,\widehat{\otimes}\,B)$. Thus suppose $\lambda\notin\sigma_{AP}(A^*)^*$; equivalently, $\overline{\lambda}\notin\sigma_{AP}(A^*)=\sigma(A^*)\backslash\sigma_{R_1}(A^*)$. Since $\lambda\in\sigma_{AP}(A)\subseteq\sigma(A)$, then $\overline{\lambda}\in\sigma(A)^*=\sigma(A^*)$, and so $\overline{\lambda}\in\sigma_{R_1}(A^*)$. That is, suppose

$$\lambda\in\sigma_{R_1}(A^*)^*=\sigma_{P_1}(A).$$

Part III. Since $\overline{\lambda}\in\sigma_{R_1}(A^*)$ and $\mu\in\sigma_{R_1}(B)$, and since $\sigma_{R_1}(\,\cdot\,)$ is an open subset of the partition $\sigma(\,\cdot\,)=\sigma_{AP}(\,\cdot\,)\cup\sigma_{R_1}(\,\cdot\,)$, there exists a real number $\tau_0\geq1$ close enough to 1 such that

$$\tau_0\overline{\lambda}\in\sigma_{R_1}(A^*)\quad\text{ and }\quad\tfrac{\mu}{\tau_0}\in\sigma_{R_1}(B).$$

Thus there exists a real number $\kappa\geq\tau_0$ such that

$$\kappa\overline{\lambda}\in\sigma_{R_1}(A^*)^-\subseteq\sigma_{AP}(A^*)\quad\text{ or }\quad\tfrac{\mu}{\kappa}\in\sigma_{R_1}(B)^-\subseteq\sigma_{AP}(B),$$

and

$$\tau\overline{\lambda}\in\sigma_{R_1}(A^*)\quad\text{ and }\quad\tfrac{\mu}{\tau}\in\sigma_{R_1}(B).\quad\text{ for every }\quad\tau\in[1,\kappa).$$

(i) First suppose $\kappa\overline{\lambda}\in\sigma_{AP}(A^*)$ and $\frac{\mu}{\kappa}\notin\sigma_{AP}(B)$, thus $\frac{\mu}{\kappa}\in\sigma_{R_1}(B)$ and hence $\frac{\overline{\mu}}{\kappa}\in\sigma_{R_1}(B)^*=\sigma_{P_1}(B^*)\subseteq\sigma_{AP}(B^*)$. Then by Claim 2(a) $\overline{\lambda\mu}=\kappa\overline{\lambda}\frac{\overline{\mu}}{\kappa}$ lies in $\sigma_{AP}(A^*\,\widehat{\otimes}\,B^*)\subseteq\sigma(A^*\,\widehat{\otimes}\,B^*)=\sigma\big((A\,\widehat{\otimes}\,B)^*\big)=\sigma(A\,\widehat{\otimes}\,B)^*$. So $\lambda\mu\in\sigma(A\,\widehat{\otimes}\,B)$.

(ii) Symmetrically, if $\frac{\mu}{\kappa}\in\sigma_{AP}(B)$ and $\kappa\overline{\lambda}\notin\sigma_{AP}(A^*)$, so that $\kappa\overline{\lambda}\in\sigma_{R_1}(A^*)=\sigma_{P_1}(A)^*\subseteq\sigma_{PAP}(A)^*$, then $\overline{\lambda\mu}\in\sigma(A\,\widehat{\otimes}\,B)^*$, and so $\lambda\mu\in\sigma(A\,\widehat{\otimes}\,B)$.

(iii) Finally, suppose $\kappa\overline{\lambda}\in\sigma_{AP}(A^*)$ and $\frac{\mu}{\kappa}\in\sigma_{AP}(B)$. Let $\{\tau_n\}$ be a real sequence such that each $\tau_n\in(1,\kappa)$ and $\tau_n\to\kappa$. Thus each τ_n is such that $\tau_n\overline{\lambda}\in\sigma_{R_1}(A^*)$, and hence $\tau_n\lambda\in\sigma_{R_1}(A^*)^*=\sigma_{P_1}(A)\subseteq\sigma_{AP}(A)$. Therefore, since $\frac{\mu}{\kappa}\in\sigma_{AP}(B)$, Claim 2(a) says that $\frac{\tau_n}{\kappa}\lambda\mu\in\sigma_{AP}(A\,\widehat{\otimes}\,B)$. Since $\frac{\tau_n}{\kappa}\to1$ and since $\sigma_{AP}(\,\cdot\,)$ is closed, we get $\lambda\mu\in\sigma_{AP}(A\,\widehat{\otimes}\,B)\subseteq\sigma(A\,\widehat{\otimes}\,B)$.

Then, by Parts I, II, III,

$$\lambda\in\sigma_{AP}(A)\quad\text{ and }\quad\mu\in\sigma_{R_1}(B)\quad\Longrightarrow\quad\lambda\mu\in\sigma(A\,\widehat{\otimes}\,B).$$

Symmetrically (since $A \mathbin{\widehat{\otimes}} B \cong B \mathbin{\widehat{\otimes}} A$ and $\sigma(\cdot)$ is invariant under $\cong$),

$$\lambda \in \sigma_{R_1}(A) \quad \text{and} \quad \mu \in \sigma_{AP}(B) \quad \Longrightarrow \quad \lambda\mu \in \sigma(A \mathbin{\widehat{\otimes}} B). \qquad \square$$

By Claims 2,3,4, $\lambda \in \sigma(A)$ and $\mu \in \sigma(B)$ implies $\lambda\mu \in \sigma(A \mathbin{\widehat{\otimes}} B)$. Hence

$$\sigma(A) \cdot \sigma(B) \subseteq \sigma(A \mathbin{\widehat{\otimes}} B),$$

concluding the proof of the theorem. $\square$

Theorem 9.12 is a classical result due to Brown and Pearcy [4] (and the above proof follows the argument in [4]). The spectral formula (SF) has been generalized, including extensions to Banach spaces by, for instance, Schechter [86], Ichinose [38], Harte [33], Reed and Simon [79], Embry and Rosenblum [24]. Thus the spectral formula holds in general for operators A in $\mathcal{B}[\mathcal{X}]$ and B in $\mathcal{B}[\mathcal{Y}]$ acting on (complex nonzero) Banach spaces $\mathcal{X}$ and $\mathcal{Y}$, and with $A \mathbin{\widehat{\otimes}} B$ in $\mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}}_\alpha \mathcal{Y}]$ acting on $\mathcal{X} \mathbin{\widehat{\otimes}}_\alpha \mathcal{Y}$ with respect to any uniform crossnorm $\|\cdot\|_\alpha$. So we will use the spectral formula (SF) for Banach-space operators as well.

Remark 9.13. *Hilbert–Schmidt*. Recall from Section 7.4 (between Propositions 7.R an 7.S) the notion of trace-class operators $T \in \mathcal{B}_1[\mathcal{X}]$ acting on a Hilbert space $\mathcal{X}$. If $|T|^2 \in \mathcal{B}_1[\mathcal{X}]$, then T is said to be a *Hilbert–Schmidt operator*, and $\mathcal{B}_2[\mathcal{X}]$ denotes the class of all Hilbert–Schmidt operators on $\mathcal{X}$. These classes are operator ideals such that $\mathcal{B}_0[\mathcal{X}] \subseteq \mathcal{B}_1[\mathcal{X}] \subseteq \mathcal{B}_2[\mathcal{X}] \subseteq \mathcal{B}_\infty[\mathcal{X}]$, and $\|T\|_2 = \||T|^2\|_1^{1/2}$ is the *Hilbert–Schmidt norm* of $T \in \mathcal{B}_2[\mathcal{X}]$, which is such that $\|T\| \le \|T\|_2 \le \|T\|_1$ if $T \in \mathcal{B}_1[\mathcal{X}]$, where $\mathcal{B}_1[\mathcal{X}]$ equipped with $\|\cdot\|_1$ is a Banach space, and $\mathcal{B}_2[\mathcal{X}]$ equipped with $\|\cdot\|_2$ is a Hilbert space (see, e.g., [52, Problems 5.64–5.69] or [58]). On the other hand, we have seen above (before Theorem 9.8) that $\mathcal{X} \mathbin{\widehat{\otimes}}_\eta \mathcal{Y}$ is referred to as the Hilbert–Schmidt tensor product, and $\|\cdot\|_\eta$ as the Hilbert–Schmidt norm. Indeed, $\mathcal{X} \mathbin{\widehat{\otimes}}_\eta \mathcal{X}$ is isometrically embedded in $\mathcal{B}_2[\mathcal{X}]$ (see, e.g., [15, Section 26.6]), which justifies the terminology. As a matter of fact, there are two proofs of the spectral formula (SF) in [4]; one of them explores such an identification (which implies that the tensor product on $\mathcal{X} \mathbin{\widehat{\otimes}}_\eta \mathcal{X}$ of two operators on the Hilbert space $\mathcal{X}$ can be identified with multiplication on the Hilbert–Schmidt class $\mathcal{B}_2[\mathcal{X}]$).

The spectral formula (SF) in Theorem 9.12 offers one of the proofs for the following *spectral radius identity*.

Corollary 9.14. *For every $A \in \mathcal{B}[\mathcal{X}]$ and $B \in \mathcal{B}[\mathcal{Y}]$ on Banach spaces $\mathcal{X}$ and $\mathcal{Y}$,*

$$r(A \mathbin{\widehat{\otimes}} B) = r(A)\, r(B),$$

whenever $\mathcal{X} \otimes \mathcal{Y}$ is equipped with a uniform crossnorm $\|\cdot\|_\alpha$.

First Proof. From Theorem 9.3(c) we get by induction $(A \mathbin{\widehat{\otimes}} B)^n = A^n \mathbin{\widehat{\otimes}} B^n$ for every integer $n \ge 1$. So $\|(A \mathbin{\widehat{\otimes}} B)^n\|^{\frac{1}{n}} = \|A^n \mathbin{\widehat{\otimes}} B^n\|^{\frac{1}{n}} = \|A^n\|^{\frac{1}{n}} \|B^n\|^{\frac{1}{n}}$ by Theorem 8.3(a). Hence $\lim_n \|(A \mathbin{\widehat{\otimes}} B)^n\|^{\frac{1}{n}} = \lim_n \|A^n\|^{\frac{1}{n}} \lim_n \|B^n\|^{\frac{1}{n}}$, and we get the spectral radius identity by the Gelfand–Beurling formula (Proposition 5.N).

Second Proof. The spectral formula (SF) of Theorem 9.12 and the definition of spectral radius (cf. Section 5.4) ensures that $r(A \mathbin{\widehat{\otimes}} B) = \sup_{\nu \in \sigma(A \mathbin{\hat{\otimes}} B)} |\nu| = \sup_{\lambda \in \sigma(A), \mu \in \sigma(B)} |\lambda\, \mu| = \sup_{\lambda \in \sigma(A)} |\lambda| \sup_{\mu \in \sigma(B)} |\mu| = r(A)\, r(B)$. □

Let $\sigma_{\rm iso}(T)$ denote the set of all isolated points of the spectrum $\sigma(T)$ of an operator $T \in \mathcal{B}[\mathcal{X}]$ on a Banach space $\mathcal{X}$. Thus set $\sigma_{\rm acc}(T) = \sigma(T) \backslash \sigma_{\rm iso}(T)$, the set of all accumulation points of $\sigma(T)$.

Corollary 9.15. *Consider again the setup of Theorem 9.9, where $A \in \mathcal{B}[\mathcal{X}]$ and $B \in \mathcal{B}[\mathcal{Y}]$ are operators acting on Banach spaces $\mathcal{X}$ and $\mathcal{Y}$.*

(a) $\sigma_{\rm iso}(A) \neq \varnothing$ *and* $\sigma_{\rm iso}(B) \neq \varnothing \quad \Longrightarrow \quad \sigma_{\rm iso}(A \mathbin{\widehat{\otimes}} B) \subsetneqq \sigma_{\rm iso}(A) \cdot \sigma_{\rm iso}(B)$.

(b) $\sigma_{\rm iso}(A) = \varnothing$ *or* $\sigma_{\rm iso}(B) = \varnothing \quad \Longrightarrow \quad \sigma_{\rm iso}(A \mathbin{\widehat{\otimes}} B) \subsetneqq \{0\}$.

(c) $\sigma_{\rm iso}(A \mathbin{\widehat{\otimes}} B) = \{0\} \quad \Longrightarrow \quad 0 \in \sigma_{\rm iso}(A) \cup \sigma_{\rm iso}(B)$.

(d) $\sigma_{\rm iso}(A) = \sigma_{\rm iso}(B) = \varnothing \quad \Longrightarrow \quad \sigma_{\rm iso}(A \mathbin{\widehat{\otimes}} B) = \varnothing$.

Proof. If $\sigma_{\rm iso}(A \mathbin{\widehat{\otimes}} B) = \varnothing$, then the inclusions in (a) and (b) hold trivially. Thus for proving (a) and (b) suppose $\sigma_{\rm iso}(A \mathbin{\widehat{\otimes}} B) \neq \varnothing$. If $\sigma(A) = \{0\}$ or $\sigma(B) = \{0\}$, then (using (SF)) the results in (a) and (b) also hold trivially (the inclusions become the identity $\sigma_{\rm iso}(A \mathbin{\widehat{\otimes}} B) = \{0\}$). Thus, again, for proving (a) and (b) suppose $\sigma(A) \neq \{0\}$ and $\sigma(B) \neq \{0\}$. If $\nu \in \sigma_{\rm iso}(A \mathbin{\widehat{\otimes}} B)$, then $\nu = \lambda\mu$ is an isolated point of $\sigma(A \mathbin{\widehat{\otimes}} B) = \sigma(A) \cdot \sigma(B)$ with $\lambda \in \sigma(A)$ and $\mu \in \sigma(B)$. So the inclusion in (a) is equivalently stated as

$$\nu \in \sigma_{\rm iso}(A \mathbin{\widehat{\otimes}} B) \quad \Longrightarrow \quad \nu = \lambda\mu \ \text{ for some } \ \lambda \in \sigma_{\rm iso}(A) \ \text{ and } \ \mu \in \sigma_{\rm iso}(B).$$

(a) Then assume $\sigma_{\rm iso}(A) \neq \varnothing$ and $\sigma_{\rm iso}(B) \neq \varnothing$, and take an arbitrary $\nu = \lambda\mu \in \sigma_{\rm iso}(A \mathbin{\widehat{\otimes}} B) \neq \varnothing$. We split the proof of (a) into three parts.

($\rm a_1$) Suppose $\lambda\mu \neq 0$ so that $0 \neq \lambda \in \sigma(A)$ and $0 \neq \mu \in \sigma(B)$. If one of them, say $\lambda \in \sigma(A)$, is not an isolated point of $\sigma(A)$, then $0 \neq \lambda \in \sigma_{\rm acc}(A)$. So there is a sequence $\{\lambda_n\}$ of distinct points in $\sigma(A) \backslash \{\lambda\}$ such that $\lambda_n \to \lambda$. Since $\mu \neq 0$, $\{\lambda_n \mu\}$ is a sequence of distinct points in $\sigma(A) \cdot \sigma(B) \backslash \{\lambda\mu\}$ such that $\lambda_n \mu \to \lambda\mu$, and so $\lambda\mu$ is an accumulation point of $\sigma(A) \cdot \sigma(B) = \sigma(A \mathbin{\widehat{\otimes}} B)$; that is, $\lambda\mu \in \sigma_{\rm acc}(A \mathbin{\widehat{\otimes}} B)$, which is a contradiction.

($\rm a_2$) Suppose $\lambda\mu = 0$ so that one of λ or μ is zero, say $\lambda = 0$. If $0 = \lambda \in \sigma_{\rm acc}(A)$, then there exists a sequence $\{\lambda_n\}$ of distinct nonzero points in $\sigma(A) \backslash \{0\}$ such that $\lambda_n \to \lambda = 0$. If $\mu \neq 0$, then $\{\lambda_n \mu\}$ is a sequence of distinct nonzero points in $\sigma(A) \cdot \sigma(B) \backslash \{0\}$ such that $\lambda_n \mu \to \lambda\mu = 0$, and so $\lambda\mu = 0$ is an accumulation point of $\sigma(A) \cdot \sigma(B) = \sigma(A \mathbin{\widehat{\otimes}} B)$, which is again a contradiction. If μ is also zero, then (as $\sigma(B) \neq \{0\}$ by assumption) take any nonzero $\mu' \in \sigma(B)$ so that $\lambda\mu = \lambda\mu' = 0$ with $\lambda = 0$ and $\mu' \neq 0$, thus reducing to the above case.

So in any case (either $\lambda\mu \neq 0$ or $\lambda\mu = 0$), if $\nu \in \sigma_{\rm iso}(A \mathbin{\widehat{\otimes}} B)$, then $\nu = \lambda\mu$ for some $\lambda \in \sigma_{\rm iso}(A)$ and some $\mu \in \sigma_{\rm iso}(B)$. This proves the inclusion in (a): $\sigma_{\rm iso}(A \mathbin{\widehat{\otimes}} B) \subseteq \sigma_{\rm iso}(A) \cdot \sigma_{\rm iso}(B)$.

(a_3) Such an inclusion may be proper. For instance, if $\sigma(A) = \{1\} \cup [2,3] \cup \{4\}$ and $\sigma(B) = \{1\} \cup \{2\}$, then $\sigma(A \mathbin{\widehat{\otimes}} B) = \sigma(A) \cdot \sigma(B)$ is such that

$$\sigma(A) \cdot \sigma(B) = \{1\} \cup [2,3] \cup \{4\} \cup \{2\} \cup [4,6] \cup \{8\} = \{1\} \cup [2,3] \cup [4,6] \cup \{8\},$$

and therefore $\{1,2,4,8\} = \sigma_{\text{iso}}(A) \cdot \sigma_{\text{iso}}(B) \not\subseteq \sigma_{\text{iso}}(A \mathbin{\widehat{\otimes}} B) = \{1,8\}$.

(b) Suppose one of the operators, say A, is such that $\sigma_{\text{iso}}(A) = \varnothing$. If there exists $0 \neq \nu = \lambda\mu \in \sigma_{\text{iso}}(A \mathbin{\widehat{\otimes}} B)$, then $0 \neq \lambda \in \sigma_{\text{acc}}(A)$ and $0 \neq \mu \in \sigma(B)$. Thus the same argument as in (a_1) leads to the conclusion that $\nu = \lambda\mu$ is an accumulation point of $\sigma(A) \cdot \sigma(B)$, which is a contradiction. Hence $\sigma_{\text{iso}}(A \mathbin{\widehat{\otimes}} B) \subseteq \{0\}$. This proves the inclusion in (b), which may be a proper inclusion. For instance, if $\sigma_{\text{iso}}(A) = \varnothing$, $0 \in \sigma_{\text{acc}}(A)$ and $\sigma(B) = \sigma_{\text{iso}}(B) = \{0,1\}$, then $\sigma(A) \cdot \sigma(B) = \sigma_{\text{acc}}(A)$ so that $\sigma_{\text{iso}}(A \mathbin{\widehat{\otimes}} B) = \varnothing$ (i.e., $\sigma(A \mathbin{\widehat{\otimes}} B) = \sigma_{\text{acc}}(A) = \sigma_{\text{acc}}(A \mathbin{\widehat{\otimes}} B)$).

(c) If $\sigma_{\text{iso}}(A \mathbin{\widehat{\otimes}} B) = \{0\}$, then $0 \in \sigma(A) \cdot \sigma(B)$, and hence 0 lies in the spectrum of one of A or B, say, $0 \in \sigma(A)$. If $\sigma_{\text{iso}}(A) = \varnothing$, then $0 \in \sigma_{\text{acc}}(A)$. So the same argument as in (a_2) leads to a contradiction, and therefore $0 \in \sigma_{\text{iso}}(A)$.

(d) If $\sigma(A) = \sigma_{\text{acc}}(A)$ and $\sigma(B) = \sigma_{\text{acc}}(B)$, then it is readily verified (applying the same argument as above) that $\sigma(A \mathbin{\widehat{\otimes}} B) = \sigma_{\text{acc}}(A \mathbin{\widehat{\otimes}} B)$. □

An operator $T \in \mathcal{B}[\mathcal{X}]$ on a Banach space $\mathcal{X}$ is called *isoloid* if the isolated points of its spectrum are eigenvalues. That is, if

$$\sigma_{\text{iso}}(T) \subseteq \sigma_P(T).$$

Corollary 9.16. *Let A and B be operators on Banach spaces $\mathcal{X}$ and $\mathcal{Y}$, and take the tensor product $A \mathbin{\widehat{\otimes}} B$ on $\mathcal{X} \mathbin{\widehat{\otimes}}_\alpha \mathcal{Y}$.*

$$A \text{ and } B \text{ are isoloid} \implies A \mathbin{\widehat{\otimes}} B \text{ is isoloid}$$

$$(\textit{i.e., } \sigma_{\text{iso}}(A) \subseteq \sigma_P(A) \text{ and } \sigma_{\text{iso}}(B) \subseteq \sigma_P(B) \implies \sigma_{\text{iso}}(A \mathbin{\widehat{\otimes}} B) \subseteq \sigma_P(A \mathbin{\widehat{\otimes}} B)).$$

Proof. If $\sigma_{\text{iso}}(A) \neq \varnothing$ and $\sigma_{\text{iso}}(B) \neq \varnothing$, then by Theorem 9.9(a,b) and Corollary 9.15(a), $\sigma_{\text{iso}}(A \mathbin{\widehat{\otimes}} B) \subseteq \sigma_{\text{iso}}(A) \cdot \sigma_{\text{iso}}(A) \subseteq \sigma_P(A) \cdot \sigma_P(B) \subseteq \sigma_P(A \mathbin{\widehat{\otimes}} B)$, and so $A \mathbin{\widehat{\otimes}} B$ is isoloid. If just one of $\sigma_{\text{iso}}(A)$ or $\sigma_{\text{iso}}(B)$ is empty, say $\sigma_{\text{iso}}(A) = \varnothing$, then $\sigma_{\text{iso}}(A \mathbin{\widehat{\otimes}} B) \subseteq \{0\}$ by Corollary 9.15(b). If $\sigma_{\text{iso}}(A \mathbin{\widehat{\otimes}} B) = \{0\}$, then $0 \in \sigma_{\text{iso}}(B)$ by Corollary 9.15(c), and so $0 \in \sigma_P(B)$, which implies by Theorem 9.9(a,d) that $0 \in \sigma_P(A \mathbin{\widehat{\otimes}} B)$, and hence $\sigma_{\text{iso}}(A \mathbin{\widehat{\otimes}} B) \subseteq \sigma_P(A \mathbin{\widehat{\otimes}} B)$, so that $A \mathbin{\widehat{\otimes}} B$ is again isoloid. If $\sigma_{\text{iso}}(A \mathbin{\widehat{\otimes}} B) = \varnothing$, then $A \mathbin{\widehat{\otimes}} B$ is trivially isoloid, which happens whenever $\sigma_{\text{iso}}(A) = \sigma_{\text{iso}}(B) = \varnothing$ by Corollary 9.15(d). □

Corollary 9.16 cannot be tightened by replacing the point spectrum with the set of eigenvalues of finite multiplicity. That is,

$$\sigma_{\text{iso}}(A) \subseteq \sigma_{PF}(A) \text{ and } \sigma_{\text{iso}}(B) \subseteq \sigma_{PF}(B) \not\implies \sigma_{\text{iso}}(A \mathbin{\widehat{\otimes}} B) \subseteq \sigma_{PF}(A \mathbin{\widehat{\otimes}} B).$$

For instance, take the diagonal $A = \text{diag}\{0, \frac{k}{k+1}\}_{k\geq 1}$ on ℓ^2_+. This is isoloid with $\sigma_{\text{iso}}(A) = \sigma_P(A) = \sigma_{PF}(A) = \{0, \frac{k}{k+1}\}_{k\geq 1}$ and $\sigma(A) = \sigma_{\text{iso}}(A) \cup \{1\}$. Now 0

lies in $\sigma_{\rm iso}(A \mathbin{\widehat{\otimes}} A)$ (as $\sigma(A \mathbin{\widehat{\otimes}} A) = \sigma(A) \cdot \sigma(A) = \{0\} \cup \{\frac{j}{j+1} \frac{k}{k+1}\}_{j,k\geq 1} \cup \{1\}$). But 0 does not lie in $\sigma_{PF}(A \mathbin{\widehat{\otimes}} A)$ by Corollary 9.11(b). However, when the eigenvalues of finite multiplicity are intersected with the isolated points of the spectrum, we get a useful result on isoloid operators stated in the next theorem. First, consider the following notation. For any $T \in \mathcal{B}[\mathcal{X}]$, let $\pi_{00}(T)$ denote the set of all isolated eigenvalues of finite multiplicity,

$$\pi_{00}(T) = \sigma_{\rm iso}(T) \cap \sigma_{PF}(T).$$

Corollary 9.17. *Let A and B be operators on Banach spaces $\mathcal{X}$ and $\mathcal{Y}$, and take the tensor product $A \mathbin{\widehat{\otimes}} B$ on $\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}$. If A and B are isoloid, then*

$$\pi_{00}(A \mathbin{\widehat{\otimes}} B) \subseteq \pi_{00}(A) \cdot \pi_{00}(B).$$

Proof. Take $A \in \mathcal{B}[\mathcal{X}]$ and $B \in \mathcal{B}[\mathcal{Y}]$ where $\mathcal{X}$ and $\mathcal{Y}$ are Banach spaces. In a finite-dimensional space, the spectrum $\sigma(\,\cdot\,)$ is a finite set and $\sigma(\,\cdot\,) = \sigma_P(\,\cdot\,) = \sigma_{PF}(\,\cdot\,) = \sigma_{\rm iso}(\,\cdot\,) = \pi_{00}(\,\cdot\,)$, so the above inclusion follows at once by Theorem 9.9(b) — actually, it is an identity by the spectral formula (SF). Thus suppose $\dim \mathcal{X} = \infty$ or $\dim \mathcal{Y} = \infty$, and take $A \mathbin{\widehat{\otimes}} B \in \mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}]$. If $\sigma_{\rm iso}(A) \subseteq \{0\}$ or $\sigma_{\rm iso}(B) \subseteq \{0\}$ (so that $\pi_{00}(A) \subseteq \{0\}$ or $\pi_{00}(B) \subseteq \{0\}$), then $\sigma_{\rm iso}(A \mathbin{\widehat{\otimes}} B) \subseteq \{0\}$ by Corollary 9.15(a,b,d), and so $\pi_{00}(A \mathbin{\widehat{\otimes}} B) = \sigma_{\rm iso}(A \mathbin{\widehat{\otimes}} B) \cap \sigma_{PF}(A \mathbin{\widehat{\otimes}} B) = \varnothing$ by Corollary 9.11(b) (since one of $\mathcal{X}$ or $\mathcal{Y}$ is infinite-dimensional), and the claimed inclusion holds. Thus suppose $\sigma_{\rm iso}(A) \not\subseteq \{0\}$ and $\sigma_{\rm iso}(B) \not\subseteq \{0\}$. Then

$$\sigma_{\rm iso}(A \mathbin{\widehat{\otimes}} B) \cap \sigma_{PF}(A \mathbin{\widehat{\otimes}} B) \subseteq \big(\sigma_{\rm iso}(A) \cdot \sigma_{\rm iso}(B)\big) \cap \sigma_{PF}(A \mathbin{\widehat{\otimes}} B)$$

by Corollary 9.15(a). Take any $\nu \in \sigma_{\rm iso}(A \mathbin{\widehat{\otimes}} B) \cap \sigma_{PF}(A \mathbin{\widehat{\otimes}} B) \subseteq \sigma(A \mathbin{\widehat{\otimes}} B)$, so that $\nu = \lambda\mu$ with $\lambda \in \sigma(A)$ and $\mu \in \sigma(B)$ by (SF). Now suppose A and B are isoloid. Then the above displayed inclusion ensures that

$$\lambda \in \sigma_{\rm iso}(A) \subseteq \sigma_P(A) \quad \text{and} \quad \mu \in \sigma_{\rm iso}(B) \subseteq \sigma_P(B).$$

Thus, according to Corollary 9.11(c),

$$\nu = \lambda\mu \in \big(\sigma_P(A) \cdot \sigma_P(B)\big) \cap \sigma_{PF}(A \mathbin{\widehat{\otimes}} B) \subseteq \sigma_{PF}(A) \cdot \sigma_{PF}(B).$$

Hence, as $\lambda \in \sigma(A)$ and $\mu \in \sigma(B)$, $\lambda \in \sigma_{PF}(A)$ and $\mu \in \sigma_{PF}(B)$. Therefore

$$\lambda \in \pi_{00}(A) = \sigma_{\rm iso}(A) \cap \sigma_{PF}(A) \quad \text{and} \quad \mu \in \pi_{00}(B) = \sigma_{\rm iso}(B) \cap \sigma_{PF}(B),$$

so that $\nu = \lambda\mu \in \pi_{00}(A) \cdot \pi_{00}(B)$. □

It is worth noticing from Corollaries 9.11(b) and 9.15(a,b,d) that

$$\sigma_{\rm iso}(A) \cdot \sigma_{\rm iso}(B) \subseteq \{0\} \implies \sigma_{\rm iso}(A \mathbin{\widehat{\otimes}} B) \subseteq \{0\} \implies \pi_{00}(A \mathbin{\widehat{\otimes}} B) = \varnothing$$

whenever one of the Banach spaces $\mathcal{X}$ or $\mathcal{Y}$ is infinite-dimensional. Also observe that if both $\mathcal{X}$ and $\mathcal{Y}$ are infinite-dimensional (otherwise the expression below is trivially empty), then (since $\sigma(A \mathbin{\widehat{\otimes}} B) = \sigma(A) \cdot \sigma(B)$)

$$\sigma_{\rm acc}(A \mathbin{\widehat{\otimes}} B) = \big(\sigma_{\rm acc}(A) \cdot \sigma(B)\big) \cup \big(\sigma(A) \cdot \sigma_{\rm acc}(B)\big).$$

9.4 Additional Propositions

Theorem 9.1, early in this chapter, extended Theorem 7.3 from bounded linear functionals (i.e., rank-one bounded linear transformations) to arbitrary bounded linear transformations. Two proofs of Theorem 9.1 were given, and both of them relied on Proposition 9.A below. Let $\mathcal{V}, \mathcal{W}, \mathcal{X}, \mathcal{Y}$ be normed spaces. Equip their tensor products with a uniform crossnorm $\|\cdot\|_\alpha$. Consider the induced uniform norm $\|\cdot\|_{[\alpha,\alpha]}$ on $\mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{V} \otimes_\alpha \mathcal{W}]$. By Theorem 8.3(c), $\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}] \subseteq \mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{V} \otimes_\alpha \mathcal{W}]$ and, as shown at the beginning of the proof of Theorem 9.1, $\mathcal{B}[\mathcal{X},\mathcal{V}]^* \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}]^* \subseteq \mathcal{L}\big[\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes_{[\alpha,\alpha]} \mathcal{B}[\mathcal{Y},\mathcal{W}], \mathbb{F}\big]$. Equip $\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}]$ with $\|\cdot\|_{[\alpha,\alpha]}$ inherited from $\mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{V} \otimes_\alpha \mathcal{W}]$. Set $\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes_{[\alpha,\alpha]} \mathcal{B}[\mathcal{Y},\mathcal{W}] = (\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}], \|\cdot\|_{[\alpha,\alpha]})$. Take the induced uniform norm $\|\cdot\|_{*[\alpha,\alpha]}$ on $\mathcal{B}\big[\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes_{[\alpha,\alpha]} \mathcal{B}[\mathcal{Y},\mathcal{W}], \mathbb{F}\big] = (\mathcal{B}[\mathcal{X},\mathcal{V}] \otimes_{[\alpha,\alpha]} \mathcal{B}[\mathcal{Y},\mathcal{W}])^*$.

Proposition 9.A. *If $\|\cdot\|_\alpha$ is a uniform crossnorm, then*

$$\|\varphi \otimes \eta\|_{*[\alpha,\alpha]} \le \|\varphi\|\,\|\eta\|$$

for every $\varphi \in \mathcal{B}[\mathcal{X},\mathcal{V}]^*$ and $\eta \in \mathcal{B}[\mathcal{Y},\mathcal{W}]^*$.

Proposition 9.B. *If $\|\cdot\|_\alpha$ is a uniform crossnorm, then*

$$\inf_{0\neq F\in\mathcal{X}\otimes\mathcal{Y}} \frac{\|F\|_\vee}{\|F\|_\alpha}\,\|\mathbb{L}\|_{[\vee,\vee]} \le \|\mathbb{L}\|_{[\alpha,\alpha]} \le \sup_{0\neq F\in\mathcal{X}\otimes\mathcal{Y}} \frac{\|F\|_\wedge}{\|F\|_\alpha}\,\|\mathbb{L}\|_{[\wedge,\wedge]}$$

for every $\mathbb{L} \in \mathcal{B}[\mathcal{X},\mathcal{V}] \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}]$.

This chapter has considered a line of action for dealing with tensor products of linear transformations acting on normed (or Banach) spaces. In the case of tensor products of Hilbert-space operators, the appropriate uniform crossnorm is that induced by the inner product defined in Section 9.2.

The converse of Theorem 9.3(d) (extending Proposition 3.L) goes as follows.

Proposition 9.C. *Let $\mathcal{V}, \mathcal{W}, \mathcal{X}, \mathcal{Y}$ be nonzero normed spaces and equip their tensor product spaces with a uniform crossnorm $\|\cdot\|_\alpha$. Take $A \in \mathcal{B}[\mathcal{X},\mathcal{V}]$ and $B \in \mathcal{B}[\mathcal{Y},\mathcal{W}]$, and suppose their tensor product*

$$A \otimes B \in \mathcal{B}[\mathcal{X},\mathcal{V}] \otimes \mathcal{B}[\mathcal{Y},\mathcal{W}] \subseteq \mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}, \mathcal{V} \otimes_\alpha \mathcal{W}]$$

is invertible. If its inverse $(A \otimes B)^{-1} \in \mathcal{B}[\mathcal{V} \otimes_\alpha \mathcal{W}, \mathcal{X} \otimes_\alpha \mathcal{Y}]$ is such that

$$(A \otimes B)^{-1} = C \otimes D \in \mathcal{B}[\mathcal{V},\mathcal{X}] \otimes \mathcal{B}[\mathcal{W},\mathcal{Y}] \subseteq \mathcal{B}[\mathcal{V} \otimes_\alpha \mathcal{W}, \mathcal{X} \otimes_\alpha \mathcal{Y}]$$

for some $C \in \mathcal{B}[\mathcal{V},\mathcal{X}]$ and $D \in \mathcal{B}[\mathcal{W},\mathcal{Y}]$ with at least one of them invertible, then $A \in \mathcal{B}[\mathcal{X},\mathcal{V}]$ and $B \in \mathcal{B}[\mathcal{Y},\mathcal{W}]$ are invertible and $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$.

Recall from Section 5.4: two vectors in an inner product space are orthogonal if their inner product is null, and orthonormal if, in addition, they have unit norms. Consider the setup of Section 9.2. For Hilbert spaces $\mathcal{X}$ and $\mathcal{Y}$,

let $\perp$ stand for orthogonality either in $(\mathcal{X}, \langle\cdot\,;\,\cdot\rangle_{\mathcal{X}})$ or in $(\mathcal{Y}, \langle\cdot\,;\,\cdot\rangle_{\mathcal{Y}})$, as usual. In addition, let $\perp^1$ denote orthonormality in either Hilbert space. Accordingly, let $\perp_\eta$ and $\perp^1_\eta$ stand for orthogonality and orthonormality, respectively, in the inner product space $(\mathcal{X} \otimes \mathcal{Y}, \langle\cdot\,;\,\cdot\rangle_\eta)$ with the inner product defined in (IP). The proposition below shows that orthonormality in $\mathcal{X}$ or in $\mathcal{Y}$ implies orthonormality for single tensors in $\mathcal{X} \otimes_\eta \mathcal{Y}$. (This has been applied in the proof of Theorem 9.6 for proving that $\|\cdot\|_\eta$ is a uniform crossnorm).

Proposition 9.D. *Let $\{x_\gamma\}$ and $\{y_\delta\}$ be families of vectors in Hilbert spaces $\mathcal{X}$ and $\mathcal{Y}$, respectively, and take single tensors in the inner product space $\mathcal{X} \otimes_\eta \mathcal{Y}$ equipped with the inner product defined in* (IP).

(a) $(x_\gamma \otimes y_\delta) \perp_\eta (x_{\gamma'} \otimes y_{\delta'}) \iff x_\gamma \perp x_{\gamma'}$ *or* $y_\delta \perp y_{\delta'}$.

(b) $(x_\gamma \otimes y_\delta) \perp^1_\eta (x_{\gamma'} \otimes y_{\delta'}) \iff$
$x_\gamma \perp x_{\gamma'}$ *or* $y_\delta \perp y_{\delta'}$ *as in (a), and both* $\|x_\gamma\|\,\|y_\delta\| = 1$ *and* $\|x_{\gamma'}\|\,\|y_{\delta'}\| = 1$.

(c) $x_\gamma \perp^1 x_{\gamma'}$ *or* $y_\delta \perp^1 y_{\delta'} \implies (x_\gamma \otimes y_\delta) \perp^1_\eta (x_{\gamma'} \otimes y_{\delta'})$.

Let $\mathcal{X}$ and $\mathcal{Y}$ be normed spaces, let $\|\cdot\|_\alpha$ be a uniform crossnorm on $\mathcal{X} \otimes \mathcal{Y}$, and take operators $A \in \mathcal{B}[\mathcal{X}]$ and $B \in \mathcal{B}[\mathcal{Y}]$. Since the tensor product $A \otimes B$ lies in $\mathcal{B}[\mathcal{X} \otimes_\alpha \mathcal{Y}]$ (Definition 8.2), the kernel $\mathcal{N}(A \otimes B)$ is closed in the normed space $\mathcal{X} \otimes_\alpha \mathcal{Y}$, which in turn is densely embedded in the Banach space $\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}$. Let $\mathcal{N}(A \otimes B)^-$ denote the closure of (the isometrically isomorphic image of) $\mathcal{N}(A \otimes B)$ in $\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}$. Since the range $\mathcal{R}(A \otimes B)$ is not necessarily closed in the normed space $\mathcal{X} \otimes_\alpha \mathcal{Y}$, use the same notation $\mathcal{R}(A \otimes B)^-$ to denote the closure of $\mathcal{R}(A \otimes B)$ in $\mathcal{X} \otimes_\alpha \mathcal{Y}$ and of (the isometrically isomorphic image of) $\mathcal{R}(A \otimes B)$ in $\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}$. Now let $\mathcal{X}$ and $\mathcal{Y}$ be Hilbert spaces, equip their tensor product with the inner product $\langle\cdot\,;\,\cdot\rangle_\eta$ in (IP) inducing the uniform crossnorm $\|\cdot\|_\eta$ in (N), and consider the Hilbert space $\mathcal{X} \mathbin{\widehat{\otimes}_\eta} \mathcal{Y}$. Under these assumptions, Proposition 9.E is a special unfoldment of Theorem 9.3(e,h).

Proposition 9.E. *Take operators $A \in \mathcal{B}[\mathcal{X}]$ and $B \in \mathcal{B}[\mathcal{Y}]$ on Hilbert spaces $\mathcal{X}$ and $\mathcal{Y}$, and consider the Hilbert space $\mathcal{X} \mathbin{\widehat{\otimes}_\eta} \mathcal{Y}$. Then*

(a) $\mathcal{N}(A \mathbin{\widehat{\otimes}} B) = \mathcal{N}(A \otimes B)^-$ (*closure in* $\mathcal{X} \mathbin{\widehat{\otimes}_\eta} \mathcal{Y}$),

(b) $\mathcal{R}(A \mathbin{\widehat{\otimes}} B)$ *is closed in* $\mathcal{X} \mathbin{\widehat{\otimes}_\eta} \mathcal{Y}$ *if and only if* $\mathcal{R}(A)$ *and* $\mathcal{R}(B)$ *are closed in* $\mathcal{X}$ *and* $\mathcal{Y}$, *respectively*; *and in this case,*

(c) $\mathcal{R}(A \mathbin{\widehat{\otimes}} B) = \mathcal{R}(A \otimes B)^-$.

We saw in Section 4.4 how the Kronecker product $A \circledast B$ of matrices representing linear transformations $A \in \mathcal{L}[\mathbb{F}^m]$ and $B \in \mathcal{L}[\mathbb{F}^n]$ on finite-dimensional linear spaces $\mathbb{F}^m$ and $\mathbb{F}^n$ is identified with (i.e., is similar to) the tensor product $A \otimes B$ of linear transformations. In fact, $A \circledast B \cong A \otimes B$ and $x \circledast y \cong x \otimes y$, with $\mathbb{F}^m \circledast \mathbb{F}^n \cong \mathbb{F}^m \otimes \mathbb{F}^n \cong \mathcal{L}[\mathbb{F}^m, \mathbb{F}^n] \cong \mathbb{F}^m(\mathbb{F}^n) = \bigoplus_{j=1}^m \mathbb{F}^n \cong \mathbb{F}^{mn}$, where $\bigoplus$ means direct sum and $\circledast$ stands for Kronecker product. This can be extended from finite-dimensional linear spaces to separable Hilbert spaces if the tensor

product $\mathcal{X} \otimes \mathcal{Y}$ is equipped with the inner product $\langle \cdot\,;\cdot \rangle_\eta$ as follows. (Recall: in a Hilbert-space setting, isometrically isomorphic spaces, and isometrically equivalent transformations, are both called unitarily equivalent.)

Let $\mathcal{X}$ and $\mathcal{Y}$ be Hilbert spaces. Suppose $\mathcal{X}$ is separable, so that $\mathcal{X} \cong \ell^2_+ = \ell^2_+(\mathbb{C})$. Let $\{e_k\}$ be the canonical orthonormal basis for ℓ^2_+, and consider a respective (countable) orthonormal basis for $\mathcal{X}$ (i.e., one made up of unitary images on $\mathcal{X}$ of $\{e_k\}$, also referred to as a canonical orthonormal basis for $\mathcal{X}$). Take arbitrary operators $A \in \mathcal{B}[\mathcal{X}]$ and $B \in \mathcal{B}[\mathcal{Y}]$, and consider the representation of A with respect to such a canonical orthonormal basis for $\mathcal{X}$, say,

$$A = \begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots \\ \alpha_{21} & \alpha_{22} & \cdots \\ \vdots & \vdots & \end{pmatrix} \quad \text{in} \quad \mathcal{B}[\mathcal{X}]$$

on the Hilbert space $\mathcal{X}$. With $\bigoplus$ now meaning orthogonal direct sum, set

$$A \circledast B = \begin{pmatrix} \alpha_{11}B & \alpha_{12}B & \cdots \\ \alpha_{21}B & \alpha_{22}B & \cdots \\ \vdots & \vdots & \end{pmatrix} \quad \text{in} \quad \mathcal{B}\big[\bigoplus\nolimits_k \mathcal{Y}\big] = \mathcal{B}[\ell^2_+(\mathcal{Y})]$$

on the Hilbert space $\ell^2_+(\mathcal{Y}) = \bigoplus_k \mathcal{Y}$ of square-summable sequences of vectors in $\mathcal{Y}$. This is the Kronecker product of A and B. Since it acts on the complete inner product space $\ell^2_+(\mathcal{Y})$, it coincides with its extension over completion:

$$A \circledast B = A \mathbin{\widehat{\circledast}} B.$$

Take an arbitrary $x \in \mathcal{X}$, identify it with its representation (via Fourier series expansion) with respect to such a canonical orthonormal basis for $\mathcal{X}$, say, $x \cong \{\xi_k\} = (\xi_1, \xi_2, \dots) \in \ell^2_+ \cong \mathcal{X}$, and define the Kronecker product of x and y as

$$x \circledast y = (\xi_1 y, \xi_2 y, \dots) \ \text{ in } \ \bigoplus\nolimits_k \mathcal{Y} = \ell^2_+(\mathcal{Y})$$

for every $y \in \mathcal{Y}$. Also, set $\mathcal{X} \circledast \mathcal{Y} = \big\{\sum_i x_i \circledast y_i \colon\, x_i \circledast y_i \in \ell^2_+(\mathcal{Y})\big\}$, the collection of all finite sums of Kronecker products $x_i \circledast y_i$ of vectors $x_i \in \mathcal{X}$ and $y_i \in \mathcal{Y}$, which is a linear manifold of $\ell^2_+(\mathcal{Y})$. So, as in Proposition 4.E, we get

$$A \circledast B\Big(\sum\nolimits_i x_i \circledast y_i\Big) = \sum\nolimits_i Ax_i \circledast By_i \quad \text{for every} \quad \sum\nolimits_i x_i \circledast y_i \in \mathcal{X} \circledast \mathcal{Y}.$$

Moreover, set $\mathcal{X} \mathbin{\widehat{\circledast}} \mathcal{Y} = (\mathcal{X} \circledast \mathcal{Y})^-$, the closure of $\mathcal{X} \circledast \mathcal{Y}$ in $\ell^2_+(\mathcal{Y})$.

Proposition 9.F. *If $\mathcal{X}$ and $\mathcal{Y}$ are Hilbert spaces, with $\mathcal{X}$ separable, then*

(a) $\mathcal{X} \circledast \mathcal{Y} \cong \mathcal{X} \otimes_\eta \mathcal{Y}$ *and* $\mathcal{X} \mathbin{\widehat{\circledast}}_\eta \mathcal{Y} \cong \mathcal{X} \mathbin{\widehat{\circledast}} \mathcal{Y} = \ell^2_+(\mathcal{Y}) = \bigoplus_k \mathcal{Y}$,

(b) $x \circledast y \cong x \otimes y$ *and* $A \mathbin{\widehat{\otimes}} B \cong A \mathbin{\widehat{\circledast}} B = A \circledast B$.

Recall that $\bigoplus_k \mathcal{Y} = \ell^2_+(\mathcal{Y})$ means the orthogonal direct sum of countably many copies of $\mathcal{Y}$, consisting of all square-summable $\mathcal{Y}$-valued sequences whose

entries are mutually orthogonal. So, by Proposition 9.F, for each pair of vectors $x \in \mathcal{X}$ and $y \in \mathcal{Y}$, the Kronecker product $x \circledast y$ in $\mathcal{X} \circledast \mathcal{Y} \subseteq \ell^2_+(\mathcal{Y})$ is unitarily equivalent to the single tensor $x \otimes y$ in $\mathcal{X} \otimes_\eta \mathcal{Y}$ — in the sense that one is a unitary image of the other. Still by Proposition 9.F, the Kronecker product $A \mathbin{\widehat{\circledast}} B$ in $\mathcal{B}[\mathcal{X} \widehat{\circledast} \mathcal{Y}]$ of operators $A \in \mathcal{B}[\mathcal{X}]$ and $B \in \mathcal{B}[\mathcal{Y}]$ is unitarily equivalent to the tensor product $A \mathbin{\widehat{\otimes}} B$ in $\mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}}_\eta \mathcal{Y}]$ — in the sense that they are unitarily equivalent operators on unitarily equivalent Hilbert spaces. Such an identification of $A \mathbin{\widehat{\otimes}} B$ with $A \mathbin{\widehat{\circledast}} B = A \circledast B$ provides a large supply of examples of tensor products of operators (e.g., it was used in the proof of Theorem 9.9(c)).

Take an arbitrary subspace $\mathcal{M}$ of a Hilbert space $\mathcal{X}$ so that $\mathcal{M}$ and its orthogonal complement $\mathcal{M}^\perp$ are Hilbert spaces. Consider the orthogonal decomposition $\mathcal{X} = \mathcal{M} \oplus \mathcal{M}^\perp$ of Proposition 5.Q. Write an operator $T \in \mathcal{B}[\mathcal{X}]$ as $T = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right) \in \mathcal{B}[\mathcal{M} \oplus \mathcal{M}^\perp]$, where the entries of the operator matrix are such that $A \in \mathcal{B}[\mathcal{M}]$, $B \in \mathcal{B}[\mathcal{M}^\perp, \mathcal{M}]$, $C \in \mathcal{B}[\mathcal{M}, \mathcal{M}^\perp]$ and $D \in \mathcal{B}[\mathcal{M}^\perp]$ (as in Proposition 1.S, now with $\oplus$ denoting orthogonal direct sum).

Proposition 9.G. *Consider the above setup, where T and S are operators on Hilbert spaces $\mathcal{X}$ and $\mathcal{Y}$, respectively. Equip the tensor product space of every pair of Hilbert spaces with the uniform crossnorm $\|\cdot\|_\eta$ induced by the inner product $\langle \cdot\,;\cdot \rangle_\eta$ in* (PI)*. Then*

$$T \mathbin{\widehat{\otimes}} S = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \mathbin{\widehat{\otimes}} S = \begin{pmatrix} A \mathbin{\widehat{\otimes}} S & B \mathbin{\widehat{\otimes}} S \\ C \mathbin{\widehat{\otimes}} S & D \mathbin{\widehat{\otimes}} S \end{pmatrix} \text{ in } \mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}} \mathcal{Y}],$$

with entries $A \mathbin{\widehat{\otimes}} S$ in $\mathcal{B}[\mathcal{M} \mathbin{\widehat{\otimes}}_\alpha \mathcal{Y}]$, $B \mathbin{\widehat{\otimes}} S$ in $\mathcal{B}[\mathcal{M}^\perp \mathbin{\widehat{\otimes}}_\alpha \mathcal{Y}, \mathcal{M} \mathbin{\widehat{\otimes}}_\alpha \mathcal{Y}]$, $C \mathbin{\widehat{\otimes}} S$ in $\mathcal{B}[\mathcal{M} \mathbin{\widehat{\otimes}}_\alpha \mathcal{Y}, \mathcal{M}^\perp \mathbin{\widehat{\otimes}}_\alpha \mathcal{Y}]$ and $D \mathbin{\widehat{\otimes}} S$ in $\mathcal{B}[\mathcal{M}^\perp \mathbin{\widehat{\otimes}}_\alpha \mathcal{Y}]$.

The problem of transferring properties from a pair of Hilbert-space operators to their tensor product is frequently concerned with classes of operators, and often with classical classes of Hilbert-space operators. By an isometry between normed spaces we always mean a linear isometry. An operator $T \in \mathcal{B}[\mathcal{X}]$ on a Hilbert space $\mathcal{X}$ is an *isometry* if (and only if) $T^*T = I$ (of course, we will be dealing with the Hilbert-space adjoint T^* of T). It is *unitary* if $T^* = T^{-1}$ (equivalently, if it is an invertible isometry, that is, an isometric isomorphism — since isometries are always injective, this means a surjective isometry). Also, T is *self-adjoint* if $T^* = T$, *nonnegative* (i.e., $O \le T$) if it is self-adjoint and $0 \le \langle Tx\,; x \rangle$ for every $x \in \mathcal{X}$, *normal* if T commutes with its adjoint (i.e., $TT^* = T^*T$), *quasinormal* if T commutes with T^*T, *subnormal* if it has a normal extension, *hyponormal* if $TT^* \le T^*T$, and *normaloid* if $r(T) = \|T\|$ (the spectral radius equals the norm; equivalently, if $\|T^n\| = \|T\|^n$ for every positive integer n). There are, of course, many relevant classes in between or beyond these. As is well known, these classes are related by proper inclusion:

$$\text{Nonnegative} \subset \text{Self-Adjoint} \subset \text{Normal}, \quad \text{Unitary} \subset \text{Normal}, \quad \text{and}$$
$$\text{Normal} \subset \text{Quasinormal} \subset \text{Subnormal} \subset \text{Hyponormal} \subset \text{Normaloid}.$$

Let $\mathcal{X}$ and $\mathcal{Y}$ be Hilbert spaces, equip their tensor product with the inner product $\langle \cdot\,;\,\cdot \rangle_\eta$ defined in (IP), and consider the Hilbert space $\mathcal{X} \mathbin{\widehat{\otimes}}_\eta \mathcal{Y}$. Under these assumptions, Propositions 9.H, 9.I and 9.J deal with the problem of transferring properties between a pair of operators and their tensor product.

Proposition 9.H. *Take $A \in \mathcal{B}[\mathcal{X}]$ and $B \in \mathcal{B}[\mathcal{Y}]$. If they are nonzero and*

(a) *nonnegative, or*

(b) *self-adjoint, or*

(c) *unitary, or*

(d) *normal, or*

(e) *quasinormal, or*

(f) *subnormal, or*

(g) *hyponormal, or*

(h) *normaloid,*

then so is $A \mathbin{\widehat{\otimes}} B \in \mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}}_\eta \mathcal{Y}]$. Conversely, if $A \mathbin{\widehat{\otimes}} B \in \mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}}_\eta \mathcal{Y}]$ is nonzero and

(a) *nonnegative, or*

(b) *self-adjoint, or*

(c) *unitary,*

then so are γA and $\gamma^{-1} B$ for some nonzero $\gamma \in \mathbb{C}$; and if it is nonzero and

(d) *normal, or*

(e) *quasinormal, or*

(f) *subnormal, or*

(g) *hyponormal, or*

(h) *normaloid,*

then so are A and B.

We first met a unilateral shift of multiplicity one in the proof of Proposition 7.S(d). When acting on a Hilbert space, it is defined as the operator that shifts an orthonormal basis; thus it is an isometry (see also proof of Theorem 9.9(c)).

Proposition 9.I. *Let $A \in \mathcal{B}[\mathcal{X}]$ and $B \in \mathcal{B}[\mathcal{Y}]$ be Hilbert-space operators.*

(a) *$A \mathbin{\widehat{\otimes}} B$ in $\mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}}_\eta \mathcal{Y}]$ is an isometry if and only if A and B are (nonzero, complex) multiples of isometries with $\|A\| = \|B\|^{-1}$.*

(b) *$A \mathbin{\widehat{\otimes}} B$ is a unilateral shift if and only if $A \mathbin{\widehat{\otimes}} B = V \mathbin{\widehat{\otimes}} U$ where $V \in \mathcal{B}[\mathcal{X}]$ and $U \in \mathcal{B}[\mathcal{Y}]$ are isometries, at least one of them being a unilateral shift.*

Proposition 9.J. *A tensor product $A \mathbin{\widehat{\otimes}} B \in \mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}}_\eta \mathcal{Y}]$ is nonzero and compact if and only if $A \in \mathcal{B}[\mathcal{X}]$ and $B \in \mathcal{B}[\mathcal{Y}]$ are both nonzero and compact.*

The notions of uniform and strong convergence (together with the particular case of uniform and strong stability) were discussed in Section 5.4. For a sequence of operators on a complex Hilbert space, the notion of weak convergence is considerably simplified. If $\mathcal{X}$ and $\mathcal{Y}$ are Hilbert spaces, and if a $\mathcal{B}[\mathcal{X},\mathcal{Y}]$-valued sequence $\{T_n\}$ is such that there exists $T \in \mathcal{B}[\mathcal{X},\mathcal{Y}]$ for which $\langle (T_n - T)x\,;y\rangle \to 0$ for every $x \in \mathcal{X}$ and $y \in \mathcal{Y}$, then the sequence $\{T_n\}$ is *weakly convergent*, or it is said to converge weakly to T (notation: $T_n \xrightarrow{w} T$). If $\mathcal{Y} = \mathcal{X}$ (so that T_n and T are operators in $B[\mathcal{X}]$), and if $\mathcal{X}$ is a *complex* Hilbert space, then the above weak convergence criterion is equivalent to saying that $\langle (T_n - T)x\,;x\rangle \to 0$ for every $x \in \mathcal{X}$. In particular, an operator $T \in \mathcal{B}[\mathcal{X}]$ on a Hilbert space $\mathcal{X}$ is *weakly stable* if the power sequence $\{T^n\}$ converges weakly to the null operator O; in other words, if $T^n \xrightarrow{w} O$.

For Propositions 9.K and 9.L, let $\mathcal{X}$ and $\mathcal{Y}$ be complex Hilbert spaces, and again equip their tensor product with the inner product $\langle \cdot\,;\cdot\rangle_\eta$ in (IP).

Proposition 9.K. *Let $\{A_n\}$ and $\{B_n\}$ be sequences of operators in $\mathcal{B}[\mathcal{X}]$ and $\mathcal{B}[\mathcal{Y}]$, respectively. If one of them converges to zero uniformly (strongly, weakly) and the other is bounded, then the sequence $\{A_n \mathbin{\widehat{\otimes}} B_n\}$ of operators in $\mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}}_\eta \mathcal{Y}]$ converges to zero uniformly (strongly, weakly).*

Let $\{n\}_{n\ge 0}$ be the naturally ordered self-indexing of the set of all nonnegative integers. A subsequence $\{n_k\}_{k\ge 0}$ of $\{n\}_{n\ge 0}$ is of *bounded increments* if $\sup_{k\ge 0}(n_{k+1} - n_k) < \infty$. A Hilbert-space operator T is *power incremented* if either the power sequence $\{T^n\}$ converges weakly to zero or there is a subsequence of bounded increments $\{n_k\}_{k\ge 0}$ of $\{n\}_{n\ge 0}$ such that $\limsup_k |\langle T^{n_k}x\,;y\rangle| > 0$ whenever $\langle T^n x\,;y\rangle \not\to 0$ for some pair of vectors x and y.

Also recall that if $A \in \mathcal{B}[\mathcal{X}]$ and $B \in \mathcal{B}[\mathcal{Y}]$ are operators on normed spaces $\mathcal{X}$ and $\mathcal{Y}$, then $(A \otimes B)^n = A^n \otimes B^n$ in $\mathcal{L}[\mathcal{X} \otimes \mathcal{Y}]$ for every nonnegative integer n according to Theorem 3.19(c), and so $(A \mathbin{\widehat{\otimes}} B)^n = A^n \mathbin{\widehat{\otimes}} B^n$ in $\mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}}_\alpha \mathcal{Y}]$ for every uniform crossnorm $\|\cdot\|_\alpha$ by Theorem 9.3(c), and hence $\|(A \mathbin{\widehat{\otimes}} B)^n\| = \|A^n\|\,\|B^n\|$ for every n by Theorem 8.3(a) (as in the proof of Corollary 9.14).

Proposition 9.L. *Let $A \in \mathcal{B}[\mathcal{X}]$ and $B \in \mathcal{B}[\mathcal{Y}]$ be operators on Hilbert spaces $\mathcal{X}$ and $\mathcal{Y}$, and take their tensor product $A \mathbin{\widehat{\otimes}} B \in \mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}}_\eta \mathcal{Y}]$. If $A \mathbin{\widehat{\otimes}} B$ is uniformly or strongly stable, then so is one of A or B. If $A \mathbin{\widehat{\otimes}} B$ is weakly stable and one of A or B is power incremented, then one of them is weakly stable.*

In other words, if the $\mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}}_\eta \mathcal{Y}]$-valued sequence $\{A^n \mathbin{\widehat{\otimes}} B^n\}$ converges to zero uniformly or strongly, then so does one of the sequences $\{A^n\}$ or $\{B^n\}$ of operators in $\mathcal{B}[\mathcal{X}]$ and $\mathcal{B}[\mathcal{Y}]$. If $\{A^n \mathbin{\widehat{\otimes}} B^n\}$ converges to zero weakly, and one of A or B is power incremented, then one of $\{A^n\}$ or $\{B^n\}$ converges to zero weakly.

Let $T \in \mathcal{B}[\mathcal{X}]$ be an operator on a (complex) Hilbert space $\mathcal{X}$. Consider the subset $\sigma_0(T)$ of the spectrum $\sigma(T)$ consisting of all $\lambda \in \mathbb{C}$ for which $\lambda I - T$ has closed range, is not surjective, is not injective, and has a finite-dimensional kernel together with its adjoint, both with the same finite dimension,

$$\sigma_0(T) = \big\{\lambda \in \sigma(T)\colon\ \mathcal{R}(\lambda I - T)^- = \mathcal{R}(\lambda I - T) \neq \mathcal{X} \text{ and } 0 < \dim \mathcal{N}(\lambda I - T) = \dim \mathcal{N}(\overline{\lambda} I - T^*) < \infty\big\},$$

so that $\sigma_0(T) \subseteq \sigma_{PF}(T) \cup \sigma_{PF}(T^*)$, and set

$$\sigma_w(T) = \sigma(T) \backslash \sigma_0(T) \qquad \text{and} \qquad \sigma_b(T) = \sigma(T) \backslash \big(\sigma_{\text{iso}}(T) \cap \sigma_0(T)\big).$$

The sets $\sigma_w(T)$ and $\sigma_b(T)$ are the so-called *Weyl spectrum* and *Browder spectrum* of T, respectively. An operator T is said to *satisfy Weyl's theorem* if

$$\sigma_0(T) = \sigma_{\text{iso}}(T) \cap \sigma_{PF}(T)$$

(i.e., if $\sigma_0(T) = \pi_{00}(T)$). Also, T is said to *satisfy Browder's theorem* if

$$\sigma_0(T) \subseteq \sigma_{\text{iso}}(T) \cap \sigma_{PF}(T)$$

(i.e., if $\sigma_0(T) \subseteq \pi_{00}(T)$), and so T satisfies Browder's theorem if and only if

$$\sigma_0(T) \subseteq \sigma_{\text{iso}}(T),$$

that is,

$$\sigma_{\text{acc}}(T) \subseteq \sigma_w(T);$$

equivalently, T satisfies Browder's theorem if and only if

$$\sigma_w(T) = \sigma_b(T).$$

Observe that if T satisfies Weyl's theorem, then it satisfies Browder's theorem.

For Propositions 9.M, 9.N, 9.O, and 9.P, consider again the following setup: $A \in \mathcal{B}[\mathcal{X}]$ and $B \in \mathcal{B}[\mathcal{Y}]$ are operators on complex Hilbert spaces $\mathcal{X}$ and $\mathcal{Y}$, and the tensor product Hilbert space $\mathcal{X} \mathbin{\widehat{\otimes}_\eta} \mathcal{Y}$ is equipped with the inner product $\langle \cdot\,;\cdot \rangle_\eta$ in (IP) so that the tensor product operator $A \mathbin{\widehat{\otimes}} B$ lies in $\mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}_\eta} \mathcal{Y}]$.

Proposition 9.M. *The Browder and Weyl spectra of* $A \mathbin{\widehat{\otimes}} B$ *are such that*

$$\sigma_b(A \mathbin{\widehat{\otimes}} B) = \sigma_b(A) \cdot \sigma(B) \cup \sigma(A) \cdot \sigma_b(B),$$
$$\sigma_w(A \mathbin{\widehat{\otimes}} B) \subseteq \sigma_w(A) \cdot \sigma(B) \cup \sigma(A) \cdot \sigma_w(B).$$

For a long time, whether the above inclusion was an identity or might become a proper inclusion, remained unknown. In fact, there are operators A and B for which $\sigma_w(A \mathbin{\widehat{\otimes}} B) \subset \sigma_w(A) \cdot \sigma(B) \cup \sigma(A) \cdot \sigma_w(B)$, as we will see below.

Proposition 9.N. *If both operators* A *and* B *satisfy Browder's theorem, then the tensor product* $A \mathbin{\widehat{\otimes}} B$ *satisfies Browder's theorem if and only if*

$$\sigma_w(A \mathbin{\widehat{\otimes}} B) = \sigma_w(A) \cdot \sigma(B) \cup \sigma(A) \cdot \sigma_w(B).$$

The main character for the next proposition is again a unilateral shift S of multiplicity one on a separable Hilbert space $\mathcal{X}$ (i.e., $\mathcal{X} \cong \ell^2_+$), where the operators A and B act on the orthogonal direct sum $\mathcal{X} \oplus \mathcal{X}$.

Proposition 9.O. *Let* S *be a unilateral shift of multiplicity one on a separable Hilbert space* $\mathcal{X}$*. If* A *and* B *are operators on the Hilbert space* $\mathcal{X} \oplus \mathcal{X}$ *given by*

$$A = (I - SS^*) \oplus (\tfrac{1}{2}S - I) \quad \text{and} \quad B = -(I - SS^*) \oplus (\tfrac{1}{2}S^* + I),$$

then A and B satisfy Browder's theorem, but their tensor product $A \mathbin{\widehat{\otimes}} B$ does not satisfy Browder's theorem.

Propositions 9.N and 9.O together show that there exist Hilbert-space operators A and B for which the inclusion in Proposition 9.M is proper:

$$\sigma_w(A \mathbin{\widehat{\otimes}} B) \subset \sigma_w(A) \cdot \sigma(B) \cup \sigma(A) \cdot \sigma_w(B).$$

If the inclusion in Proposition 9.M becomes an identity for isoloid operators that satisfy Weyl's theorem, then the property of satisfying Weyl's theorem is transferred from the operators to their tensor product.

Proposition 9.P. *If the operators A and B are* (i) *isoloid,* (ii) *satisfy Weyl's theorem, and if* (iii) $\sigma_w(A \mathbin{\widehat{\otimes}} B) = \sigma_w(A) \cdot \sigma(B) \cup \sigma(A) \cdot \sigma_w(B)$*, then their tensor product $A \mathbin{\widehat{\otimes}} B$ satisfies Weyl's theorem (and is isoloid).*

Let $\mathcal{X}, \mathcal{Y}, \mathcal{V}, \mathcal{W}$ be linear spaces. Take $A \in \mathcal{L}[\mathcal{X}, \mathcal{V}]$ and $B \in \mathcal{L}[\mathcal{Y}, \mathcal{W}]$. The range and kernel of a tensor product transformation $A \otimes B$ in $\mathcal{L}[\mathcal{X} \otimes \mathcal{Y}, \mathcal{V} \otimes \mathcal{W}]$ naturally play an essential role in the theory of linear transformations, mainly in the spectral theory on Banach spaces. The expressions for kernel and range discussed in this book are summarized below, where linear transformations are also assumed to be bounded when acting between normed spaces.

Remark 9.Q. *A Summary of Kernel and Range of Tensor Products.*

LINEAR SPACE RESULTS.

- $\mathcal{R}(A) \otimes \mathcal{R}(B) = \mathcal{R}(A \otimes B)$ (Theorem 3.19(e)).
- $\mathcal{R}(A \otimes B) = \mathcal{V} \otimes \mathcal{W} \iff \mathcal{R}(A) = \mathcal{V}$ and $\mathcal{R}(B) = \mathcal{W}$ (Theorem 3.19(f)).
- $\mathcal{N}(A \otimes B) = \{0\} \iff \mathcal{N}(A) = \{0\}$ and $\mathcal{N}(B) = \{0\}$ (Theorem 3.19(g)).
- $\mathcal{N}(A) \otimes \mathcal{N}(B) \subseteq \mathcal{N}(A \otimes B) = \big(\mathcal{N}(A) \otimes \mathcal{Y}\big) + \big(\mathcal{X} \otimes \mathcal{N}(B)\big)$ (Theorem 3.19(h)).
- $\big(\mathcal{N}(A) \otimes \mathcal{Y}\big) \cup \big(\mathcal{X} \otimes \mathcal{N}(B)\big) \subseteq \mathcal{N}(A \otimes B)$ (Proposition 3.M(a)).
- $\mathcal{N}(\lambda I - A) \otimes \mathcal{N}(\mu I - B) \subseteq \mathcal{N}(\lambda\,\mu I - A \otimes B)$ (Proposition 3.M(c)).

NORMED SPACE RESULTS.

- $\mathcal{R}(A) \mathbin{\widehat{\otimes}_\alpha} \mathcal{R}(B) = \mathcal{R}(A \mathbin{\widehat{\otimes}} B)^-$ (Theorem 9.3(e)).
- $\mathcal{R}(A \mathbin{\widehat{\otimes}} B)^- = \mathcal{V} \mathbin{\widehat{\otimes}_\alpha} \mathcal{W} \iff \mathcal{R}(A) = \mathcal{V}$ and $\mathcal{R}(B) = \mathcal{W}$ (Theorem 9.3(f)).
- $\mathcal{N}(A) \mathbin{\widehat{\otimes}_\alpha} \mathcal{N}(B) \subseteq \mathcal{N}(A \mathbin{\widehat{\otimes}} B) \subseteq \big(\mathcal{N}(A) \mathbin{\widehat{\otimes}_\alpha} \mathcal{Y}\big) + \big(\mathcal{X} \mathbin{\widehat{\otimes}_\alpha} \mathcal{N}(B)\big)$ (Theorem 9.3(g)).
- $\mathcal{N}(A \mathbin{\widehat{\otimes}} B) = \{0\} \iff \mathcal{N}(A) = \{0\}$ and $\mathcal{N}(B) = \{0\}$ (Theorem 9.3(h)).
- $\mathcal{N}(A \mathbin{\widehat{\otimes}} B) = \mathcal{N}(A \otimes B)^-$ in $\mathcal{X} \mathbin{\widehat{\otimes}_\eta} \mathcal{Y}$ (Proposition 9.E(a)).
- $\mathcal{R}(A \mathbin{\widehat{\otimes}} B)$ closed in $\mathcal{X} \mathbin{\widehat{\otimes}_\eta} \mathcal{Y} \Leftrightarrow \mathcal{R}(A)$ and $\mathcal{R}(B)$ closed (Proposition 9.E(b)).
- $\mathcal{R}(A \mathbin{\widehat{\otimes}} B) = \mathcal{R}(A \otimes B)^-$ in $\mathcal{X} \mathbin{\widehat{\otimes}_\eta} \mathcal{Y}$ if the above holds (Proposition 9.E(c)).

Notes: Proposition 9.A was proved in [88, Theorem 3]. Actually, a special case of Proposition 9.A was proved in [88, Theorem 3] for the particular case of $\mathcal{V} = \mathcal{X}$ and $\mathcal{W} = \mathcal{Y}$ with these being Banach spaces, but the same argument suffices to show the extension of that result for arbitrary normed spaces $\mathcal{X}, \mathcal{Y}, \mathcal{V}, \mathcal{W}$ as stated in Proposition 9.A.

Proof of Proposition 9.B. Since $\|\cdot\|_\vee$, $\|\cdot\|_\alpha$, and $\|\cdot\|_\wedge$ are uniform crossnorms (Theorem 8.4), they may equip any tensor product space (Definition 8.2); in particular, they are uniform crossnorms on $\mathcal{X} \otimes \mathcal{Y}$ and on $\mathcal{V} \otimes \mathcal{W}$. According to Theorem 7.4, $\|\cdot\|_\vee \le \|\cdot\|_\alpha \le \|\cdot\|_\wedge$. Consider the setup in the proof of Theorem 9.1. First, take the inequality $\|\cdot\|_\vee \le \|\cdot\|_\alpha$. Since $\|F\|_\vee \le \|F\|_\alpha$ and $\|\mathbb{L}(F)\|_\vee \le \|\mathbb{L}(F)\|_\alpha$ for every $F \in \mathcal{X} \otimes \mathcal{Y}$, Proposition 5.Z ensures that

$$\inf_{F \notin \mathcal{N}(\mathbb{L})} \frac{\|\mathbb{L}(F)\|_\vee}{\|\mathbb{L}(F)\|_\alpha} \, \|\mathbb{L}\|_{[\alpha,\alpha]} \le \|\mathbb{L}\|_{[\vee,\vee]} \le \sup_{F \neq 0} \frac{\|F\|_\alpha}{\|F\|_\vee} \, \|\mathbb{L}\|_{\alpha,\alpha]}.$$

Next, consider the inequality $\|\cdot\|_\alpha \le \|\cdot\|_\wedge$. Again, since $\|F\|_\alpha \le \|F\|_\wedge$ and $\|\mathbb{L}(F)\|_\alpha \le \|\mathbb{L}(F)\|_\wedge$ for every $F \in \mathcal{X} \otimes \mathcal{Y}$, we get by Proposition 5.Z

$$\inf_{F \notin \mathcal{N}(\mathbb{L})} \frac{\|\mathbb{L}(F)\|_\alpha}{\|\mathbb{L}(F)\|_\wedge} \, \|\mathbb{L}\|_{[\wedge,\wedge]} \le \|\mathbb{L}\|_{[\alpha,\alpha]} \le \sup_{F \neq 0} \frac{\|F\|_\wedge}{\|F\|_\alpha} \, \|\mathbb{L}\|_{[\wedge,\wedge]}.$$

Just the second inequalities in the above displayed chains of inequalities are needed to prove Proposition 9.B:

$$\left(\sup_{F \neq 0} \frac{\|F\|_\alpha}{\|F\|_\vee} \right)^{-1} \|\mathbb{L}\|_{[\vee,\vee]} \le \|\mathbb{L}\|_{[\alpha,\alpha]} \le \sup_{F \neq 0} \frac{\|F\|_\wedge}{\|F\|_\alpha} \, \|\mathbb{L}\|_{[\wedge,\wedge]}. \qquad \square$$

This section focused on some aspects of the problem of transferring properties from a pair of operators to their tensor product, especially for Hilbert-space operators. As one may expect, Remark 4.11 *on multiple tensor products* still holds for n-tuples of operators. For further problems on a variety of questions related to tensor product of operators acting on a large collection of Banach spaces, see, for instance, [15, 17, 83, 85]. For extensions far beyond the scope of the present introduction, the reader is referred to [18, 76, 78].

Proof of Proposition 9.C. Recall from the Inverse Mapping Theorem (Proposition 5.B) that, in a Banach-space setting, $T \in \mathcal{B}[\mathcal{X}, \mathcal{Y}]$ is invertible if and only if it has a bounded inverse (i.e., an inverse in $\mathcal{B}[\mathcal{Y}, \mathcal{X}]$). So the proof follows the same argument as in the proof of Proposition 3.L in Section 3.4. $\square$

Proof of Proposition 9.D. The equivalence for orthogonal single tensors in (a) is trivially verified by $(\text{IP})_1$: $\langle x_\gamma \otimes y_\delta \,;\, x_{\gamma'} \otimes y_{\delta'} \rangle_\eta = \langle x_\gamma \,;\, y_\delta \rangle \, \langle x_{\gamma'} \,;\, y_{\delta'} \rangle$. Also by $(\text{N})_1$, $\|x_\gamma \otimes y_\delta\|_\eta = \|x_\gamma\| \, \|y_\delta\|$. So (b) and (c) are consequences of (a). $\square$

Proposition 9.E was proved in [41, Theorems 1.2, 1.3, 1.7] (In fact, part of the results in Proposition 9.E were proved in [41] for Banach spaces under some complementation assumption; and so the complete set of results holds in a Hilbert-space setting.) For Proposition 9.F, see, for instance, [48, Remark 5]. Proposition 9.G has been motivated by Proposition 9.F, but dismisses the

separability assumption — see, e.g., [48, Eq. (14)]. For the direct implications in Proposition 9.H — viz., if A and B are nonnegative, or self-adjoint, or unitary, or normal, or quasinormal, or subnormal, or hyponormal, then so is $A \mathbin{\widehat{\otimes}} B \in \mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}}_\eta \mathcal{Y}]$ — see, e.g., [48, Section 5]. For the normaloid case, use the spectral radius identity of Corollary 9.14, $r(A \mathbin{\widehat{\otimes}} B) = r(A)\, r(B)$, and recall that $\|A \mathbin{\widehat{\otimes}} B\| = \|A\|\, \|B\|$ by using Theorem 8.3(a) again. Thus

$$r(A) = \|A\| \ \text{ and } \ r(B) = \|B\| \ \implies \ r(A \mathbin{\widehat{\otimes}} B) = \|A \mathbin{\widehat{\otimes}} B\|.$$

For the converse, since $r(T) \le \|T\|$ for every operator T on a Banach space (cf. Proposition 5.M), if $r(A)\, r(B) = r(A \mathbin{\widehat{\otimes}} B) = \|A \mathbin{\widehat{\otimes}} B\| = \|A\|\, \|B\|$, then

$$\|A\| \le \|A\| \, \tfrac{\|B\|}{r(B)} = r(A) \le \|A\|$$

(for the nonzero case), so that $r(A) = \|A\|$; symmetrically, $r(B) = \|B\|$. For the other converses in Proposition 9.H (i.e., in the case of $A \mathbin{\widehat{\otimes}} B \in \mathcal{B}[\mathcal{X} \mathbin{\widehat{\otimes}}_\eta \mathcal{Y}]$ being nonnegative, self-adjoint, unitary, normal, quasinormal, subnormal, or hyponormal), see [90, Theorem 2.4]. There are, however, some well-known classes of operators that do not behave so nicely. One such is the class of *paranormal* operators. A normed-space operator $T \in \mathcal{B}[\mathcal{X}]$ is paranormal if $\|Tx\|^2 \le \|T^2 x\|\, \|x\|$ for every $x \in \mathcal{X}$. If $\mathcal{X}$ is a Hilbert space, then the class of paranormal operators is a class properly including the hyponormal operators and properly included in the class of normaloid operators; that is,

$$\text{Hyponormal} \subset \text{Paranormal} \subset \text{Normaloid}.$$

An example of a Hilbert-space paranormal operator A for which $A \mathbin{\widehat{\otimes}} A$ is not paranormal was given in [84, Example 6.4]. On the other hand, the transfer problem is extended to further classes of Hilbert-space operators along this line: see, for instance, [20, 49, 50, 64, 90] and the references therein. Proposition 9.I(a) was considered in [90, Theorem 2.4], and 9.I(b) in [65, Lemma 1]. For Proposition 9.J, see [96, Corollary 1]. For Propositions 9.K and 9.L, see [65, Theorems 2 and 3] (see also [20, Theorem 1]). Proposition 9.M was considered in [39, Theorem 4.2(a,f)], where the inclusion

$$\sigma_w(A \mathbin{\widehat{\otimes}} B) \subseteq \sigma_w(A) \cdot \sigma(B) \cup \sigma(A) \cdot \sigma_w(B)$$

was proved (for a follow-up article, see [40]). The equivalence in Proposition 9.N was established in [61, Corollary 6]. (For extensions of Proposition 9.N, see, e.g., [21, Theorem 1].) The example in Proposition 9.O was given in [47, Sections 2 and 3]. Proposition 9.P is a consequence of Proposition 9.N and Corollaries 9.11(b,c), 9.15(a) and 9.16 (for a detailed proof of Proposition 9.P, see [62, Theorem 1]); for other consequences of Corollary 9.16 along this line, see, e.g., [35, Theorem 7]).

If the inclusion in Proposition 9.M(b) (namely, the one displayed above) becomes an identity, that is, if A and B are such that

$$\sigma_w(A \mathbin{\widehat{\otimes}} B) = \sigma_w(A) \cdot \sigma(B) \cup \sigma(A) \cdot \sigma_w(B),$$

then it is said that the *Weyl spectral identity* holds. Thus Propositions 9.N and 9.P say: (i) *if A and B satisfy Browder's theorem, then $A \widehat{\otimes} B$ satisfies Browder's theorem if and only if the Weyl spectral identity holds*, and (ii) *if A and B are isoloid, satisfy Weyl's theorem, and the Weyl spectral identity holds, then $A \widehat{\otimes} B$ satisfies Weyl's theorem*, respectively. Also, Propositions 9.N and 9.O together ensure that *there exist operators for which the Weyl spectral identity does not hold*. For accounts on some recent research on classes of operators for which the Weyl spectral identity holds, see, for instance, [22, 63].

Proposition 9.P suggests the question: *do there exist isoloid operators satisfying Weyl's theorem for which the Weyl spectral identity fails*?

Suggested Readings

Defant and Floret [15]
Diestel, Fourie and Swart [17]
Ryan [83]
Schatten [85]

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List of Symbols

$\to$, 1
$\in$, 1
range$(\cdot)$, 1, 3
$=$, 1
$\neq$, 1
$G|_S$, 1
$\circ$, 1
I, 2, 3
$\varnothing$, 2
$\prod$, 2, 5
$\{x_\gamma\}_{\gamma\in\Gamma}$, 2
X^Γ, 2
I_n, 2, 5
$\leq$, 2
$\mathbb{N}$, 2
X^n, 2
X^{I_n}, 2
$\times$, 2, 5
$\mathbb{F}$, 2
$\mathbb{C}$, 2
$\mathbb{R}$, 2
$(\cdot+\cdot)$, 2
$(\cdot\centerdot\cdot)$, 2
$\mathcal{L}at\,(\cdot)$, 2
span S, 2
dim $\mathcal{X}$, 2
$\mathbb{N}_0$, 2
$\mathcal{L}[\mathcal{X},\mathcal{Y}]$, 3
$\mathcal{L}[\mathcal{X}]$, 3
$\mathcal{R}(\cdot)$, 3, 80
$\mathcal{N}(\cdot)$, 3, 80
kernel$\,(\cdot)$, 3
$\{0\}$, 3
O, 3
$\mathcal{X}^\sharp$, 3
$\langle x\,;f\rangle$, 3
$L^\sharp$, 3
$\cong$, 3, 4, 79, 82
$\widetilde{\mathcal{X}}$, 3, 81
$\widetilde{L}$, 4
$\widetilde{x}$, 4, 85
$\hookrightarrow$, 4
$\cap$, 5
codim $\mathcal{M}$, 5
$\oplus$, 5
$\bigoplus$, 5
$\mathcal{X}^n$, 5
$\mathbb{F}^n(\mathcal{X})$, 5, 168, 229
$\mathcal{X}^{I_n}$, 5
F^n, 5, 53, 73
$\Longleftrightarrow$, 5
$\#$, 6
$\setminus$, 6
$<$, 6
∞, 6
$\cup$, 6
$\notin$, 6
χ_S, 6
$\approx$, 6
$\sum$, 6
$\widehat{L}$, 7
$\bigcup$, 7

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C. S. Kubrusly, *Bilinear Maps and Tensor Products in Operator Theory*, Universitext, https://doi.org/10.1007/978-3-031-34093-2

$\bigvee$, 7
$\mathcal{D}(\cdot)$, 7
$\sim$, 8
$/$, 8, 9
$[\cdot]$, 8, 9
$\widehat{\ }$, 8
$\equiv$, 9
mod $\mathcal{M}$, 9
π, 10, 93
$\Longrightarrow$, 11
$\not\Longrightarrow$, 11
$\ell^{\#}$, 15
$\mathcal{P}[0,1]$, 15
$\subset$, 18
$\subseteq$, 18
$\subsetneqq$, 18
$\not\subseteq$, 18
ϕ^{y}, 19
ϕ_{x}, 19
$b[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$, 19
$b[\mathcal{X}\oplus\mathcal{Y},\mathcal{Z}]$, 21
$\langle\cdot\,;\cdot\rangle$, 22, 106, 171
$\|\cdot\|^2$, 22, 106
$\mathit{\Pi}$, 27, 144
$\mathit{\Pi}_X$, 28
$\mathit{\Pi}_Y$, 28
$\ominus$, 29, 106
$\mapsto$, 29
$\star$, 30
$\bar{\ }$, 32
$(\mathcal{T},\theta)$, 35
$\otimes$, 36, 38, 45
$\top_{E,D}$, 40
$\boxplus$, 52
$\not\cong$, 52
$(\mathcal{S}/\mathcal{M},\theta)$, 59
$(\operatorname{span}\top_{\mathcal{X},\mathcal{Y},\mathcal{F}},\theta)$, 63
$\top_{\mathcal{X},\mathcal{Y},F}$, 65
$(\operatorname{span}\top'_{\mathcal{X},\mathcal{Y}},\theta')$, 66
$(\operatorname{span}\top''_{\mathcal{X},\mathcal{Y}},\theta'')$, 67
$(\operatorname{span}\top'''_{\mathcal{X},\mathcal{Y}},\theta''')$, 68
$\bigotimes$, 72
$\circledast$, 73, 230
$\mathbb{F}^m(\mathbb{F}^n)$, 73
$\mathbb{F}_{m,n}$, 75
$\mathcal{P}_n[0,1]$, 75
$p(\cdot)$, 76, 111
$\sigma(\cdot)$, 76, 104, 218
$\operatorname{tr}(\cdot)$, 76, 172
$\det(\cdot)$, 76
$\|\cdot\|$, 79
$(\mathcal{X},\|\cdot\|)$, 79
$\mathcal{M}^{-}$, 79
$d_p(\cdot,\cdot)$, 80, 115
$d_\infty(\cdot,\cdot)$, 80, 115
$\|\cdot\|_p$, 80
$\|\cdot\|_\infty$, 80, 166, 169
$\mathcal{B}[\mathcal{X},\mathcal{Y}]$, 80
$\|\cdot\|_{\mathcal{B}[\mathcal{X},\mathcal{Y}]}$, 80
$\mathcal{B}[\mathcal{X}]$, 80
$\mathcal{X}^*$, 81
$|\cdot|$, 81, 165
T^*, 81, 108
$\widetilde{T}$, 81, 155
$\widehat{\mathcal{X}}$, 82
$\widehat{x}$, 82
$\widehat{T}$, 83, 155
$\overline{\subseteq}$, 84
$\widehat{f}$, 88, 158
$\|\cdot\|_2$, 88, 224
$\mathit{\Gamma}_R[\mathcal{X}]$, 92
$\mathit{\Gamma}_N[\mathcal{X}]$, 92
$\mathit{\Gamma}[\mathcal{X}]$, 92
$|\!|\!|\cdot|\!|\!|$, 93
G_L, 101, 168
$\xrightarrow{u}$, 102
$\xrightarrow{s}$, 102
$\mathcal{B}_0[\mathcal{X},\mathcal{Y}]$, 103
$\mathcal{B}_\infty[\mathcal{X},\mathcal{Y}]$, 103
$\mathcal{B}_0[\mathcal{X}]$, 103
$\mathcal{B}_\infty[\mathcal{X}]$, 103
$\rho(\cdot)$, 104
$\sigma_P(\cdot)$, 104, 218
$\sigma_C(\cdot)$, 105, 218
$\sigma_R(\cdot)$, 105, 218
$r(\cdot)$, 105
$\perp$, 106, 229
T^{**}, 108
$\mathit{\Lambda}^*$, 109
ℓ^p_+, 112
ℓ^∞_+, 112
$\boldsymbol{b}[\mathcal{X}\times\mathcal{Y},\mathcal{Z}]$, 113
$\hat{\phi}$, 118, 119, 158
$L^1[0,1]$, 118
c_0, 119, 123, 168, 169
L^1, 123, 168
$\widetilde{\phi}$, 125
G_ϕ, 129

$\mathcal{X}^{**}$, 130, 137
ℓ_+^2, 132, 230
ϕ^*, 133
$\|\cdot\|_\alpha$, 135
$\otimes_\alpha$, 135
$\|\cdot\|_{*\alpha}$, 135
$\otimes_{*\alpha}$, 137
$\boldsymbol{B}_{\mathcal{X}}$, 138, 167
$\boldsymbol{B}_{\mathcal{X}^{**}}$, 138
$\|\cdot\|_\vee$, 139
$\|\cdot\|_\wedge$, 139
$\|\cdot\|_{*\vee}$, 140
$\|\cdot\|_{*\wedge}$, 140
$\otimes_\vee$, 141
$\otimes_\wedge$, 141
$\boldsymbol{B}_\vee$, 141
$\boldsymbol{B}_\alpha$, 141
$\boldsymbol{B}_\wedge$, 141
$\widehat{\otimes}_\alpha$, 141
$\widetilde{\otimes}_\alpha$, 142
$\widehat{\otimes}_\vee$, 142
$\widehat{\otimes}_\wedge$, 142
$\|\cdot\|_{t_\alpha}$, 144
${}^tF$, 144
${}^{tt}F$, 144
$\|\cdot\|_{{}^{tt}\alpha}$, 144
$\otimes_{t_\alpha}$, 144
$\|\cdot\|_{t_\wedge}$, 145
$\|\cdot\|_{t_\vee}$, 145
$\widehat{\otimes}$, 148, 182, 208
$\widehat{\otimes}_{*\alpha}$, 153
$\widehat{\otimes}_{*\wedge}$, 154
$\otimes_{*\wedge}$, 154
$\overline{T}$, 155
$\widetilde{f}$, 158
$\overline{f}$, 158
$\boldsymbol{B}_{\mathcal{X}^*}$, 164
$C[X,\mathbb{F}]$, 165
$\mathcal{X}_{\mathcal{T}}$, 165
$\boldsymbol{X}$, 165
ℓ_+^1, 168, 173
$\ell_+^1(\mathcal{Z})$, 168
$\ell_+^1(\mathbb{F})$, 168
$L^1(\mathcal{Z})$, 168
$L^1(\mu,\mathcal{Z})$, 168
$\|\cdot\|_1$, 168, 172
$L^1(\mu)$, 168
$c_0(\mathcal{Z})$, 168
$c_0(\mathbb{F})$, 168
$C(K)$, 169
$C(K,\mathcal{Z})$, 169
$\|\cdot\|_N$, 169, 170, 199
$\boldsymbol{b}_N[\mathcal{X}\times\mathcal{Y},\mathbb{F}]$, 170, 199
$\mathcal{B}_N[\mathcal{X},\mathcal{Y}]$, 170, 199
$\mathcal{B}_N[\mathcal{X}]$, 171
$|T|$, 172
$\mathcal{B}_1[\mathcal{X}]$, 172
$\|\cdot\|_{\widehat{\alpha}}$, 188
$\otimes_{\widehat{\alpha}}$, 188
$\|\cdot\|_{*\widehat{\alpha}}$, 189
$\|\cdot\|_{\alpha^*}$, 192, 197
$\|\cdot\|_{\alpha^\sharp}$, 193
$\otimes_{\alpha^\sharp}$, 193
$\|\cdot\|_{\sharp\alpha}$, 193
$\mathcal{M}^{\sharp\sharp}$, 193
$\|\cdot\|_{(\alpha^\sharp)^\sharp}$, 194
$\|\cdot\|_{\alpha^{\sharp\sharp}}$, 194
$\|\cdot\|_{({}^t\alpha)^\sharp}$, 194
$\|\cdot\|_{{}^t(\alpha^\sharp)}$, 194
$\|\cdot\|_{\vee^\sharp}$, 194
$\|\cdot\|_{\wedge^\sharp}$, 194
$\otimes_{\alpha^*}$, 197
$\|\cdot\|_{\wedge^*}$, 200
$\|\cdot\|_{\vee^*}$, 200
$\|\cdot\|_{[\alpha,\alpha]}$, 203
$\otimes_{[\alpha,\alpha]}$, 204
$\|\cdot\|_{*[\alpha,\alpha]}$, 204
$\otimes_{[\alpha,\alpha]}$, 204
$\|\cdot\|_{[\vee,\vee]}$, 206
$\|\cdot\|_{[\wedge,\wedge]}$, 206
$\langle\cdot\,;\cdot\rangle_\eta$, 211
$\|\cdot\|_\eta$, 211
$\otimes_\eta$, 212
$\|\cdot\|_{*\eta}$, 214
$\otimes_{*\eta}$, 214
$\langle\cdot\,;\cdot\rangle_\eta$, 216
$\|\cdot\|_*$, 216
$\widehat{\otimes}_\eta$, 216
$\sigma_{PF}(\cdot)$, 219
$\sigma_{P_1}(\cdot)$, 222
$\sigma_{P_2}(\cdot)$, 222
$\sigma_{P_3}(\cdot)$, 222
$\sigma_{P_4}(\cdot)$, 222
$\sigma_{R_1}(\cdot)$, 222
$\sigma_{R_2}(\cdot)$, 222
$\sigma_{AP}(\cdot)$, 222
$\mathcal{B}_2[\mathcal{X}]$, 224
$\sigma_{\text{iso}}(\cdot)$, 225

$\sigma_{\mathrm{acc}}(\,\cdot\,)$, 225
$\pi_{00}(\,\cdot\,)$, 227
$\perp^1$, 229
$\perp_\eta$, 229
$\perp^1_\eta$, 229
$\ell^2_+(\mathbb{C})$, 230
$\ell^2_+(\mathcal{Y})$, 230
$\xrightarrow{w}$, 233
$\{n\}_{n\geq 0}$, 233
$\sigma_0(\cdot)$, 234
$\sigma_w(\cdot)$, 234
$\sigma_b(\cdot)$, 234

Index

A
accumulation points of spectrum, 225
adjoint, 3, 81, 108, 133
adjoint norm, 194
adjoint of a bounded bilinear map, 133
adjoint of a bounded linear transformation, 81, 108
adjoint of a linear transformation, 3
affine space, 9
algebraic adjoint, 3
algebraic complement, 4
algebraic concepts, 1
algebraic direct sum, 5
algebraic disjoint, 4
algebraic dual, 3
algebraic embedding, 4
algebraic isomorphism, 3
algebraic tensor product, 35, 58, 77
approximable transformation, 104
approximate point spectrum, 222
approximation property, 104, 190
automorphism, 3

B
Banach–Alaoglu Theorem, 165
Banach algebra, 80
Banach space, 79
Banach–Steinhaus Theorem, 102, 129
bidual, 130, 132
bijective function, 1
bilinear form, 19
bilinear functional, 19
bilinear map, 19
Bochner integrable function, 168
Borel σ-algebra, 164
bounded approximation property, 190
bounded below transformation, 101
bounded bilinear extension, 119, 124
bounded bilinear map, 113
bounded increments, 233
bounded inverse, 101
Bounded Inverse Theorem, 101
bounded linear extension, 82, 83, 90
bounded linear functional, 81
bounded linear transformation, 80, 101
bounded sequence, 103, 112
bounded tensor product transformation, 179
bounded transformation, 80, 101
boundedness, 103
Browder spectrum, 234
Browder's theorem, 234

C
cardinality, 6
Cartesian product, 2
characteristic function, 6
classical partition, 105
Closed Graph Theorem, 102, 129, 168
closed unit ball, 138, 141, 164
closure, 79
codimension, 5

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codomain, 1
compact linear transformation, 103
compact set, 165
complementary linear manifold, 4
complementary projection, 3
complementary subspace, 89
complementation, 89
complemented normed space, 89
complemented subspace, 89
complete lattice, 197
complete normed space, 79, 89
completion, 82
complex Banach space, 104
complex conjugate, 32
complex field, 2
complex Hilbert space, 106
complex polarization identity, 32
composition, 1, 21
composition map, 30
composition of bilinear map, 21
congruent, 9
contraction, 80
continuous bilinear map, 115, 130
continuous linear transformation, 80
continuous map, 101, 129
continuous projection, 90
continuous spectrum, 105
contragradient norm, 194
convolution map, 30
coset, 9

D
decomposable element, 36, 59
decomposition, 16
determinant, 76
dimension, 2, 14–18
direct product, 54
direct sum of linear spaces, 5
direct sum of linear transformations, 16
domain, 1
dual norm, 192, 197
dual space, 3, 81
dual tensor product space, 42, 47

E
eigenvalue, 76, 104
eigenvalue of finite multiplicity, 219
elementary tensor, 36
embedding, 4
embedding of tensor product, 68
endomorphism, 3
epimorphism, 3
equivalence class, 8, 9
equivalence relation, 3, 8, 9
equivalent transformation, 81
evaluation map, 30
extension, 1
extension by continuity, 83, 87, 119, 122
extension of bilinear map, 23
extension of bounded bilinear map, 119
extension of bounded linear transformation, 82
extension of linear transformation, 7, 87
extension operation, 8
extension over completion, 83, 87, 119, 130
extension property, 89
external direct sum, 5

F
finite-dimensional space, 2, 42
finite-dimensional uniform crossnorm, 180
finite-rank linear transformation, 103
finitely generated uniform crossnorm, 185
First Isomorphism Theorem, 12, 17, 95
formal linear combination, 5
formal product, 54
Fourier coefficients, 107
Fourier series expansion, 107
Fourier Series Theorem, 107
free linear space, 6
full algebraic direct sum, 5
$\mathbb{F}$-valued function, 3
$\mathbb{F}$-valued measure, 164

G
Gelfand–Beurling Formula, 105
Goldstine Theorem, 138
graph, 101, 129, 168
graph of a bilinear map, 128
graph of a linear transformation, 101

Grothendieck Theorem, 148

H
Hahn–Banach extension, 88
Hahn–Banach Theorem, 88
Hamel basis, 2, 14
Hausdorff space, 165
Hermitian symmetric sesquilinear form, 32
Hilbert–Schmidt class, 224
Hilbert–Schmidt operator, 224
Hilbert–Schmidt tensor norm, 216, 224
Hilbert–Schmidt tensor product, 216, 224
Hilbert space, 66, 79, 106, 171, 210
Hilbert-space adjoint, 108
homeomorphism, 79
homomorphism, 3
hyponormal operator, 231

I
ideal, 171, 172
idempotent function, 2
identity transformation, 3
image, 1
image space, 3
induced uniform norm, 80, 113
infinite-dimensional space, 2
injective function, 1
injective norm, 141, 145, 182, 185, 186
injective tensor product, 142, 159
injective uniform crossnorm, 183
inner product, 22, 33, 106
inner product space, 106
internal direct sum, 5, 106
interpretation, 57
inverse, 104
Inverse Mapping Theorem, 101
invertible bounded linear transformation, 101
invertible function, 1
invertible isometry, 80, 231
involution, 3
irregular linear manifold, 44
isolated points of the spectrum, 225
isoloid operator, 226
isometric embedding, 83
isometric isomorphism, 80
isometrically equivalent transformations, 81
isometrically isomorphic spaces, 80
isometry, 80, 231
isomorphic equivalence, 3
isomorphic linear spaces, 3
isomorphic linear transformations, 3
isomorphically equivalent transformations, 4
isomorphism, 3

K
kernel, 3, 235
Kronecker product, 73
Kronecker product of matrices, 75
Kronecker product of operators, 230
Kronecker product space, 74

L
ℓ^2_+-norm, 87
lattice of linear manifolds, 2
lattice of tensor norm, 196
left inverse, 104
linear-bilinear approach, 64
linear equivalence relation, 9
linear extension, 6, 87
linear form, 3
linear functional, 3
linear isometry, 80
linear manifold, 2, 79, 142
linear space, 2
linear-space embedding, 4
linear-space isomorphism, 3
linear subspace, 2
linear transformation, 2
linear variety, 9
linearization of bilinear maps, 39, 63
linearly independent, 14

M
matrix, 16
metric approximation property, 190, 199
minimal projection, 107
modulo, 9
monomorphism, 3
monotone Schauder basis, 190
morphism, 3
multiple tensor product, 72, 236

N
n-tuples, 72, 73, 236
natural bilinear map, 36
natural embedding, 132, 138
natural identification, 6
natural map, 36
natural permutation isomorphism, 50, 144
natural quotient isomorphism, 13
natural quotient map, 94
Neumann expansion, 102
nonnegative operator, 231
norm, 79
norm on tensor product, 135
norm squared, 22
normal operator, 231
normaloid operator, 231
normed quotient space, 93
normed space, 79
normed-space adjoint, 81, 108
nuclear bilinear form, 169
nuclear norm, 171
nuclear operator, 171
nuclear representation, 169, 170
nuclear transformation, 170
null space, 3
null transformation, 3

O
one-complemented, 107
one-to-one, 1
open map, 101
Open Mapping Theorem, 101
operator, 80
operator norm, 179
operator norm property, 179
ordinary sum, 4
origin, 3
orthogonal complement, 106
orthogonal decomposition, 106
orthogonal direct sum, 106
orthogonal linear manifolds, 106
orthogonal projection, 106
orthogonal set, 107
orthogonal subspace, 106
orthogonal vectors, 106
orthonormal basis, 107
orthonormal sequence, 176
orthonormal set, 107

P
paranormal operator, 237
permutation map, 27, 50
Pisier space, 173
plain uniform crossnorm, 180
point spectrum, 105
pointwise convergence, 102, 129
polar decomposition, 176, 177
polarization identity, 32, 133
polynomial, 15, 75, 105, 113
positive quadratic form, 33
power bounded operator, 103
power incremented operator, 233
power sequence, 103
product metric, 80, 115
product topology, 115
projection, 3
Projection Theorem, 106
projective norm, 141, 145, 182, 185, 186
projective tensor product, 142, 146
projective uniform crossnorm, 183
p-summable sequence, 112
Pythagorean Theorem, 106, 215

Q
quadratic form, 22, 32, 106
quasinormal operator, 231
quotient map, 9
quotient norm, 93, 94
quotient of a space, 96
quotient space, 8, 9, 93
quotient space approach, 61
quotient topology, 100
quotient transformation, 96, 183

R
range, 1, 3, 235
rank and nullity identity, 14
real field, 2
real Hilbert space, 106, 132
real inner product space, 22
real polarization identity, 32
real-valued function, 30
reasonable crossnorm, 135
reflexive relation, 3
reflexivity, 9
regular linear manifold, 44

regular subspace, 142
residual spectrum, 105
resolvent set, 104
restriction, 1
restriction of bilinear map, 20
restriction of linear transformation, 3
Riesz representation, 107
Riesz Representation Theorem, 107, 165

S
scalar, 9
scalar field, 9
scalar multiplication, 2, 128
scalar-valued bilinear map, 19
scalar-valued function, 5
Schauder basis, 103, 200
Schwartz inequality, 106, 213
second dual, 130, 132
Second Isomorphism Theorem, 17
self-adjoint operator, 231
separable space, 103
separable Hilbert space, 107
sequentially continuous function, 129
sesquilinear form, 32
set product, 76, 217
similar linear transformations, 4
simple tensor, 36
single tensor, 36, 59, 61
single tensor in Hilbert space, 66
single tensor product, 36
span, 2
spectral formula, 221
Spectral Mapping Theorem for Polynomials, 105
spectral radius, 105, 224
spectral radius identity, 224
spectrum, 76, 104
square-summable sequences, 132
square ordering, 200
standard norms, 80
strong convergence, 102, 233
strongly continuous functional, 130
strongly convergent sequence, 102, 233
strongly stable operator, 103, 233
subnormal operator, 231
subspace, 79
sum, 4
sup-norm, 166, 169
surjective function, 1
surjective isometry, 80, 231
symmetric bilinear form, 32
symmetric reasonable crossnorms, 145
symmetric relation, 3
symmetry, 9

T
tensor basis, 40, 200
tensor norm, 185
tensor product, 35
tensor product of bounded linear transformations, 179, 203
tensor product of linear spaces, 35
tensor product of linear transformations, 44, 45
tensor product operator, 218
tensor product space, 35
tensor product transformation, 45, 179, 203
tensor sum of transformations, 52
Third Isomorphism Theorem, 17
Three-Space Property, 94
topological isomorphism, 79
topological quotient map, 100
topological vector space, 165
topologically isomorphic spaces, 79
topology of Cartesian product, 80
topology of direct sum, 80
total variation, 165
trace, 76, 172
trace-class, 172, 224
trace-norm, 172
transitive relation, 3
transitivity, 9
translation, 9
transpose, 144
transpose norm, 144, 186
Tychonoff Theorem, 165

U
uniform continuity, 87
uniform convergence, 102, 233
uniform crossnorm, 179, 180
uniform crossnorm on finite-dimensional spaces, 180
uniform stability, 105
uniformly convergent sequence, 102, 233

uniformly stable operator, 103, 233
unilateral shift, 178, 219, 232, 234
unit ball, 97, 138, 141, 144, 165, 167
unital complex Banach algebra, 104
unitary equivalence, 106
unitary operator, 231
unitary transformation, 106
unitarily equivalent Hilbert spaces, 231
unitarily equivalent operators, 231
unitarily equivalent spaces, 106
unitarily equivalent transformations, 230
universal mapping principle, 151
universal mapping property, 151
universal property, 11, 95

V
vector addition, 2

W
weak convergence, 233
weakly continuous functional, 130
weakly convergent sequence, 233
weakly stable operator, 233
weak* compact, 165
weak* topology, 164
Weierstrass (little) Theorem, 165
Weyl spectrum, 234
Weyl spectral identity, 238
Weyl's theorem, 234

X
x-section, 19

Z
y-section, 19
$\mathcal{Y}$-valued functions, 2

Z
Zorn's Lemma, 7, 24
$\mathcal{Z}$-valued bilinear map, 20
$\mathcal{Z}$-valued functions, 20